(b) Measure the clearance aft of the actuator rod end [56].

NOTE: A larger or smaller clearance is permitted at other locations along the flight spoiler [54].

SUBTASK 27-61-00-860-031

(2) If the clearances (Dimension X) between the spoiler panels and flaps are not in the limits, adjust the clearance:

(a) Make a record of the clearance.

(b) Lift the flight spoiler [54] and extend the flaps to get access to the flight spoiler actuator [53]:

1) Do this task: Apply Pressure to the Speed Brake Hydraulic Systems A and B, TASK 27-62-00-800-801.

2) Do this task: Extend the Trailing Edge Flaps, TASK 27-51-00-860-803.

3) Move the speed brake lever to the UP position to lift the flight spoilers [54].

4) Do this task: Depressurization of the Speed Brake Hydraulic Systems A and B, TASK 27-62-00-800-802.

5) Move the speed brake lever to the DOWN position.

NOTE: This will release the hydraulic pressure from the spoiler actuators.

6) Move the speed brake lever to the UP position.

NOTE: The speedbrake lever is set to the UP position as a safety precaution.

(c) Loosen the check nut [55] on the actuator rod end [56].

(d) Turn the piston rod on the flight spoiler actuator [53] to adjust the clearance between the flight spoilers [54] and flaps.

NOTE: Turning the piston rod by one serration of the locking device will change the spoiler gap by approximately 0.006 in. (0.152 mm).

(e) Tighten the check nut [55] to 100 in-lb (11.3 N·m) - 200 in-lb (22.6 N·m) temporarily.

(f) Do a check of the adjustment:

1) Do this task: Apply Pressure to the Speed Brake Hydraulic Systems A and B, TASK 27-62-00-800-801.

2) Do this task: Retract the Trailing Edge Flaps, TASK 27-51-00-860-804.

3) Slowly move the speed brake lever to the DOWN position.

4) Make sure the clearances between the flight spoilers [54] and flaps are within the tolerances.

a) If the clearances do not meet the values shown in Table 501, repeat the adjustment steps above.

5) Turn the control wheel fully in the clockwise direction.

a) Make sure that flight spoilers [54] No. 2, 3, 4, and 5 do not move down.

b) If the spoilers move down, go back to the start of the task.

NOTE: You do not need to do a check of the fit of the rig pins a second time.

6) Turn the control wheel back to neutral to retract the flight spoilers [54].

7) Turn the control wheel fully in the counterclockwise direction.

a) Make sure that flight spoilers [54] No. 8, 9, 10, and 11 do not move down.

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b) If the flight spoilers [54] move down, go back to the start of the task.

NOTE: You do not need to do a check of the fit of the rig pins a second time.

(g) Do this task: Extend the Trailing Edge Flaps, TASK 27-51-00-860-803.

(h) Move the speed brake lever to the full up position to lift the flight spoilers [54].

(i) Clean all grease from the piston rod and actuator rod end [56] threads with solvent, B00062, or solvent, B00184.

(j) Apply a layer of the corrosion preventive D5026NS or ZC-026 compound, G50171, to the threads of the actuator rod end [56].

(k) Tighten the check nut [55] to 275 ±25 in-lb (31.1 ±2.9 N·m) .

(l) Install the MS20995NC32 lockwire, G01912, on the check nut [55] with double twist method (TASK 20-10-44-400-801).

(m) Fully seal the check nut [55] with sealant, A00247.

SUBTASK 27-61-00-820-006

(3) Do this task: Flight Spoiler System Test, TASK 27-61-00-730-801.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 27-61-00-860-032

(1) If it is necessary, move the speed brake lever to the DOWN position to lower the spoilers.

SUBTASK 27-61-00-860-033

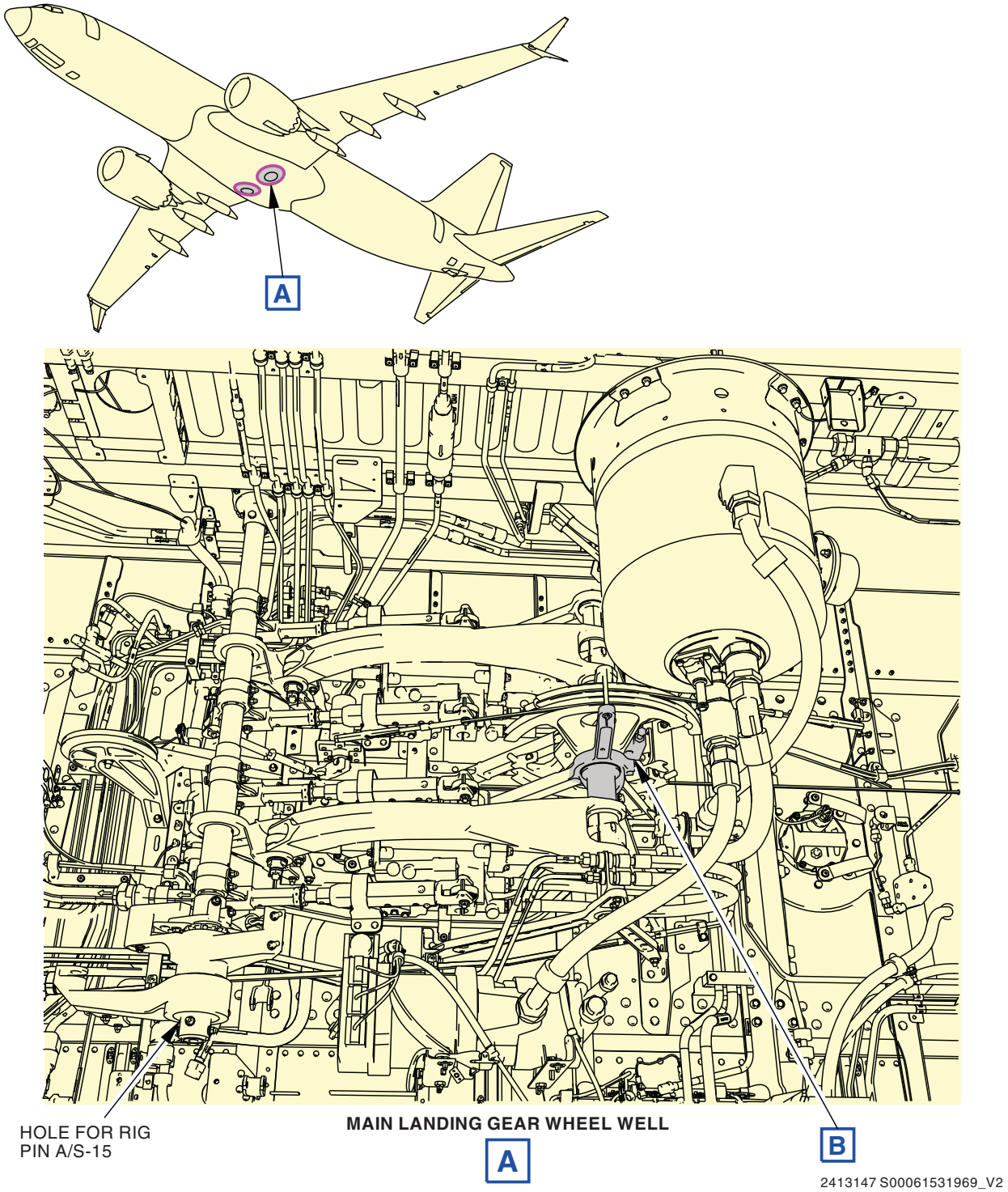
(2) Do this task: Retract the Trailing Edge Flaps, TASK 27-51-00-860-804.

SUBTASK 27-61-00-860-035

(3) Do this task: Shut Down the Speed Brake Hydraulic Systems A and B, TASK 27-62-00-840-801.

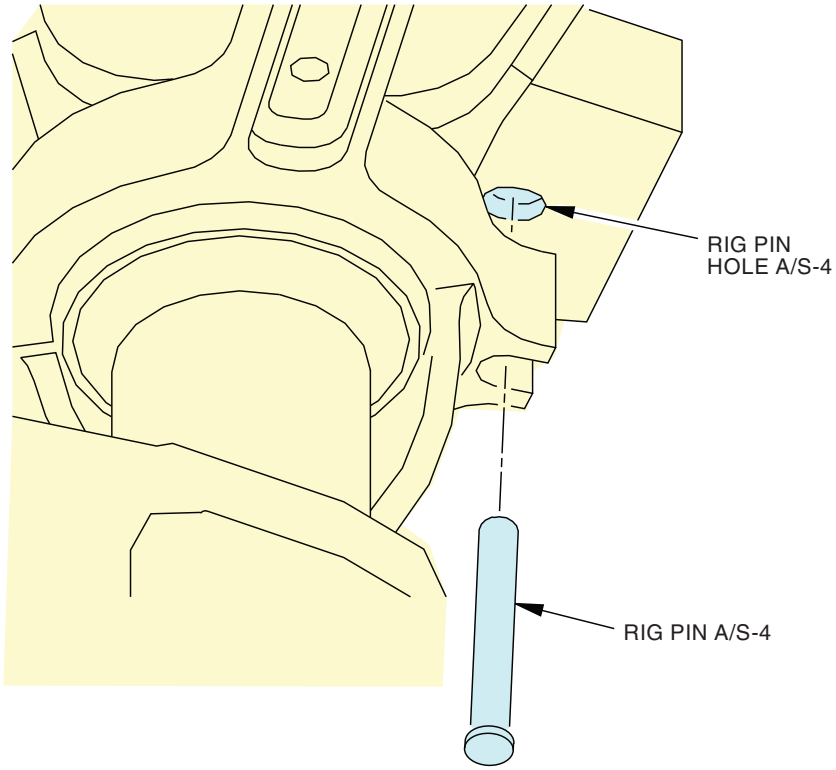
——— END OF TASK ———

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Flight Spoiler Adjustment - Rig Pin Installation
Figure 501/27-61-00-990-806 (Sheet 1 of 2)

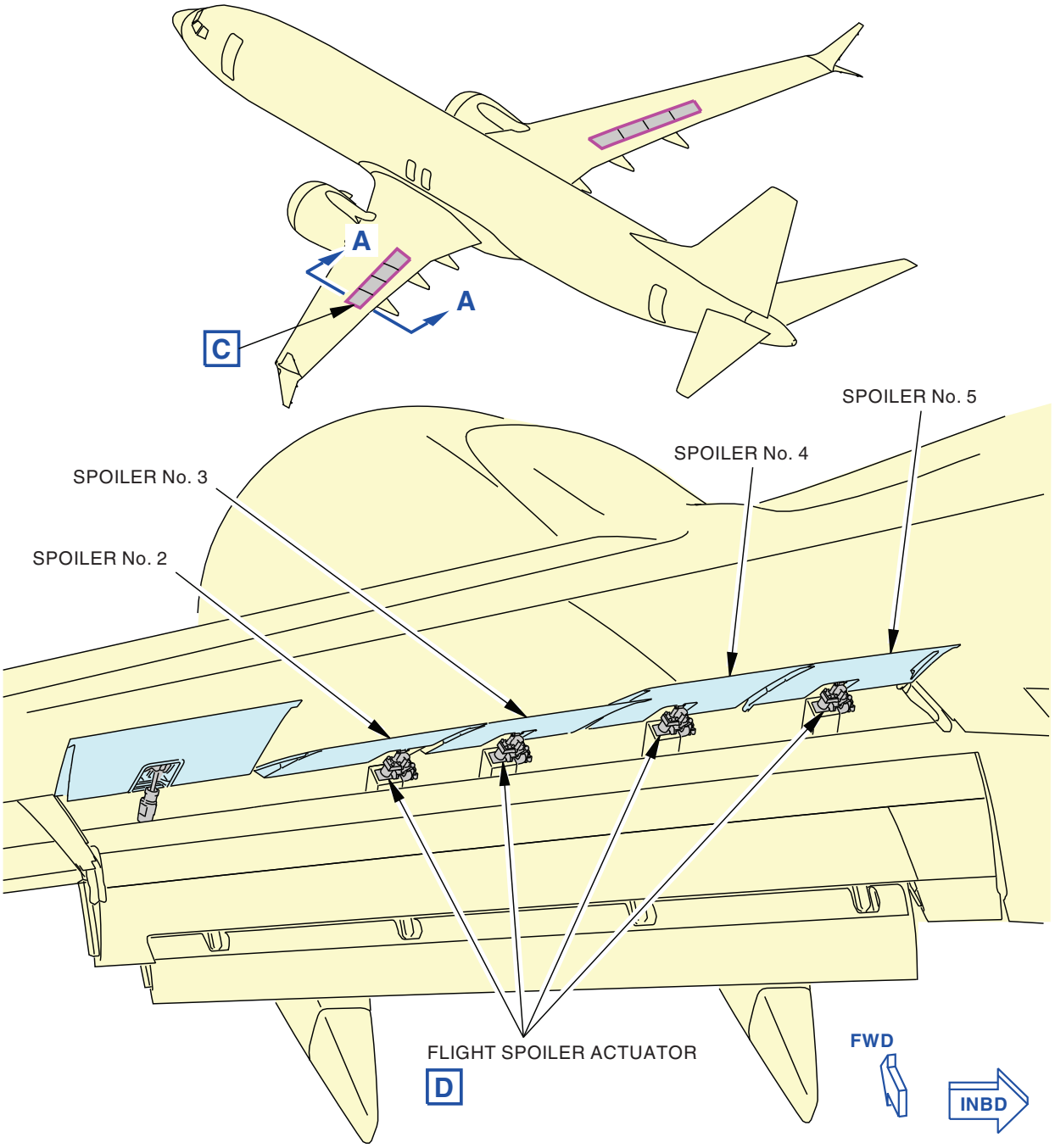
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2413148 S00061531970_V2

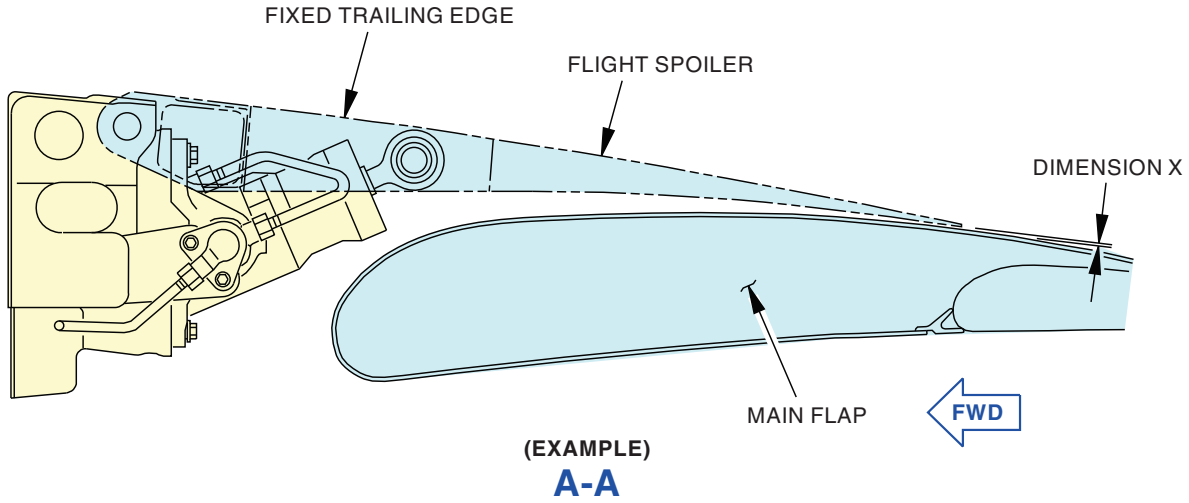
Flight Spoiler Adjustment - Rig Pin Installation
Figure 501/27-61-00-990-806 (Sheet 2 of 2)

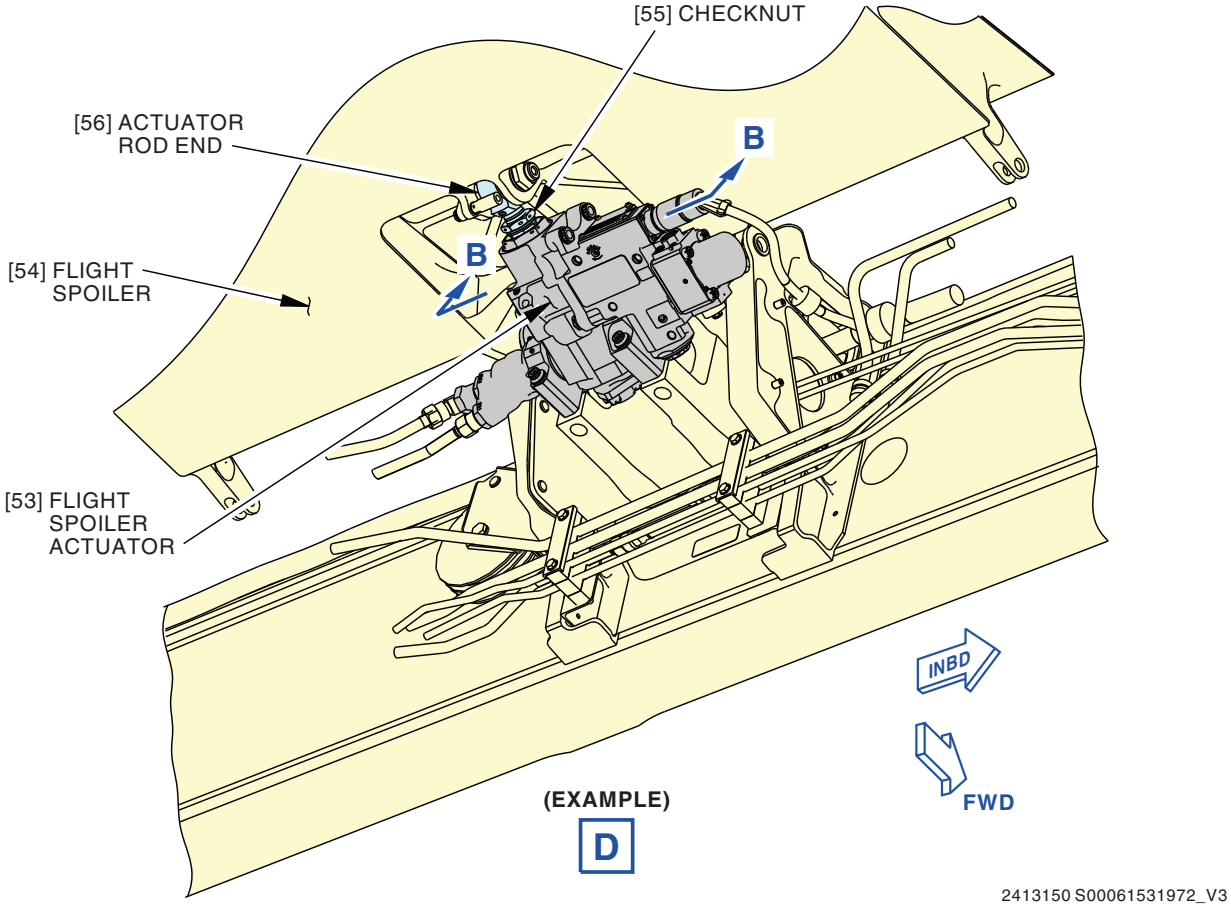
ECCN 9E991 BOEING PROPRIETARY - See title page for details



Flight Spoiler Adjustment
Figure 502/27-61-00-990-812 (Sheet 1 of 3)

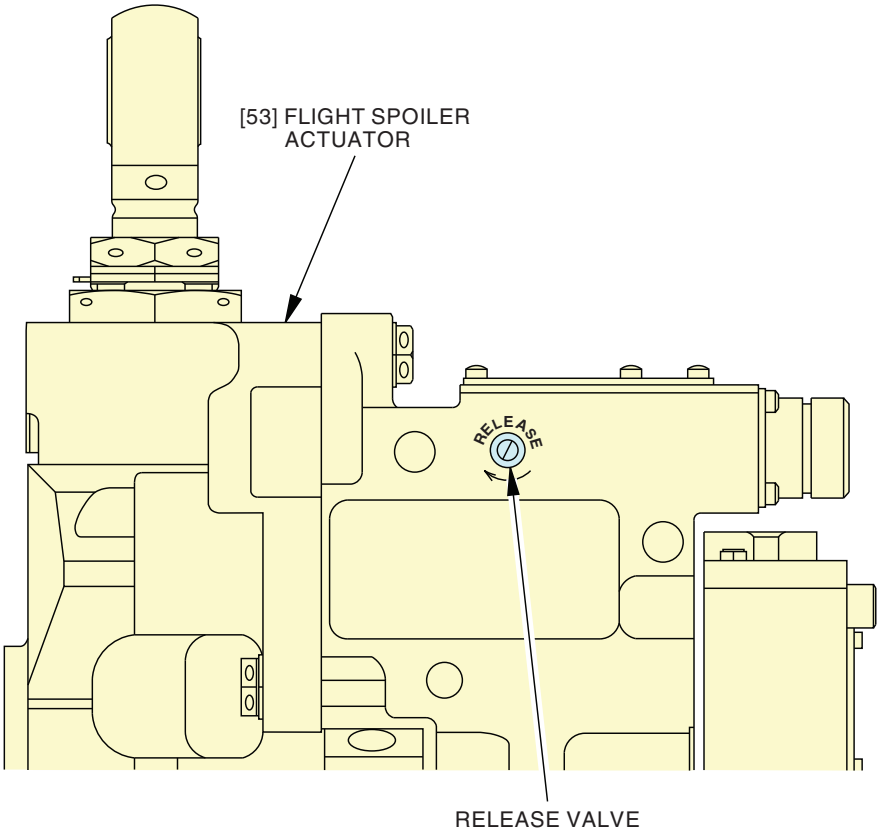
ECCN 9E991 BOEING PROPRIETARY - See title page for details





Flight Spoiler Adjustment
Figure 502/27-61-00-990-812 (Sheet 2 of 3)

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B-B

2485111 S0000584770_V1

Flight Spoiler Adjustment
Figure 502/27-61-00-990-812 (Sheet 3 of 3)

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TASK 27-61-00-730-801

4. Flight Spoiler System Test

A. References

Reference	Title
27-11-00-990-803	Figure: Rig Pin Location (P/B 501)
27-61-00-800-801	Apply Pressure to the Spoiler Hydraulic Systems A and B (P/B 201)
27-61-00-840-801	Shut Down the Spoiler Hydraulic Systems A and B (P/B 201)
27-61-45-730-805	Flight Spoiler Test (P/B 501)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1691	Protractor - Digital, Vertical or Horizontal Part #: 21465A82 Supplier: 3A054 Part #: 31-038-3 (Model Pro 360) Supplier: 1X822 Part #: 31-040-9 (Model Pro 3600) Supplier: 1X822 Part #: 950-317 (Model Pro 360) Supplier: 01EP7 Part #: 950-318 (Model Pro 3600) Supplier: 01EP7 Part #: KS5549 Supplier: 75245 Part #: KS6005 Supplier: 75245
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205
SPL-1670	Mount - Control Wheel Protractor Part #: C27060-22 Supplier: 81205 Part #: F72790 Supplier: 81205 Opt Part #: C27060-1 Supplier: 81205
SPL-1680	Protractor Assembly - Control Column Part #: A27021-29 Supplier: 81205 Part #: G76002-42 Supplier: 81205 Opt Part #: 4MIT65B80307-1 Supplier: 81205 Opt Part #: F52485-500 Supplier: 81205 Opt Part #: G76002-19 Supplier: 81205 Opt Part #: G76002-37 Supplier: 81205

C. Location Zones

Zone	Area
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50
562	Left Wing - Spoiler No. 5
563	Left Wing - Spoiler No. 4
564	Left Wing - Spoiler No. 3
565	Left Wing - Spoiler No. 2
662	Right Wing - Spoiler No. 8
663	Right Wing - Spoiler No. 9
664	Right Wing - Spoiler No. 10
665	Right Wing - Spoiler No. 11

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D. Prepare for the Test

NOTE: This test may be done for all actuators or for one or more symmetrical pairs (2 & 11, 3 & 10, 4 & 9, or 5 & 8). If a single actuator or panel was replaced, do the test for both actuators in the symmetrical pair. It is not necessary to make measurements or observations for the actuators not being tested.

SUBTASK 27-61-00-480-005

(1) Make sure that the rig pin A/S-15 from the rigging pin kit, SPL-1585, fits freely in the aileron trim actuator located in the Main Landing Gear wheel well (Figure 27-11-00-990-803).

(a) Remove the rig pin A/S-15 from the aileron trim actuator.

SUBTASK 27-61-00-730-001

(2) Do this task: Flight Spoiler Test, TASK 27-61-45-730-805.

SUBTASK 27-61-00-860-037

(3) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	9	C00440	FLIGHT CONTROL AUTO SPEED BRAKE
B	11	C04040	GROUND SPOILER CONTROL
B	12	C00204	FLIGHT CONTROL SHUTOFF VALVES SPOILER

WARNING

KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT SPOILERS. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU OPEN/CLOSE THESE CIRCUIT BREAKERS. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(4) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
C	10	C04032	SPOILER CONTROL ELEC PWR A
C	12	C04033	SPOILER CONTROL ELEC PWR B

SUBTASK 27-61-00-860-038

(5) Make sure that the speedbrake lever is in the DOWN position.

SUBTASK 27-61-00-020-015

(6) Remove the map holder assembly from the captain's control wheel.

SUBTASK 27-61-00-820-007

(7) Install the mount, SPL-1670, and the protractor, SPL-1680, on the captain's aileron control wheel.

NOTE: As an alternative to the protractor, SPL-1680, the CAPT CTRL WHL or SCE FO CTRL WHL can be used as readings for the Control Wheel position throughout this test. They are located on the FLIGHT CONTROLS page from the MAINT DATA PGS on the forward display.

SUBTASK 27-61-00-860-039

(8) Adjust the protractor, SPL-1680, to zero.

(a) In addition to the protractor, SPL-1680, use the readings of the FLT SPLRS on the FLIGHT CONTROLS page on the forward display In the subsequent test.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-61-00-480-006

WARNING MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. AILERONS, RUDDERS, ELEVATORS, FLAPS, SLATS, SPOILERS, LANDING GEAR, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(9) Do this task: Apply Pressure to the Spoiler Hydraulic Systems A and B, TASK 27-61-00-800-801.

SUBTASK 27-61-00-730-002

(10) Cycle the Captain's control wheel through its full travel at least six times.

(a) Center the Captain's control wheel.

SUBTASK 27-61-00-820-008

CAUTION INSTALL THE PROTRACTOR ON THE SPOILER TOP SURFACE. IF THE PROTRACTOR INSTALLED ON THE BOTTOM OF THE SPOILER, DAMAGE TO AIRCRAFT CAN OCCUR.

(11) Install the protractor, COM-1691, on the upper surface of flight spoilers 2, 3, 4, 5, 8, 9, 10, and 11.

NOTE: This test may be completed with multiple protractors used to measure the spoiler panel movement. If multiple protractors are used for the spoiler panels, the "zero" measurement must be verified with the spoiler panels DOWN before each measurement.

SUBTASK 27-61-00-820-009

(12) Adjust the protractors, COM-1691, to zero.

E. Flight Spoiler System Test

SUBTASK 27-61-00-820-010

(1) Turn the control wheel to 87 (+10, -1) degrees in the counterclockwise direction and hold.

(a) Make sure that the spoilers No. 4 and No. 5 are up 35-39 degrees.

(b) Make sure that spoiler No. 3 is up 30-34 degrees.

(c) Make sure that the spoiler No. 2 is up 30-34 degrees.

(d) Make sure that the spoilers No. 8, No. 9, No. 10 and No. 11 are in the fully down position.

(e) If the spoiler panel is not within the spoiler position limits stated, adjust the gap for spoiler No. 2, 3, 4 and 5 (TASK 27-61-00-820-802).

SUBTASK 27-61-00-860-040

(2) Do these steps to center the Captain's control wheel:

(a) Turn the Captain's control wheel in the counterclockwise direction to more than 87 degrees.

(b) Slowly turn the Captain's control wheel in the clockwise direction until it is in the neutral or zero position.

(c) Make sure that all flight spoilers are down.

(d) Shake-out and center the Captain's control wheel.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-61-00-820-022

(3) Turn the control wheel to 87 (+10, -1) degrees in the clockwise direction and hold.

(a) Make sure that the spoilers No. 8 and No. 9 are up 35-39 degrees.

(b) Make sure that spoiler No. 10 is up 30-34 degrees.

(c) Make sure that the spoiler No. 11 is up 30-34 degrees.

(d) Make sure that the spoilers No. 2, No. 3, No. 4, and No. 5 are in the fully down position.

(e) If the spoiler panel is not within the spoiler position limits stated, adjust the gap for spoiler No. 2, 3, 4 and 5 (Flight Spoiler Adjustment, TASK 27-61-00-820-802).

SUBTASK 27-61-00-820-023

(4) Do these steps to center the Captain's control wheel:

(a) Turn the Captain's control wheel in the clockwise direction to more than 87 degrees.

(b) Slowly turn the Captain's control wheel in the counterclockwise direction until it is in the neutral or zero position.

(c) Make sure that all flight spoilers are down.

(d) Shake-out and center the Captain's control wheel.

SUBTASK 27-61-00-730-003

(5) Do these steps to do a check of the spoiler operation with one system hydraulic power:

(a) Put the SPOILER A switch on the forward overhead panel to the OFF position.

NOTE: This will turn off the pressure from Hydraulic System A.

(b) Move the speedbrake lever to the FULL UP position.

(c) Make sure that the flight spoiler panels No. 3, 5, 8, and 10 are FULL UP.

NOTE: If the Hydraulic System A is still pressurized, the ground spoiler panels No. 1, 6, 7, and 12 will also be in the FULL UP position.

(d) Make sure that the spoiler panels No. 2, 4, 9, and 11 are in the FULL DOWN position.

(e) Move the speedbrake lever to the DOWN position.

(f) On the forward overhead panel, put the SPOILER A switch ON.

(g) On the forward overhead panel, put the SPOILER B switch to OFF.

(h) Move the speedbrake lever to the FULL UP position.

(i) Make sure that the flight spoiler panels No. 2, 4, 9, and 11 are in the FULL UP position.

NOTE: If the Hydraulic System A is still pressurized, the ground spoiler panels No. 1, 6, 7, and 12 will also be in the FULL UP position.

(j) Make sure that the flight spoiler panels No. 3, 5, 8, and 10 are FULL DOWN.

(k) Move the speedbrake lever to the DOWN position.

(l) Make sure that all spoiler panels are in the FULL DOWN position.

(m) On the forward overhead panel, put the SPOILER B switch to ON.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 27-61-00-860-041

(1) Do this task: Shut Down the Spoiler Hydraulic Systems A and B, TASK 27-61-00-840-801.

SUBTASK 27-61-00-420-004

(2) Install the map holder assembly to the captain's control wheel.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-61-00-020-008

(3) Remove the mount, SPL-1670, and the protractor, SPL-1680, from the captain's aileron control wheel.

SUBTASK 27-61-00-020-009

(4) Remove the protractor, COM-1691, from the upper surface of spoilers No. 2, 3, 4, 5, 8, 9, 10, and 11.

——— END OF TASK ———

TASK 27-61-00-000-801

5. Spoiler Manual Operation Test

A. References

Reference	Title
27-61-00-800-801	Apply Pressure to the Spoiler Hydraulic Systems A and B (P/B 201)
27-61-00-840-801	Shut Down the Spoiler Hydraulic Systems A and B (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1691	Protractor - Digital, Vertical or Horizontal Part #: 21465A82 Supplier: 3A054 Part #: 31-038-3 (Model Pro 360) Supplier: 1X822 Part #: 31-040-9 (Model Pro 3600) Supplier: 1X822 Part #: 950-317 (Model Pro 360) Supplier: 01EP7 Part #: 950-318 (Model Pro 3600) Supplier: 01EP7 Part #: KS5549 Supplier: 75245 Part #: KS6005 Supplier: 75245

C. Location Zones

Zone	Area
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50
562	Left Wing - Spoiler No. 5
563	Left Wing - Spoiler No. 4
564	Left Wing - Spoiler No. 3
565	Left Wing - Spoiler No. 2
662	Right Wing - Spoiler No. 8
663	Right Wing - Spoiler No. 9
664	Right Wing - Spoiler No. 10
665	Right Wing - Spoiler No. 11

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D. Prepare for the Test

SUBTASK 27-61-00-860-049

> **WARNING** KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Do this task: Apply Pressure to the Spoiler Hydraulic Systems A and B, TASK 27-61-00-800-801.

SUBTASK 27-61-00-860-051

(2) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	9	C00440	FLIGHT CONTROL AUTO SPEED BRAKE
B	11	C04040	GROUND SPOILER CONTROL
B	12	C00204	FLIGHT CONTROL SHUTOFF VALVES SPOILER

> **WARNING** KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT SPOILERS. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU OPEN/CLOSE THESE CIRCUIT BREAKERS. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(3) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
C	10	C04032	SPOILER CONTROL ELEC PWR A
C	12	C04033	SPOILER CONTROL ELEC PWR B

SUBTASK 27-61-00-860-052

(4) Move the speed brake lever to the DOWN position.

SUBTASK 27-61-00-860-053

(5) Put the ailerons in the neutral position.

SUBTASK 27-61-00-480-011

> **CAUTION** INSTALL THE PROTRACTOR ON THE SPOILER TOP SURFACE. IF THE PROTRACTOR INSTALLED ON THE BOTTOM OF THE SPOILER, DAMAGE TO AIRCRAFT CAN OCCUR.

(6) Install the protractor, COM-1691, on the upper surface of spoilers 1, 6, 7, and 12.

SUBTASK 27-61-00-820-013

(7) Adjust the protractor, COM-1691, to zero.

SUBTASK 27-61-00-860-054

(8) Make sure spoilers No. 1, 6, 7, and 12 are in the fully down position.

E. Spoiler Manual Operation Test

SUBTASK 27-61-00-020-010

(1) Move the speed brake lever aft until spoilers No. 1, 6, 7 and 12 move to the up position.

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(a) Make sure the speed lever is 29 +/-2 degrees (2.78 in. (70.61 mm) +/- 0.20 in. (5.08 mm)), measured along the indicator surface aft of the DOWN position.

SUBTASK 27-61-00-020-011

(2) Move the speed brake lever forward until spoilers No. 1, 6, 7 and 12 retract.

(a) Make sure the speed lever is 21 +/-4 degrees (2 in. (50.8 mm) +/- 0.4 in. (10.2 mm)), measured along the indicator surface aft of the DOWN position.

SUBTASK 27-61-00-710-001

(3) Move the speed brake control lever to the DOWN and LOCKED position.

(a) Adjust the protractor, COM-1691, to zero degrees on the spoiler No. 1.

1) Note the position of the protractor on the spoiler panel.

NOTE: The protractor will need to be in the same location when you measure the spoiler panel in the UP position.

(b) Do not adjust the protractor, COM-1691, measure and record the DOWN position of the spoilers No. 6, 7 and 12.

1) Record the position of the protractor on each spoiler panel.

NOTE: The protractor will need to be in the same location when you measure the spoiler panel in the UP position.

2) Record the +/- value measurement on the protractor, COM-1691.

NOTE: This value will be used to calculate total spoiler panel travel.

SUBTASK 27-61-00-020-012

(4) Move the speed brake control lever to the UP position.

(a) Do not adjust the protractor, COM-1691, measure and record the UP position of the spoilers No. 1, 6, 7 and 12.

1) Place the protractor, COM-1691, on the same location that was used when the DOWN position was measured.

2) Record the +/- value measurement on the protractor, COM-1691.

NOTE: This value will be used to calculate total spoiler panel travel.

(b) To calculate the total travel for spoiler panels No. 1, 6, 7 and 12, do this step:

1) Subtract the DOWN position measurement from the FULL UP measurement.

(c) Make sure that the spoiler No. 1 is up 60 +/-2 degrees.

(d) Make sure that the spoiler No. 6 is up 60 +/-2 degrees.

(e) Make sure that the spoiler No. 7 is up 60 +/-2 degrees.

(f) Make sure that the spoiler No. 12 is up 60 +/-2 degrees.

(g) Make sure that there is no more than 2 degrees difference between spoiler panel No. 1 and 12.

(h) Make sure that there is no more than 2 degrees difference between spoiler panel No. 6 and 7.

(i) Put the speedbrake lever to the DOWN and LOCKED position.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 27-61-00-860-055

(1) Do this task: Shut Down the Spoiler Hydraulic Systems A and B, TASK 27-61-00-840-801.

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SUBTASK 27-61-00-080-007

(2) Remove the protractor, COM-1691, from the upper surface of spoilers No. 1, 6, 7, and 12.

END OF TASK

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SPOILER CONTROL SYSTEM - INSPECTION/CHECK

1. General

A. This procedure has this task:

(1) A spoiler PCU and wiring integrity verification with insulation resistance check.

TASK 27-61-00-700-801

2. Spoiler PCU and Wiring Integrity Verification - Insulation Resistance Check

(Figure 601, Figure 602)

A. General

(1) This is an optional task and the results are for evaluation purposes only. No action is required as a result of this inspection. Recommendations made reflect Boeing's best practices to ensure maximum aircraft reliability and are optional so long as a flight deck effect or other noted system defect does not exist. If a flight deck effect or other noted system defect exists, it must be corrected in accordance with established processes.

(2) This task contains the procedure to perform an integrity verification of wiring between the Spoiler Control Electronics (SCE) to the Spoiler Power Control Unit (PCU) with the insulation resistance check.

(3) Insulation resistance testing must only be performed on de-energized circuits.

(4) The test will determine the condition of the wiring from the SCE to the PCU as well as the condition of the PCU.

(5) Insulation resistance testing provides a qualitative value on the condition of conductor insulation and the internal insulation of various components.

(6) To properly evaluate instructions, refer to Table 601.

(7) Before the performing this task, refer to Insulation Resistance Testing - Special Detail Inspection, TASK 20-10-39-700-801.

WARNING

OBEY ALL WARNINGS AND CAUTIONS REFLECTED IN MM TASK 27-30-00.

(8) Obey all WARNINGS and CAUTIONS during this task.

B. References

Reference	Title
20-10-39-700-801	Insulation Resistance Testing - Special Detail Inspection (P/B 501)
24-22-00-860-801	Supply Electrical Power (P/B 201)
24-22-00-860-802	Remove Electrical Power (P/B 201)
27-51-00-860-803	Extend the Trailing Edge Flaps (P/B 201)
27-51-00-860-804	Retract the Trailing Edge Flaps (P/B 201)
27-61-45-000-801	Spoiler Control Electronics Unit Removal (P/B 401)
27-61-45-400-801	Spoiler Control Electronics Unit Installation (P/B 401)
WDM 27-61-48	Wiring Diagram Manual

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C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-17798	Multimeter - Digital, Insulation Resistance Part #: 1004-745 Supplier: U0146 Part #: 1507 Supplier: 4U744 Part #: 1587 FC Supplier: 4U744 Part #: MIT410/2 Supplier: U0146 Opt Part #: 1587 Supplier: 89536

D. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50
562	Left Wing - Spoiler No. 5
563	Left Wing - Spoiler No. 4
564	Left Wing - Spoiler No. 3
565	Left Wing - Spoiler No. 2
662	Right Wing - Spoiler No. 8
663	Right Wing - Spoiler No. 9
664	Right Wing - Spoiler No. 10
665	Right Wing - Spoiler No. 11

E. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

F. Prepare for the Check

SUBTASK 27-61-00-860-100

(1) Extend the trailing edge flaps, do this task: Extend the Trailing Edge Flaps, TASK 27-51-00-860-803.

SUBTASK 27-61-00-860-101

CAUTION

MAKE SURE THAT YOU REMOVE ELECTRICAL POWER FROM THE AIRPLANE. IF YOU SUPPLY ELECTRICAL POWER TO THE ELECTRICAL/ELECTRONIC EQUIPMENT WHEN THE LAVATORY/GALLEY VENTILATION SYSTEM IS NOT IN OPERATION, THE ELECTRICAL/ELECTRONIC EQUIPMENT CAN BECOME TOO HOT. THIS CAN CAUSE DAMAGE TO THE ELECTRICAL/ELECTRONIC EQUIPMENT.

(2) Remove aircraft electrical power, do this task: Remove Electrical Power, TASK 24-22-00-860-802.

SUBTASK 27-61-00-860-102

(3) Make sure that the white AVAIL light in the EXT PWR switch, on the P5 panel, is off.

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SUBTASK 27-61-00-860-103

(4) Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 27-61-00-020-016

(5) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	9	C00440	FLIGHT CONTROL AUTO SPEED BRAKE
B	11	C04040	GROUND SPOILER CONTROL

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
C	10	C04032	SPOILER CONTROL ELEC PWR A
C	12	C04033	SPOILER CONTROL ELEC PWR B

SUBTASK 27-61-00-020-017

(6) Remove the spoiler control electronics unit, do this task: Spoiler Control Electronics Unit Removal, TASK 27-61-45-000-801.

G. Spoiler PCU and Wiring Integrity Verification - Insulation Resistance Check

SUBTASK 27-61-00-200-001

(1) Do a wiring health check between connectors D15998A, D15998B, D15998D, D15998E at the SCE, M02940 and Structure Ground (WDM 27-61-48).

(a) Use an insulation resistance multimeter, COM-17798, to connect the applicable connector.

(b) Connect the negative probe of the multimeter to the most accessible ground point.

(c) Identify SPLR XX LVDT EXCT, SPLR XX SERVO CMD, and SPLR XX, LVDT POSN wiring (WDM 27-61-48).

(d) Test each wire to structure ground with meter positive lead.

(e) Record resulting values to the Table 601.

Table 601/27-61-00-993-809 Spoiler Control Electronics E3-2 Rack w/ Power Control Unit (Connected)

D15998A	Pin to Ground Ohm@ 500VDC	D15998D	Pin to Ground Ohm@ 500VDC	D15998B	Pin to Ground Ohm@ 500VDC	D15998E	Pin to Ground Ohm@ 500VDC
B4 or A4		J4 or K4		B4 or A4		J4 or K4	
D4 or C4		G4 or H4		D4 or C4		G4 or H4	
A5 or C5 or B5		K5 or H5 or J5		A5 or C5 or B5		K5 or H5 or J5	
B6 or A6		J6 or K6		B6 or A6		J6 or K6	
D6 or C6		G6 or H6		D6 or C6		G6 or H6	
A7 or C7 or B7		K7 or H7 or J7		A7 or C7 or B7		K7 or H7 or J7	

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(f) If any resistance is less than 50 Mohms (50000000 ohm), indicating a short, disconnect at associate PCU connector.

(g) Repeat wire health check to ground to check the short.

H. Test Evaluation

SUBTASK 27-61-00-200-002

(1) If any resistance is less than 50 Mohms (50000000 ohm), indicating a short, disconnect at associate PCU connector.

SUBTASK 27-61-00-200-003

(2) Repeat wire health check to ground from the PCU connector for any values less than 50 Mohms (50000000 ohm).

SUBTASK 27-61-00-200-004

(3) With PCU still disconnected, perform a wire to wire check of wires of PCU connector.

(a) Identify SPLR XX LVDT EXCT, SPLR XX SERVO CMD, and SPLR XX, LVDT POSN wiring (WDM 27-61-48).

(b) Test each wire to structure ground with meter positive lead.

SUBTASK 27-61-00-860-104

(4) If it is necessary, isolate suspect circuit.

SUBTASK 27-61-00-970-001

(5) Record values to the Table 602.

Table 602/27-61-00-993-810 Power Control Unit (Disconnected)

PCU Connector DXXXXX	Pin to Ground Ohm@ 500VDC	Pin to Pin Ohm@ 500VDC							
		5	1	10	11	12	7	8	9
5		-							
1		-	-						
10		-	-	-					
11		-	-	-	-				
12		-	-	-	-	-			
7		-	-	-	-	-	-		
8		-	-	-	-	-	-	-	
9		-	-	-	-	-	-	-	-

I. Put the Airplane Back to Its Usual Condition

SUBTASK 27-61-00-420-005

(1) Install the spoiler control electronics unit, do this task: Spoiler Control Electronics Unit Installation, TASK 27-61-45-400-801.

SUBTASK 27-61-00-860-105

(2) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	9	C00440	FLIGHT CONTROL AUTO SPEED BRAKE

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(Continued)

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	11	C04040	GROUND SPOILER CONTROL

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
C	10	C04032	SPOILER CONTROL ELEC PWR A
C	12	C04033	SPOILER CONTROL ELEC PWR B

SUBTASK 27-61-00-860-106

(3) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 27-61-00-860-107

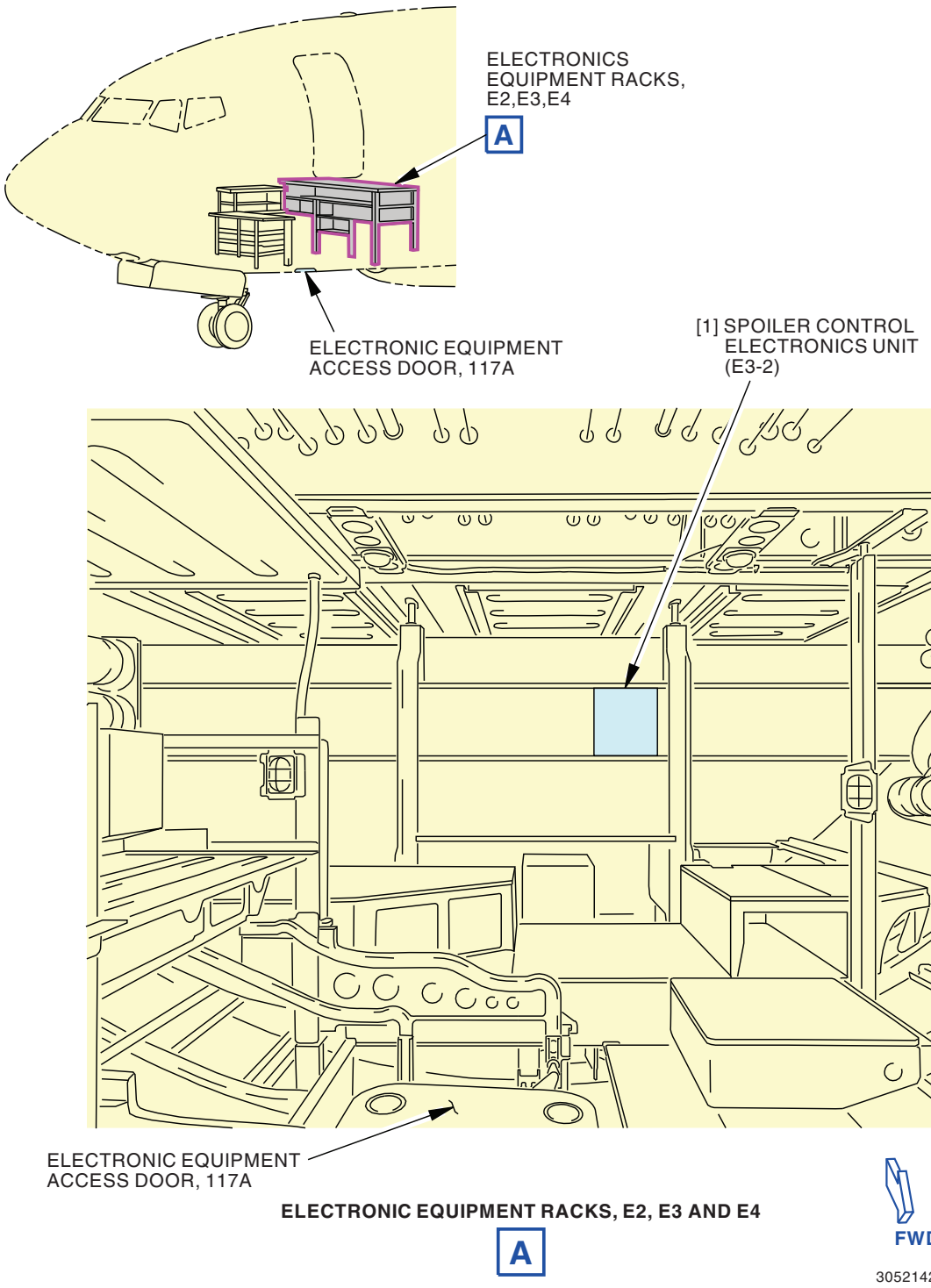
(4) Supply aircraft electrical power, do this task: Supply Electrical Power, TASK 24-22-00-860-801.

SUBTASK 27-61-00-860-108

(5) Retract the trailing edge flaps, do this task: Retract the Trailing Edge Flaps, TASK 27-51-00-860-804.

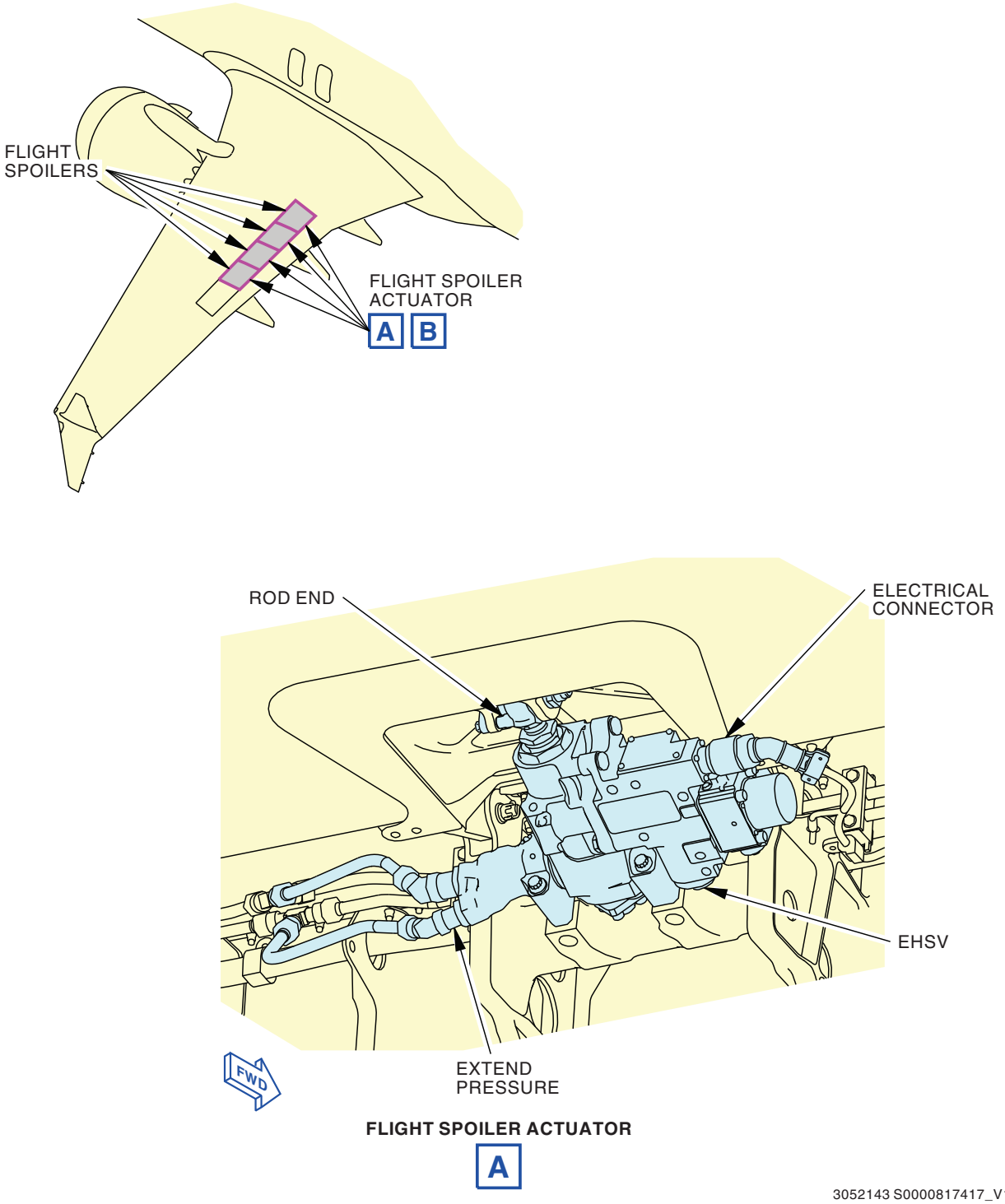
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



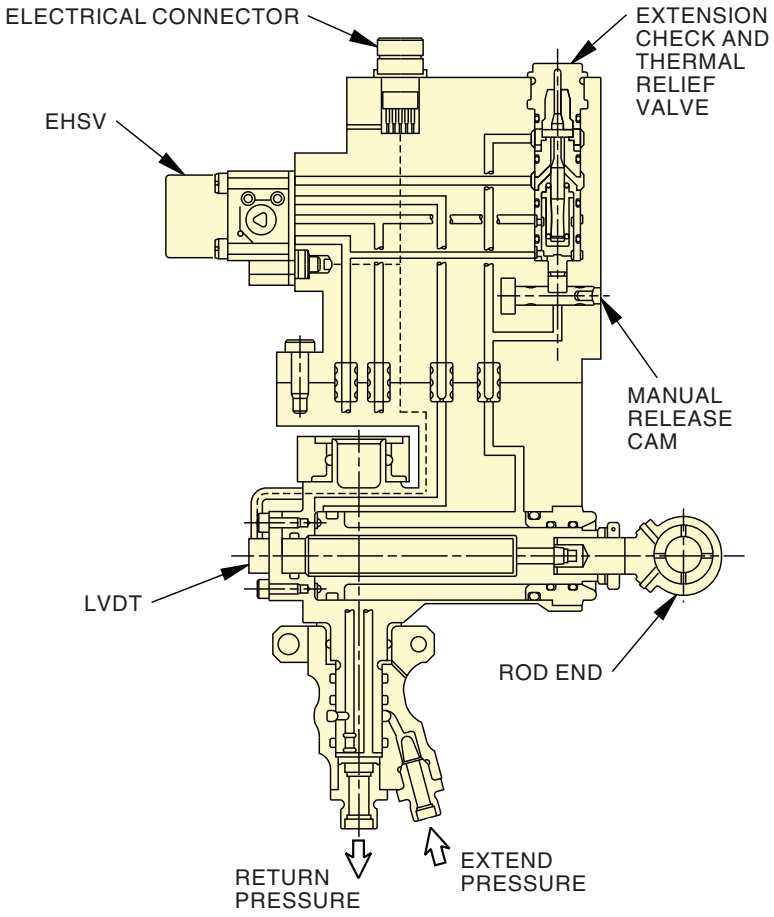
Electronics Equipment Racks - Insulation Resistance Check
Figure 601/27-61-00-990-814

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Flight Spoiler Actuator - Insulation Resistance Check
Figure 602/27-61-00-990-815 (Sheet 1 of 2)

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B

FLIGHT SPOILER ACTUATOR (FUNCTIONAL SCHEMATIC)

3052145 S0000817418_V1

Flight Spoiler Actuator - Insulation Resistance Check
Figure 602/27-61-00-990-815 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FLIGHT SPOILER - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the flight spoilers

(2) An installation of the flight spoilers.

TASK 27-61-11-000-801

2. Flight Spoiler Removal

(Figure 401)

A. General

(1) This task for the left wing spoilers 2, 3, 4, and 5 is applicable to the right wing spoilers 8, 9, 10, and 11.

B. References

Reference	Title
27-51-00-860-803	Extend the Trailing Edge Flaps (P/B 201)
27-61-51-000-801	Flight Spoiler Actuator - Removal (P/B 401)
27-62-00-800-801	Apply Pressure to the Speed Brake Hydraulic Systems A and B (P/B 201)
27-62-00-800-802	Depressurization of the Speed Brake Hydraulic Systems A and B (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-14632	Lock Set - Flight Spoiler Actuator Part #: C27107-1 Supplier: 81205

D. Location Zones

Zone	Area
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50
562	Left Wing - Spoiler No. 5
563	Left Wing - Spoiler No. 4
564	Left Wing - Spoiler No. 3
565	Left Wing - Spoiler No. 2
662	Right Wing - Spoiler No. 8
663	Right Wing - Spoiler No. 9
664	Right Wing - Spoiler No. 10
665	Right Wing - Spoiler No. 11

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E. Prepare for the Removal

SUBTASK 27-61-11-860-002

WARNING

MAKE SURE THAT ALL PERSONS AND EQUIPMENT ARE CLEAR OF CONTROL SURFACES, CABLES, AND LANDING GEAR AREAS. PRESSURIZATION OF A HYDRAULIC SYSTEM CAN MOVE CONTROL SURFACES, CABLES, LANDING GEAR, AND LANDING GEAR DOORS. MOVEMENT OF CONTROL SURFACES, CABLES, LANDING GEAR, AND LANDING GEAR DOORS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Do this task: Apply Pressure to the Speed Brake Hydraulic Systems A and B, TASK 27-62-00-800-801.

SUBTASK 27-61-11-860-003

(2) Do this task: Extend the Trailing Edge Flaps, TASK 27-51-00-860-803.

SUBTASK 27-61-11-860-004

(3) Put the speed brake lever in the UP position to lift the spoilers.

SUBTASK 27-61-11-860-005

(4) Do this task: Depressurization of the Speed Brake Hydraulic Systems A and B, TASK 27-62-00-800-802.

SUBTASK 27-61-11-480-001

(5) Install the lock set, SPL-14632, on the actuator for the applicable flight spoiler.

F. Flight Spoiler Removal

SUBTASK 27-61-11-020-001

(1) For the flight spoiler no. 2 [1], do these steps:

(a) Remove the nut [7] and washers [6] to disconnect the bonding jumper [14] from the structure.

1) Discard the nut [7] and washers [6].

(b) Remove the nut [23] and washer [22] to disconnect the bonding jumper [21] from the structure.

1) Discard the nut [23] and washer [22].

(c) Disconnect the spoiler actuator [5] from the spoiler no. 2 [1] (TASK 27-61-51-000-801).

(d) Remove the cotter pin [13], nut [12], washer [11], bushing [10], washer [9], and bolt [8].

(e) Remove the cotter pin [29], nut [28], washer [27], bushings [26], washer [25], and bolt [24].

(f) Remove the cotter pin [20], nut [19], washer [18], bushing [17], washer [16], and bolt [15].

(g) Remove the flight spoiler no. 2 [1] from the airplane.

SUBTASK 27-61-11-020-002

(2) For the flight spoiler no. 3 [2], spoiler no. 4 [3], or spoiler no. 5 [4], do these steps:

(a) Remove the nut [32] and washers [31] to disconnect the bonding jumper [39] from the structure.

1) Discard the nut [32] and washers [31].

(b) Remove the nut [60] and washers [59] to disconnect the bonding jumper [52] from the structure.

1) Discard the nut [60] and washers [59].

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(c) Disconnect the spoiler actuator [30] from the spoiler no. 3 [2], spoiler no. 4 [3], or spoiler no. 5 [4] (TASK 27-61-51-000-801).

(d) Remove the cotter pin [38], nut [37], washer [36], bushing [35], washer [34], and bolt [33].

(e) Remove the cotter pin [58], nut [57], washer [56], bushing [55], washer [54], and bolt [53].

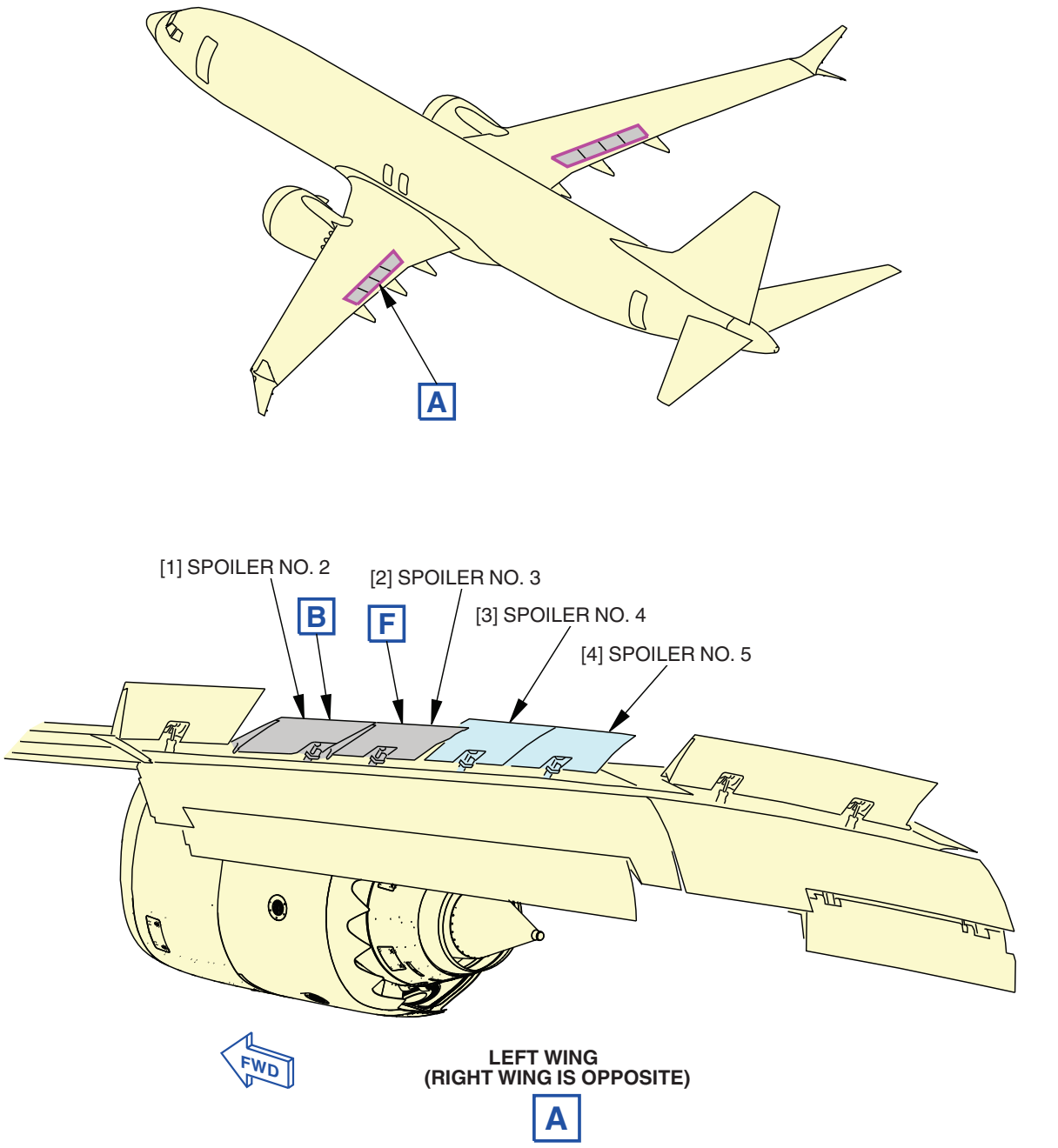
(f) Remove the cotter pin [45], nut [44], washer [43], bushing [42], washer [41], and bolt [40].

(g) Remove the cotter pin [51], nut [50], washer [49], bushings [48], washer [47], and bolt [46].

(h) Remove the flight spoiler no. 3 [2], spoiler no. 4 [3], or spoiler no. 5 [4] from the airplane.

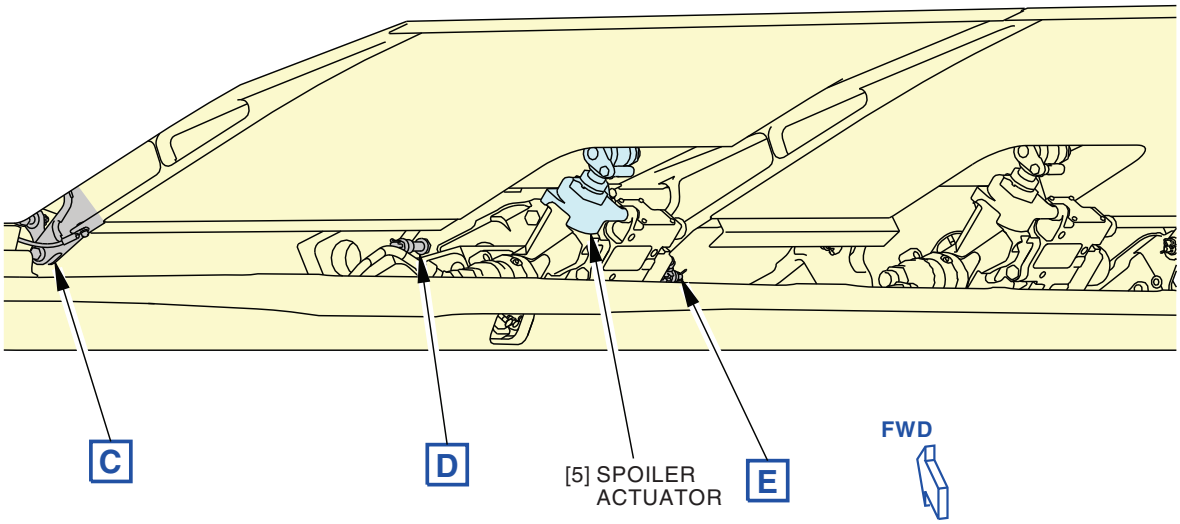
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Flight Spoiler Installation
Figure 401/27-61-11-990-801 (Sheet 1 of 7)

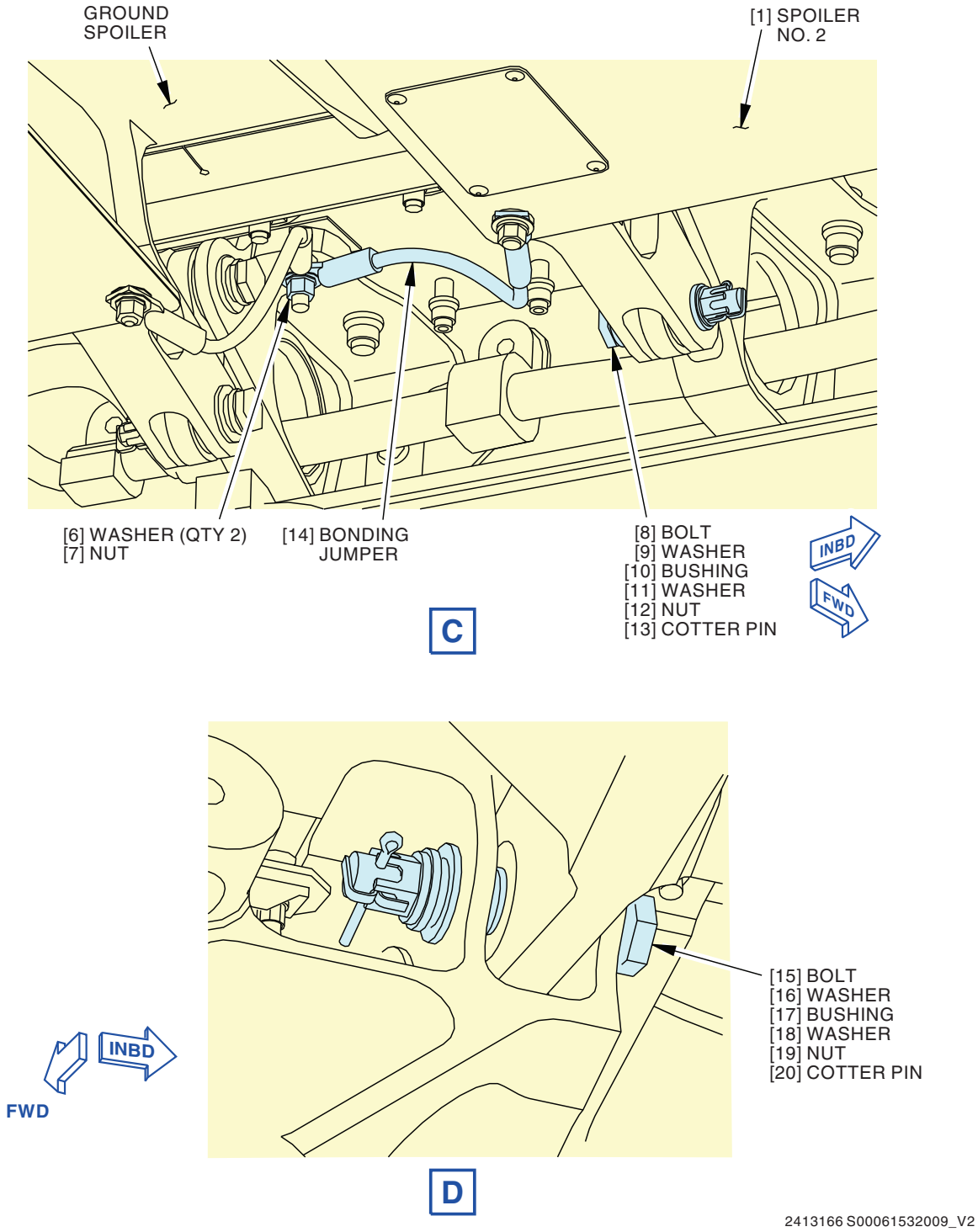
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2413165 S00061532008_V2

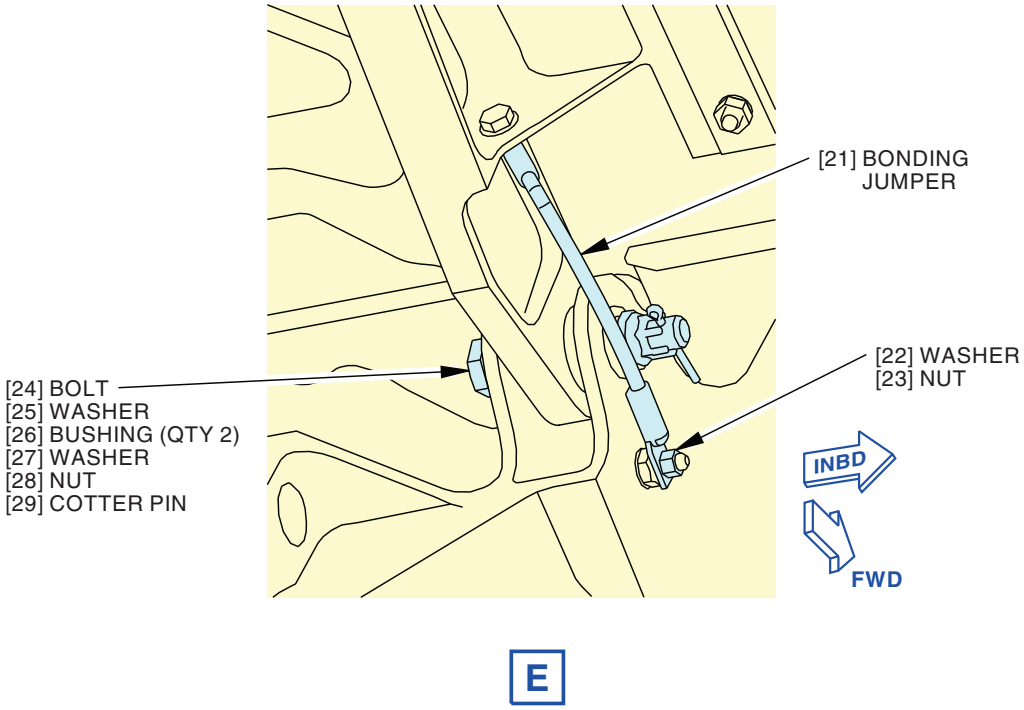
Flight Spoiler Installation
Figure 401/27-61-11-990-801 (Sheet 2 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Flight Spoiler Installation
Figure 401/27-61-11-990-801 (Sheet 3 of 7)

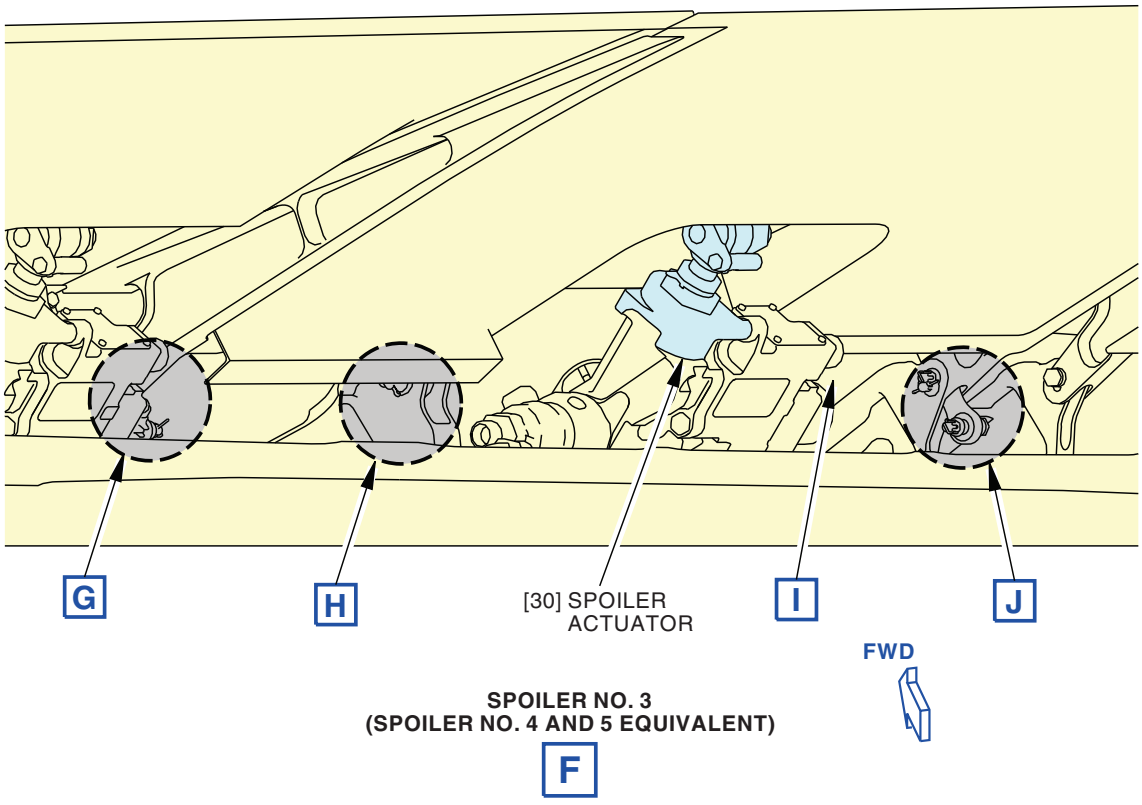
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2413167 S00061532010_V4

Flight Spoiler Installation
Figure 401/27-61-11-990-801 (Sheet 4 of 7)

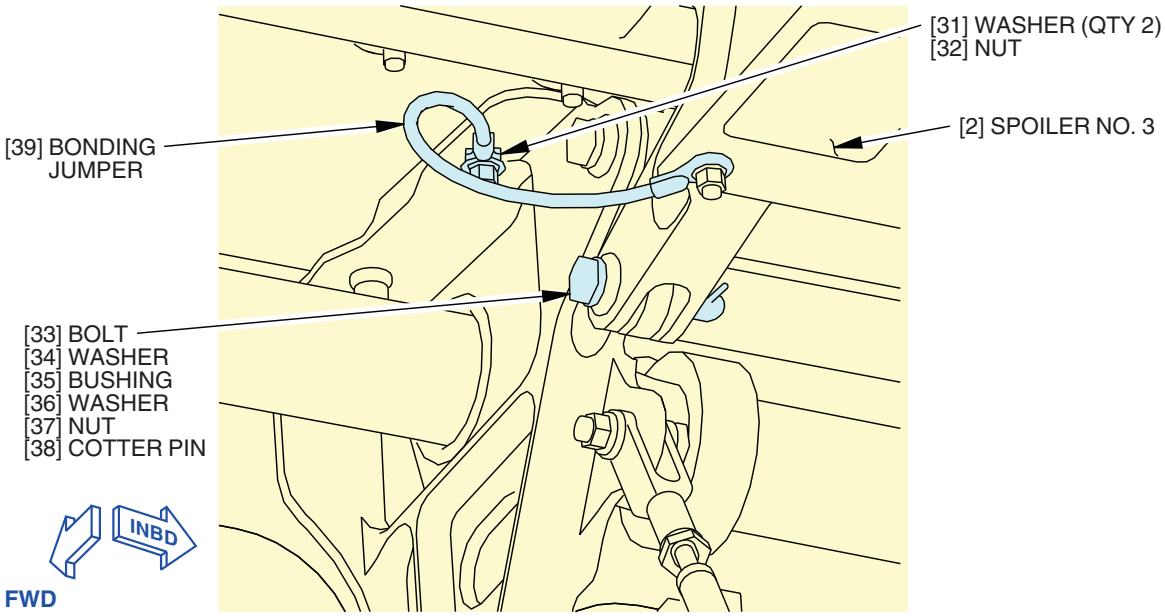
ECCN 9E991 BOEING PROPRIETARY - See title page for details



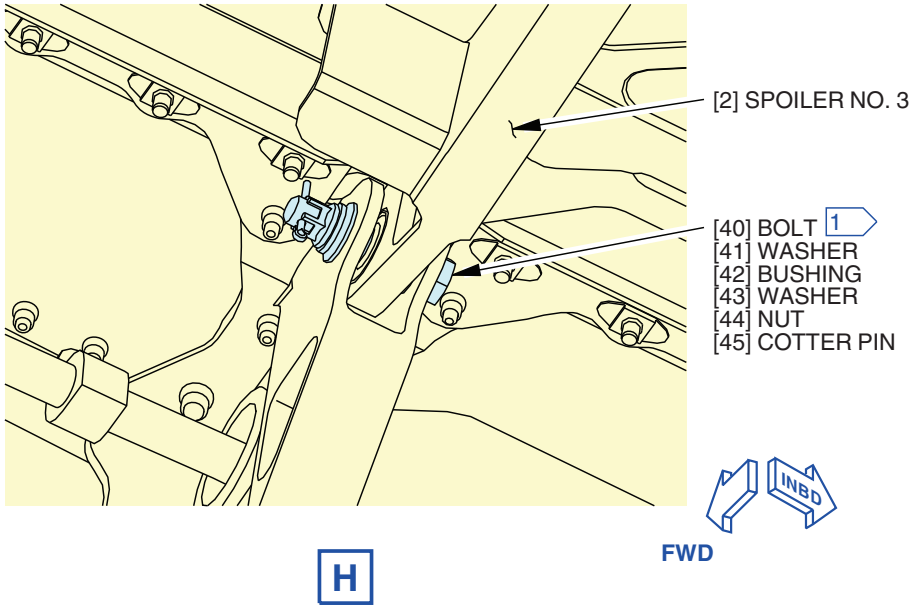
2413168 S00061532011_V2

Flight Spoiler Installation
Figure 401/27-61-11-990-801 (Sheet 5 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



G



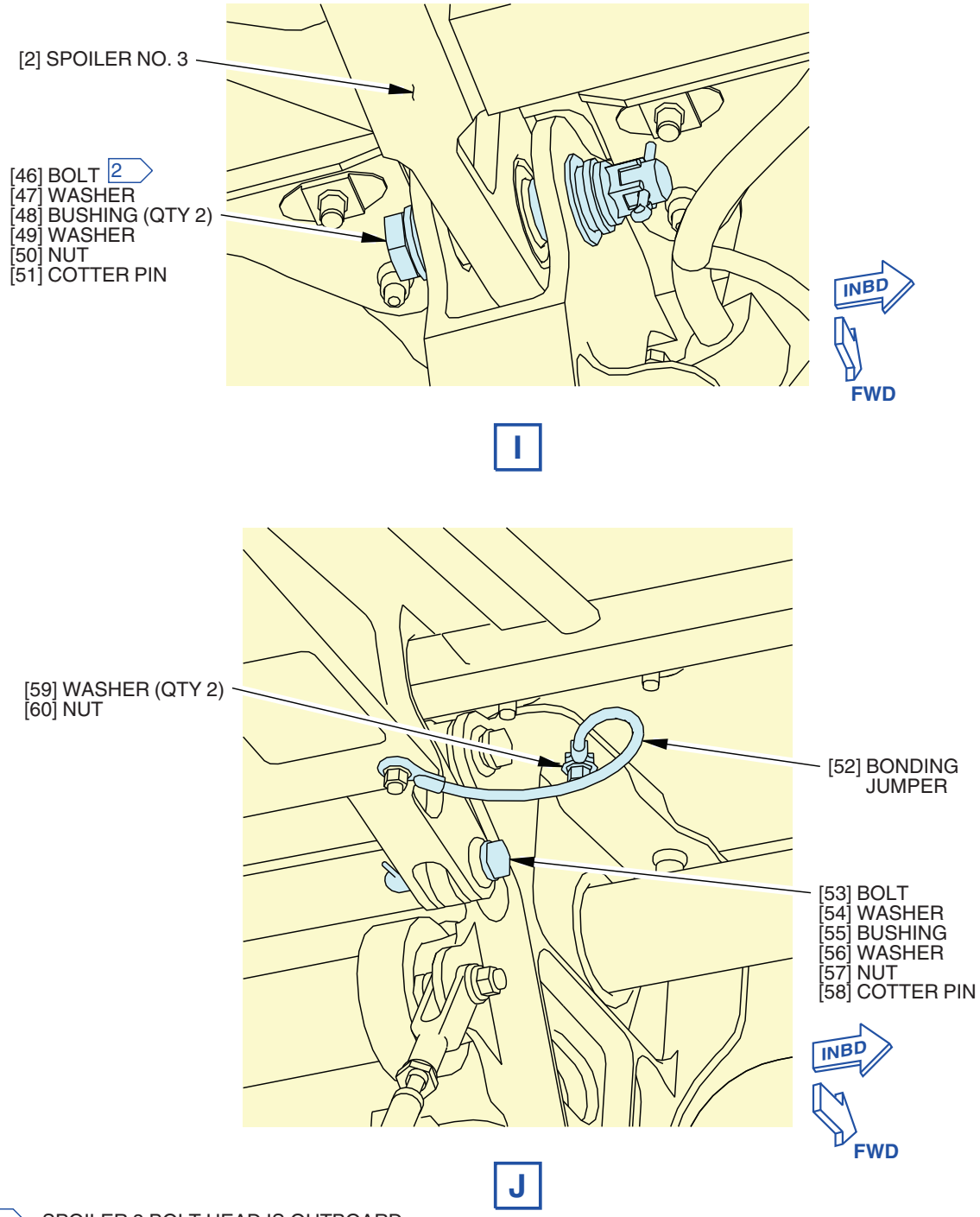
H

1 SPOILER 3 BOLT HEAD IS INBOARD.
SPOILERS 4 AND 5 BOLT HEAD IS OUTBOARD.

2413169 S00061532012_V5

Flight Spoiler Installation
Figure 401/27-61-11-990-801 (Sheet 6 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2 SPOILER 3 BOLT HEAD IS OUTBOARD.
SPOILERS 4 AND 5 BOLT HEAD IS INBOARD.

2413170 S00061532013_V4

Flight Spoiler Installation
Figure 401/27-61-11-990-801 (Sheet 7 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-61-11-400-801

3. Flight Spoiler Installation

(Figure 401)

A. General

(1) This task for the left wing spoilers 2, 3, 4 and, 5 is applicable to the right wing spoilers 8, 9, 10, and 11.

B. References

Reference	Title
20-10-44-400-801	Lockwire, Cotter Pins, and Lockrings - Installation (P/B 401)
27-51-00-860-804	Retract the Trailing Edge Flaps (P/B 201)
27-61-00-730-801	Flight Spoiler System Test (P/B 501)
27-61-00-800-801	Apply Pressure to the Spoiler Hydraulic Systems A and B (P/B 201)
27-61-00-820-802	Flight Spoiler Adjustment (P/B 501)
27-61-00-840-801	Shut Down the Spoiler Hydraulic Systems A and B (P/B 201)
27-61-00-840-802	Pressurization of the Spoiler Hydraulic Systems A and B to its Usual Condition (P/B 201)
27-61-45-730-805	Flight Spoiler Test (P/B 501)
27-61-51-400-801	Flight Spoiler Actuator - Installation (P/B 401)
SWPM 20-20-00	ELECTRICAL BONDING PROCESSES

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-614	Bonding Meters - Non-Intrinsically Safe (For use in outside Class I, Divisions I & II non-hazardous locations. For hazardous locations, use COM-1550). Part #: 247000 Supplier: 00426 Part #: 620LK Supplier: 1CRL2 Part #: BLR-0003-XX Supplier: KC432 Part #: BT51 Supplier: 00426 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659 Opt Part #: 247001 Supplier: 00426
SPL-14632	Lock Set - Flight Spoiler Actuator Part #: C27107-1 Supplier: 81205

D. Consumable Materials

Reference	Description	Specification
A50110	Sealant - Fuel Tank	BMS5-45 Class B-2
A50155	Sealant - Fuel Tank	BMS5-45 Class C
A50231	Sealant - Pressure And Environmental - Chromate Type	BMS5-95 Class B
A50296	Sealant - Pressure And Environmental - Chromate Type	BMS5-95 Class C

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Reference	Description	Specification
A50337	Sealant - Fuel Tank	BMS5-45 Class B
A50359	Sealant - Low Density, Non-Chromate Type	BMS5-142 Type II Class B
B00184	Solvent - Presealing, Cleaning Solvent	BMS11-7
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Spoiler no. 2	27-61-11-01-100	PTF ALL
2	Spoiler no. 3	27-61-11-10-100	PTF ALL
3	Spoiler no. 4	27-61-11-20-100	PTF ALL
4	Spoiler no. 5	27-61-11-30-100	PTF ALL
6	Washer	27-62-91-01-020	PTF ALL
7	Nut	27-62-91-01-025	PTF ALL
13	Cotter pin	27-61-11-01-030	PTF ALL
20	Cotter pin	27-61-11-01-035	PTF ALL
22	Washer	27-61-11-01-020	PTF ALL
23	Nut	27-61-11-01-025	PTF ALL
29	Cotter pin	27-61-11-01-035	PTF ALL
31	Washer	27-61-11-10-020	PTF ALL
		27-61-11-20-020	PTF ALL
		27-61-11-30-020	PTF ALL
32	Nut	27-61-11-10-025	PTF ALL
		27-61-11-20-025	PTF ALL
		27-61-11-30-025	PTF ALL
38	Cotter pin	27-61-11-10-030	PTF ALL
45	Cotter pin	27-61-11-10-035	PTF ALL
51	Cotter pin	27-61-11-10-035	PTF ALL
58	Cotter pin	27-61-11-10-030	PTF ALL
59	Washer	27-61-11-10-020	PTF ALL
		27-61-11-20-020	PTF ALL
		27-61-11-30-020	PTF ALL
60	Nut	27-61-11-10-025	PTF ALL
		27-61-11-20-025	PTF ALL
		27-61-11-30-025	PTF ALL

F. Location Zones

Zone	Area
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50
562	Left Wing - Spoiler No. 5

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Zone	Area
563	Left Wing - Spoiler No. 4
564	Left Wing - Spoiler No. 3
565	Left Wing - Spoiler No. 2
662	Right Wing - Spoiler No. 8
663	Right Wing - Spoiler No. 9
664	Right Wing - Spoiler No. 10
665	Right Wing - Spoiler No. 11

G. Flight Spoiler Installation

SUBTASK 27-61-11-420-001

(1) To install the flight spoiler no. 2 [1], do these steps (View B, Figure 401), (TASK 20-10-44-400-801):

(a) Put the flight spoiler no. 2 [1] in its position on the airplane.

(b) Install the bolt [15] with head inboard, washer [16], bushing [17], washer [18], nut [19], and cotter pin [20].

(c) Install the bolt [24] with head outboard, washer [25], bushings [26], washer [27], nut [28], and cotter pin [29].

(d) Install the bolt [8] with head outboard, washer [9], bushing [10], washer [11], nut [12], and cotter pin [13].

(e) Connect the rod end of the spoiler actuator [5] to flight spoiler no. 2 [1] (TASK 27-61-51-400-801).

(f) Install the new nut [7] and new washers [6] to connect the bonding jumper [14] to the structure.

1) Use cotton wiper, G00034, moist in solvent, B00184, to clean the mating surfaces of the airplane structure, nut [7], washers [6], and bonding jumper [14].

2) Install the nut [7], washers [6], and bonding jumper [14].

3) Tighten the nut [7] to 28 in-lb (3 N·m) - 35 in-lb (4 N·m).

4) Measure the electrical resistance between bonding jumper [14] and airplane structure (SWPM 20-20-00).

a) Use a non-intrinsically safe bonding meter, COM-614.

b) Make sure that the electrical bonding resistance is 0.001 ohm (1.0 milliohm) or less.

5) Apply a fillet seal with sealant, A50337, sealant, A50231, or sealant, A50359, to cover the nut [7], washers [6], and bonding jumper [14].

(g) Install the new nut [23] and new washer [22] to connect the bonding jumper [21] to the structure.

1) Use cotton wiper, G00034, moist in solvent, B00184, to clean the mating surfaces of the airplane structure, nut [23], washer [22], and bonding jumper [21].

2) Apply sealant, A50155, sealant, A50110, sealant, A50296, or sealant, A50231, to the mating surfaces of the ground stud, nut [23], washer [22], and bonding jumper [21].

3) Install the nut [23], washer [22], and bonding jumper [21].

4) Tighten the nut [23] to 28 in-lb (3 N·m) - 35 in-lb (4 N·m).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

a) Make sure that there is squeeze out around all the interfaces of each component.

5) Measure the electrical resistance between bonding jumper [21] and airplane structure (SWPM 20-20-00).

a) Use a non-intrinsically safe bonding meter, COM-614.

b) Make sure that the electrical bonding resistance is 0.001 ohm (1.0 milliohm) or less.

SUBTASK 27-61-11-420-002

(2) To install the flight spoiler no. 3 [2], spoiler no. 4 [3], or spoiler no. 5 [4], do these steps (View F, Figure 401), (TASK 20-10-44-400-801):

(a) Put the flight spoiler in its position on the airplane.

(b) Install the bolt [46], washer [47], bushings [48], washer [49], nut [50], and cotter pin [51].

NOTE: For the spoiler no. 3 [2], the bolt [46] head is outboard.

NOTE: For the spoiler no. 4 [3] and spoiler no. 5 [4], the bolt [46] head is inboard.

(c) Install the bolt [40], washer [41], bushing [42], washer [43], nut [44], and cotter pin [45].

NOTE: For the spoiler no. 3 [2], the bolt [40] head is inboard.

NOTE: For the spoiler no. 4 [3] and spoiler no. 5 [4], the bolt [40] head is outboard.

(d) Install the bolt [53] (with head inboard), washer [54], bushing [55], washer [56], nut [57], and cotter pin [58].

(e) Install the bolt [33] (with head outboard), washer [34], bushing [35], washer [36], nut [37], and cotter pin [38].

(f) Connect the rod end of the spoiler actuator [30] to flight spoiler no. 3 [2] (TASK 27-61-51-400-801).

(g) Install the new nut [32] and new washers [31] to connect the bonding jumper [39] to the structure.

1) Use cotton wiper, G00034, moist in solvent, B00184, to clean the mating surfaces of the airplane structure, nut [32], washers [31], and bonding jumper [39].

2) Install the nut [32], washers [31], and bonding jumper [39].

3) Tighten the nut [32] to 28 in-lb (3 N·m) - 35 in-lb (4 N·m).

4) Measure the electrical resistance between bonding jumper [39] and airplane structure (SWPM 20-20-00).

a) Use a non-intrinsically safe bonding meter, COM-614.

b) Make sure that the electrical bonding resistance is 0.001 ohm (1.0 milliohm) or less.

5) Apply a fillet seal with sealant, A50337, sealant, A50231, or sealant, A50359, to cover the nut [32], washers [31], and bonding jumper [39].

(h) Install the new nut [60] and new washers [59] to connect the bonding jumper [52] to the structure.

1) Use cotton wiper, G00034, moist in solvent, B00184, to clean the mating surfaces of the airplane structure, nut [60], washers [59], and bonding jumper [52].

2) Install the nut [60], washers [31], and bonding jumper [39].

3) Tighten the nut [60] to 28 in-lb (3 N·m) - 35 in-lb (4 N·m).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

4) Measure the electrical resistance between bonding jumper [52] and airplane structure (SWPM 20-20-00).
   a) Use a non-intrinsically safe bonding meter, COM-614.
   b) Make sure that the electrical bonding resistance is 0.001 ohm (1.0 milliohm) or less.
5) Apply a fillet seal with sealant, A50337, sealant, A50231, or sealant, A50359, to cover the mating surface of the nut [60], washers [59], and bonding jumper [52].

SUBTASK 27-61-11-080-001

(3) Remove the lock set, SPL-14632, from the applicable flight spoiler actuator.

SUBTASK 27-61-11-720-002

(4) To unlatch the SPOILER ACTUATOR status message, do this task: Flight Spoiler Test, TASK 27-61-45-730-805.

NOTE: The SPOILER ACTUATOR status message can be latched when the spoilers are locked in the up position.

H. Flight Spoiler Installation Test

SUBTASK 27-61-11-860-006

(1) Do this task: Pressurization of the Spoiler Hydraulic Systems A and B to its Usual Condition, TASK 27-61-00-840-802.

SUBTASK 27-61-11-860-007

WARNING

MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. THE AILERONS, RUDDER, ELEVATORS, FLAPS, SLATS, SPOILERS, LANDING GEAR, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(2) Do this task: Apply Pressure to the Spoiler Hydraulic Systems A and B, TASK 27-61-00-800-801.

SUBTASK 27-61-11-860-008

(3) Move the speed brake lever to the DOWN position to lower the spoilers.

SUBTASK 27-61-11-860-009

(4) Do this task: Retract the Trailing Edge Flaps, TASK 27-51-00-860-804.

SUBTASK 27-61-11-860-010

(5) Do this task: Shut Down the Spoiler Hydraulic Systems A and B, TASK 27-61-00-840-801.

SUBTASK 27-61-11-820-001

(6) Do this task: Flight Spoiler Adjustment, TASK 27-61-00-820-802.

SUBTASK 27-61-11-720-001

(7) Do this task: Flight Spoiler System Test, TASK 27-61-00-730-801.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SPOILER SHUTOFF VALVE - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the spoiler shutoff valve

(2) An installation of the spoiler shutoff valve

(3) A removal of the spoiler shutoff valve motor

(4) An installation of the spoiler shutoff valve motor.

B. The spoiler shutoff valve and the flight controls shutoff valve are parts of the flight controls hydraulic module.

TASK 27-61-12-000-801

2. Spoiler Shutoff Valve Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove the spoiler shutoff valve assembly.

(2) The spoiler shutoff valve assembly contains the spoiler shutoff valve motor [7] and the spoiler shutoff valve [12].

B. References

Reference	Title
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Tools/Equipment

Reference	Description
STD-1154	Container - 5 Gallon (19 Liter)

D. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

E. Prepare for the Spoiler Shutoff Valve Removal

SUBTASK 27-61-12-860-013

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-61-12-860-003

(2) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	12	C00204	FLIGHT CONTROL SHUTOFF VALVES SPOILER

F. Spoiler Shutoff Valve Removal

SUBTASK 27-61-12-020-001

(1) Do these steps to remove the spoiler shutoff valve assembly:

(a) Disconnect the electrical connector [1] from the spoiler shutoff valve motor [7].

(b) Remove the lockwire from the bolts [4].

(c) Put a 5 gallon (19 liter) container, STD-1154, below the spoiler shutoff valve assembly to receive hydraulic fluid.

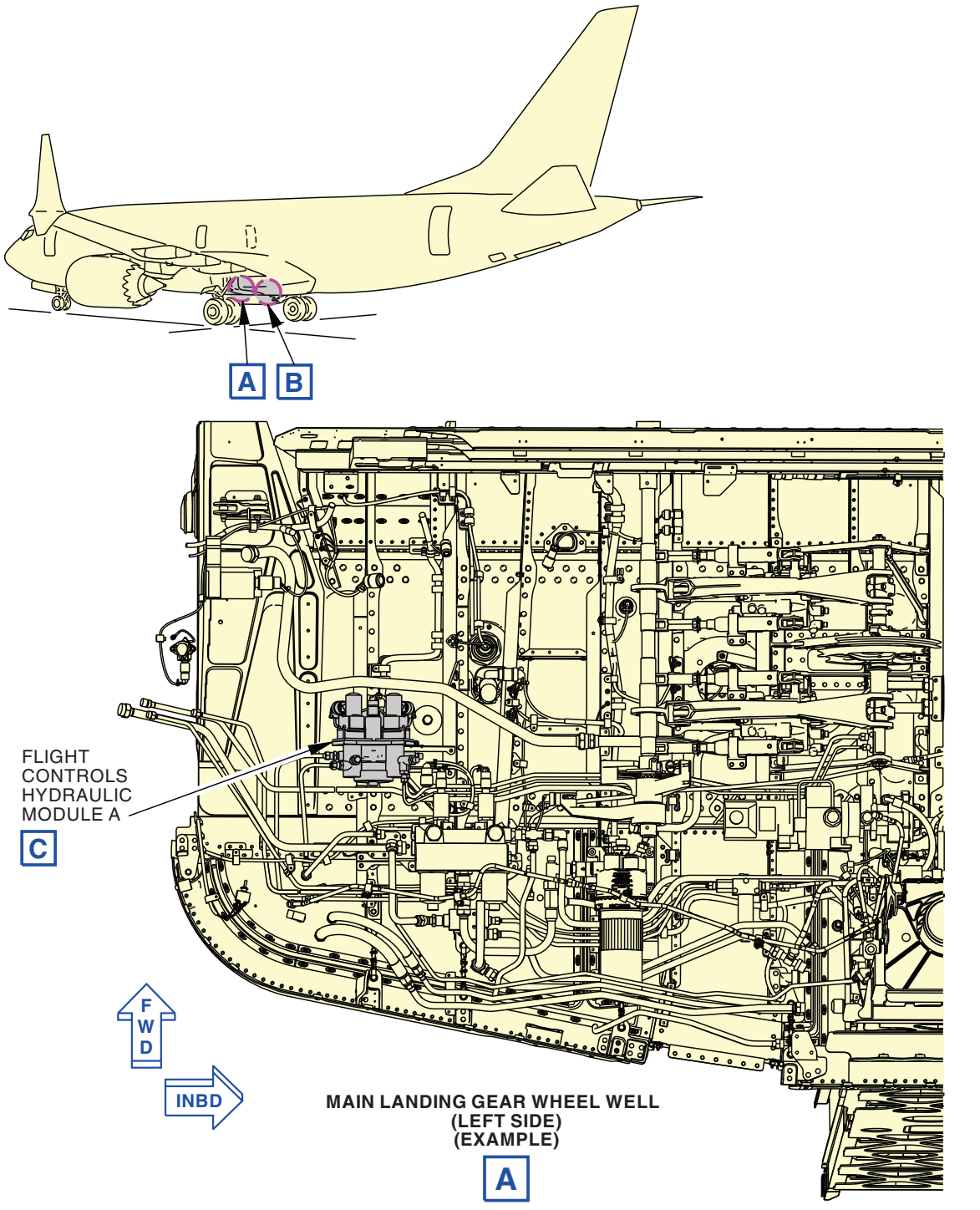
(d) Remove the bolts [4] and washers [5] that attach the spoiler shutoff valve assembly to the flight controls hydraulic module housing [2].

(e) Carefully lift the spoiler shutoff valve assembly directly up to remove it from the flight controls hydraulic module housing [2].

(f) Put a cover on the flight controls hydraulic module housing [2] to keep contamination out of the open valve hole.

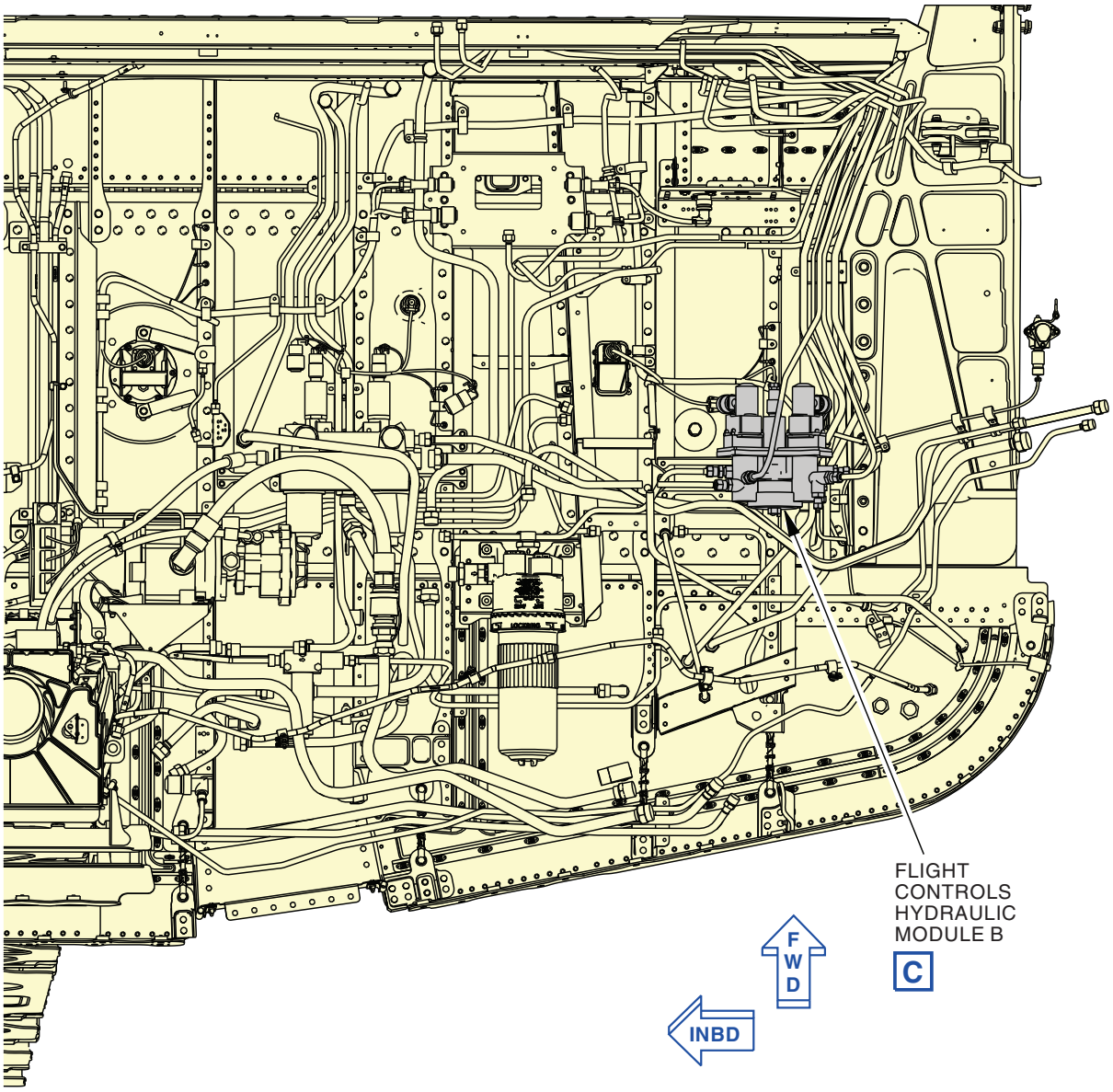
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Spoiler Shutoff Valve Installation
Figure 401/27-61-12-990-801 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



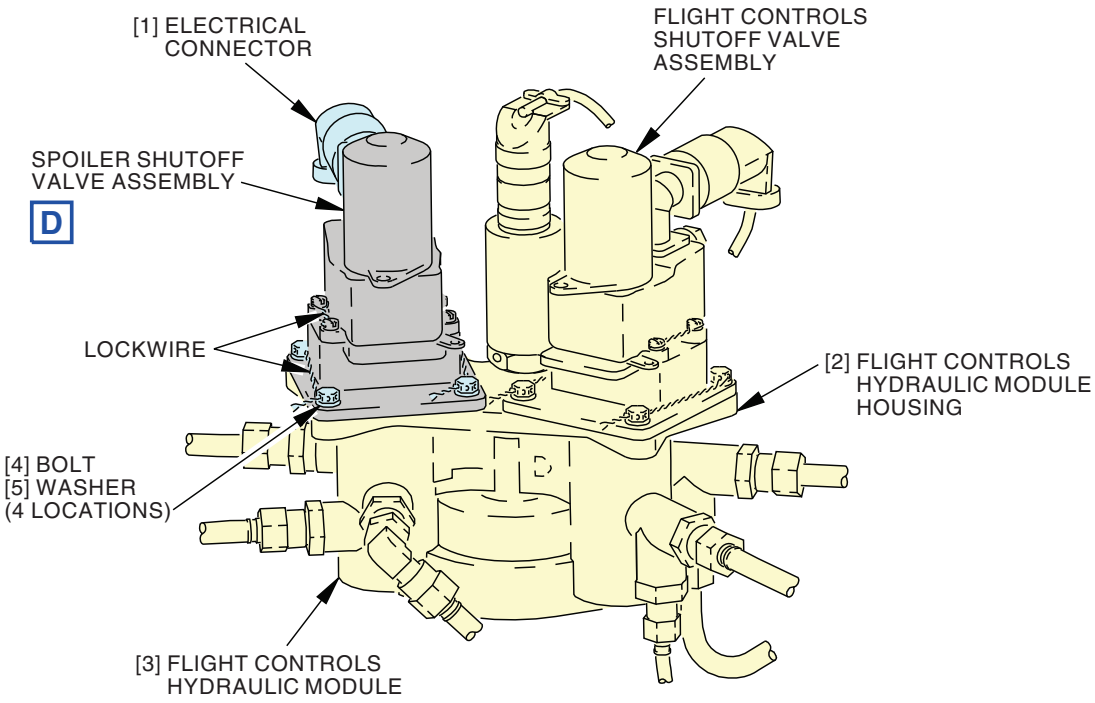
MAIN LANDING GEAR WHEEL WELL
(RIGHT SIDE)
(EXAMPLE)

B

2413173 S00061532020_V3

Spoiler Shutoff Valve Installation
Figure 401/27-61-12-990-801 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



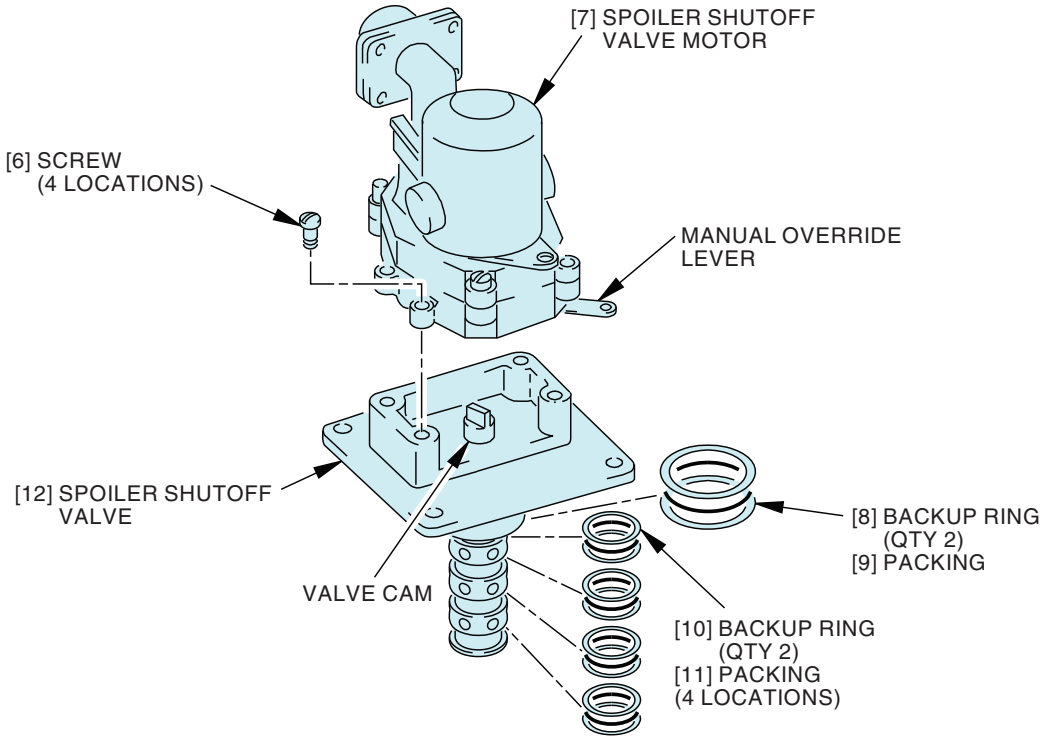
FLIGHT CONTROLS HYDRAULIC MODULE A
(MODULE B IS EQUIVALENT)

C

2413174 S00061532021_V2

Spoiler Shutoff Valve Installation
Figure 401/27-61-12-990-801 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



SPOILER SHUTOFF VALVE ASSEMBLY

D

2413175 S00061532022_V3

Spoiler Shutoff Valve Installation
Figure 401/27-61-12-990-801 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-61-12-400-801

3. Spoiler Shutoff Valve Installation

(Figure 401)

A. General

(1) This task gives the instructions to install the spoiler shutoff valve assembly.

(2) The spoiler shutoff valve assembly contains the spoiler shutoff valve motor [7] and the spoiler shutoff valve [12].

B. References

Reference	Title
12-12-00-610-801	Hydraulic Reservoir Servicing (P/B 301)
20-60-07-910-802	Inspection, Cleaning and Sealing of Electrical Connectors (P/B 201)
27-61-12-400-803	Spoiler Shutoff Valve Test (P/B 501)

C. Consumable Materials

Reference	Description	Specification
D00153	Fluid - Hydraulic Fluid, Fire Resistant (Interchangeable And Intermixable With BMS 3-11 Type V)	BMS3-11 Type IV
G50347	Lockwire - MS20995NC32, Monel - 0.032 Inch (0.8128 mm) Diameter	NASM20995

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
7	Spoiler shutoff valve motor	27-61-12-01-065	PTF ALL
9	Packing	27-61-12-01-055	PTF ALL
11	Packing	27-61-12-01-060	PTF ALL
12	Spoiler shutoff valve	27-09-21-01-125	PTF ALL

E. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

F. Spoiler Shutoff Valve Installation

SUBTASK 27-61-12-420-001

(1) If the spoiler shutoff valve [12] does not have the packing [9], the two backup rings [8], the packing [11], and the two backup rings [10] installed, then do these steps:

(a) Apply hydraulic fluid, D00153, to the packing [9], the two backup rings [8], the packing [11], and the two backup rings [10].

(b) Install the packing [9], the two backup rings [8], the packing [11], and the two backup rings [10] to the spoiler shutoff valve [12].

SUBTASK 27-61-12-420-002

(2) Carefully install the spoiler shutoff valve assembly in the flight controls hydraulic module housing [2].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Install the four bolts [4] and the four washers [5] to attach the spoiler shutoff valve assembly to the flight controls hydraulic module housing [2].

(b) Tighten each bolt [4] from 30 in-lb (3 N·m) to 40 in-lb (5 N·m).

(c) Install the new MS20995NC32 lockwire, G50347, on the four bolts [4].

(d) Do this task for the electrical connector [1]: Inspection, Cleaning and Sealing of Electrical Connectors, TASK 20-60-07-910-802.

(e) Connect the electrical connector [1] to the spoiler shutoff valve motor [7].

G. Put the Airplane Back to Its Usual Condition

SUBTASK 27-61-12-860-004

(1) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	12	C00204	FLIGHT CONTROL SHUTOFF VALVES SPOILER

SUBTASK 27-61-12-610-001

(2) If it is necessary to do the servicing of the hydraulic reservoirs, then do this task: Hydraulic Reservoir Servicing, TASK 12-12-00-610-801.

SUBTASK 27-61-12-730-001

(3) Do this task: Spoiler Shutoff Valve Test, TASK 27-61-12-400-803.

——— END OF TASK ———

TASK 27-61-12-000-802

4. Spoiler Shutoff Valve Motor Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove the spoiler shutoff valve motor.

B. References

Reference	Title
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Prepare for the Spoiler Shutoff Valve Motor Removal

SUBTASK 27-61-12-480-002

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed on all of the landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-61-12-860-005

(2) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	12	C00204	FLIGHT CONTROL SHUTOFF VALVES SPOILER

E. Spoiler Shutoff Valve Motor Removal

SUBTASK 27-61-12-020-002

(1) Do these steps to remove the spoiler shutoff valve motor [7]:

(a) Disconnect the electrical connector [1] from the spoiler shutoff valve motor [7].

(b) Set the manual override lever on the spoiler shutoff valve motor [7] to position 2.

(c) Remove the lockwire from the screws [6].

(d) Remove the screws [6] that attach the spoiler shutoff valve motor [7] to the spoiler shutoff valve [12].

(e) Remove the spoiler shutoff valve motor [7] from the spoiler shutoff valve [12].

——— END OF TASK ———

TASK 27-61-12-400-802

5. Spoiler Shutoff Valve Motor Installation

(Figure 401)

A. General

(1) This task gives the instructions to install the spoiler shutoff valve motor.

B. References

Reference	Title
20-60-07-910-802	Inspection, Cleaning and Sealing of Electrical Connectors (P/B 201)
27-61-12-400-803	Spoiler Shutoff Valve Test (P/B 501)

C. Consumable Materials

Reference	Description	Specification
G50347	Lockwire - MS20995NC32, Monel - 0.032 Inch (0.8128 mm) Diameter	NASM20995

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
7	Spoiler shutoff valve motor	27-61-12-01-065	PTF ALL
12	Spoiler shutoff valve	27-09-21-01-125	PTF ALL

E. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Spoiler Shutoff Valve Motor Installation

SUBTASK 27-61-12-420-003

(1) Do these steps to install the spoiler shutoff valve motor [7]:

(a) Make sure that the manual override lever that is on the spoiler shutoff valve motor [7] is set to position 2.

(b) Put the spoiler shutoff valve motor [7] on the spoiler shutoff valve [12].

(c) Engage the spoiler shutoff valve motor [7] with the spoiler shutoff valve [12] cam.

(d) Install the screws [6].

(e) Install the MS20995NC32 lockwire, G50347, on the screws [6].

(f) Do this task for the electrical connector [1]: Inspection, Cleaning and Sealing of Electrical Connectors, TASK 20-60-07-910-802.

(g) Connect the electrical connector [1] to the spoiler shutoff valve motor [7].

G. Put the Airplane Back to Its Usual Condition

SUBTASK 27-61-12-860-006

(1) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	12	C00204	FLIGHT CONTROL SHUTOFF VALVES SPOILER

SUBTASK 27-61-12-730-002

(2) Do this task: Spoiler Shutoff Valve Test, TASK 27-61-12-400-803.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SPOILER SHUTOFF VALVE - ADJUSTMENT/TEST

1. General

A. This procedure has this task:

(1) A test of the spoiler shutoff valve.

TASK 27-61-12-400-803

2. Spoiler Shutoff Valve Test

(Figure 501, Figure 502)

A. References

Reference	Title
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)

B. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Prepare for the Operational Test

SUBTASK 27-61-12-860-008

(1) Make sure that this circuit breaker is closed:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	12	C00204	FLIGHT CONTROL SHUTOFF VALVES SPOILER

SUBTASK 27-61-12-860-009

(2) To pressurize hydraulic systems A and B, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

D. Spoiler Shutoff Valve Test

SUBTASK 27-61-12-720-001

(1) Do the test of the spoiler shutoff valve [1]:

(a) Move the two SPOILER switches, on the P5-3 flights control panel, to the OFF position and then to the ON position (Figure 501).

(b) Make sure that the manual override levers [2] of the two spoiler shutoff valves [1] are in position 1 (Figure 502).

(c) Move one of the two aileron control wheels to bleed the hydraulic system.

(d) Make sure that there is no leakage of hydraulic fluid at the spoiler shutoff valves [1].

(e) Move the aileron control wheel to make sure that the flight spoilers operate correctly.

(f) Set the two SPOILER switches to the OFF position.

(g) Make sure that the manual override levers [2] of the two spoiler shutoff valves [1] are in position 2.

(h) Move the aileron control wheel to make sure that the flight spoilers do not operate.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Put the Airplane Back to Its Usual Condition

SUBTASK 27-61-12-860-010

(1) Set the SPOILER switches to the ON position.

SUBTASK 27-61-12-820-001

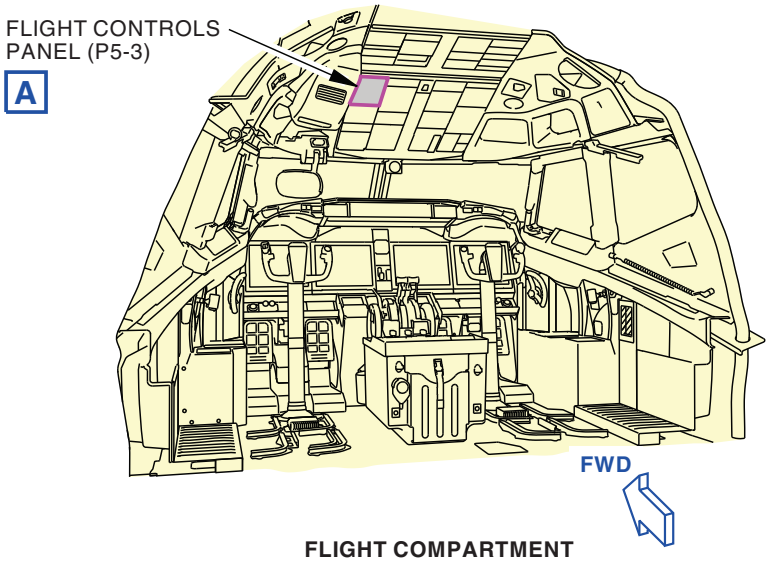
(2) Make sure that the manual override levers [2] of the two spoiler shutoff valves [1] are in position 1 (Figure 502).

SUBTASK 27-61-12-860-011

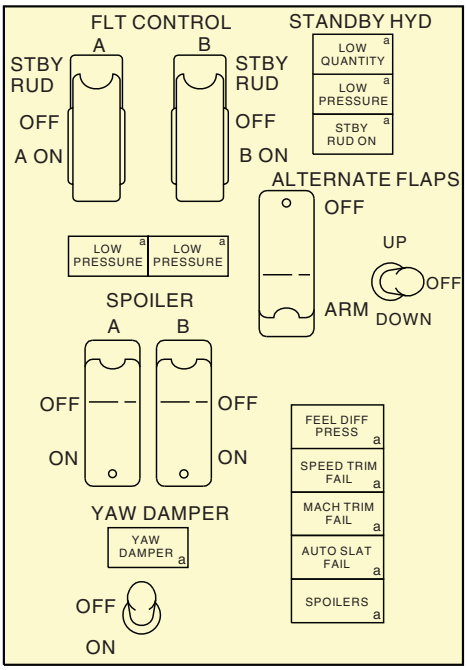
(3) To remove power from hydraulic systems A and B, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



FLIGHT COMPARTMENT



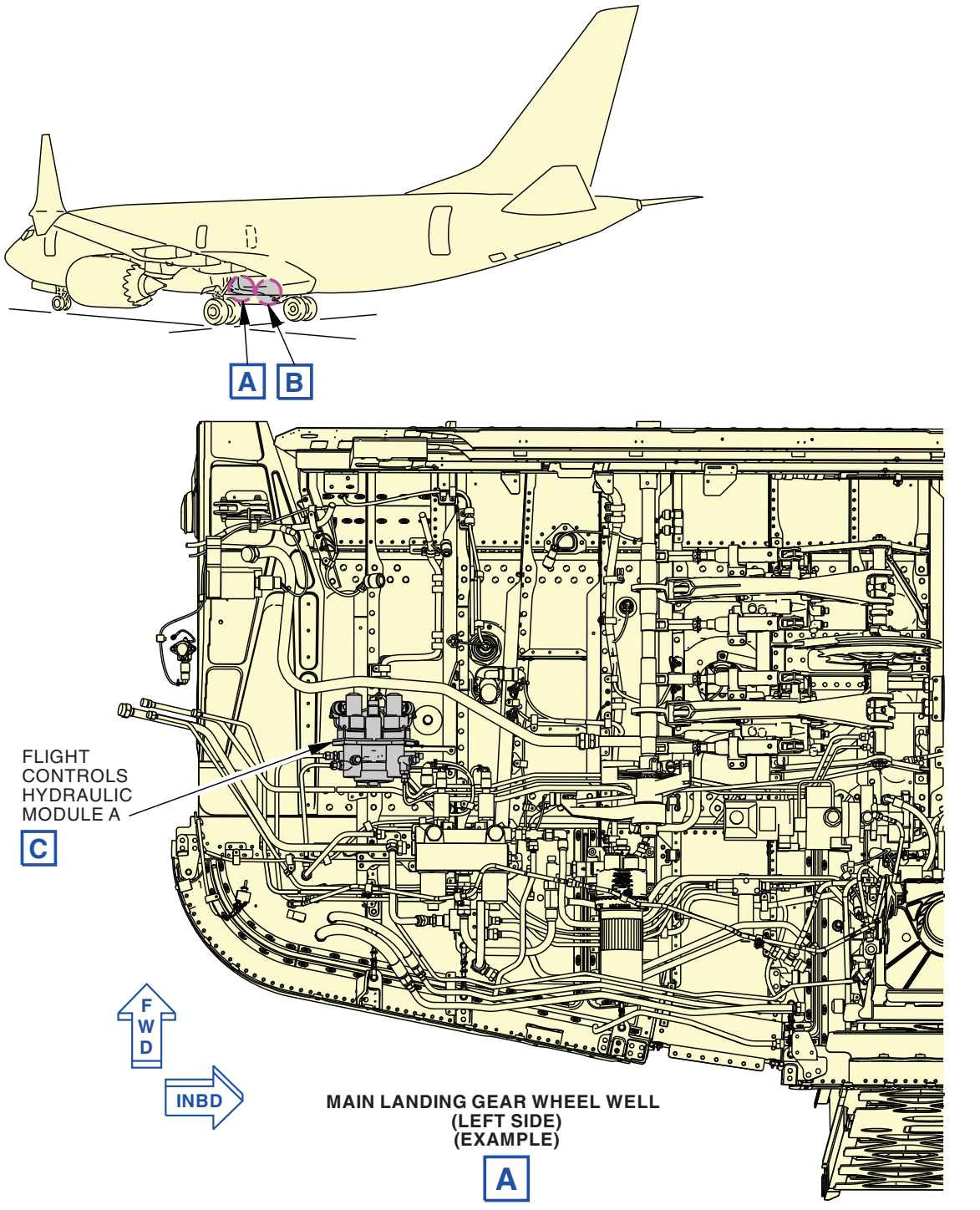
FLIGHT CONTROLS PANEL (P5-3)

A

2413176 S00061532028_V2

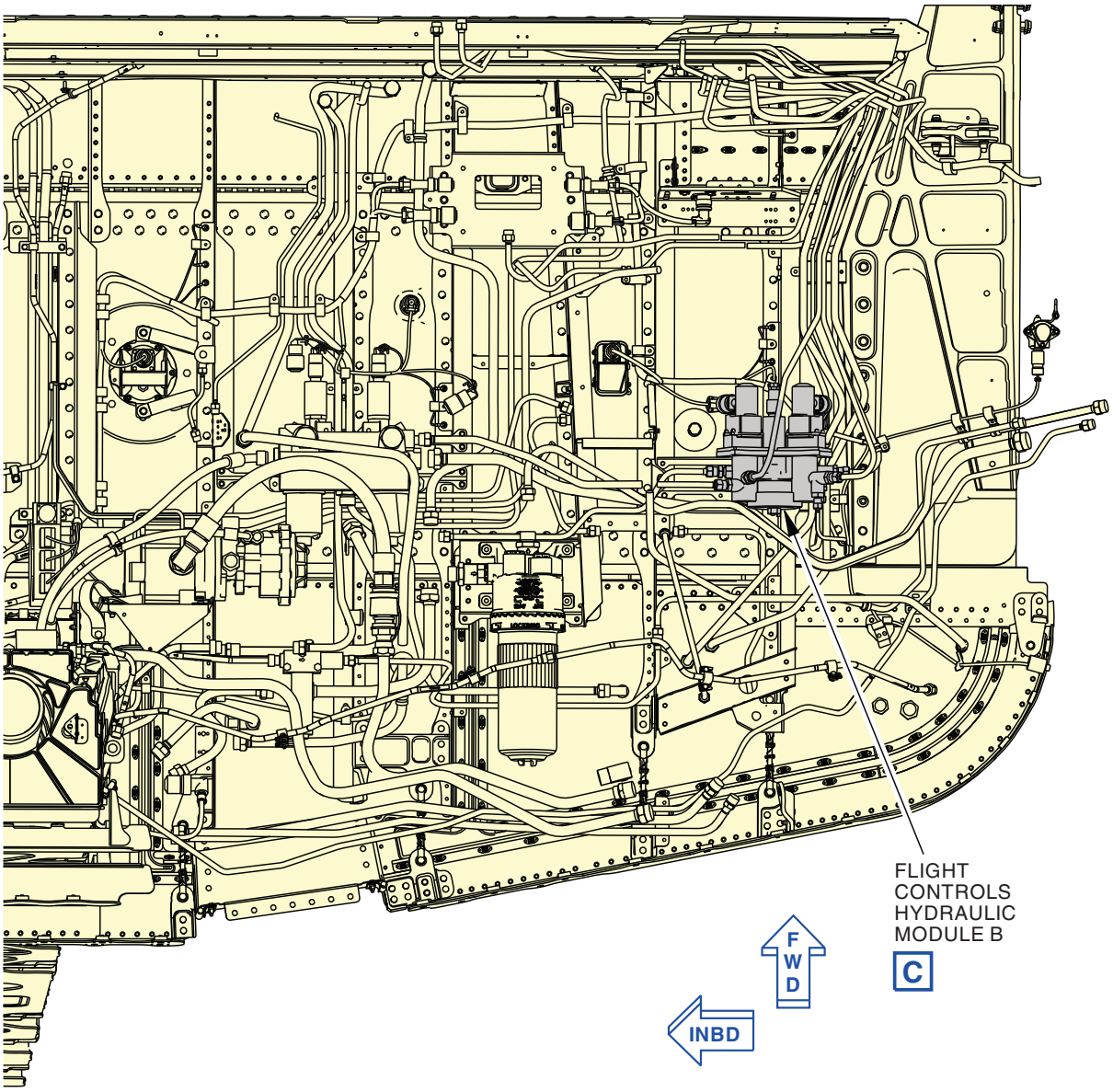
Flight Controls Panel
Figure 501/27-61-12-990-802

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Spoiler Shutoff Valve Test
Figure 502/27-61-12-990-803 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



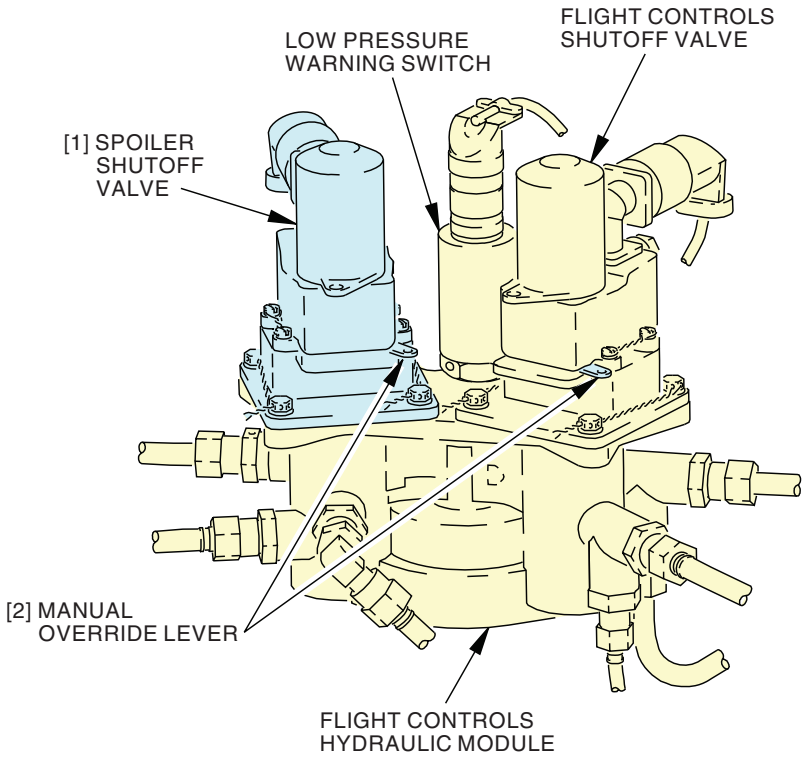
**MAIN LANDING GEAR WHEEL WELL
(RIGHT SIDE)
(EXAMPLE)**

B

2413173 S00061532020_V3

**Spoiler Shutoff Valve Test
Figure 502/27-61-12-990-803 (Sheet 2 of 3)**

ECCN 9E991 BOEING PROPRIETARY - See title page for details



FLIGHT CONTROLS HYDRAULIC MODULE A
(MODULE B IS EQUIVALENT)

C

2413179 S00061532031_V2

Spoiler Shutoff Valve Test
Figure 502/27-61-12-990-803 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FLIGHT SPOILER HINGE LINK - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the flight spoiler hinge links.

(2) An installation of the flight spoiler hinge links.

B. This procedure is applicable to the left and right flight spoiler hinge links.

TASK 27-61-15-000-801

2. Flight Spoiler Hinge Link Removal

(Figure 401)

A. References

Reference	Title
27-51-00-860-803	Extend the Trailing Edge Flaps (P/B 201)
27-61-00-800-801	Apply Pressure to the Spoiler Hydraulic Systems A and B (P/B 201)
27-61-00-800-802	Depressurization of the Spoiler Hydraulic Systems A and B (P/B 201)
27-61-11-000-801	Flight Spoiler Removal (P/B 401)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-14632	Lock Set - Flight Spoiler Actuator Part #: C27107-1 Supplier: 81205

C. Location Zones

Zone	Area
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50
562	Left Wing - Spoiler No. 5
563	Left Wing - Spoiler No. 4
564	Left Wing - Spoiler No. 3
565	Left Wing - Spoiler No. 2
662	Right Wing - Spoiler No. 8
663	Right Wing - Spoiler No. 9
664	Right Wing - Spoiler No. 10
665	Right Wing - Spoiler No. 11

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. Prepare for the Removal

SUBTASK 27-61-15-860-002

WARNING

KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Do this task: Apply Pressure to the Spoiler Hydraulic Systems A and B, TASK 27-61-00-800-801.

SUBTASK 27-61-15-860-003

(2) Do this task: Extend the Trailing Edge Flaps, TASK 27-51-00-860-803.

SUBTASK 27-61-15-860-004

(3) Put the speed brake lever in the UP position to lift the spoilers.

SUBTASK 27-61-15-480-001

(4) Install the lock set, SPL-14632 on the actuator for the applicable flight spoiler.

SUBTASK 27-61-15-860-005

(5) Do this task: Depressurization of the Spoiler Hydraulic Systems A and B, TASK 27-61-00-800-802.

SUBTASK 27-61-15-020-001

(6) Do this task: Flight Spoiler Removal, TASK 27-61-11-000-801.

E. Flight Spoiler Hinge Link Removal

SUBTASK 27-61-15-020-002

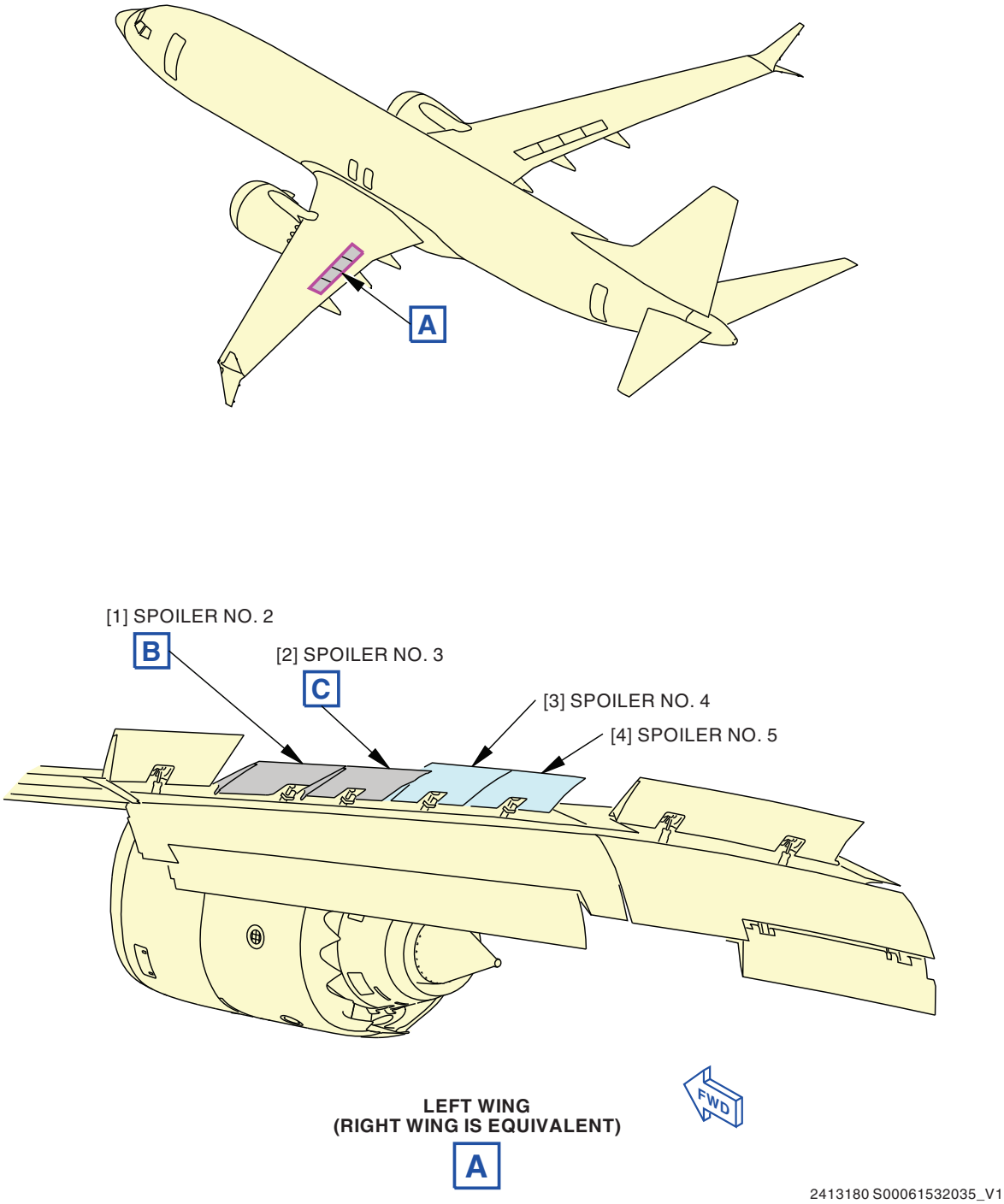
(1) Do these steps to remove the spoiler hinge link [5] from spoiler no. 2 [1], spoiler no. 3 [2], spoiler no. 4 [3] or spoiler no. 5 [4]:

(a) Remove pins [11], nuts [10], washers [9], bushings [8], washers [7] and bolts [6].

(b) Remove the spoiler hinge link [5] from the airplane.

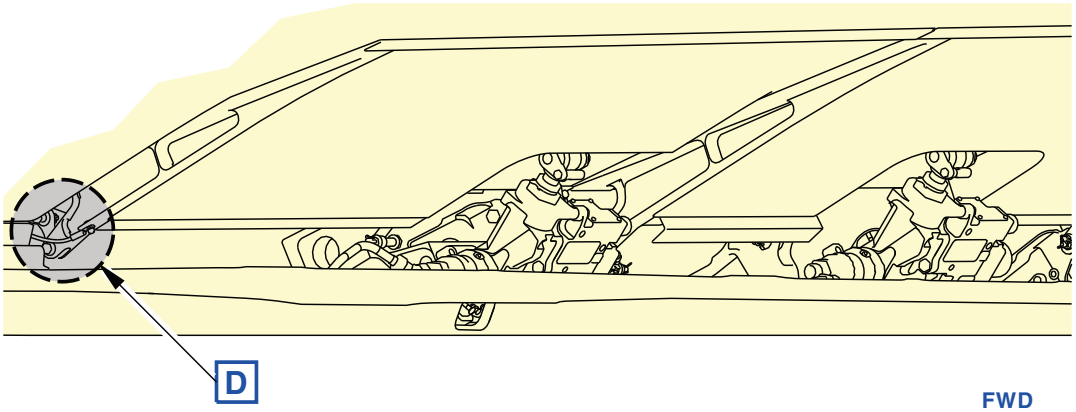
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Flight Spoilers Hinge Link Installation
Figure 401/27-61-15-990-801 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



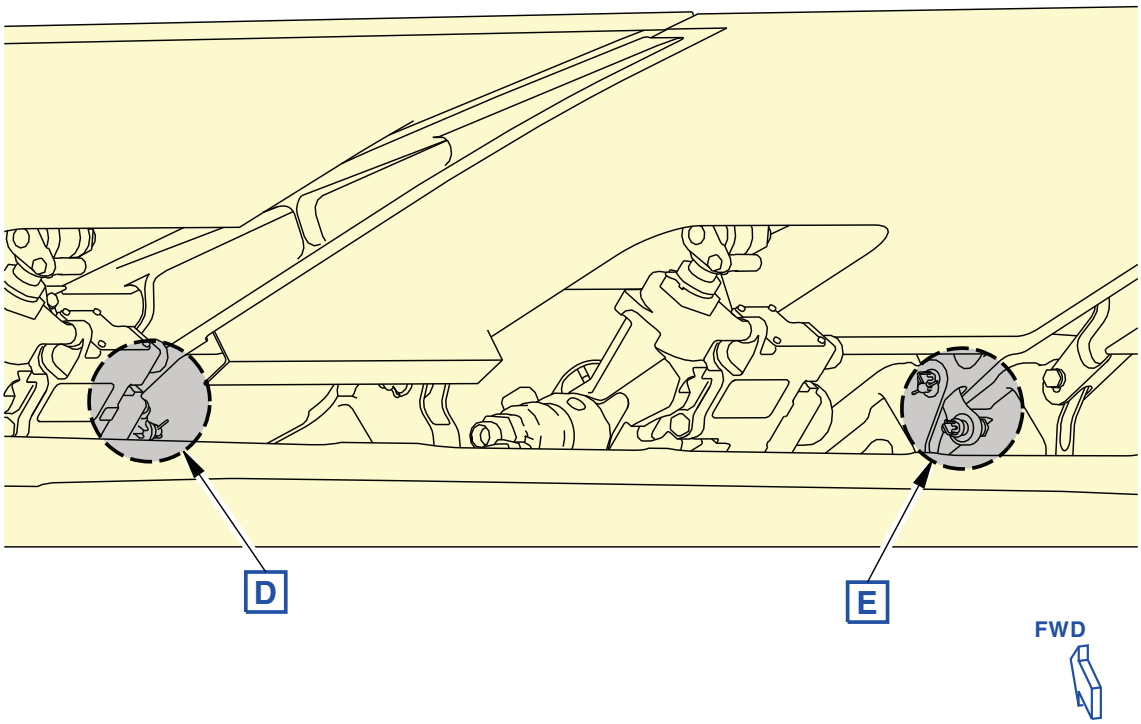
SPOILER NO. 2

B

2413181 S00061532036_V1

Flight Spoilers Hinge Link Installation
Figure 401/27-61-15-990-801 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



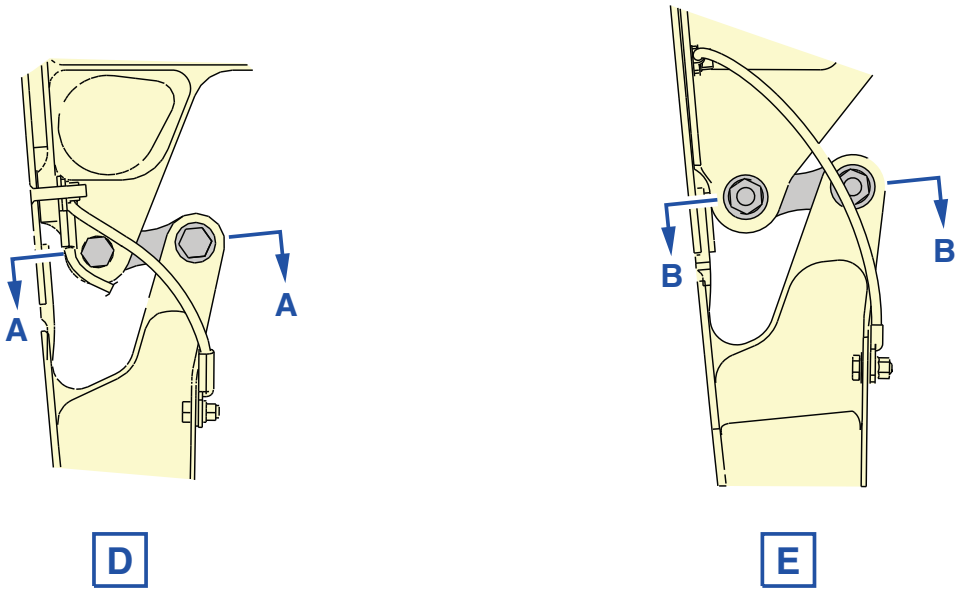
SPOILER NO. 3
(SPOILER NO. 4 AND 5 ARE EQUIVALENT)

C

2413182 S00061532037_V1

Flight Spoilers Hinge Link Installation
Figure 401/27-61-15-990-801 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2413183 S00061532038_V1

Flight Spoilers Hinge Link Installation
Figure 401/27-61-15-990-801 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-61-15-400-801

3. Flight Spoiler Hinge Link Installation

(Figure 401)

A. References

Reference	Title
27-51-00-860-804	Retract the Trailing Edge Flaps (P/B 201)
27-61-00-730-801	Flight Spoiler System Test (P/B 501)
27-61-00-800-801	Apply Pressure to the Spoiler Hydraulic Systems A and B (P/B 201)
27-61-00-820-802	Flight Spoiler Adjustment (P/B 501)
27-61-00-840-801	Shut Down the Spoiler Hydraulic Systems A and B (P/B 201)
27-61-00-840-802	Pressurization of the Spoiler Hydraulic Systems A and B to its Usual Condition (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-14632	Lock Set - Flight Spoiler Actuator Part #: C27107-1 Supplier: 81205

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
5	Spoiler hinge link	27-61-15-01-050	PTF ALL
11	Pin	27-61-15-01-020	PTF ALL

D. Location Zones

Zone	Area
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50
562	Left Wing - Spoiler No. 5
563	Left Wing - Spoiler No. 4
564	Left Wing - Spoiler No. 3
565	Left Wing - Spoiler No. 2
662	Right Wing - Spoiler No. 8
663	Right Wing - Spoiler No. 9
664	Right Wing - Spoiler No. 10
665	Right Wing - Spoiler No. 11

E. Flight Spoiler Hinge Link Installation

SUBTASK 27-61-15-420-001

(1) Install the bolts [6], washers [7], bushings [8], washers [9], nuts [10] and new pins [11] to install the spoiler hinge link [5].

SUBTASK 27-61-15-860-006

(2) Make sure hydraulic power is not on. To remove it, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-61-15-080-001

(3) Remove the lock set, SPL-14632 from the applicable flight spoiler actuator.

SUBTASK 27-61-15-860-007

(4) Do this task: Pressurization of the Spoiler Hydraulic Systems A and B to its Usual Condition, TASK 27-61-00-840-802.

SUBTASK 27-61-15-860-008

WARNING

KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(5) Do this task: Apply Pressure to the Spoiler Hydraulic Systems A and B, TASK 27-61-00-800-801.

SUBTASK 27-61-15-860-009

(6) Move the speed brake lever to the DOWN position to lower the spoilers.

SUBTASK 27-61-15-860-010

(7) Do this task: Retract the Trailing Edge Flaps, TASK 27-51-00-860-804.

SUBTASK 27-61-15-860-011

(8) Do this task: Shut Down the Spoiler Hydraulic Systems A and B, TASK 27-61-00-840-801.

SUBTASK 27-61-15-820-001

(9) Do this task: Flight Spoiler Adjustment, TASK 27-61-00-820-802.

SUBTASK 27-61-15-720-001

(10) Do this task: Flight Spoiler System Test, TASK 27-61-00-730-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CONTROL WHEEL POSITION TRANSDUCER - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the control wheel position transducer.

(2) An installation of the control wheel position transducer.

TASK 27-61-44-000-801

2. Control Wheel Position Transducer Removal

(Figure 401)

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
24-22-00-860-802	Remove Electrical Power (P/B 201)
27-11-00-860-801	Pressure from the Aileron Hydraulic Systems A and B - Deactivation (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Access Panels

Number	Name/Location
112A	Forward Access Door

D. Prepare for the Removal

SUBTASK 27-61-44-480-001

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed on all of the landing gear, then do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 27-61-44-860-001

(2) Do this task: Pressure from the Aileron Hydraulic Systems A and B - Deactivation, TASK 27-11-00-860-801.

SUBTASK 27-61-44-860-002

(3) Do this task: Remove Electrical Power, TASK 24-22-00-860-802.

SUBTASK 27-61-44-860-003

(4) Open this access panel:

Number	Name/Location
112A	Forward Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Control Wheel Position Transducer Removal

SUBTASK 27-61-44-020-001

(1) Do these steps to remove the control wheel position transducer [1]:

(a) Remove the bolt [3], the bolt [4], the four washers [5], the two nuts [6], and the three clamps [2].

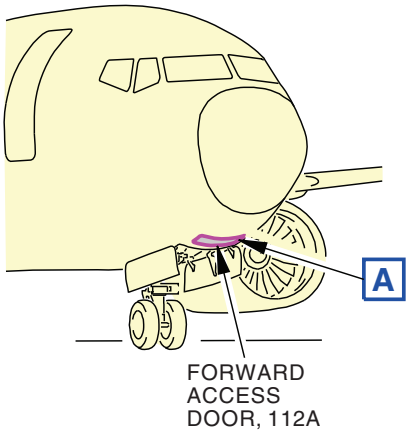
(b) Disconnect the P1 connector, the P2 connector, and the P3 connector.

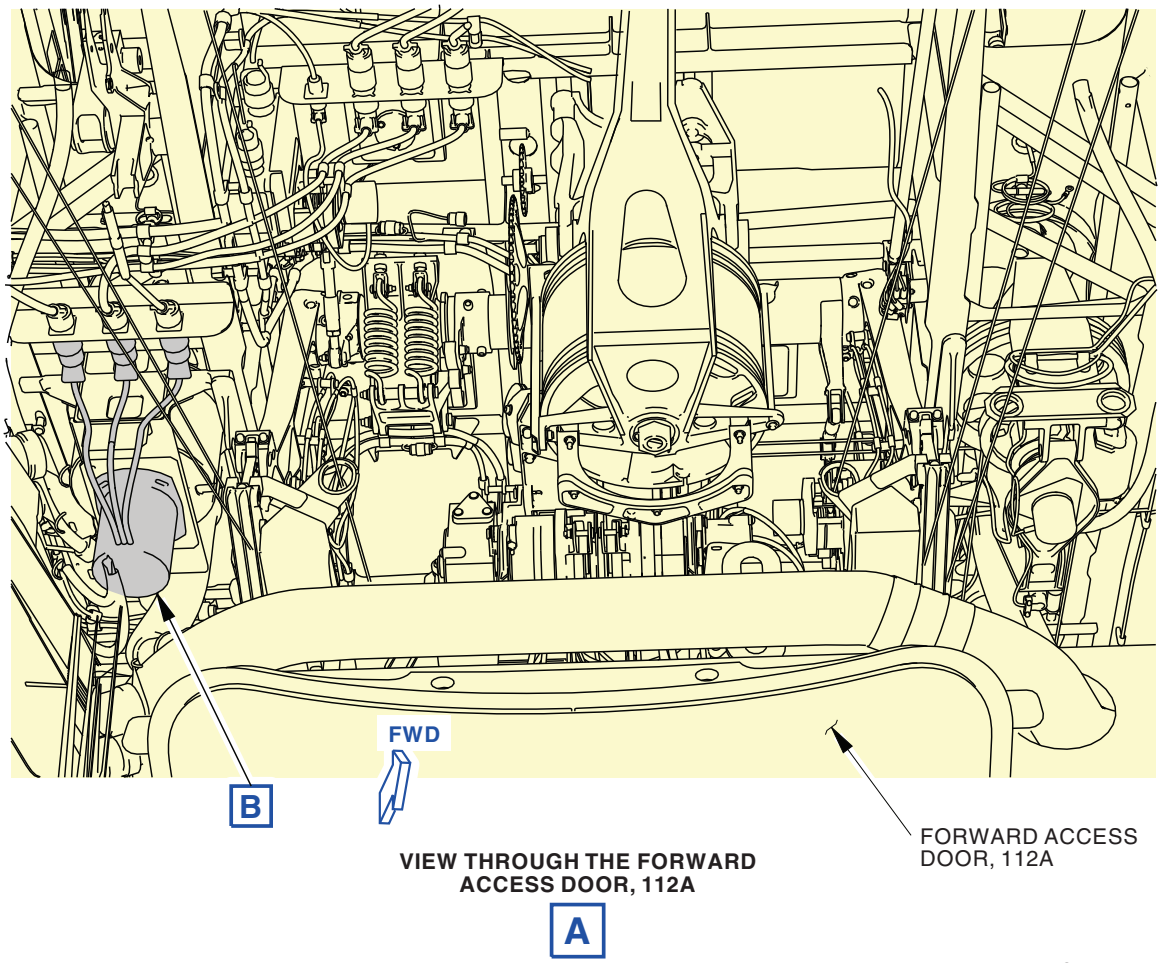
(c) Remove the four nuts [7] and the four washers [8].

(d) Remove the control wheel position transducer [1].

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

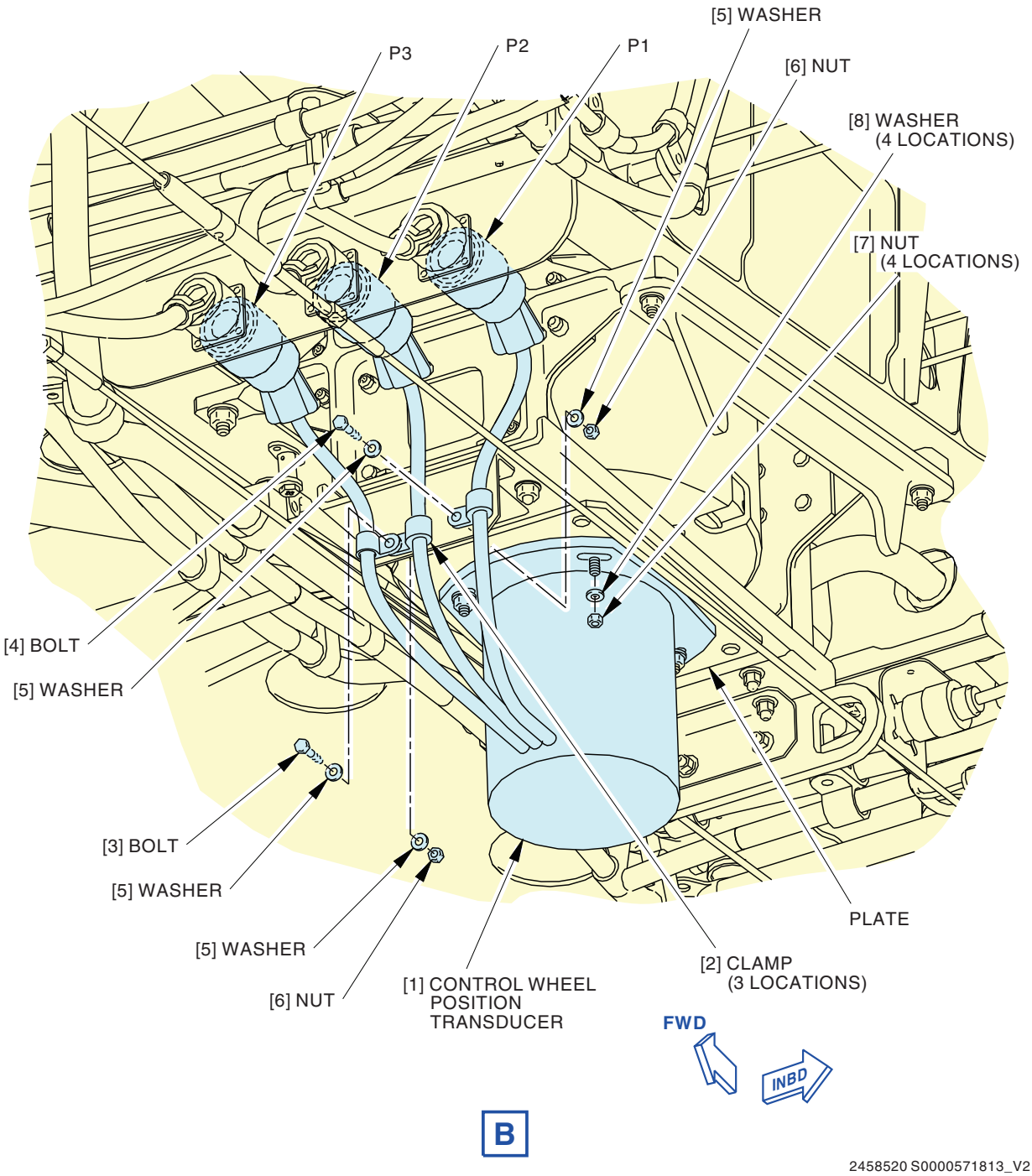




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Control Wheel Position Transducer Installation
Figure 401/27-61-44-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Control Wheel Position Transducer Installation
Figure 401/27-61-44-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-61-44-400-801

3. Control Wheel Position Transducer Installation

(Figure 401)

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
20-60-07-910-802	Inspection, Cleaning and Sealing of Electrical Connectors (P/B 201)
27-11-00-820-801	Aileron Cables Adjustment (P/B 501)
27-61-45-730-803	First Officer's Wheel Transducer Test (P/B 501)

B. Consumable Materials

Reference	Description	Specification
D00633	Grease - Aircraft General Purpose	BMS3-33

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Control wheel position transducer	27-61-44-01-040	PTF ALL

D. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
211	Flight Compartment - Left
212	Flight Compartment - Right

E. Access Panels

Number	Name/Location
112A	Forward Access Door

F. Control Wheel Position Transducer Installation

SUBTASK 27-61-44-420-001

(1) Do these steps to install the control wheel position transducer [1] as follows:

(a) Apply a thin layer of grease, D00633, to the splines of the control wheel position transducer [1].

(b) Align the spline indexes and put the control wheel position transducer [1] in its position.

(c) Install the four washers [8] and the four nuts [7] that attach the control wheel position transducer [1] to the plate.

(d) Tighten the four nuts [7] to 35 ±1 in-lb (4.0 ±0.2 N·m).

(e) For the P1, P2, and P3 connectors, do this task: Inspection, Cleaning and Sealing of Electrical Connectors, TASK 20-60-07-910-802.

(f) Connect the P1, P2, and P3 connectors to their applicable ports.

(g) Install the bolt [3], the bolt [4], the four washers [5], the two nuts [6], and the three clamps [2].

G. Control Wheel Position Transducer Adjustment

SUBTASK 27-61-44-820-002

(1) Do these steps to adjust the control wheel position transducer [1]:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Make sure that there are no faults that are related to the first officer wheel position transducer.

1) If there is fault, do this task: First Officer's Wheel Transducer Test, TASK 27-61-45-730-803.

(b) Use the Onboard Maintenance System (OMS) to show the Flight Controls Maintenance Data page (MAINT DATA PGS).

(c) Find the data indications for the SCE FO CTRL WHL, RVDT A, B, and C.

(d) To adjust the aileron system, do this task: Aileron Cables Adjustment, TASK 27-11-00-820-801.

(e) Loosen the four nuts [7] that attach the control wheel position transducer [1] to the plate.

(f) Make sure that you can turn the control wheel position transducer [1] clockwise and counterclockwise.

(g) Turn the control wheel position transducer [1] until the data indications for all three RVDT channels are 0.00 +/-0.5 degrees.

(h) Tighten the four nuts [7] that attach the control wheel position transducer [1] to the plate to 35 ±1 in-lb (4.0 ±0.2 N·m).

(i) Make sure that all three RVDT channels continue to be 0.00 +/-0.5 degrees.

(j) Turn the control wheel clockwise.

1) Make sure that the data indications of the RVDT channels are positive values.

(k) Turn the control wheel counterclockwise.

1) Make sure that the data indications of the RVDT channels are negative values.

(l) Turn the control wheel clockwise and counterclockwise many times.

(m) Put the control wheel in the neutral position.

(n) To adjust the aileron system, do this task: Aileron Cables Adjustment, TASK 27-11-00-820-801.

(o) Make sure that all three RVDT channels are 0.00 +/-0.5 degrees.

H. Control Wheel Position Transducer Test

SUBTASK 27-61-44-730-001

(1) Do this task: First Officer's Wheel Transducer Test, TASK 27-61-45-730-803.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 27-61-44-410-001

(1) Close this access panel:

Number	Name/Location
112A	Forward Access Door

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SPOILER CONTROL ELECTRONICS - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) Spoiler Control Electronics (SCE) software installation with the Onboard Network System (ONS)

(2) SCE software configuration check.

TASK 27-61-45-470-801

2. SCE Software Installation with Onboard Network System

A. General

(1) This task gives the steps to install the Operational Program Software (OPS) into the Spoiler Control Electronics (SCE), M02940.

(2) Loading the software to the SCE on the airplane is accomplished in the flight deck by using the Portable Maintenance Device (PMD) or the multifunction display (MFD) to access the Onboard Network System (ONS) load function.

(3) The SCE is made up of 3 independent Digital Processing Lanes (DPL). Each lane (DPL 1, DPL 2 and DPL 3) will require independent enabling in order to load software, only one lane can be loaded at a time.

(4) The airplane must be on the ground with the engines shutdown before you can do the software installation.

B. References

Reference	Title
46-13-00-470-801	LRU Software Parts Installation (P/B 201)
46-13-02-710-801	Onboard Maintenance Function Ground Test (P/B 201)

C. Location Zones

Zone	Area
212	Flight Compartment - Right

D. SCE Software Installation with Onboard Network System

SUBTASK 27-61-45-860-034

(1) Make sure that the SYSTEM SELECT switch on the P61 panel is set to NORMAL.

SUBTASK 27-61-45-470-001

(2) Load the required software into SCE Lanes 1, 2 and 3. Do this task: LRU Software Parts Installation, TASK 46-13-00-470-801.

NOTE: Do not cycle circuit breakers C04032 and C04033 before loading software into Lanes 1, 2 and 3. Cycling of circuit breakers C04032 and C04033 before loading software in all three lanes can result in appearance of an incompatible software fault and SCE removal after three load attempts.

(a) One software part is loaded at a time:

1) For SCE Lane 1, load DPL 1 OPS

2) For SCE Lane 2, load DPL 2 OPS

3) For SCE Lane 3, load DPL 3 OPS

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-61-45-710-005

CAUTION

MAKE SURE THAT YOU USE A RELIABLE POWER SOURCE DURING SCE SOFTWARE INSTALLATION. DO NOT DO THE OPS LOAD AT THE SAME TIME WITH OTHER TASKS. SOME TASKS CAN CAUSE A POWER INTERRUPT AND CAN CAUSE OPS LOAD FAILURE. IF THE OPS LOAD TO A SINGLE-LANE FAILS MORE THAN TWICE, REPLACE THE SCE.

(3) If the OPS load to a lane fails twice, look in the display unit for SCE or NSF faults (TASK 46-13-02-710-801).

NOTE: SCE software installation is power sensitive. Power interrupts can cause the OPS load to fail. A failed SCE OPS load attempt occurs when the software upload is in progress and a power interrupt occurs during specific times.

(a) If it is necessary, refer to the IFIM and do the applicable procedure(s) to troubleshoot and correct any SCE or NSF faults that you find.

SUBTASK 27-61-45-700-001

(4) Do this task to verify the software was loaded: SCE Configuration Check, TASK 27-61-45-750-801.

END OF TASK

TASK 27-61-45-750-801

3. SCE Configuration Check

A. General

(1) This procedure gives the steps to verify the Operational Program Software (OPS) software installed in each of 3 SCE Lanes.

(2) The aircraft must be on the ground before you can perform this task.

B. References

Reference	Title
46-13-00-710-802	System Configuration Check (P/B 201)

C. Location Zones

Zone	Area
212	Flight Compartment - Right

D. Procedure

SUBTASK 27-61-45-700-002

(1) Do a check of the each SCE Lane software. Do this task: System Configuration Check, TASK 46-13-00-710-802.

(a) Record and verify that each software part number is correct.

1) Check SCE Lane 1, SCE Lane 2 and SCE Lane 3.

2) If a software part number is not correct, software installation is required. Do this task: SCE Software Installation with Onboard Network System, TASK 27-61-45-470-801.

3) If all software part numbers are correct, then continue.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SPOILER CONTROL ELECTRONICS - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) The removal of the spoiler control electronics unit.

(2) The installation of the spoiler control electronics unit.

TASK 27-61-45-000-801

2. Spoiler Control Electronics Unit Removal

A. References

Reference	Title
20-10-07-000-801	E/E Box Removal (P/B 401)
20-40-12-000-802	ESDS Handling for Metal Encased Unit Removal (P/B 201)

B. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left

C. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

D. Spoiler Control Electronics (SCE) Unit Removal Procedure

SUBTASK 27-61-45-860-001

(1) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	9	C00440	FLIGHT CONTROL AUTO SPEED BRAKE
B	11	C04040	GROUND SPOILER CONTROL

WARNING

KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT SPOILERS. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU OPEN/CLOSE THESE CIRCUIT BREAKERS. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(2) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
C	10	C04032	SPOILER CONTROL ELEC PWR A
C	12	C04033	SPOILER CONTROL ELEC PWR B

SUBTASK 27-61-45-010-001

(3) Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-61-45-020-001

CAUTION

DO NOT TOUCH THE CONNECTOR PINS, OR OTHER CONDUCTORS. IF YOU TOUCH THESE CONDUCTORS, ELECTROSTATIC DISCHARGE CAN CAUSE DAMAGE TO THE COMPONENTS.

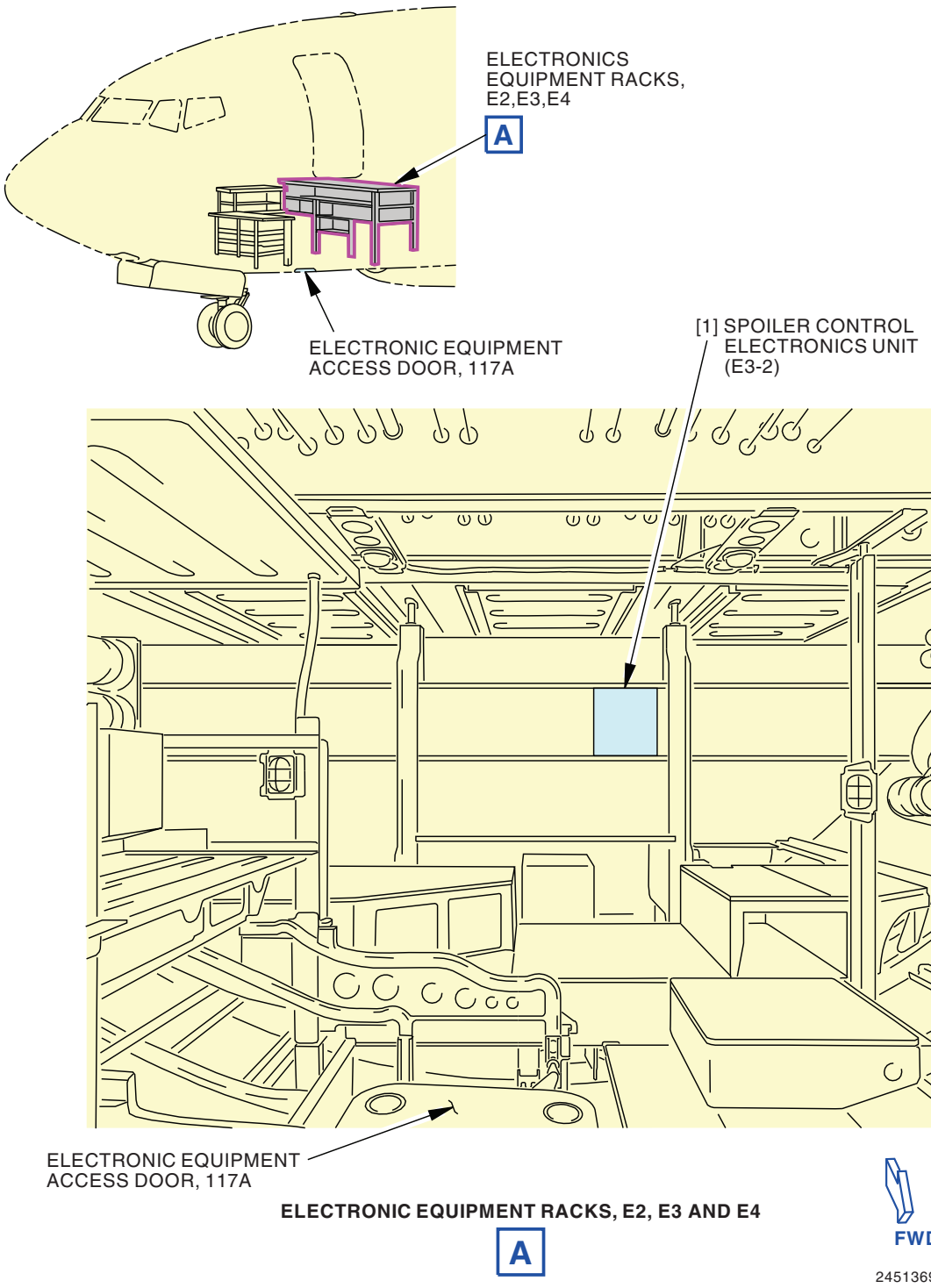
(4) Before you touch the spoiler control electronics unit [1], do this task: ESDS Handling for Metal Encased Unit Removal, TASK 20-40-12-000-802.

SUBTASK 27-61-45-020-002

(5) To remove the spoiler control electronics unit [1], do this task: E/E Box Removal, TASK 20-10-07-000-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Spoiler Control Electronics Unit
Figure 401/27-61-45-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-61-45-400-801

3. Spoiler Control Electronics Unit Installation

A. General

(1) The installation test at the end of this procedure makes sure that the spoiler control electronics unit is installed correctly.

B. References

Reference	Title
20-10-07-400-801	E/E Box Installation (P/B 401)
20-40-12-400-802	ESDS Handling for Metal Encased Unit Installation (P/B 201)
27-61-45-730-801	Spoiler Control Electronics Self Test (P/B 501)
27-61-45-730-802	Speedbrake Lever Transducer Test (P/B 501)
27-61-45-730-803	First Officer's Wheel Transducer Test (P/B 501)
27-61-45-730-804	Ground Spoiler Control Module Replacement Test (P/B 501)
27-61-45-730-805	Flight Spoiler Test (P/B 501)
27-61-45-750-801	SCE Configuration Check (P/B 201)

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Spoiler control electronics unit	27-61-45-01-010	PTF ALL

D. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
211	Flight Compartment - Left
212	Flight Compartment - Right

E. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

F. Spoiler Control Electronics (SCE) Unit Installation

SUBTASK 27-61-45-020-003

CAUTION DO NOT TOUCH THE CONNECTOR PINS, OR OTHER CONDUCTORS. IF YOU TOUCH THESE CONDUCTORS, ELECTROSTATIC DISCHARGE CAN CAUSE DAMAGE TO THE COMPONENTS.

(1) Before you touch the spoiler control electronics unit [1], do this task: ESDS Handling for Metal Encased Unit Installation, TASK 20-40-12-400-802.

SUBTASK 27-61-45-420-001

(2) To install the spoiler control electronics unit [1], do this task: E/E Box Installation, TASK 20-10-07-400-801.

SUBTASK 27-61-45-860-002

(3) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	9	C00440	FLIGHT CONTROL AUTO SPEED BRAKE
B	11	C04040	GROUND SPOILER CONTROL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT SPOILERS. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU OPEN/CLOSE THESE CIRCUIT BREAKERS. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(4) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
C	10	C04032	SPOILER CONTROL ELEC PWR A
C	12	C04033	SPOILER CONTROL ELEC PWR B

SUBTASK 27-61-45-700-003

(5) Do this task to verify the software was loaded: SCE Configuration Check, TASK 27-61-45-750-801.

SUBTASK 27-61-45-410-001

(6) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

G. Spoiler Control Electronics (SCE) Unit Installation Test

SUBTASK 27-61-45-710-002

WARNING KEEP PERSONNEL AND EQUIPMENT AWAY FROM THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(1) Do this task: Spoiler Control Electronics Self Test, TASK 27-61-45-730-801.

SUBTASK 27-61-45-710-003

(2) Do this task: Speedbrake Lever Transducer Test, TASK 27-61-45-730-802.

SUBTASK 27-61-45-710-004

(3) Do this task: First Officer's Wheel Transducer Test, TASK 27-61-45-730-803.

SUBTASK 27-61-45-730-005

(4) Do this task: Ground Spoiler Control Module Replacement Test, TASK 27-61-45-730-804.

SUBTASK 27-61-45-730-006

(5) Do this task: Flight Spoiler Test, TASK 27-61-45-730-805.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 27-61-45-860-004

(1) Set the GND TEST switch to NORM.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SPOILER CONTROL ELECTRONICS - ADJUSTMENT/TEST

1. General

A. This procedure has these tasks:

(1) Spoiler Control Electronics Self Test

(2) Speedbrake Lever Transducer Test

(3) First Officer's Wheel Transducer Test

(4) Ground Spoiler Control Module Replacement Test

(5) Flight Spoiler Test.

TASK 27-61-45-730-801

2. Spoiler Control Electronics Self Test

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
34-21-00-710-801	Air Data Inertial Reference System - Operational Test (P/B 501)
46-13-02-710-801	Onboard Maintenance Function Ground Test (P/B 201)

B. Location Zones

Zone	Area
212	Flight Compartment - Right

C. Prepare for the Test

SUBTASK 27-61-45-860-008

(1) Make sure that the Air Data Inertial Reference Unit (ADIRU) is on and operational (TASK 34-21-00-710-801).

NOTE: Some faults can occur and prevent the test from operating. The ADIRU must be on and operational to reset these faults.

SUBTASK 27-61-45-860-046

(2) Set the LIGHTS switch, on the P1 captain's main instrument panel, to the BRT position.

NOTE: If the LIGHTS switch is not in the BRT position, the Spoiler Control Electronics (SCE) self test can fail.

D. Spoiler Control Electronics (SCE) Self Test

SUBTASK 27-61-45-740-002

(1) Do these steps to do the SCE self test (TASK 46-13-02-710-801):

NOTE: When the Onboard Maintenance Function (OMF) instructions call for hydraulic power, hydraulic power must be controlled at the source, i.e. pumps or ground service. Flight controls A and B, and spoilers A and B switches, on the P5 panel, should remain in the ON position.

NOTE: When the OMF instructions call for the removal of hydraulic power, monitor the A and B system pressures, and wait until system A and B pressures drop below 200 psi (1379 kPa) before continuing.

(a) Set the GND TEST switch, on the P61-4 panel, to the ENABLE position.

(b) Push the SYS button in the Multi Function Display (MFD) section of the Multi Function Panel (MFP).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Use the SELECTOR, on the MFP, to make these selections:

NOTE: If the speedbrake lever is in the ARMED, FLIGHT DETENT, or UP positions, the speedbrake lever can retract to the DOWN position when starting the ground test.

- MAINT
- ONBD MAINT
- LINE MAINT
- GROUND TESTS
- 27 - Spoiler Control
- Operational Test
- SCE Test
- CONTINUE.

(d) Follow the on-screen instructions.

(e) Select the rectangular CONTINUE button.

(f) Use the SELECTOR, on the MFP, to make this selection:

- START TEST.

(g) Make sure that when the test is complete, there are no faults and you see the TEST CONDITION: PASS message.

1) If you see TEST CONDITION: FAILED, refer to the IFIM and do the applicable procedure(s).

2) If you do additional tests, on the MFP, select the NEW TEST button.

E. Put the Airplane Back to Its Usual Condition

SUBTASK 27-61-45-860-009

(1) Set the GND TEST switch to the NORM position.

SUBTASK 27-61-45-860-010

(2) Set the IRS R and IRS L switches on the Inertial Reference System (IRS) Mode Select Unit (MSU), on the P5 panel, to the OFF position.

——— END OF TASK ———

TASK 27-61-45-730-802

3. Speedbrake Lever Transducer Test

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
34-21-00-710-801	Air Data Inertial Reference System - Operational Test (P/B 501)
46-13-02-710-801	Onboard Maintenance Function Ground Test (P/B 201)

B. Location Zones

Zone	Area
212	Flight Compartment - Right

C. Prepare for the Test

SUBTASK 27-61-45-860-048

(1) Make sure that the airplane has weight on wheels.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-61-45-860-013

(2) Make sure that the Air Data Inertial Reference Unit (ADIRU) is on and operational (TASK 34-21-00-710-801).

NOTE: Some faults may occur and prevent the test from running. The ADIRU must be on and operational to reset these faults.

D. Speedbrake Lever Transducer Test

SUBTASK 27-61-45-730-002

(1) Do these steps to do the speedbrake lever transducer test (TASK 46-13-02-710-801):

NOTE: When the Onboard Maintenance Function (OMF) instructions call for Hydraulic Power, Hydraulic Power must be controlled at the source, i.e. Pumps or Ground Service. Flight Controls A and B, and Spoilers A and B switches on the P5 panel should remain ON.

NOTE: When the OMF instructions call for the removal of Hydraulic Power, monitor the A & B system pressures, and wait until System A and B pressures drop below 200 psi (1379 kPa) before continuing.

(a) Set the GND TEST switch on the P61-4 panel to ENABLE.

(b) Push the SYS button in the Multi Function Display (MFD) section of the Multi Function Panel.

(c) Use the SELECTOR on the Multi Function Panel to make these selections:

NOTE: If the speedbrake lever is in the ARMED, FLIGHT DETENT, or UP positions, the speedbrake lever may retract to the DOWN position when starting the ground test.

- MAINT
- ONBD MAINT
- LINE MAINT
- GROUND TESTS
- 27 - Spoiler Control
- Operational Test
- Speedbrake Lever
- CONTINUE

(d) Follow the on-screen instructions and then select the rectangular CONTINUE button.

(e) Use the SELECTOR on the Multi Function Panel to make this selection:

- CONTINUE
- START TEST

(f) Make sure that when the test is complete, there are no faults and you see the TEST CONDITION: PASS message.

1) If you see TEST CONDITION: FAILED, refer to the IFIM and do the applicable procedure(s).

2) If you do additional tests on the Multi Function Panel, select the NEW TEST button.

E. Put the Airplane Back to Its Usual Condition

SUBTASK 27-61-45-860-014

(1) Set the GND TEST switch to NORM.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-61-45-860-015

(2) Set the IRS R and IRS L switches on the IRS Mode Select Unit, located on the P5 panel, to the OFF position.

——— END OF TASK ———

TASK 27-61-45-730-803

4. First Officer's Wheel Transducer Test

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
34-21-00-710-801	Air Data Inertial Reference System - Operational Test (P/B 501)
46-13-02-710-801	Onboard Maintenance Function Ground Test (P/B 201)

B. Location Zones

Zone	Area
212	Flight Compartment - Right

C. Prepare for the Test

SUBTASK 27-61-45-860-047

(1) Make sure that the airplane has weight on wheels.

SUBTASK 27-61-45-860-019

(2) Make sure that the Air Data Inertial Reference Unit (ADIRU) is on and operational (TASK 34-21-00-710-801).

NOTE: Some faults may occur and prevent the test from running. The ADIRU must be on and operational to reset these faults.

D. First Officer's Wheel Transducer Test

SUBTASK 27-61-45-710-001

(1) Do these steps to do the first officers wheel transducer test (TASK 46-13-02-710-801):

NOTE: When the Onboard Maintenance Function (OMF) instructions call for Hydraulic Power, Hydraulic Power must be controlled at the source, i.e. Pumps or Ground Service. Flight Controls A and B, and Spoilers A and B switches on the P5 panel should remain ON.

NOTE: When the OMF instructions call for the removal of Hydraulic Power, monitor the A & B system pressures, and wait until System A and B pressures drop below 200 psi (1379 kPa) before continuing.

(a) Set the GND TEST switch on the P61-4 panel to ENABLE.

(b) Push the SYS button in the Multi Function Display (MFD) section of the Multi Function Panel.

(c) Use the SELECTOR on the Multi Function Panel to make these selections:

NOTE: If the speedbrake lever is in the ARMED, FLIGHT DETENT, or UP positions, the speedbrake lever may retract to the DOWN position when starting the ground test.

- MAINT
- ONBD MAINT
- LINE MAINT

ECCN 9E991 BOEING PROPRIETARY - See title page for details

- GROUND TESTS
- 27 - Spoiler Control
- Operational Test
- Control Wheel
- CONTINUE

(d) Follow the on-screen instructions and then select the rectangular CONTINUE button.

NOTE: During the ground test, you only have to release the control wheel to allow the control wheel transducers to return to the neutral position. The control wheel may not be centred.

(e) Use the SELECTOR on the Multi Function Panel to make these selections:

- START TEST
- CONTINUE

(f) Make sure that when the test is complete, there are no faults and you see the TEST CONDITION: PASS message.

1) If you see TEST CONDITION: FAILED, refer to the IFIM and do the applicable procedure(s).

2) If you do additional tests on the Multi Function Panel, select the NEW TEST button.

E. Put the Airplane Back to Its Usual Condition

SUBTASK 27-61-45-860-020

(1) Set the GND TEST switch to NORM.

SUBTASK 27-61-45-860-021

(2) Set the IRS R and IRS L switches on the IRS Mode Select Unit, located on the P5 panel, to the OFF position.

——— END OF TASK ———

TASK 27-61-45-730-804

5. Ground Spoiler Control Module Replacement Test

A. References

Reference	Title
34-21-00-710-801	Air Data Inertial Reference System - Operational Test (P/B 501)
46-13-02-710-801	Onboard Maintenance Function Ground Test (P/B 201)

B. Location Zones

Zone	Area
212	Flight Compartment - Right

C. Prepare for the Test

SUBTASK 27-61-45-860-025

(1) Make sure that the Air Data Inertial Reference Unit (ADIRU) is on and operational (TASK 34-21-00-710-801).

NOTE: Some faults may occur and prevent the test from running. The ADIRU must be on and operational to reset these faults.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. Ground Spoiler Control Module Replacement Test

SUBTASK 27-61-45-730-003

(1) Do these steps to do the ground spoiler control module replacement test (TASK 46-13-02-710-801):

NOTE: When the Onboard Maintenance Function (OMF) instructions call for Hydraulic Power, Hydraulic Power must be controlled at the source, i.e. Pumps or Ground Service. Flight Controls A and B, and Spoilers A and B switches on the P5 panel should remain ON.

NOTE: When the OMF instructions call for the removal of Hydraulic Power, monitor the A & B system pressures, and wait until System A and B pressures drop below 200 psi (1379 kPa) before continuing.

(a) Set the GND TEST switch on the P61-4 panel to ENABLE.

(b) Push the SYS button in the Multi Function Display (MFD) section of the Multi Function Panel.

(c) Use the SELECTOR on the Multi Function Panel to make these selections:

NOTE: If the speedbrake lever is in the ARMED, FLIGHT DETENT, or UP positions, the speedbrake lever may retract to the DOWN position when starting the ground test.

- MAINT
- ONBD MAINT
- LINE MAINT
- GROUND TESTS
- 27 - Spoiler Control
- System Test
- Ground Spoilers
- CONTINUE

(d) Follow the on-screen instructions and then select the rectangular CONTINUE button.

(e) Use the SELECTOR on the Multi Function Panel to make these selections:

- CONTINUE
- START TEST

(f) Follow the second set of on-screen instructions and then select the rectangular CONTINUE button.

(g) Make sure that when the test is complete, there are no faults and you see the TEST CONDITION: PASS message.

1) If you see TEST CONDITION: FAILED, refer to the IFIM and do the applicable procedure(s).

2) If you do additional tests on the Multi Function Panel, select the NEW TEST button.

E. Put the Airplane Back to Its Usual Condition

SUBTASK 27-61-45-860-026

(1) Set the GND TEST switch to NORM.

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SUBTASK 27-61-45-860-027

(2) Set the IRS R and IRS L switches on the IRS Mode Select Unit, located on the P5 panel, to the OFF position.

——— END OF TASK ———

TASK 27-61-45-730-805

6. Flight Spoiler Test

A. References

Reference	Title
34-21-00-710-801	Air Data Inertial Reference System - Operational Test (P/B 501)
46-13-02-710-801	Onboard Maintenance Function Ground Test (P/B 201)

B. Location Zones

Zone	Area
212	Flight Compartment - Right

C. Prepare for the Test

SUBTASK 27-61-45-860-031

(1) Make sure that the Air Data Inertial Reference Unit (ADIRU) is on and operational (TASK 34-21-00-710-801).

NOTE: Some faults may occur and prevent the test from running. The ADIRU must be on and operational to reset these faults.

D. Flight Spoiler Test

SUBTASK 27-61-45-730-004

(1) Do these steps to do the flight spoiler test (TASK 46-13-02-710-801):

NOTE: When the Onboard Maintenance Function (OMF) instructions call for Hydraulic Power, Hydraulic Power must be controlled at the source, i.e. Pumps or Ground Service. Flight Controls A and B, and Spoilers A and B switches on the P5 panel should remain ON.

NOTE: When the OMF instructions call for the removal of Hydraulic Power, monitor the A & B system pressures, and wait until System A and B pressures drop below 200 psi (1379 kPa) before continuing.

(a) Set the GND TEST switch on the P61-4 panel to ENABLE.

(b) Push the SYS button in the Multi Function Display (MFD) section of the Multi Function Panel.

(c) Use the SELECTOR on the Multi Function Panel to make these selections:

NOTE: If the speedbrake lever is in the ARMED, FLIGHT DETENT, or UP positions, the speedbrake lever may retract to the DOWN position when starting the ground test.

- MAINT
- ONBD MAINT
- LINE MAINT
- GROUND TESTS
- 27 - Spoiler Control
- System Test

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- Flight Spoilers
- CONTINUE

(d) Follow the on-screen instructions and then select the rectangular CONTINUE button.

(e) Use the SELECTOR on the Multi Function Panel to make these selections:

- CONTINUE
- START TEST

(f) Make sure that when the test is complete, there are no faults and you see the TEST CONDITION: PASS message.

1) If you see TEST CONDITION: FAILED, refer to the IFIM and do the applicable procedure(s).

2) If you do additional tests on the Multi Function Panel, select the NEW TEST button.

E. Put the Airplane Back to Its Usual Condition

SUBTASK 27-61-45-860-032

(1) Set the GND TEST switch to NORM.

SUBTASK 27-61-45-860-033

(2) Set the IRS R and IRS L switches on the IRS Mode Select Unit, located on the P5 panel, to the OFF position.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FLIGHT SPOILER ACTUATOR - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the flight spoiler actuator.

(2) An installation of the flight spoiler actuator.

(3) A removal of the electrohydraulic servo valve.

(4) An installation of the electrohydraulic servo valve.

B. In this procedure the flight spoiler actuator is referred to as the "actuator".

TASK 27-61-51-000-801

2. Flight Spoiler Actuator - Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove the flight spoiler actuator.

B. References

Reference	Title
27-51-00-040-801	Trailing Edge Flap System Deactivation (P/B 201)
27-51-00-860-803	Extend the Trailing Edge Flaps (P/B 201)
27-61-00-800-802	Depressurization of the Spoiler Hydraulic Systems A and B (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-14632	Lock Set - Flight Spoiler Actuator Part #: C27107-1 Supplier: 81205
STD-1110	Container - Hydraulic Fluid Resistant, 5 Gallon (19 Liter)
STD-14686	Fittings - Hydraulic, Caps/Plugs (ref. AMM 20-10-XX)

D. Location Zones

Zone	Area
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50
562	Left Wing - Spoiler No. 5
563	Left Wing - Spoiler No. 4
564	Left Wing - Spoiler No. 3
565	Left Wing - Spoiler No. 2
662	Right Wing - Spoiler No. 8
663	Right Wing - Spoiler No. 9
664	Right Wing - Spoiler No. 10
665	Right Wing - Spoiler No. 11

E. Prepare for the Removal

SUBTASK 27-61-51-860-002

(1) Do this task: Extend the Trailing Edge Flaps, TASK 27-51-00-860-803.

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SUBTASK 27-61-51-040-001

(2) Do this task: Trailing Edge Flap System Deactivation, TASK 27-51-00-040-801.

SUBTASK 27-61-51-860-021

(3) Put the speed brake lever in the UP position to lift the flight spoilers [16].

SUBTASK 27-61-51-480-002

(4) Install the lock set, SPL-14632, on the actuator [15] for the applicable flight spoiler [16].

SUBTASK 27-61-51-860-003

(5) Do this task: Depressurization of the Spoiler Hydraulic Systems A and B, TASK 27-61-00-800-802.

SUBTASK 27-61-51-862-001

(6) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	11	C04040	GROUND SPOILER CONTROL
B	12	C00204	FLIGHT CONTROL SHUTOFF VALVES SPOILER

SUBTASK 27-61-51-860-023

WARNING

KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT SPOILERS. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU OPEN/CLOSE THESE CIRCUIT BREAKERS. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(7) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
C	10	C04032	SPOILER CONTROL ELEC PWR A
C	12	C04033	SPOILER CONTROL ELEC PWR B

F. Actuator Removal

SUBTASK 27-61-51-030-001

WARNING

KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT SPOILERS. SUDDEN MOVEMENT CAN OCCUR WHEN YOU DISCONNECT THE OPPOSITE SPOILERS PCU-ELECTRICAL-CONNECTOR. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Disconnect the electrical connector [25] from the actuator [15].

SUBTASK 27-61-51-020-019

(2) For the spoilers No.3, 4, and 8, disconnect the elbow [35] and hydraulic line [32] from the actuator [15], do these steps:

(a) Put a 5 gallon (19 liter) hydraulic fluid resistant container, STD-1110, below the actuator [15] to catch the hydraulic fluid.

(b) Disconnect the elbow [35] and hydraulic line [32] from the actuator [15].

(c) If it is necessary, disconnect the elbow [35] from the hydraulic line [12].

SUBTASK 27-61-51-020-020

(3) For the spoilers No.2, 5, 10, and 11, disconnect the elbow [35] and elbow [36] from the actuator [15], do these steps:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Put a 5 gallon (19 liter) hydraulic fluid resistant container, STD-1110, below the actuator [15] to catch the hydraulic fluid.

(b) Disconnect the elbow [35] and elbow [36] from the actuator [15].

(c) If it is necessary, disconnect the elbow [35] from the hydraulic line [41].

(d) If it is necessary, disconnect the elbow [36] from the hydraulic line [42].

SUBTASK 27-61-51-020-021

(4) For the spoiler No.9, disconnect the tee fitting [40] and hydraulic line [38] from the actuator [15], do these steps:

(a) Put a 5 gallon (19 liter) hydraulic fluid resistant container, STD-1110, below the actuator [15] to catch the hydraulic fluid.

(b) Disconnect the tee fitting [40] and hydraulic line [38] from the actuator [15].

(c) If it is necessary, disconnection the tee fitting [40] from the hydraulic line [37] and hydraulic line [39].

SUBTASK 27-61-51-020-002

(5) Disconnect the actuator rod end [17] from the flight spoiler [16], do these steps:

(a) Remove the lockwire [23], bolt [1], washer [2], washer [4], and pin [5] from the rod end [17].

(b) Remove the nut [7], washer [6], and pin [3] that attach the actuator rod end [17] to the flight spoiler [16].

SUBTASK 27-61-51-020-003

(6) Disconnect the actuator [15] from the structure, do these steps:

(a) Remove the bolts [20] and washers [21].

(b) Remove the bolts [18] and washers [19].

SUBTASK 27-61-51-020-004

(7) Remove the actuator [15] from the airplane.

SUBTASK 27-61-51-020-024

(8) If it is necessary, remove the union [33] and O-ring [34].

SUBTASK 27-61-51-020-025

(9) If it is necessary, remove the union [13] and O-ring [14].

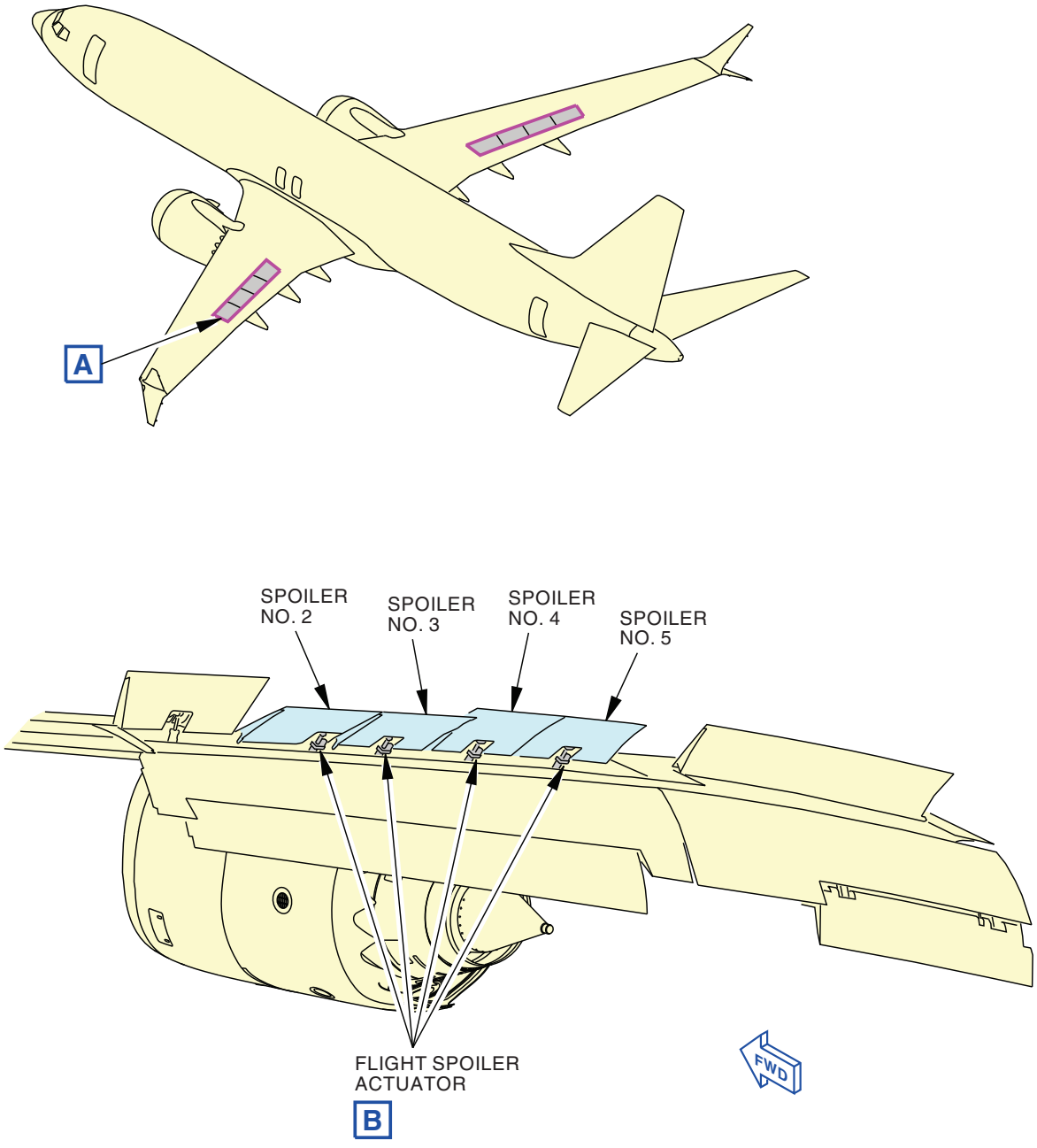
(a) If it is necessary, remove the washer [30] and filter [29].

SUBTASK 27-61-51-480-005

(10) Install the caps/plugs, STD-14686, to the hydraulic lines and ports.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



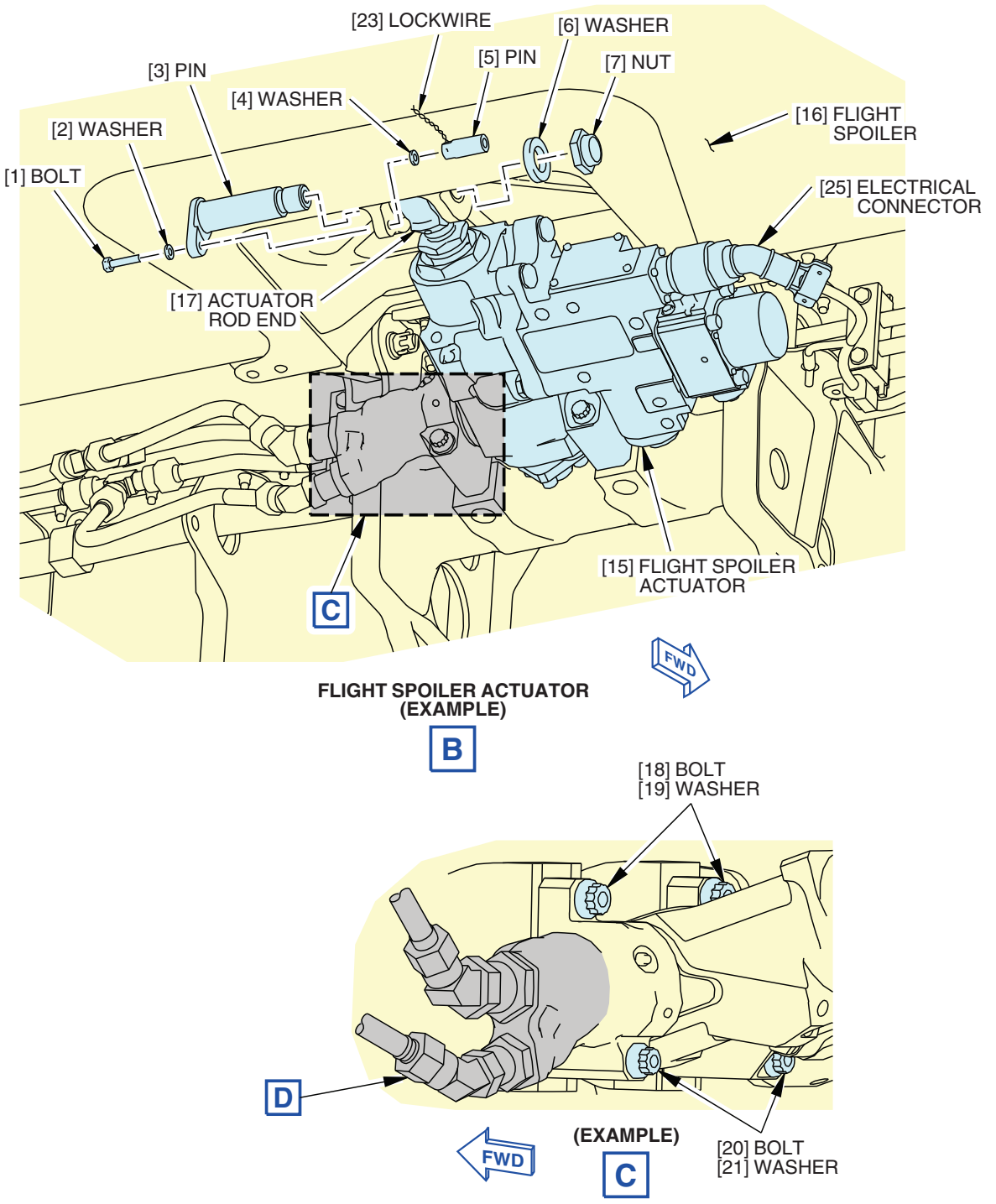
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A

2413203 S00061532081_V1

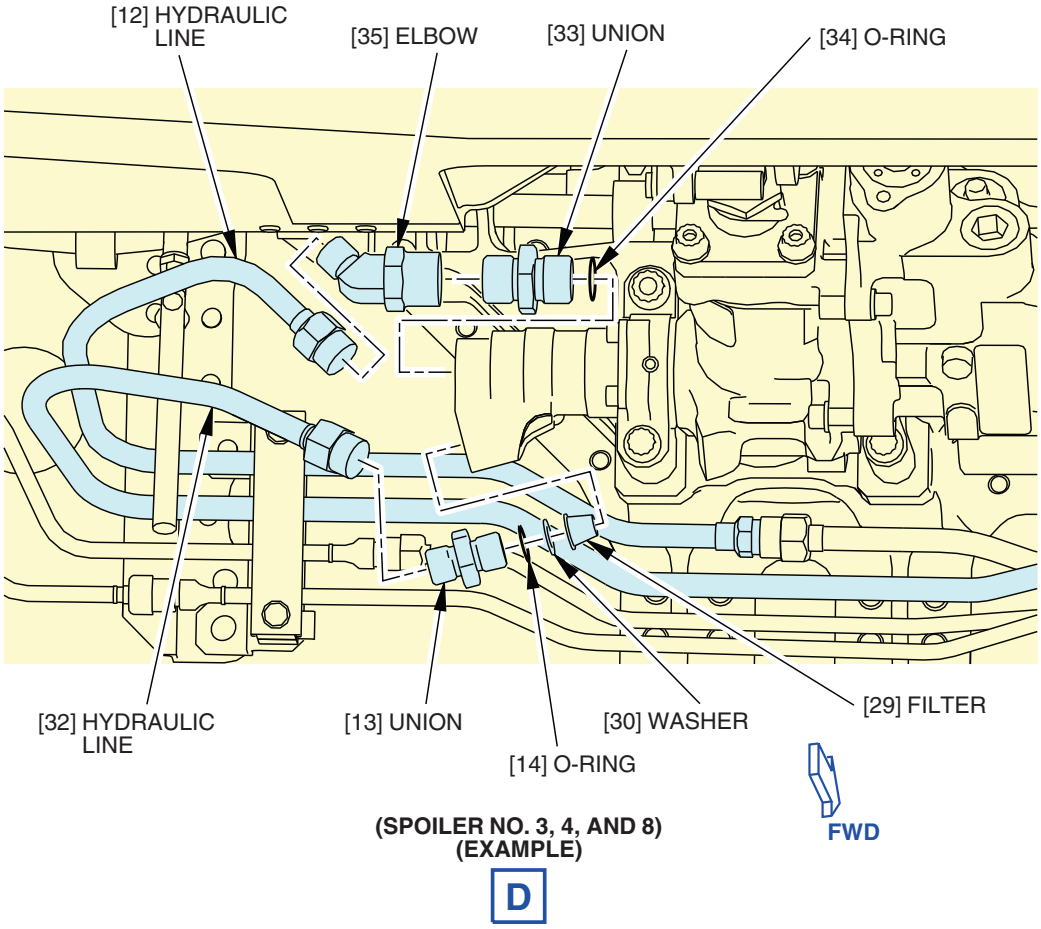
Flight Spoiler Actuator Installation
Figure 401/27-61-51-990-801 (Sheet 1 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Flight Spoiler Actuator Installation
Figure 401/27-61-51-990-801 (Sheet 2 of 5)

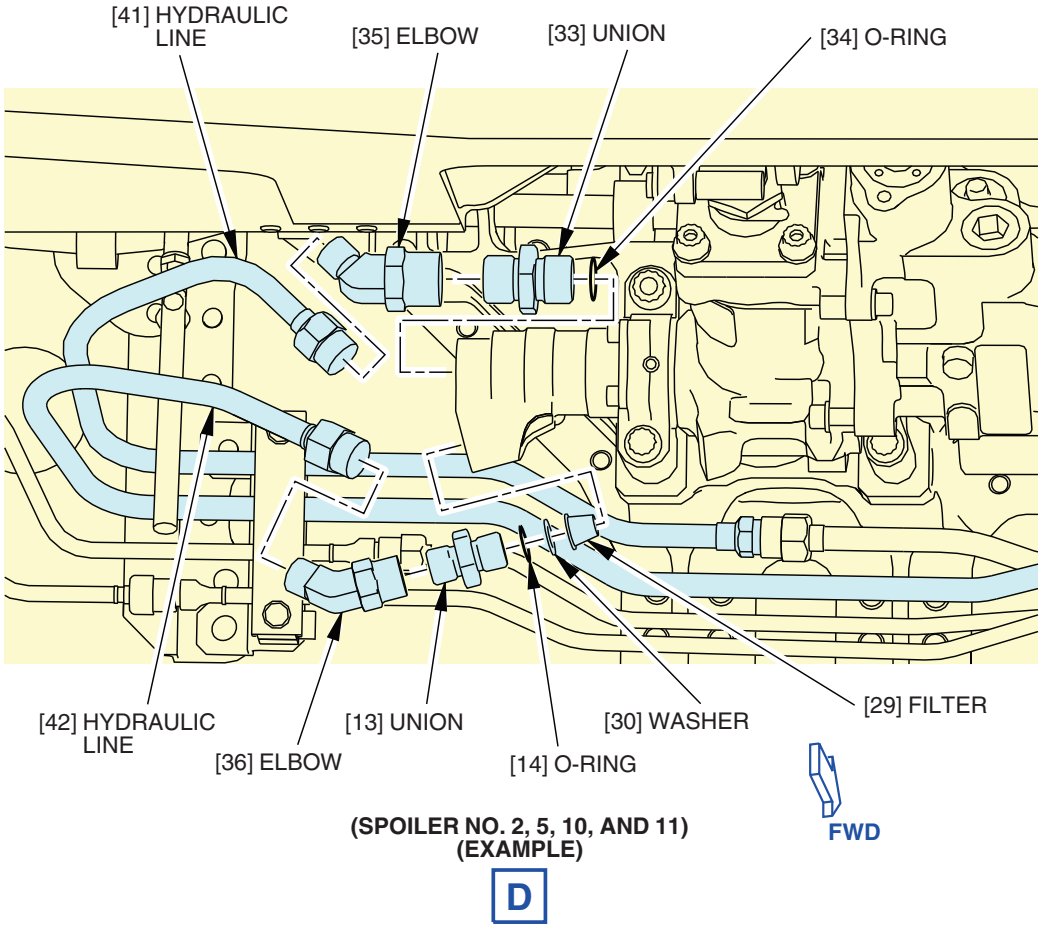
ECCN 9E991 BOEING PROPRIETARY - See title page for details



3097323 S0000844744_V1

Flight Spoiler Actuator Installation
Figure 401/27-61-51-990-801 (Sheet 3 of 5)

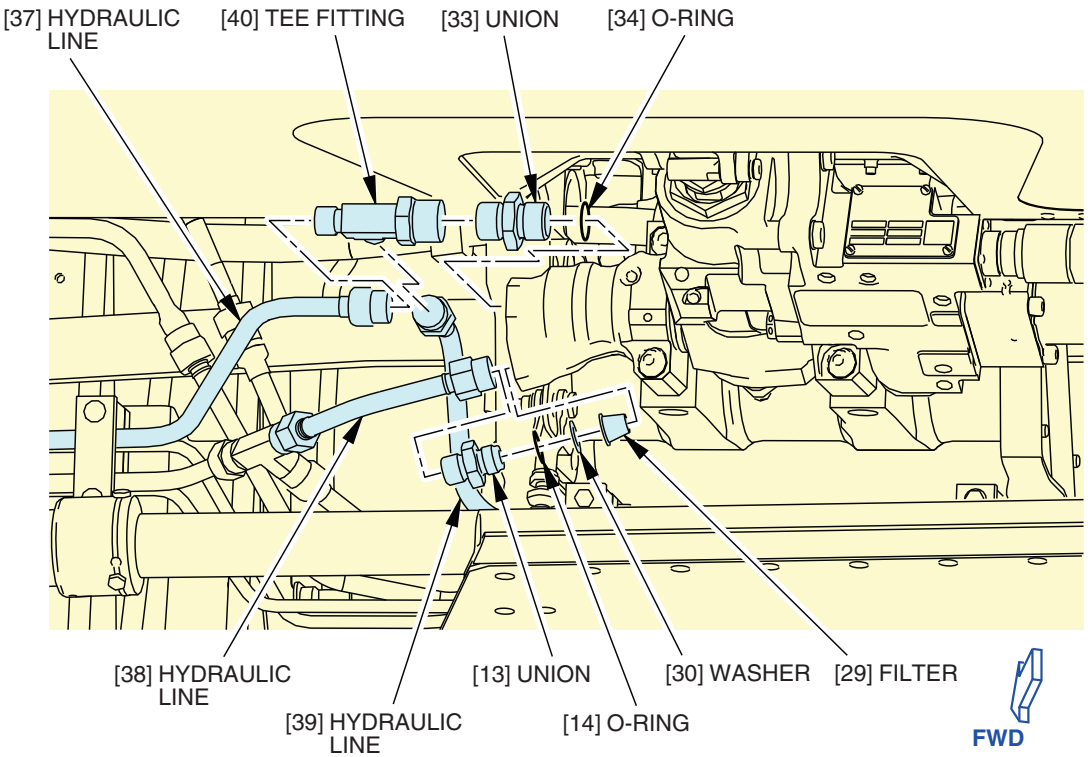
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2985126 S0000763087_V2

Flight Spoiler Actuator Installation
Figure 401/27-61-51-990-801 (Sheet 4 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



(SPOILER NO. 9)
(EXAMPLE)

D

2985136 S0000763088_V2

Flight Spoiler Actuator Installation
Figure 401/27-61-51-990-801 (Sheet 5 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-61-51-400-801

3. Flight Spoiler Actuator - Installation

(Figure 401)

A. General

(1) This task gives the instructions to install the flight spoiler actuator.

B. References

Reference	Title
12-12-00-610-801	Hydraulic Reservoir Servicing (P/B 301)
12-22-61-600-802	Flight Spoiler Actuator and Rod End Lubrication (P/B 301)
20-60-07-910-802	Inspection, Cleaning and Sealing of Electrical Connectors (P/B 201)
27-51-00-440-801	Trailing Edge Flap System Reactivation (P/B 201)
27-51-00-860-804	Retract the Trailing Edge Flaps (P/B 201)
27-61-00-730-801	Flight Spoiler System Test (P/B 501)
27-61-00-800-801	Apply Pressure to the Spoiler Hydraulic Systems A and B (P/B 201)
27-61-00-820-802	Flight Spoiler Adjustment (P/B 501)
27-61-00-840-801	Shut Down the Spoiler Hydraulic Systems A and B (P/B 201)
27-61-00-840-802	Pressurization of the Spoiler Hydraulic Systems A and B to its Usual Condition (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-14632	Lock Set - Flight Spoiler Actuator Part #: C27107-1 Supplier: 81205
STD-14686	Fittings - Hydraulic, Caps/Plugs (ref. AMM 20-10-XX)

D. Consumable Materials

Reference	Description	Specification
D00153	Fluid - Hydraulic Fluid, Fire Resistant (Interchangeable And Intermixable With BMS 3-11 Type V)	BMS3-11 Type IV
D00633	Grease - Aircraft General Purpose	BMS3-33
G50136	Compound - Corrosion Inhibiting, Non-drying Paste	BMS3-38
G50347	Lockwire - MS20995NC32, Monel - 0.032 Inch (0.8128 mm) Diameter	NASM20995

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
14	O-ring	27-61-51-50-325	PTF ALL
15	Actuator	27-61-51-01-105	PTF ALL
29	Filter	27-61-51-50-315	PTF ALL
34	O-ring	27-61-51-50-340	PTF ALL

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F. Location Zones

Zone	Area
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50
562	Left Wing - Spoiler No. 5
563	Left Wing - Spoiler No. 4
564	Left Wing - Spoiler No. 3
565	Left Wing - Spoiler No. 2
662	Right Wing - Spoiler No. 8
663	Right Wing - Spoiler No. 9
664	Right Wing - Spoiler No. 10
665	Right Wing - Spoiler No. 11

G. Actuator Installation

SUBTASK 27-61-51-080-003

(1) Remove the cap/plug, STD-14686, from the hydraulic lines and ports.

SUBTASK 27-61-51-420-016

(2) If it is necessary, install the union [13].

(a) If it is necessary, install the new filter [29] and washer [30].

(b) Apply hydraulic fluid, D00153, to the new O-ring [14].

(c) Install the O-ring [14] and union [13] in the actuator [15].

1) Tighten the union [13] to 215 in-lb (24.3 N·m) - 225 in-lb (25.4 N·m) more than the run-on torque.

SUBTASK 27-61-51-420-017

(3) If it is necessary, install the union [33].

(a) Apply hydraulic fluid, D00153, to the new O-ring [34].

(b) Install the O-ring [34] and union [33] in the actuator [15].

1) Tighten the union [33] to 350 in-lb (39.5 N·m) - 370 in-lb (41.8 N·m) more than the run-on torque.

SUBTASK 27-61-51-420-018

(4) For the spoilers No.3, 4, and 8, if the elbow [35] was removed, loosely install the elbow [35] to the hydraulic line [12].

SUBTASK 27-61-51-420-019

(5) For the spoilers No.2, 5, 10, and 11, if the elbow [35] was removed, loosely install the elbow [35] to the hydraulic line [41].

SUBTASK 27-61-51-420-020

(6) For the spoilers No.2, 5, 10, and 11, if the elbow [36] was removed, loosely install the elbow [36] to the hydraulic line [42].

SUBTASK 27-61-51-420-021

(7) For the spoilers No.9, if the tee fitting [40] was removed, loosely install the tee fitting [40] to the hydraulic line [37] and hydraulic line [39].

SUBTASK 27-61-51-420-028

(8) Put the actuator [15] in its position.

(a) For the spoilers No.3, 4, and 8, make sure that the elbow [35] and hydraulic line [32] align with the fittings.

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1) Loosely install the elbow [35] and hydraulic line [32] to the actuator [15].

(b) For the spoilers No.2, 5, 10, and 11, make sure that the elbow [35] and elbow [36] align with the fittings.

1) Loosely install the elbow [35] and elbow [36] to the actuator [15].

(c) For the spoilers No.9, make sure that the tee fitting [40] and hydraulic line [38] align with the fittings.

1) Loosely install the tee fitting [40] and hydraulic line [38] to the actuator [15].

SUBTASK 27-61-51-420-029

(9) Connect the actuator [15] to the structure.

(a) Install the bolts [20] and washers [21].

(b) Install the bolts [18] and washers [19].

(c) Tighten the bolts [20] to 225 in-lb (25.4 N·m) - 275 in-lb (31.1 N·m).

(d) Tighten the bolts [18] to 450 in-lb (50.8 N·m) - 550 in-lb (62.1 N·m).

SUBTASK 27-61-51-420-030

(10) For the spoilers No.3, 4, and 8, do these steps:

(a) Tighten the B-nut on the elbow [35] to 500 ±25 in-lb (56.5 ±2.8 N·m).

1) If it is necessary, tighten the B-nut on the hydraulic line [12] to 170 ±9 in-lb (19.2 ±1.0 N·m).

(b) Tighten the B-nut on the hydraulic line [32] to 270 ±14 in-lb (30.5 ±1.6 N·m).

SUBTASK 27-61-51-420-031

(11) For the spoilers No.2, 5, 10, and 11, do these steps:

(a) Tighten the B-nut on the elbow [35] to 500 ±25 in-lb (56.5 ±2.8 N·m).

1) If it is necessary, tighten the B-nut on the hydraulic line [41] to 170 ±9 in-lb (19.2 ±1.0 N·m).

(b) Tighten the B-nut on the elbow [36] to 270 ±14 in-lb (30.5 ±1.6 N·m).

1) If it is necessary, tighten the B-nut on the hydraulic line [42] to 270 ±14 in-lb (30.5 ±1.6 N·m).

SUBTASK 27-61-51-420-032

(12) For the spoilers No.9, do these steps:

(a) Tighten the B-nut on the tee fitting [40] to 500 ±25 in-lb (56.5 ±2.8 N·m).

1) If it is necessary, tighten the B-nut on the hydraulic line [37] to 170 ±9 in-lb (19.2 ±1.0 N·m).

2) If it is necessary, tighten the B-nut on the hydraulic line [39] to 170 ±9 in-lb (19.2 ±1.0 N·m).

(b) Tighten the B-nut on the hydraulic line [38] to 270 ±14 in-lb (30.5 ±1.6 N·m).

SUBTASK 27-61-51-420-037

(13) Connect the actuator rod end [17] to the flight spoiler [16]:

(a) Apply a layer of grease, D00633, to the outer diameter of pin [3].

(b) Install the pin [3], washer [6], and nut [7].

(c) Tighten the nut [7] to 540 in-lb (61.0 N·m) to 660 in-lb (74.6 N·m).

(d) Apply a thin coat of corrosion inhibiting material, G50136, to the shank of the bolt [1].

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1) Make sure that the threads of the bolt [1] are free of corrosion inhibiting material, G50136.

(e) Install the bolt [1], washer [2], washer [4], pin [5].

1) Tighten the bolt [1] to 30 in-lb (3.4 N·m) - 35 in-lb (4.0 N·m).

(f) Install the lockwire [23] (MS20995NC32 lockwire, G50347) on pin [5] with double twist method.

SUBTASK 27-61-51-640-001

(14) Do this task: Flight Spoiler Actuator and Rod End Lubrication, TASK 12-22-61-600-802.

SUBTASK 27-61-51-910-001

(15) Do this task: Inspection, Cleaning and Sealing of Electrical Connectors, TASK 20-60-07-910-802.

SUBTASK 27-61-51-430-001

(16) Connect the electrical connector [25] to the receptacle on the actuator [15].

H. Flight Spoiler Test

SUBTASK 27-61-51-080-001

(1) Remove the lock set, SPL-14632, from the actuator [15] for the applicable flight spoiler [16].

SUBTASK 27-61-51-860-004

(2) Do this task: Pressurization of the Spoiler Hydraulic Systems A and B to its Usual Condition, TASK 27-61-00-840-802.

SUBTASK 27-61-51-860-005

(3) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	11	C04040	GROUND SPOILER CONTROL
B	12	C00204	FLIGHT CONTROL SHUTOFF VALVES SPOILER

SUBTASK 27-61-51-860-024

WARNING KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT SPOILERS. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU OPEN/CLOSE THESE CIRCUIT BREAKERS. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(4) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
C	10	C04032	SPOILER CONTROL ELEC PWR A
C	12	C04033	SPOILER CONTROL ELEC PWR B

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-61-51-860-006

WARNING KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(5) Do this task: Apply Pressure to the Spoiler Hydraulic Systems A and B, TASK 27-61-00-800-801.

SUBTASK 27-61-51-870-001

(6) Move the speed brake control lever through its full cycle to bleed the spoiler actuators.

SUBTASK 27-61-51-700-001

(7) Make sure that there are no leaks at the hydraulic lines.

SUBTASK 27-61-51-820-001

(8) Do this task: Flight Spoiler Adjustment, TASK 27-61-00-820-802.

SUBTASK 27-61-51-730-001

(9) Do this task: Flight Spoiler System Test, TASK 27-61-00-730-801.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 27-61-51-440-001

(1) Do this task: Trailing Edge Flap System Reactivation, TASK 27-51-00-440-801.

SUBTASK 27-61-51-860-007

(2) Do this task: Retract the Trailing Edge Flaps, TASK 27-51-00-860-804.

SUBTASK 27-61-51-860-008

(3) Do this task: Shut Down the Spoiler Hydraulic Systems A and B, TASK 27-61-00-840-801.

SUBTASK 27-61-51-610-001

(4) If it is necessary, service the hydraulic reservoirs (TASK 12-12-00-610-801).

——— END OF TASK ———

TASK 27-61-51-000-802

4. Electrohydraulic Servo Valve (EHSV) - Removal

(Figure 402)

A. References

Reference	Title
27-51-00-040-801	Trailing Edge Flap System Deactivation (P/B 201)
27-51-00-860-803	Extend the Trailing Edge Flaps (P/B 201)
27-61-00-800-802	Depressurization of the Spoiler Hydraulic Systems A and B (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-14632	Lock Set - Flight Spoiler Actuator Part #: C27107-1 Supplier: 81205

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. Consumable Materials

Reference	Description	Specification
G01048	Lockwire - MS20995C32, Corrosion Resistant Steel - 0.032 Inch (0.8128 mm) Diameter	NASM20995

D. Location Zones

Zone	Area
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50
562	Left Wing - Spoiler No. 5
563	Left Wing - Spoiler No. 4
564	Left Wing - Spoiler No. 3
565	Left Wing - Spoiler No. 2
662	Right Wing - Spoiler No. 8
663	Right Wing - Spoiler No. 9
664	Right Wing - Spoiler No. 10
665	Right Wing - Spoiler No. 11

E. Prepare for the Removal

SUBTASK 27-61-51-860-013

(1) Do this task: Extend the Trailing Edge Flaps, TASK 27-51-00-860-803.

SUBTASK 27-61-51-040-002

(2) Do this task: Trailing Edge Flap System Deactivation, TASK 27-51-00-040-801.

SUBTASK 27-61-51-860-022

(3) Put the speed brake lever in the UP position to lift the flight spoilers.

SUBTASK 27-61-51-480-003

(4) Install the lock set, SPL-14632, on the actuator [15] for the applicable flight spoiler.

SUBTASK 27-61-51-860-014

(5) Do this task: Depressurization of the Spoiler Hydraulic Systems A and B, TASK 27-61-00-800-802.

SUBTASK 27-61-51-860-015

(6) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	11	C04040	GROUND SPOILER CONTROL
B	12	C00204	FLIGHT CONTROL SHUTOFF VALVES SPOILER

SUBTASK 27-61-51-860-026



WARNING KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT SPOILERS. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU OPEN/CLOSE THESE CIRCUIT BREAKERS. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(7) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
C	10	C04032	SPOILER CONTROL ELEC PWR A
C	12	C04033	SPOILER CONTROL ELEC PWR B

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Electrohydraulic Servo Valve (EHSV) Removal

SUBTASK 27-61-51-020-015

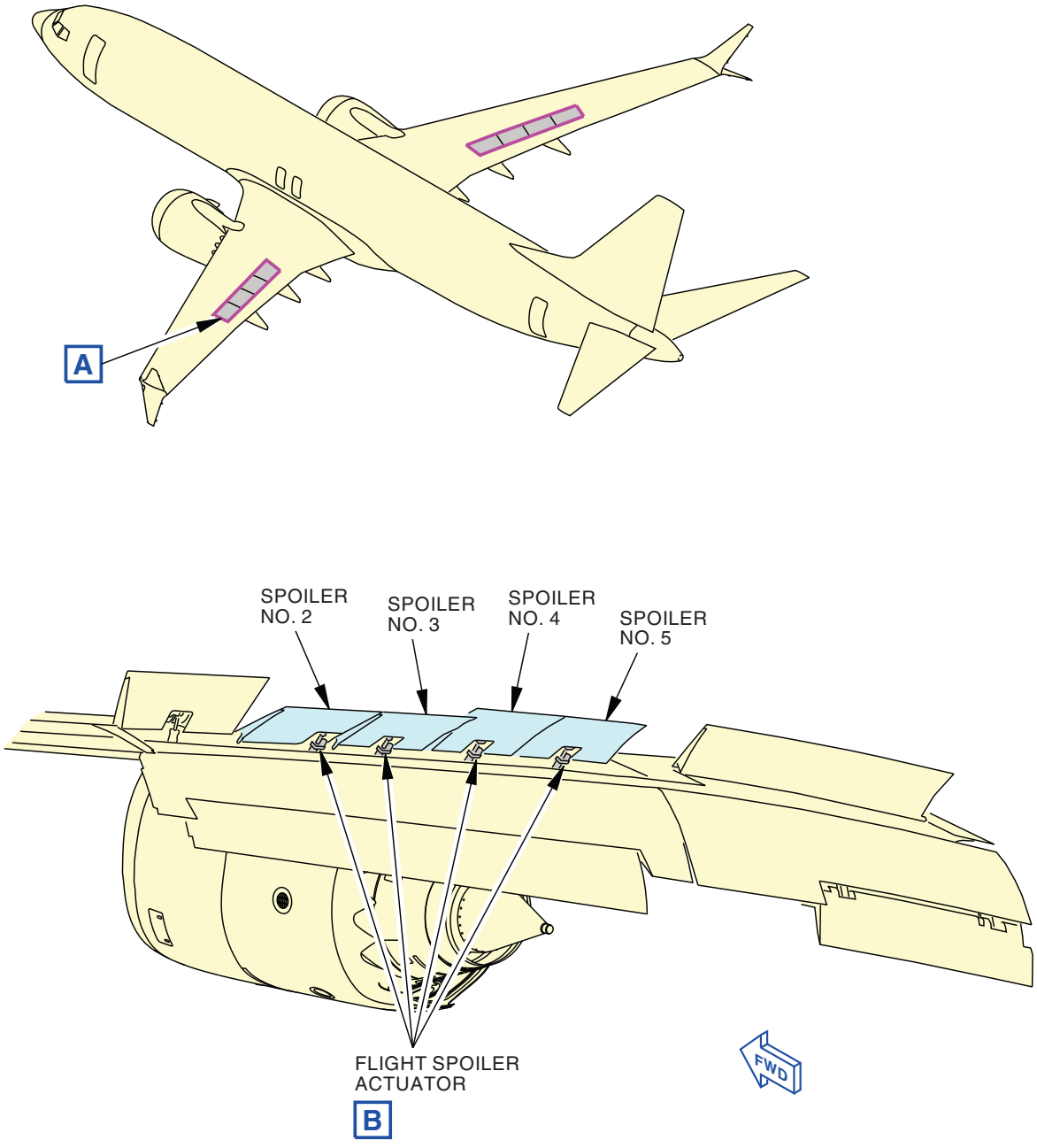
(1) Remove the MS20995C32 lockwire, G01048, and screws [27].

SUBTASK 27-61-51-020-016

(2) Remove the electrohydraulic servo valve [26], packing [31], and gasket [28].

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



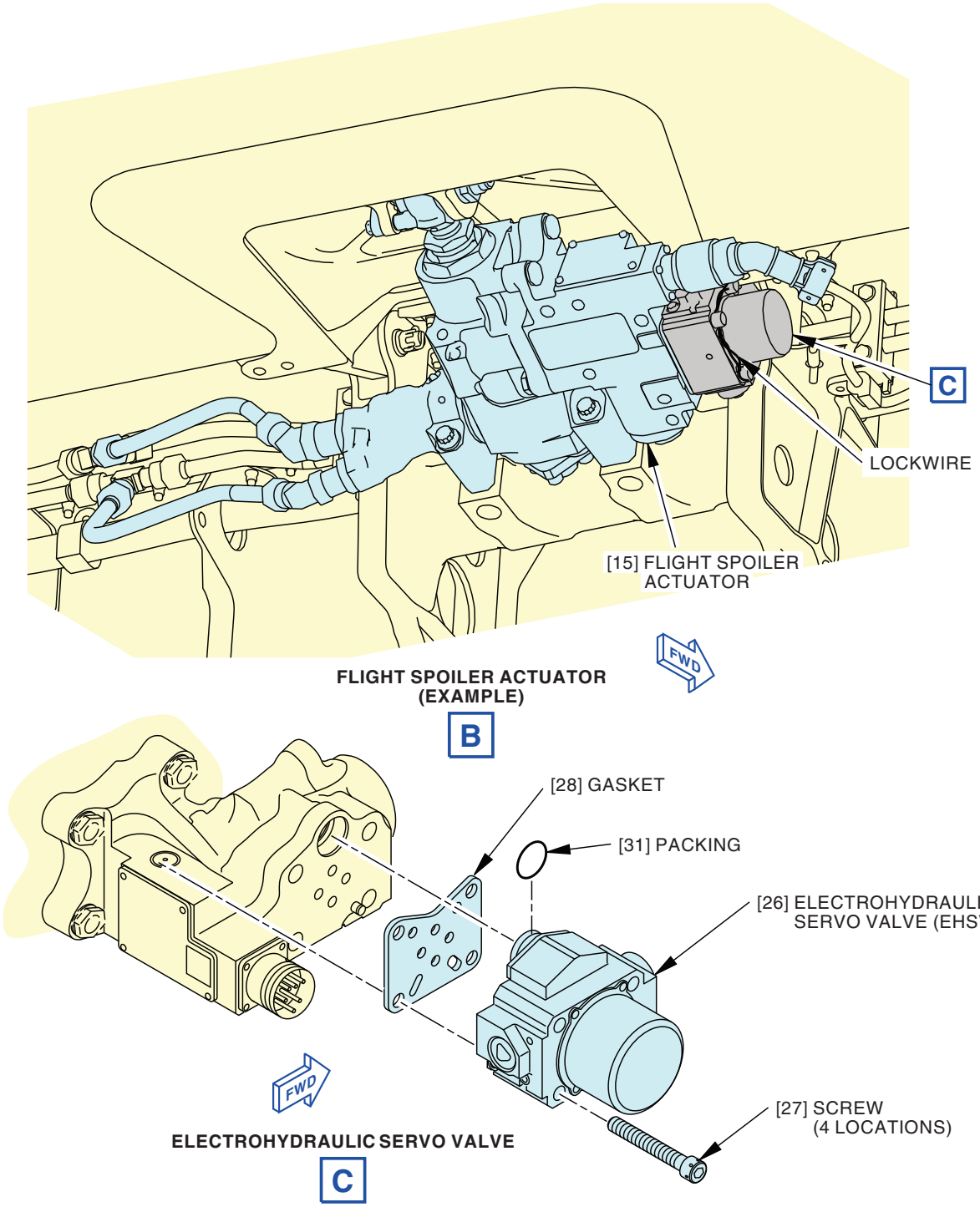
(LEFT WING IS SHOWN, RIGHT WING IS EQUIVALENT)

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Electrohydraulic Servo Valve Installation
Figure 402/27-61-51-990-803 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Electrohydraulic Servo Valve Installation
Figure 402/27-61-51-990-803 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-61-51-400-802

5. Electrohydraulic Servo Valve (EHSV) - Installation

(Figure 402)

A. References

Reference	Title
12-12-00-610-801	Hydraulic Reservoir Servicing (P/B 301)
20-10-44-400-801	Lockwire, Cotter Pins, and Lockrings - Installation (P/B 401)
27-51-00-440-801	Trailing Edge Flap System Reactivation (P/B 201)
27-51-00-860-804	Retract the Trailing Edge Flaps (P/B 201)
27-61-00-730-801	Flight Spoiler System Test (P/B 501)
27-61-00-800-801	Apply Pressure to the Spoiler Hydraulic Systems A and B (P/B 201)
27-61-00-840-801	Shut Down the Spoiler Hydraulic Systems A and B (P/B 201)
27-61-00-840-802	Pressurization of the Spoiler Hydraulic Systems A and B to its Usual Condition (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-14632	Lock Set - Flight Spoiler Actuator Part #: C27107-1 Supplier: 81205

C. Consumable Materials

Reference	Description	Specification
G01048	Lockwire - MS20995C32, Corrosion Resistant Steel - 0.032 Inch (0.8128 mm) Diameter	NASM20995

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
26	Electrohydraulic servo valve	27-61-51-50-160	PTF ALL
28	Gasket	27-61-51-50-150	PTF ALL
31	Packing	27-61-51-50-155	PTF ALL

E. Location Zones

Zone	Area
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50
562	Left Wing - Spoiler No. 5
563	Left Wing - Spoiler No. 4
564	Left Wing - Spoiler No. 3
565	Left Wing - Spoiler No. 2
662	Right Wing - Spoiler No. 8
663	Right Wing - Spoiler No. 9
664	Right Wing - Spoiler No. 10
665	Right Wing - Spoiler No. 11

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Electrohydraulic Servo Valve (EHSV) Installation

SUBTASK 27-61-51-420-010

(1) Install the electrohydraulic servo valve [26], new packing [31], and new gasket [28].

SUBTASK 27-61-51-420-011

(2) Install the screws [27] that attach the electrohydraulic servo valve [26].

(a) Tighten the screws [27] to 60 in-lb (6.8 N·m) to 70 in-lb (7.9 N·m).

SUBTASK 27-61-51-420-012

(3) Install the MS20995C32 lockwire, G01048, to the screws [27] (TASK 20-10-44-400-801).

G. Flight Spoiler Test

SUBTASK 27-61-51-080-002

(1) Remove the lock set, SPL-14632, from the actuator [15] for the applicable flight spoiler.

SUBTASK 27-61-51-860-016

(2) Do this task: Pressurization of the Spoiler Hydraulic Systems A and B to its Usual Condition, TASK 27-61-00-840-802.

SUBTASK 27-61-51-860-017

(3) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	11	C04040	GROUND SPOILER CONTROL
B	12	C00204	FLIGHT CONTROL SHUTOFF VALVES SPOILER

SUBTASK 27-61-51-860-025

WARNING

KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT SPOILERS. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU OPEN/CLOSE THESE CIRCUIT BREAKERS. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(4) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
C	10	C04032	SPOILER CONTROL ELEC PWR A
C	12	C04033	SPOILER CONTROL ELEC PWR B

SUBTASK 27-61-51-860-018

WARNING

KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(5) Do this task: Apply Pressure to the Spoiler Hydraulic Systems A and B, TASK 27-61-00-800-801.

SUBTASK 27-61-51-870-004

(6) Move the speed brake control lever through its full cycle to bleed the spoiler actuators.

SUBTASK 27-61-51-700-002

(7) Make sure that there are no leaks at the electrohydraulic servo valve [26].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-61-51-730-004

(8) Do this task: Flight Spoiler System Test, TASK 27-61-00-730-801.

H. Put the Airplane Back to Its Usual Condition.

SUBTASK 27-61-51-440-002

(1) Do this task: Trailing Edge Flap System Reactivation, TASK 27-51-00-440-801.

SUBTASK 27-61-51-860-019

(2) Do this task: Retract the Trailing Edge Flaps, TASK 27-51-00-860-804.

SUBTASK 27-61-51-860-020

(3) Do this task: Shut Down the Spoiler Hydraulic Systems A and B, TASK 27-61-00-840-801.

SUBTASK 27-61-51-610-002

(4) Service the hydraulic reservoirs if it is necessary (TASK 12-12-00-610-801).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FLIGHT SPOILER ACTUATOR - INSPECTION/CHECK

1. General

A. This task supplies the data for a dimensional inspection of the flight spoiler actuator.

B. This task does not remove or install the flight spoiler actuator, refer to the Flight Spoiler Actuator - Removal/Installation for procedures to remove and install the flight spoiler actuator.

TASK 27-61-51-200-801

2. Flight Spoiler Actuator Inspection

(Figure 601)

A. General

(1) This procedure gives wear limits and repair data for the flight spoiler actuators.

B. References

Reference	Title
27-61-51-000-801	Flight Spoiler Actuator - Removal (P/B 401)
27-61-51-400-801	Flight Spoiler Actuator - Installation (P/B 401)

C. Tools/Equipment

Reference	Description
STD-1096	Micrometer - Depth, 0-1 inch, Readable to 1/1000 inch
STD-1097	Caliper - Vernier, 0-6 inch, Readable to 1/1000 inch

D. Location Zones

Zone	Area
562	Left Wing - Spoiler No. 5
563	Left Wing - Spoiler No. 4
564	Left Wing - Spoiler No. 3
565	Left Wing - Spoiler No. 2
662	Right Wing - Spoiler No. 8
663	Right Wing - Spoiler No. 9
664	Right Wing - Spoiler No. 10
665	Right Wing - Spoiler No. 11

E. Flight Spoiler Actuator Inspection

SUBTASK 27-61-51-020-005

(1) Remove the flight spoiler actuator to inspect it (TASK 27-61-51-000-801).

SUBTASK 27-61-51-020-006

(2) Use these tools to do a check for wear on the parts of the flight spoiler actuator:

(a) micrometer (0-1 inch, readable to 1/1000 inch), STD-1096

(b) vernier caliper (0 - 6 inch, readable to 1/1000 inch), STD-1097.

SUBTASK 27-61-51-020-007

(3) Examine the parts of the flight spoiler actuator for worn areas:

(a) Remove the parts to be checked.

(b) Measure the parts as necessary with a micrometer (0-1 inch, readable to 1/1000 inch), STD-1096, or a vernier caliper (0 - 6 inch, readable to 1/1000 inch), STD-1097.

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(c) Compare the dimensions you measured with the permitted wear dimensions in Table 601.

(d) If the parts are not in the tolerance, replace or repair the parts.

Table 601/27-61-51-993-801

NAME OF THE PARTS THAT ARE IN CONTACT	DIM.	DIAMETER DESIGN LIMITS		WEAR LIMITS	
		MINIMUM	MAXIMUM	PERMITTED MAXIMUM I.D./ MINIMUM O.D. OF WORN PART	MAXIMUM SERVICE DIAMETRIC CLEARANCE
#1 BUSHING	I.D.	0.7500 in. (19.050 mm)	0.7515 in. (19.088 mm)	0.7520 in. (19.101 mm)	0.0057 in. (0.145 mm)
#1 PIN	O.D.	0.7475 in. (18.987 mm)	0.7480 in. (18.999 mm)	0.7458 in. (18.943 mm)	
#2 BEARING	I.D.	0.7495 in. (19.037 mm)	0.7500 in. (19.050 mm)	0.7505 in. (19.063 mm)	0.0042 in. (0.107 mm)
#2 PIN	O.D.	0.7475 in. (18.987 mm)	0.7480 in. (18.999 mm)	0.7458 in. (18.943 mm)	

SUBTASK 27-61-51-020-008

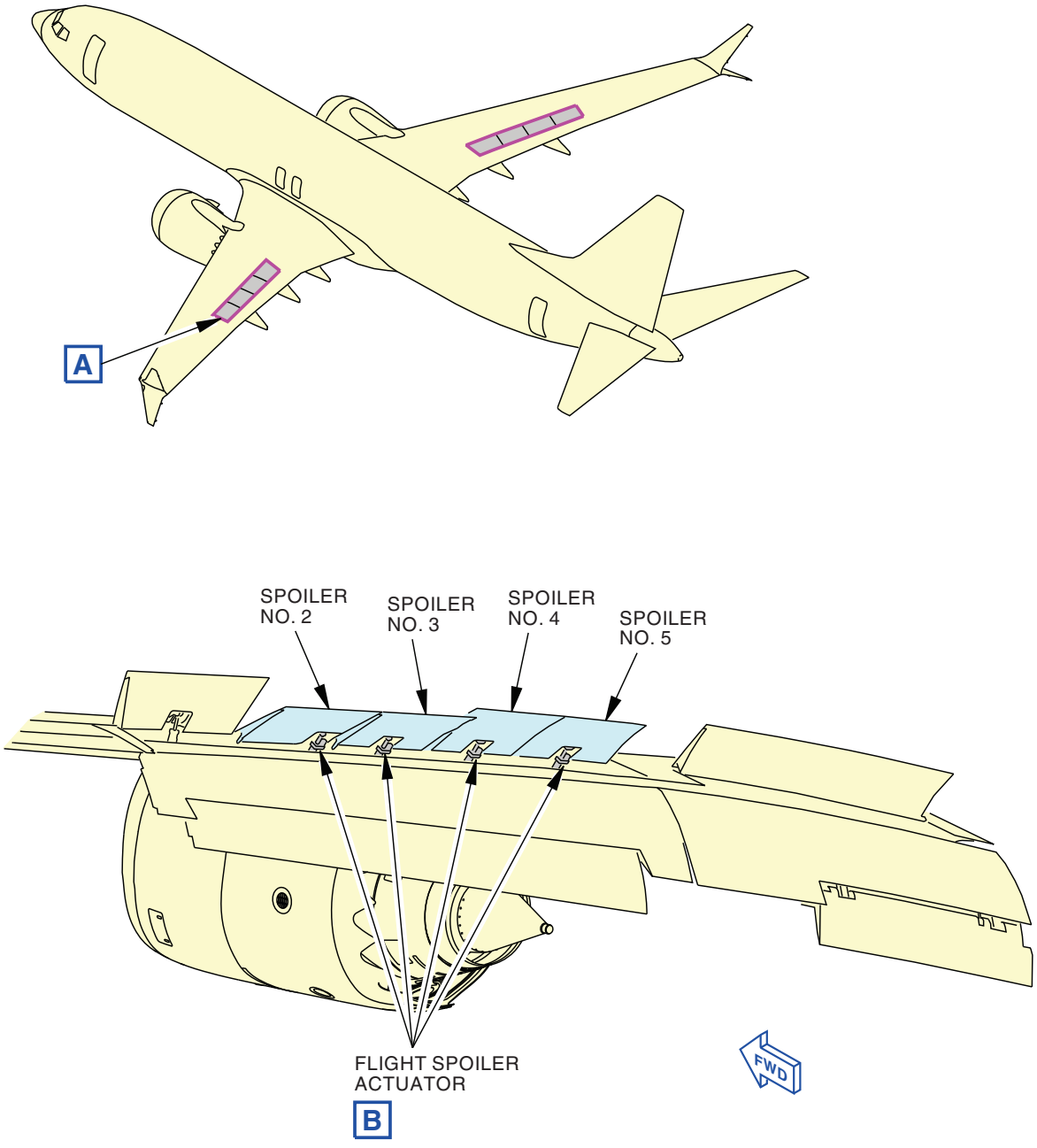
(4) Reassemble the flight spoiler actuator, if it is necessary.

SUBTASK 27-61-51-020-009

(5) Install the flight spoiler actuator (TASK 27-61-51-400-801).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



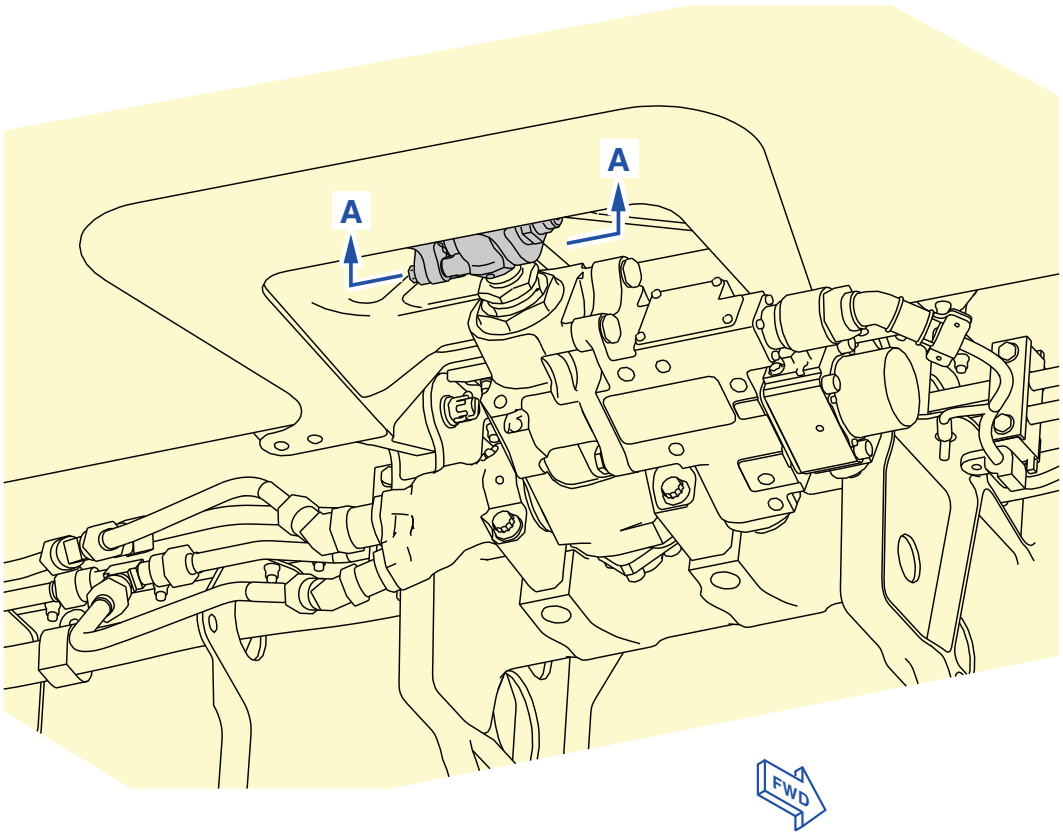
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Flight Spoiler Actuator Wear Limits
Figure 601/27-61-51-990-802 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



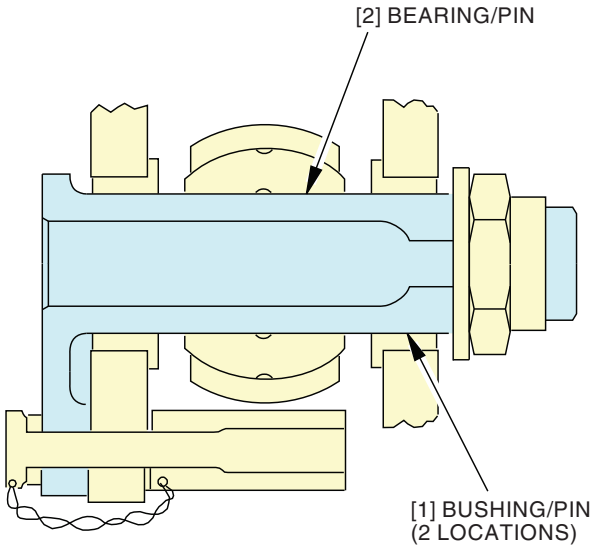
FLIGHT SPOILER ACTUATOR
(EXAMPLE)

B

2413206 S00061532087_V1

Flight Spoiler Actuator Wear Limits
Figure 601/27-61-51-990-802 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



A-A

2413207 S00061532088_V2

Flight Spoiler Actuator Wear Limits
Figure 601/27-61-51-990-802 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SPEED BRAKE CONTROL SYSTEM - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) Apply Pressure to the Speed Brake Hydraulic Systems A and B

(2) Shut Down the Speed Brake Hydraulic Systems A and B

(3) Depressurization of the Speed Brake Hydraulic Systems A and B

(4) Pressurization of the Speed Brake Hydraulic Systems A and B to its Usual Condition

TASK 27-62-00-800-801

2. Apply Pressure to the Speed Brake Hydraulic Systems A and B

(Figure 201)

A. General

(1) This procedure starts the speed brake hydraulic systems A and B and applies pressure to the speed brake.

B. References

Reference	Title
12-12-00-610-801	Hydraulic Reservoir Servicing (P/B 301)
29-11-00-860-802	Hydraulic System A or B Pressurization with a Portable Hydraulic Cart (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50

D. Procedure

SUBTASK 27-62-00-860-010

CAUTION

DO NOT OPERATE SYSTEM A OR B HYDRAULIC PUMPS FOR MORE THAN TWO MINUTES WITHOUT FUEL IN THE FUEL TANKS. THE NO. 1 (FOR SYSTEM A) AND NO. 2 (FOR SYSTEM B) FUEL TANKS MUST HAVE A MINIMUM OF 250 GALLONS (1675 POUNDS/760 KILOGRAMS) OF FUEL IN THEM. IF THERE IS NOT SUFFICIENT FUEL IN THE FUEL TANKS, THE HYDRAULIC FLUID WILL BECOME TOO HOT.

(1) Make sure the No. 1 and No. 2 fuel tanks each have 250 gal (946 l) (1675 lb (760 kg)) of fuel in them.

SUBTASK 27-62-00-210-004

(2) Make sure the hydraulic reservoirs are full (TASK 12-12-00-610-801).

SUBTASK 27-62-00-860-011

(3) Make sure that this circuit breaker is closed:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	12	C00204	FLIGHT CONTROL SHUTOFF VALVES SPOILER

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-62-00-860-012

(4) Set the parking brake.

SUBTASK 27-62-00-480-002

WARNING MAKE SURE THAT THE GROUND LOCKS ARE INSTALLED IN ALL THE LANDING GEARS BEFORE YOU PRESSURIZE THE HYDRAULIC SYSTEM. WITHOUT THE GROUND LOCKS, THE LANDING GEARS CAN RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(5) Install the ground lock assemblies in the landing gear (TASK 32-00-01-480-801).

SUBTASK 27-62-00-860-013

WARNING KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(6) Pressurize the speed brake hydraulic systems A and B with one of these methods:

(a) Method 1:

1) Set the ELEC 2, HYD PUMPS A or ELEC 1, HYD PUMPS B switch on the forward overhead panel to the ON position.

2) To pressurize the hydraulic system A, set the SPOILER A switch on the forward overhead panel to the ON position.

3) To pressurize the hydraulic system B, set the SPOILER B switch on the forward overhead panel to the ON position.

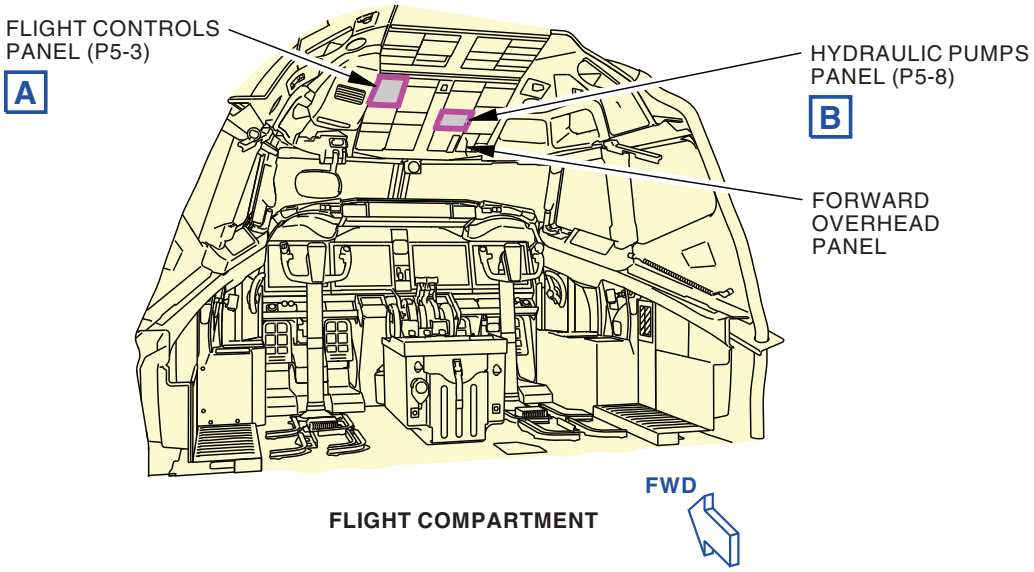
(b) Method 2:

NOTE: The hydraulic cart is a more flexible alternative. It can vary the pressure of the hydraulic fluid. Also, it is not necessary to have a specified quantity of fuel for cooling in tanks.

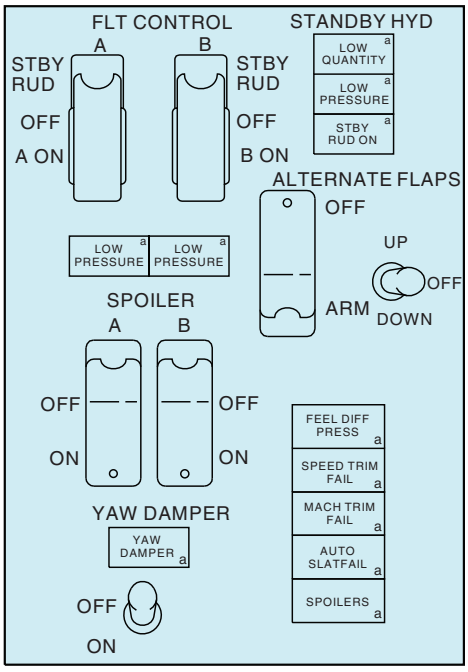
1) Do this task to use a portable hydraulic cart to supply hydraulic pressure: Hydraulic System A or B Pressurization with a Portable Hydraulic Cart, TASK 29-11-00-860-802.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

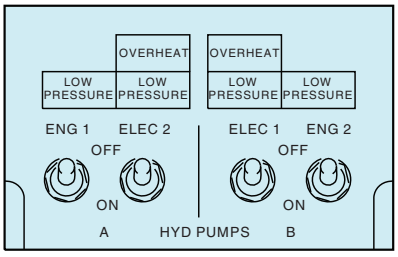


FLIGHT COMPARTMENT



FLIGHT CONTROLS PANEL (P5-3)

A



HYDRAULIC PUMPS PANEL (P5-8)

B

2413138 S00061531952_V1

Spoiler Power Control Switches
Figure 201/27-62-00-990-816

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-62-00-840-801

3. Shut Down the Speed Brake Hydraulic Systems A and B

(Figure 201)

A. General

(1) This procedure shut downs the speed brake hydraulic systems A and B but does not remove the head pressure.

B. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50

C. Procedure

SUBTASK 27-62-00-860-014

(1) Set the ELEC 2, HYD PUMPS A and the ELEC 1, HYD PUMPS B switches on the forward overhead panel to the OFF position.

SUBTASK 27-62-00-860-015

(2) Set the SPOILER A and B switches on the forward overhead panel to the OFF position.

——— END OF TASK ———

TASK 27-62-00-800-802

4. Depressurization of the Speed Brake Hydraulic Systems A and B

(Figure 201, Figure 202)

A. General

(1) This procedure removes the head pressure from the speed brake hydraulic systems A and B and locks the system out.

B. References

Reference	Title
20-60-07-910-802	Inspection, Cleaning and Sealing of Electrical Connectors (P/B 201)
29-09-00-860-802	Hydraulic Reservoirs Depressurization (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1675	Lock Assembly - Flight Controls (used on valve p/n 10-60581-2 and 10-60581-3) Part #: F80049-83 Supplier: 81205 Opt Part #: F80049-56 Supplier: 81205 Opt Part #: F80049-65 Supplier: 81205
STD-858	Tag - DO NOT OPERATE

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50

E. Procedure

SUBTASK 27-62-00-860-016

(1) Set the SPOILER A and B switches on the forward overhead panel to the OFF position.

(a) Install the DO NOT OPERATE tag, STD-858.

SUBTASK 27-62-00-860-017

(2) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	12	C00204	FLIGHT CONTROL SHUTOFF VALVES SPOILER

SUBTASK 27-62-00-860-018

(3) Set the ELEC 2, HYD PUMPS A and the ELEC 1, HYD PUMPS B switches on the forward overhead panel to the OFF position.

SUBTASK 27-62-00-870-002

(4) Operate the rudder pedals to decrease the remaining hydraulic system pressure.

SUBTASK 27-62-00-860-019

(5) If it is necessary to open a hydraulic line, you must release pressure from the reservoir for the applicable hydraulic system. To release it, do this task: Hydraulic Reservoirs Depressurization, TASK 29-09-00-860-802.

SUBTASK 27-62-00-480-003

(6) Install the flight controls hydraulic module locks (Figure 202):

(a) Disconnect the electrical connector from the spoiler shutoff valve at the flight controls hydraulic modules A and B.

(b) After you disconnect the electrical connector(s), examine the connector(s) for corrosion (TASK 20-60-07-910-802).

NOTE: The inspection of the electrical connector(s) can be done after the removal step or before the installation step of the electrical connector(s). This is an operator option.

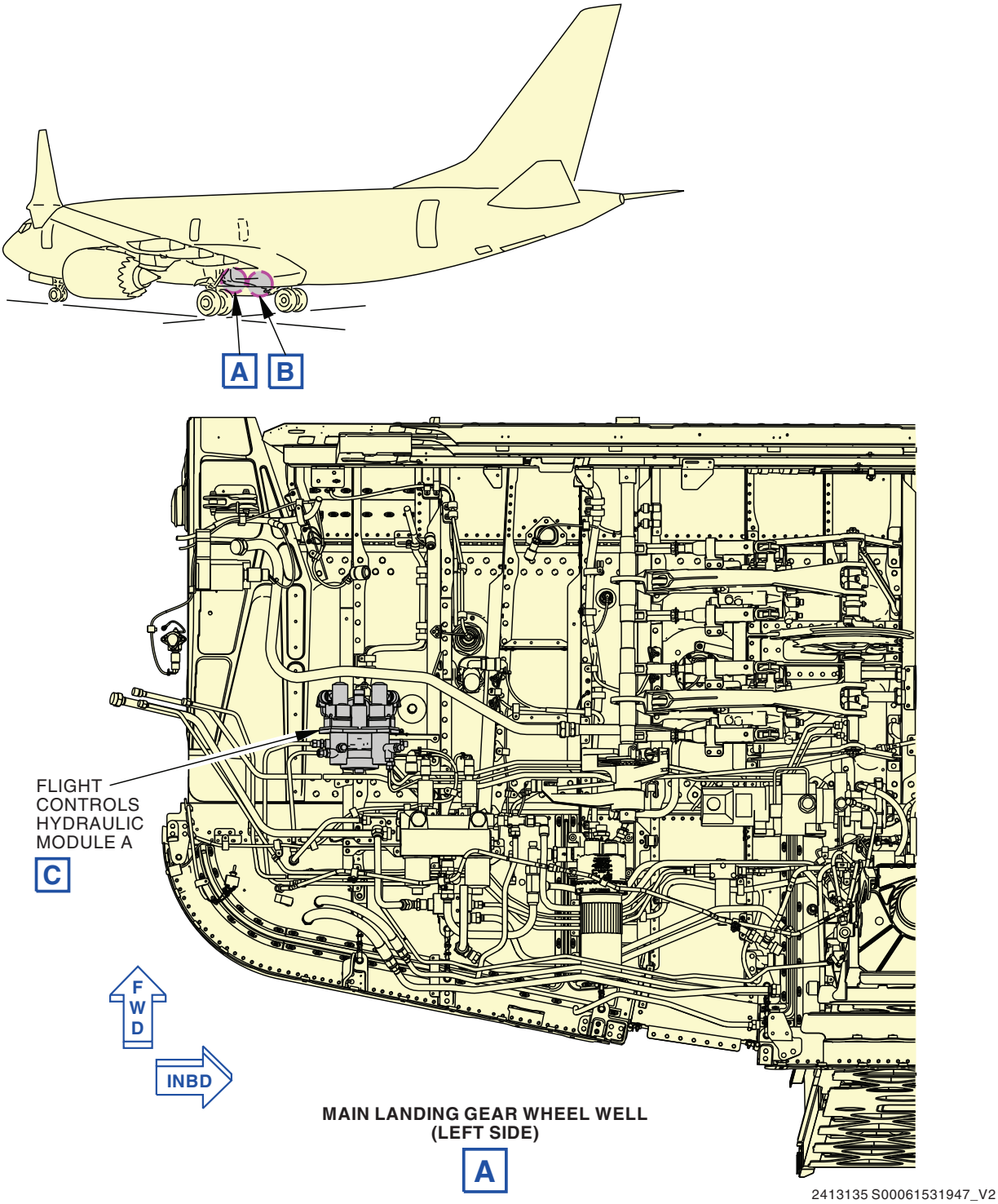
(c) Put the manual override lever at the two spoiler shutoff valves in the position 2 (OFF position).

(d) Install the lock, SPL-1675 on the two spoiler shutoff valves.

1) Install the detent pins.

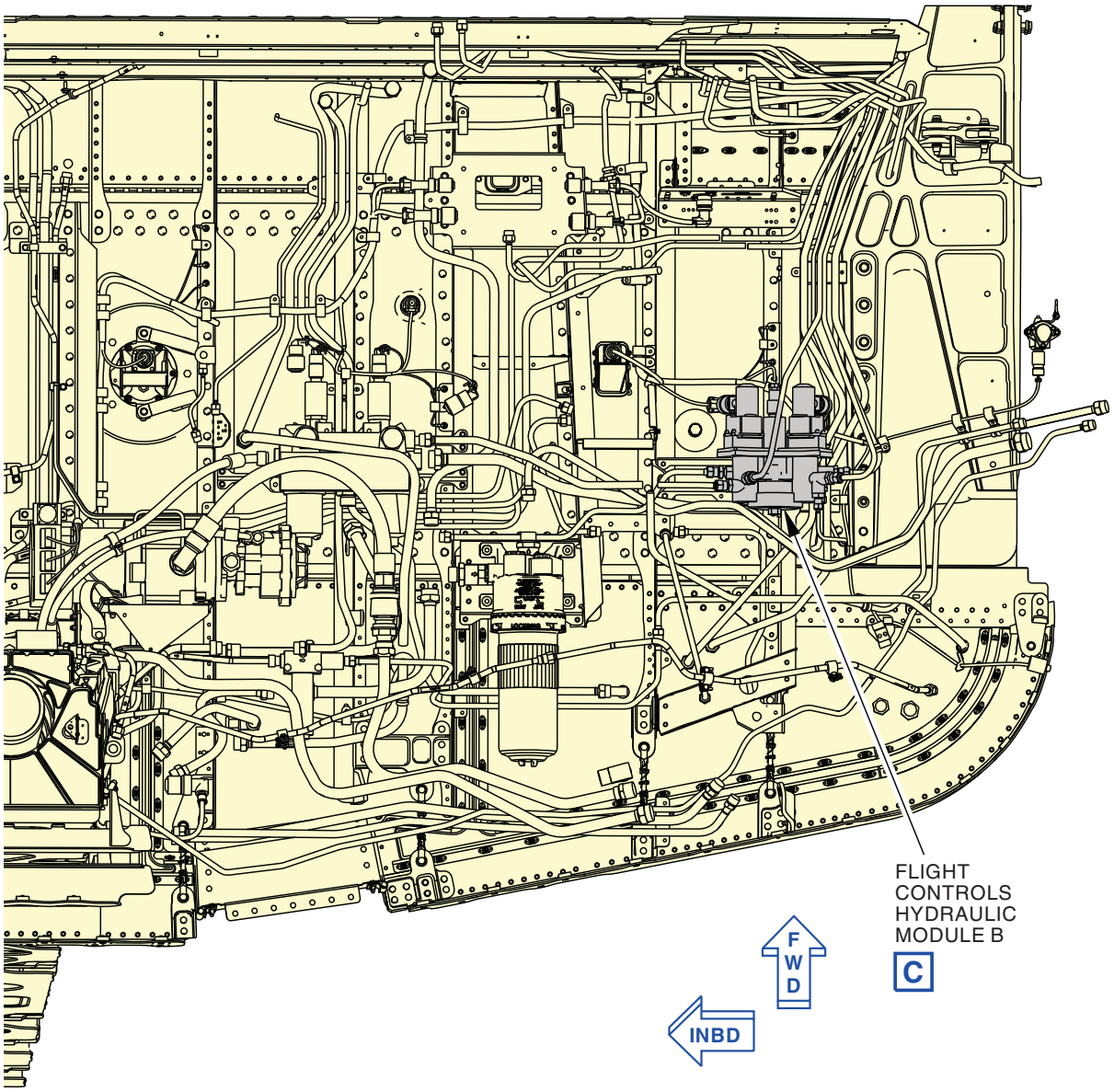
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



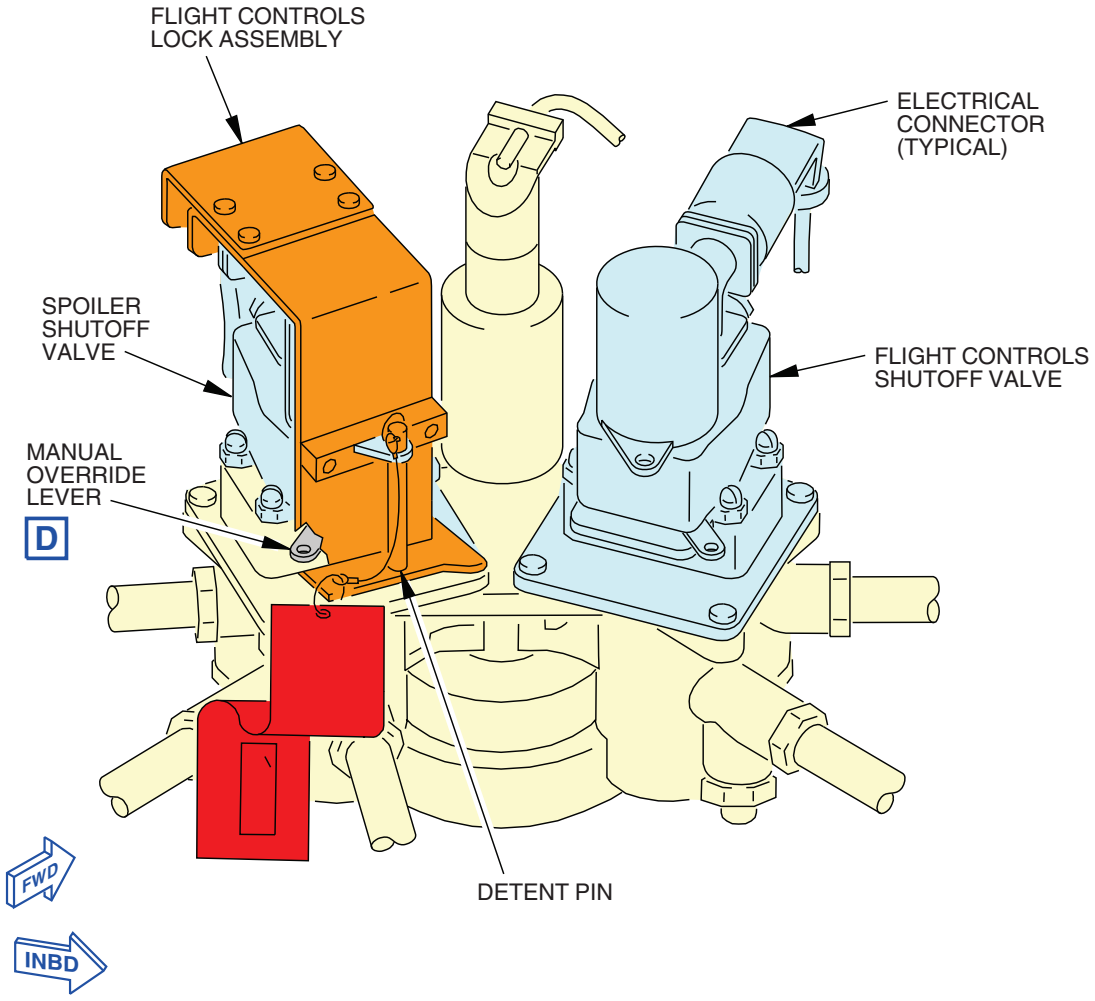
Flight Controls Hydraulic Module Lock Installation
Figure 202/27-62-00-990-804 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Flight Controls Hydraulic Module Lock Installation
Figure 202/27-62-00-990-804 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



FLIGHT CONTROLS HYDRAULIC MODULE A
(MODULE B IS EQUIVALENT)

C

2413137 S00061531949_V4

Flight Controls Hydraulic Module Lock Installation
Figure 202/27-62-00-990-804 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-62-00-840-802

5. Pressurization of the Speed Brake Hydraulic Systems A and B to its Usual Condition

(Figure 201, Figure 202)

A. General

(1) This procedure unlocks the speed brake hydraulic systems A and B and applies the hydraulic pressure back to the system.

B. References

Reference	Title
20-60-07-910-802	Inspection, Cleaning and Sealing of Electrical Connectors (P/B 201)
29-09-00-860-801	Hydraulic Reservoirs Pressurization (P/B 201)
SWPM 20-60-03	Special Protection of Electrical Connectors

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1675	Lock Assembly - Flight Controls (used on valve p/n 10-60581-2 and 10-60581-3) Part #: F80049-83 Supplier: 81205 Opt Part #: F80049-56 Supplier: 81205 Opt Part #: F80049-65 Supplier: 81205
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
G50171	Compound - Corrosion Inhibiting Compound, Interior Application - D5026NS or ZC-026	

E. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50

F. Procedure

SUBTASK 27-62-00-080-004

(1) Remove the flight controls hydraulic module locks from the spoiler shutoff valves.

(a) Remove the lockpins.

(b) Remove the lock, SPL-1675 from the two spoiler shutoff valves.

(c) Put the manual override lever at the two spoiler shutoff valves in the position 1 (ON position).

(d) Apply the D5026NS or ZC-026 compound, G50171 to the connector (SWPM 20-60-03).

(e) Connect the electrical connector to the spoiler shutoff valve at the spoiler hydraulic modules A and B. (Inspection, Cleaning and Sealing of Electrical Connectors, TASK 20-60-07-910-802)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-62-00-860-020

(2) Set the SPOILER A and B switches on the forward overhead panel to the ON position.

(a) Remove the DO NOT OPERATE tag, STD-858.

SUBTASK 27-62-00-860-021

(3) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	12	C00204	FLIGHT CONTROL SHUTOFF VALVES SPOILER

SUBTASK 27-62-00-860-022

(4) Pressurize the hydraulic reservoirs. Hydraulic Reservoirs Pressurization, TASK 29-09-00-860-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SPEED BRAKE CONTROL SYSTEM - ADJUSTMENT/TEST

1. General

A. This procedure has these tasks:

- Speed Brake Control System Operational Test
- Speed Brake Control Lever Adjustment
- Ground Spoiler Adjustment
- Speed Brake Lever Friction Test
- Spoiler Operation Time Test
- Speed Brake Throttle Test
- Speed Brake Antiskid Inoperative Test
- Speed Brake RTO Test
- Speed Brake Extended Light Test

TASK 27-62-00-710-801

2. Speed Brake Control System Operational Test

A. General

(1) Do this test to quickly give the status of the speed brake control system. Special tools or equipment are not necessary.

B. References

Reference	Title
27-62-00-800-801	Apply Pressure to the Speed Brake Hydraulic Systems A and B (P/B 201)
27-62-00-840-801	Shut Down the Speed Brake Hydraulic Systems A and B (P/B 201)

C. Location Zones

Zone	Area
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50

D. Procedure

SUBTASK 27-62-00-710-002

(1) Do the speed brake control system operational test:

WARNING MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. THE AILERONS, RUDDER, ELEVATORS, FLAPS, SLATS, SPOILERS, LANDING GEAR, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(a) Do this task: Apply Pressure to the Speed Brake Hydraulic Systems A and B, TASK 27-62-00-800-801.

NOTE: When operating the speedbrakes on the ground with only the system A and B Electric Motor-Driven Pump (EMDP)s supplying power, the pump low pressure light may momentarily illuminate. This illumination and associated pressure drop should be considered normal as long as the pressure recovers within 10 seconds of initiating movement of the speedbrakes.

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(b) Move the speed brake lever from the DOWN position to the UP position.

1) Make sure that the speed brakes move to the UP position.

(c) Move the speed brake lever from the UP position to the DOWN position.

1) Make sure that the speed brakes move to the retracted position.

(d) If necessary, move the flap lever to the FLAPS UP position.

(e) Do this task: Shut Down the Speed Brake Hydraulic Systems A and B, TASK 27-62-00-840-801.

——— END OF TASK ———

TASK 27-62-00-820-801

3. Speed Brake Control Lever Adjustment

(Figure 501, Figure 502)

A. General

(1) Use this procedure to adjust the speed brake control lever.

B. References

Reference	Title
27-61-45-730-802	Speedbrake Lever Transducer Test (P/B 501)
46-13-02-700-805	Existing Faults Check (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-14768	Rigging Lock - Four Degree Speedbrake Lever Part #: C27095-1 Supplier: 81205

D. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50

E. Access Panels

Number	Name/Location
112A	Forward Access Door

F. Prepare for the Speed Brake Control Lever Adjustment

SUBTASK 27-62-00-730-006

(1) Use the Onboard Maintenance Function (OMF) to make sure that there are no faults associated with the Speed Brake Lever Transducer (TASK 27-61-45-730-802).

(a) If there are faults, clear them with the Speed Brake Lever Test from the OMF system.

SUBTASK 27-62-00-730-007

(2) Use the MAX Display System (MDS) to bring up the Flight Controls Maintenance Data page (Figure 501).

NOTE: The Flight Controls Maintenance Data page will be used to do a check of the settings of the RVDT channels A, B and C.

(a) Find the readings for the SCE SPDBRK LVR POSN and the RVDT A, B and C channels.

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G. Speed Brake Control Lever Adjustment

SUBTASK 27-62-00-860-201

(1) Do these steps to adjust the speed brake lever:

(a) Put the speed brake lever in the DOWN and LOCKED position.

(b) Make sure that the SPEEDBRAKES EXTENDED and SPEEDBRAKE DO NOT ARM lights do not come on.

(c) Put the speed brake lever to the ARMED position (Figure 502).

NOTE: The ARMED position is approximately 4 degrees aft of the DOWN detent position.

1) Install the rigging lock, SPL-14768, to hold the speed brake lever in the ARMED detent position.

2) Moving the speed brake lever without hydraulic power can latch the GROUND SPOILER CONTROL Status Message.

a) Do this task if the Status Message occurs: Existing Faults Check, TASK 46-13-02-700-805.

(d) Open this access door:

Number	Name/Location
112A	Forward Access Door

(e) Disconnect the actuator rod end bolt from the speed brake mechanical assembly.

(f) Make sure that the cam roller is centered in the ARMED detent position.

(g) Make sure that the speed brake lever is still in the ARMED position.

NOTE: The ARMED position is approximately 4 degrees aft of the DOWN detent position.

(h) Remove the lower cover plate on the control stand to get access to the lower control rod.

1) Remove the screws [2] and washers [3].

(i) Adjust the lower control rod so that all three RVDT channels are within 3.3 degrees and 4.3 degrees.

1) Tighten the checknut on the lower control rod to 45 in-lb (5 N·m) - 50 in-lb (6 N·m).

2) Make sure that the lower control rod end is visible and covers at least 50 percent of the inspection hole.

(j) Make sure that the cam roller is centered in the ARMED detent position.

(k) Make sure that the speed brake lever is still in the ARMED position.

NOTE: The ARMED position is approximately 4 degrees aft of the DOWN detent position.

(l) Make sure that the actuator assembly is in the retracted position.

(m) Adjust the rod end from the actuator assembly until it aligns with the clevis from the speed brake mechanical assembly.

1) Make sure that the rod end bolt can be freely inserted in the rod end and the clevis.

(n) Install and tighten the actuator rod end bolt with the bolt head facing inboard.

(o) Make sure that the cam roller is centered in the ARMED detent position.

(p) Remove the rigging lock, SPL-14768, from the speed brake lever.

(q) Put the speed brake lever in the DOWN and LOCKED position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(r) Make sure that the roller is not centered in the ARMED detent position.

(s) Make sure that RVDT A, RVDT B and RVDT C channels are within 0.0 ± 1.0 degrees.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 27-62-00-410-004

(1) Close this access door:

Number	Name/Location
112A	Forward Access Door

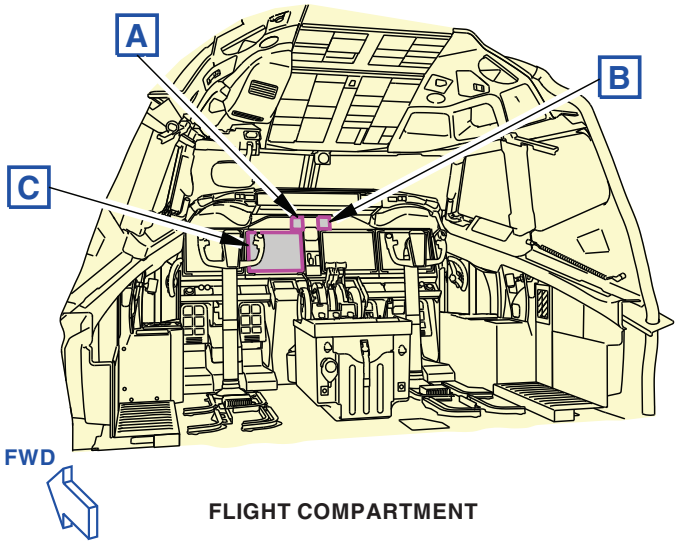
SUBTASK 27-62-00-410-005

(2) Install the lower cover plate on the control stand.

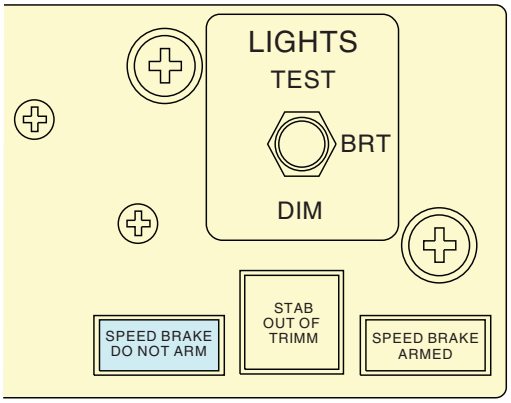
(a) Install the screws [2] and washers [3].

——— END OF TASK ———

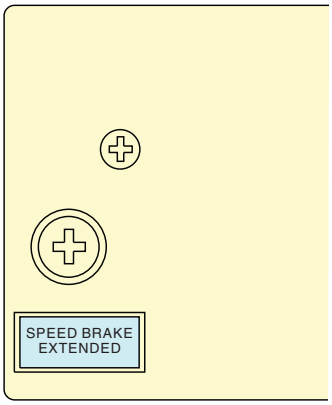
ECCN 9E991 BOEING PROPRIETARY - See title page for details



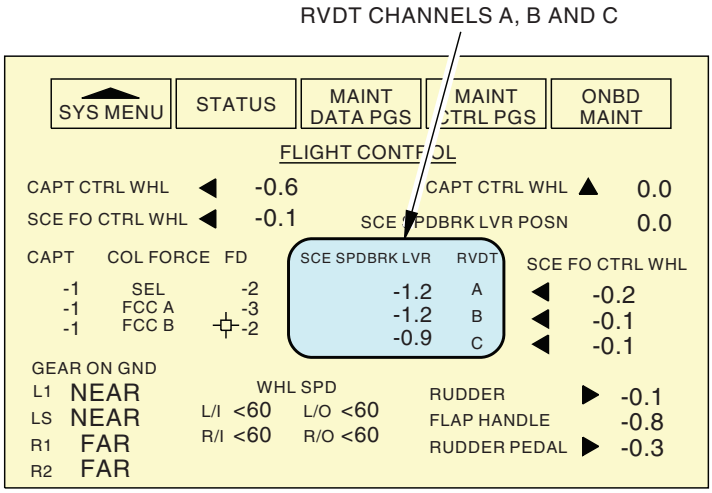
FLIGHT COMPARTMENT



A



B



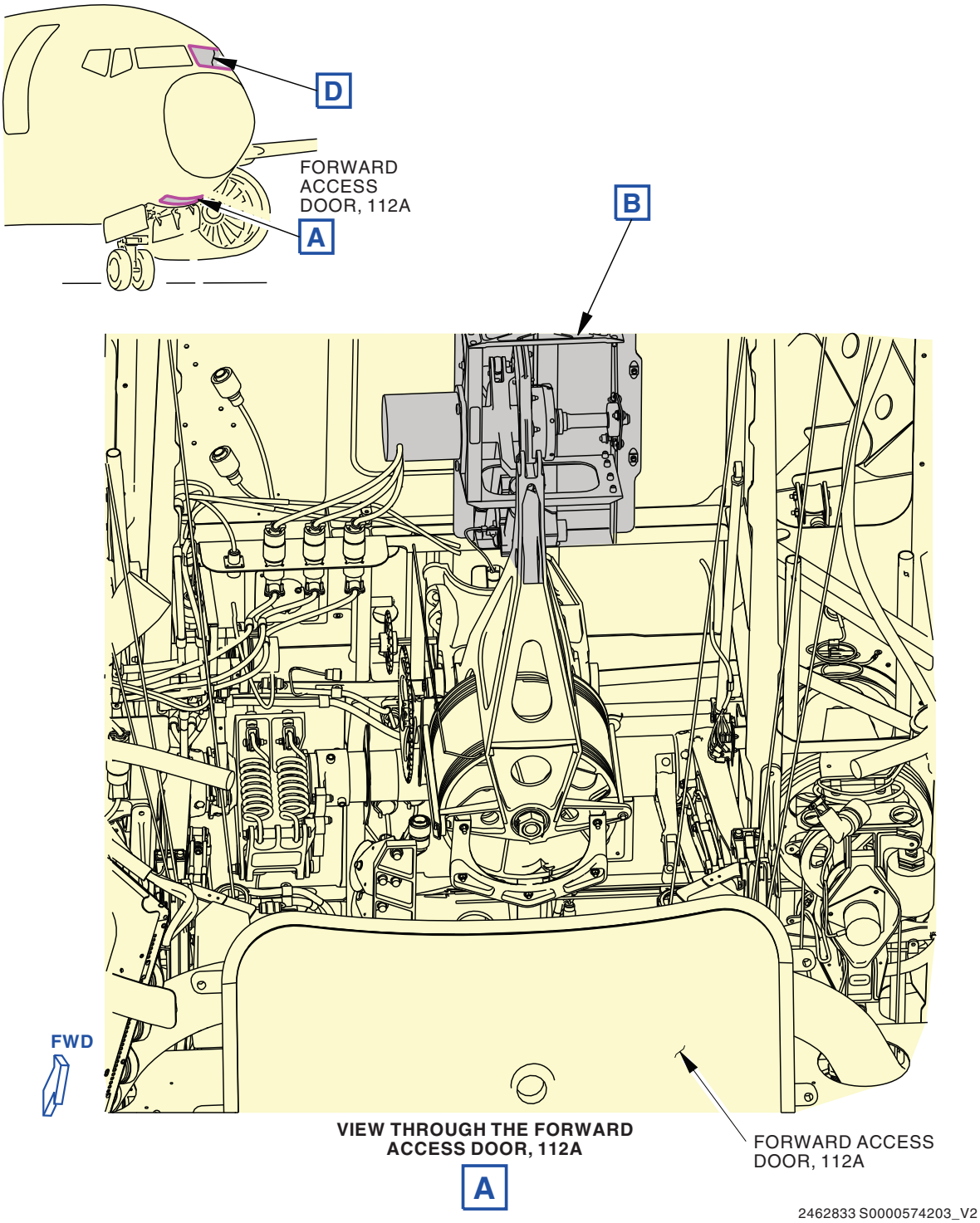
FLIGHT CONTROLS MAINTENANCE DATA PAGE

C

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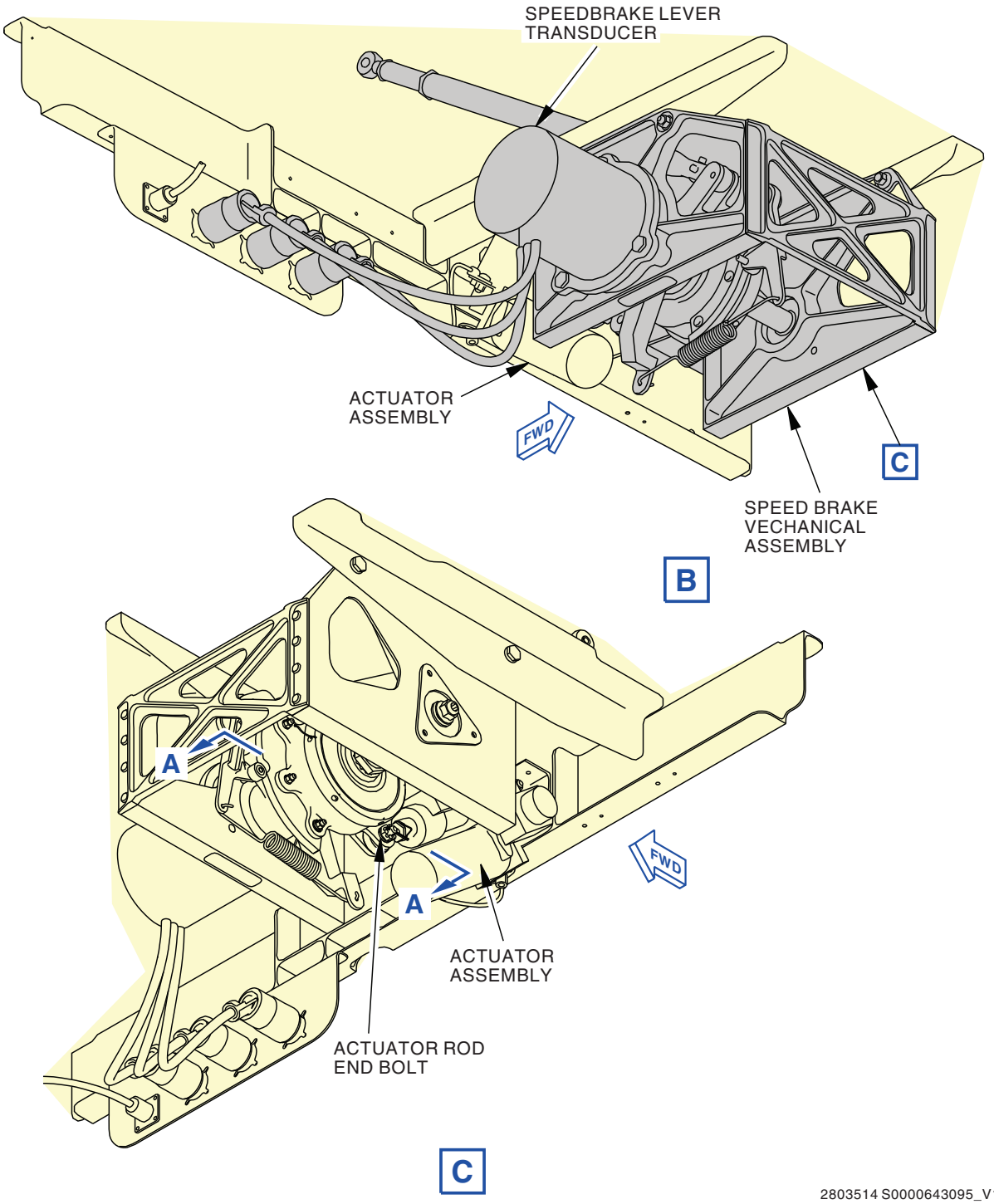
Speedbrake Control Lever Adjustment
Figure 501/27-62-00-990-805

ECCN 9E991 BOEING PROPRIETARY - See title page for details



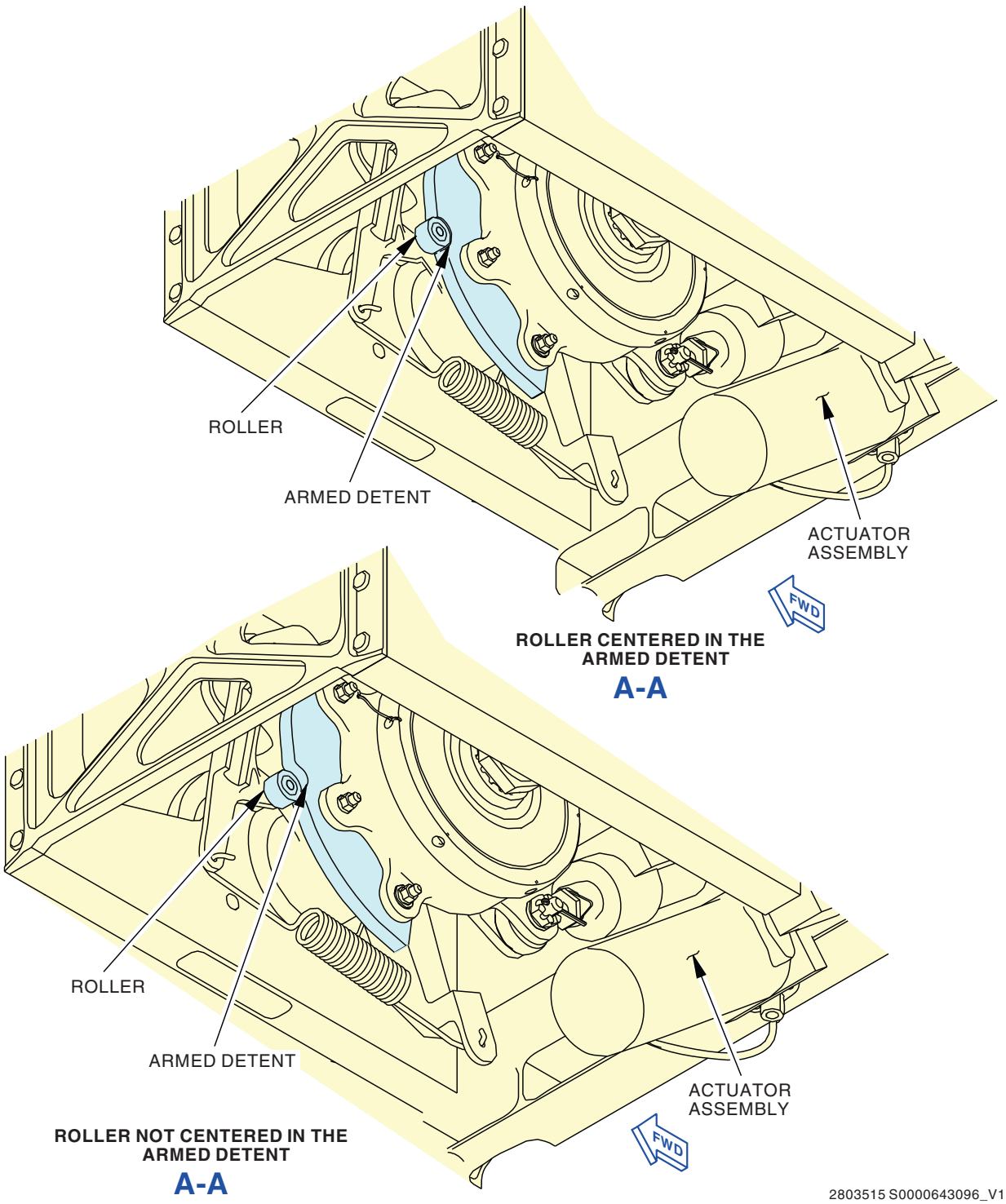
Speedbrake Control Lever Adjustment
Figure 502/27-62-00-990-818 (Sheet 1 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



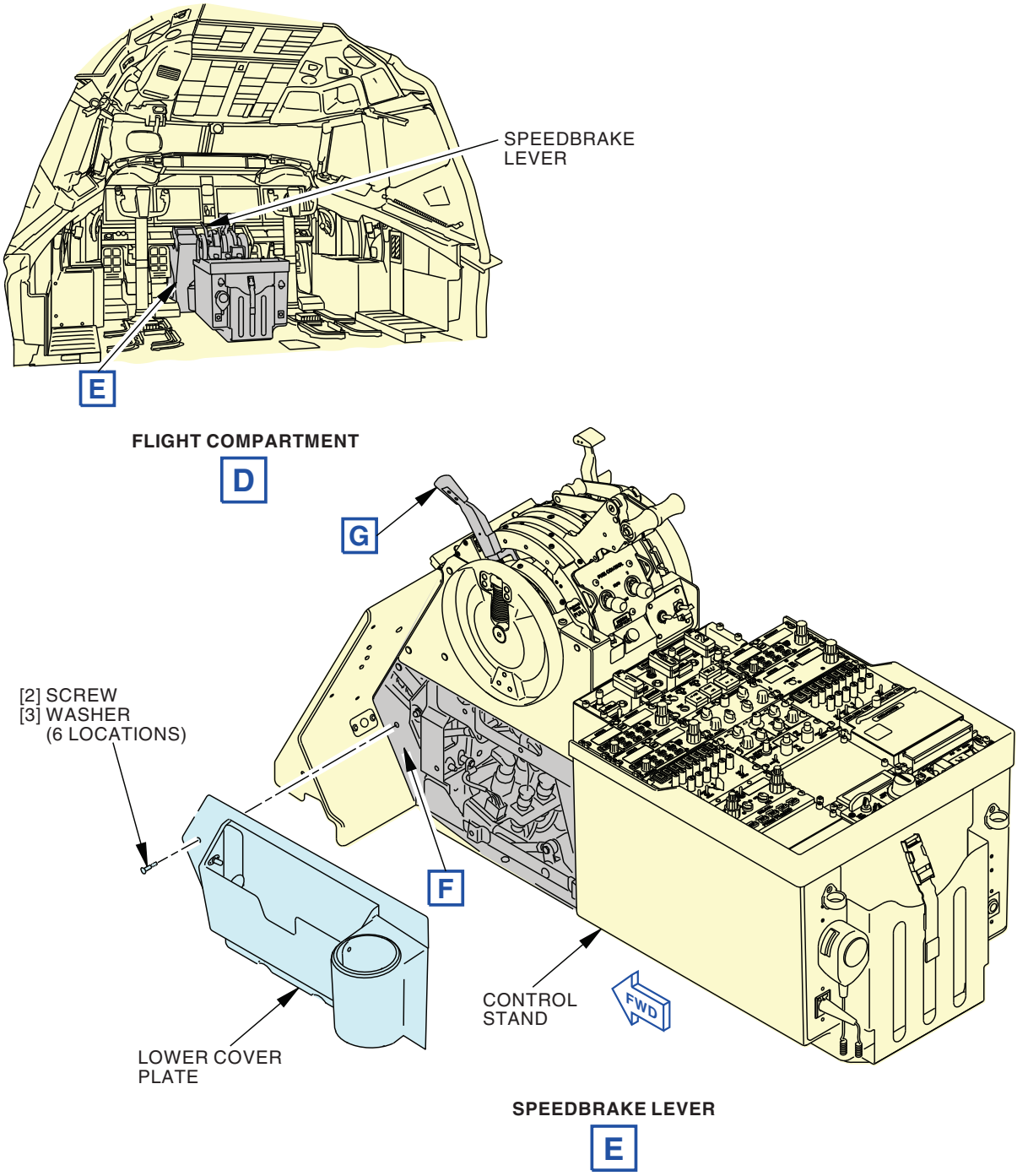
Speedbrake Control Lever Adjustment
Figure 502/27-62-00-990-818 (Sheet 2 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Speedbrake Control Lever Adjustment
Figure 502/27-62-00-990-818 (Sheet 3 of 6)

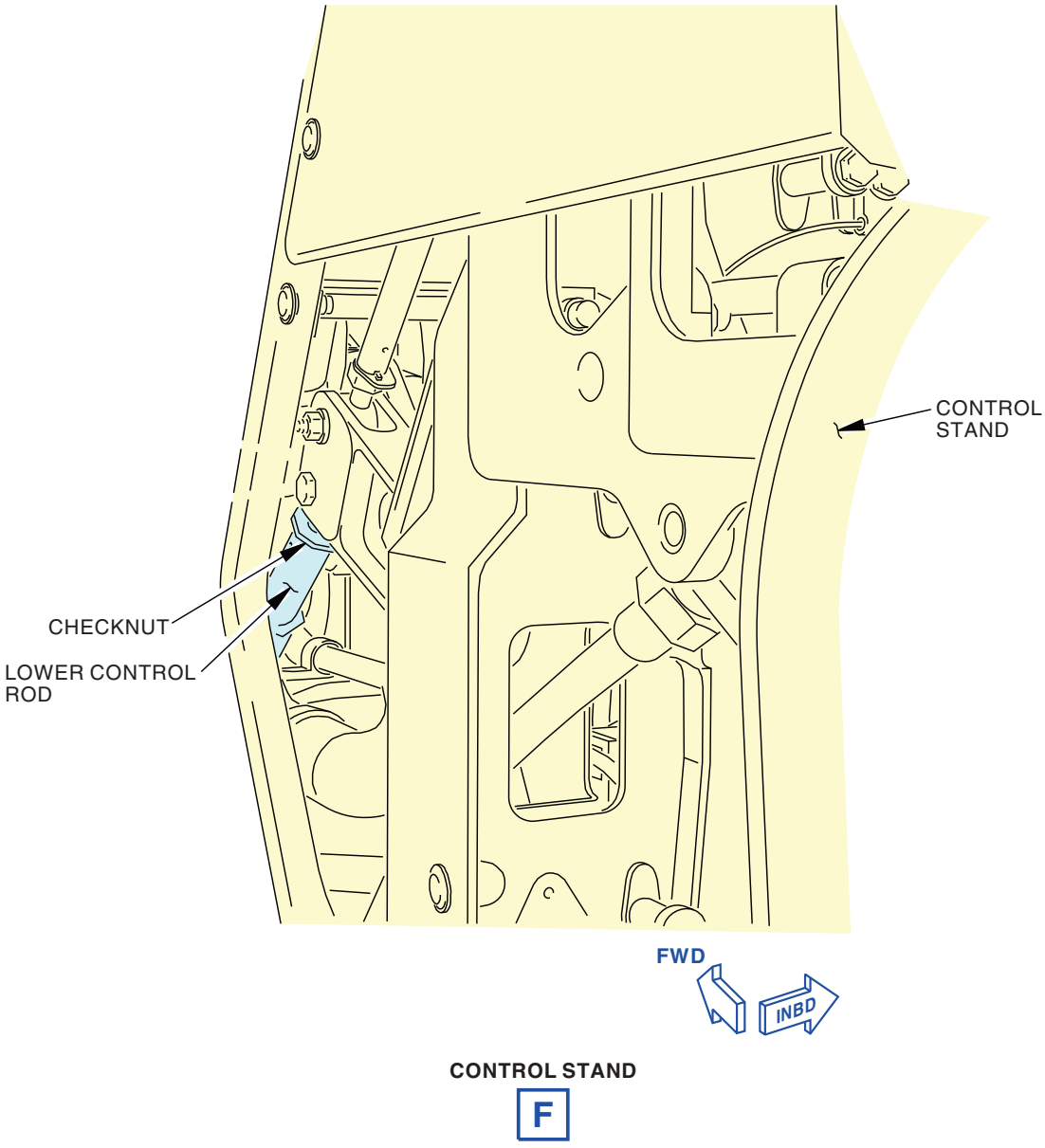
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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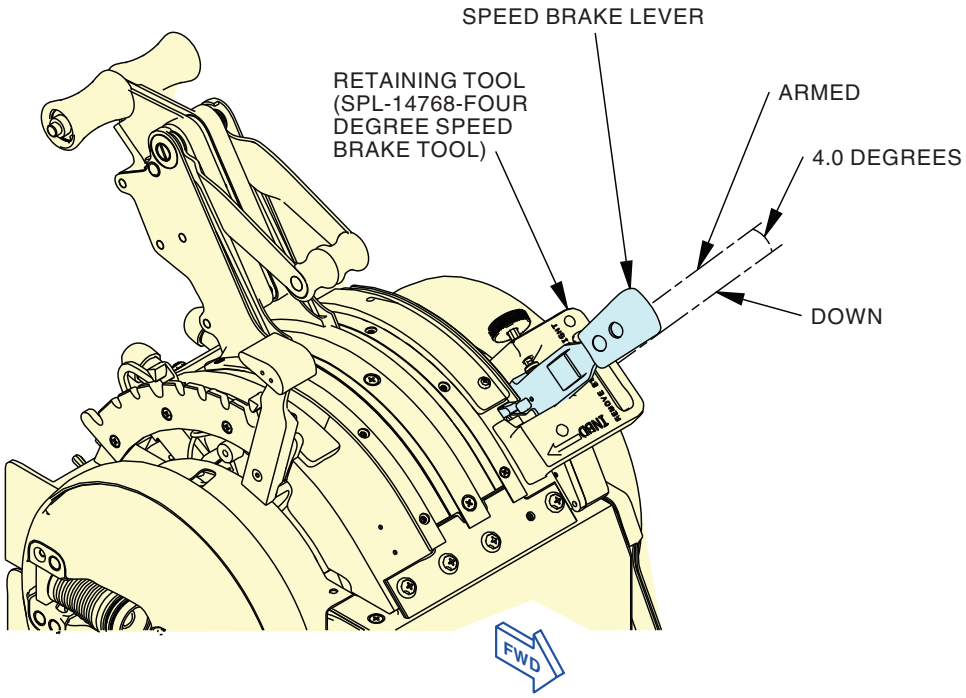
Speedbrake Control Lever Adjustment
Figure 502/27-62-00-990-818 (Sheet 4 of 6)

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Speedbrake Control Lever Adjustment
Figure 502/27-62-00-990-818 (Sheet 5 of 6)

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Speedbrake Control Lever Adjustment
Figure 502/27-62-00-990-818 (Sheet 6 of 6)

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TASK 27-62-00-820-802

4. Ground Spoiler Adjustment

(Figure 503)

A. General

(1) There are two different sets of values for the spoiler gaps.

(a) One set - standard gaps.

(b) The other set - reduced gaps. Reduced gaps are for airplane drag reduction.

(2) You may use either the standard or the reduced gaps for any airplane. However, for each airplane, you must choose which set of values (standard gaps or reduced gaps) to use to adjust the spoilers.

(3) You must use only one set of values for each airplane.

(4) Do not use standard gaps on some spoilers and reduced gaps on other spoilers on the same airplane.

B. References

Reference	Title
27-51-00-860-803	Extend the Trailing Edge Flaps (P/B 201)
27-51-00-860-804	Retract the Trailing Edge Flaps (P/B 201)
27-62-00-800-801	Apply Pressure to the Speed Brake Hydraulic Systems A and B (P/B 201)
27-62-00-800-802	Depressurization of the Speed Brake Hydraulic Systems A and B (P/B 201)
27-62-00-840-802	Pressurization of the Speed Brake Hydraulic Systems A and B to its Usual Condition (P/B 201)
46-13-02-700-805	Existing Faults Check (P/B 201)

C. Consumable Materials

Reference	Description	Specification
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95
B00184	Solvent - Presealing, Cleaning Solvent	BMS11-7
G01912	Lockwire - MS20995NC32, Monel - 0.032 Inch (0.8128 mm) Diameter	NASM20995
G50171	Compound - Corrosion Inhibiting Compound, Interior Application - D5026NS or ZC-026	

D. Location Zones

Zone	Area
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50
552	Left Wing - Spoiler No. 6
566	Left Wing - Spoiler No. 1
652	Right Wing - Spoiler No. 7
666	Right Wing - Spoiler No. 12

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E. Prepare for the Adjustment

SUBTASK 27-62-00-860-032

WARNING

MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. THE AILERONS, RUDDER, ELEVATORS, FLAPS, SLATS, SPOILERS, LANDING GEAR, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Do this task: Apply Pressure to the Speed Brake Hydraulic Systems A and B, TASK 27-62-00-800-801.

SUBTASK 27-62-00-866-003

(2) Do this task: Extend the Trailing Edge Flaps, TASK 27-51-00-860-803.

SUBTASK 27-62-00-860-033

(3) Move the speed brake lever to the UP position to lift the spoilers.

SUBTASK 27-62-00-860-034

(4) Do this task: Retract the Trailing Edge Flaps, TASK 27-51-00-860-804.

SUBTASK 27-62-00-860-035

(5) Move the speed brake lever to the DOWN position to lower the ground spoilers.

F. Ground Spoiler Adjustment

SUBTASK 27-62-00-860-036

(1) Measure the clearance (Dimension X) between the trailing edge of the spoiler panels and the flaps near the centerline of the spoiler actuators.

(2) Make sure the clearance (Dimension X) between the spoilers and the flaps are within the limits specified in the Spoilers Gaps Table.

(a) You do not need to check the clearance at the inboard actuator.

(b) Adjust the spoilers for one airplane to the values in only one table.

1) Do not adjust some spoilers to one table and other spoilers on the same airplane to a different table.

STANDARD SPOILERS GAPS		
DIMENSION X Figure 503		
SPOILERS	MINIMUM	MAXIMUM
1, 12	0.020 in. (0.508 mm)	0.120 in. (3.048 mm)
6, 7	0.030 in. (0.762 mm)	0.125 in. (3.175 mm)

REDUCED SPOILERS GAPS		
DIMENSION X Figure 503		
SPOILERS	MINIMUM	MAXIMUM
1, 12	0.030 in. (0.762 mm)	0.130 in. (3.302 mm)
6, 7	0.070 in. (1.78 mm)	0.090 in. (2.286 mm)

(3) Make sure there is clearance between the trailing edge of the spoiler panels and the flaps for the full span of the spoiler.

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SUBTASK 27-62-00-860-037

(4) If the clearances (Dimension X) between the ground spoiler panels and the flaps are not in the limits, adjust the clearance.

(a) Make a record of the clearance.

(b) Lift the spoiler or extend the flaps to get access to the spoiler actuator:

1) Do this task: Extend the Trailing Edge Flaps, TASK 27-51-00-860-803.

2) Move the speed brake lever to the UP position to lift the spoilers.

3) Do this task: Depressurization of the Speed Brake Hydraulic Systems A and B, TASK 27-62-00-800-802.

(c) Loosen the checknut on the rod end of the ground spoiler actuator.

(d) Remove the locking key from the piston rod.

(e) Do these steps to adjust the Outboard Ground Spoilers (No. 1 and 12):

1) Turn the piston rod on the spoiler actuator to adjust the clearance between the spoiler and the flaps to the correct tolerance.

2) Temporarily torque the actuator check nut to 100 in-lb (11.3 N·m) to 200 in-lb (22.6 N·m).

3) Move the speedbrake lever to the UP position.

WARNING

MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. THE AILERONS, RUDDER, ELEVATORS, FLAPS, SLATS, SPOILERS, LANDING GEAR, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

4) Do this task: Apply Pressure to the Speed Brake Hydraulic Systems A and B, TASK 27-62-00-800-801.

5) Do this task: Retract the Trailing Edge Flaps, TASK 27-51-00-860-804.

6) Move the speedbrake lever to the DOWN position.

7) Check the clearance at the trailing edge of the ground spoilers.

a) Make sure the clearance is within the limits (Figure 503).

8) If the clearance of the ground spoiler is not in the limits, do these steps to adjust the actuator.

a) Do this task: Extend the Trailing Edge Flaps, TASK 27-51-00-860-803.

b) Do this task: Depressurization of the Speed Brake Hydraulic Systems A and B, TASK 27-62-00-800-802.

c) Adjust the spoiler actuator again, as necessary.

9) If the clearance of the ground spoiler is in the limits, do these steps:

a) Do this task: Extend the Trailing Edge Flaps, TASK 27-51-00-860-803.

b) Move the speed brake lever to the UP position to extend the spoilers.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. THE AILERONS, RUDDER, ELEVATORS, FLAPS, SLATS, SPOILERS, LANDING GEAR, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

c) Do this task: Depressurization of the Speed Brake Hydraulic Systems A and B, TASK 27-62-00-800-802.

d) Clean all grease from the piston rod and the rod end threads with solvent, B00184.

e) Apply a layer of the corrosion preventive D5026NS or ZC-026 compound, G50171, to the threads of the piston rod end.

f) Install the locking key in the actuator rod end.

g) Torque the checknut to 250 in-lb (28.25 N·m) - 300 in-lb (33.90 N·m).

h) Install the MS20995NC32 lockwire, G01912, on the checknut with double twist method.

i) Apply a bead of sealant, A00247, to the rod end checknut.

(f) Do these steps to adjust the Inboard Ground Spoilers (No. 6 and 7):

1) Turn the piston rod on the outboard spoiler actuator to adjust the clearance between the spoiler and the flaps to the correct tolerance.

2) Turn the piston rod on the inboard spoiler actuator until the pin that connects the rod end to the spoiler panel moves freely or is loose.

NOTE: Make sure there is no force fight between the inboard and outboard actuators.

3) Temporarily torque the actuator check nut to 100 in-lb (11.3 N·m) to 200 in-lb (22.6 N·m).

4) Move the speed brake lever to the UP position.

a) Moving the speed brake lever without hydraulic power can latch the GROUND SPOILER CONTROL Status Message.

<1> Do this task if the Status Message occurs: Existing Faults Check, TASK 46-13-02-700-805.

WARNING MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. THE AILERONS, RUDDER, ELEVATORS, FLAPS, SLATS, SPOILERS, LANDING GEAR, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

5) Do this task: Apply Pressure to the Speed Brake Hydraulic Systems A and B, TASK 27-62-00-800-801.

6) Do this task: Retract the Trailing Edge Flaps, TASK 27-51-00-860-804.

7) Move the speedbrake lever to the DOWN position.

8) Check the clearance at the trailing edge of the ground spoilers.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

a) Make sure the clearance is within the limits (Figure 503).

9) If the clearance of the ground spoiler is not in the limits, do these steps to adjust the actuator:

a) Do this task: Extend the Trailing Edge Flaps, TASK 27-51-00-860-803.

b) Do this task: Depressurization of the Speed Brake Hydraulic Systems A and B, TASK 27-62-00-800-802.

c) Adjust the spoiler actuator again, as necessary.

10) If the clearance of the ground spoiler is in the limits, do these steps:

a) Do this task: Extend the Trailing Edge Flaps, TASK 27-51-00-860-803.

b) Move the speed brake lever to the UP position to extend the spoilers.

<1> Moving the speed brake lever without hydraulic power can latch the GROUND SPOILER CONTROL Status Message.

<a> Do this task if the Status Message occurs: Existing Faults Check, TASK 46-13-02-700-805.

WARNING

MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. THE AILERONS, RUDDER, ELEVATORS, FLAPS, SLATS, SPOILERS, LANDING GEAR, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

c) Do this task: Depressurization of the Speed Brake Hydraulic Systems A and B, TASK 27-62-00-800-802.

d) Clean all grease from the piston rod and the rod end threads with solvent, B00184.

e) Apply a layer of the corrosion preventive D5026NS or ZC-026 compound, G50171, to the threads of the piston rod end.

f) Torque the checknut to 600 in-lb (67.79 N·m) - 720 in-lb (81.35 N·m).

g) Install the MS20995NC32 lockwire, G01912, on the checknut with double twist method.

h) Apply a bead of sealant, A00247, to the rod end checknut.

SUBTASK 27-62-00-860-038

(5) If the clearances (Dimension X) between the ground spoiler panels and the flaps are within the limits, then continue.

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G. Put the Airplane Back to Its Usual Condition

SUBTASK 27-62-00-860-039

WARNING

MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. THE AILERONS, RUDDER, ELEVATORS, FLAPS, SLATS, SPOILERS, LANDING GEAR, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Do this task: Apply Pressure to the Speed Brake Hydraulic Systems A and B, TASK 27-62-00-800-801.

SUBTASK 27-62-00-860-040

(2) Move the speed brake lever to the DOWN position to lower the spoilers if it is necessary.

SUBTASK 27-62-00-860-041

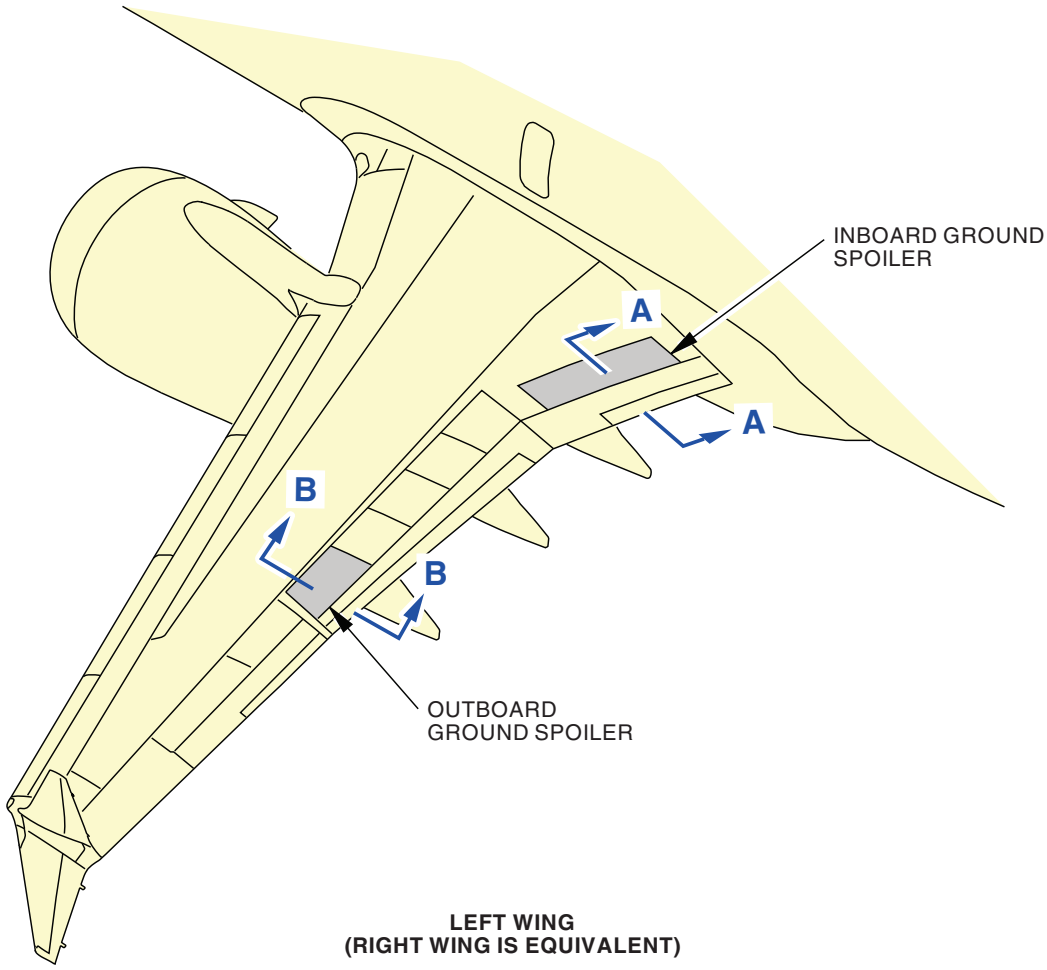
(3) Do this task: Retract the Trailing Edge Flaps, TASK 27-51-00-860-804.

SUBTASK 27-62-00-860-042

(4) Do this task: Pressurization of the Speed Brake Hydraulic Systems A and B to its Usual Condition, TASK 27-62-00-840-802.

——— END OF TASK ———

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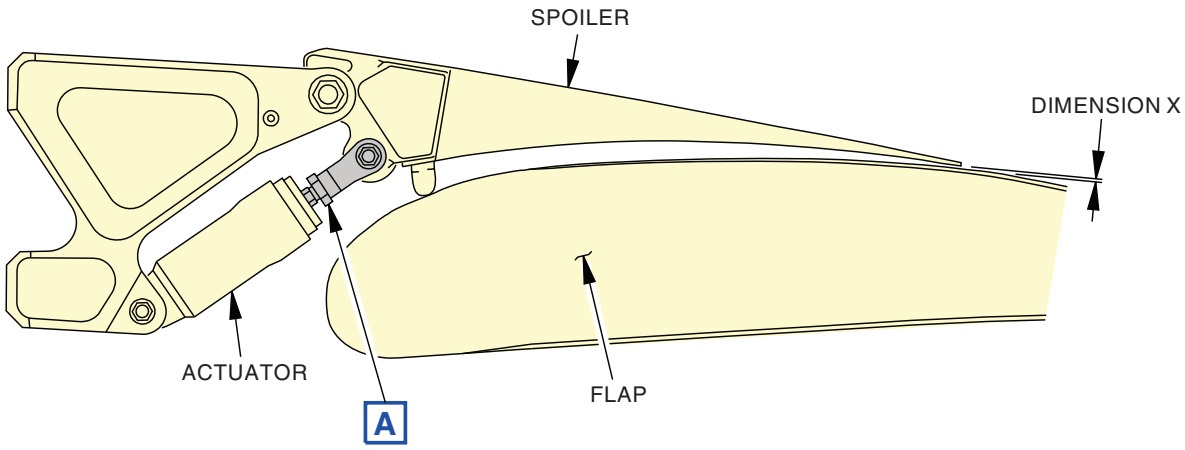


LEFT WING
(RIGHT WING IS EQUIVALENT)

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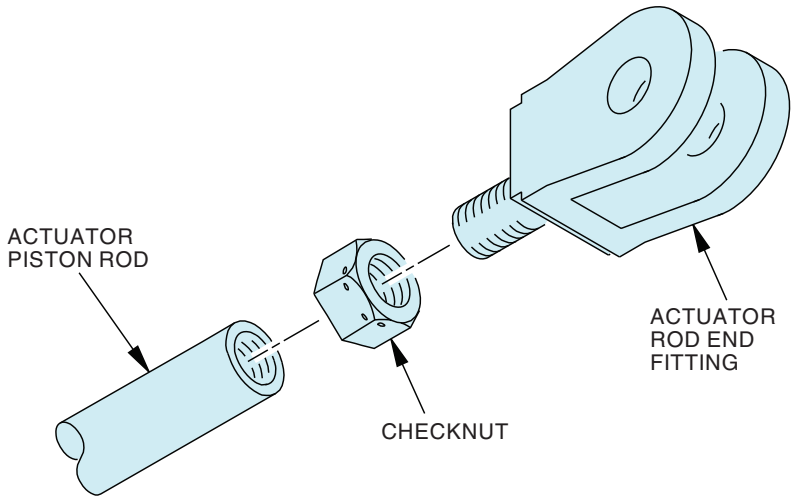
Ground Spoiler Adjustment
Figure 503/27-62-00-990-810 (Sheet 1 of 3)

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**INBOARD GROUND SPOILER
(OUTBOARD ACTUATOR)**

A-A



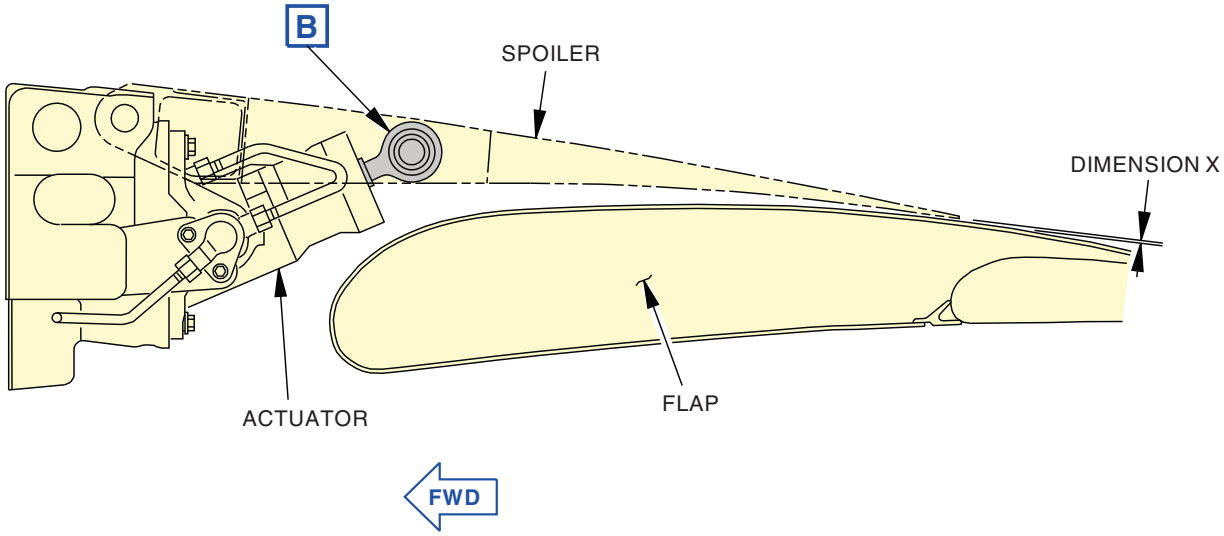
ACTUATOR ROD END

A

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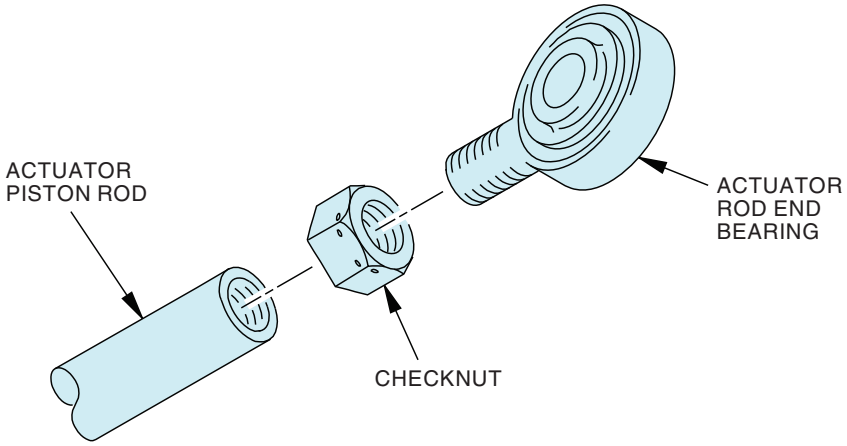
**Ground Spoiler Adjustment
Figure 503/27-62-00-990-810 (Sheet 2 of 3)**

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OUTBOARD GROUND SPOILER

B-B



ACTUATOR ROD END

B

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Ground Spoiler Adjustment
Figure 503/27-62-00-990-810 (Sheet 3 of 3)

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TASK 27-62-00-800-803

5. Speed Brake Lever Friction Test

A. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1557	Gauge - Force Part #: DG-200 Supplier: 92456 Part #: FDIX 100 Supplier: 0BFD9 Part #: FDIX 50 Supplier: 0BFD9 Part #: FG-60KBM Supplier: 45KA8 Part #: LG-050 Supplier: 92456 Part #: LG-100 Supplier: 92456 Opt Part #: DPP-500G Supplier: 92456 Opt Part #: DPPH-150 Supplier: 92456 Opt Part #: DPPH-200 Supplier: 92456 Opt Part #: DPPH-50 Supplier: 92456 Opt Part #: FDI 100 Supplier: 0BFD9 Opt Part #: FDI 50 Supplier: 0BFD9 Opt Part #: FDV 100 Supplier: 0BFD9 Opt Part #: FDV 50 Supplier: 0BFD9

B. Consumable Materials

Reference	Description	Specification
D50116	Lubricant - Bonded Solid Film	BAC5811 Type 7 Class 1
D50117	Lubricant - Solid Film, Air Cured, Corrosion Inhibiting	MIL-L-23398 Type II

C. Location Zones

Zone	Area
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50

D. Prepare for the Speed Brake Lever Friction Test

SUBTASK 27-62-00-860-200

WARNING KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT SPOILERS. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU OPEN/CLOSE THESE CIRCUIT BREAKERS. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
C	10	C04032	SPOILER CONTROL ELEC PWR A
C	12	C04033	SPOILER CONTROL ELEC PWR B

E. Speed Brake Lever Friction Test

SUBTASK 27-62-00-710-008

(1) Do these steps for the Speed Brake Lever Friction Test:

(a) Move the speed brake lever through the full range of travel in both directions.

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(b) Make sure that the clearance between the inboard face of the speed brake lever and edge of the light plate is between 0.01 in. (0.25 mm) to 0.10 in. (2.54 mm).

(c) Move the speed brake lever several times through the full range of travel in both directions.

(d) Make sure that the speed brake lever moves freely and there is no sign of binding or roughness between the detent positions.

SUBTASK 27-62-00-866-004

(2) Move the flaps to the FLAPS 1 or greater detent position.

SUBTASK 27-62-00-710-003

(3) Use the force gauge, COM-1557, to measure the fore and aft forces necessary to move the speed brake lever.

(a) Apply the force at the tip of the speed brake lever.

SUBTASK 27-62-00-710-004

(4) Move the speed brake lever to the ARMED DETENT position.

(5) Make sure that the forces necessary to move the speed brake lever are less than these force limits:

(a) From the ARMED DETENT position move the speed brake lever forward - 20 lbf (89 N).

(b) From the ARMED DETENT position move the speed brake lever aft - 20 lbf (89 N).

SUBTASK 27-62-00-710-005

(6) Move the speed brake lever to the FLIGHT DETENT position and make sure that the forces necessary to move the speed brake lever are less than these force limits:

(a) From the FLIGHT DETENT position move the speed brake lever forward - 26 lbf (116 N).

(b) From the FLIGHT DETENT position move the speed brake lever aft - 26 lbf (116 N).

SUBTASK 27-62-00-710-006

(7) Move the speed brake lever to the ARMED DETENT position.

(a) Move the speed brake lever aft to the FLIGHT DETENT position.

1) Make sure that the force necessary to move the speed brake lever does not exceed 20 lbf (89 N).

NOTE: Skewed-roller brake creates speed brake lever forces within the speed brake mechanism assembly. There is no on-airplane adjustment. Supplier must accomplish any adjustment.

SUBTASK 27-62-00-710-007

(8) Move the speed brake lever to the FLIGHT DETENT position.

(a) Move the speed brake lever aft to the UP position.

1) Make sure that the force necessary to move the speed brake lever does not exceed 28 lbf (125 N).

NOTE: Skewed-roller brake creates speed brake lever forces within the speed brake mechanism assembly. There is no on-airplane adjustment. Supplier must accomplish any adjustment.

SUBTASK 27-62-00-640-001

(9) Lubricate the speed brake lever if the forces required to move the speed brake lever are excessive.

(a) Fully extend the affected handle.

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CAUTION IF YOU DO NOT MASK THE HANDLES CORRECTLY, CONTAMINATION OF ADJACENT COMPONENTS CAN OCCUR. CONTAMINATION CAN CAUSE DAMAGE TO EQUIPMENT.

(b) Mask off the surrounding area to protect from over spray.

(c) Apply lubricant, D50116, or lubricant, D50117, to the sliding part of the speed brake lever.

1) Use an extension straw to allow spray in hard to reach areas.

NOTE: Allow to air dry if possible. The lubricant will dry within minutes; however, the coating will not be cured for several hours depending on the ambient temperature. The lubricant will cure within 6 hours at 77 ±3°F (25 ±2°C). 24 hours gives the best wear and corrosion protection.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 27-62-00-866-005

(1) Move the speed brake lever to the DOWN position.

SUBTASK 27-62-00-080-005

(2) Remove the force gauge, COM-1557, from the speed brake lever.

SUBTASK 27-62-00-860-199

WARNING KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT SPOILERS. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU OPEN/CLOSE THESE CIRCUIT BREAKERS. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(3) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
C	10	C04032	SPOILER CONTROL ELEC PWR A
C	12	C04033	SPOILER CONTROL ELEC PWR B

END OF TASK

TASK 27-62-00-820-803

6. Spoiler Operation Time Test

(Figure 504)

A. References

Reference	Title
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-09-00-860-802	Return the Airplane to the Ground Mode (P/B 201)
73-21-00-800-801-G00	EEC Maintenance Power Selection (P/B 201)

B. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50

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C. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

D. Prepare for the Test

SUBTASK 27-62-00-860-048

WARNING

MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. THE AILERONS, RUDDER, ELEVATORS, FLAPS, SLATS, SPOILERS, LANDING GEAR, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Use the Engine Driven Pump (EDP) or a ground service cart to pressurize the Speed Brake Hydraulic Systems A and B (TASK 29-11-00-860-801).

NOTE: The Electric Motor-Driven Pump (EMDP) flow rate is too low for the spoilers to operate in the permitted time.

SUBTASK 27-62-00-860-229

(2) Move the speedbrake lever several times through the complete range of travel at least six times.

SUBTASK 27-62-00-860-230

(3) Move the forward thrust levers No. 1 and 2 to the IDLE position.

SUBTASK 27-62-00-860-231

(4) Make sure that the ground spoilers are in the DOWN position.

SUBTASK 27-62-00-860-232

(5) Make sure that the speed brake lever is in the DOWN and LOCKED position.

SUBTASK 27-62-00-860-233

(6) Move one of the thrust levers to the FULL FORWARD position to retract the auto speed brake actuator.

SUBTASK 27-62-00-860-234

(7) Move the thrust lever to the idle position.

SUBTASK 27-62-00-860-235

(8) Make sure that the ANTI-SKID INOP light, on the P9 multifunction panel, is not on.

SUBTASK 27-62-00-860-236

(9) Set the autobrake select switch, on the P9 multifunction panel, to 1 position.

SUBTASK 27-62-00-860-237

(10) Make sure that the parking brake is released.

SUBTASK 27-62-00-010-007

(11) Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

E. Spoiler Operation Time Test

SUBTASK 27-62-00-860-247

(1) Set GND TEST switch, on the P61-4 maintenance bite panel, to the ENABLE position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-62-00-860-238

(2) Set the EEC MAINT ENG 1 POWER switch and EEC MAINT ENG 2 POWER switch to the TEST position.

(a) Do this task: EEC Maintenance Power Selection, TASK 73-21-00-800-801-G00.

SUBTASK 27-62-00-860-248

(3) Make sure that the airplane is in ground mode (TASK 32-09-00-860-802).

SUBTASK 27-62-00-860-239

(4) Move the speedbrake lever to the ARMED position.

SUBTASK 27-62-00-860-240

(5) Move the TEST SELECTOR switch to the OUTBD position on the Antiskid/Autobrake Control Unit (AACU).

SUBTASK 27-62-00-860-241

(6) At the same time that you push and hold the ENABLE/VERIFY button, move the PRESS/TEST toggle switch up to the TEST position on the AACU.

(a) Make sure that the speed brake lever moves to the FULL UP position.

(b) Make sure that the all flight spoilers move from FULL DOWN to FULL UP position in 1.2 seconds or less.

(c) Release the ENABLE/VERIFY button and put the PRESS/TEST toggle switch to the OFF position.

SUBTASK 27-62-00-860-242

(7) Move the TEST SELECTOR switch to the NORM position on the AACU.

SUBTASK 27-62-00-860-243

(8) Move the No. 1 thrust lever to the FULL FORWARD position.

(a) Make sure that speed brake lever moves to the DOWN and LOCKED position.

(b) Make sure that all flight spoilers move from FULL UP to FULL DOWN position in one second or less.

SUBTASK 27-62-00-860-244

(9) Return the No. 1 thrust lever to the IDLE position.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 27-62-00-860-245

(1) Set the EEC MAINT ENG 1 POWER switch and EEC MAINT ENG 2 POWER switch to the NORM position.

(a) Do this task: EEC Maintenance Power Selection, TASK 73-21-00-800-801-G00.

SUBTASK 27-62-00-860-249

(2) Set the GND TEST switch, on the P61-4 maintenance bite panel, to the NORM position.

SUBTASK 27-62-00-410-006

(3) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 27-62-00-860-246

(4) Move the autobrake select switch, on the P9 multifunction panel, to OFF position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-62-00-866-007

(5) If necessary, move the flap lever to the FLAPS UP position.

SUBTASK 27-62-00-860-054

(6) Do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

——— END OF TASK ———

TASK 27-62-00-820-804

7. Speed Brake Throttle Test

(Figure 504)

A. References

Reference	Title
27-62-00-800-801	Apply Pressure to the Speed Brake Hydraulic Systems A and B (P/B 201)
27-62-00-840-801	Shut Down the Speed Brake Hydraulic Systems A and B (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
73-21-00-800-801-G00	EEC Maintenance Power Selection (P/B 201)

B. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50

C. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

D. Prepare for the Test

SUBTASK 27-62-00-860-209

(1) Apply maintenance power to the Electronic Engine Control (EEC) system. Do this task: EEC Maintenance Power Selection, TASK 73-21-00-800-801-G00.

(a) Make sure that the GND TEST switch on the P61-4 MAINT/BITE panel is set to the ENABLE position.

SUBTASK 27-62-00-480-005

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(2) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-62-00-860-056

WARNING MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. THE AILERONS, RUDDER, ELEVATORS, FLAPS, SLATS, SPOILERS, LANDING GEAR, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(3) Do this task: Apply Pressure to the Speed Brake Hydraulic Systems A and B, TASK 27-62-00-800-801.

SUBTASK 27-62-00-860-207

(4) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	14	C01070	FLIGHT CONTROL AUTOSLAT DC 2
C	14	C01068	FLIGHT CONTROL AUTOSLAT DC 1

SUBTASK 27-62-00-860-058

(5) Move the thrust levers No. 1 and 2 to the idle position.

SUBTASK 27-62-00-860-059

(6) Check the Flight Controls Maint Data Page (on the MAX Display System) and make sure all four GEAR ON GND say “NEAR”.

SUBTASK 27-62-00-860-208

(7) Make sure that the Radio Altitude (RA) Transceiver's circuit breakers are in and that there are valid RA values on the Primary Flight Display.

(a) Make sure that these circuit breakers are closed:

CAPT Electrical System Panel, P18-1

Row	Col	Number	Name
B	4	C01384	RADIO NAVIGATION RADIO ALTM 1

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
A	16	C01385	RADIO NAVIGATION RADIO ALTM 2

SUBTASK 27-62-00-860-060

(8) Make sure that the ground spoilers are in the DOWN position.

SUBTASK 27-62-00-860-061

(9) Make sure that the speed brake lever is in the DOWN position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-62-00-760-001

WARNING

MAKE SURE THAT YOU OPEN THE CIRCUIT BREAKER FOR THE WEATHER RADAR SYSTEM. THE FORWARD MOVEMENT OF A THRUST LEVER CAN CAUSE THE AUTOMATIC OPERATION OF THE SYSTEM. THE OPERATION OF THIS SYSTEM CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT IN THE AREA OF THE NOSE RADOME.

(10) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	13	C00120	WEATHER RADAR RT

SUBTASK 27-62-00-860-062

(11) Move one of the thrust levers to the full forward position to retract the speed brake actuator.

SUBTASK 27-62-00-860-063

(12) Move the thrust lever to the idle position.

SUBTASK 27-62-00-860-202

(13) Put the autobrake select switch located on the P9 multifunction panel to 1.

SUBTASK 27-62-00-860-064

(14) Make sure that the parking brake is released.

SUBTASK 27-62-00-010-001

(15) Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

E. Speed Brake Throttle Switch Test

SUBTASK 27-62-00-860-065

NOTE: Throughout this test, the Speed Brake “ARMED” and “DO NOT ARM” lights indicate opposing conditions. They should never illuminate simultaneously. Therefore, only one light shall be specified unless both lights are off.

(1) Move the speed brake lever to the ARMED position.

(a) Make sure that the SPEED BRAKE ARMED light comes on.

SUBTASK 27-62-00-860-066

(2) Move the TEST SELECTOR switch to the INBD position on the Antiskid/Autobrake Control Unit (AACU).

SUBTASK 27-62-00-860-067

(3) Push the ENABLE/VERIFY button and move the toggle switch to the PRESS/TEST position on the AACU.

(a) Make sure that the spoilers and speed brake lever move to the UP position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-62-00-860-068

WARNING

MAKE SURE THAT YOU OPEN THE CIRCUIT BREAKER FOR THE WEATHER RADAR SYSTEM. THE FORWARD MOVEMENT OF A THRUST LEVER CAN CAUSE THE AUTOMATIC OPERATION OF THE SYSTEM. THE OPERATION OF THIS SYSTEM CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT IN THE AREA OF THE NOSE RADOME.

(4) Make sure that this circuit breaker is open and has safety tag:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	13	C00120	WEATHER RADAR RT

SUBTASK 27-62-00-860-069

(5) Move the No. 1 thrust lever to the full forward position.

(a) Make sure that all spoilers move to the DOWN position.

(b) Make sure that the speed brake lever moves to the DOWN position.

(c) Make sure that the SPEED BRAKE ARMED and SPEED BRAKE DO NOT ARM lights are off.

SUBTASK 27-62-00-860-070

(6) Move the No. 1 thrust lever back to the idle position.

SUBTASK 27-62-00-860-071

(7) Release the ENABLE/VERIFY button and the PRESS/TEST switch on the AACU.

SUBTASK 27-62-00-860-072

(8) Do the test of the throttle switches again except move the TEST SELECTOR switch to the OUTBD position on the AACU and use the No. 2 thrust lever.

(a) Make sure that results are the same.

SUBTASK 27-62-00-860-073

(9) Move the TEST SELECTOR switch to the NORM position on the AACU.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 27-62-00-860-210

(1) Remove maintenance power to the EEC system (TASK 73-21-00-800-801-G00).

SUBTASK 27-62-00-860-213

(2) If necessary, move the flap lever to the FLAPS UP position.

SUBTASK 27-62-00-860-074

(3) Do this task: Shut Down the Speed Brake Hydraulic Systems A and B, TASK 27-62-00-840-801.

SUBTASK 27-62-00-860-075

(4) Close these circuit breakers:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	13	C00120	WEATHER RADAR RT

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	14	C01070	FLIGHT CONTROL AUTOSLAT DC 2
C	14	C01068	FLIGHT CONTROL AUTOSLAT DC 1

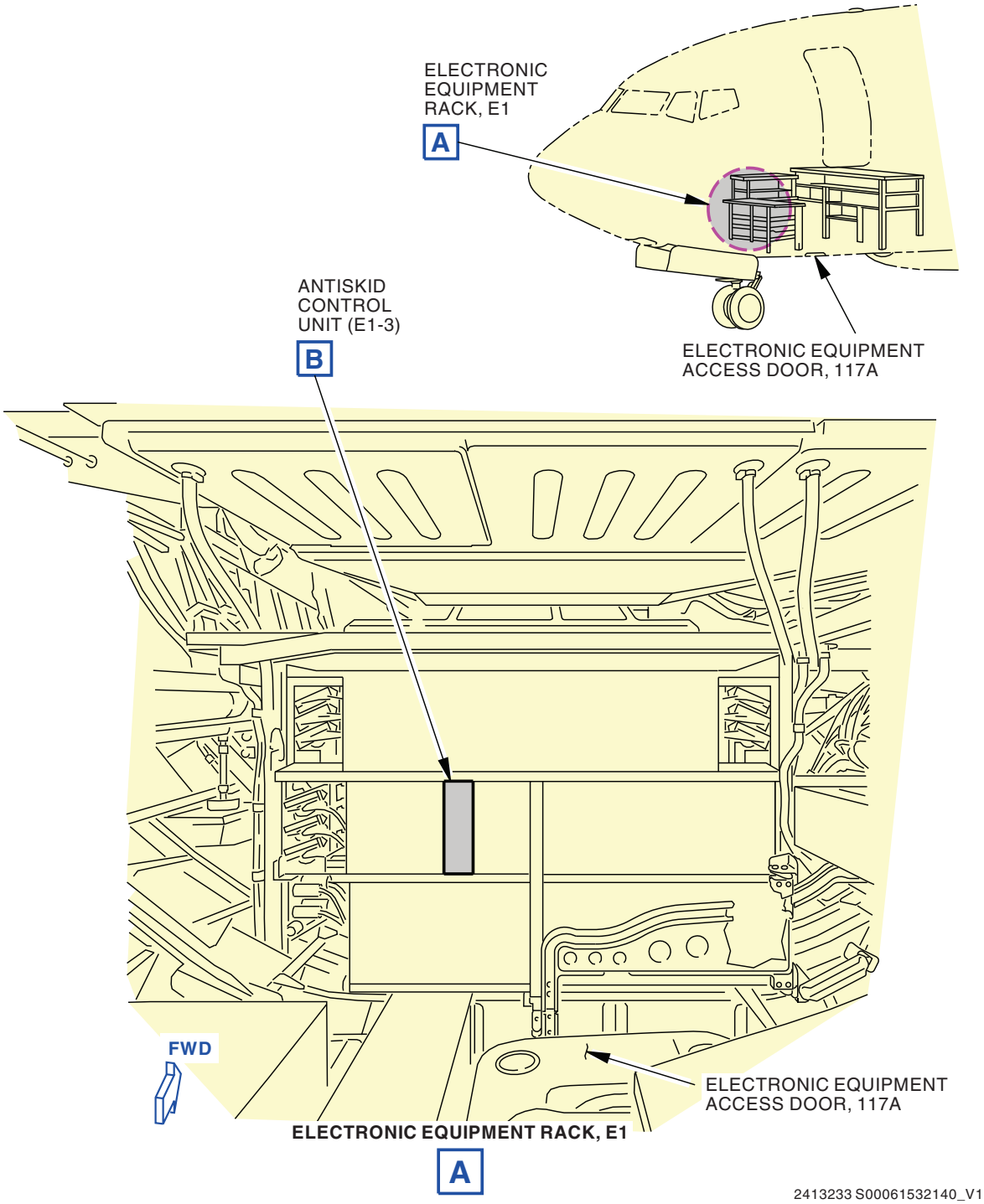
SUBTASK 27-62-00-410-001

(5) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

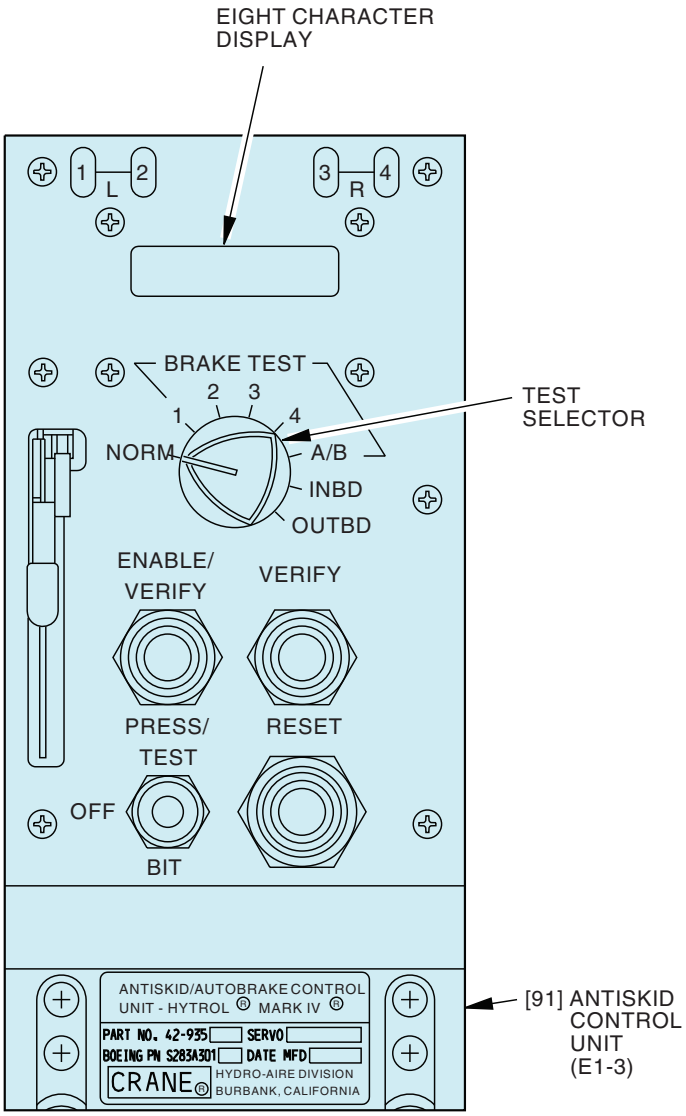
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Antiskid/Autobrake Control Unit Adjustment
Figure 504/27-62-00-990-811 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



ANTISKID CONTROL UNIT (E1-3)

B

2413234 S00061532141_V1

Antiskid/Autobrake Control Unit Adjustment
Figure 504/27-62-00-990-811 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-62-00-820-805

8. Speed Brake Antiskid Inoperative Test

(Figure 504)

A. References

Reference	Title
27-62-00-800-801	Apply Pressure to the Speed Brake Hydraulic Systems A and B (P/B 201)
27-62-00-840-801	Shut Down the Speed Brake Hydraulic Systems A and B (P/B 201)

B. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50

C. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

D. Prepare for the Test

SUBTASK 27-62-00-860-093

WARNING

MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. THE AILERONS, RUDDER, ELEVATORS, FLAPS, SLATS, SPOILERS, LANDING GEAR, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Do this task: Apply Pressure to the Speed Brake Hydraulic Systems A and B, TASK 27-62-00-800-801.

SUBTASK 27-62-00-860-095

(2) Move the thrust levers No. 1 and 2 to the IDLE position.

SUBTASK 27-62-00-860-096

(3) Make sure that the airplane is on its landing gear or compress the right main gear strut a minimum of 5 in. (127 mm).

SUBTASK 27-62-00-860-097

(4) Make sure that the ground spoilers are in the DOWN position.

SUBTASK 27-62-00-860-098

(5) Make sure that the speed brake lever is in the DOWN position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-62-00-860-099

WARNING MAKE SURE THAT YOU OPEN THE CIRCUIT BREAKER FOR THE WEATHER RADAR SYSTEM. THE FORWARD MOVEMENT OF A THRUST LEVER CAN CAUSE THE AUTOMATIC OPERATION OF THE SYSTEM. THE OPERATION OF THIS SYSTEM CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT IN THE AREA OF THE NOSE RADOME.

(6) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	13	C00120	WEATHER RADAR RT

SUBTASK 27-62-00-860-100

(7) Move one of the thrust levers to the full forward position to retract the speed brake actuator.

SUBTASK 27-62-00-860-101

(8) Move the thrust lever to the IDLE position.

SUBTASK 27-62-00-010-002

(9) Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 27-62-00-860-102

(10) Make sure that the parking brake is released.

E. Speed Brake Antiskid Inoperative Test

SUBTASK 27-62-00-860-103

(1) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	16	C00196	LANDING GEAR ANTISKID INBD

SUBTASK 27-62-00-860-104

NOTE: Throughout this test, the Speed Brake “ARMED” and “DO NOT ARM” lights indicate opposing conditions. They should never illuminate simultaneously. Therefore, only one light shall be specified unless both lights are off.

(2) Move the speed brake lever to the ARMED position.

(a) Make sure that the SPEED BRAKE ARMED light comes on.

SUBTASK 27-62-00-860-105

(3) Move the TEST SELECTOR switch to the INBD position on the Antiskid/Autobrake Control Unit (AACU).

SUBTASK 27-62-00-860-106

(4) On the AACU, push the ENABLE/VERIFY button and move the toggle switch to the TEST position and hold for a few seconds.

(a) Make sure that the SPEED BRAKE ARMED light stays on.

(b) Make sure that the spoilers do not move.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-62-00-860-108

(5) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	16	C00196	LANDING GEAR ANTISKID INBD

SUBTASK 27-62-00-860-109

(6) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	18	C00195	LANDING GEAR ANTISKID OUTBD

SUBTASK 27-62-00-860-110

(7) Make sure that the speed brake lever is in the ARMED position.

(a) Make sure that the SPEED BRAKE ARMED light comes on.

SUBTASK 27-62-00-860-111

(8) Move the TEST SELECTOR switch to the OUTBOARD position on the AACU.

SUBTASK 27-62-00-860-204

(9) On the AACU, push the ENABLE/VERIFY button and move the toggle switch to the TEST position and hold for a few seconds.

(a) Make sure that the SPEED BRAKE ARMED light stays on.

(b) Make sure that the spoilers do not move.

SUBTASK 27-62-00-860-114

(10) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	18	C00195	LANDING GEAR ANTISKID OUTBD

SUBTASK 27-62-00-860-115

(11) Move the speed brake lever to the DOWN position.

SUBTASK 27-62-00-860-116

(12) Move the TEST SELECTOR switch to the NORM position on the AACU.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 27-62-00-860-117

(1) Do this task: Shut Down the Speed Brake Hydraulic Systems A and B, TASK 27-62-00-840-801.

SUBTASK 27-62-00-860-118

(2) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	13	C00120	WEATHER RADAR RT

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-62-00-410-002

(3) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

——— END OF TASK ———

TASK 27-62-00-820-807

9. Speed Brake RTO Test

(Figure 504)

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
27-62-00-800-801	Apply Pressure to the Speed Brake Hydraulic Systems A and B (P/B 201)
27-62-00-840-801	Shut Down the Speed Brake Hydraulic Systems A and B (P/B 201)
32-09-00-860-802	Return the Airplane to the Ground Mode (P/B 201)
73-21-00-800-801-G00	EEC Maintenance Power Selection (P/B 201)

B. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50

C. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

D. Prepare for the Test

SUBTASK 27-62-00-860-224

(1) Supply maintenance power to the Engine Electronic Controller (EEC) (TASK 73-21-00-800-801-G00).

SUBTASK 27-62-00-860-142

WARNING

MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. THE AILERONS, RUDDER, ELEVATORS, FLAPS, SLATS, SPOILERS, LANDING GEAR, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(2) Do this task: Apply Pressure to the Speed Brake Hydraulic Systems A and B, TASK 27-62-00-800-801.

SUBTASK 27-62-00-860-144

(3) Move the engine 1 and 2 forward thrust levers to the idle position.

SUBTASK 27-62-00-860-145

(4) Make sure that the airplane is in ground mode (TASK 32-09-00-860-802).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-62-00-860-223

(5) Set GND TEST switch, on the P61-4 maintenance bite panel, to the ENABLE position.

SUBTASK 27-62-00-860-146

(6) Make sure that the ground spoilers are in the down position.

SUBTASK 27-62-00-860-147

(7) Make sure the speed brake lever is in the DOWN position.

SUBTASK 27-62-00-860-148

WARNING

MAKE SURE THAT YOU OPEN THE CIRCUIT BREAKER FOR THE WEATHER RADAR SYSTEM. THE FORWARD MOVEMENT OF A THRUST LEVER CAN CAUSE THE AUTOMATIC OPERATION OF THE SYSTEM. THE OPERATION OF THIS SYSTEM CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT IN THE AREA OF THE NOSE RADOME.

(8) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	13	C00120	WEATHER RADAR RT

SUBTASK 27-62-00-860-149

(9) Move one of the engine forward thrust levers to the full forward position to retract the auto speed brake actuator.

SUBTASK 27-62-00-860-150

(10) Move the engine forward thrust lever to the idle position.

SUBTASK 27-62-00-860-206

(11) Set the autobrake select switch, on the P9 multifunction panel, to the 1 position.

SUBTASK 27-62-00-210-007

(12) Make sure that the ANTI-SKID INOP light, on the P9 multifunction panel, is not on.

SUBTASK 27-62-00-860-151

(13) Make sure that the parking brake is released.

SUBTASK 27-62-00-010-003

(14) Open this access door:

Number	Name/Location
117A	Electronic Equipment Access Door

E. Speed Brake RTO Test

NOTE: Throughout this test, the Speed Brake ARMED and DO NOT ARM lights show opposing conditions. They should never come on simultaneously. Therefore, only one light shall be specified unless both lights are off.

SUBTASK 27-62-00-860-154

WARNING

KEEP PERSONNEL AND EQUIPMENT AWAY FROM THE AREA AROUND THE THRUST REVERSERS. THE THRUST REVERSERS EXTEND AND RETRACT QUICKLY. INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Move the engine 1 reverse thrust lever to the reverse thrust position.

(a) Make sure that the SPEED BRAKE ARMED light comes on.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-62-00-860-155

(2) Set the TEST SELECTOR switch to the INBD position on the antiskid/autobrake control unit (AACU).

SUBTASK 27-62-00-860-156

(3) At the same time that you push and hold the ENABLE/VERIFY button, set the PRESS/TEST switch to the TEST position on the AACU.

NOTE: The SPEED BRAKE DO NOT ARM light will come on momentarily if you do not push the buttons at the same time.

(a) Make sure that the speed brake lever moves to the UP position.

(b) Make sure that the ground and flight spoilers move to the up position.

(c) Make sure that the SPEED BRAKE ARMED light stays on.

SUBTASK 27-62-00-860-250

(4) Release the ENABLE/VERIFY button.

SUBTASK 27-62-00-860-251

(5) Set the PRESS/TEST switch to the OFF position.

SUBTASK 27-62-00-860-160

(6) Move the engine 1 reverse thrust lever to the retract (stow) position.

SUBTASK 27-62-00-860-205

(7) Move the engine 1 forward thrust lever forward.

(a) Make sure that all the ground spoilers, flight spoilers, and speed brake lever move to the down position.

SUBTASK 27-62-00-860-252

(8) Move the engine 1 forward thrust lever to the idle position.

SUBTASK 27-62-00-860-161

WARNING KEEP PERSONNEL AND EQUIPMENT AWAY FROM THE AREA AROUND THE THRUST REVERSERS. THE THRUST REVERSERS EXTEND AND RETRACT QUICKLY. INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT CAN OCCUR.

(9) Move the engine 2 reverse thrust lever to the reverse thrust position.

(a) Make sure that the SPEED BRAKE ARMED light comes on.

SUBTASK 27-62-00-860-253

(10) Make sure that the TEST SELECTOR switch is set to the OUTBD position on the AACU.

SUBTASK 27-62-00-860-254

(11) At the same time that you push and hold the ENABLE/VERIFY button, set the PRESS/TEST switch to the TEST position on the AACU.

NOTE: The SPEED BRAKE DO NOT ARM light will come on momentarily if you do not push the buttons at the same time.

(a) Make sure that the speed brake lever moves to the UP position.

(b) Make sure that the ground and flight spoilers move to the up position.

(c) Make sure that the SPEED BRAKE ARMED light stays on.

SUBTASK 27-62-00-860-255

(12) Release the ENABLE/VERIFY button.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-62-00-860-256

(13) Set the PRESS/TEST switch to the OFF position.

SUBTASK 27-62-00-860-257

(14) Move the engine 2 reverse thrust lever to the retract (stow) position.

SUBTASK 27-62-00-860-258

(15) Move the engine 2 forward thrust lever forward.

(a) Make sure that all the ground spoilers, flight spoilers, and speed brake lever move to the down position.

SUBTASK 27-62-00-860-259

(16) Move the engine 2 forward thrust lever to the idle position.

SUBTASK 27-62-00-860-162

(17) Set the TEST SELECTOR switch to the NORM position on the AACU.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 27-62-00-860-228

(1) Set the autobrake select switch, on the P9 multifunction panel, to the OFF position.

SUBTASK 27-62-00-860-164

(2) Do this task: Shut Down the Speed Brake Hydraulic Systems A and B, TASK 27-62-00-840-801.

SUBTASK 27-62-00-860-226

(3) Remove maintenance power from the EEC (TASK 73-21-00-800-801-G00).

SUBTASK 27-62-00-860-165

(4) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	13	C00120	WEATHER RADAR RT

SUBTASK 27-62-00-010-004

(5) Close this access door:

Number	Name/Location
117A	Electronic Equipment Access Door

——— END OF TASK ———

TASK 27-62-00-820-808

10. Speed Brake Extended Light Test

A. General

(1) Ignore the SPEED BRAKE ARMED and SPEED BRAKE DO NOT ARM lights during this test. They may come on.

B. References

Reference	Title
27-51-00-860-804	Retract the Trailing Edge Flaps (P/B 201)
27-62-00-800-801	Apply Pressure to the Speed Brake Hydraulic Systems A and B (P/B 201)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Reference	Title
27-62-00-840-801	Shut Down the Speed Brake Hydraulic Systems A and B (P/B 201)

C. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50

D. Prepare for the Test

SUBTASK 27-62-00-860-167

WARNING

MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. THE AILERONS, RUDDER, ELEVATORS, FLAPS, SLATS, SPOILERS, LANDING GEAR, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Do this task: Apply Pressure to the Speed Brake Hydraulic Systems A and B, TASK 27-62-00-800-801.

SUBTASK 27-62-00-860-169

(2) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P6-1

Row	Col	Number	Name
B	4	C01384	RADIO NAVIGATION RADIO ALTM 1
E	5	C01009	ADIRU LEFT DC
E	7	C01007	ADIRU LEFT AC

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
A	16	C01385	RADIO NAVIGATION RADIO ALTM 2
C	14	C01008	ADIRU RIGHT AC
C	17	C01010	ADIRU RIGHT DC

E. Speed Brakes Extended Light Test

SUBTASK 27-62-00-860-170

(1) Move the flaps to the 15 unit position.

SUBTASK 27-62-00-860-171

(2) Move the speed brake lever to the ARMED position.

NOTE: The spoilers can move up during this step.

SUBTASK 27-62-00-860-172

(3) Make sure the SPEEDBRAKES EXTENDED light is off.

SUBTASK 27-62-00-860-173

(4) Move the speed brake lever 0.6 in. (15.2 mm) aft from the ARMED detent position .

NOTE: The spoilers can move up during this step.

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SUBTASK 27-62-00-860-174

(5) Make sure the SPEEDBRAKES EXTENDED light turns on.

SUBTASK 27-62-00-860-175

(6) Do this task: Retract the Trailing Edge Flaps, TASK 27-51-00-860-804.

SUBTASK 27-62-00-860-176

(7) Make sure the SPEEDBRAKES EXTENDED light is off.

SUBTASK 27-62-00-860-185

(8) Move the speed brake lever to the DOWN position.

NOTE: The spoilers can move down during this step.

SUBTASK 27-62-00-860-186

(9) Do this task: Retract the Trailing Edge Flaps, TASK 27-51-00-860-804.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 27-62-00-860-187

(1) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-1

Row	Col	Number	Name
B	4	C01384	RADIO NAVIGATION RADIO ALTM 1
E	5	C01009	ADIRU LEFT DC
E	7	C01007	ADIRU LEFT AC

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
A	16	C01385	RADIO NAVIGATION RADIO ALTM 2
C	14	C01008	ADIRU RIGHT AC
C	17	C01010	ADIRU RIGHT DC

SUBTASK 27-62-00-860-188

(2) Do this task: Shut Down the Speed Brake Hydraulic Systems A and B, TASK 27-62-00-840-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SPEED BRAKE CONTROL SYSTEM - INSPECTION/CHECK

1. General

A. This procedure has these inspection tasks:

(1) Speed Brake System Components

(2) Flight Spoiler and Ground Spoiler Actuators

TASK 27-62-00-210-801

2. Speedbrake System Components - Inspection

(Figure 601)

A. References

Reference	Title
27-62-22-000-801	Speedbrake Lever - Removal (P/B 401)
27-62-22-400-801	Speedbrake Lever - Installation (P/B 401)

B. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50

C. Access Panels

Number	Name/Location
112A	Forward Access Door

D. Speedbrake System Components Inspection

SUBTASK 27-62-00-010-005

(1) To get access to the speedbrake control system components,

Open this access panel:

Number	Name/Location
112A	Forward Access Door

SUBTASK 27-62-00-210-005

(2) Do a general visual inspection of these items:

(a) Speedbrake Lever Assembly

1) Examine the speedbrake control lever for a loose fastener.

2) Remove the left upper side panel and the left lower side panel (Speedbrake Lever - Removal, TASK 27-62-22-000-801).

3) Examine the link assembly for loose fasteners.

4) Examine the attach points of the link assembly for signs that they do not move freely.

5) Make sure that cables or lines do not touch the speedbrake control lever or link assembly during movement.

6) Install the left upper side panel and the left lower side panel (Speedbrake Lever - Installation, TASK 27-62-22-400-801).

(b) Auto Speedbrake Actuator

1) Examine the auto speedbrake actuator and actuator rod end for loose fasteners.

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2) Examine the attach points of the auto speedbrake actuator and actuator rod end for signs that they do not move freely.

(c) Lower Control Rod

1) Examine the lower control rod for loose fasteners.

2) Examine the attach points of the lower control rod for signs that they do not move freely.

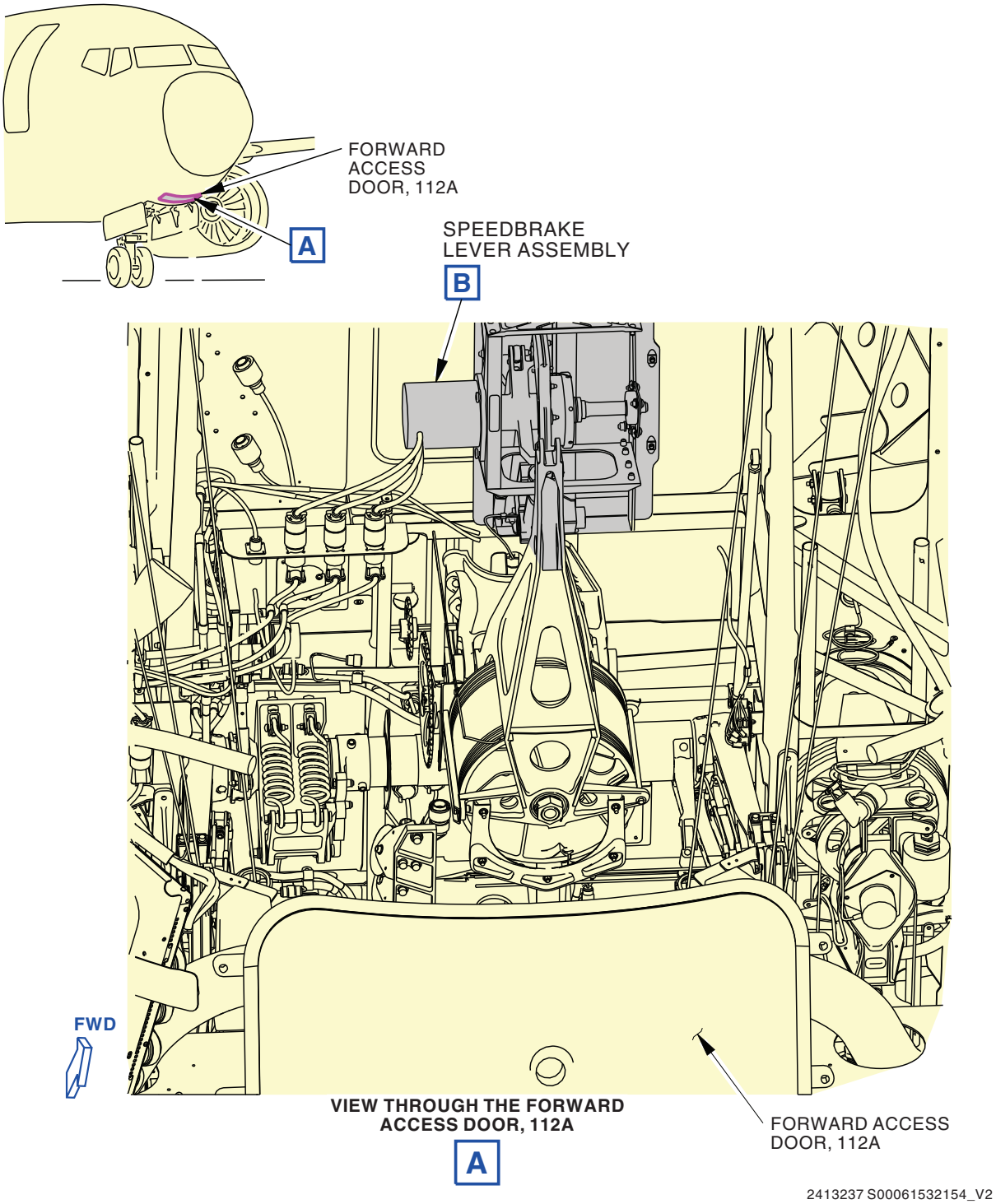
SUBTASK 27-62-00-410-003

(3) Close this access panel:

Number	Name/Location
112A	Forward Access Door

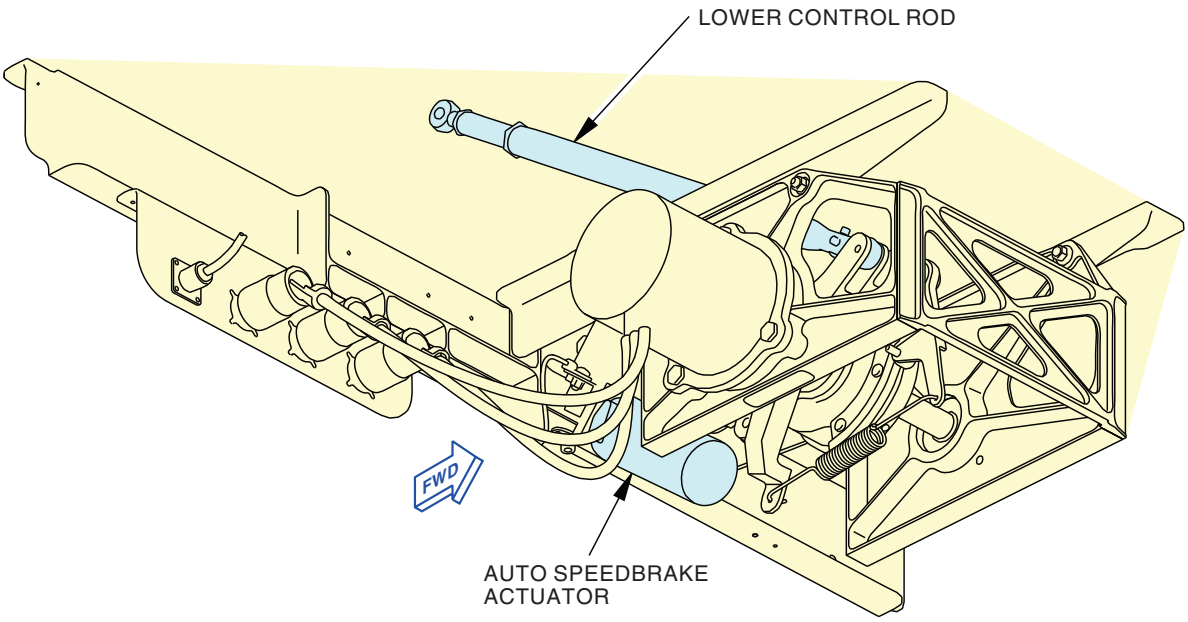
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Speedbrake System Components Inspection
Figure 601/27-62-00-990-814 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



SPEEDBRAKE LEVER ASSEMBLY

B

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Speedbrake System Components Inspection
Figure 601/27-62-00-990-814 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-62-00-210-802

3. Spoiler Actuator - Inspection

A. References

Reference	Title
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)

B. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

C. Location Zones

Zone	Area
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50
552	Left Wing - Spoiler No. 6
562	Left Wing - Spoiler No. 5
563	Left Wing - Spoiler No. 4
564	Left Wing - Spoiler No. 3
565	Left Wing - Spoiler No. 2
566	Left Wing - Spoiler No. 1
652	Right Wing - Spoiler No. 7
662	Right Wing - Spoiler No. 8
663	Right Wing - Spoiler No. 9
664	Right Wing - Spoiler No. 10
665	Right Wing - Spoiler No. 11
666	Right Wing - Spoiler No. 12

D. Spoiler Actuator Inspection

SUBTASK 27-62-00-010-006

WARNING KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 27-62-00-860-190

(2) Move the flap control lever to the 30-unit position to extend the flaps.

SUBTASK 27-62-00-860-191

(3) Move the speedbrake lever from the DOWN position to the UP position.

SUBTASK 27-62-00-860-192

(4) Attach a DO NOT OPERATE tag, STD-858, on the flap control lever and speedbrake lever.

SUBTASK 27-62-00-860-193

(5) Do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 27-62-00-210-006

(6) Do a general visual inspection of the spoiler actuators (Figure 602).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-62-00-860-194

(7) Do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 27-62-00-860-195

(8) Remove a DO NOT OPERATE tag, STD-858, from the flap control lever and speedbrake lever.

SUBTASK 27-62-00-860-196

(9) Move the flap control lever to the UP position to retract the flaps.

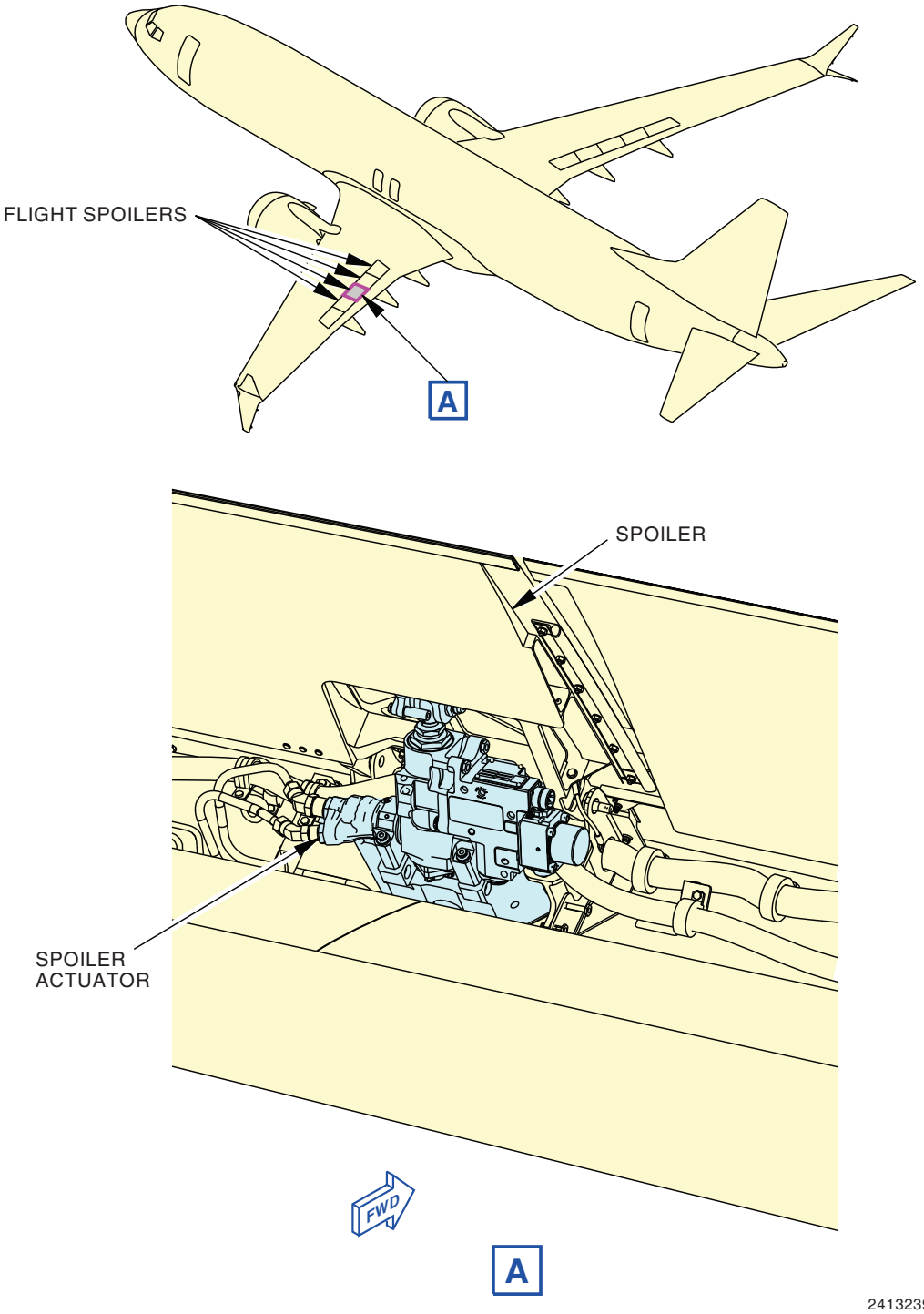
SUBTASK 27-62-00-860-197

(10) Move the speedbrake lever from the UP position to the DOWN position.

(11) If the hydraulic power is not needed, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

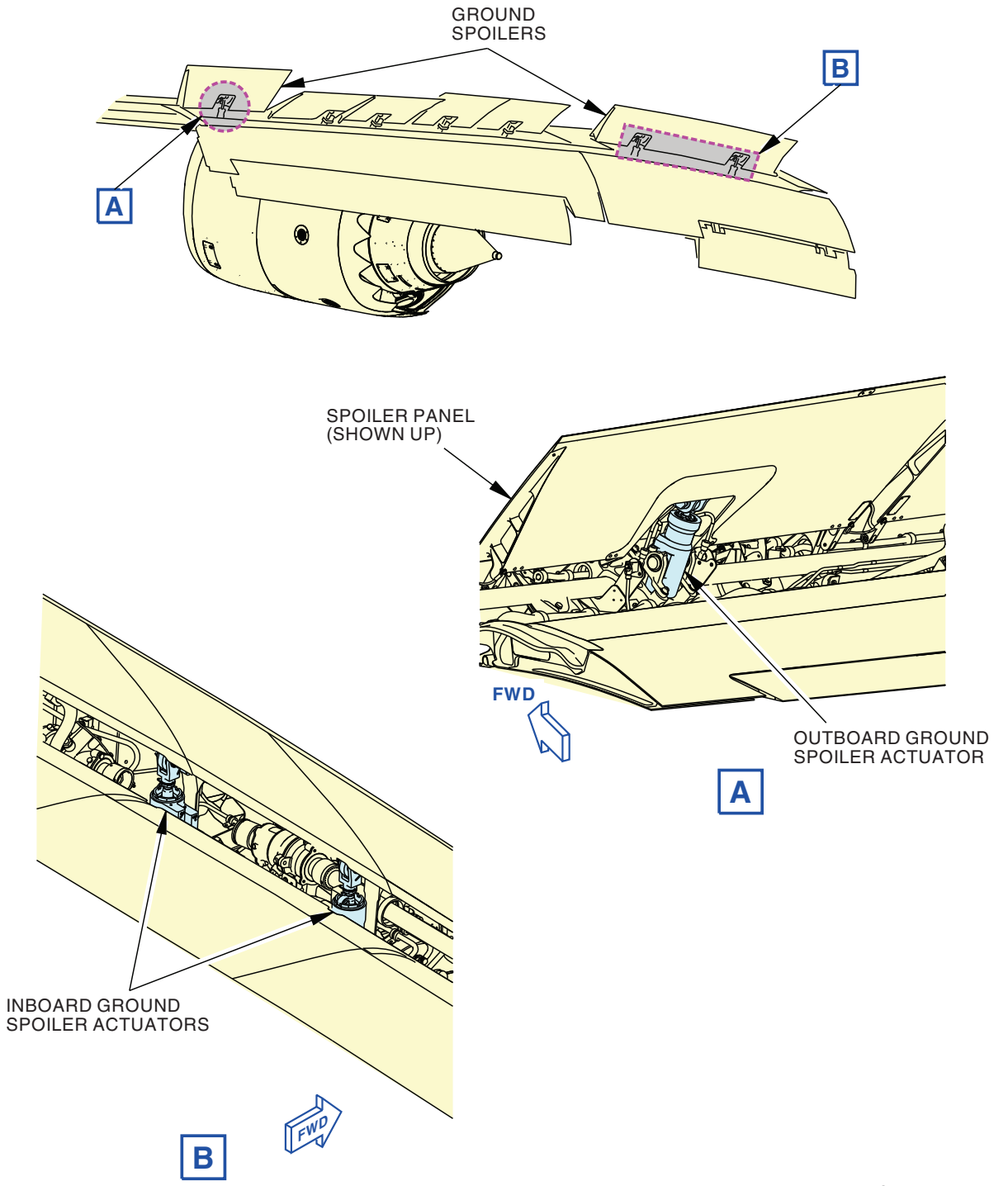
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Flight Spoiler Actuator
Figure 602/27-62-00-990-815 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Flight Spoiler Actuator
Figure 602/27-62-00-990-815 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

INBOARD GROUND SPOILER - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the inboard ground spoiler

(2) An installation of the inboard ground spoiler

(3) A functional test of the inboard ground spoiler.

B. This procedure is applicable to the left and right inboard ground spoilers.

TASK 27-62-12-000-801

2. Inboard Ground Spoiler Removal

(Figure 401)

A. References

Reference	Title
27-51-00-860-803	Extend the Trailing Edge Flaps (P/B 201)
27-62-00-800-801	Apply Pressure to the Speed Brake Hydraulic Systems A and B (P/B 201)
27-62-00-800-802	Depressurization of the Speed Brake Hydraulic Systems A and B (P/B 201)
27-62-71-000-801	Inboard Ground Spoiler Actuator Removal (P/B 401)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1744	Lock Set - Inboard Ground Spoiler Actuator (Contains 4 Lock Assemblies) Part #: C27046-13 Supplier: 81205 Opt Part #: C27046-1 Supplier: 81205

C. Location Zones

Zone	Area
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50
552	Left Wing - Spoiler No. 6
652	Right Wing - Spoiler No. 7

D. Prepare for the Removal

SUBTASK 27-62-12-860-001



WARNING

KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Do this task: Apply Pressure to the Speed Brake Hydraulic Systems A and B, TASK 27-62-00-800-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-62-12-860-002

(2) Do this task: Extend the Trailing Edge Flaps, TASK 27-51-00-860-803.

SUBTASK 27-62-12-860-003

(3) Put the SPOILER A and SPOILER B switches to the OFF position to remove the hydraulic power from the flight spoilers.

NOTE: SPOILER A AND SPOILER B switches are on the P5-3 flight control panel.

SUBTASK 27-62-12-860-004

(4) Put the speed brake lever in the UP position to lift the ground spoilers.

SUBTASK 27-62-12-480-001

(5) Install the lock set, SPL-1744, for the applicable ground spoiler.

SUBTASK 27-62-12-860-005

(6) Do this task: Depressurization of the Speed Brake Hydraulic Systems A and B, TASK 27-62-00-800-802.

E. Inboard Ground Spoiler Removal

SUBTASK 27-62-12-020-001

(1) Disconnect the left inboard actuator [17] from the inboard ground spoiler [1] (TASK 27-62-71-000-801).

SUBTASK 27-62-12-020-002

(2) Disconnect the right inboard actuator [24] from the inboard ground spoiler [1] (TASK 27-62-71-000-801).

SUBTASK 27-62-12-020-003

(3) Remove the nut [10] and washer [8].

(a) Disconnect the bonding jumper [9] from the structure.

SUBTASK 27-62-12-020-004

(4) Remove the nut [31], washer [32], washer [33], washer [35], and bolt [36].

(a) Disconnect the bonding jumper [34].

SUBTASK 27-62-12-020-005

(5) Remove the inboard ground spoiler [1]:

(a) Remove the cotter pin [7], nut [6], washer [5], bushing [4], washer [3] and bolt [2] to disconnect the outboard hinge fitting.

(b) Remove the cotter pin [30], nut [29], washer [28], bushing [27], washer [26], and bolt [25] to disconnect the inboard hinge fitting.

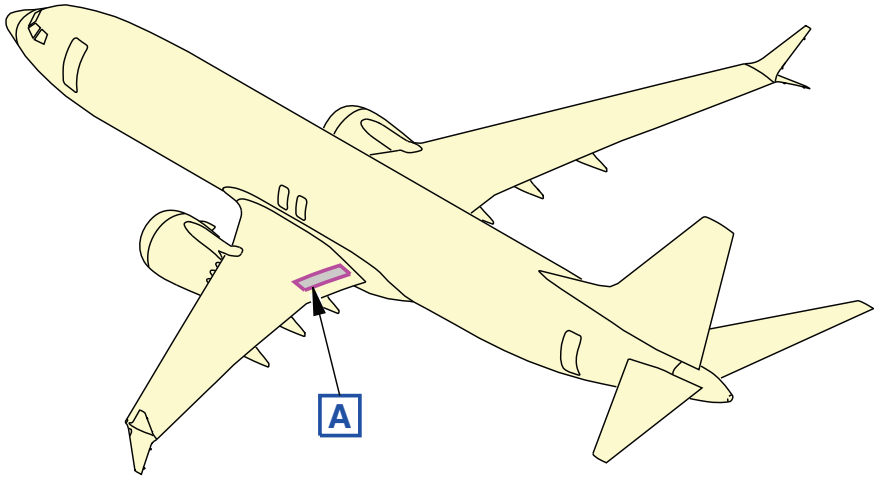
(c) Remove the cotter pin [16], nut [15], washer [14], bushings [13], washer [12], and bolt [11] to disconnect the center left hinge fitting.

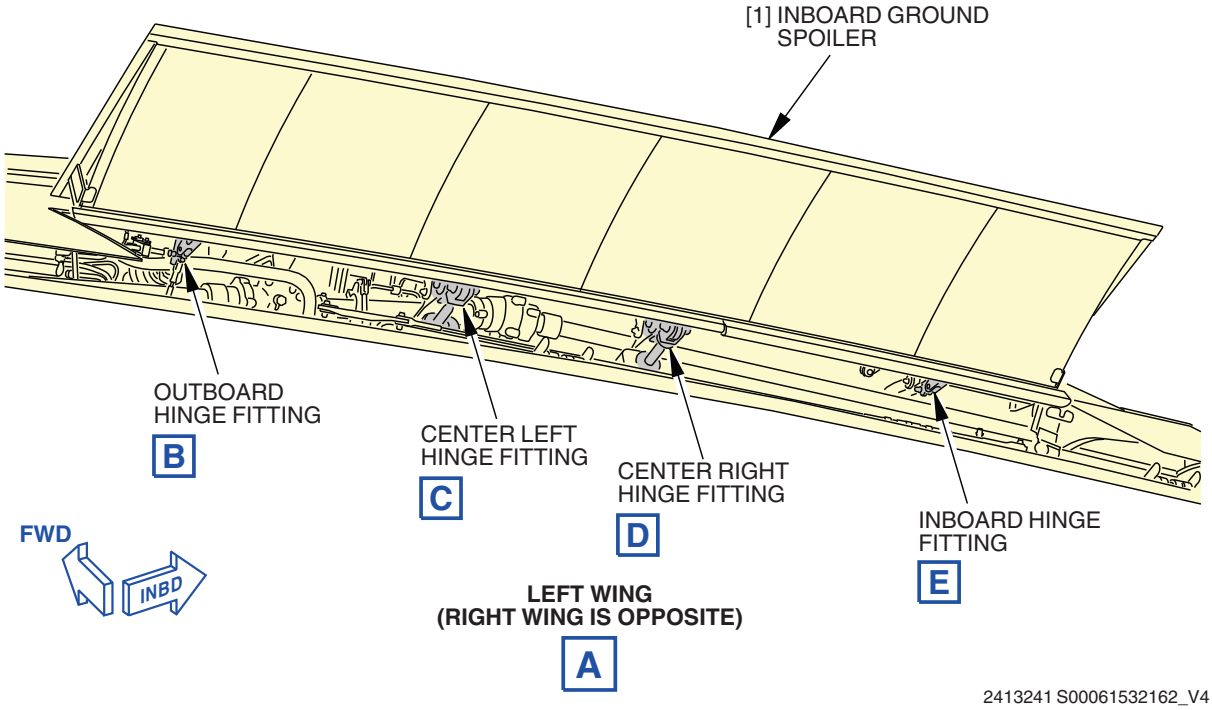
(d) Remove the cotter pin [23], nut [22], washer [21], bushing [20], washer [19], and bolt [18] to disconnect the center right hinge fitting.

(e) Remove the inboard ground spoiler [1] from the airplane.

END OF TASK

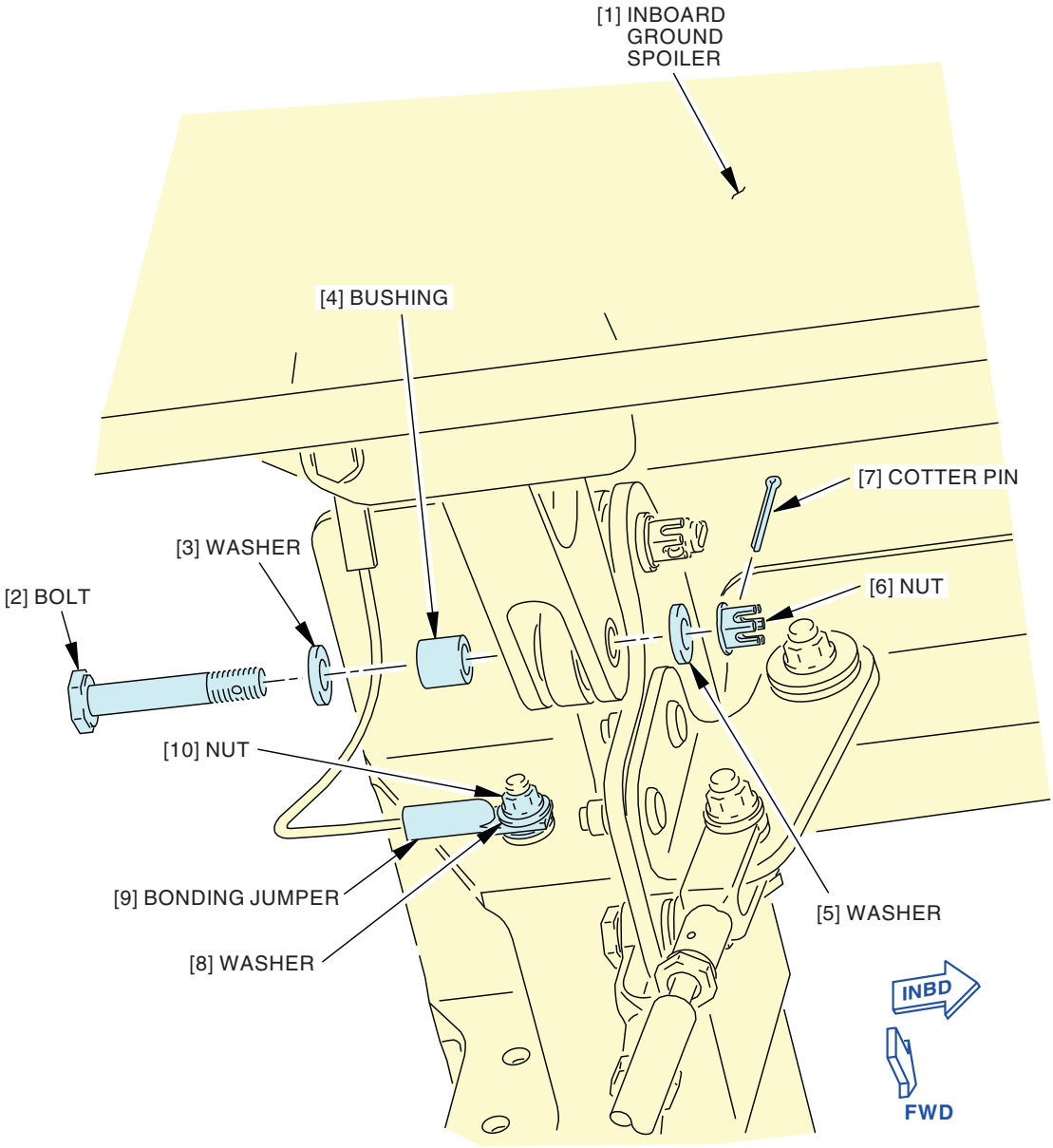
ECCN 9E991 BOEING PROPRIETARY - See title page for details





Inboard Ground Spoiler Installation
Figure 401/27-62-12-990-801 (Sheet 1 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



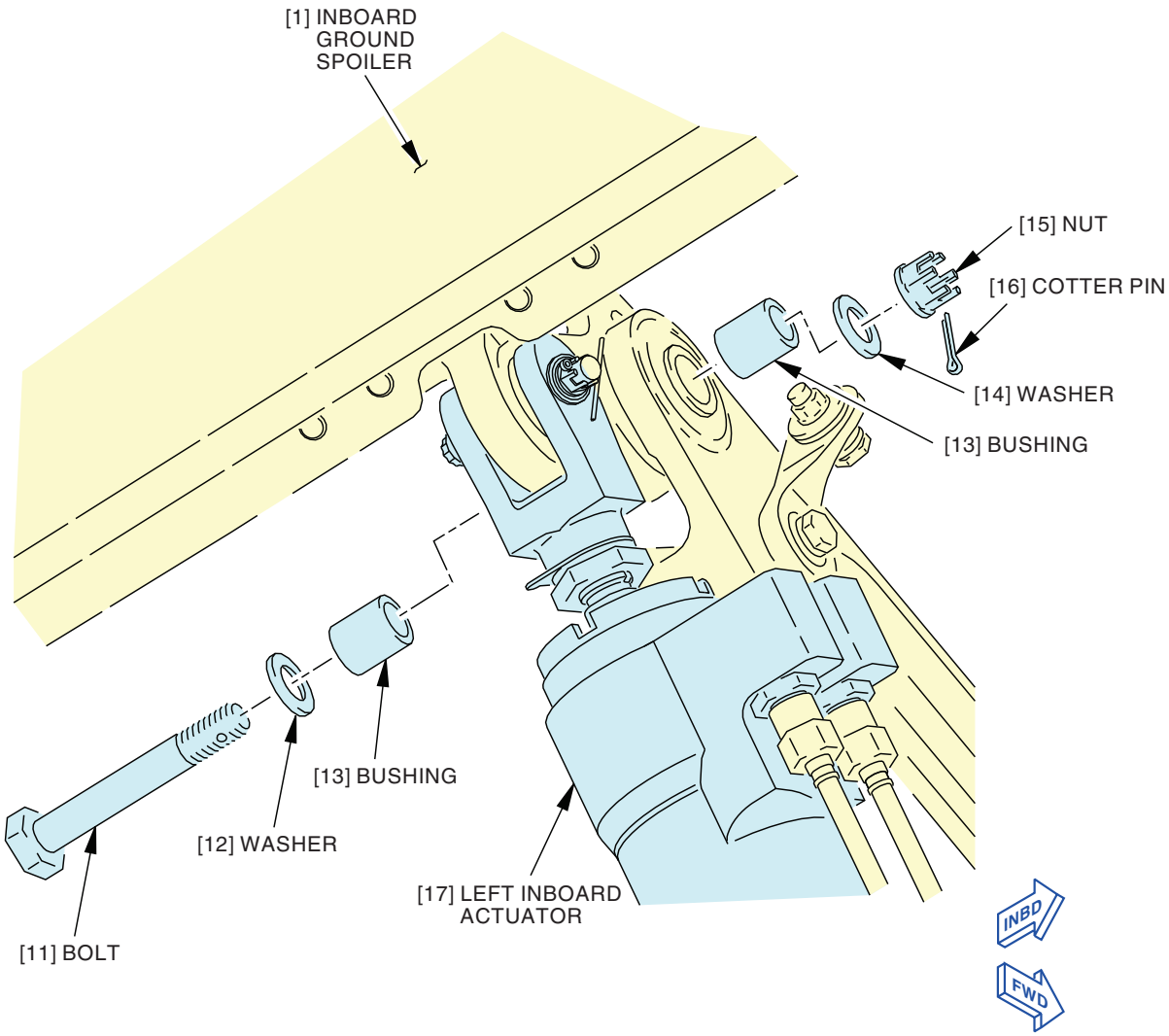
OUTBOARD HINGE FITTING

B

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Inboard Ground Spoiler Installation
Figure 401/27-62-12-990-801 (Sheet 2 of 6)

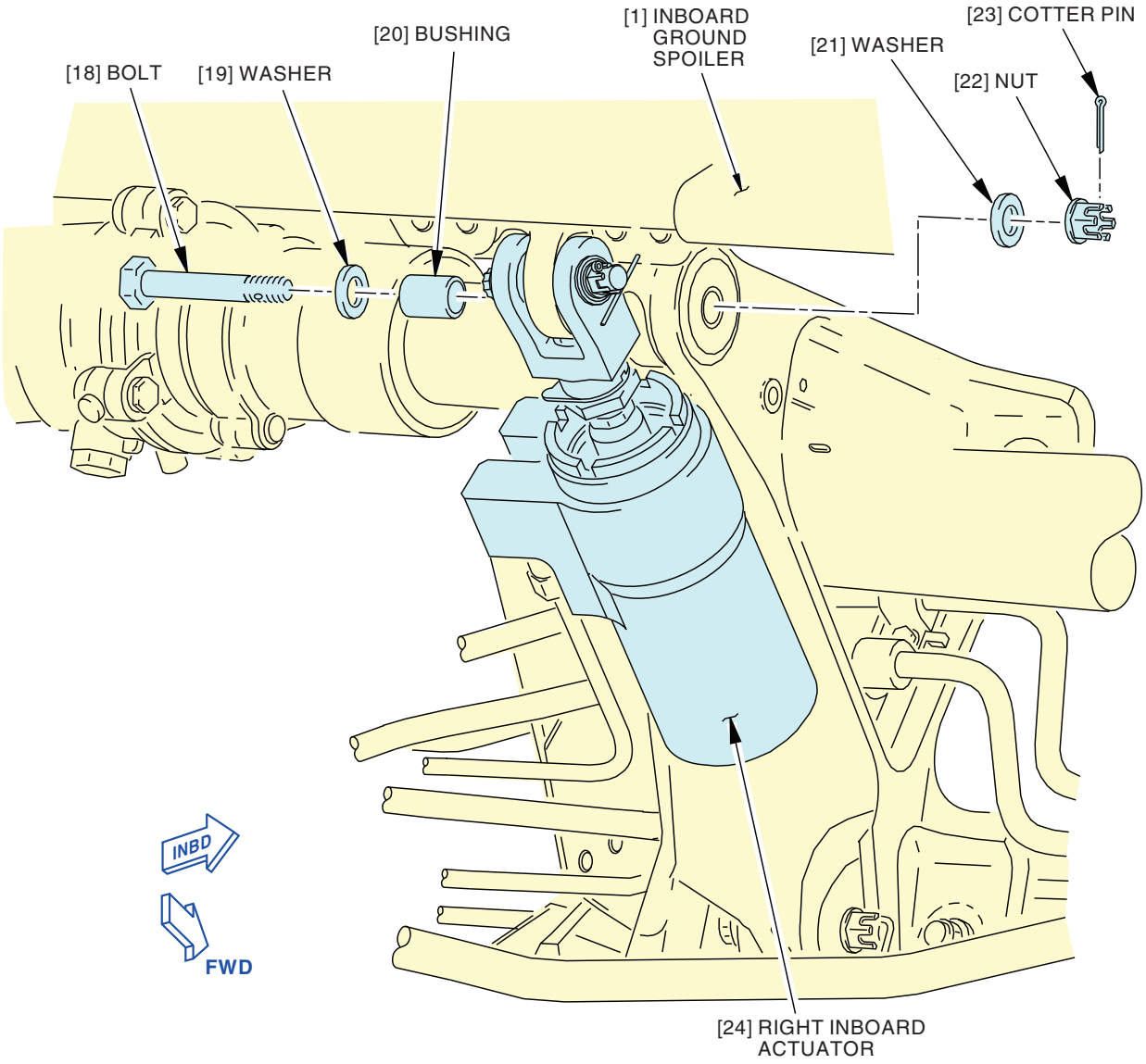
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Inboard Ground Spoiler Installation
Figure 401/27-62-12-990-801 (Sheet 3 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



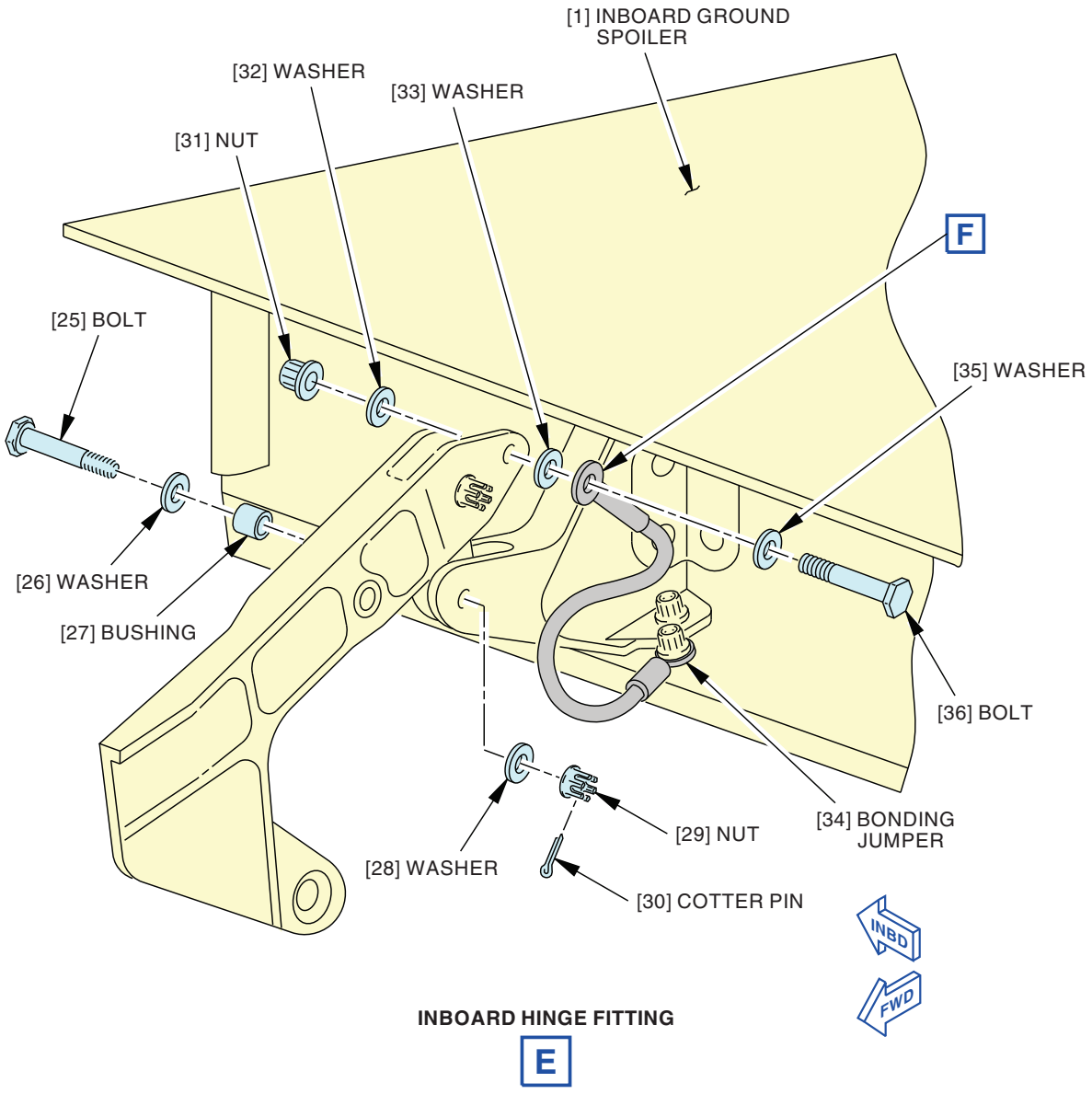
CENTER RIGHT HINGE FITTING

D

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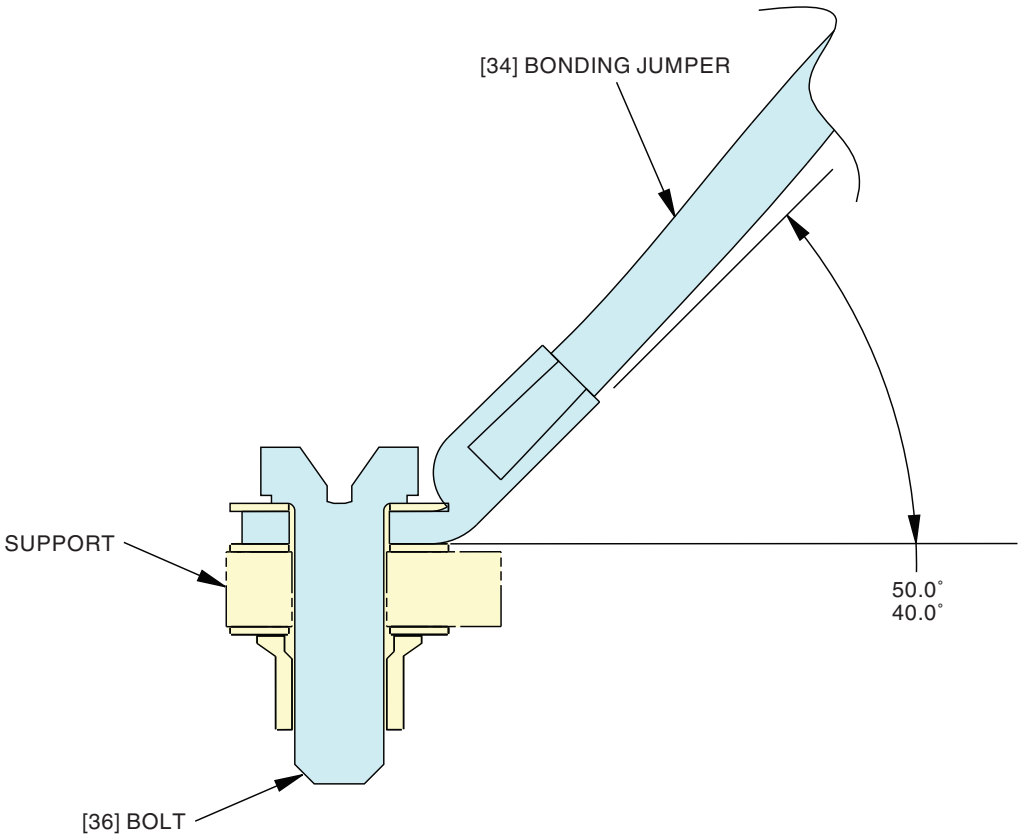
Inboard Ground Spoiler Installation
Figure 401/27-62-12-990-801 (Sheet 4 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Inboard Ground Spoiler Installation
Figure 401/27-62-12-990-801 (Sheet 5 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



F

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Inboard Ground Spoiler Installation
Figure 401/27-62-12-990-801 (Sheet 6 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-62-12-400-801

3. Inboard Ground Spoiler Installation

(Figure 401)

A. References

Reference	Title
27-51-00-860-804	Retract the Trailing Edge Flaps (P/B 201)
27-62-00-800-801	Apply Pressure to the Speed Brake Hydraulic Systems A and B (P/B 201)
27-62-00-820-802	Ground Spoiler Adjustment (P/B 501)
27-62-00-840-801	Shut Down the Speed Brake Hydraulic Systems A and B (P/B 201)
27-62-00-840-802	Pressurization of the Speed Brake Hydraulic Systems A and B to its Usual Condition (P/B 201)
27-62-41-720-801	Ground Spoiler Control Module Functional Test (P/B 401)
27-62-71-400-801	Inboard Ground Spoiler Actuator Installation (P/B 401)
SWPM 20-20-00	ELECTRICAL BONDING PROCESSES
SWPM 20-20-00, Paragraph 2A	Cleaning Procedure 1
SWPM 20-20-10 Paragraph 4.C	Category 1 Unsealed and Category 6 Fillet Sealed Pre-Installed Ground Stud Assembly
SWPM 20-20-10 Paragraph 4.D	Category 2 Fay Sealed Pre-Installed Ground Stud Assembly
SWPM 20-20-10 Paragraph 4.E	Category 1 Unsealed and Category 6 Fillet Sealed Direct Ground Stud Assembly
SWPM 20-20-10 Paragraph 4.F	Category 2 Fay Sealed Direct Ground Stud Installation

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-614	Bonding Meters - Non-Intrinsically Safe (For use in outside Class I, Divisions I & II non-hazardous locations. For hazardous locations, use COM-1550). Part #: 247000 Supplier: 00426 Part #: 620LK Supplier: 1CRL2 Part #: BLR-0003-XX Supplier: KC432 Part #: BT51 Supplier: 00426 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659 Opt Part #: 247001 Supplier: 00426
SPL-1744	Lock Set - Inboard Ground Spoiler Actuator (Contains 4 Lock Assemblies) Part #: C27046-13 Supplier: 81205 Opt Part #: C27046-1 Supplier: 81205

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. Consumable Materials

Reference	Description	Specification
A50110	Sealant - Fuel Tank	BMS5-45 Class B-2
A50155	Sealant - Fuel Tank	BMS5-45 Class C
A50231	Sealant - Pressure And Environmental - Chromate Type	BMS5-95 Class B
A50296	Sealant - Pressure And Environmental - Chromate Type	BMS5-95 Class C
A50337	Sealant - Fuel Tank	BMS5-45 Class B
A50359	Sealant - Low Density, Non-Chromate Type	BMS5-142 Type II Class B
B00184	Solvent - Presealing, Cleaning Solvent	BMS11-7
C00064	Coating - Aluminum Chemical Conversion	BAC5719 Type II Class A (MIL-DTL-5541 Class 1A)
C00175	Primer - Urethane Compatible, Corrosion Resistant	BMS10-79 Type III
C50075	Coating - Protective Enamel (BAC 707 Gray Color)	BMS10-60 Type II
C50315	Coating - Chemical Conversion - Bonderite M-CR 600 Aero (Formerly Alodine 600)	BAC5719 Class A, C or D, MIL-DTL-81706 Type I Class 1A or 3
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Inboard ground spoiler	27-62-12-01-100	PTF ALL
		27-62-12-01-105	PTF ALL
7	Cotter pin	27-62-12-01-035	PTF ALL
16	Cotter pin	27-62-12-01-040	PTF ALL
23	Cotter pin	27-62-12-01-040	PTF ALL
30	Cotter pin	27-62-12-01-035	PTF ALL

E. Location Zones

Zone	Area
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50
552	Left Wing - Spoiler No. 6
652	Right Wing - Spoiler No. 7

F. Inboard Ground Spoiler Installation

SUBTASK 27-62-12-420-001

(1) Install the inboard ground spoiler [1].

(a) Put the inboard ground spoiler [1] in its position on the airplane.

(b) Install the bolt [18] with the head outboard, washer [19], bushing [20], washer [21], nut [22], and new cotter pin [23] in the center right hinge fitting.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Install the bolt [11] with the head outboard, washer [12], bushings [13], washer [14], nut [15], and new cotter pin [16] in the center left hinge fitting.

(d) Install the bolt [25] with the head inboard, washer [26], bushing [27], washer [28], nut [29], and new cotter pin [30] in the inboard hinge fitting.

(e) Install the bolt [2] with the head outboard, washer [3], bushing [4], washer [5], nut [6], and new cotter pin [7] in the outboard hinge fitting.

(f) Install the washer [8] and nut [10] to connect the bonding jumper [9] to the structure.

1) Use cotton wiper, G00034, moist in solvent, B00184, to clean the mating surfaces of the airplane structure, nut [10], bonding jumper [9], and washer [8].

2) Method 1 (preferred). Apply sealant, A50110, or sealant, A50155, or sealant, A50231, or sealant, A50296, to the mating surfaces of the ground stud, nut [10], washer [8], and bonding jumper [9] (SWPM 20-20-10 Paragraph 4.D).

3) Install the bonding jumper [9], washer [8], and nut [10].

4) Tighten the nut [10] to 28 in-lb (3 N·m) - 35 in-lb (4 N·m).

a) Method 1. Make sure that there is squeeze out around all the interfaces of each component.

5) Measure the electrical resistance between the bonding jumper [9] and airplane structure (SWPM 20-20-00).

a) Use a non-intrinsically safe bonding meter, COM-614.

b) Make sure that the electrical bonding resistance is 0.001 ohm (1.0 milliohm) or less.

6) Method 2 (optional). Apply a fillet seal with sealant, A50337, or sealant, A50231, or sealant, A50359, to cover the nut [10], washer [8], and bonding jumper [9] (SWPM 20-20-10 Paragraph 4.C).

(g) Install the bolt [36], washer [35], washer [33], washer [32], and nut [31] to connect the bonding jumper [34] to the structure.

1) Clean the surface of the bolt [36], washer [35], bonding jumper [34], washer [33], washer [32], and nut [31] (SWPM 20-20-00, Paragraph 2A).

2) Use cotton wiper, G00034, moist in solvent, B00184, to final clean the mating surfaces of the airplane structure, bolt [36], washer [35], bonding jumper [34], washer [33], washer [32], and nut [31].

3) Method 1 (preferred). Apply sealant, A50110, or sealant, A50155, or sealant, A50231, or sealant, A50296, to the mating surfaces of the bolt [36], washer [35], bonding jumper [34], washer [33], washer [32], and nut [31] (SWPM 20-20-10 Paragraph 4.F).

4) Install the bolt [36], washer [35], bonding jumper [34], washer [33], washer [32], and nut [31] .

5) Tighten the nut [31] to 28 in-lb (3 N·m) - 35 in-lb (4 N·m).

a) Method 1. Make sure that there is squeeze out around all the interfaces of each component.

6) Measure the electrical resistance between the bonding jumper [34] and airplane structure (SWPM 20-20-00).

a) Use a non-intrinsically safe bonding meter, COM-614.

b) Make sure that the electrical bonding resistance is 0.001 ohm (1.0 milliohm) or less.

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7) Bend the bonding jumper [34] 45 ±5°arc (0.8 ±0.1 rad) away from the support (View F, Figure 401).

8) Method 2 (optional). Apply a fillet seal with sealant, A50337, or sealant, A50231, or sealant, A50359, to cover the bolt [36], washer [35], bonding jumper [34], washer [33], washer [32], and nut [31] (SWPM 20-20-10 Paragraph 4.E).

9) Manually apply finish with coating, C00064, or Bonderite M-CR 600 Aero coating, C50315.

a) Apply primer, C00175, and coating, C50075.

SUBTASK 27-62-12-980-001

(2) Move the inboard ground spoiler [1] through its full travel.

(a) Make sure that the inboard ground spoiler [1] moves freely.

SUBTASK 27-62-12-420-002

(3) Connect the rod end of the left inboard actuator [17] to the inboard ground spoiler [1] (TASK 27-62-71-400-801).

SUBTASK 27-62-12-420-003

(4) Connect the rod end of the right inboard actuator [24] to the inboard ground spoiler [1] (TASK 27-62-71-400-801).

SUBTASK 27-62-12-080-001

(5) Remove the lock set, SPL-1744.

SUBTASK 27-62-12-720-002

(6) To unlatch the SPOILER ACTUATOR status message, do this task: Ground Spoiler Control Module Functional Test, TASK 27-62-41-720-801.

NOTE: The SPOILER ACTUATOR status message can be latched when the spoilers are locked in the up position.

SUBTASK 27-62-12-860-006

(7) Do this task: Pressurization of the Speed Brake Hydraulic Systems A and B to its Usual Condition, TASK 27-62-00-840-802.

SUBTASK 27-62-12-860-007

WARNING

KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(8) Do this task: Apply Pressure to the Speed Brake Hydraulic Systems A and B, TASK 27-62-00-800-801.

SUBTASK 27-62-12-860-008

(9) Move the speed brake lever to the DOWN position to lower the spoilers.

SUBTASK 27-62-12-860-013

(10) Do this task: Retract the Trailing Edge Flaps, TASK 27-51-00-860-804.

SUBTASK 27-62-12-860-009

(11) Do this task: Shut Down the Speed Brake Hydraulic Systems A and B, TASK 27-62-00-840-801.

SUBTASK 27-62-12-820-001

(12) Adjust the inboard ground spoilers (TASK 27-62-00-820-802).

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SUBTASK 27-62-12-720-001

(13) Do this task: Inboard Ground Spoiler Functional Test, TASK 27-62-12-720-801.

END OF TASK

TASK 27-62-12-720-801

4. Inboard Ground Spoiler Functional Test

A. References

Reference	Title
24-22-00-860-801	Supply Electrical Power (P/B 201)
27-61-00-840-801	Shut Down the Spoiler Hydraulic Systems A and B (P/B 201)
27-62-00-800-801	Apply Pressure to the Speed Brake Hydraulic Systems A and B (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1691	Protractor - Digital, Vertical or Horizontal Part #: 21465A82 Supplier: 3A054 Part #: 31-038-3 (Model Pro 360) Supplier: 1X822 Part #: 31-040-9 (Model Pro 3600) Supplier: 1X822 Part #: 950-317 (Model Pro 360) Supplier: 01EP7 Part #: 950-318 (Model Pro 3600) Supplier: 01EP7 Part #: KS5549 Supplier: 75245 Part #: KS6005 Supplier: 75245

C. Location Zones

Zone	Area
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50
552	Left Wing - Spoiler No. 6
652	Right Wing - Spoiler No. 7

D. Prepare for the Test

SUBTASK 27-62-12-860-010

(1) Do this task: Supply Electrical Power, TASK 24-22-00-860-801.

SUBTASK 27-62-12-860-011

WARNING

MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. THE AILERONS, RUDDER, ELEVATORS, FLAPS, SLATS, SPOILERS, LANDING GEAR, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(2) Do this task: Apply Pressure to the Speed Brake Hydraulic Systems A and B, TASK 27-62-00-800-801.

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SUBTASK 27-62-12-480-002

CAUTION

INSTALL THE PROTRACTOR ON THE SPOILER TOP SURFACE. IF THE PROTRACTOR INSTALLED ON THE BOTTOM OF THE SPOILER, DAMAGE TO AIRCRAFT CAN OCCUR.

(3) Install the protractor, COM-1691, on the upper surface of inboard ground spoilers 6 and 7.

SUBTASK 27-62-12-820-002

(4) Adjust the protractor, COM-1691, to 0°arc (0 rad).

E. Inboard Ground Spoiler Functional Test

SUBTASK 27-62-12-020-006

(1) Move the speed brake lever to the UP position to lift the spoilers.

SUBTASK 27-62-12-020-007

(2) Make sure the spoilers No. 6 and 7 are 60 ±2°arc (1 ±0 rad) up from the retracted position.

SUBTASK 27-62-12-020-008

(3) Make sure the ground spoilers No. 6 and 7 align in the limit of 2°arc (0 rad).

SUBTASK 27-62-12-020-009

(4) Move the speed brake lever to the DOWN position to lower the spoilers.

SUBTASK 27-62-12-820-003

(5) Make sure the spoilers move to the fully retracted position.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 27-62-12-860-012

(1) Do this task: Shut Down the Spoiler Hydraulic Systems A and B, TASK 27-61-00-840-801.

SUBTASK 27-62-12-080-002

(2) Remove the protractor, COM-1691, from the upper surface of spoilers No. 6 and 7.

——— END OF TASK ———

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GROUND SPOILER HINGE LINK - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) The ground spoiler hinge links removal

(2) The ground spoiler hinge links installation.

B. This procedure is applicable to the left and right ground spoiler hinge links.

TASK 27-62-15-000-801

2. Ground Spoiler Hinge Link Removal

(Figure 401)

A. References

Reference	Title
27-51-00-860-803	Extend the Trailing Edge Flaps (P/B 201)
27-61-00-800-801	Apply Pressure to the Spoiler Hydraulic Systems A and B (P/B 201)
27-61-00-800-802	Depressurization of the Spoiler Hydraulic Systems A and B (P/B 201)
27-62-12-000-801	Inboard Ground Spoiler Removal (P/B 401)
27-62-91-000-801	Outboard Ground Spoiler Removal (P/B 401)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1743	Ground Lock Set - Outboard Spoiler Actuators Part #: C27001-51 Supplier: 81205
SPL-1744	Lock Set - Inboard Ground Spoiler Actuator (Contains 4 Lock Assemblies) Part #: C27046-13 Supplier: 81205 Opt Part #: C27046-1 Supplier: 81205

C. Location Zones

Zone	Area
552	Left Wing - Spoiler No. 6
566	Left Wing - Spoiler No. 1
652	Right Wing - Spoiler No. 7
666	Right Wing - Spoiler No. 12

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D. Prepare for the Removal

SUBTASK 27-62-15-860-002

WARNING

KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Do this task: Apply Pressure to the Spoiler Hydraulic Systems A and B, TASK 27-61-00-800-801.

SUBTASK 27-62-15-860-003

(2) Do this task: Extend the Trailing Edge Flaps, TASK 27-51-00-860-803.

SUBTASK 27-62-15-860-004

WARNING

MAKE SURE THE SPOILER WILL REMAIN IN THE OPEN POSITION DURING INSTALLATION OF THE LOCK-SET. PUT ONE PERSON IN THE FLIGHT COMPARTMENT TO MONITOR THE SPOILER CONTROL LEVER. IF THE SPOILER CONTROL LEVER CHANGES POSITION, THE SPOILER PANEL COULD MOVE AND CAUSE INJURY TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(3) Put the speed brake lever in the UP position to lift the spoilers.

SUBTASK 27-62-15-480-001

(4) Install the lock set, SPL-1744 or the ground lock set, SPL-1743 on the actuator for the applicable ground spoiler.

SUBTASK 27-62-15-860-005

(5) Do this task: Depressurization of the Spoiler Hydraulic Systems A and B, TASK 27-61-00-800-802.

SUBTASK 27-62-15-020-001

(6) To remove the outboard ground spoiler hinge link, do this task: Outboard Ground Spoiler Removal, TASK 27-62-91-000-801.

SUBTASK 27-62-15-020-002

(7) To remove the inboard ground spoiler hinge links, do this task: Inboard Ground Spoiler Removal, TASK 27-62-12-000-801.

E. Ground Spoiler Hinge Link Removal

SUBTASK 27-62-15-020-003

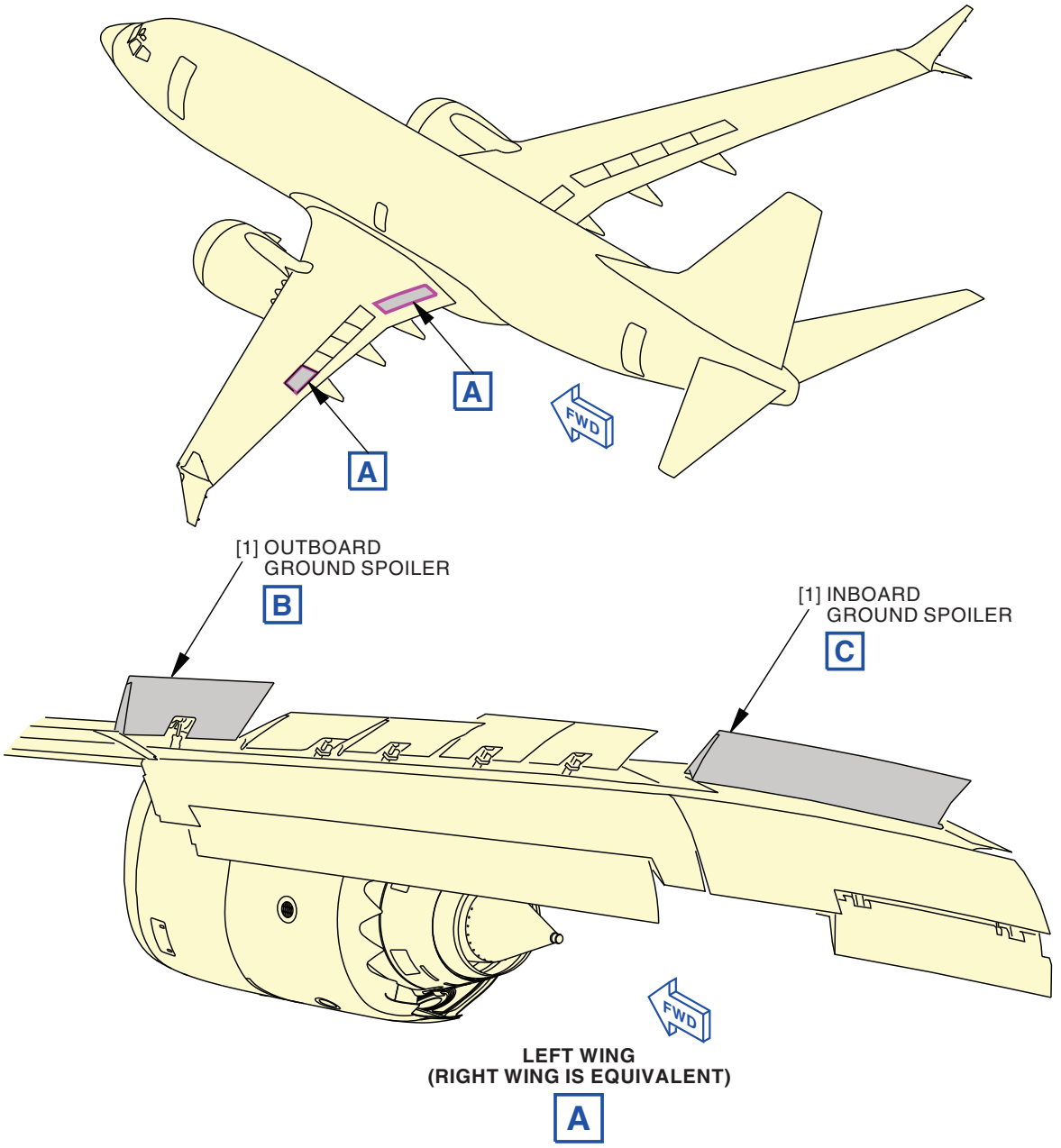
(1) To remove the spoiler hinge link [5], spoiler hinge link [5A], spoiler hinge link [5B] or spoiler hinge link [5C], do these steps:

(a) Remove pins [11], nuts [10], washers [9], bushings [8], washers [7] and bolts [6].

(b) Remove ground spoiler hinge link from the airplane.

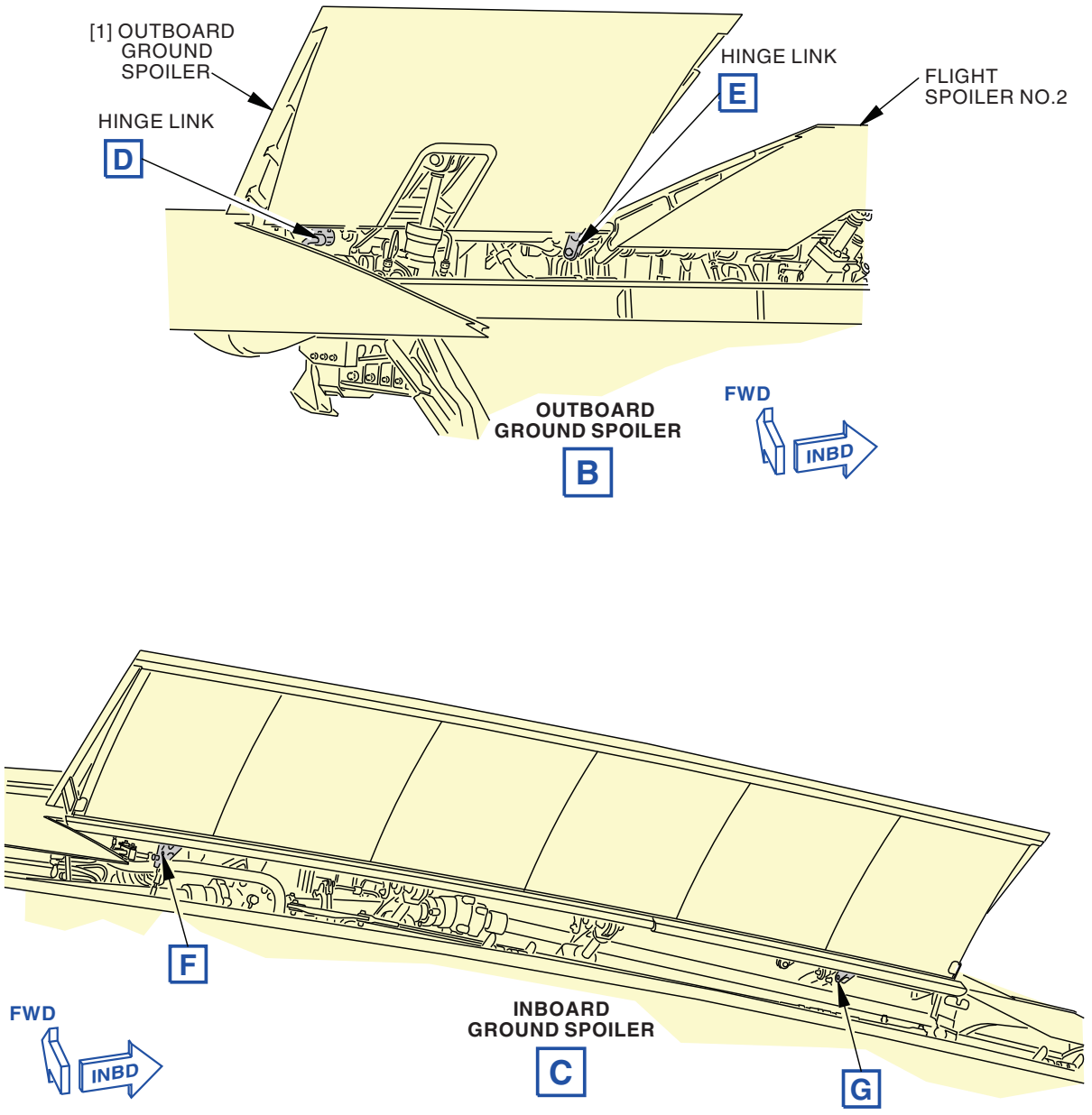
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Ground Spoiler Hinge Link Installation
Figure 401/27-62-15-990-801 (Sheet 1 of 4)

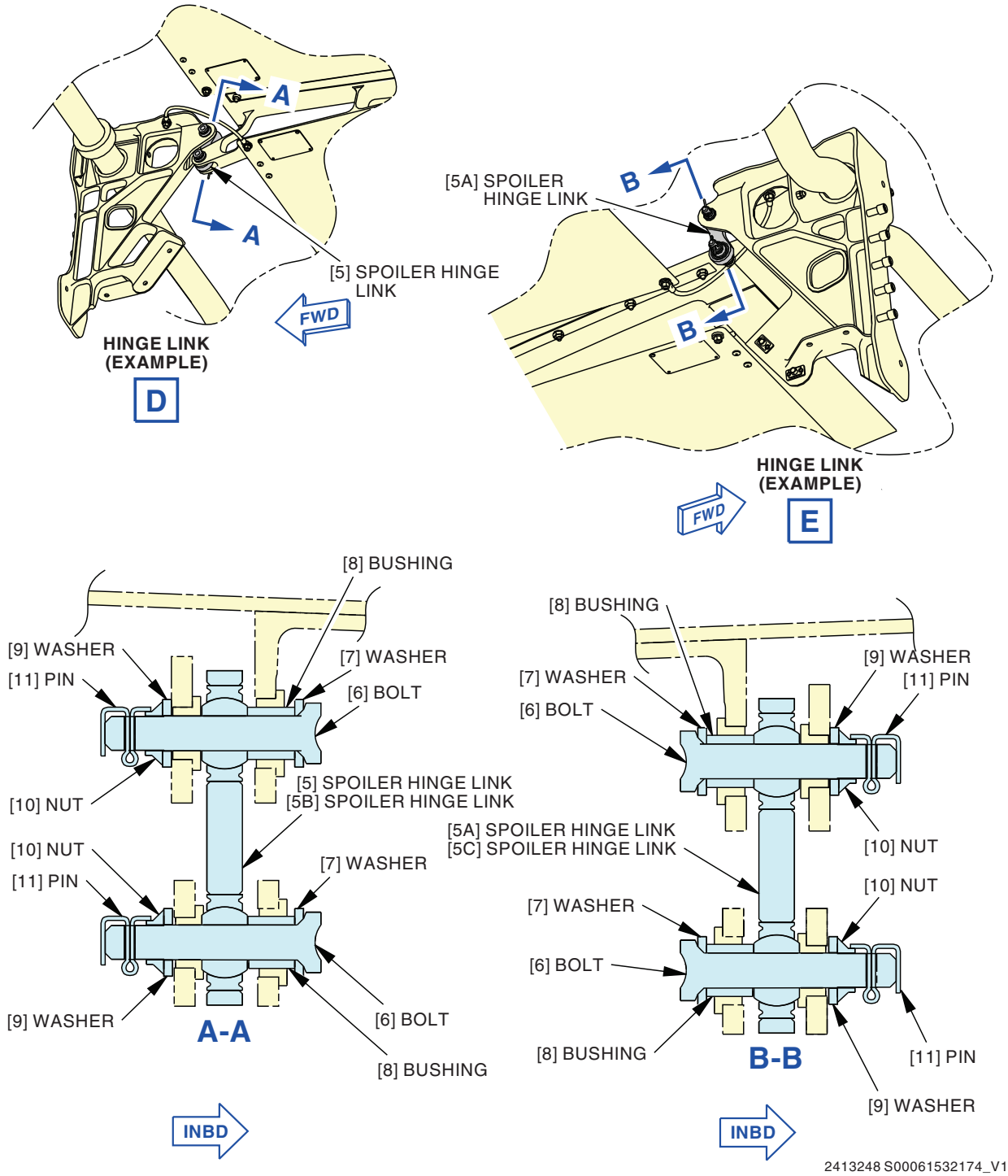
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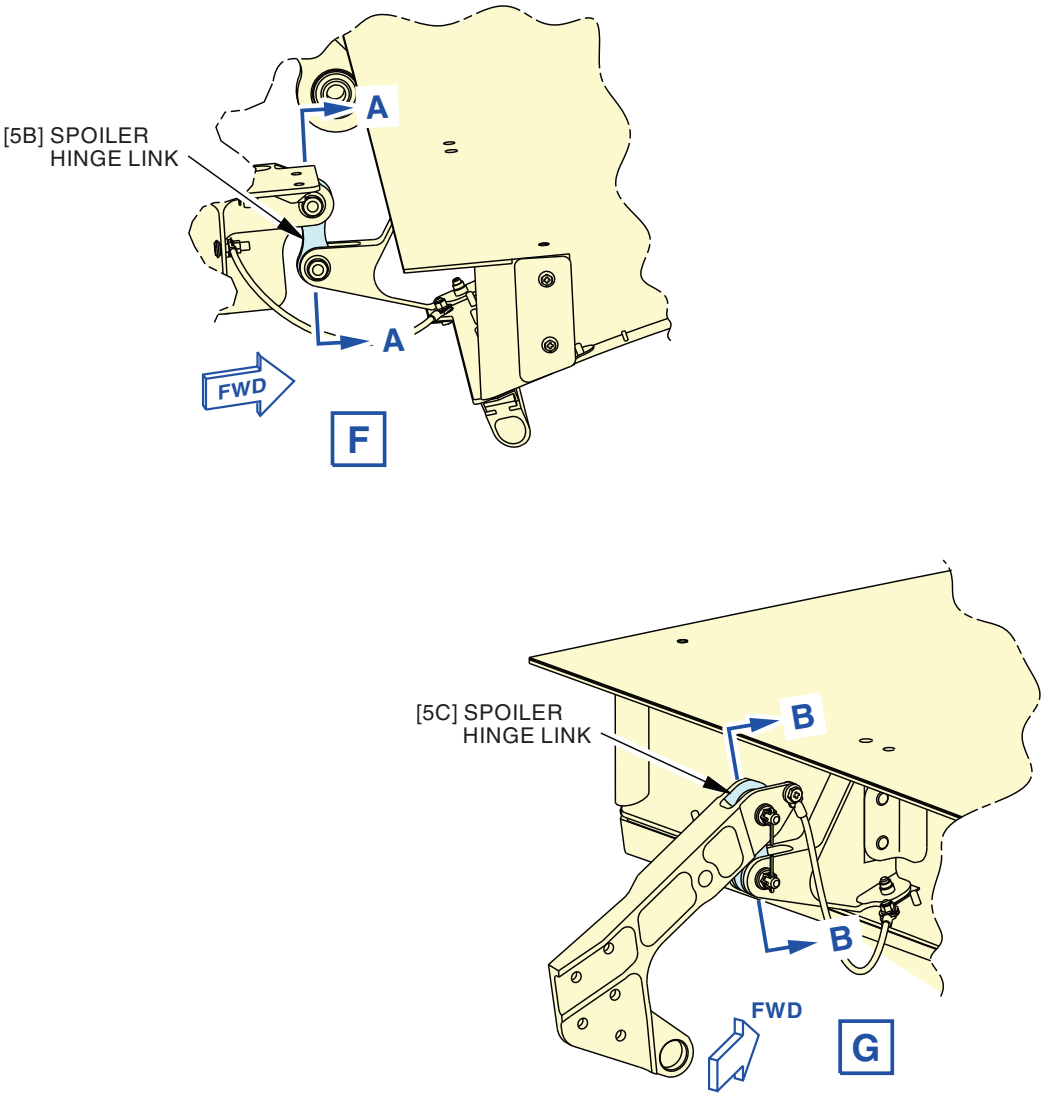
Ground Spoiler Hinge Link Installation
Figure 401/27-62-15-990-801 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Ground Spoiler Hinge Link Installation
Figure 401/27-62-15-990-801 (Sheet 3 of 4)

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Ground Spoiler Hinge Link Installation
Figure 401/27-62-15-990-801 (Sheet 4 of 4)

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TASK 27-62-15-400-801

3. Ground Spoiler Hinge Link Installation

(Figure 401)

A. References

Reference	Title
27-51-00-860-804	Retract the Trailing Edge Flaps (P/B 201)
27-61-00-800-801	Apply Pressure to the Spoiler Hydraulic Systems A and B (P/B 201)
27-61-00-840-801	Shut Down the Spoiler Hydraulic Systems A and B (P/B 201)
27-62-00-820-802	Ground Spoiler Adjustment (P/B 501)
27-62-12-400-801	Inboard Ground Spoiler Installation (P/B 401)
27-62-41-720-801	Ground Spoiler Control Module Functional Test (P/B 401)
27-62-91-000-802	Outboard Ground Spoiler Test (P/B 401)
27-62-91-400-801	Outboard Ground Spoiler Installation (P/B 401)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1743	Ground Lock Set - Outboard Spoiler Actuators Part #: C27001-51 Supplier: 81205
SPL-1744	Lock Set - Inboard Ground Spoiler Actuator (Contains 4 Lock Assemblies) Part #: C27046-13 Supplier: 81205 Opt Part #: C27046-1 Supplier: 81205

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
5	Spoiler hinge link	27-62-15-01-050	PTF ALL
5A	Spoiler hinge link	27-62-15-01-050	PTF ALL
5B	Spoiler hinge link	27-62-15-01-050	PTF ALL
5C	Spoiler hinge link	27-62-15-01-050	PTF ALL
11	Pin	27-62-15-01-020	PTF ALL

D. Location Zones

Zone	Area
552	Left Wing - Spoiler No. 6
566	Left Wing - Spoiler No. 1
652	Right Wing - Spoiler No. 7
666	Right Wing - Spoiler No. 12

E. Ground Spoiler Hinge Link Installation

SUBTASK 27-62-15-020-004

(1) To install the spoiler hinge link [5], spoiler hinge link [5A], spoiler hinge link [5B] or spoiler hinge link [5C], do these steps:

(a) Install the bolts [6], washers [7], bushings [8], washers [9], nuts [10] and new pins [11].

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SUBTASK 27-62-15-400-001

(2) If you removed the outboard ground spoiler hinge link, do this task: Outboard Ground Spoiler Installation, TASK 27-62-91-400-801.

SUBTASK 27-62-15-400-002

(3) If you removed the inboard ground spoiler hinge links, do this task: Inboard Ground Spoiler Installation, TASK 27-62-12-400-801.

SUBTASK 27-62-15-860-006

(4) Make sure that hydraulic power is not on.

SUBTASK 27-62-15-080-001

(5) Remove lock set, SPL-1744, or the ground lock set, SPL-1743, from the applicable spoiler actuator.

SUBTASK 27-62-15-720-002

(6) Do this task to unlatch the SPOILER ACTUATOR status message: Ground Spoiler Control Module Functional Test, TASK 27-62-41-720-801.

NOTE: The SPOILER ACTUATOR status message can be latched when the spoilers are locked in the up position.

SUBTASK 27-62-15-860-007

WARNING

KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(7) Do this task: Apply Pressure to the Spoiler Hydraulic Systems A and B, TASK 27-61-00-800-801.

SUBTASK 27-62-15-860-008

(8) Move the speed brake lever to the DOWN position to lower the spoilers.

SUBTASK 27-62-15-860-009

(9) Do this task: Retract the Trailing Edge Flaps, TASK 27-51-00-860-804.

SUBTASK 27-62-15-860-010

(10) Do this task: Shut Down the Spoiler Hydraulic Systems A and B, TASK 27-61-00-840-801.

SUBTASK 27-62-15-820-001

(11) Do this task: Ground Spoiler Adjustment, TASK 27-62-00-820-802.

SUBTASK 27-62-15-720-001

(12) Do this task: Outboard Ground Spoiler Test, TASK 27-62-91-000-802.

END OF TASK

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SPEEDBRAKE MECHANISM - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Speedbrake mechanism removal

(2) Speedbrake mechanism installation.

TASK 27-62-21-000-801

2. Speedbrake Mechanism Removal

(Figure 401)

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
27-62-37-000-801	Speedbrake Lever Transducer Removal (P/B 401)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Access Panels

Number	Name/Location
112A	Forward Access Door

D. Prepare for the Removal

SUBTASK 27-62-21-860-001

(1) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	9	C00440	FLIGHT CONTROL AUTO SPEED BRAKE



WARNING

KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT SPOILERS. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU OPEN/CLOSE THESE CIRCUIT BREAKERS. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(2) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
C	10	C04032	SPOILER CONTROL ELEC PWR A
C	12	C04033	SPOILER CONTROL ELEC PWR B

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SUBTASK 27-62-21-860-002

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(3) If the downlock pins are not installed on all landing gears, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 27-62-21-010-001

(4) Open this access panel:

Number	Name/Location
112A	Forward Access Door

E. Removal Procedure

SUBTASK 27-62-21-020-001

(1) Remove the speedbrake lever transducer [2] (Speedbrake Lever Transducer Removal, TASK 27-62-37-000-801).

SUBTASK 27-62-21-020-003

(2) Remove the pin [4], nut [5], washer [6] and bolt [7] that attach the actuator assembly [3] to the speedbrake mechanism [1].

SUBTASK 27-62-21-020-004

(3) Remove the nut [9], washer [10] and bolt [11] that attach the control rod [8] to the speedbrake mechanism [1].

SUBTASK 27-62-21-020-002

(4) Do these steps to remove the turnbuckle [12]:

(a) Remove the lockwire, nut [13], washer [14], bushing [15], washer [16] and bolt [17].

(b) Remove the nut [18], washer [19], washer [20] and bolt [21].

(c) Remove the turnbuckle [12].

SUBTASK 27-62-21-020-005

(5) Remove the speedbrake mechanism [1]:

(a) Remove the nuts [22], washers [23], washers [24] and bolts [25] that attach the speedbrake mechanism [1] to the structure.

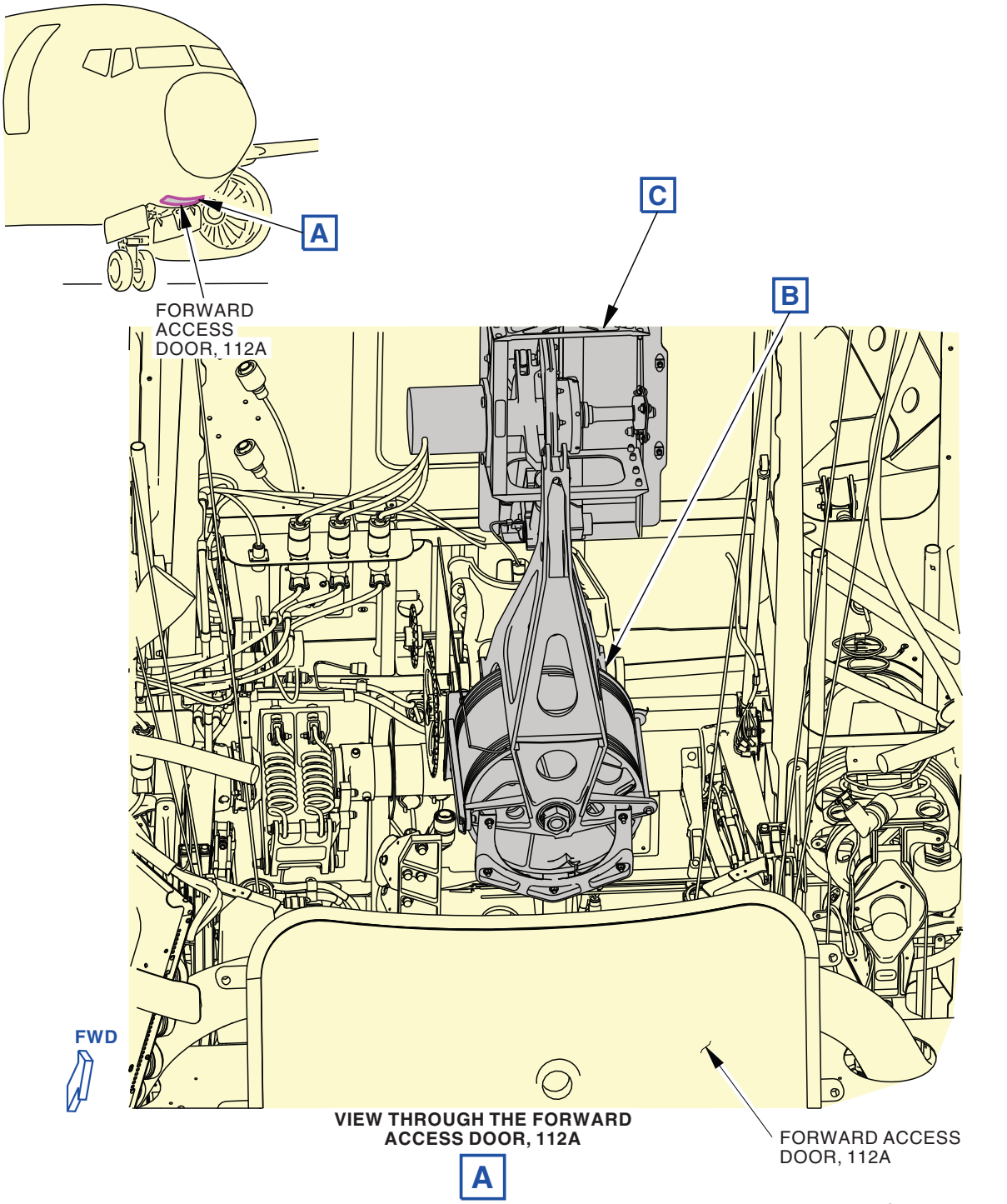
(b) Remove the nuts [26], bushings [27], washers [28] and bolts [29] that attach the speedbrake mechanism [1] to the structure.

(c) Remove the speedbrake mechanism [1].

NOTE: The speedbrake mechanism [1] can be removed on the left side of the cable drum.

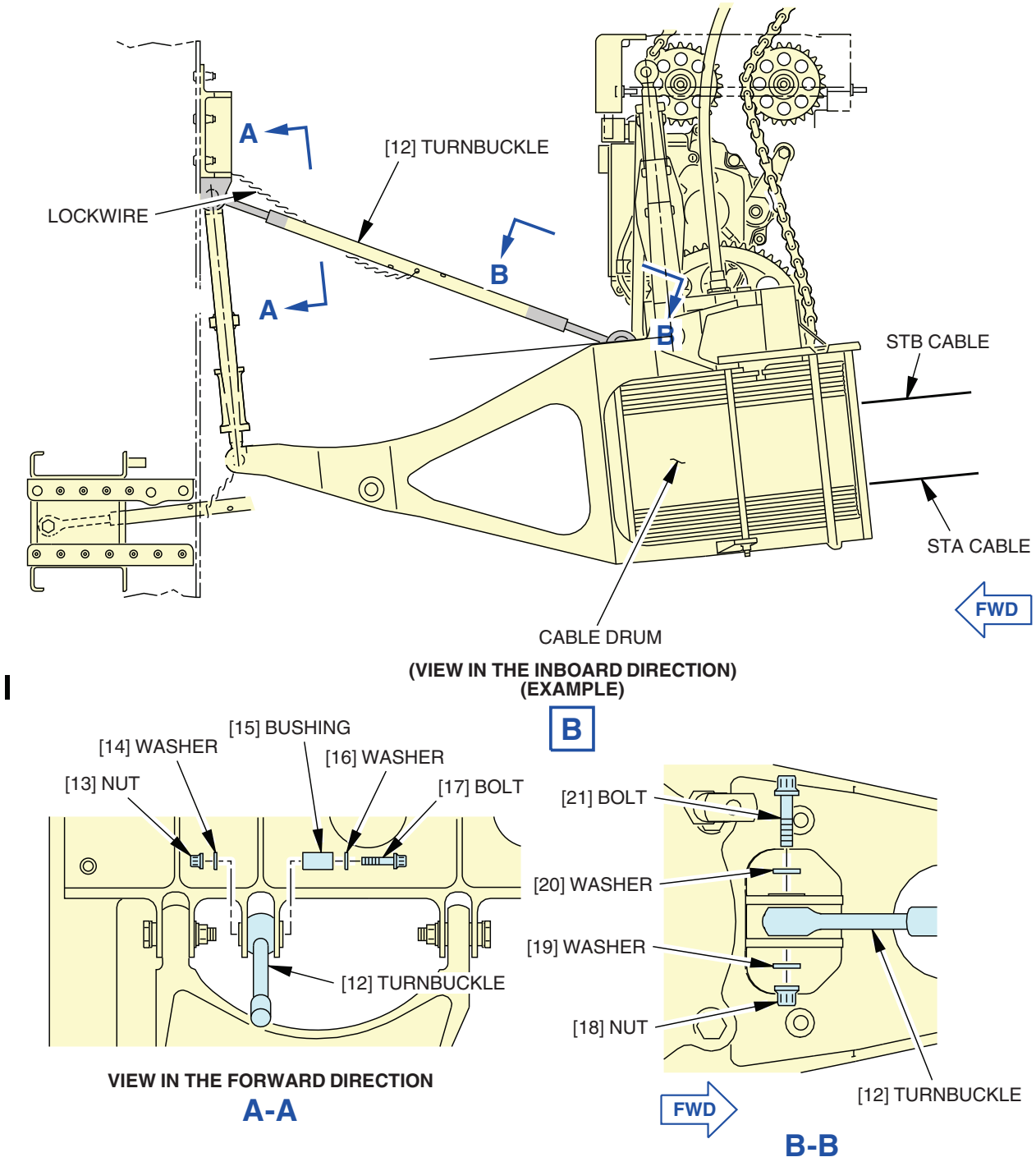
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



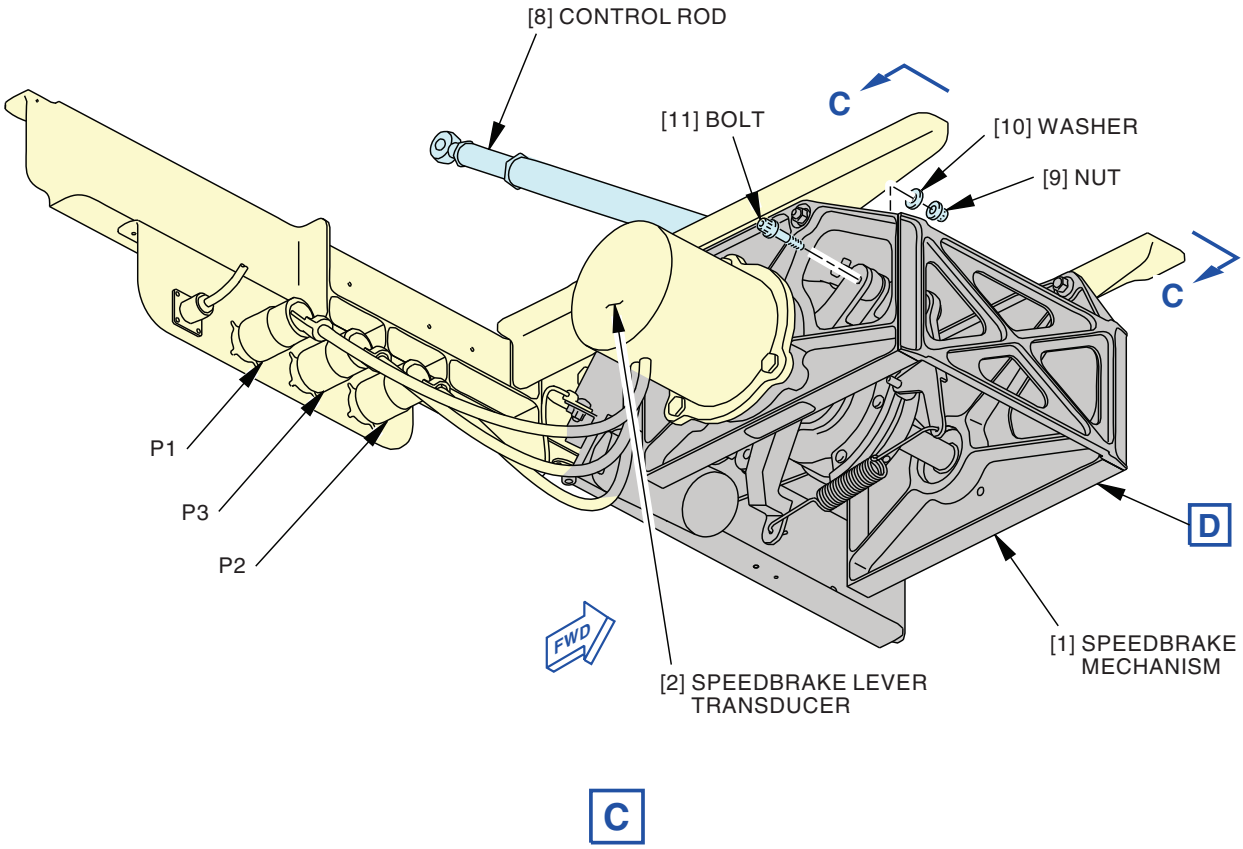
Speedbrake Mechanism Installation
Figure 401/27-62-21-990-801 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Speedbrake Mechanism Installation
Figure 401/27-62-21-990-801 (Sheet 2 of 4)

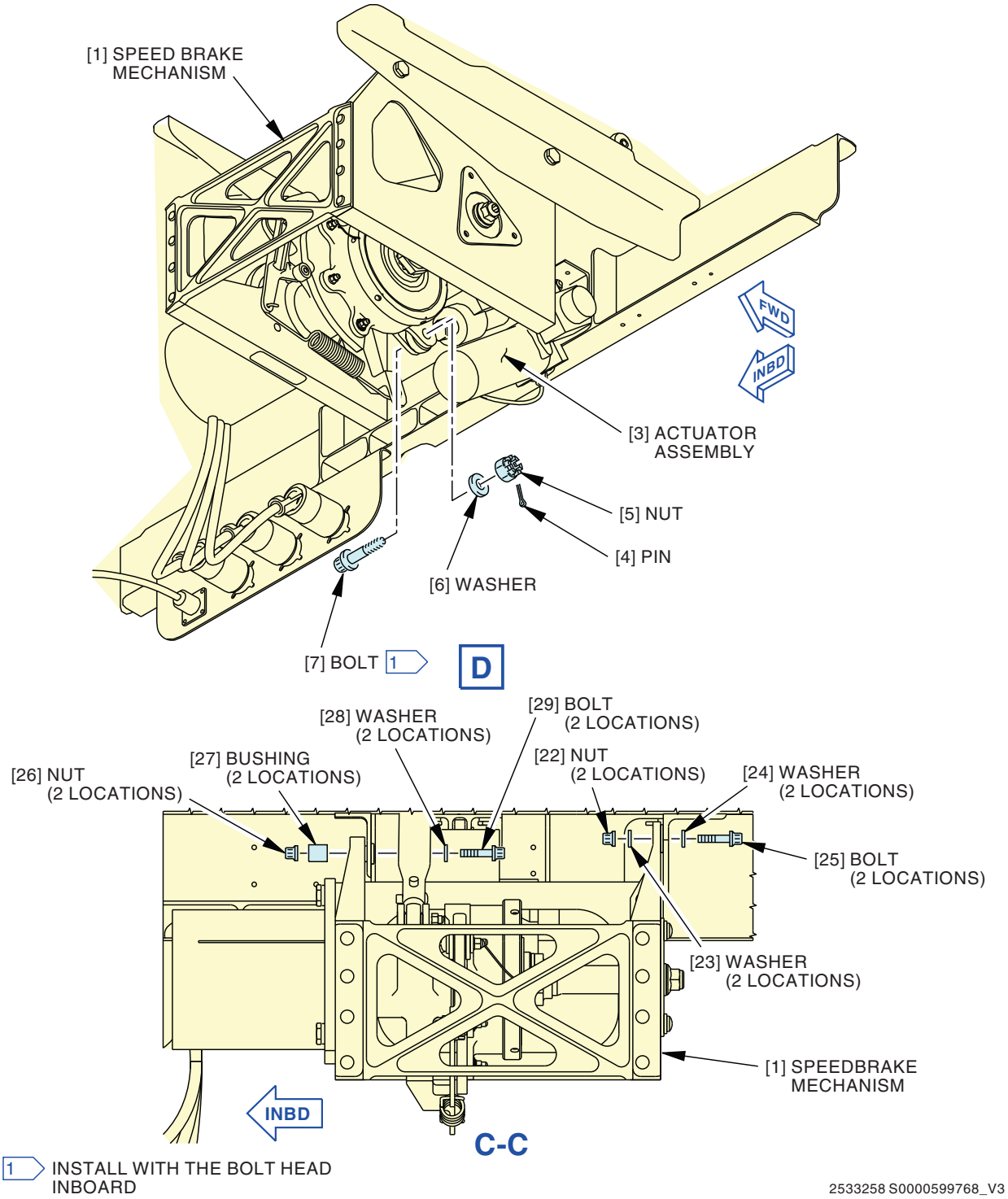
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Speedbrake Mechanism Installation
Figure 401/27-62-21-990-801 (Sheet 3 of 4)

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Speedbrake Mechanism Installation
Figure 401/27-62-21-990-801 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-62-21-400-801

3. Speedbrake Mechanism Installation

(Figure 401)

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
27-62-37-400-801	Speedbrake Lever Transducer Installation (P/B 401)

B. Consumable Materials

Reference	Description	Specification
G01912	Lockwire - MS20995NC32, Monel - 0.032 Inch (0.8128 mm) Diameter	NASM20995

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Speedbrake mechanism	27-62-31-01-165	PTF ALL
4	Pin	27-62-31-01-115	PTF ALL

D. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
211	Flight Compartment - Left
212	Flight Compartment - Right

E. Access Panels

Number	Name/Location
112A	Forward Access Door

F. Speedbrake Mechanism Installation

SUBTASK 27-62-21-420-002

(1) Install the speedbrake mechanism [1]:

(a) Put the speedbrake mechanism [1] in position.

(b) Install the nuts [26], bushings [27], washers [28] and bolts [29] that attach the speedbrake mechanism [1] to the structure.

(c) Install the nuts [22], washers [23], washers [24] and bolts [25] that attach the speedbrake mechanism [1] to the structure.

SUBTASK 27-62-21-420-003

(2) Do these steps to attach the actuator assembly [3] to the speedbrake mechanism [1]:

(a) Put the actuator assembly [3] in its position on the speedbrake mechanism [1].

(b) Install the new pin [4], nut [5], washer [6], and bolt [7] that attach the actuator assembly [3] to the speedbrake mechanism [1].

1) Install bolt [7] with the bolt head inboard.

SUBTASK 27-62-21-420-004

(3) Do these steps to attach the control rod [8] to the speedbrake mechanism [1]:

(a) Put the control rod [8] in its position on the speedbrake mechanism [1].

(b) Install the nut [9], washer [10] and bolt [11] that attach the control rod [8] to the speedbrake mechanism [1].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-62-21-420-005

(4) Install the turnbuckle [12]:

(a) Put the aft end of the turnbuckle [12] in its position.

1) Install the bolt [21], washer [20], washer [19], and nut [18].

(b) Put the forward end of the turnbuckle [12] in its position.

1) Install the bolt [17], washer [16], bushing [15], washer [14], and nut [13].

2) Install the MS20995NC32 lockwire, G01912, with the double twist method.

SUBTASK 27-62-21-420-006

(5) Install the speedbrake lever transducer [2] (TASK 27-62-37-400-801).

G. Put the Airplane Back to Its Usual Condition

SUBTASK 27-62-21-860-004

WARNING KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT SPOILERS. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU OPEN/CLOSE THESE CIRCUIT BREAKERS. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
C	10	C04032	SPOILER CONTROL ELEC PWR A
C	12	C04033	SPOILER CONTROL ELEC PWR B

(2) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	9	C00440	FLIGHT CONTROL AUTO SPEED BRAKE

SUBTASK 27-62-21-410-001

(3) Close this access panel:

Number	Name/Location
112A	Forward Access Door

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SPEEDBRAKE LEVER - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the speedbrake lever

(2) An installation of the speedbrake lever.

TASK 27-62-22-000-801

2. Speedbrake Lever - Removal

A. References

Reference	Title
25-11-01-000-801	Captain and First Officer Seat Removal (P/B 401)
27-41-00-820-801	Stabilizer Control Cable and Chain - Adjustment (P/B 501)
27-51-00-040-801	Trailing Edge Flap System Deactivation (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205
SPL-2409	Dowel Set - Thrust Lever, Control Stand Part #: F80195-1 Supplier: 81205
SPL-2411	Tool Set - Control Stand Disassembly Part #: C76002-26 Supplier: 81205
STD-589	Mat - Protective Cover
STD-858	Tag - DO NOT OPERATE

C. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Prepare for the Removal

SUBTASK 27-62-22-040-001

(1) Do this task: Trailing Edge Flap System Deactivation, TASK 27-51-00-040-801.

SUBTASK 27-62-22-040-002

(2) Make sure that the left and right engine start switches are off and install DO NOT OPERATE tags, STD-858.

SUBTASK 27-62-22-040-003

(3) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	1	C00458	ENGINE 1 IGNITION RIGHT

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	3	C00153	ENGINE 1 IGNITION LEFT
A	4	C04008	ENGINE 1 ALTN PWR CHAN B
A	5	C04007	ENGINE 1 ALTN PWR CHAN A
B	1	C01316	ENGINE 1 START LEVER CHAN A
B	2	C01317	ENGINE 1 START LEVER CHAN B
B	3	C01312	ENGINE 1 RUN/PWR
B	5	C00276	ENGINE 1 THRUST REVERSER CONT
B	6	C01412	ENGINE 1 THRUST REVERSER INTLK
B	7	C01266	ENGINE 1 THRUST REVERSER SYNC LOCK
B	8	C01103	ENGINE 1 START VALVE

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	5	C01320	ENGINE FUEL ENGINE 1 HPSOV CONT
E	6	C01395	ENGINE FUEL ENGINE 1 HPSOV IND

(4) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	5	C01313	ENGINE 2 RUN/PWR
B	6	C01318	ENGINE 2 START LEVER CHAN A
B	7	C01319	ENGINE 2 START LEVER CHAN B
C	4	C00154	ENGINE 2 START VALVE
C	5	C01267	ENGINE 2 THRUST REVERSER SYNC LOCK
C	6	C01413	ENGINE 2 THRUST REVERSER INTLK
C	7	C00277	ENGINE 2 THRUST REVERSER CONT
D	4	C00459	ENGINE 2 IGNITION RIGHT
D	6	C00151	ENGINE 2 IGNITION LEFT
D	7	C04010	ENGINE 2 ALTN PWR CHAN B
D	8	C04009	ENGINE 2 ALTN PWR CHAN A

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	3	C01321	ENGINE FUEL ENGINE 2 HPSOV CONT
E	4	C01396	ENGINE FUEL ENGINE 2 HPSOV IND

SUBTASK 27-62-22-040-004

(5) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-1

Row	Col	Number	Name
E	1	C00721	AUTOTHROTTLE DC 1

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	9	C00440	FLIGHT CONTROL AUTO SPEED BRAKE

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-62-22-860-001

WARNING MAKE SURE THAT YOU OPEN THE CIRCUIT BREAKERS FOR THE WEATHER RADAR SYSTEM. THE FORWARD MOVEMENT OF A THRUST LEVER CAN CAUSE THE AUTOMATIC OPERATION OF THE SYSTEM. THE OPERATION OF THIS SYSTEM CAN CAUSE INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT.

(6) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	13	C00120	WEATHER RADAR RT

SUBTASK 27-62-22-040-005

(7) Deactivate the speedbrakes and control lever on the control stand.

(a) Move the speedbrake control lever [22] to the DOWN position.

(b) Move to the forward bay below the flight compartment.

1) Install the rig pin S/B-1 from the rigging pin kit, SPL-1585, into the forward drum of the speed brake mechanism.

SUBTASK 27-62-22-010-001

(8) Do this task: Captain and First Officer Seat Removal, TASK 25-11-01-000-801.

SUBTASK 27-62-22-010-002

(9) Put a protective cover mat, STD-589, on the aft electronics panel P8.

NOTE: This will prevent damage to the switches and glass surfaces of the indicators and displays.

SUBTASK 27-62-22-010-003

(10) Remove control stand lightplates to get access to the speed brake lever (Figure 406).

(a) Remove the first officer stabilizer trim lightplate [60].

1) Remove the screws [64] that attach the first officer stabilizer trim lightplate [60].
2) Remove the nut [68] and lockwasher [69].
3) Remove the terminal lug and washer [70] from the stud on the lightplate.
4) Remove the lightplate.

(b) Remove the captain stabilizer trim lightplate [61].

1) Remove the screws [65] that attach the captain stabilizer trim lightplate [61].
2) Remove the nut [68] and lockwasher [69].
3) Remove the terminal lug and washer [70] from the stud on the lightplate.
4) Remove the lightplate.

(c) Remove the flap lever indicator lightplate [62].

1) Remove the screws [66] that attach the flap lever indicator lightplate [62].
2) Remove the nut [68] and lockwasher [69].
3) Remove the terminal lug and washer [70] from the stud on the lightplate.
4) Remove the lightplate.

(d) Remove the speed brake lightplate [63].

1) Remove the screws [67] that attach the speed brake lightplate [63].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) Remove the nut [68] and lockwasher [69].
3) Remove the terminal lug and washer [70] from the stud on the lightplate.
4) Remove the lightplate.

SUBTASK 27-62-22-010-004

(11) Remove seal retainers and seals to get access to the speedbrake lever (Figure 407).
(a) To remove the left seal [80] and seal retainer [82], do the following steps:
1) Remove the screws [83].
2) Remove the seal retainer [82].
3) Remove the left seal [80].
4) Remove the spacer [81].
(b) To remove the right seal [84] and seal retainer [86], do the following steps:
1) Remove the screws [87].
2) Remove the seal retainer [86].
3) Remove the right seal [84].
4) Remove the spacer [85].

SUBTASK 27-62-22-020-001

(12) Remove covers and stops to get access to the control shaft components (Figure 408).
(a) To remove the forward thrust stop [90] and aft thrust stop [91], do the following steps:
1) Remove the fuel control knobs [6] and nuts [98].
2) Remove the screws [92].
3) Remove the forward thrust stop [90].
4) Remove the screws [93].
5) Remove the aft thrust stop [91].
(b) To remove the right side cover assembly [94], do the following steps:
1) Remove the screws [95].
2) Remove the right side cover assembly [94].
(c) To remove the left side cover assembly [96], do the following steps:
1) Remove the screws [97].
2) Remove the left side cover assembly [96].

SUBTASK 27-62-22-010-005

(13) To get access to the speedbrake control lever [22] on the control stand, do the following steps (Figure 401).
(a) Remove the bolt [8], washer [8A], and the right stabilizer trim wheel [7].
1) Keep the left stabilizer trim wheel [11] attached at this time, so you can turn the assembled levers during removal.
(b) Remove the screws [51] and upper right cover [21].
(c) Remove the screws [2] and left upper side panel [1].
(d) Remove the screws [2] and the left lower side panel [5].
(e) Remove the screws [2] and the right upper side panel [3].
(f) Remove the screws [2] and the right lower side panel [4].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(g) To move the stabilizer trim switch panel [10] from the control stand, do the following steps:

1) Remove screws [9].

2) Apply masking tape to prevent damage to the surface of the stabilizer trim switch panel [10].

3) Move the stabilizer trim switch panel [10] on to the protective cover mat, STD-589.

E. Control Stand Components Removal

SUBTASK 27-62-22-020-002

(1) Disconnect the stabilizer trim indicator assembly.

(a) Go to the right side of the control stand.

(b) Remove the nut [16], washer [14], and bolt [17] from the lower end of the link assembly [13].

(c) Safety the link assembly [13] to the control stand.

SUBTASK 27-62-22-020-003

(2) Disconnect the flap lever position synchro assembly.

(a) Remove the nut [18] and bolt [19] from the link assembly [20].

(b) Safety the link assembly [13] to the control stand.

SUBTASK 27-62-22-020-004

(3) Disconnect the stabilizer trim controls.

(a) Turn the left trim wheel until you find the link in the trim chain [31] (Figure 402).

NOTE: This will help if it becomes necessary to disconnect the trim chain [31] during installation.

(b) Safety the trim chain [31] to the stabilizer trim sprocket [32].

1) Do not remove the trim chain [31] from the stabilizer trim sprocket [32].

(c) Move to the forward bay below the flight compartment.

1) Release the tension from the stabilizer control chain (TASK 27-41-00-820-801).

SUBTASK 27-62-22-020-005

(4) Do these steps to remove the control shaft components (Figure 401):

(a) Remove the bolt [8], washer [8A], and left stabilizer trim wheel [11].

(b) Remove screws [50] and upper left cover [12].

(c) Carefully move the stabilizer trim stabilizer trim shaft [27] to the left side of the control stand and do these steps at the same time (Figure 402).

1) Remove the bearing [35].

2) Remove the clamp up bushing [36].

3) Catch and remove the bushing [33] and inside bearing [34] as the stabilizer trim shaft [27] is moved to the left.

NOTE: The bushing is between the inside bearing [34] and the stabilizer trim sprocket.

4) Catch and move the stabilizer trim sprocket [32] and trim chain [31].

5) Move the stabilizer trim shaft [27] to the left side until the stabilizer trim sprocket [32] is clear of the stabilizer trim shaft [27].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

6) Move the stabilizer trim sprocket [32] and trim chain [31] to the bottom of the control stand.

(d) If the inside bearing [34] did not fall out with the bushing [33], then reach inside the short shaft [37] from the left side and remove the inside bearing [34].

(e) Remove the stabilizer trim stabilizer trim shaft [27] from the control stand.

1) Remove the bearing [28].

2) Remove the long bushing [29].

3) Remove the clamp up bushing [30].

SUBTASK 27-62-22-020-006

(5) To remove the stabilizer trim chain guard [45], do the following steps (Figure 403):

(a) Remove the flap control switches [99], screws [100], and washers [103].

(b) Remove the clamps [104], screw [101], screw [102], and washers [103].

(c) Go to the opening where the stabilizer trim switch panel [10] for the stabilizer trim cutout is found.

(d) Use the 90 degree screwdriver from the tool set, SPL-2411, to hold the screw [44].

(e) Remove the nut [42], washer [43], and screw [44].

(f) Remove the upper center screw [39] and washer [38].

(g) Remove the forward screw [41] and washer [40].

(h) Remove the stabilizer trim chain guard [45].

F. Speedbrake Lever Removal

SUBTASK 27-62-22-030-001

(1) To disconnect the speedbrake control lever [22] from the link assembly [23], do the following steps (Figure 401):

(a) Remove the bolt [24], washer [25], and nut [26] from the speedbrake control lever [22].

SUBTASK 27-62-22-020-007

(2) To disconnect the lockwasher [47] and nut [48] from the long control shaft [46], do the following steps (Figure 404):

(a) Find the bent tab on the lockwasher [47].

(b) Use the lockwasher removal tool from the tool set, SPL-2411, to bend the tab.

(c) Bend the tab out of the notch in the nut [48].

(d) Put one control shaft wrench from the tool set, SPL-2411, on the left side of the long control shaft [46] to hold it in position.

(e) Put one control shaft nut wrench from the tool set, SPL-2411, in position and remove the nut [48] from the long control shaft [46].

(f) Remove and discard the lockwasher [47].

SUBTASK 27-62-22-020-008

(3) To remove the speedbrake control lever [22], do the following steps (Figure 404):

(a) Put the dowel set, SPL-2409, against the right side end of the long control shaft [46].

(b) Hit lightly (carefully tap) on the dowel.

NOTE: This will move the long control shaft off the control levers.

1) Move the dowel until the speedbrake control lever [22] is on the dowel.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Do not move the long control shaft [46] from this position.

NOTE: More movement will let the speedbrake control lever fall.

(d) Hold the speedbrake control lever [22] and move the dowel from the speedbrake control lever [22].

1) Stop the dowel movement when you can lift up the speedbrake control lever [22].

2) Make sure that the washers [49] are still held by the dowel and long control shaft [46].

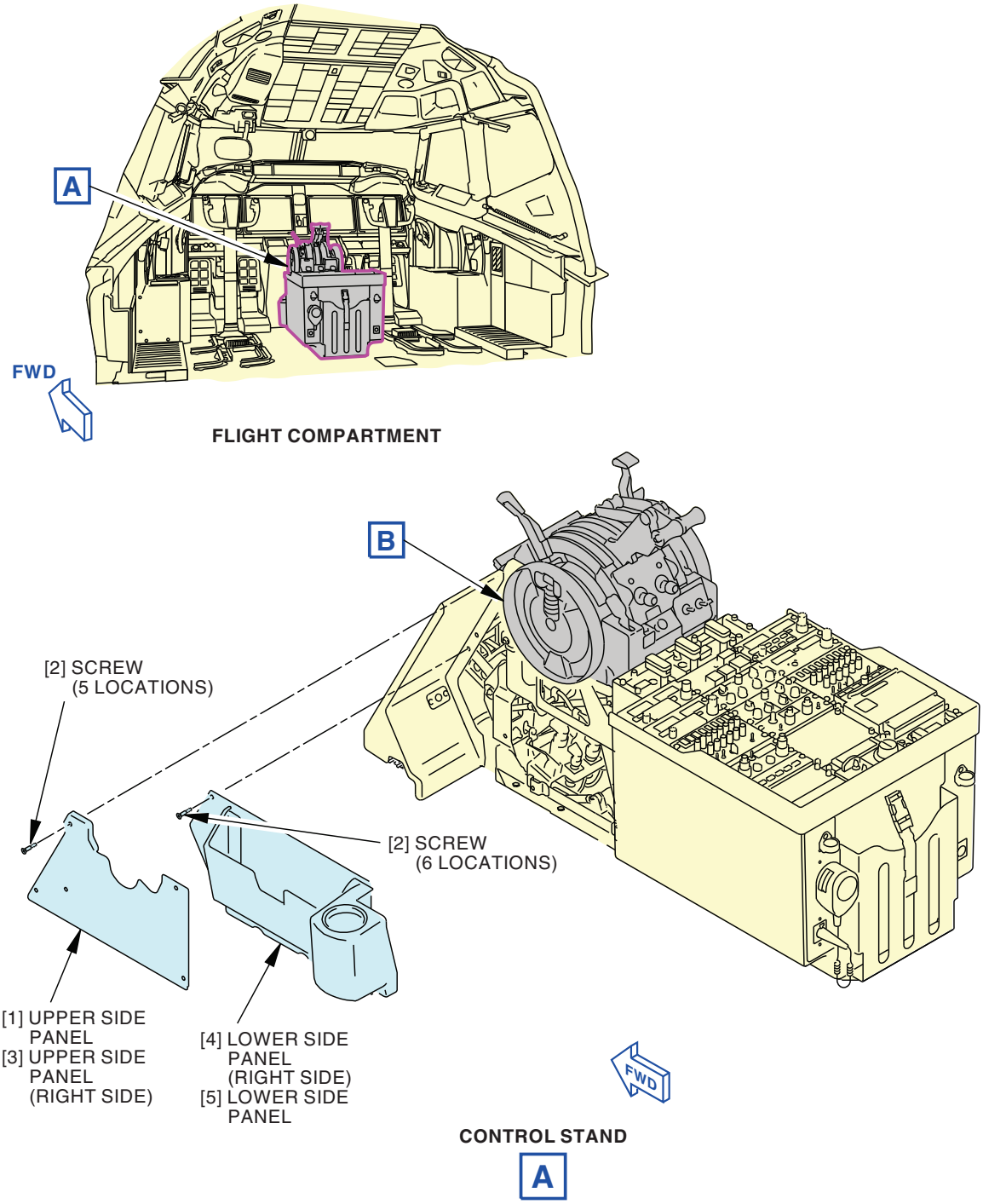
(e) Remove the speedbrake control lever [22] through the top of the control stand.

(f) Make sure that the washers [49] do not fall into the control stand.

(g) Do not move the dowel from this position.

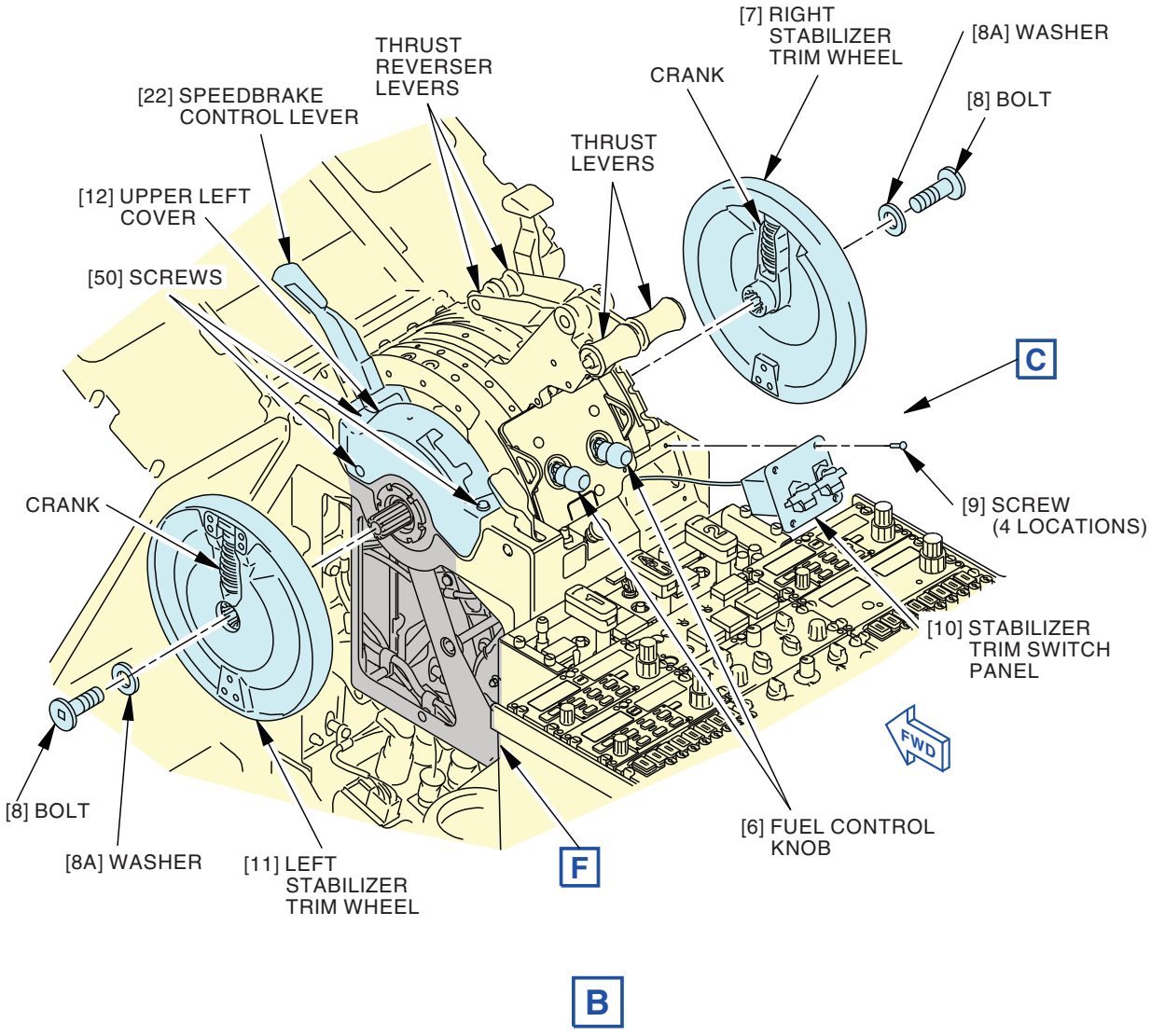
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Control Stand Installation
Figure 401/27-62-22-990-801 (Sheet 1 of 4)

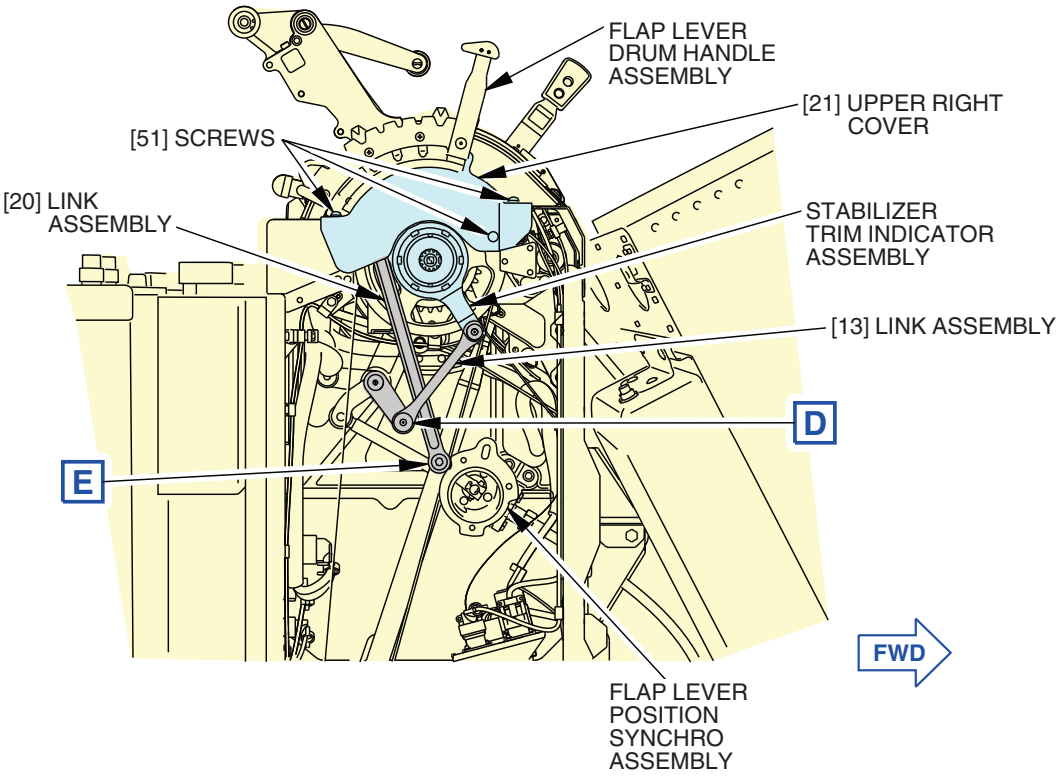
ECCN 9E991 BOEING PROPRIETARY - See title page for details



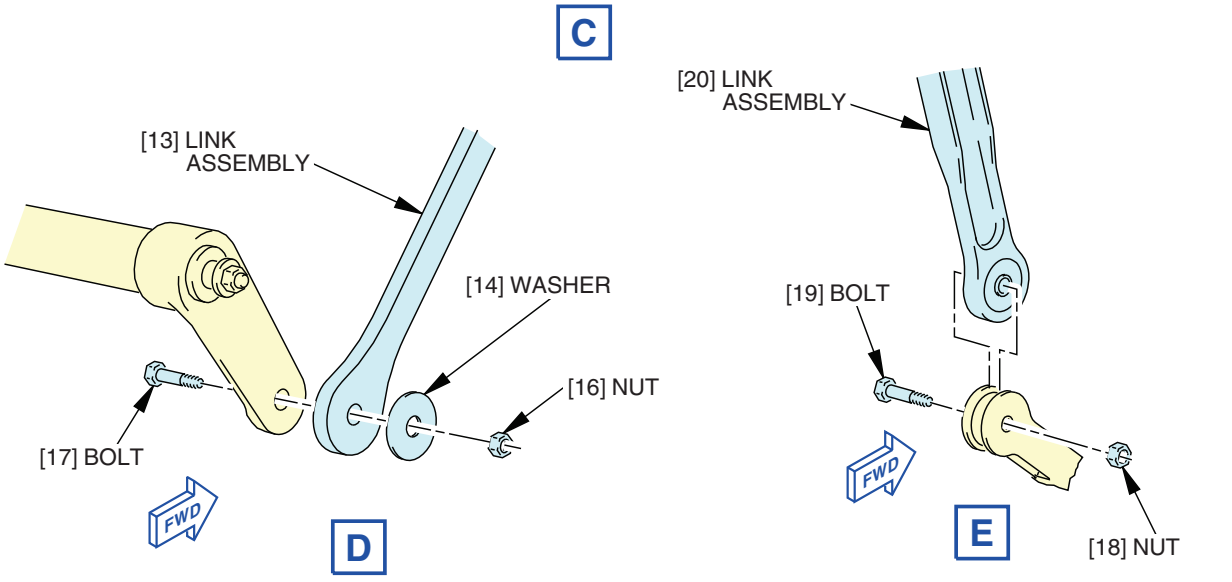
2413252 S00061532186_V2

Control Stand Installation
Figure 401/27-62-22-990-801 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



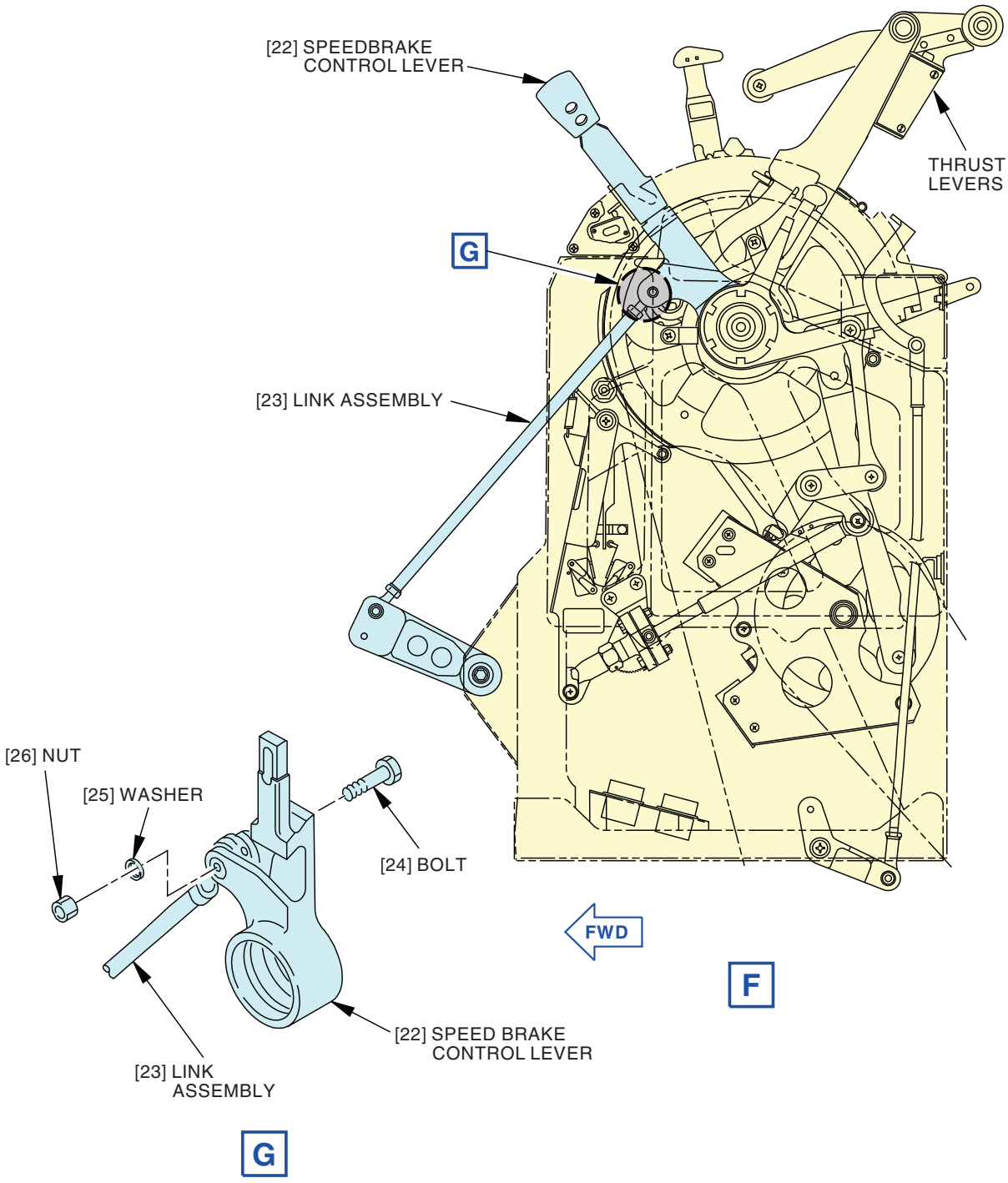
(REMOVED A PORTION OF THE CONTROL STAND STRUCTURE FOR CLARITY)



2413253 S00061532187_V2

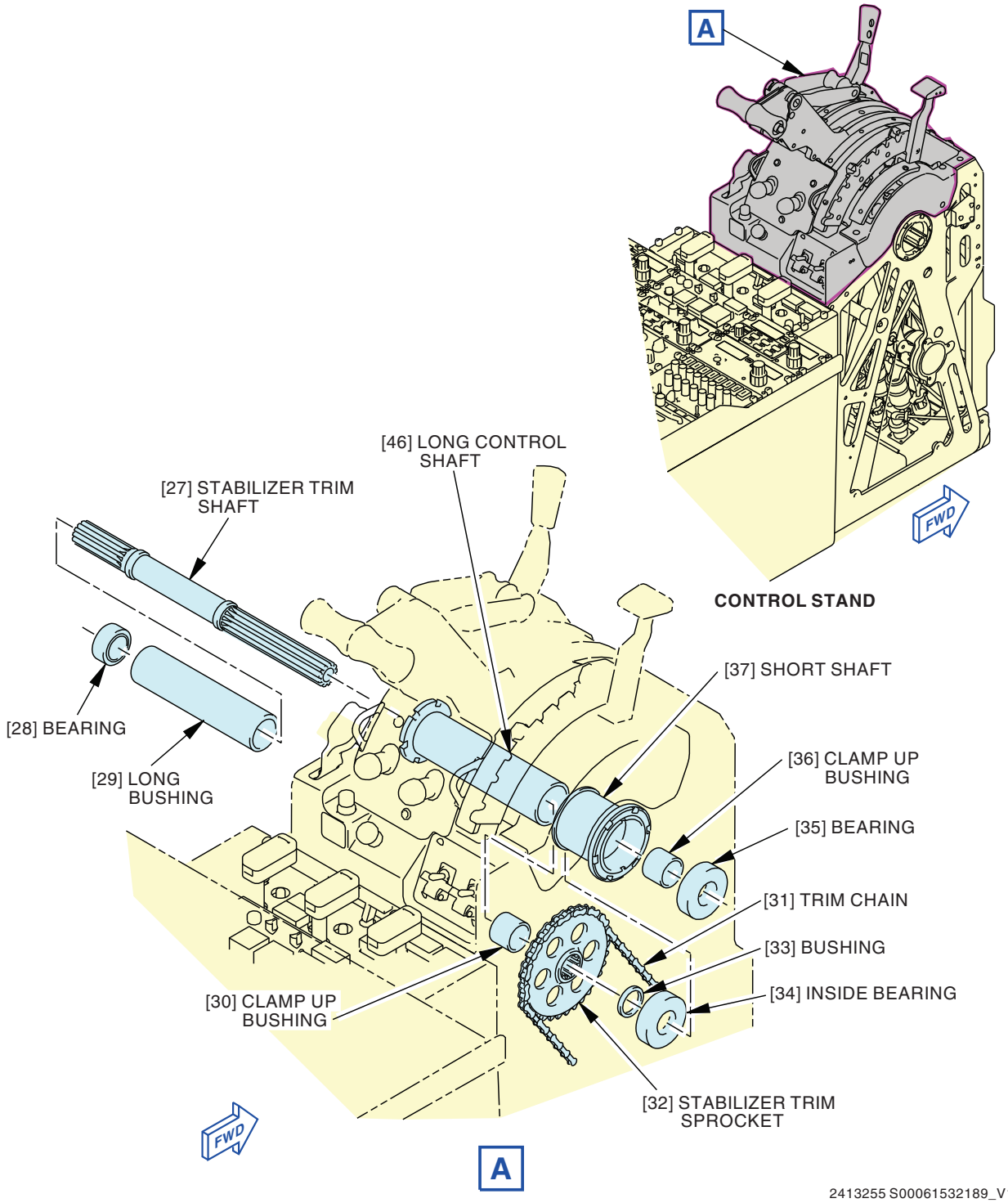
Control Stand Installation
Figure 401/27-62-22-990-801 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



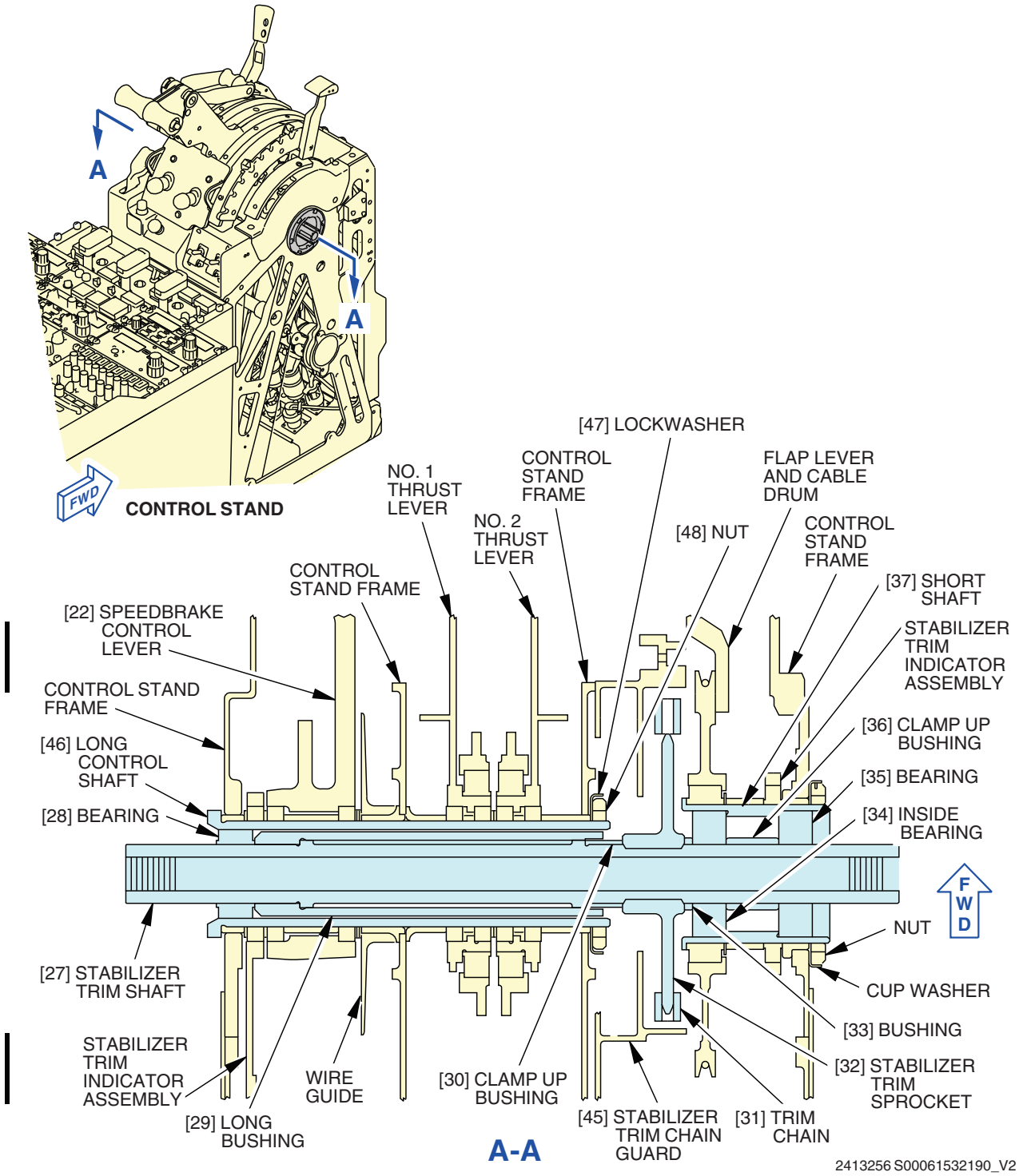
Control Stand Installation
Figure 401/27-62-22-990-801 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



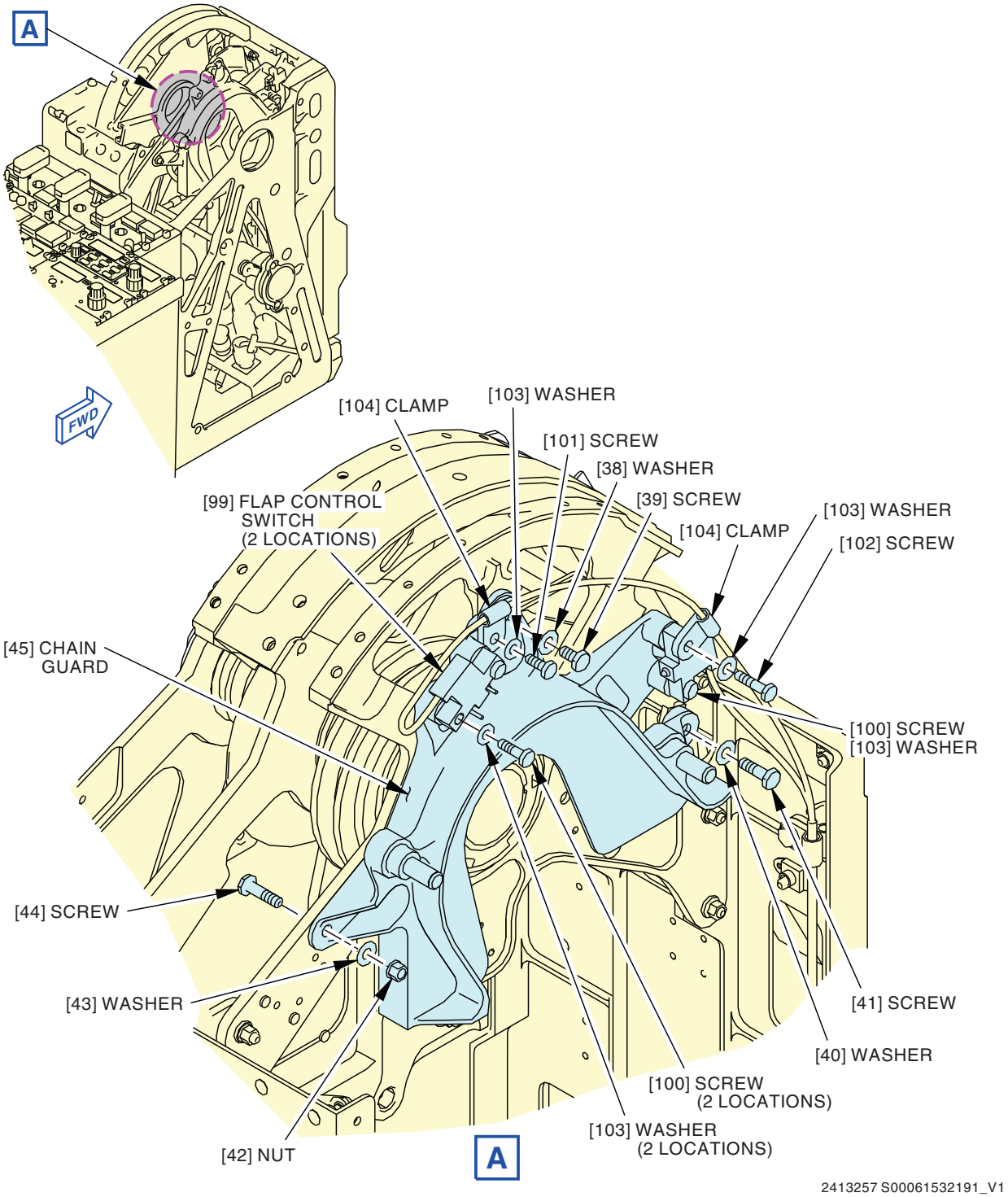
Control Shaft Components Installation
Figure 402/27-62-22-990-802 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



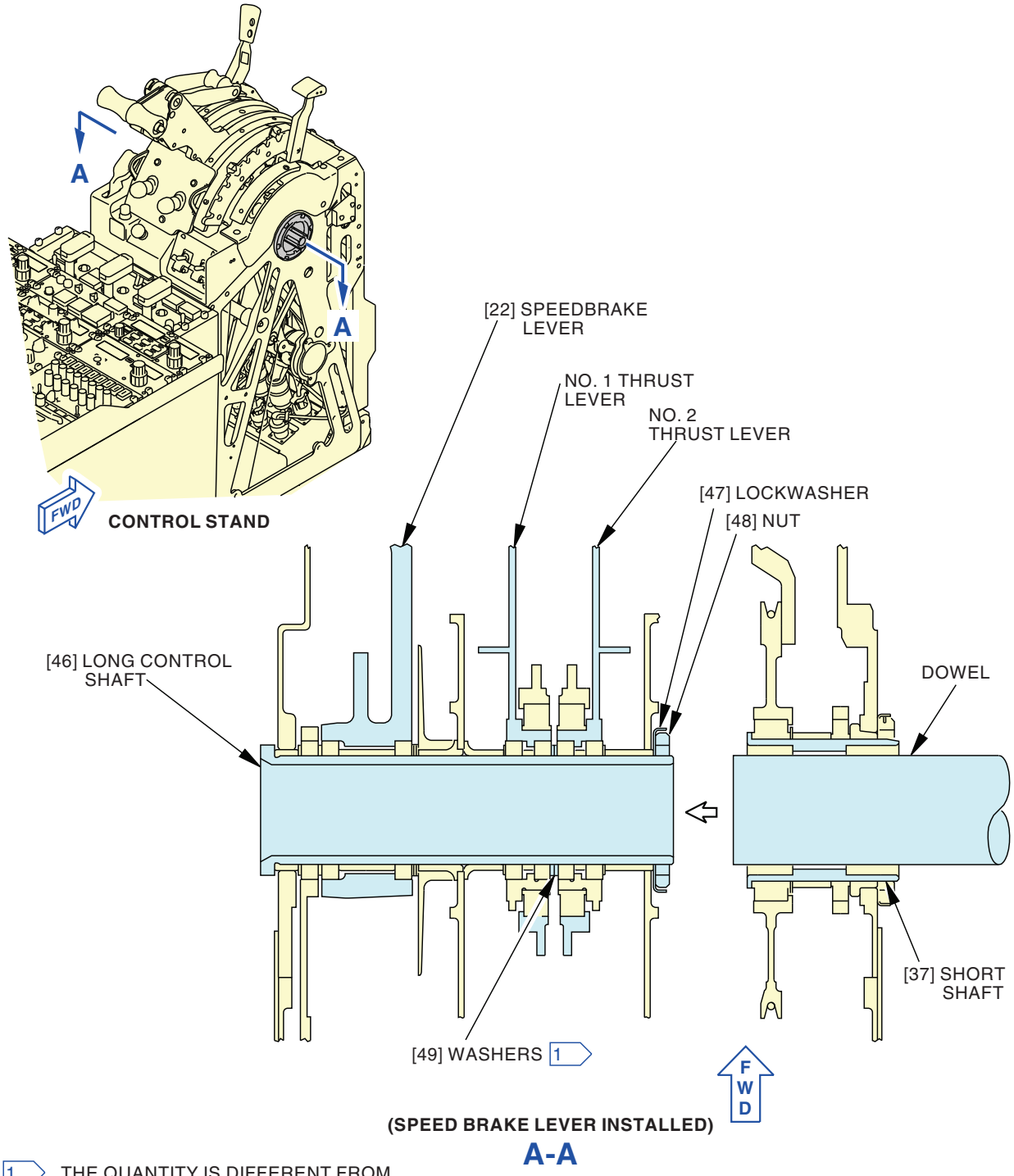
Control Shaft Components Installation
Figure 402/27-62-22-990-802 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



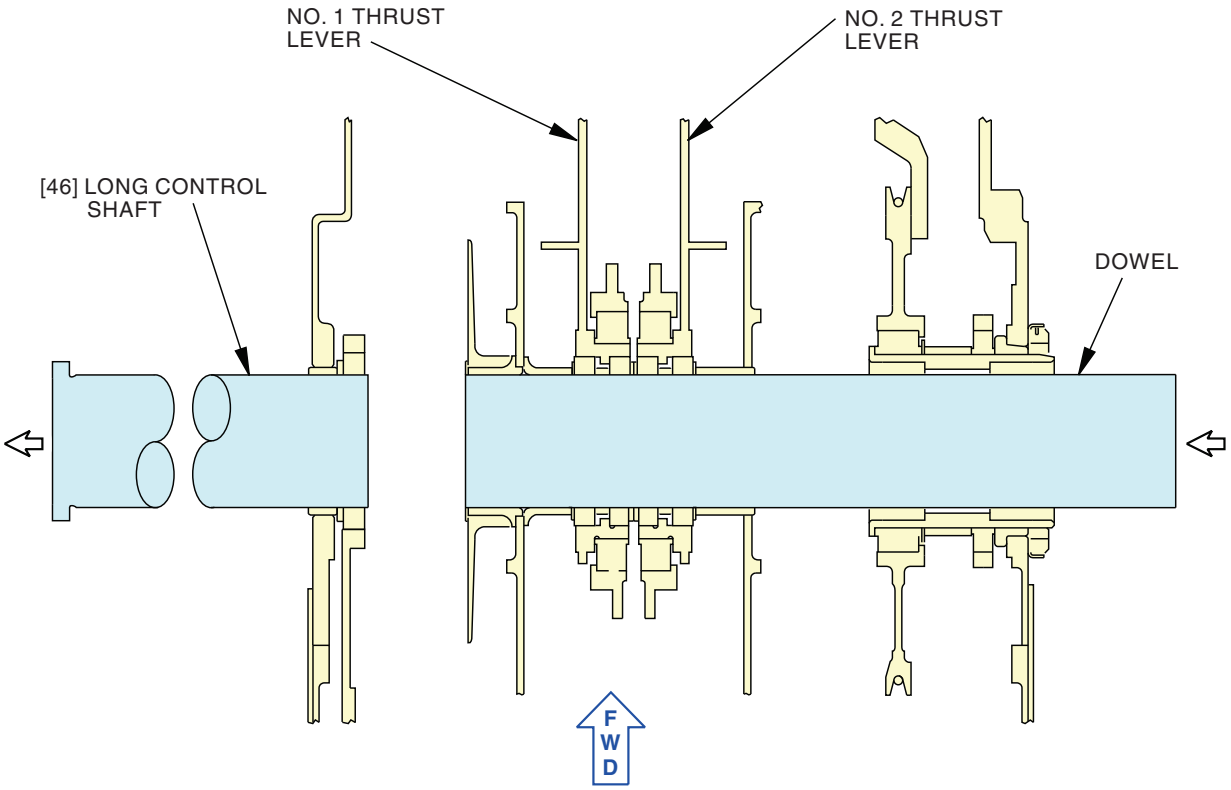
Chain Guard Assembly Installation
Figure 403/27-62-22-990-803

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Speed Brake Lever Installation
Figure 404/27-62-22-990-804 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



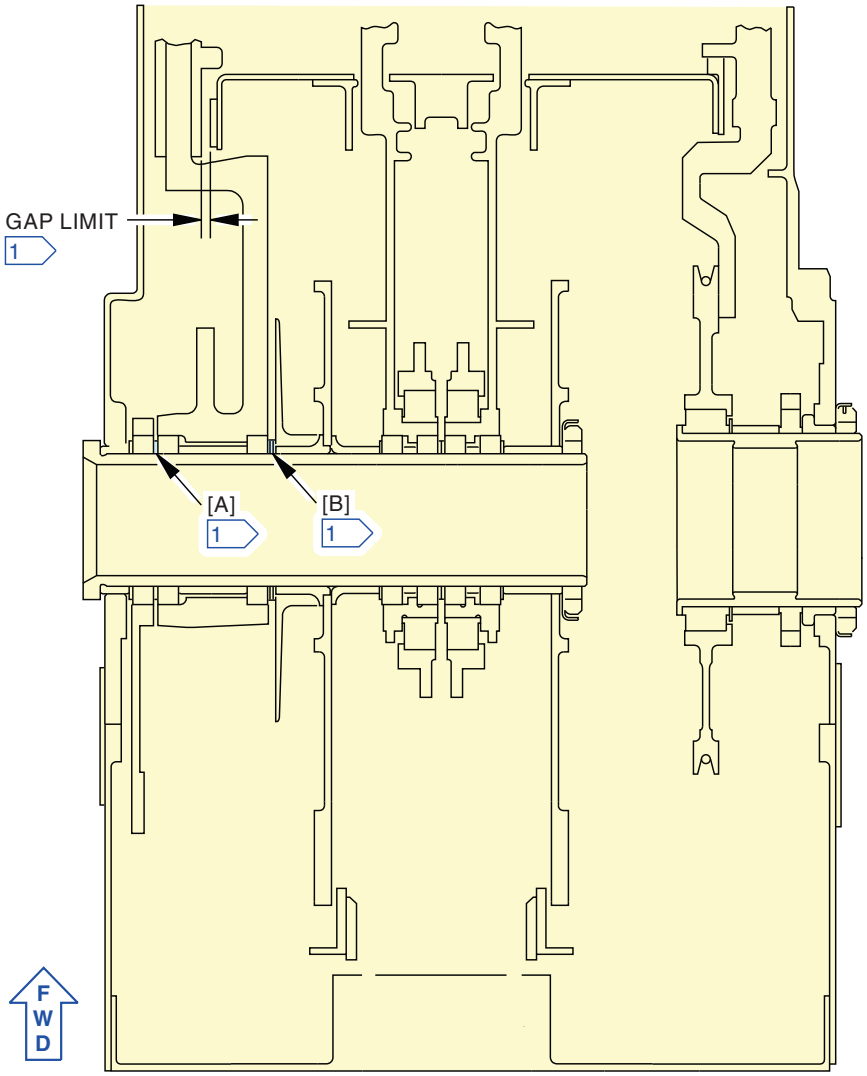
(SPEEDBRAKE LEVER REMOVED)

A-A

2413259 S00061532193_V1

Speed Brake Lever Installation
Figure 404/27-62-22-990-804 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



1 SEE TABLE A - CONTROL SHAFT ADJUSTMENT GAP LIMITS

2413260 S00061532194_V1

Control Shaft Washer (Shim) Limits
Figure 405/27-62-22-990-805 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

GAP LIMIT INCH (mm)	ACTUAL MEASUREMENT INCH (mm)	ADJUSTMENT NECESSARY TO GET GAP LIMIT (REFER TO TABLE B FOR SHIM QUANTITY)
0.030-0.060 (0.762-1.524)	> 0.100 (2.54)	MOVE THREE WASHERS FROM [B] TO [A] 2
	0.080-0.100 (2.032-2.54)	MOVE TWO WASHERS FROM [B] TO [A] 2
	0.060-0.080 (1.524-2.032)	MOVE ONE WASHER FROM [B] TO [A] 2
	< 0.030 (0.762)	MOVE ONE WASHER FROM [A] TO [B] 2

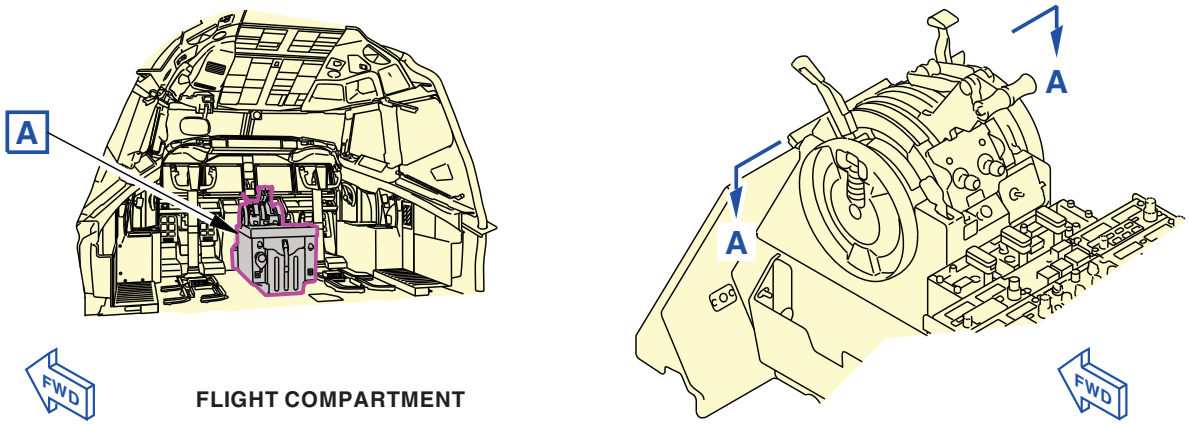
CONTROL SHAFT ADJUSTMENT GAP LIMITS
TABLE A

2 THE MAXIMUM NUMBER OF WASHERS IN LOCATIONS A AND B IS 4.

2413261 S00061532195_V1

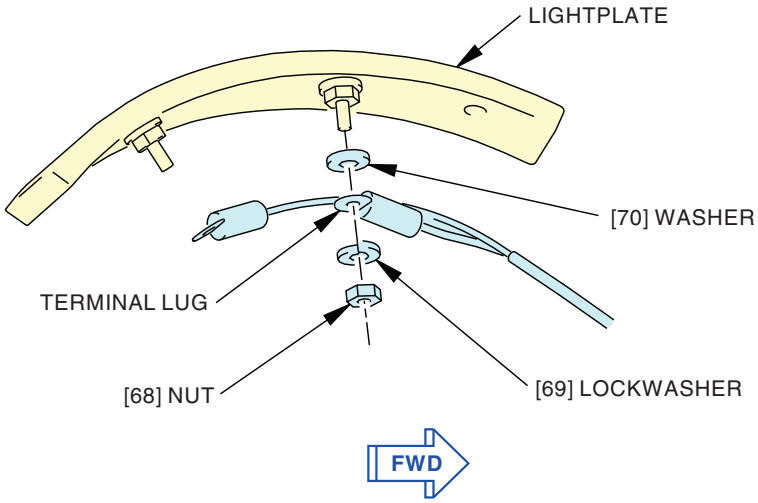
Control Shaft Washer (Shim) Limits
Figure 405/27-62-22-990-805 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Control Stand Lightplate Installation
Figure 406/27-62-22-990-806 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



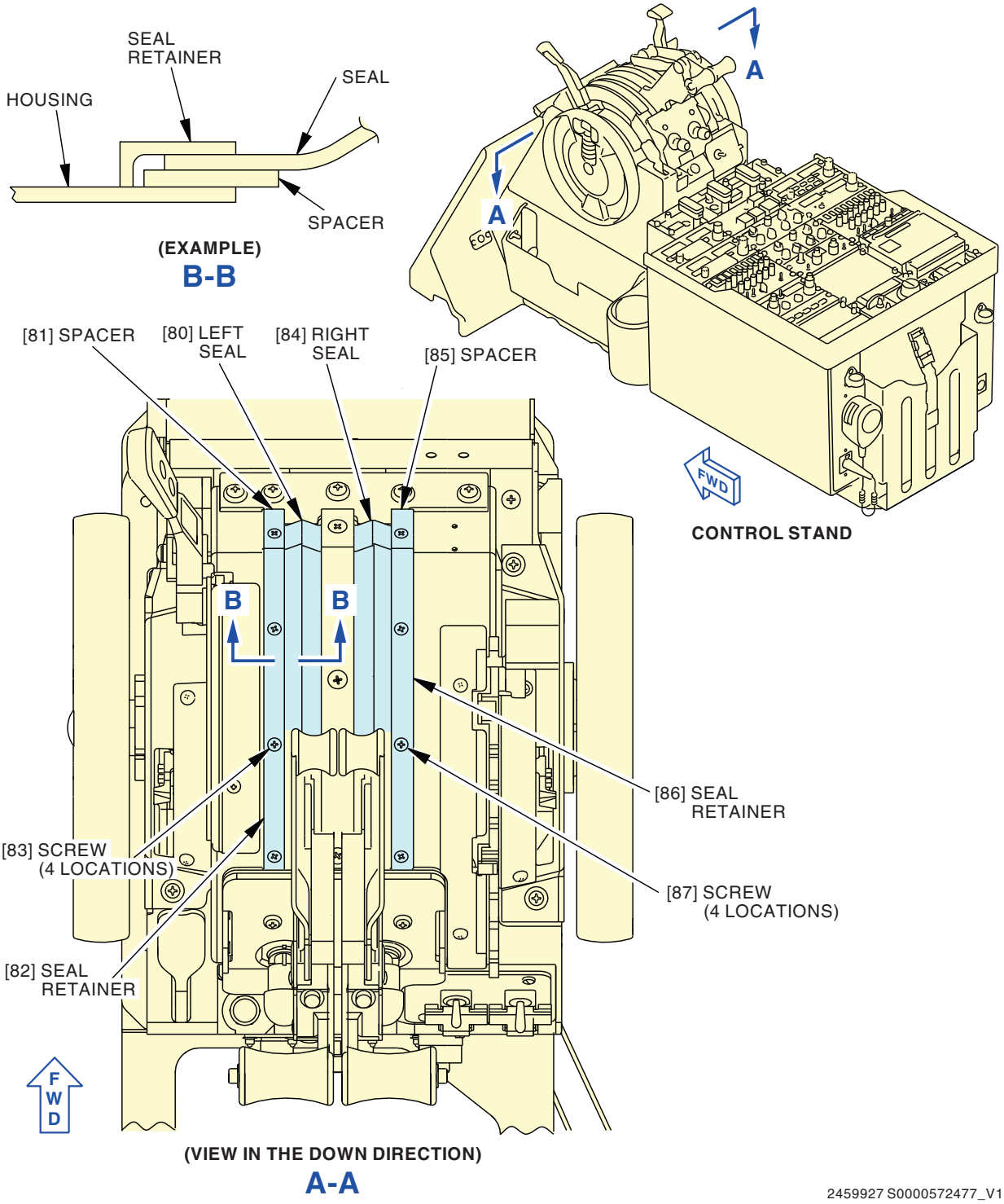
(EXAMPLE - TWO LOCATIONS ON EACH LIGHTPLATE)

B

2459802 S0000572473_V1

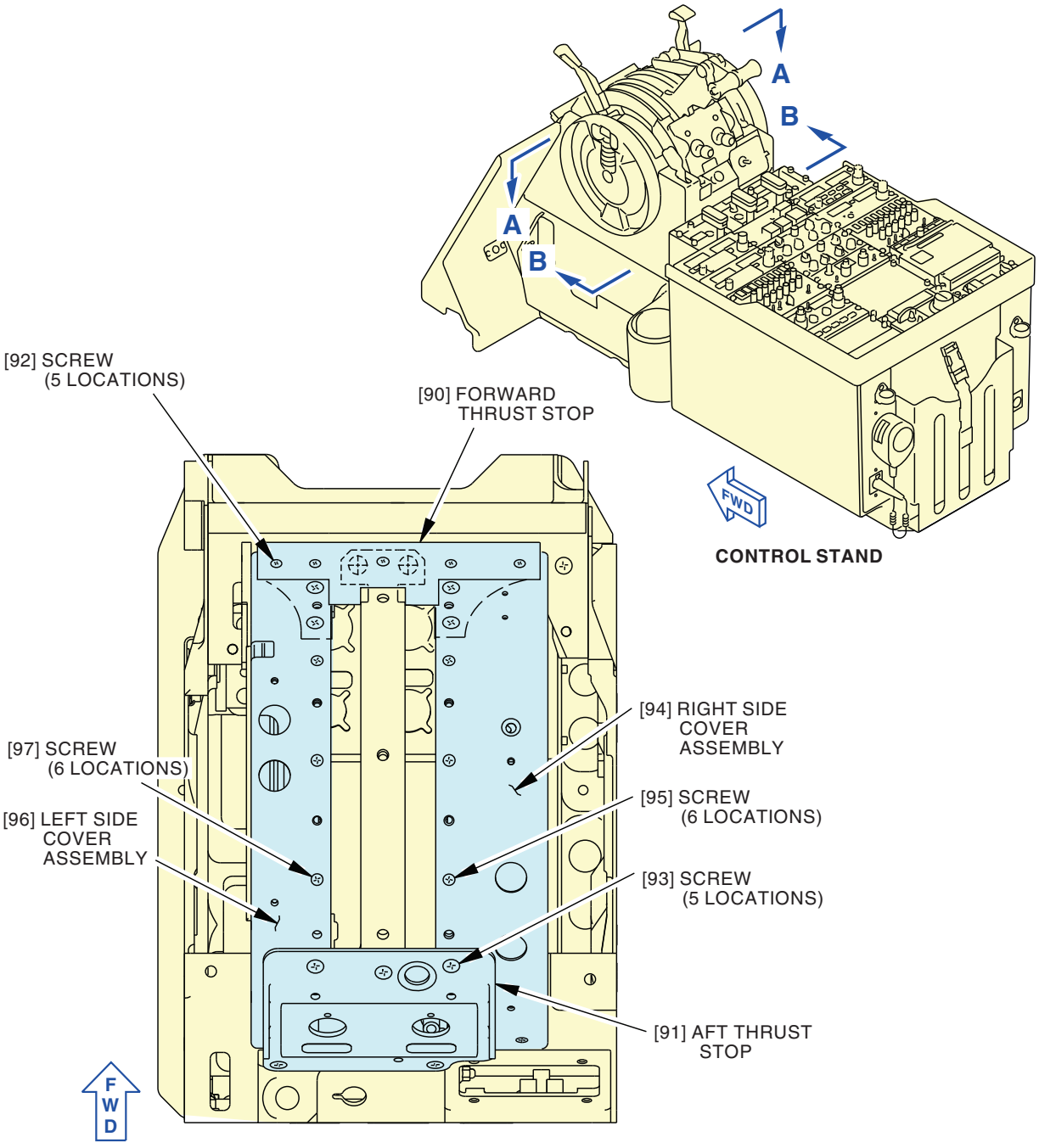
Control Stand Lightplate Installation
Figure 406/27-62-22-990-806 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



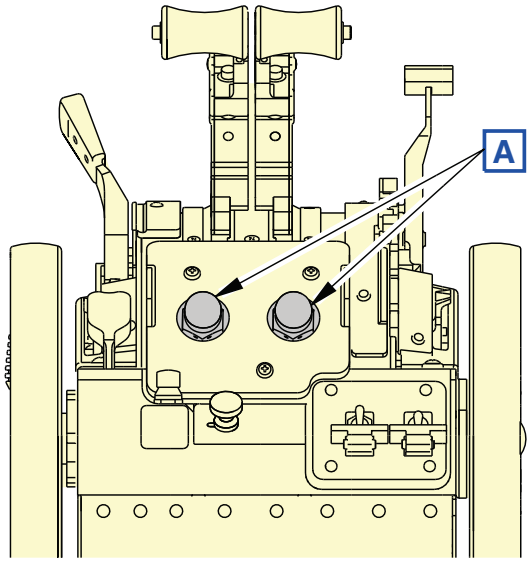
Control Stand Seal, Spacer and Retainer Installation
Figure 407/27-62-22-990-807

ECCN 9E991 BOEING PROPRIETARY - See title page for details

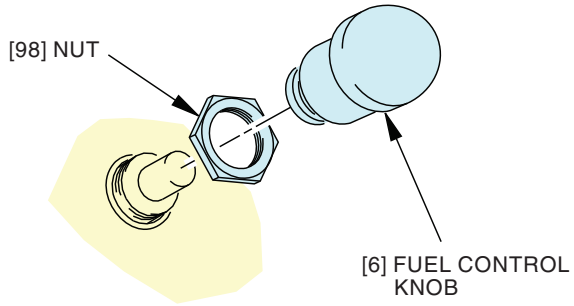


Control Stand Cover and Stop Installation
Figure 408/27-62-22-990-808 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



B-B



(EXAMPLE)

A

2479326 S0000581419_V1

Control Stand Cover and Stop Installation
Figure 408/27-62-22-990-808 (Sheet 2 of 2)

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TASK 27-62-22-400-801

3. Speedbrake Lever - Installation

A. References

Reference	Title
25-11-01-400-801	Captain and First Officer Seat Installation (P/B 401)
27-41-00-820-801	Stabilizer Control Cable and Chain - Adjustment (P/B 501)
27-51-00-440-801	Trailing Edge Flap System Reactivation (P/B 201)
27-62-00-820-801	Speed Brake Control Lever Adjustment (P/B 501)
27-62-12-720-801	Inboard Ground Spoiler Functional Test (P/B 401)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205
SPL-2409	Dowel Set - Thrust Lever, Control Stand Part #: F80195-1 Supplier: 81205
SPL-2411	Tool Set - Control Stand Disassembly Part #: C76002-26 Supplier: 81205
STD-589	Mat - Protective Cover
STD-858	Tag - DO NOT OPERATE

C. Consumable Materials

Reference	Description	Specification
D00013	Grease - Aircraft And Instrument Grease	MIL-PRF-23827 (NATO G-354) (Supersedes MIL-G-23827)

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
22	Speedbrake control lever	27-62-22-01-050	PTF ALL

E. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
211	Flight Compartment - Left
212	Flight Compartment - Right

F. Access Panels

Number	Name/Location
112A	Forward Access Door

G. Speedbrake Lever Installation

SUBTASK 27-62-22-420-001

(1) Do these steps to install the speedbrake control lever [22] on the control stand (Figure 404):

(a) Apply grease, D00013, to the inside of the speedbrake control lever [22].

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(b) Apply a thin layer of grease, D00013, to each side of the washers [49].

NOTE: The application of grease, D00013, will help hold the washers to the speedbrake control lever.

(c) Hold the speedbrake control lever [22] and washers [49] in the correct position.

(d) Move the long control shaft [46] through the speedbrake control lever [22] and washers [49].

(e) Put the long control shaft [46] against the right side end of the dowel set, SPL-2409.

(f) Move the speedbrake control lever [22] forward and aft.

NOTE: The lever and washers must move freely when you move them from the dowel to the long control shaft.

(g) Move the long control shaft [46] and the dowel to the right until all the lever assemblies and washers [49] are on the long control shaft [46].

(h) Remove the dowel.

SUBTASK 27-62-22-420-002

(2) Do these steps to lock the speedbrake control lever [22]:

(a) Set the keyway on the long control shaft [46] at the top.

(b) Install a new lockwasher [47], but do not bend the tab at this time.

1) Align the key tab on the lockwasher [47] with the keyway on the long control shaft [46].

(c) Put one control shaft wrench from the tool set, SPL-2411, on the left side of the long control shaft [46] to hold it in position.

(d) Install the nut [48] as follows:

1) Use one control shaft nut wrench from the tool set, SPL-2411, to tighten the nut [48].

2) Tighten the nut [48] to 100 in-lb (11.3 N·m) - 150 in-lb (16.9 N·m).

SUBTASK 27-62-22-020-009

(3) Measure the gap after the lever is installed as follows (Figure 405):

(a) If the gap is not in the limits, follow the instructions in Table A to get the correct limits:

1) Use the applicable steps in the removal and installation tasks to adjust the washer [49] on the long control shaft [46].

SUBTASK 27-62-22-420-003

(4) After the speedbrake control lever [22] is set correctly, lock the assembly in its position as follows (Figure 404):

(a) Make sure that the nut [48] is tightened to 100 in-lb (11.3 N·m) - 150 in-lb (16.9 N·m).

(b) Bend the rim of the lockwasher [47] into one of the notches on the nut [48].

1) Use a chisel or the lockwasher break tool from the tool set, SPL-2411, to bend the rim of the lockwasher [47].

2) Make sure that the bend in the lockwasher [47] is not more than 0.15 in. (3.8 mm) into the notch on the nut [48].

SUBTASK 27-62-22-420-004

(5) Connect the speedbrake control lever [22] to the link assembly [23] as follows (Figure 401):

(a) Install the bolt [24], washer [25], and nut [26] in the speedbrake control lever [22] and link assembly [23].

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H. Control Stand Components Installation

SUBTASK 27-62-22-020-010

(1) Do these steps to install the stabilizer trim chain guard [45] (Figure 403):

(a) Put the stabilizer trim chain guard [45] in its position.

(b) Install the upper center screw [39] and washer [38].

(c) Install the forward screw [41] and washer [40].

(d) Install the screw [44] through the thrust lever opening.

(e) Use the 90 degree screwdriver from the tool set, SPL-2411, to hold the screw [44].

(f) Install the nut [42] and washer [43].

(g) Install the clamps [104], screw [101], screw [102], and washers [103].

(h) Install the flap control switches [99], screws [100], and washers [103].

SUBTASK 27-62-22-020-011

(2) Do these steps to install the control shaft components (Figure 402):

(a) Install the clamp up bushing [30] and long bushing [29] on the stabilizer trim shaft [27].

1) Put the clamp up bushing [30] on the end with the longest splines.

(b) Put the inside bearing [34] into the inside end of the short shaft [37].

(c) Move the stabilizer trim shaft [27] into the control stand from the left side.

1) Put the end of the stabilizer trim shaft [27] with the longest splines in first.

(d) Move the stabilizer trim sprocket [32] and trim chain [31] up from the bottom of the control stand as follows:

1) Put the stabilizer trim sprocket [32] on the splined end of the stabilizer trim shaft [27].

2) Move the stabilizer trim shaft [27] through the stabilizer trim sprocket [32].

(e) Put the bushing [33] on the splined end of the stabilizer trim shaft [27].

1) Move the stabilizer trim shaft [27] through the bushing [33].

(f) Move the stabilizer trim shaft [27] through the inside bearing [34] until the end is moved out through the short shaft [37].

(g) Install the bearing [28] on the left side of the stabilizer trim shaft [27].

(h) Install the clamp up bushing [36] on the right side of the stabilizer trim shaft [27].

(i) Install the bearing [35] on the right side of the stabilizer trim shaft [27].

(j) Lightly tap the stabilizer trim shaft [27] from the left side.

1) Make sure that the assembled parts are tight and in the correct position.

SUBTASK 27-62-22-420-005

(3) Do these steps to activate and adjust the stabilizer trim assembly:

(a) Remove the temporary wire tie from the trim chain [31] and stabilizer trim sprocket [32].

(b) Adjust the stabilizer control chain (TASK 27-41-00-820-801).

SUBTASK 27-62-22-410-001

(4) Do these steps to connect the flap lever position synchro assembly (View E, Figure 401):

(a) Put the link assembly [20] in the clevis for the flap lever position synchro assembly.

(b) Install the nut [18] and bolt [19].

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SUBTASK 27-62-22-410-002

(5) Do this step to connect the stabilizer trim indicator assembly (View D, Figure 401):

(a) Put the link assembly [13] to the arm.

(b) Install the bolt [17], washer [14], and nut [16].

SUBTASK 27-62-22-440-001

(6) Activate the speed brake lever as follows:

(a) Go into the access area under the forward flight compartment floor.

(b) Remove the rig pin from the rigging pin kit, SPL-1585, from the forward drum of the speed break assembly.

(c) Do this task: Speed Brake Control Lever Adjustment, TASK 27-62-00-820-801.

SUBTASK 27-62-22-420-006

(7) Install these covers on the control stand.

(a) Install the upper left cover [12].

(b) Install the screws [50].

(c) Install the upper right cover [21].

(d) Install the screws [51].

SUBTASK 27-62-22-010-006

(8) Install the control stand access covers.

(a) Install the left upper side panel [1] with the screws [2].

(b) Install the left lower side panel [5] with the screws [2].

(c) Install the right upper side panel [3] with the screws [2].

(d) Install the right lower side panel [4] with the screws [2].

SUBTASK 27-62-22-420-007

(9) Install the stabilizer trim wheel assemblies.

(a) Install the right stabilizer trim wheel [7] as follows.

1) Install the right bolt [8] and washer [8A].

2) Tighten the bolt [8] to 150 in-lb (16.9 N·m) - 170 in-lb (19.2 N·m)

(b) Install the left stabilizer trim wheel [11] as follows:

1) Set the crank of the left stabilizer trim wheel [11] 75 to 105 degrees away from the setting of the crank on the right stabilizer trim wheel [7].

NOTE: This can be set in either direction.

2) Install the left bolt [8] and washer [8A].

3) Tighten the bolt [8] to 150 in-lb (16.9 N·m) - 170 in-lb (19.2 N·m).

SUBTASK 27-62-22-200-001

(10) Do a visual check of the assembly as follows:

(a) Move each control through its usual range.

(b) Make sure that the electrical bundles do not touch parts.

SUBTASK 27-62-22-410-003

(11) Do these steps to install the stabilizer trim switch panel [10] for the stabilizer trim:

(a) Remove the masking tape from the stabilizer trim switch panel [10].

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(b) Put the stabilizer trim switch panel [10] on the control stand.

(c) Install the screws [9].

SUBTASK 27-62-22-410-004

(12) Install these covers and stops on the control stand as follows (Figure 408):

(a) Do these steps to install the left side cover assembly [96]:

1) Install the left side cover assembly [96].

2) Install the screws [97].

(b) Do these steps to install the right side cover assembly [94]:

1) Install the right side cover assembly [94].

2) Install the screws [95].

(c) Do these steps to install the forward thrust stop [90] and aft thrust stop [91]:

1) Install the aft thrust stop [91].

2) Install the screws [93].

3) Install the fuel control knobs [6] and nuts [98].

4) Install the forward thrust stop [90].

5) Install the screws [92].

SUBTASK 27-62-22-410-005

(13) Install these seal retainers and seals on the control stand as follows (Figure 407):

(a) Do these steps to install the left seal [80] and seal retainer [82]:

1) Install the spacer [81].

2) Install the left seal [80].

3) Install the seal retainer [82].

4) Install the screws [83].

(b) Do these steps to install the right seal [84] and seal retainer [86]:

1) Install the spacer [85].

2) Install the right seal [84].

3) Install the seal retainer [86].

4) Install the screws [87].

SUBTASK 27-62-22-410-006

(14) Install these control stand lightplates as follows (Figure 406):

(a) Do these steps to install the first officer stabilizer trim lightplate [60]:

1) Install the washer [70].

2) Connect the terminal lug on the stud attached to the first officer stabilizer trim lightplate [60].

3) Install the nut [68] and lockwasher [69].

4) Install the first officer stabilizer trim lightplate [60] on the control stand.

5) Install the screws [64] that attach the first officer stabilizer trim lightplate [60] to the control stand.

(b) Do these steps to install the captain stabilizer trim lightplate [61]:

1) Install the washer [70].

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2) Connect the terminal lug on the stud attached to the captain stabilizer trim lightplate [61].

3) Install the nut [68] and lockwasher [69].

4) Install the captain stabilizer trim lightplate [61] on the control stand.

5) Install the screws [65] that attach the captain stabilizer trim lightplate [61] to the control stand.

(c) Do these steps to install the flap lever indicator lightplate [62]:

1) Install the washer [70].

2) Connect the terminal lug on the stud attached to the flap lever indicator lightplate [62].

3) Install the nut [68] and lockwasher [69].

4) Install the flap lever indicator lightplate [62] on the control stand.

5) Install the screws [66] that attach the flap lever indicator lightplate [62] to the control stand.

(d) Do these steps to install the speed brake lightplate [63]:

1) Install the washer [70].

2) Connect the terminal lug on the stud attached to the speed brake lightplate [63].

3) Install the nut [68] and lockwasher [69].

4) Install the speed brake lightplate [63] on the control stand.

5) Install the screws [67] that attach the speed brake lightplate [63] to the control stand.

SUBTASK 27-62-22-820-001

(15) Do this task: Speed Brake Control Lever Adjustment, TASK 27-62-00-820-801.

SUBTASK 27-62-22-480-001

(16) Remove the protective cover mat, STD-589, from the aft electronics panel P8.

SUBTASK 27-62-22-410-007

(17) Do this task: Captain and First Officer Seat Installation, TASK 25-11-01-400-801.

I. Speedbrake Lever Installation Test

SUBTASK 27-62-22-720-001

(1) Do this task: Inboard Ground Spoiler Functional Test, TASK 27-62-12-720-801.

J. Put the Airplane Back to Its Usual Condition

SUBTASK 27-62-22-860-002

(1) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	13	C00120	WEATHER RADAR RT

SUBTASK 27-62-22-860-003

(2) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-1

Row	Col	Number	Name
E	1	C00721	AUTOTHROTTLE DC 1

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CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	1	C00458	ENGINE 1 IGNITION RIGHT
A	3	C00153	ENGINE 1 IGNITION LEFT
A	4	C04008	ENGINE 1 ALTN PWR CHAN B
A	5	C04007	ENGINE 1 ALTN PWR CHAN A
B	1	C01316	ENGINE 1 START LEVER CHAN A
B	2	C01317	ENGINE 1 START LEVER CHAN B
B	3	C01312	ENGINE 1 RUN/PWR
B	5	C00276	ENGINE 1 THRUST REVERSER CONT
B	6	C01412	ENGINE 1 THRUST REVERSER INTLK
B	7	C01266	ENGINE 1 THRUST REVERSER SYNC LOCK
B	8	C01103	ENGINE 1 START VALVE

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	9	C00440	FLIGHT CONTROL AUTO SPEED BRAKE

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	5	C01320	ENGINE FUEL ENGINE 1 HPSOV CONT
E	6	C01395	ENGINE FUEL ENGINE 1 HPSOV IND

SUBTASK 27-62-22-860-004

(3) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	5	C01313	ENGINE 2 RUN/PWR
B	6	C01318	ENGINE 2 START LEVER CHAN A
B	7	C01319	ENGINE 2 START LEVER CHAN B
C	4	C00154	ENGINE 2 START VALVE
C	5	C01267	ENGINE 2 THRUST REVERSER SYNC LOCK
C	6	C01413	ENGINE 2 THRUST REVERSER INTLK
C	7	C00277	ENGINE 2 THRUST REVERSER CONT
D	4	C00459	ENGINE 2 IGNITION RIGHT
D	6	C00151	ENGINE 2 IGNITION LEFT
D	7	C04010	ENGINE 2 ALTN PWR CHAN B
D	8	C04009	ENGINE 2 ALTN PWR CHAN A

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	3	C01321	ENGINE FUEL ENGINE 2 HPSOV CONT
E	4	C01396	ENGINE FUEL ENGINE 2 HPSOV IND

SUBTASK 27-62-22-040-006

(4) Do this task: Trailing Edge Flap System Reactivation, TASK 27-51-00-440-801.

SUBTASK 27-62-22-860-005

(5) Remove the DO NOT OPERATE tags, STD-858, from the engine start panel.

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SUBTASK 27-62-22-410-008

(6) Close this access door:

Number	Name/Location
112A	Forward Access Door

——— END OF TASK ———

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AUTO SPEED BRAKE ACTUATOR - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the auto speed brake actuator

(2) An installation of the auto speed brake actuator.

TASK 27-62-31-000-801

2. Auto Speed Brake Actuator Removal

(Figure 401)

A. References

Reference	Title
27-62-00-800-802	Depressurization of the Speed Brake Hydraulic Systems A and B (P/B 201)

B. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

C. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Access Panels

Number	Name/Location
112A	Forward Access Door

E. Prepare for the Removal

SUBTASK 27-62-31-860-012

(1) Do this task: Depressurization of the Speed Brake Hydraulic Systems A and B, TASK 27-62-00-800-802.

SUBTASK 27-62-31-860-013

(2) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	9	C00440	FLIGHT CONTROL AUTO SPEED BRAKE
B	10	C00207	FLIGHT CONTROL STAB TRIM CONT
D	10	C00840	FLIGHT CONTROL STAB TRIM ACTUATOR

SUBTASK 27-62-31-860-014

(3) Put the speed brake lever [18] in the DOWN position.

SUBTASK 27-62-31-860-015

(4) Set the STAB TRIM PRI Switch, S272, on the aft area of the control stand, to the CUTOUT position.

(a) Attach a DO NOT OPERATE tag, STD-858, to the switch.

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SUBTASK 27-62-31-860-016

(5) Attach DO NOT OPERATE tags, STD-858, to the stabilizer trim wheels.

SUBTASK 27-62-31-010-002

(6) To get access the auto speed brake actuator assembly [16].

Open this access panel:

Number	Name/Location
112A	Forward Access Door

F. Auto Speed Brake Actuator Removal

SUBTASK 27-62-31-020-006

(1) Disconnect the electrical connector [17] from the actuator assembly [16].

SUBTASK 27-62-31-020-007

(2) Remove the bolt [5], washers [6], washer [4], and nut [8] to disconnect the bonding jumper [7].

SUBTASK 27-62-31-020-008

(3) Remove the pin [13], nut [12], washer [11], and bolt [10] that connect the actuator assembly [16] rod end to the brake housing clevis in the speed brake mechanical assembly [15].

SUBTASK 27-62-31-020-009

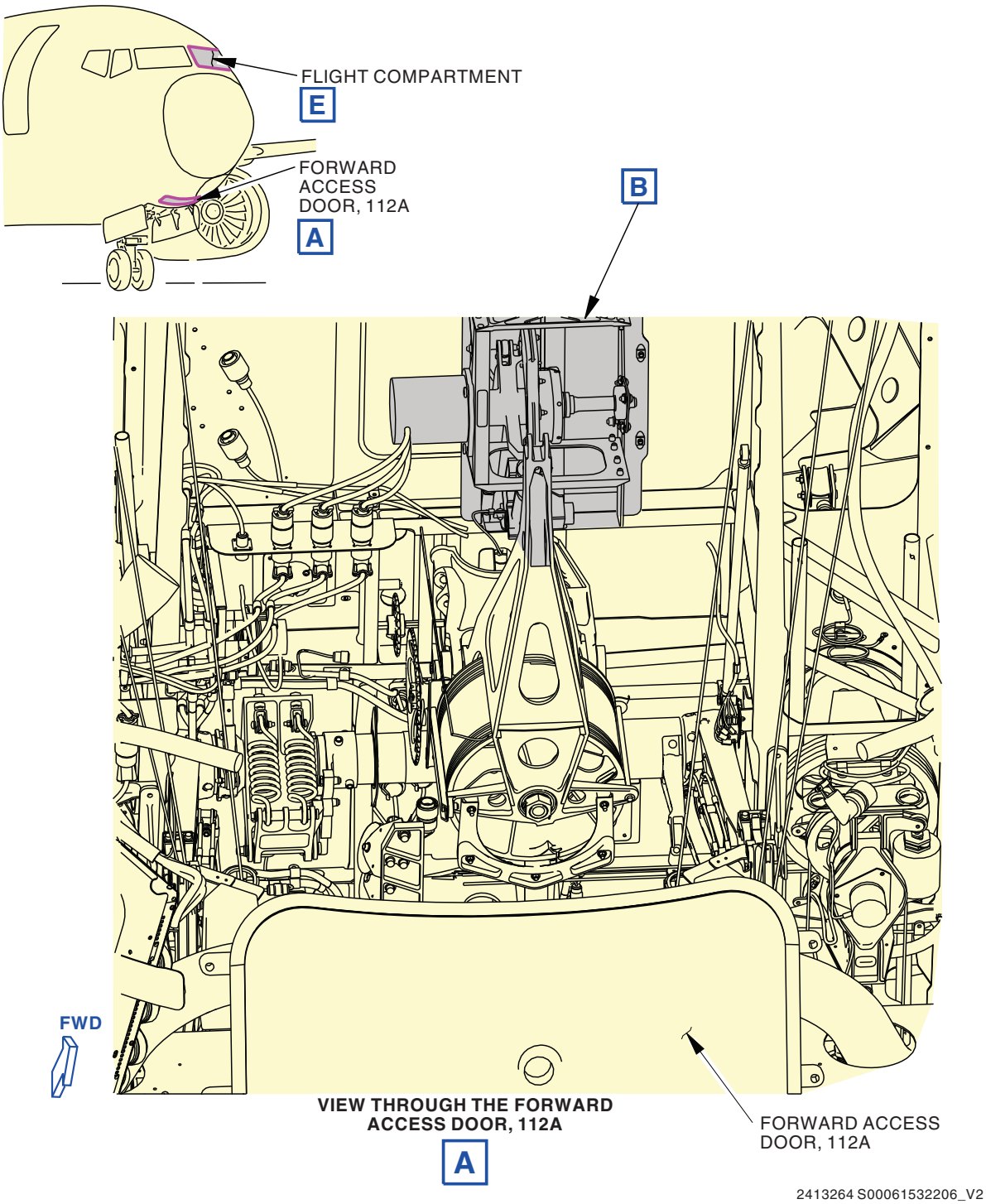
(4) Remove the lockwire or safety cable [3], washer [2], and bolt [1] that connect the actuator assembly [16] to the structure.

SUBTASK 27-62-31-020-010

(5) Remove the actuator assembly [16] from the airplane.

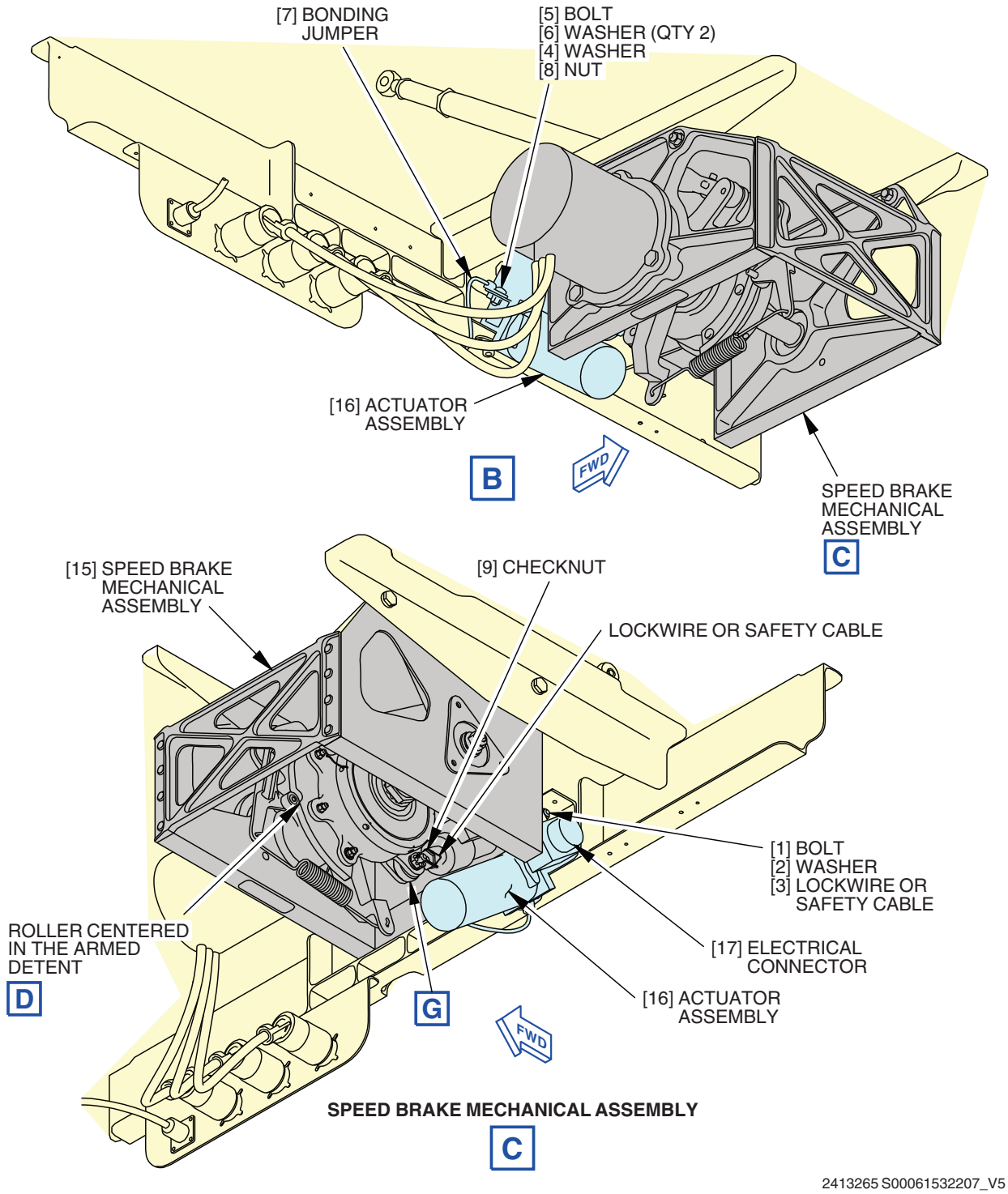
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



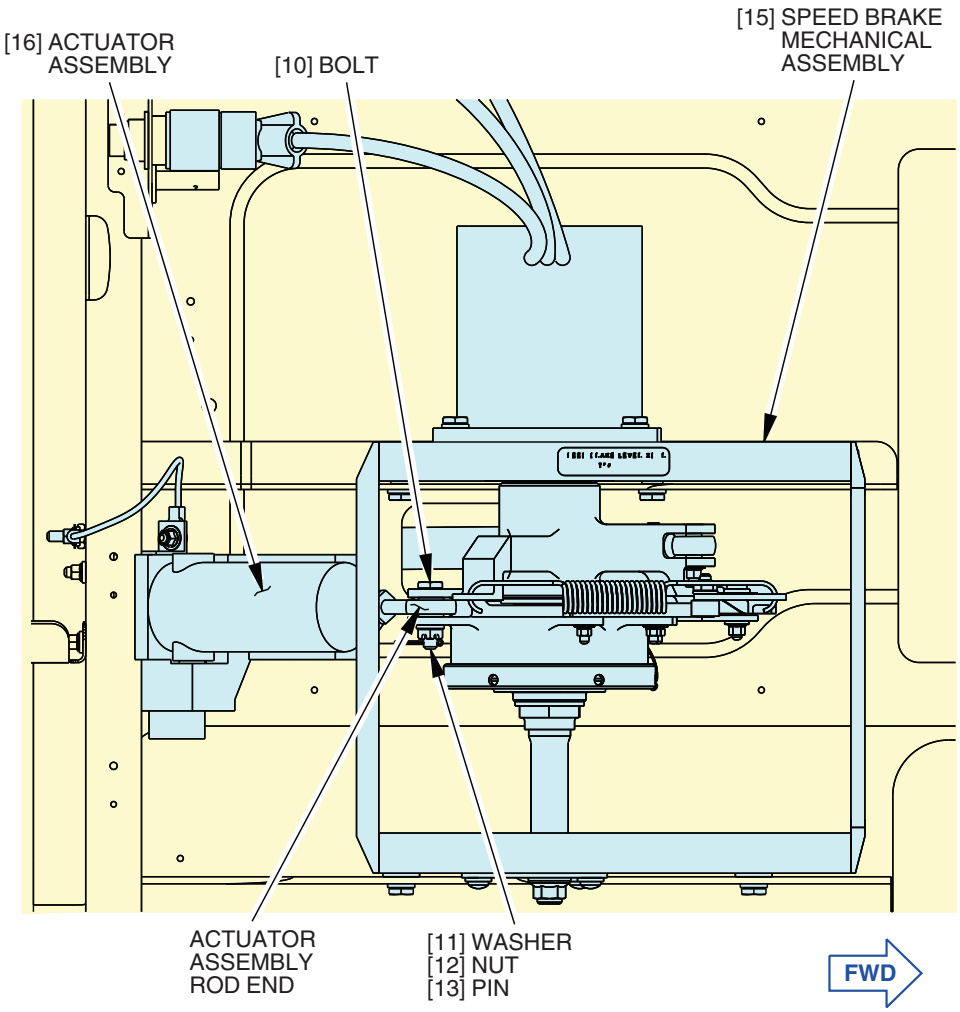
Auto Speedbrake Actuator Installation
Figure 401/27-62-31-990-802 (Sheet 1 of 6)

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Auto Speedbrake Actuator Installation
Figure 401/27-62-31-990-802 (Sheet 2 of 6)

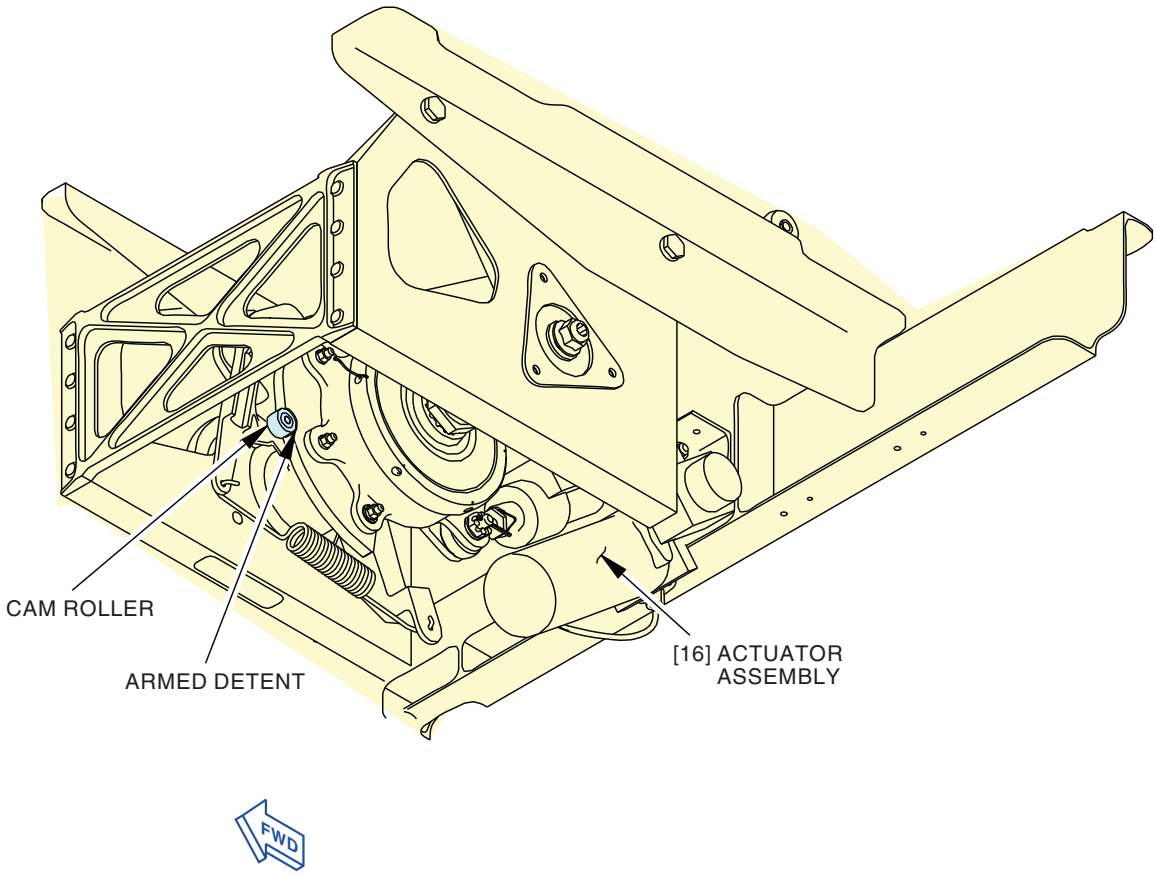
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Auto Speedbrake Actuator Installation
Figure 401/27-62-31-990-802 (Sheet 3 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



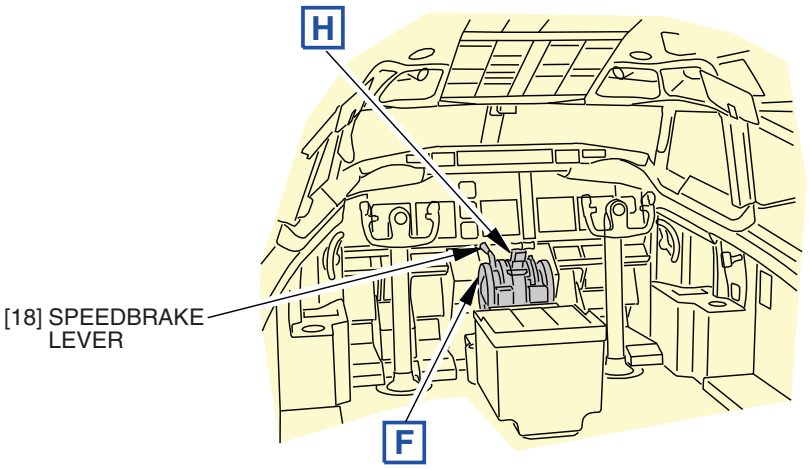
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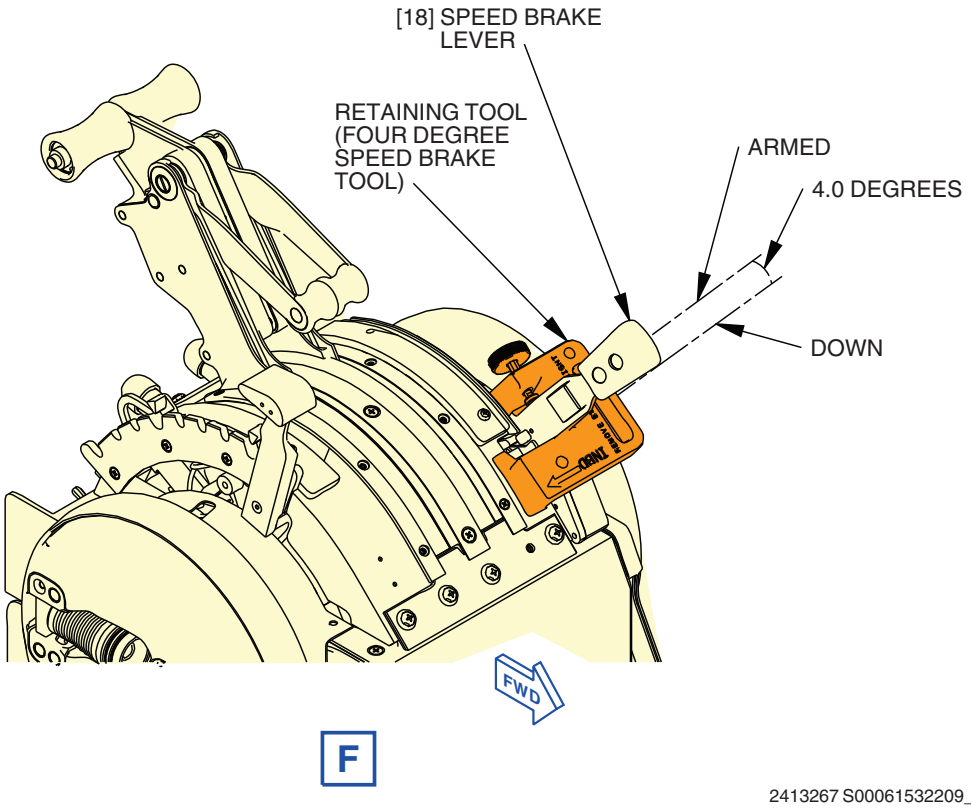
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Auto Speedbrake Actuator Installation
Figure 401/27-62-31-990-802 (Sheet 4 of 6)

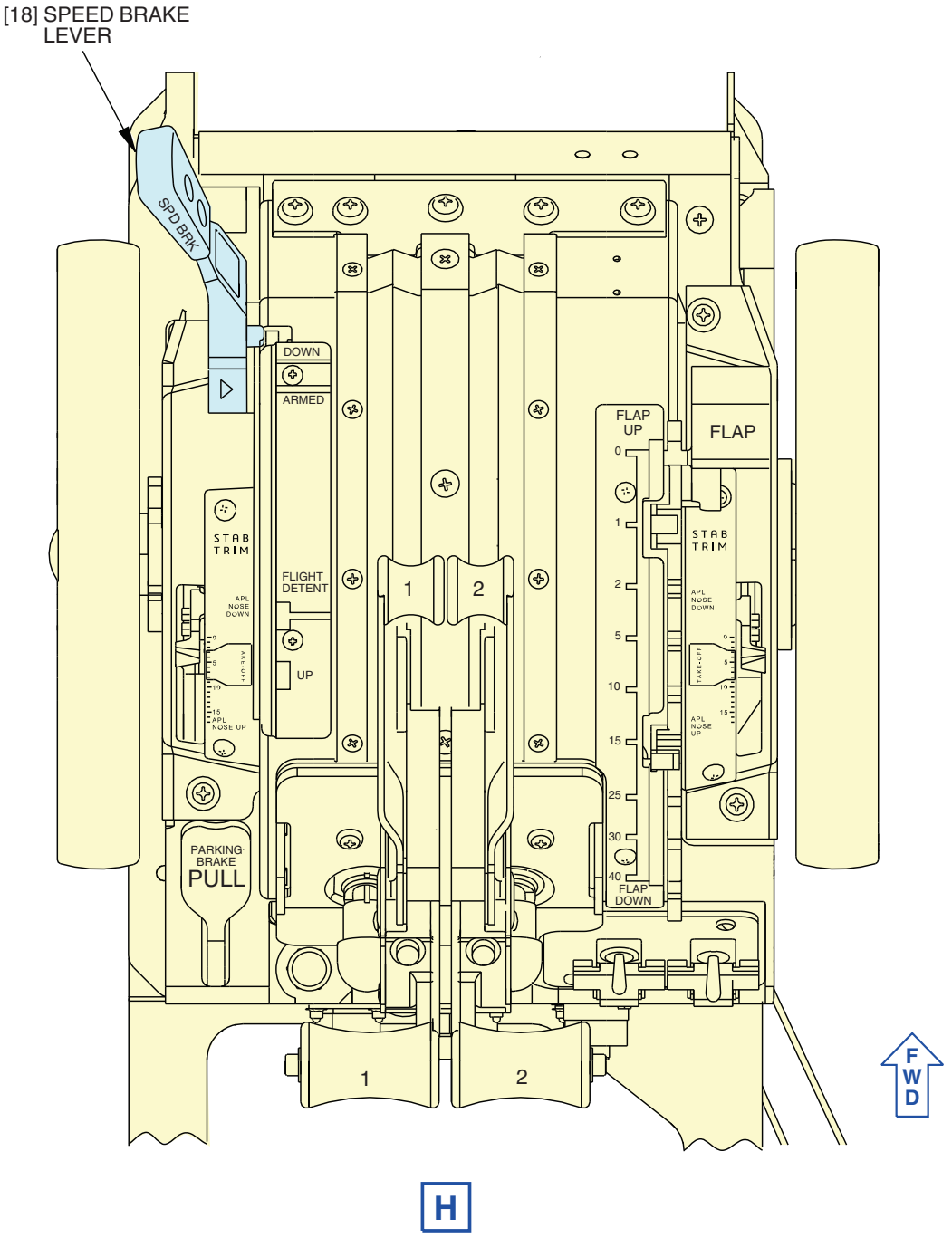
ECCN 9E991 BOEING PROPRIETARY - See title page for details





Auto Speedbrake Actuator Installation
Figure 401/27-62-31-990-802 (Sheet 5 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Auto Speedbrake Actuator Installation
Figure 401/27-62-31-990-802 (Sheet 6 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-62-31-400-801

3. Auto Speed Brake Actuator Installation

(Figure 401)

A. References

Reference	Title
27-62-00-820-801	Speed Brake Control Lever Adjustment (P/B 501)
27-62-00-820-804	Speed Brake Throttle Test (P/B 501)
27-62-00-840-802	Pressurization of the Speed Brake Hydraulic Systems A and B to its Usual Condition (P/B 201)
SWPM 20-20-00	ELECTRICAL BONDING PROCESSES

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-614	Bonding Meters - Non-Intrinsically Safe (For use in outside Class I, Divisions I & II non-hazardous locations. For hazardous locations, use COM-1550). Part #: 247000 Supplier: 00426 Part #: 620LK Supplier: 1CRL2 Part #: BLR-0003-XX Supplier: KC432 Part #: BT51 Supplier: 00426 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659 Opt Part #: 247001 Supplier: 00426
SPL-14768	Rigging Lock - Four Degree Speedbrake Lever Part #: C27095-1 Supplier: 81205
STD-686	Power Supply - 28 VDC

C. Consumable Materials

Reference	Description	Specification
B00184	Solvent - Presealing, Cleaning Solvent	BMS11-7
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
13	Pin	27-62-31-01-115	PTF ALL
16	Actuator assembly	27-62-31-01-145	PTF ALL

E. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
211	Flight Compartment - Left
212	Flight Compartment - Right

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Access Panels

Number	Name/Location
112A	Forward Access Door

G. Auto Speed Brake Actuator Installation

SUBTASK 27-62-31-860-006

(1) Move the speed brake lever [18] to the ARMED detent position (View H, Figure 401).

(a) Make sure that the cam roller is centered in the ARMED detent.

SUBTASK 27-62-31-480-001

(2) Install the rigging lock, SPL-14768, to hold the speed brake lever [18] at 4 degrees aft of the DOWN position.

SUBTASK 27-62-31-420-001

(3) Install the auto speed brake actuator assembly [16]:

(a) Place the auto speed brake actuator assembly [16] in position.

(b) Install the bolt [1], washer [2], and lockwire or safety cable [3] that attaches the actuator assembly [16] to the structure.

(c) Do these steps to put the actuator assembly [16] in the retracted position:

1) Use 28 VDC power supply, STD-686, or equivalent, to apply 28 VDC, positive to Pin 3 and a ground to Pin 1 in the socket used for electrical connector [17].

NOTE: The recommended electrical current limit value is 10A.

a) Use pin 1 as the ground.

NOTE: The pins on the actuator have numbers.

(d) Adjust the rod end fitting on the actuator assembly [16].

1) Remove the lockwire or safety cable on checknut [9].

2) Loosen the checknut [9] on the actuator assembly [16] rod end.

3) Adjust the actuator assembly [16] rod end until it aligns with the brake housing clevis in the speed brake mechanical assembly [15] to insert the bolt [10] freely.

4) Make sure that the cam roller is centered in the ARMED detent.

WARNING MAKE SURE THAT THE BOLT HEAD POSITION IS CORRECT. THE INCORRECT POSITION OF THE BOLT CAN PREVENT THE MOVEMENT OF THE SPOILERS AND SPEEDBRAKE LEVER. IF YOU DO NOT OBEY, INJURY TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

5) Install the bolt [10], washer [11], nut [12], and new pin [13] to attach the actuator assembly [16] rod end.

a) Make sure that the head of the bolt [10] is to the inboard side, towards the speed brake lever position transducer (View G, Figure 401).

6) Tighten the checknut [9] on the actuator assembly [16] rod end to 45 in-lb (5.1 N·m) - 55 in-lb (6.2 N·m).

7) Install the lockwire or safety cable on checknut [9].

SUBTASK 27-62-31-080-001

(4) Remove the rigging lock, SPL-14768, from the speed brake lever [18].

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SUBTASK 27-62-31-980-001

(5) Manually move the speed brake lever [18] through the full travel to make sure it moves freely.

SUBTASK 27-62-31-420-002

(6) Install the bonding jumper [7], do these steps:

(a) Use Cleaning Procedure 1 to manually clean the mating surfaces of the aluminum bracket (SWPM 20-20-00).

(b) Use cotton wiper, G00034, moist in solvent, B00184, to clean the mating surfaces of the aluminum bracket, bolt [5], washers [6], washer [4], nut [8], and bonding jumper [7].

(c) Install the bolt [5], washers [6], washer [4], and nut [8] to attach the bonding jumper [7].

1) Tighten the nut [8] to 28 in-lb (3.2 N·m) - 35 in-lb (4.0 N·m).

(d) Measure the electrical resistance between the actuator assembly [16] and structure (SWPM 20-20-00, Paragraph 21).

1) Use a non-intrinsically safe bonding meter, COM-614.

2) Make sure that the resistance is 0.0025 ohm (2.5 milliohms) or less.

SUBTASK 27-62-31-080-002

(7) Disconnect the 28 VDC power supply, STD-686, from the actuator assembly [16].

SUBTASK 27-62-31-420-003

(8) Install the electrical connector [17].

H. Auto Speed Brake Actuator Installation Test

SUBTASK 27-62-31-860-007

(1) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	9	C00440	FLIGHT CONTROL AUTO SPEED BRAKE
B	10	C00207	FLIGHT CONTROL STAB TRIM CONT
D	10	C00840	FLIGHT CONTROL STAB TRIM ACTUATOR

SUBTASK 27-62-31-860-008

(2) Do this task: Pressurization of the Speed Brake Hydraulic Systems A and B to its Usual Condition, TASK 27-62-00-840-802.

SUBTASK 27-62-31-860-017

(3) Do this task: Speed Brake Control Lever Adjustment, TASK 27-62-00-820-801.

SUBTASK 27-62-31-720-001

(4) Do this task: Speed Brake Throttle Test, TASK 27-62-00-820-804.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 27-62-31-860-009

(1) Set the STAB TRIM PRI Switch, S272, on the aft of the control stand to the NORMAL position.

(a) Remove the DO NOT OPERATE tag from the switch.

SUBTASK 27-62-31-860-010

(2) Remove the DO NOT OPERATE tags from the stabilizer trim wheels.

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SUBTASK 27-62-31-410-001

(3) Close this access panel:

Number	Name/Location
112A	Forward Access Door

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SPEED BRAKE REFUSED TAKEOFF (RTO) SWITCH - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the speed brake refused takeoff (RTO) switch S650.

(2) An installation of the speed brake refused takeoff (RTO) switch S650.

B. The speed brake refused takeoff switch (RTO) is installed in the left forward part of the control stand.

C. The speed brake refused takeoff switch (RTO) is operated by the movement of the reverse thrust levers in a reverse thrust direction. The RTO switch operates with an arm which lifts the speed brake lever out of the down detent to permit automatic extension of the speed brakes.

D. In this procedure the speed brake refused takeoff (RTO) switch is referred to as the "RTO switch".

TASK 27-62-32-000-801

2. Speed Brake Refused Takeoff (RTO) Switch Removal

(Figure 401)

A. Location Zones

Zone	Area
211	Flight Compartment - Left

B. Prepare for the Removal

SUBTASK 27-62-32-860-001

(1) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	9	C00440	FLIGHT CONTROL AUTO SPEED BRAKE

C. Speed Brake Refused Takeoff (RTO) Switch Removal

SUBTASK 27-62-32-020-001

(1) Remove screws [1], washers [2] and the cover plate [3] on the control stand to get access to the RTO switch [13].

SUBTASK 27-62-32-020-003

(2) Remove the screw [9] and screw [10] that attach the RTO switch [13] to the structure.

SUBTASK 27-62-32-020-002

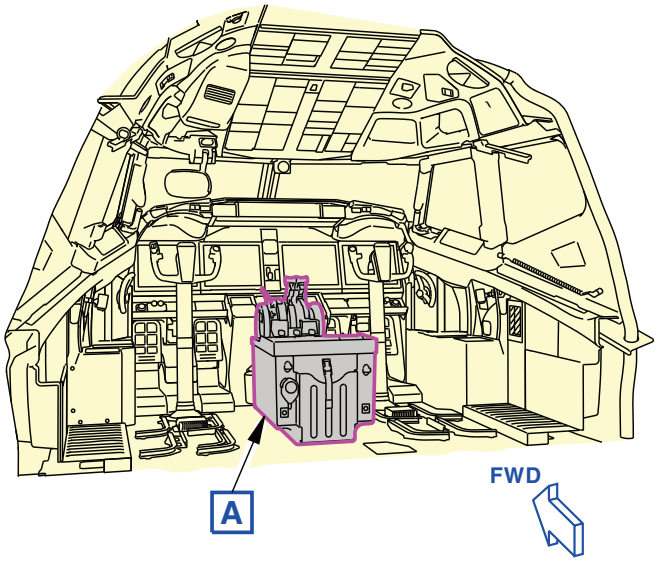
(3) Remove the screws [6] and washers [7] to disconnect the electrical wires [8] from the RTO switch [13].

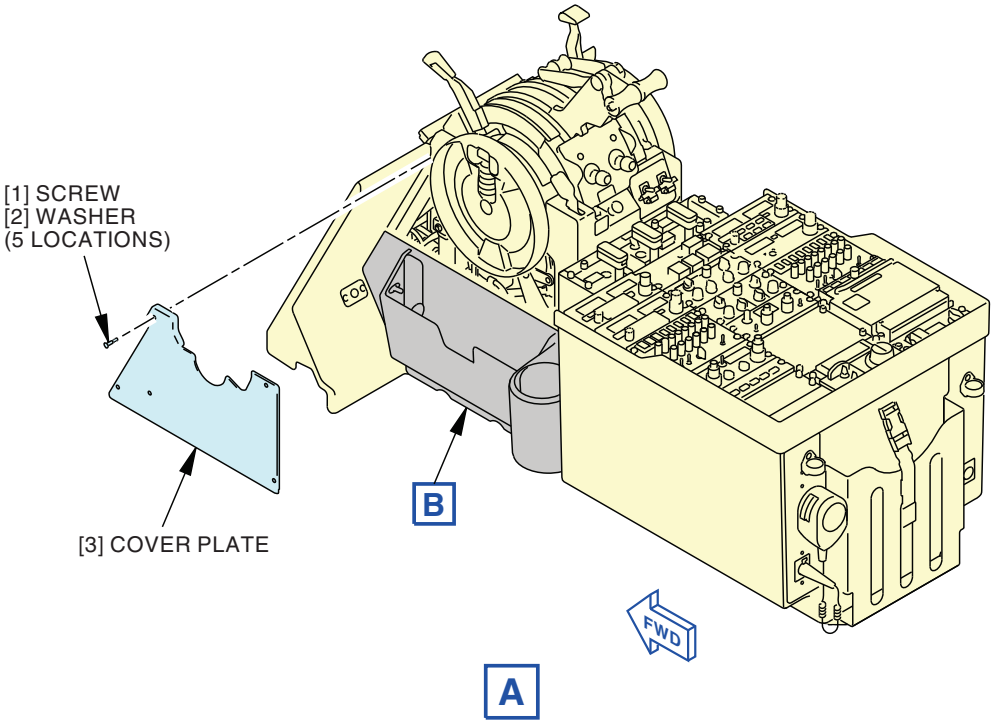
SUBTASK 27-62-32-020-004

(4) Remove the RTO switch [13].

——— END OF TASK ———

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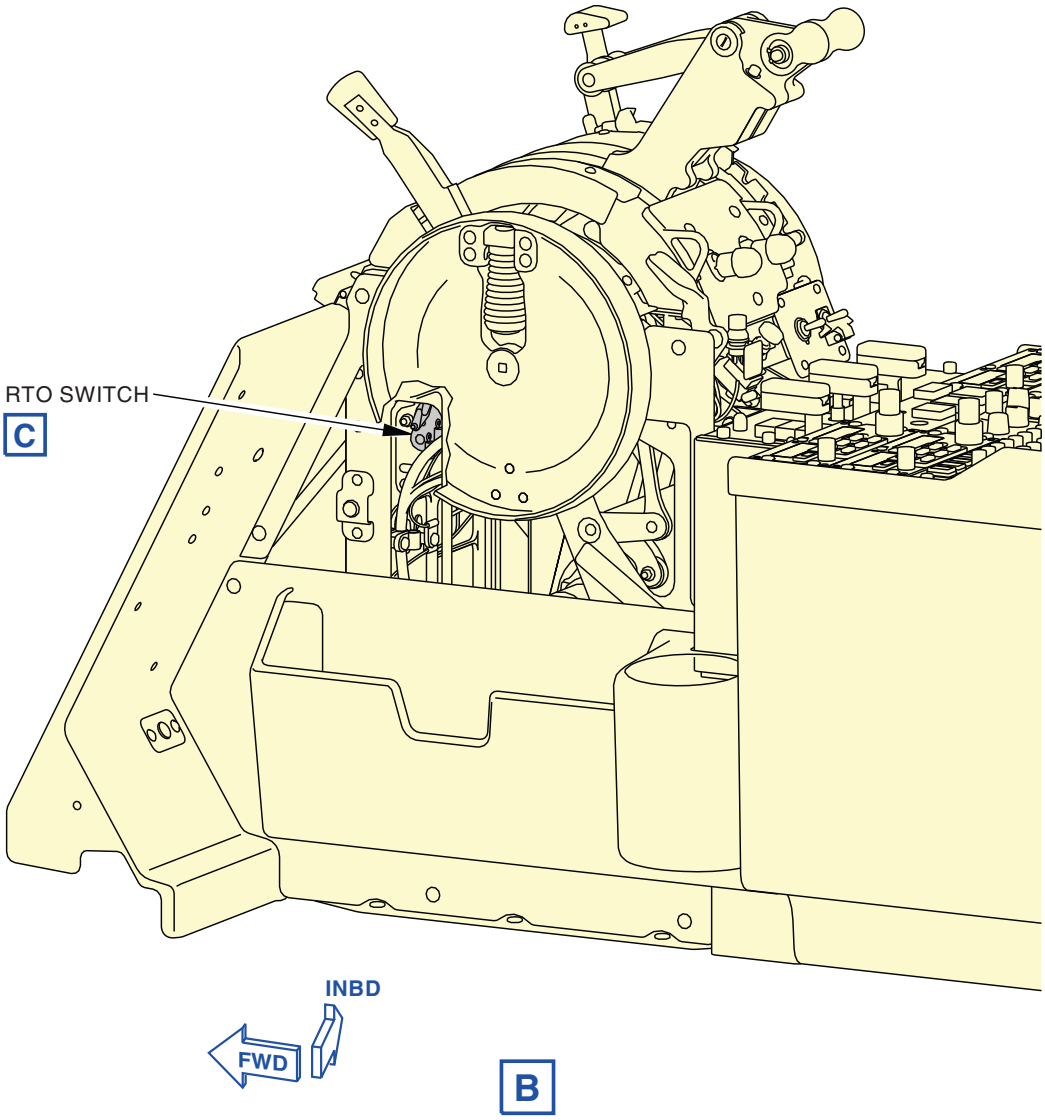




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Speed Brake Refused Takeoff (RTO) Switch Installation
Figure 401/27-62-32-990-801 (Sheet 1 of 3)

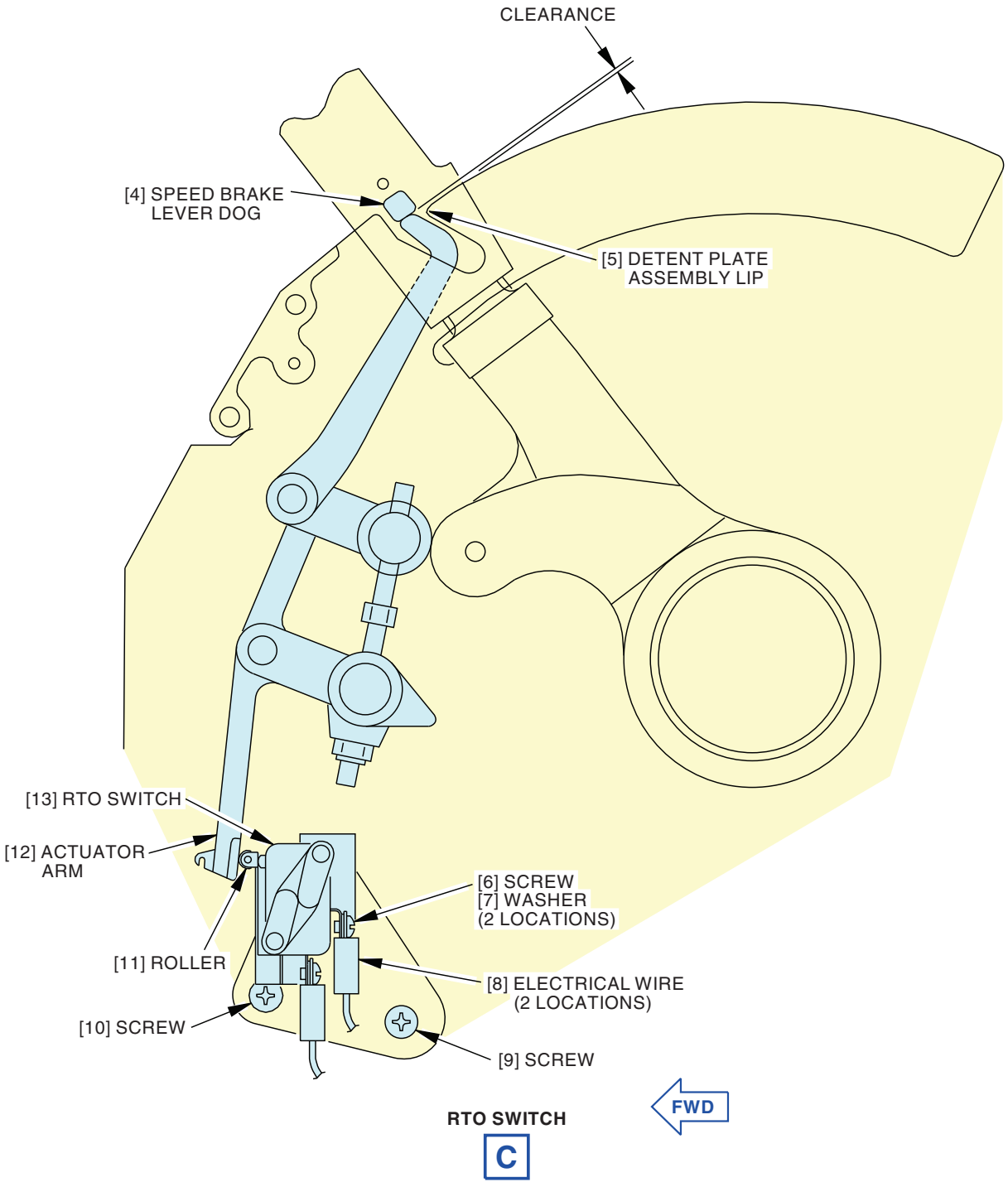
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Speed Brake Refused Takeoff (RTO) Switch Installation
Figure 401/27-62-32-990-801 (Sheet 2 of 3)

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Speed Brake Refused Takeoff (RTO) Switch Installation
Figure 401/27-62-32-990-801 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-62-32-400-801

3. Speed Brake Refused Takeoff (RTO) Switch Installation

(Figure 401)

A. References

Reference	Title
27-62-00-820-807	Speed Brake RTO Test (P/B 501)
27-62-32-820-801	Speed Brake Refused Takeoff (RTO) Switch Adjustment (P/B 501)

B. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
13	RTO switch	27-62-32-01-005	PTF ALL

C. Location Zones

Zone	Area
211	Flight Compartment - Left

D. Speed Brake Refused Takeoff (RTO) Switch Installation

SUBTASK 27-62-32-420-001

(1) Install the RTO switch [13]:

(a) Put the RTO switch [13] in its position on the structure.

(b) Install the screw [9] and screw [10] to attach the RTO switch [13] to the structure.

(c) Install the screws [6] and washers [7] to connect the electrical wires [8] to the RTO switch [13].

SUBTASK 27-62-32-860-002

(2) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	9	C00440	FLIGHT CONTROL AUTO SPEED BRAKE

SUBTASK 27-62-32-820-001

(3) Do this task: Speed Brake Refused Takeoff (RTO) Switch Adjustment, TASK 27-62-32-820-801.

SUBTASK 27-62-32-420-003

(4) Install the screws [1] and washers [2] to attach the cover plate [3] on the control stand.

E. Speed Brake Refused Takeoff (RTO) Switch Installation Test

SUBTASK 27-62-32-700-001

(1) Do this task: Speed Brake RTO Test, TASK 27-62-00-820-807.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SPEED BRAKE REFUSED TAKEOFF (RTO) SWITCH - ADJUSTMENT/TEST

1. General

A. This procedure has this task:

(1) An adjustment of the speed brake refused takeoff (RTO) switch S650.

TASK 27-62-32-820-801

2. Speed Brake Refused Takeoff (RTO) Switch Adjustment

(Figure 501)

A. Location Zones

Zone	Area
211	Flight Compartment - Left

B. Speed Brake Refused Takeoff (RTO) Switch Adjustment

SUBTASK 27-62-32-820-002

(1) Adjust the RTO switch [13]:

(a) Make sure that the thrust levers are at idle and the reverse thrust levers are down and stowed.

(b) Make sure that the speed brake lever is in the DOWN position.

(c) Loosen screw [9] and screw [10].

(d) Turn the RTO switch [13] as far aft as possible.

(e) Make sure that there is no continuity between pin 8 and 9 of the D5148J connector.

(f) Move the left reverse thrust lever aft until there is a 0.0100 in. (0.2540 mm) to 0.0300 in. (0.7620 mm) clearance between the speed brake lever dog [4] and the detent plate assembly lip [5].

(g) Turn the RTO switch [13] until the roller [11] touches the actuator arm [12] and there is continuity between pins 8 and 9 of the D5148J connector.

(h) Tighten the screw [9] and screw [10].

(i) Slowly move the left thrust lever back to the down and stowed position.

1) Make sure that there is no continuity between pins 8 and 9.

(j) Slowly move the left thrust lever aft until there is continuity between pins 8 and 9.

1) Make sure that the clearance between the speed brake lever dog [4] and the detent plate assembly lip [5] is 0.0100 in. (0.2540 mm) to 0.0300 in. (0.7620 mm).

(k) Slowly move the left reverse thrust lever forward until there is no continuity between pins 8 and 9.

1) Make sure that the speed brake lever dog [4] is 0.0100 in. (0.2540 mm) above the detent plate assembly lip [5].

(l) Move the left reverse thrust lever to the down and stowed position.

1) Make sure that there is no continuity between pins 8 and 9.

(m) Slowly move and the right reverse thrust lever aft until there is continuity between pins 8 and 9.

1) Make sure that the speed brake lever dog [4] is 0.0100 in. (0.2540 mm) to 0.0300 in. (0.7620 mm) above the detent plate assembly lip [5].

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(n) Slowly move the right reverse thrust lever forward until there is no continuity between pins 8 and 9.

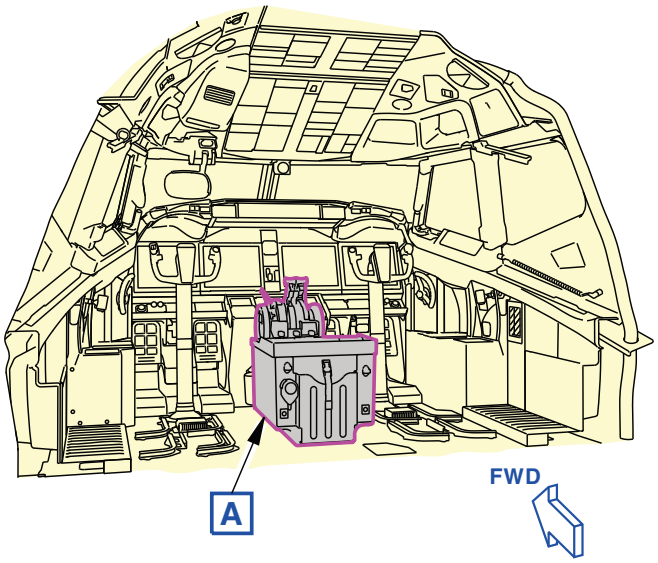
1) Make sure that the speed brake lever dog [4] is 0.0100 in. (0.2540 mm) above the detent plate assembly lip [5].

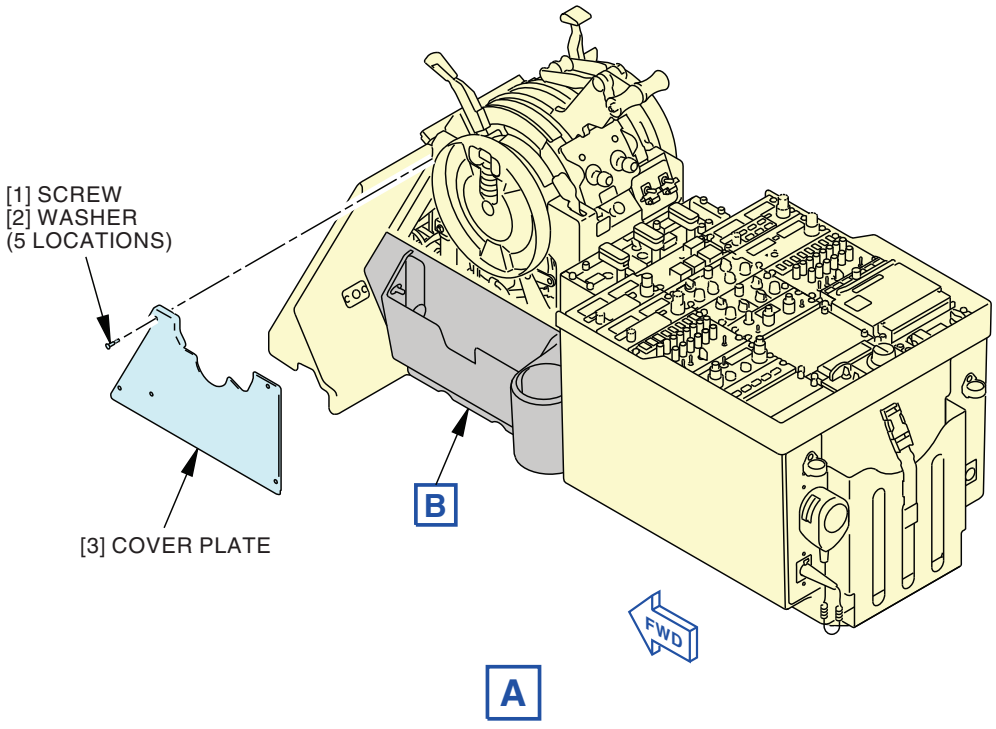
(o) Move the right reverse thrust lever to the down and stowed position.

1) Make sure that there is no continuity between pins 8 and 9.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

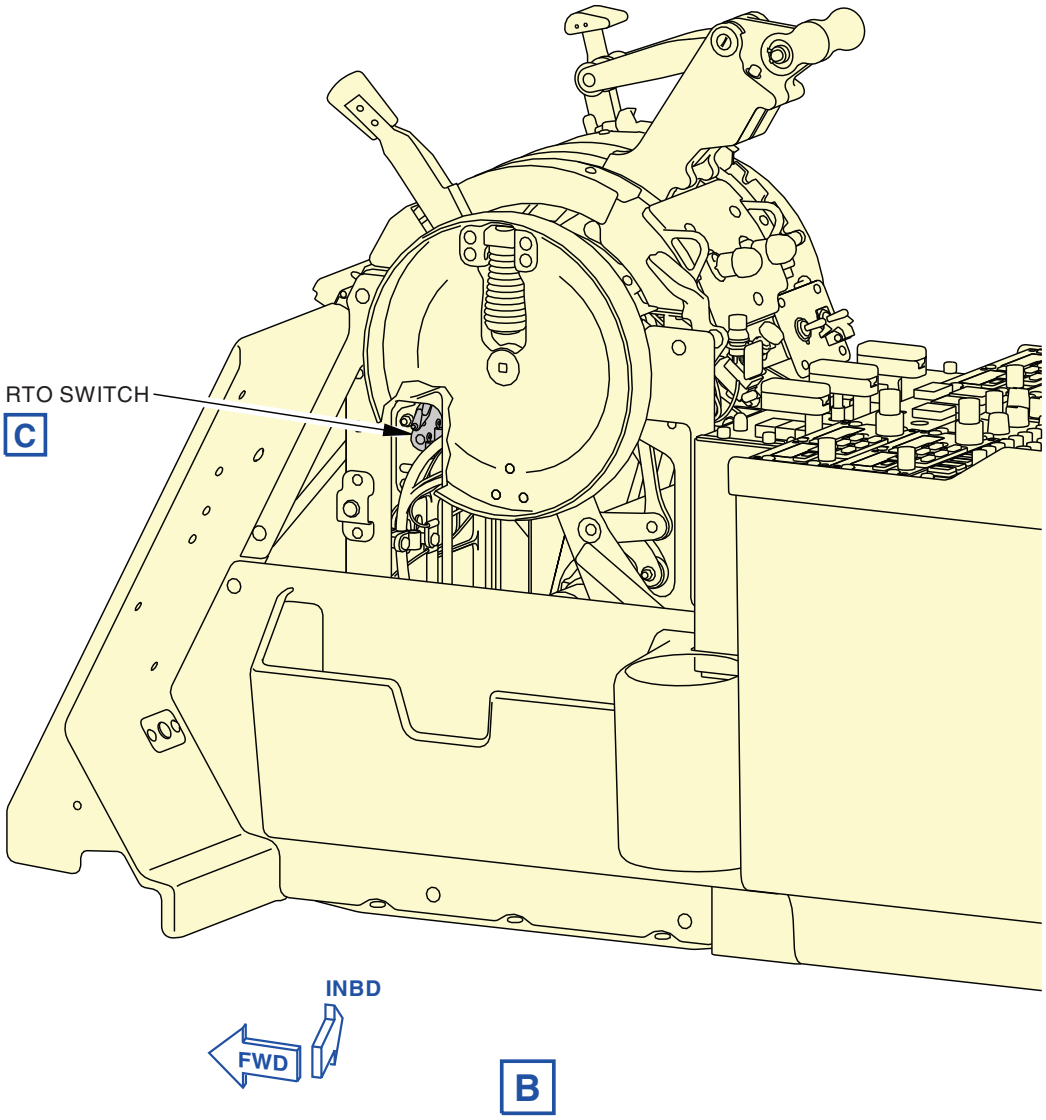




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Speed Brake Refused Takeoff (RTO) Switch Installation
Figure 501/27-62-32-990-802 (Sheet 1 of 3)

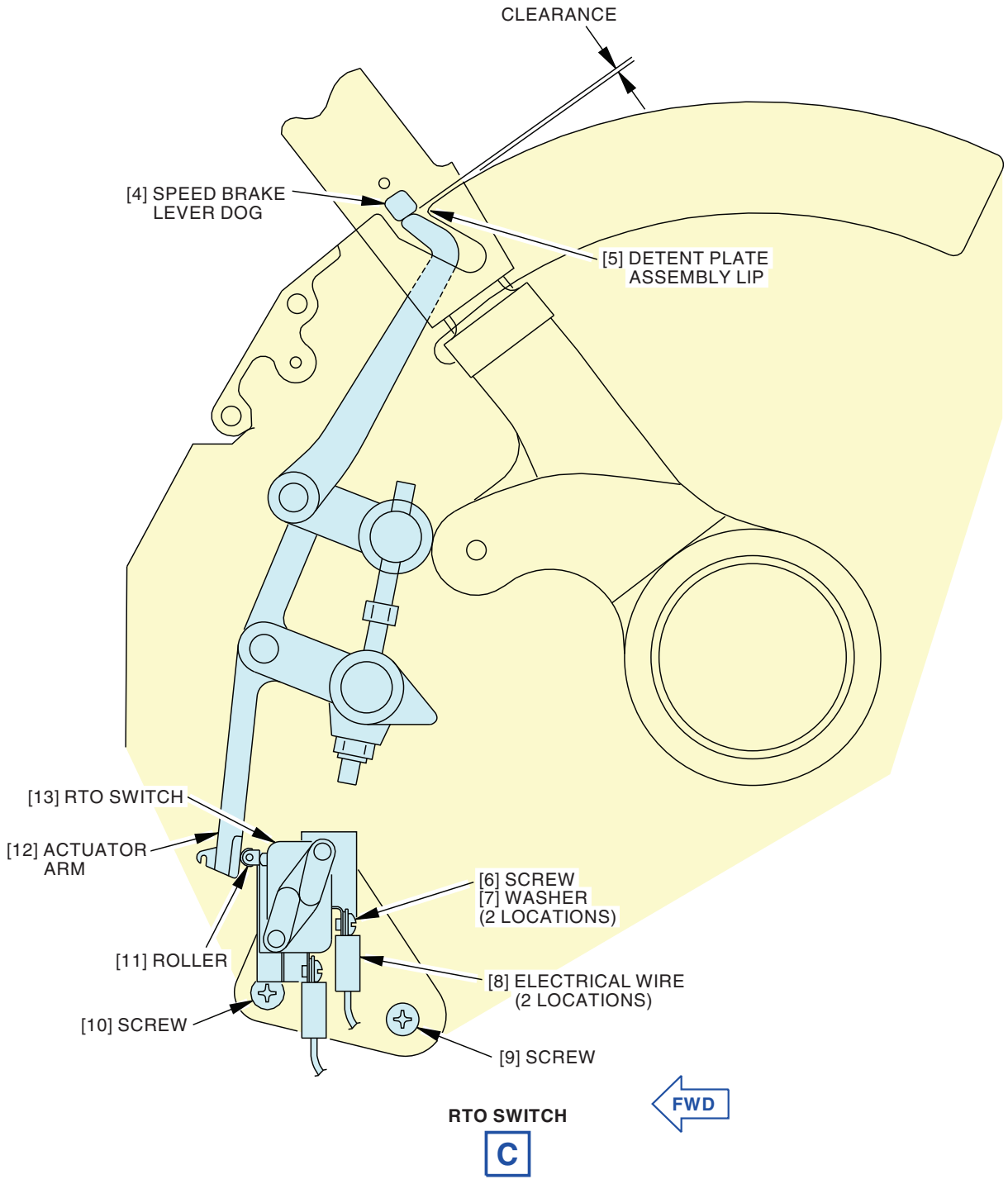
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2413270 S00061532216_V2

Speed Brake Refused Takeoff (RTO) Switch Installation
Figure 501/27-62-32-990-802 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Speed Brake Refused Takeoff (RTO) Switch Installation
Figure 501/27-62-32-990-802 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SPEEDBRAKE LEVER TRANSDUCER - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Speedbrake lever transducer removal.

(2) Speedbrake lever transducer installation.

(3) Speedbrake lever transducer adjustment.

TASK 27-62-37-000-801

2. Speedbrake Lever Transducer Removal

(Figure 401)

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
20-40-12-400-803	Conductive Dust Cap and Connector Cover Installation (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Access Panels

Number	Name/Location
112A	Forward Access Door

D. Prepare for the Removal

SUBTASK 27-62-37-860-001

WARNING KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT SPOILERS. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU OPEN/CLOSE THESE CIRCUIT BREAKERS. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
C	10	C04032	SPOILER CONTROL ELEC PWR A
C	12	C04033	SPOILER CONTROL ELEC PWR B

SUBTASK 27-62-37-860-002

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(2) If the downlock pins are not installed on all landing gears, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-62-37-010-001

(3) Open this access door:

Number	Name/Location
112A	Forward Access Door

E. Speedbrake Lever Transducer Removal

SUBTASK 27-62-37-020-001

(1) Do these steps to remove the speedbrake lever transducer [1]:

(a) Disconnect the P1 connector, P2 connector and P3 connector.

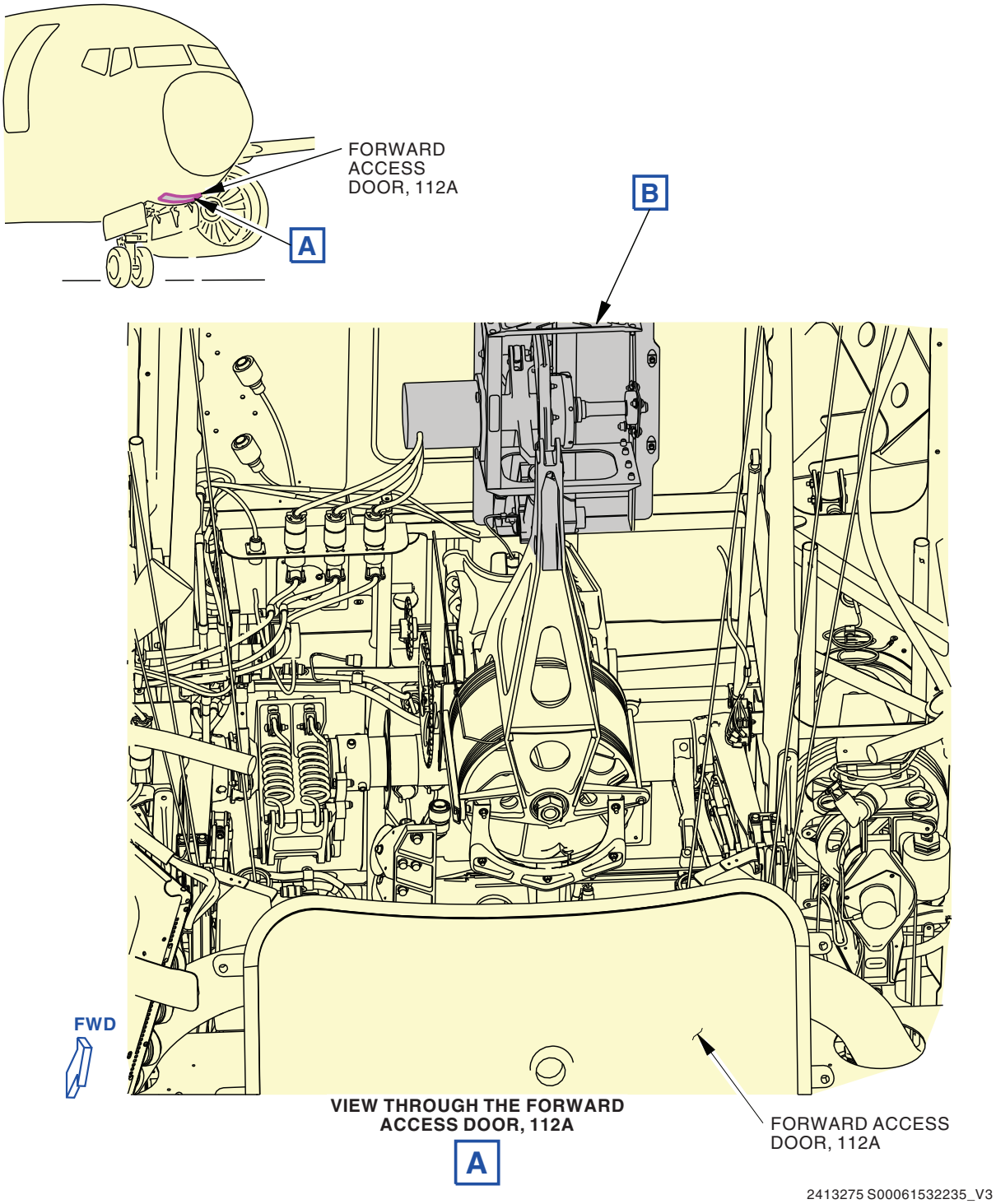
(b) Put dustcaps on the connectors and receptacles (TASK 20-40-12-400-803).

(c) Remove the bolts [2] and washers [3].

(d) Remove the speedbrake lever transducer [1].

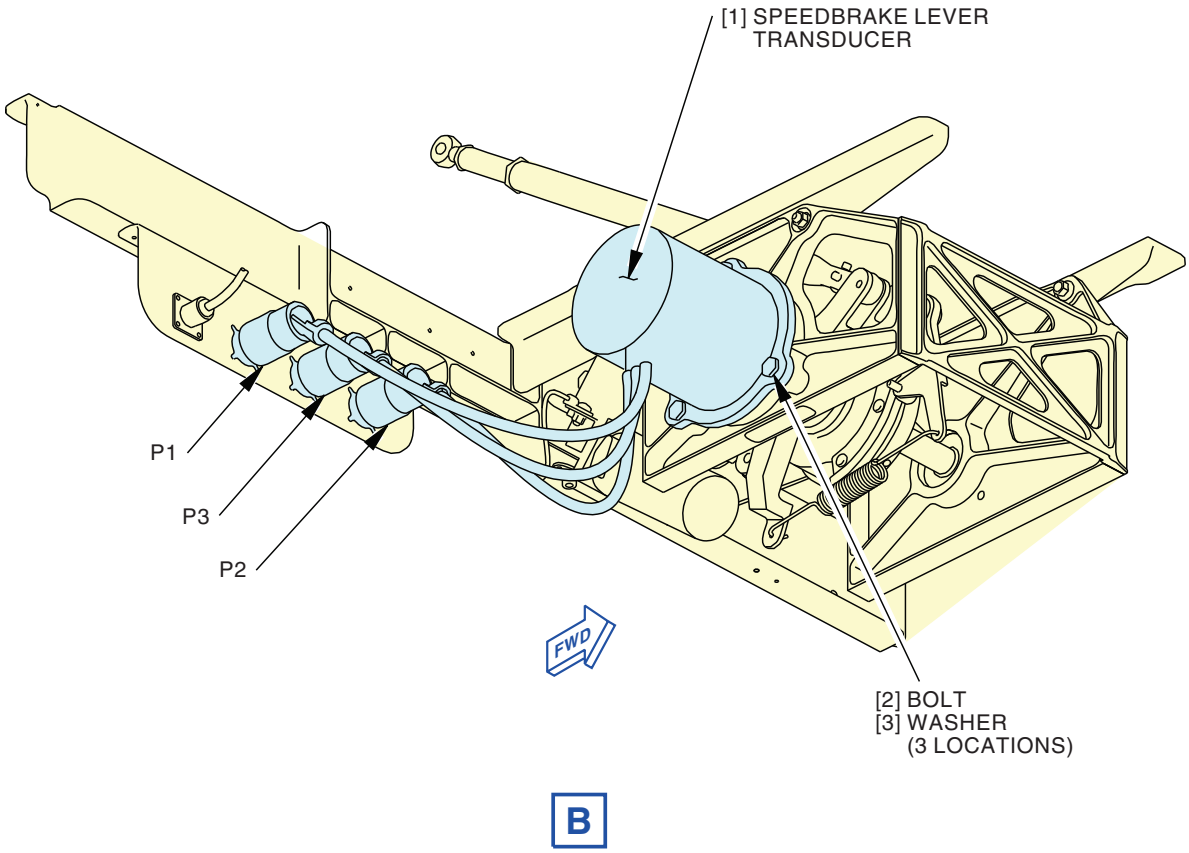
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Speedbrake Lever Transducer Installation
Figure 401/27-62-37-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Speedbrake Lever Transducer Installation
Figure 401/27-62-37-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-62-37-400-801

3. Speedbrake Lever Transducer Installation

(Figure 401, Figure 402)

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
20-40-12-000-803	Conductive Dust Cap and Connector Cover Removal (P/B 201)
20-50-11-910-801	Standard Torque Values (P/B 201)

B. Consumable Materials

Reference	Description	Specification
D00633	Grease - Aircraft General Purpose	BMS3-33

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Speedbrake lever transducer	27-62-37-01-010	PTF ALL

D. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
211	Flight Compartment - Left
212	Flight Compartment - Right

E. Access Panels

Number	Name/Location
112A	Forward Access Door

F. Speedbrake Lever Transducer Installation

SUBTASK 27-62-37-420-001

(1) Do these steps to install the speedbrake lever transducer [1]:

(a) Put the speedbrake lever [4] in the DOWN and LOCKED position.

(b) Put a thin film of grease, D00633, on the splines of the speedbrake lever transducer [1].

(c) Align the spline indexes and put the speedbrake lever transducer [1] in position.

(d) Install the bolts [2] and washers [3] that attach the speedbrake lever transducer [1] (TASK 20-50-11-910-801).

(e) Remove the dustcaps from the connectors and receptacles (TASK 20-40-12-000-803).

(f) Connect the P1 connector, P2 connector and P3 connector.

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G. Speedbrake Lever Transducer Installation Test

SUBTASK 27-62-37-860-013

WARNING KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT SPOILERS. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU OPEN/CLOSE THESE CIRCUIT BREAKERS. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
C	10	C04032	SPOILER CONTROL ELEC PWR A
C	12	C04033	SPOILER CONTROL ELEC PWR B

SUBTASK 27-62-37-820-001

(2) Do this task: Speedbrake Lever Transducer Adjustment, TASK 27-62-37-200-801.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 27-62-37-410-001

(1) If it is open, close this access panel:

Number	Name/Location
112A	Forward Access Door

——— END OF TASK ———

TASK 27-62-37-200-801

4. Speedbrake Lever Transducer Adjustment

(Figure 402)

A. References

Reference	Title
27-61-45-730-802	Speedbrake Lever Transducer Test (P/B 501)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
34-21-00-820-801	Air Data Inertial Reference System - Alignment from the FMC CDU (P/B 201)
34-21-00-820-802	Air Data Inertial Reference System - Alignment from the ISDU (P/B 201)
34-21-00-820-803	Air Data Inertial Reference System - Automatic Alignment from the IRS MSU (P/B 201)
46-13-02-700-805	Existing Faults Check (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-14768	Rigging Lock - Four Degree Speedbrake Lever Part #: C27095-1 Supplier: 81205

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Access Panels

Number	Name/Location
112A	Forward Access Door

E. Prepare for the Speedbrake Lever Transducer Adjustment

SUBTASK 27-62-37-860-006

(1) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	9	C00440	FLIGHT CONTROL AUTO SPEED BRAKE

SUBTASK 27-62-37-860-007

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(2) If the downlock pins are not installed on all landing gears, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 27-62-37-010-002

(3) Open this access door:

Number	Name/Location
112A	Forward Access Door

F. Speedbrake Lever Transducer Adjustment

SUBTASK 27-62-37-480-001

(1) Do these steps to retract the actuator assembly:

(a) Place the Speedbrake Lever in the DOWN and LOCKED position.

(b) Make sure that the SPEEDBRAKES EXTENDED light does not come on.

(c) Make sure that the SPEEDBRAKE DO NOT ARM light does not come on.

(d) Move the speedbrake lever to the ARMED position.

NOTE: The ARMED position is approximately 4 degrees aft of the DOWN position.

1) Moving the speedbrake lever without hydraulic power can latch the GROUND SPOILER CONTROL Status Message. Do this task if the Status Message occurs: Existing Faults Check, TASK 46-13-02-700-805.

SUBTASK 27-62-37-480-002

(2) Install the rigging lock, SPL-14768, to hold the speedbrake lever at 4 degrees aft of the DOWN position.

SUBTASK 27-62-37-200-001

(3) Make sure that the roller [10] is centered in the ARMED detent (Figure 402).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-62-37-820-003

(4) Do these steps to adjust the lower control rod [8] and the three Rotary Variable Displacement Transducer (RVDT) channels:

(a) Use the SEL knob on the Multi Function Panel to make the following selections:

1) MAINT DATA PGS

FLIGHT CONTROL

2) Monitor the RVDT A, B, C readings of the SCE SPDBRK LVR.

3) Adjust the lower control rod [8] so that all three RVDT channels are within 3.8 ± 0.5 degrees:

a) Remove the screws [5], washers [6] and lower cover plate [7] to get access to the lower control rod [8].

b) Loosen the checknut [9] and adjust the length of the lower control rod [8] so that all three RVDT channels are within 3.8 ± 0.5 degrees.

c) Tighten the checknut [9] to 45 in-lb (5 N·m) - 50 in-lb (6 N·m).

4) Make sure that the roller [10] is centered in the ARMED detent.

5) Remove the rigging lock, SPL-14768, from the speedbrake lever [4].

6) Move the speedbrake lever [4] to the Down and locked position.

7) Make sure that the roller [10] is not in the ARMED detent (Figure 402).

8) Make sure that the RVDT A, RVDT B, and RVDT C channels are within 0.0 ± 1.0 degrees.

9) Install the screws [5], washers [6] and lower cover plate [7].

G. Speedbrake Lever Transducer Adjustment Test

SUBTASK 27-62-37-820-004

(1) Do this task to align the IRU: Air Data Inertial Reference System - Alignment from the ISDU, TASK 34-21-00-820-802 or Air Data Inertial Reference System - Alignment from the FMC CDU, TASK 34-21-00-820-801 or Air Data Inertial Reference System - Automatic Alignment from the IRS MSU, TASK 34-21-00-820-803.

SUBTASK 27-62-37-750-001

(2) Do the Speedbrake Lever Transducer Test (TASK 27-61-45-730-802).

H. Put the Airplane Back to Its Usual Condition

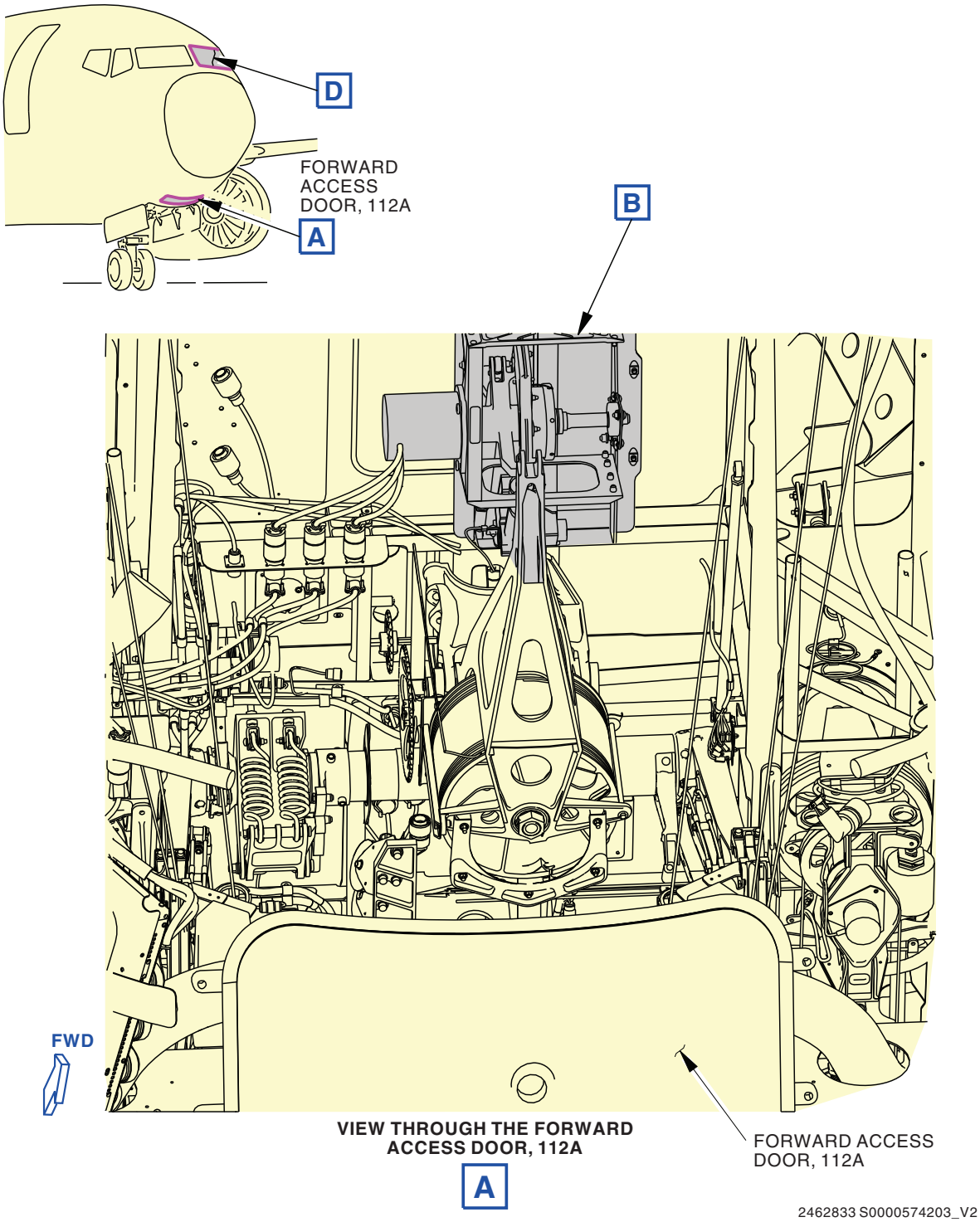
SUBTASK 27-62-37-410-003

(1) If it is open, close this access door:

Number	Name/Location
112A	Forward Access Door

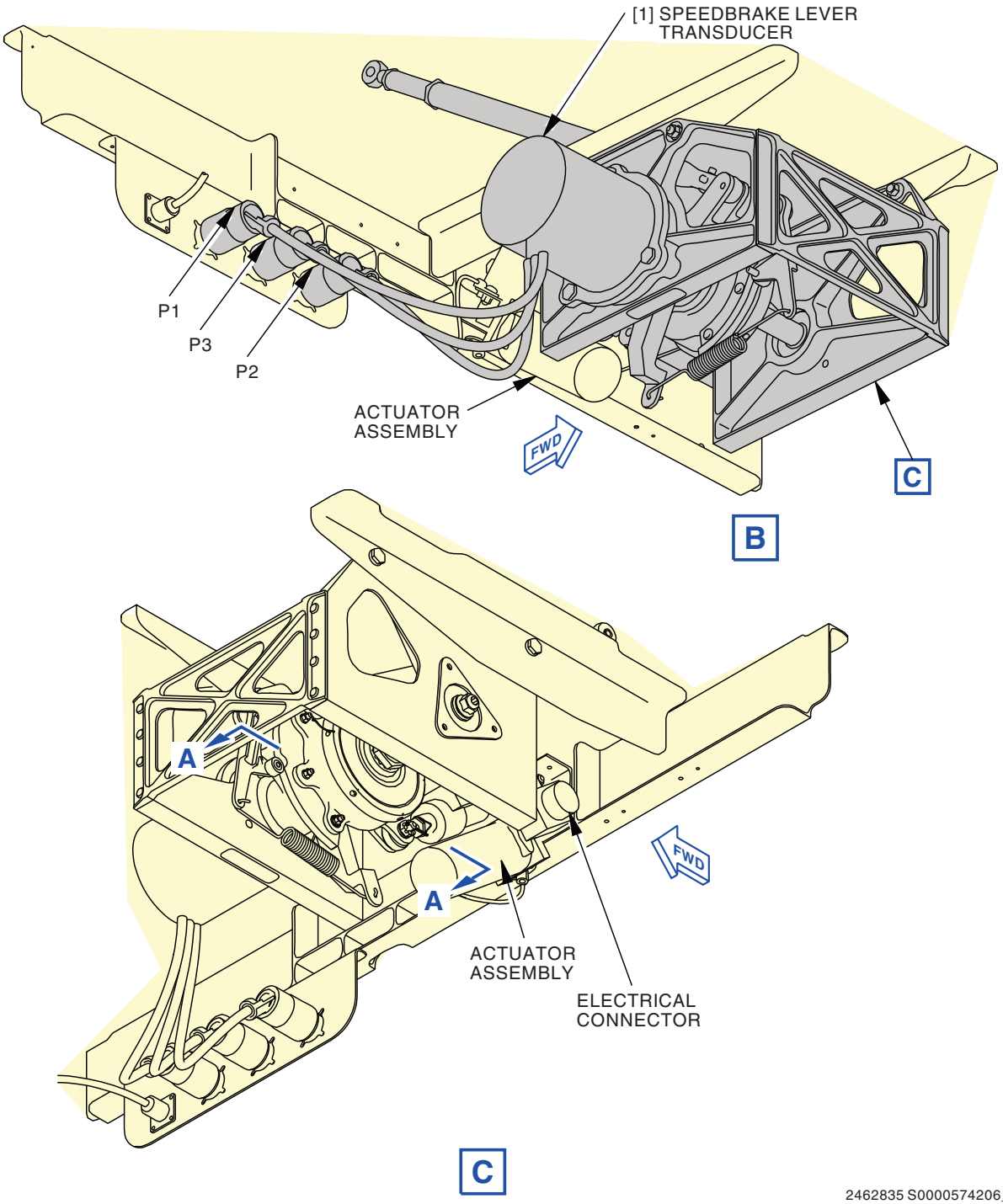
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



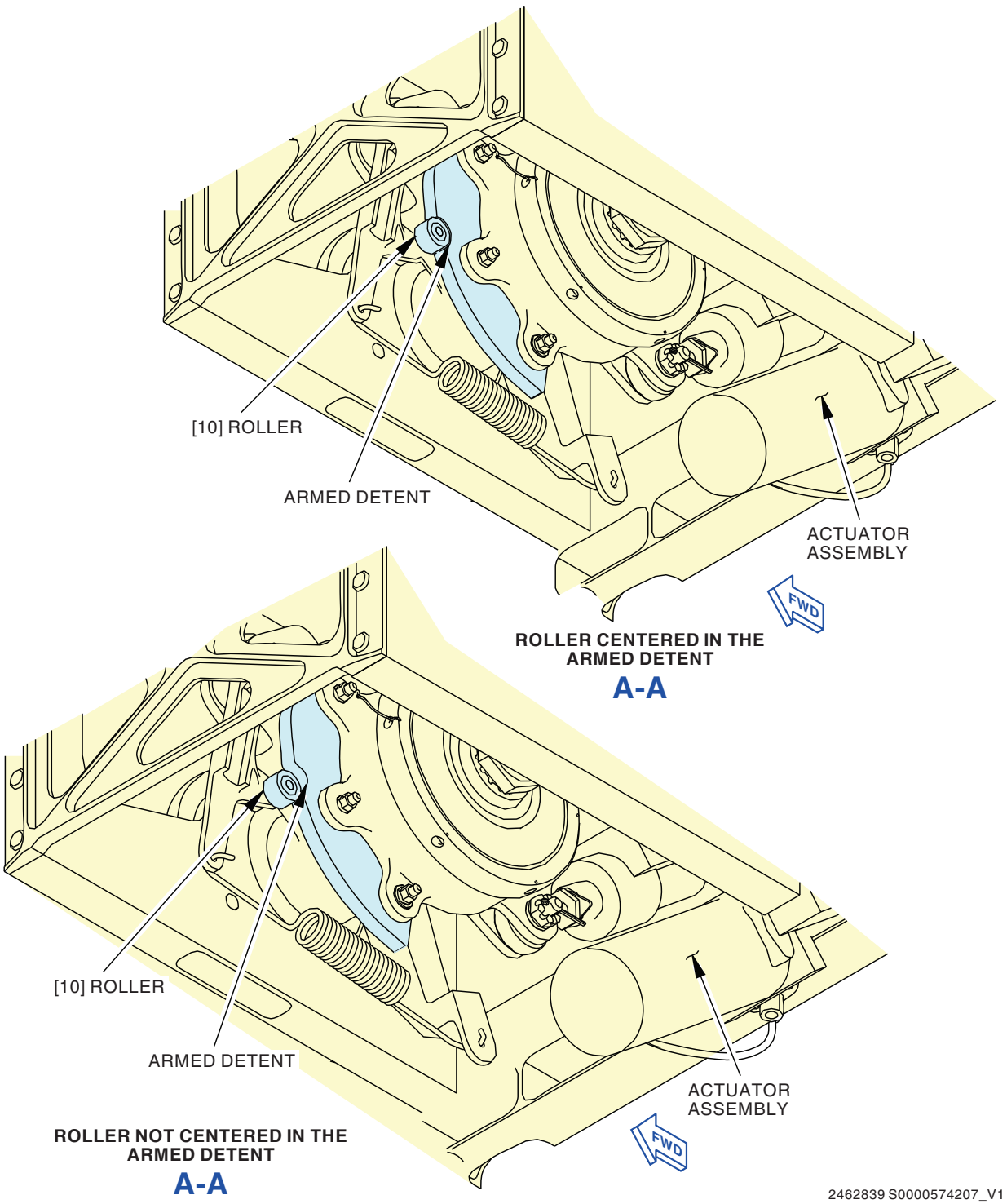
Speedbrake Lever Transducer - Adjustment
Figure 402/27-62-37-990-802 (Sheet 1 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



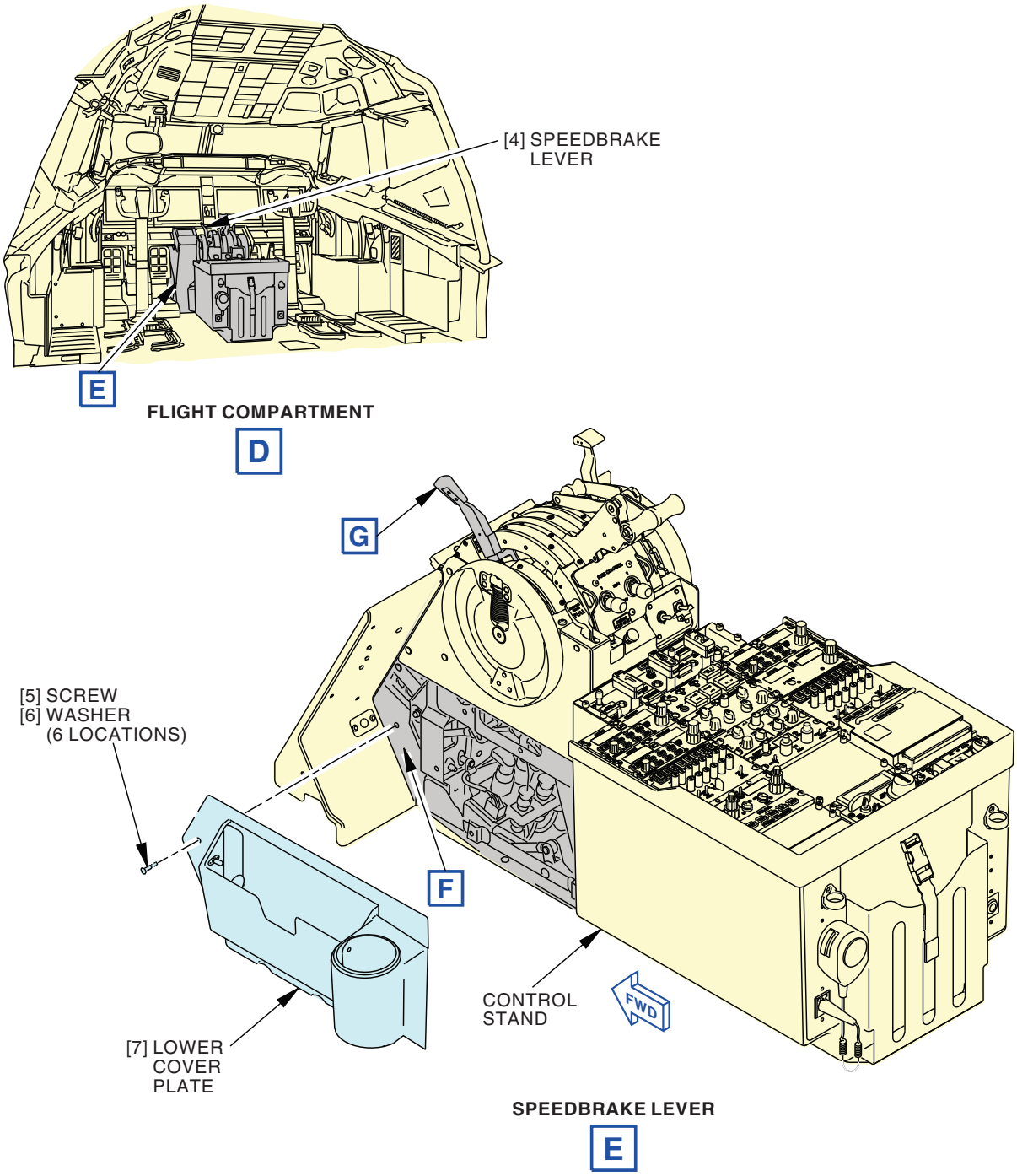
Speedbrake Lever Transducer - Adjustment
Figure 402/27-62-37-990-802 (Sheet 2 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Speedbrake Lever Transducer - Adjustment
Figure 402/27-62-37-990-802 (Sheet 3 of 6)

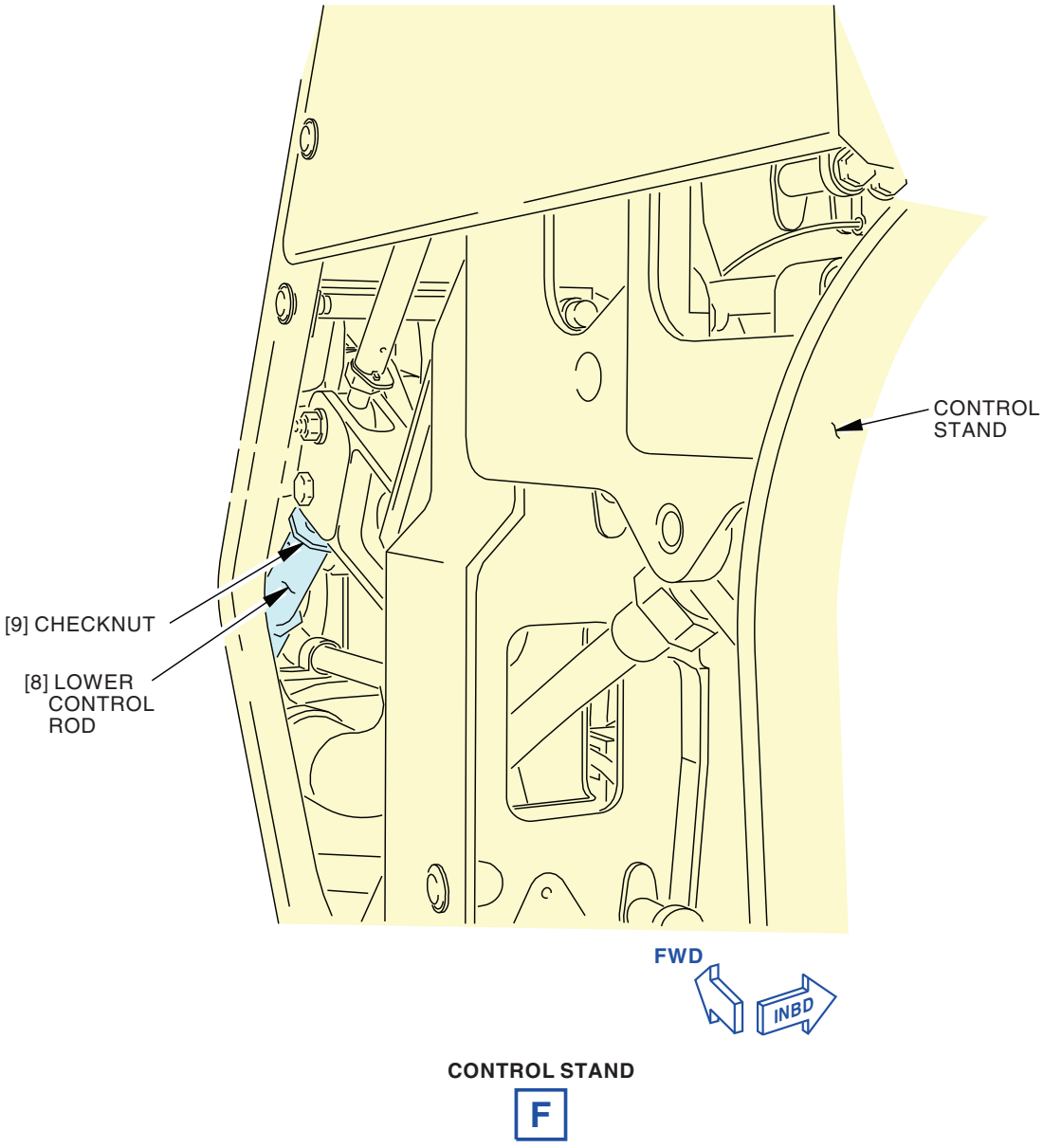
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2462841 S0000574208_V2

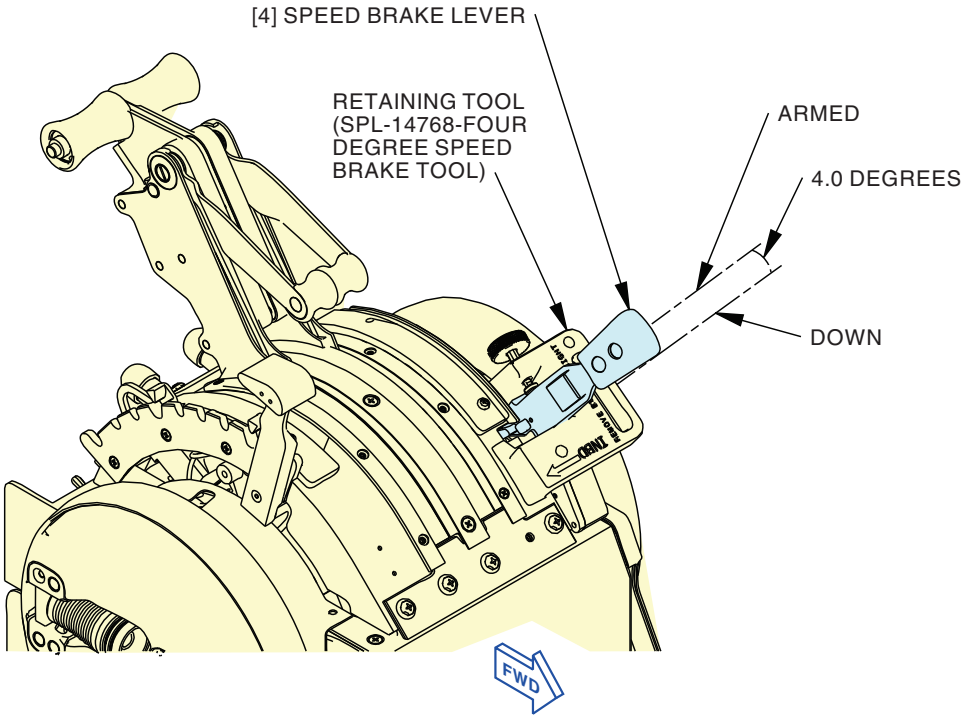
Speedbrake Lever Transducer - Adjustment
Figure 402/27-62-37-990-802 (Sheet 4 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Speedbrake Lever Transducer - Adjustment
Figure 402/27-62-37-990-802 (Sheet 5 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



G

2463032 S0000574210_V2

Speedbrake Lever Transducer - Adjustment
Figure 402/27-62-37-990-802 (Sheet 6 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

GROUND SPOILER CONTROL MODULE- REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the ground spoiler control module

(2) An installation of the ground spoiler control module

(3) A functional test of the ground spoiler control module.

TASK 27-62-41-000-801

2. Ground Spoiler Control Module Removal

(Figure 401)

A. References

Reference	Title
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)

B. Tools/Equipment

Reference	Description
STD-1110	Container - Hydraulic Fluid Resistant, 5 Gallon (19 Liter)

C. Location Zones

Zone	Area
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right

D. Prepare for the Removal

SUBTASK 27-62-41-860-001

(1) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	9	C00440	FLIGHT CONTROL AUTO SPEED BRAKE
B	11	C04040	GROUND SPOILER CONTROL

SUBTASK 27-62-41-860-011

WARNING

KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT SPOILERS. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU OPEN/CLOSE THESE CIRCUIT BREAKERS. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(2) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
C	10	C04032	SPOILER CONTROL ELEC PWR A
C	12	C04033	SPOILER CONTROL ELEC PWR B

SUBTASK 27-62-41-860-003

(3) Remove the pressure from the speed brake hydraulic system A (TASK 29-11-00-860-805).

(a) Make sure that the hydraulic pressure has been removed from the hydraulic pumps and/or portable hydraulic cart.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Ground Spoiler Control Module Removal

SUBTASK 27-62-41-020-001

(1) Do these steps to disconnect the hydraulic line [2], hydraulic line [3], hydraulc line [4], and hydraulic line [5] from the ground spoiler control module [1]:

(a) Put a 5 gallon (19 liter) hydraulic fluid resistant container, STD-1110, below the ground spoiler control module [1] to catch the hydraulic fluid.

(b) Disconnect the hydraulic line [2], hydraulic line [3], hydraulc line [4], and hydraulic line [5] from the ground spoiler control module [1].

(c) Install the plugs on the hydraulic line [2], hydraulic line [3], hydraulc line [4], and hydraulic line [5].

(d) Install the caps on the applicable hydraulic ports.

SUBTASK 27-62-41-020-002

(2) Disconnect connector [6], connector [7], connector [8], and connector [9].

SUBTASK 27-62-41-020-003

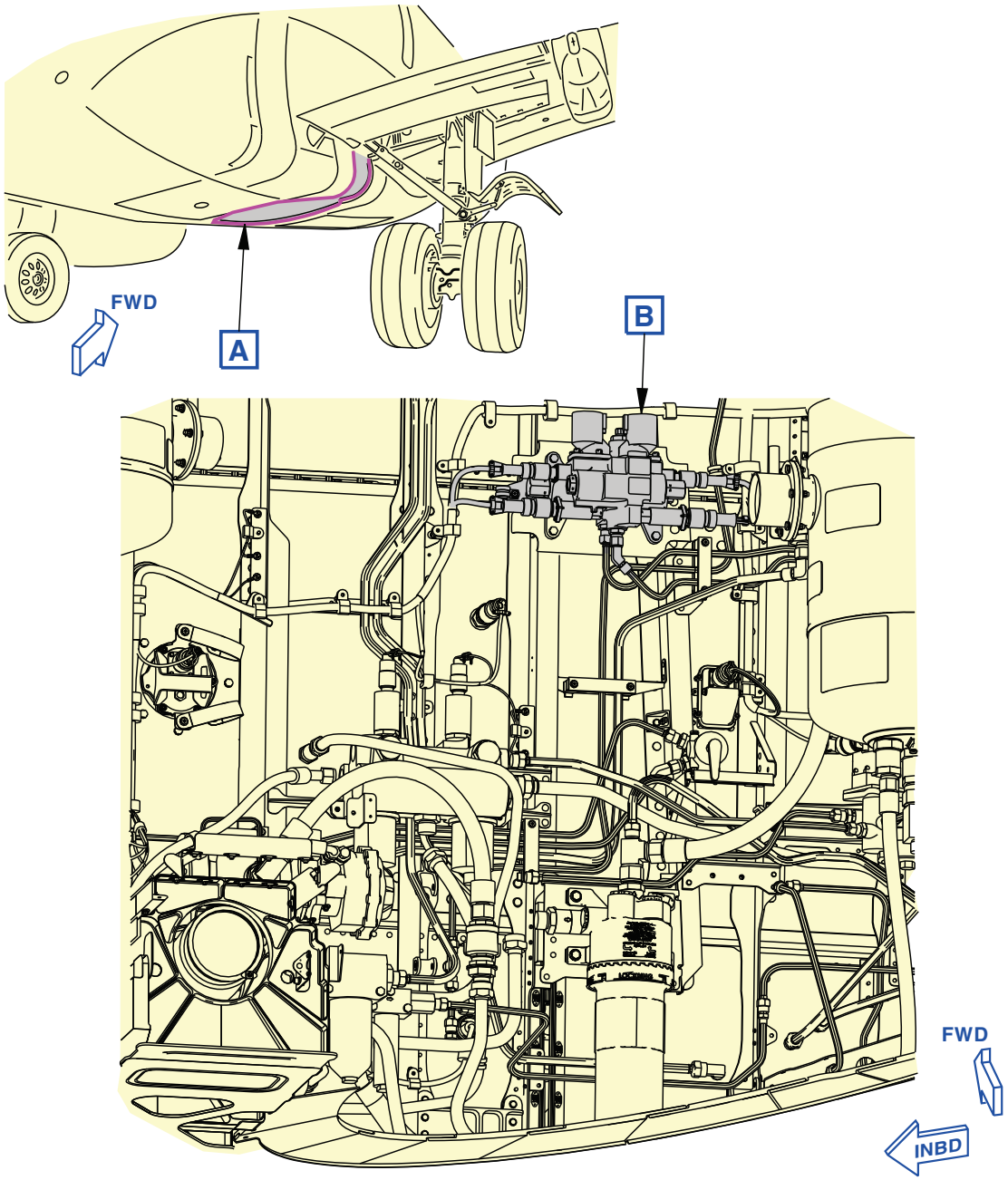
(3) Remove the bolts [12] and washers [13] that attach the ground spoiler control module [1].

SUBTASK 27-62-41-020-004

(4) Remove the ground spoiler control module [1] from the airplane.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



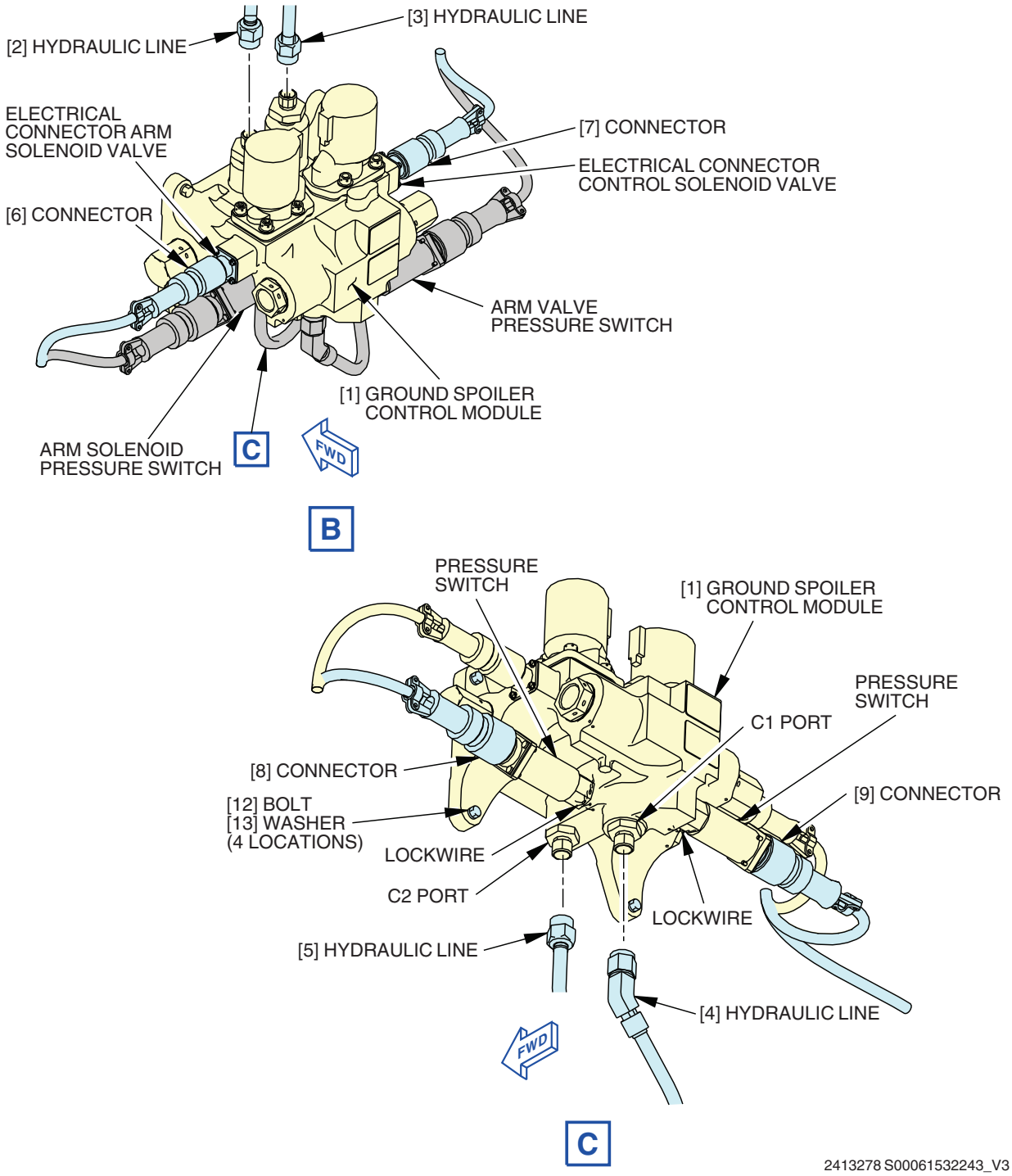
MAIN LANDING GEAR WHEEL WELL
(RIGHT SIDE)

A

2413277 S00061532242_V1

Ground Spoiler Control Module Installation
Figure 401/27-62-41-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Ground Spoiler Control Module Installation
Figure 401/27-62-41-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-62-41-400-801

3. Ground Spoiler Control Module Installation

(Figure 401)

A. References

Reference	Title
27-62-00-840-802	Pressurization of the Speed Brake Hydraulic Systems A and B to its Usual Condition (P/B 201)
SWPM 20-20-00	ELECTRICAL BONDING PROCESSES
SWPM 20-20-00, Paragraph 2A	Cleaning Procedure 1

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-614	Bonding Meters - Non-Intrinsically Safe (For use in outside Class I, Divisions I & II non-hazardous locations. For hazardous locations, use COM-1550). Part #: 247000 Supplier: 00426 Part #: 620LK Supplier: 1CRL2 Part #: BLR-0003-XX Supplier: KC432 Part #: BT51 Supplier: 00426 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659 Opt Part #: 247001 Supplier: 00426

C. Consumable Materials

Reference	Description	Specification
A50110	Sealant - Fuel Tank	BMS5-45 Class B-2
A50155	Sealant - Fuel Tank	BMS5-45 Class C
A50231	Sealant - Pressure And Environmental - Chromate Type	BMS5-95 Class B
A50296	Sealant - Pressure And Environmental - Chromate Type	BMS5-95 Class C
B00184	Solvent - Presealing, Cleaning Solvent	BMS11-7
C00259	Coating - Chemical And Solvent Resistant Finish, Corrosion Inhibiting Primer	BMS10-11 Type I
C50315	Coating - Chemical Conversion - Bonderite M-CR 600 Aero (Formerly Alodine 600)	BAC5719 Class A, C or D, MIL-DTL-81706 Type I Class 1A or 3
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Ground spoiler control module	27-62-41-01-005	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Location Zones

Zone	Area
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right

F. Ground Spoiler Control Module Installation

SUBTASK 27-62-41-420-001

(1) Do these step to install the ground spoiler control module [1]:

(a) Clean the surface of the mating surfaces of the airplane structure, bolts [12], washers [13], and ground spoiler control module [1] (SWPM 20-20-00, Paragraph 2A).

(b) Use cotton wiper, G00034, moist in solvent, B00184, to final clean the mating surfaces of the airplane structure, bolts [12], washers [13], and ground spoiler control module [1].

(c) Apply sealant, A50110, or sealant, A50155, or sealant, A50231, or sealant, A50296, to the mating surfaces of the bolts [12], washers [13], and ground spoiler control module [1].

(d) Put the ground spoiler control module [1] in its position.

(e) Install the bolts [12] and washers [13].

1) Make sure that there is squeeze out around all the interfaces of each component.

(f) Measure the electrical resistance between the ground spoiler control module [1] and airplane structure (SWPM 20-20-00).

1) Use a non-intrinsically safe bonding meter, COM-614.

2) Make sure that the electrical bonding resistance is 0.0005 ohms (0.5 milliohms) or less.

(g) Manually apply finish with Bonderite M-CR 600 Aero coating, C50315.

1) Apply two coats of primer, C00259.

(h) Make sure that all components are covered by sealant or finishes.

(i) Remove the caps from the ports in the ground spoiler control module [1].

(j) Remove the plugs from the hydraulic line [2], hydraulic line [3], hydraulc line [4], and hydraulic line [5].

(k) Install the hydraulic line [2], hydraulic line [3], hydraulc line [4], and hydraulic line [5] to the ground spoiler control module [1].

1) Put the bundle/wire to get 0.5 in. (12.7 mm) minimum separation between wire and hydraulic tubing.

NOTE: The 0.5 in. (12.7 mm) minimum separation will prevent wear and fraying between the wire/bundle and hydraulic tubing.

(l) Tighten the B-nuts on hydraulic line [2], hydraulic line [3], hydraulc line [4], and hydraulic line [5] to 270 ±14 in-lb (31 ±2 N·m).

(m) Connect the connector [6], connector [7], connector [8], and connector [9] to the ground spoiler control module [1].

G. Ground Spoiler Control Module Installation Test

SUBTASK 27-62-41-720-002

(1) Do this task: Ground Spoiler Control Module Functional Test, TASK 27-62-41-720-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

H. Put the Airplane Back to Its Usual Condition

SUBTASK 27-62-41-860-004

(1) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	9	C00440	FLIGHT CONTROL AUTO SPEED BRAKE
B	11	C04040	GROUND SPOILER CONTROL

WARNING

KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT SPOILERS. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU OPEN/CLOSE THESE CIRCUIT BREAKERS. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(2) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
C	10	C04032	SPOILER CONTROL ELEC PWR A
C	12	C04033	SPOILER CONTROL ELEC PWR B

SUBTASK 27-62-41-860-005

(3) Put the speed brake system A back to the condition before the pressure removal (TASK 27-62-00-840-802).

END OF TASK

TASK 27-62-41-720-801

4. Ground Spoiler Control Module Functional Test

(Figure 401)

A. References

Reference	Title
12-12-00-610-801	Hydraulic Reservoir Servicing (P/B 301)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
46-13-02-710-801	Onboard Maintenance Function Ground Test (P/B 201)

B. Location Zones

Zone	Area
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50

C. Ground Spoiler Control Module Functional Test

SUBTASK 27-62-41-860-008

(1) Pressurize speed brake hydraulic system A (TASK 29-11-00-860-801).

SUBTASK 27-62-41-210-001

(2) Examine the hydraulic connections on the ground spoiler control module [1] for leakage.

SUBTASK 27-62-41-860-009

(3) Remove the pressure from the speed brake hydraulic system A (TASK 29-11-00-860-805).

SUBTASK 27-62-41-730-001

(4) Do these steps for the ground spoiler test (TASK 46-13-02-710-801):

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Put the GND TEST switch on the P61-4 panel to ENABLE.

(b) Push the SYS button in the MFD section of the Multi Function Panel.

(c) Use the SELECTOR on the Multi Function Panel to make these selections:

- MAINT
- ONBD MAINT
- LINE MAINT
- GROUND TESTS
- 27 - Spoiler Control
- System Test
- Ground Spoilers
- CONTINUE
- CONTINUE
- START TEST
- CONTINUE

(d) Follow the on-screen instructions and then select the CONTINUE button.

(e) Follow the second set of on-screen instructions and then select the CONTINUE button to complete the test.

SUBTASK 27-62-41-210-002

(5) Examine the hydraulic connections on the ground spoiler control module [1] for leakage.

SUBTASK 27-62-41-610-001

(6) Make sure that the hydraulic reservoirs are full (TASK 12-12-00-610-801).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

INBOARD GROUND SPOILER ACTUATOR - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Inboard ground spoiler actuator removal

(2) Inboard ground spoiler actuator installation

(3) Inboard ground spoiler actuator functional test.

B. In this procedure, the inboard ground spoiler actuator is referred to as the "actuator".

C. This procedure is applicable to the left and right inboard ground spoiler actuators.

TASK 27-62-71-000-801

2. Inboard Ground Spoiler Actuator Removal

(Figure 401)

A. General

(1) This task has the steps to remove the inboard ground spoiler actuator.

B. References

Reference	Title
27-51-00-860-803	Extend the Trailing Edge Flaps (P/B 201)
27-62-00-800-801	Apply Pressure to the Speed Brake Hydraulic Systems A and B (P/B 201)
27-62-00-800-802	Depressurization of the Speed Brake Hydraulic Systems A and B (P/B 201)

C. Tools/Equipment

Reference	Description
STD-1110	Container - Hydraulic Fluid Resistant, 5 Gallon (19 Liter)

D. Location Zones

Zone	Area
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50
552	Left Wing - Spoiler No. 6
652	Right Wing - Spoiler No. 7

E. Prepare for the Removal

SUBTASK 27-62-71-860-002

WARNING

KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Do this task: Apply Pressure to the Speed Brake Hydraulic Systems A and B, TASK 27-62-00-800-801.

SUBTASK 27-62-71-860-003

(2) Put the SPOILER A and B switches in the OFF position to remove the hydraulic power from the flight spoilers.

NOTE: SPOILER A and B switches are on the flight control panel (P5-3).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-62-71-860-004

(3) Do this task: Extend the Trailing Edge Flaps, TASK 27-51-00-860-803.

SUBTASK 27-62-71-860-005

(4) Do this task: Depressurization of the Speed Brake Hydraulic Systems A and B, TASK 27-62-00-800-802.

F. Inboard Ground Spoiler Actuator Removal

SUBTASK 27-62-71-020-001

(1) Do these steps to disconnect the hydraulic lines [10] from the actuator [17]:

(a) Put a 5 gallon (19 liter) hydraulic fluid resistant container, STD-1110 below the actuator [17] to catch the hydraulic fluid.

(b) Disconnect the hydraulic lines [10] from the actuator [17].

(c) Install the plugs in the hydraulic lines [10].

SUBTASK 27-62-71-020-002

(2) Remove the pin [7], nut [6], washer [3], bushing [4], washer [5] and bolt [2] that attach the actuator [17] to the inboard ground spoiler [1].

SUBTASK 27-62-71-020-003

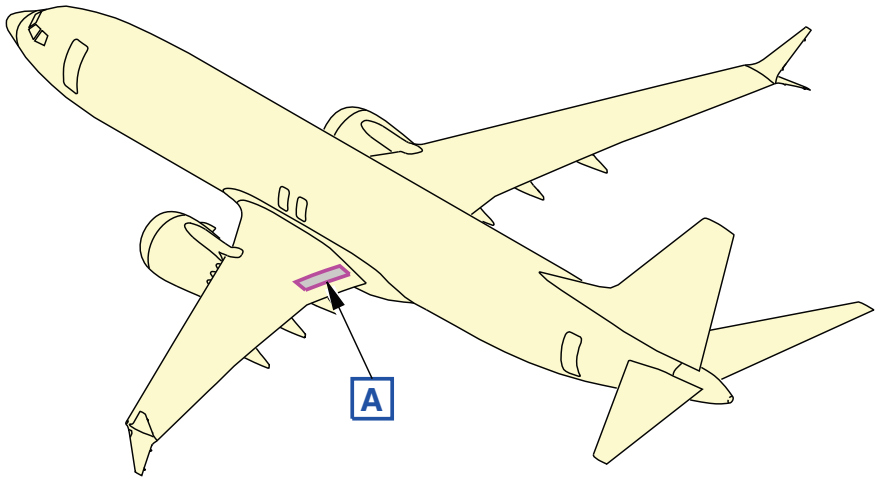
(3) Remove the pin [16], nut [15], washer [14], bushing [13], washer [12] and bolt [11] that attaches the actuator [17] to the structure.

SUBTASK 27-62-71-480-001

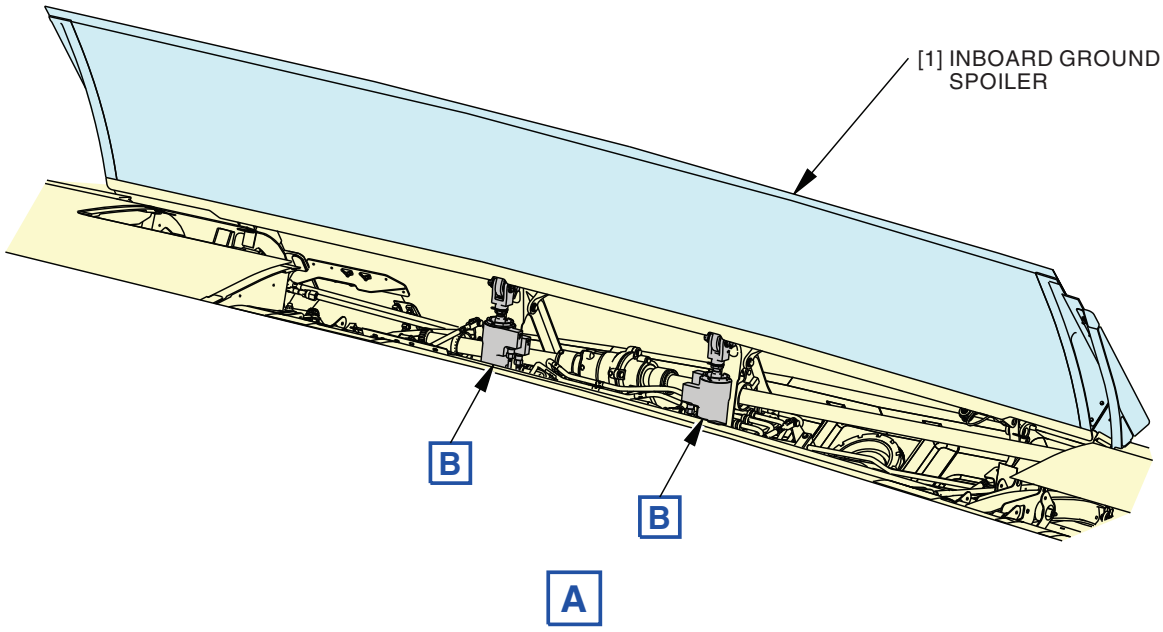
(4) Install caps in the hydraulic ports of the actuator [17].

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



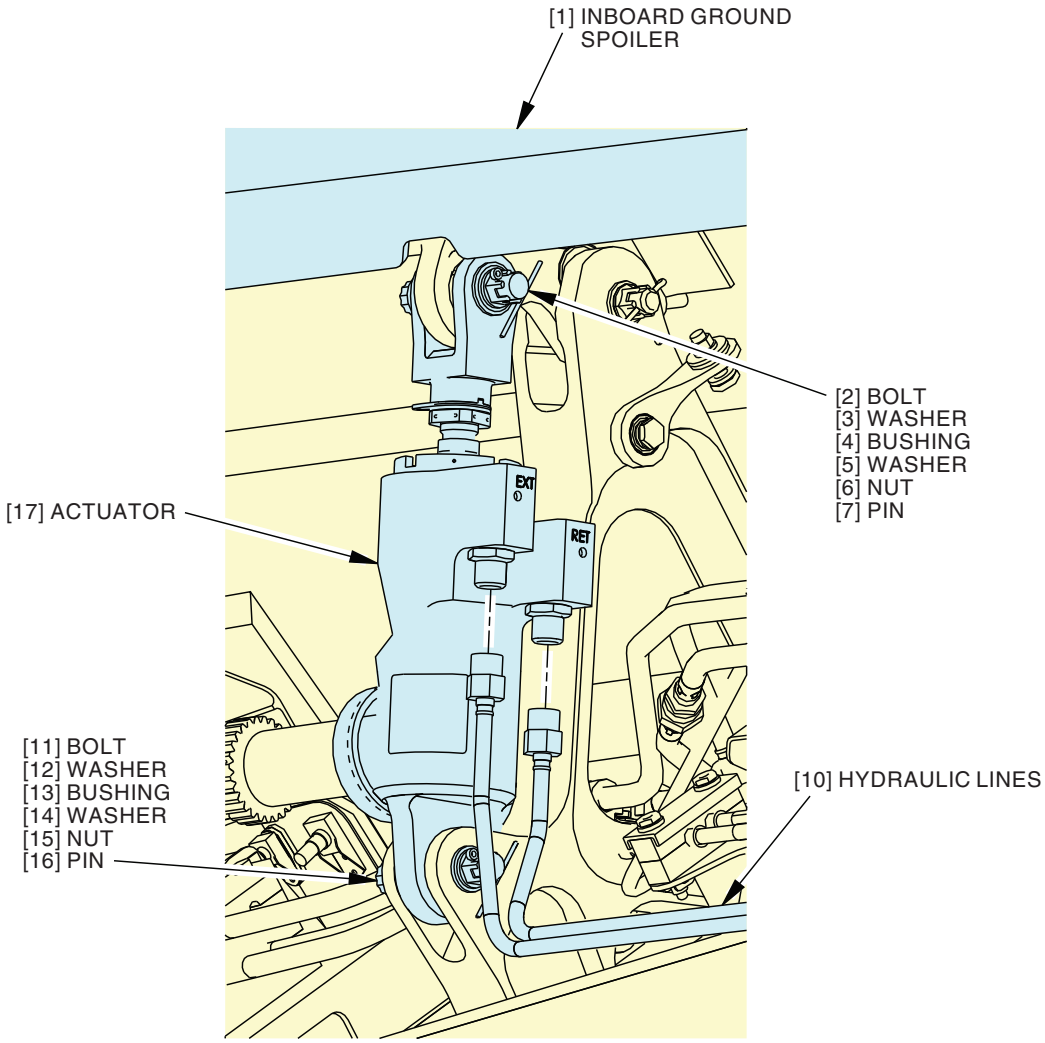
LEFT WING
(RIGHT WING IS EQUIVALENT)



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Inboard Ground Spoiler Actuator Installation
Figure 401/27-62-71-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



B

2413295 S00061532287_V1

Inboard Ground Spoiler Actuator Installation
Figure 401/27-62-71-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-62-71-400-801

3. Inboard Ground Spoiler Actuator Installation

(Figure 401)

A. General

(1) This task has the steps to install the inboard ground spoiler actuator.

B. References

Reference	Title
12-12-00-610-801	Hydraulic Reservoir Servicing (P/B 301)
27-51-00-860-804	Retract the Trailing Edge Flaps (P/B 201)
27-62-00-800-801	Apply Pressure to the Speed Brake Hydraulic Systems A and B (P/B 201)
27-62-00-820-802	Ground Spoiler Adjustment (P/B 501)
27-62-00-840-801	Shut Down the Speed Brake Hydraulic Systems A and B (P/B 201)
27-62-00-840-802	Pressurization of the Speed Brake Hydraulic Systems A and B to its Usual Condition (P/B 201)

C. Consumable Materials

Reference	Description	Specification
D00633	Grease - Aircraft General Purpose	BMS3-33

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
7	Pin	27-62-71-01-195	PTF ALL
16	Pin	27-62-71-01-195	PTF ALL
17	Actuator	27-62-71-01-225	PTF ALL

E. Location Zones

Zone	Area
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50
552	Left Wing - Spoiler No. 6
652	Right Wing - Spoiler No. 7

F. Inboard Ground Spoiler Actuator Installation

SUBTASK 27-62-71-420-001

(1) Install the actuator [17]:

(a) Remove the caps from the hydraulic ports of the actuator [17].

(b) Put the actuator [17] in its position and make sure that the hydraulic lines [10] align with the fittings.

(c) Connect the actuator [17] rod end to the inboard ground spoiler [1]:

1) Apply grease, D00633 to the bolt [2].

2) Install bolt [2] with the head outboard, washer [3], bushing [4], washer [5], nut [6] and pin [7].

3) Tighten nut to 450 in-lb (51 N·m) to 690 in-lb (78 N·m).

(d) Connect the actuator [17] to the structure:

1) Apply grease, D00633 to the bolt [11].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) Install bolt [11] with the head outboard, washer [12], bushing [13], washer [14], nut [15] and pin [16].

3) Tighten nut to 450 in-lb (51 N·m) to 690 in-lb (78 N·m).

SUBTASK 27-62-71-420-002

(2) Connect the hydraulic lines [10] to the actuator [17]:

(a) Remove the plugs from the hydraulic lines [10].

(b) Connect the hydraulic lines [10] to the actuator [17].

(c) Tighten the B-nuts on the hydraulic lines [10] to 140 ±7 in-lb (16 ±1 N·m).

SUBTASK 27-62-71-860-006

(3) Do this task: Pressurization of the Speed Brake Hydraulic Systems A and B to its Usual Condition, TASK 27-62-00-840-802.

SUBTASK 27-62-71-860-007

WARNING

KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(4) Do this task: Apply Pressure to the Speed Brake Hydraulic Systems A and B, TASK 27-62-00-800-801.

SUBTASK 27-62-71-870-001

(5) Move the speed brake control lever through its full cycle to bleed the spoiler actuators.

SUBTASK 27-62-71-700-001

(6) Make sure there are no leaks at the hydraulic lines.

SUBTASK 27-62-71-860-008

(7) Do the adjustment of the inboard ground spoiler actuator: Ground Spoiler Adjustment, TASK 27-62-00-820-802.

G. Inboard Ground Spoiler Actuator Installation Test

SUBTASK 27-62-71-820-001

(1) Do this task: Inboard Ground Spoiler Actuator Functional Test, TASK 27-62-71-700-801.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 27-62-71-860-009

(1) Do this task: Retract the Trailing Edge Flaps, TASK 27-51-00-860-804.

SUBTASK 27-62-71-860-010

(2) Do this task: Shut Down the Speed Brake Hydraulic Systems A and B, TASK 27-62-00-840-801.

SUBTASK 27-62-71-860-011

(3) Put the SPOILER A and B switches in the ON position.

SUBTASK 27-62-71-610-001

(4) Fill the hydraulic reservoirs if it is necessary. To fill, do this task: Hydraulic Reservoir Servicing, TASK 12-12-00-610-801.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-62-71-700-801

4. Inboard Ground Spoiler Actuator Functional Test

A. General

(1) This task has the steps to do a functional test of the inboard ground spoiler actuator.

B. References

Reference	Title
27-61-00-840-801	Shut Down the Spoiler Hydraulic Systems A and B (P/B 201)
27-62-00-800-801	Apply Pressure to the Speed Brake Hydraulic Systems A and B (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1691	Protractor - Digital, Vertical or Horizontal Part #: 21465A82 Supplier: 3A054 Part #: 31-038-3 (Model Pro 360) Supplier: 1X822 Part #: 31-040-9 (Model Pro 3600) Supplier: 1X822 Part #: 950-317 (Model Pro 360) Supplier: 01EP7 Part #: 950-318 (Model Pro 3600) Supplier: 01EP7 Part #: KS5549 Supplier: 75245 Part #: KS6005 Supplier: 75245

D. Location Zones

Zone	Area
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50
552	Left Wing - Spoiler No. 6
652	Right Wing - Spoiler No. 7

E. Prepare for the Test

SUBTASK 27-62-71-860-013

WARNING

KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Do this task: Apply Pressure to the Speed Brake Hydraulic Systems A and B, TASK 27-62-00-800-801.

SUBTASK 27-62-71-480-002

CAUTION

INSTALL THE PROTRACTOR ON THE SPOILER TOP SURFACE. IF THE PROTRACTOR INSTALLED ON THE BOTTOM OF THE SPOILER, DAMAGE TO AIRCRAFT CAN OCCUR.

(2) Install the protractor, COM-1691, on the upper surface of the outboard edge of the inboard ground spoilers 6 and 7.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-62-71-820-002

(3) Adjust the protractor, COM-1691, to zero.

F. Inboard Ground Spoiler Actuator Functional Test

SUBTASK 27-62-71-020-004

(1) Move the speed brake lever to the UP position to lift the spoilers.

SUBTASK 27-62-71-020-005

(2) Make sure the spoilers No. 6 and 7 are 60 +/-2 degrees up from the retracted position.

SUBTASK 27-62-71-020-006

(3) Make sure the ground spoilers No. 6 and 7 align in the limit of 2 degrees.

SUBTASK 27-62-71-020-007

(4) Move the speed brake lever to the DOWN position to lower the spoilers.

SUBTASK 27-62-71-820-003

(5) Make sure the spoilers move to the fully retracted position.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 27-62-71-860-014

(1) Do this task: Shut Down the Spoiler Hydraulic Systems A and B, TASK 27-61-00-840-801.

SUBTASK 27-62-71-080-001

(2) Remove the protractor, COM-1691, from the upper surface of spoilers No. 6 and 7.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OUTBOARD GROUND SPOILER - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the outboard ground spoilers

(2) An installation of the outboard ground spoilers

(3) A functional test of the outboard ground spoilers.

B. This procedure is applicable to the left and right outboard ground spoilers.

TASK 27-62-91-000-801

2. Outboard Ground Spoiler Removal

(Figure 401)

A. References

Reference	Title
27-51-00-860-803	Extend the Trailing Edge Flaps (P/B 201)
27-62-00-800-801	Apply Pressure to the Speed Brake Hydraulic Systems A and B (P/B 201)
27-62-00-800-802	Depressurization of the Speed Brake Hydraulic Systems A and B (P/B 201)
27-62-94-000-801	Outboard Ground Spoiler Actuator Removal (P/B 401)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1743	Ground Lock Set - Outboard Spoiler Actuators Part #: C27001-51 Supplier: 81205

C. Location Zones

Zone	Area
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50
566	Left Wing - Spoiler No. 1
666	Right Wing - Spoiler No. 12

D. Prepare for the Removal

SUBTASK 27-62-91-860-002



WARNING

KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Do this task: Apply Pressure to the Speed Brake Hydraulic Systems A and B, TASK 27-62-00-800-801.

SUBTASK 27-62-91-860-003

(2) Do this task: Extend the Trailing Edge Flaps, TASK 27-51-00-860-803.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-62-91-860-004

(3) Put the SPOILER A and SPOILER B switches to the OFF position to remove the hydraulic power from the flight spoilers.

NOTE: SPOILER A AND SPOILER B switches are on the P5-3 flight control panel.

SUBTASK 27-62-91-860-005

(4) Put the speed brake lever in the UP position to lift the ground spoilers.

SUBTASK 27-62-91-480-001

(5) Install the ground lock set, SPL-1743, for the applicable ground spoiler.

SUBTASK 27-62-91-860-006

(6) Do this task: Depressurization of the Speed Brake Hydraulic Systems A and B, TASK 27-62-00-800-802.

E. Outboard Ground Spoiler Removal

SUBTASK 27-62-91-020-001

(1) Disconnect the rod end of the spoiler actuator assembly [33] from the outboard ground spoiler [1] (TASK 27-62-94-000-801).

SUBTASK 27-62-91-020-002

(2) Remove the nut [2] and washer [3] to disconnect the bonding jumper [10] from the structure.

SUBTASK 27-62-91-020-003

(3) Remove the nut [30], washers [29], and bonding jumper [31] to disconnect the bonding jumper [32] from the structure.

(a) Discard the nut [30] and washers [29].

SUBTASK 27-62-91-020-009

(4) If it is necessary, loosen the skin panel fasteners above the center hinge bolt [11] and bolt [17].

SUBTASK 27-62-91-020-004

(5) Remove the outboard ground spoiler [1]:

(a) Remove the cotter pin [9], nut [8], washer [7], bushing [6], washer [5], and bolt [4] to disconnect the outboard hinge fitting.

(b) Remove the cotter pin [28], nut [27], washer [26], bushing [25], washer [24], and bolt [23] to disconnect the inboard hinge fitting.

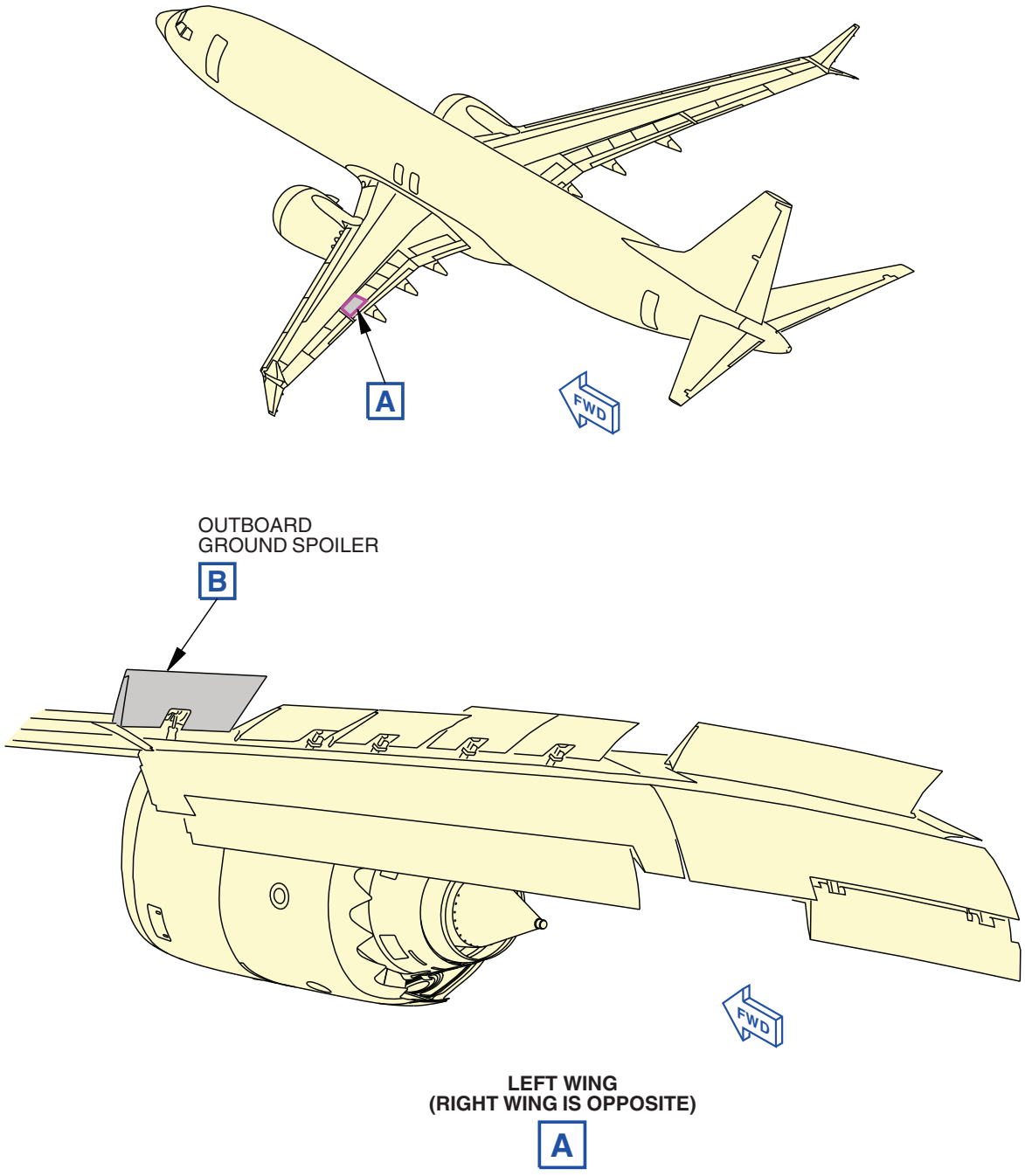
(c) Remove the cotter pin [16], nut [15], washer [14], bushing [13], washer [12], and bolt [11] to disconnect the center outboard hinge fitting.

(d) Remove the cotter pin [22], nut [21], washer [20], bushings [19], washer [18], and bolt [17] to disconnect the center inboard hinge fitting.

(e) Remove the outboard ground spoiler [1] from the airplane.

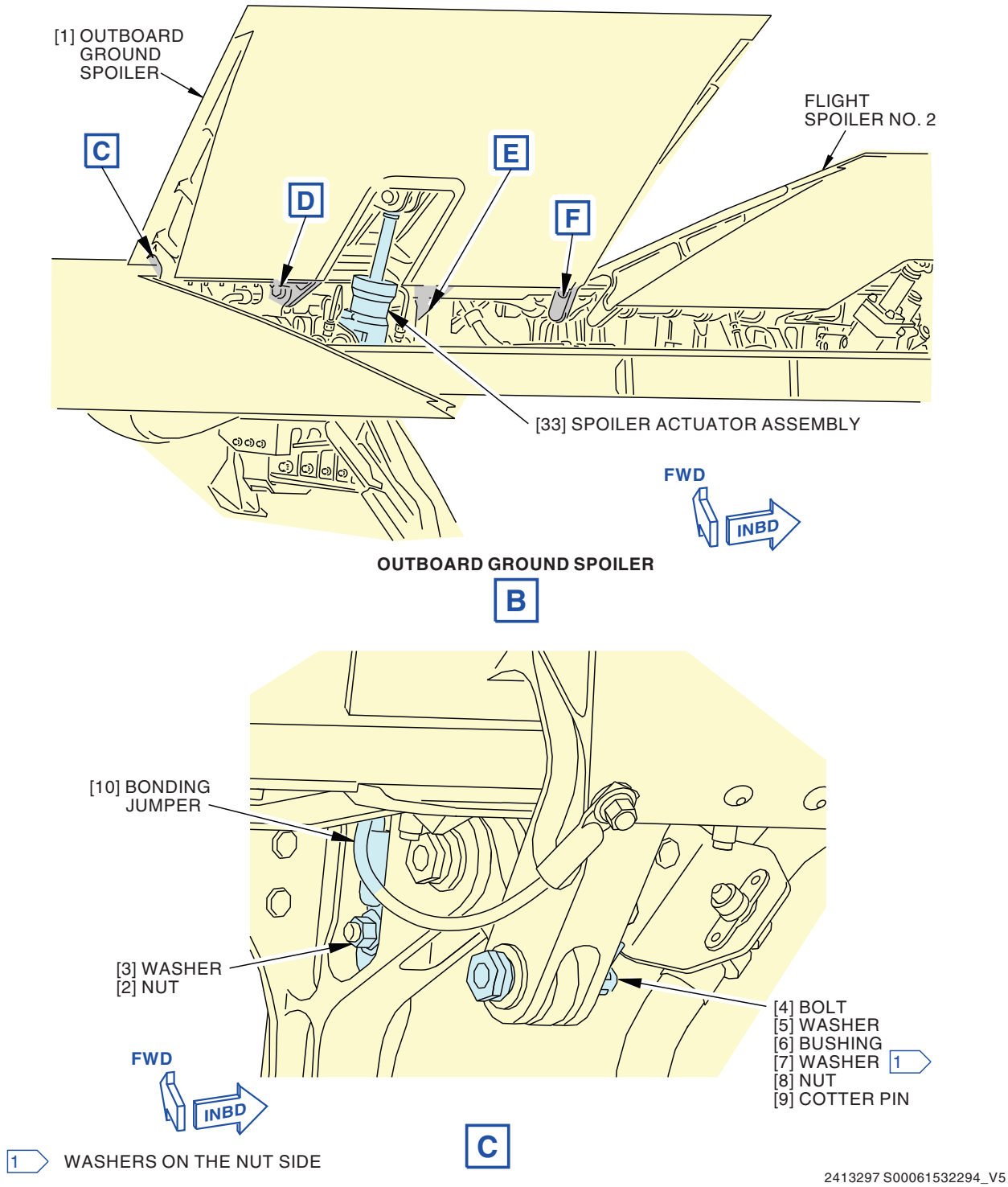
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



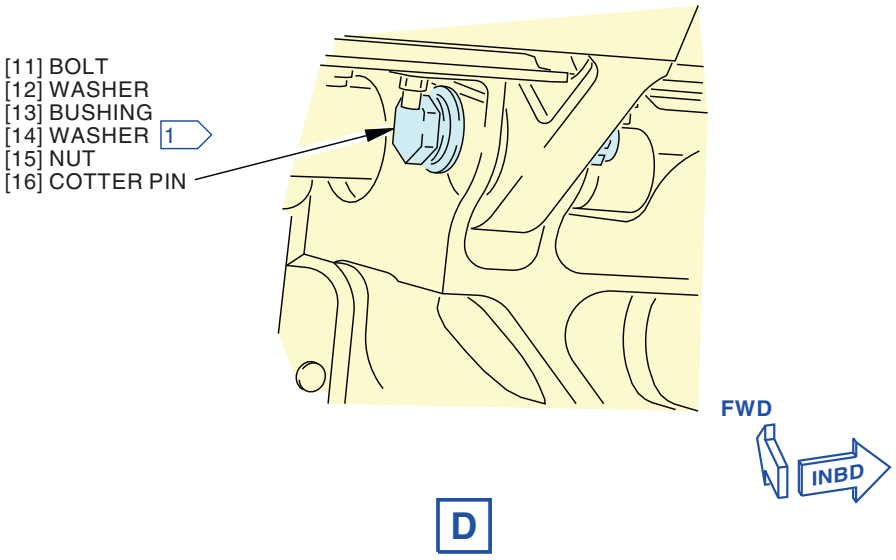
Outboard Ground Spoiler Installation
Figure 401/27-62-91-990-801 (Sheet 1 of 4)

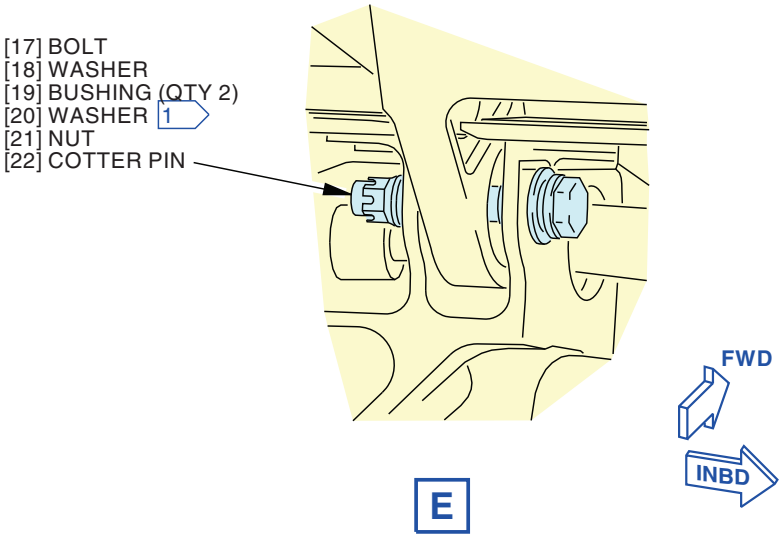
ECCN 9E991 BOEING PROPRIETARY - See title page for details



Outboard Ground Spoiler Installation
Figure 401/27-62-91-990-801 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

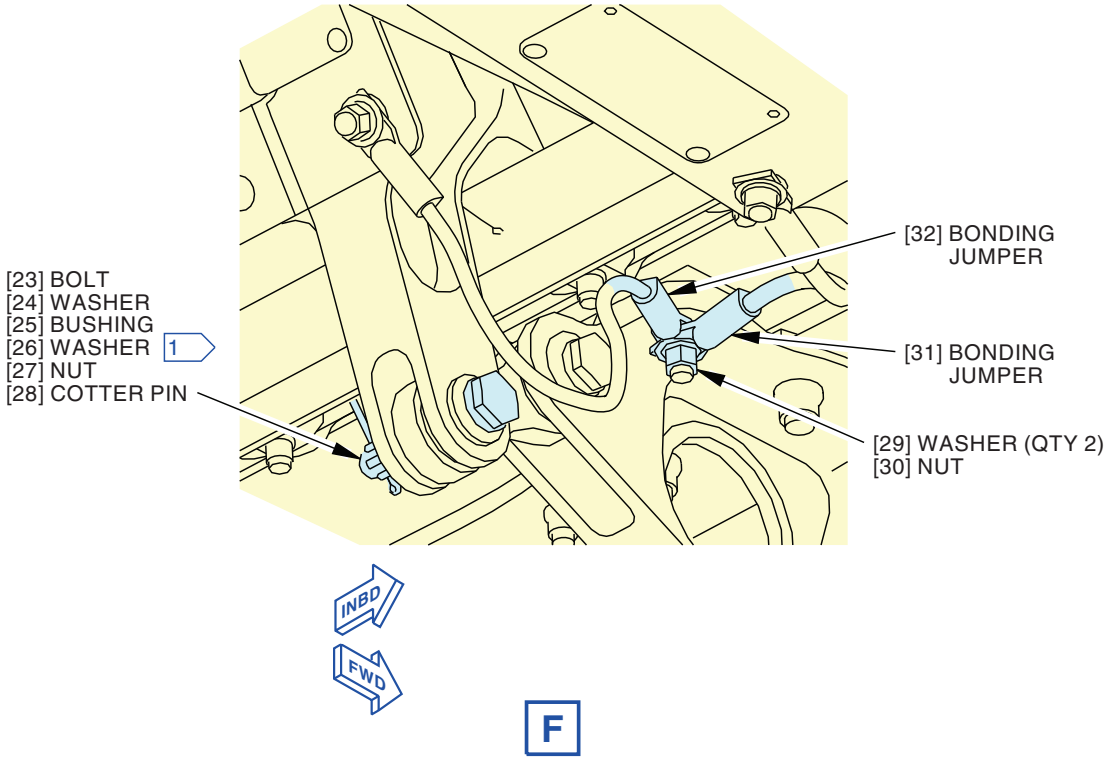




2413298 S00061532295_V5

Outboard Ground Spoiler Installation
Figure 401/27-62-91-990-801 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2413299 S00061532296_V5

Outboard Ground Spoiler Installation
Figure 401/27-62-91-990-801 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-62-91-400-801

3. Outboard Ground Spoiler Installation

(Figure 401)

A. References

Reference	Title
20-10-44-400-801	Lockwire, Cotter Pins, and Lockrings - Installation (P/B 401)
27-51-00-860-804	Retract the Trailing Edge Flaps (P/B 201)
27-62-00-800-801	Apply Pressure to the Speed Brake Hydraulic Systems A and B (P/B 201)
27-62-00-820-802	Ground Spoiler Adjustment (P/B 501)
27-62-00-840-801	Shut Down the Speed Brake Hydraulic Systems A and B (P/B 201)
27-62-00-840-802	Pressurization of the Speed Brake Hydraulic Systems A and B to its Usual Condition (P/B 201)
27-62-41-720-801	Ground Spoiler Control Module Functional Test (P/B 401)
27-62-94-400-801	Outboard Ground Spoiler Actuator Installation (P/B 401)
SWPM 20-20-00	ELECTRICAL BONDING PROCESSES
SWPM 20-20-10 Paragraph 4.C	Category 1 Unsealed and Category 6 Fillet Sealed Pre-Installed Ground Stud Assembly
SWPM 20-20-10 Paragraph 4.D	Category 2 Fay Sealed Pre-Installed Ground Stud Assembly
SWPM 20-20-10 Paragraph 4.E	Category 1 Unsealed and Category 6 Fillet Sealed Direct Ground Stud Assembly

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-614	Bonding Meters - Non-Intrinsically Safe (For use in outside Class I, Divisions I & II non-hazardous locations. For hazardous locations, use COM-1550). Part #: 247000 Supplier: 00426 Part #: 620LK Supplier: 1CRL2 Part #: BLR-0003-XX Supplier: KC432 Part #: BT51 Supplier: 00426 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659 Opt Part #: 247001 Supplier: 00426
SPL-1743	Ground Lock Set - Outboard Spoiler Actuators Part #: C27001-51 Supplier: 81205

C. Consumable Materials

Reference	Description	Specification
A50110	Sealant - Fuel Tank	BMS5-45 Class B-2
A50155	Sealant - Fuel Tank	BMS5-45 Class C
A50231	Sealant - Pressure And Environmental - Chromate Type	BMS5-95 Class B

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(Continued)

Reference	Description	Specification
A50296	Sealant - Pressure And Environmental - Chromate Type	BMS5-95 Class C
A50337	Sealant - Fuel Tank	BMS5-45 Class B
A50359	Sealant - Low Density, Non-Chromate Type	BMS5-142 Type II Class B
B00184	Solvent - Presealing, Cleaning Solvent	BMS11-7
C00175	Primer - Urethane Compatible, Corrosion Resistant	BMS10-79 Type III
C50075	Coating - Protective Enamel (BAC 707 Gray Color)	BMS10-60 Type II
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Outboard ground spoiler	27-62-91-01-100	PTF ALL
		27-62-91-01-105	PTF ALL
9	Cotter pin	27-62-91-01-030	PTF ALL
16	Cotter pin	27-62-91-01-035	PTF ALL
22	Cotter pin	27-62-91-01-035	PTF ALL
28	Cotter pin	27-62-91-01-030	PTF ALL
29	Washer	27-62-91-01-020	PTF ALL
30	Nut	27-62-91-01-025	PTF ALL

E. Location Zones

Zone	Area
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50
566	Left Wing - Spoiler No. 1
666	Right Wing - Spoiler No. 12

F. Outboard Ground Spoiler Installation

SUBTASK 27-62-91-420-001

(1) Install the outboard ground spoiler [1]:

(a) If it is necessary, use this task during installation: Lockwire, Cotter Pins, and Lockrings - Installation, TASK 20-10-44-400-801.

(b) Put the outboard ground spoiler [1] in its position on the airplane.

(c) Install the bolt [17] with the head inboard, washer [18], bushings [19], washer [20], nut [21], and cotter pin [22] in the center inboard hinge fitting.

(d) Install the bolt [11] with the head outboard, washer [12], bushing [13], washer [14], nut [15], and cotter pin [16] in the center outboard hinge fitting.

(e) Install the bolt [23] with the head inboard, washer [24], bushing [25], washer [26], nut [27], and cotter pin [28] in the inboard hinge.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(f) Install the bolt [4] with the head outboard, washer [5], bushing [6], washer [7], nut [8], and cotter pin [9] in the outboard hinge fitting.

(g) Install the washer [3] and nut [2] to connect the bonding jumper [10] to the structure.

1) Use cotton wiper, G00034, moist in solvent, B00184, to clean the mating surfaces of the airplane structure, nut [2], washer [3], and bonding jumper [10].

2) Method 1 (preferred). Apply sealant, A50110, sealant, A50155, sealant, A50231, or sealant, A50296, to the mating surfaces of the ground stud, nut [2], washer [3], and bonding jumper [10] (SWPM 20-20-10 Paragraph 4.D).

3) Install the bonding jumper [10], washer [3], and nut [2].

4) Tighten the nut [2] to 28 in-lb (3 N·m) - 35 in-lb (4 N·m).

a) Method 1. Make sure that there is squeeze out around all the interfaces of each component.

5) Measure the electrical resistance between the bonding jumper [10] and airplane structure (SWPM 20-20-00).

a) Use a non-intrinsically safe bonding meter, COM-614.

b) Make sure that the electrical bonding resistance is 0.001 ohm (1.0 milliohm) or less.

6) Method 2 (optional). Apply a fillet seal with sealant, A50337, sealant, A50231, or sealant, A50359, to cover the nut [2], washer [3], and bonding jumper [10] (SWPM 20-20-10 Paragraph 4.C).

(h) Install the new washers [29] and new nut [30] to connect the bonding jumper [31] and bonding jumper [32] to the structure.

1) Use cotton wiper, G00034, moist in solvent, B00184, to final clean the mating surfaces of the airplane structure, nut [30], washers [29], bonding jumper [31], and bonding jumper [32].

2) Install the nut [30], washers [29], bonding jumper [31], and bonding jumper [32].

3) Tighten the nut [30] to 28 in-lb (3 N·m) - 35 in-lb (4 N·m).

4) Measure the electrical resistance between the bonding jumper [32] and airplane structure (SWPM 20-20-00).

a) Use a non-intrinsically safe bonding meter, COM-614.

b) Make sure that the electrical bonding resistance is 0.001 ohm (1.0 milliohm) or less.

5) Apply a fillet seal with sealant, A50337, sealant, A50231, or sealant, A50359, to cover the nut [30], washers [29], bonding jumper [31], and bonding jumper [32] (SWPM 20-20-10 Paragraph 4.E).

6) Apply finish with primer, C00175, and coating, C50075.

SUBTASK 27-62-91-420-003

(2) If it is necessary, tighten the skin panel fasteners above the center hinge bolt [11] and bolt [17].

(a) Rotate the bolt [11] and bolt [17] to check the clearances between the cotter pin [16], cotter pin [22], and skin panel fasteners.

(b) If the cotter pin [16] and cotter pin [22] contact skin panel fasteners, flatten the head of the cotter pin [16] and cotter pin [22] as required to achieve the clearances.

NOTE: Visible defects in the cotter pins are acceptable.

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SUBTASK 27-62-91-980-001

(3) Move the outboard ground spoiler [1] through its full travel.

(a) Make sure that the outboard ground spoiler [1] moves freely.

SUBTASK 27-62-91-420-002

(4) Connect the rod end for the spoiler actuator assembly [33] to the outboard ground spoiler [1] (TASK 27-62-94-400-801).

SUBTASK 27-62-91-860-007

(5) Make sure that hydraulic power is off.

SUBTASK 27-62-91-080-001

(6) Remove the ground lock set, SPL-1743.

(a) Inspect the lockwire and locking key for damage after the spoiler tool is removed.

SUBTASK 27-62-91-720-002

(7) Unlatch the SPOILER ACTUATOR status message, do this task: Ground Spoiler Control Module Functional Test, TASK 27-62-41-720-801.

NOTE: The SPOILER ACTUATOR status message can be latched when the spoilers are locked in the up position.

SUBTASK 27-62-91-860-008

WARNING

KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(8) Do this task: Pressurization of the Speed Brake Hydraulic Systems A and B to its Usual Condition, TASK 27-62-00-840-802.

SUBTASK 27-62-91-860-014

(9) Do this task: Apply Pressure to the Speed Brake Hydraulic Systems A and B, TASK 27-62-00-800-801.

SUBTASK 27-62-91-860-009

(10) Move the speed brake lever to the DOWN position to lower the spoilers.

SUBTASK 27-62-91-860-013

(11) Do this task: Retract the Trailing Edge Flaps, TASK 27-51-00-860-804.

SUBTASK 27-62-91-860-010

(12) Do this task: Shut Down the Speed Brake Hydraulic Systems A and B, TASK 27-62-00-840-801.

SUBTASK 27-62-91-820-001

(13) Do this task: Ground Spoiler Adjustment, TASK 27-62-00-820-802.

SUBTASK 27-62-91-720-001

(14) Do this task: Outboard Ground Spoiler Test, TASK 27-62-91-000-802.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-62-91-000-802

4. Outboard Ground Spoiler Test

A. References

Reference	Title
27-61-00-840-801	Shut Down the Spoiler Hydraulic Systems A and B (P/B 201)
27-62-00-800-801	Apply Pressure to the Speed Brake Hydraulic Systems A and B (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1691	Protractor - Digital, Vertical or Horizontal Part #: 21465A82 Supplier: 3A054 Part #: 31-038-3 (Model Pro 360) Supplier: 1X822 Part #: 31-040-9 (Model Pro 3600) Supplier: 1X822 Part #: 950-317 (Model Pro 360) Supplier: 01EP7 Part #: 950-318 (Model Pro 3600) Supplier: 01EP7 Part #: KS5549 Supplier: 75245 Part #: KS6005 Supplier: 75245

C. Location Zones

Zone	Area
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50
566	Left Wing - Spoiler No. 1
666	Right Wing - Spoiler No. 12

D. Prepare for the Test

SUBTASK 27-62-91-860-012

WARNING MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. THE AILERONS, RUDDER, ELEVATORS, FLAPS, SLATS, SPOILERS, LANDING GEAR, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Do this task: Apply Pressure to the Speed Brake Hydraulic Systems A and B, TASK 27-62-00-800-801.

SUBTASK 27-62-91-480-002

CAUTION INSTALL THE PROTRACTOR ON THE SPOILER TOP SURFACE. IF THE PROTRACTOR INSTALLED ON THE BOTTOM OF THE SPOILER, DAMAGE TO AIRCRAFT CAN OCCUR.

(2) Install the protractor, COM-1691, on the upper surface of outboard ground spoilers 1 and 12.

SUBTASK 27-62-91-820-002

(3) Adjust the protractor, COM-1691, to 0°arc (0 rad).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Outboard Ground Spoiler Functional Test

SUBTASK 27-62-91-020-005

(1) Move the speed brake lever to the UP position to lift the spoilers.

SUBTASK 27-62-91-020-006

(2) Make sure that the spoilers No. 1 and 12 are 60 ±2°arc (1 ±0 rad) up from the retracted position.

SUBTASK 27-62-91-020-007

(3) Make sure that the ground spoilers No. 1 and 12 align in the limit of 2°arc (0 rad).

SUBTASK 27-62-91-020-008

(4) Move the speed brake lever to the DOWN position to lower the spoilers.

SUBTASK 27-62-91-820-003

(5) Make sure that the spoilers move to the fully retracted position.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 27-62-91-840-001

(1) Do this task: Shut Down the Spoiler Hydraulic Systems A and B, TASK 27-61-00-840-801.

SUBTASK 27-62-91-080-002

(2) Remove the protractor, COM-1691, from the upper surface of spoilers No. 1 and 12.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OUTBOARD GROUND SPOILER ACTUATOR - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the outboard ground spoiler actuator

(2) An installation of the outboard ground spoiler actuator.

B. In this procedure the outboard ground spoiler actuator is referred to as the "actuator".

TASK 27-62-94-000-801

2. Outboard Ground Spoiler Actuator Removal

(Figure 401)

A. References

Reference	Title
27-51-00-860-803	Extend the Trailing Edge Flaps (P/B 201)
27-62-00-800-801	Apply Pressure to the Speed Brake Hydraulic Systems A and B (P/B 201)
27-62-00-800-802	Depressurization of the Speed Brake Hydraulic Systems A and B (P/B 201)

B. Tools/Equipment

Reference	Description
STD-1110	Container - Hydraulic Fluid Resistant, 5 Gallon (19 Liter)

C. Location Zones

Zone	Area
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50
566	Left Wing - Spoiler No. 1
666	Right Wing - Spoiler No. 12

D. Prepare for the Removal

SUBTASK 27-62-94-860-002

WARNING

MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. AILERONS, RUDDER, ELEVATORS, FLAPS, SPOILERS, SLATS, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Do this task: Apply Pressure to the Speed Brake Hydraulic Systems A and B, TASK 27-62-00-800-801.

SUBTASK 27-62-94-860-003

(2) Put the SPOILER A and B switches to the OFF position to remove the hydraulic power from the flight spoilers.

NOTE: SPOILER A and B switches are on the flight control panel (P5-3).

SUBTASK 27-62-94-860-004

(3) Put the speed brake lever in the UP position to lift the ground spoilers.

SUBTASK 27-62-94-860-005

(4) Do this task: Extend the Trailing Edge Flaps, TASK 27-51-00-860-803.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-62-94-860-006

(5) Do this task: Depressurization of the Speed Brake Hydraulic Systems A and B, TASK 27-62-00-800-802.

E. Actuator Removal

SUBTASK 27-62-94-020-001

(1) Disconnect the piston rod end of the actuator [10] from the outboard ground spoiler [1]:

(a) Remove the pin [8], nut [7], washer [6], bushing [5], bushing [4], washer [19], washer [3], and bolt [2] from the piston rod end.

(b) Discard the pin [8].

SUBTASK 27-62-94-020-002

(2) Disconnect the hydraulic lines [9] from the actuator [10].

(a) Put a 5 gallon (19 liter) hydraulic fluid resistant container, STD-1110, below the actuator [10] to catch the hydraulic fluid.

(b) Disconnect the hydraulic lines [9] from the actuator [10].

(c) Turn the hydraulic lines [9] at the swivel joints to access the actuator [10].

(d) Install the plugs in the hydraulic lines [9].

SUBTASK 27-62-94-020-003

(3) Remove the bolts [13], washers [14], pillow blocks [11], and split ball bearings [12] from the structure.

SUBTASK 27-62-94-020-004

(4) Remove the actuator [10] from the airplane.

SUBTASK 27-62-94-860-001

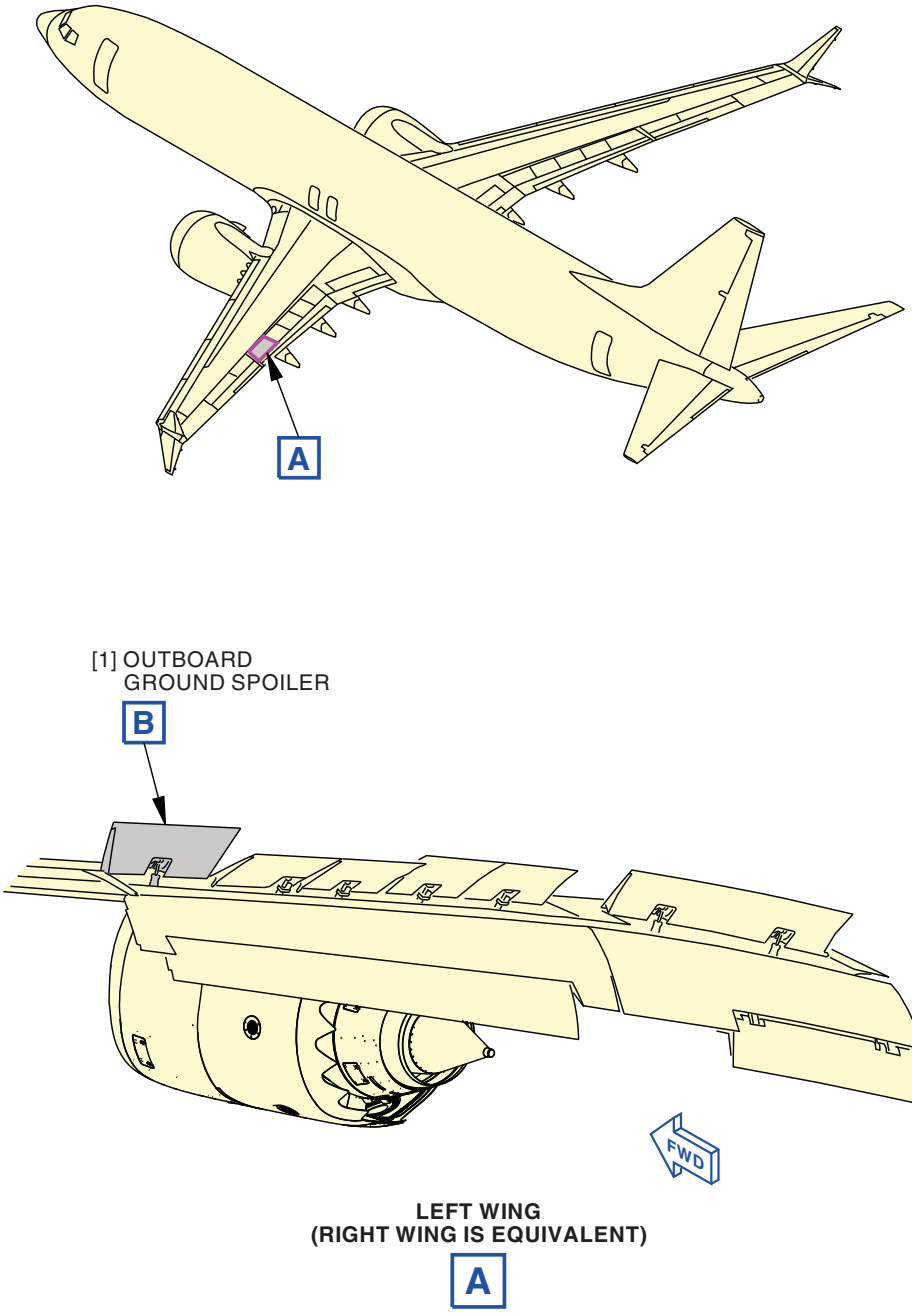
(5) Remove the restrictor [18], packing [17], union [15], and packing [16] from the hydraulic ports of the actuator [10].

SUBTASK 27-62-94-480-001

(6) Install caps in the hydraulic ports of the actuator [10].

——— END OF TASK ———

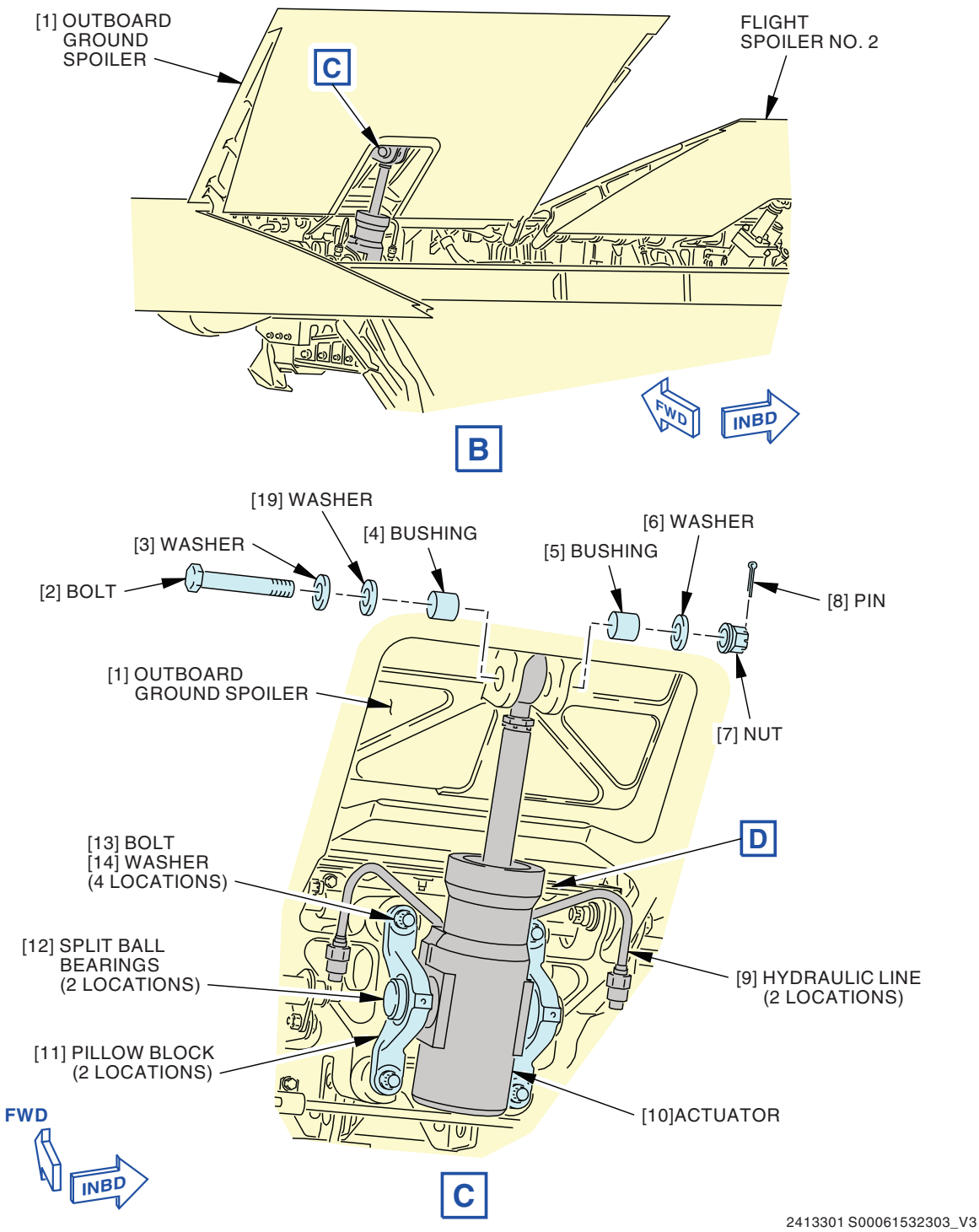
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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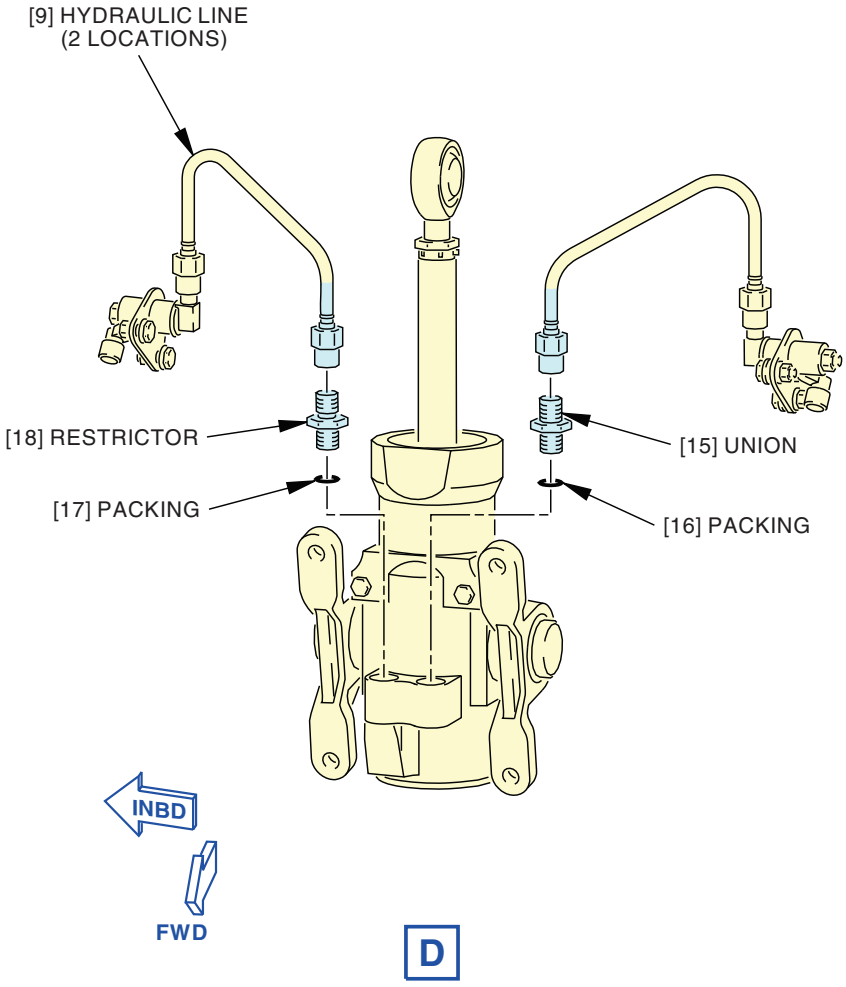
Outboard Ground Spoiler Actuator Installation
Figure 401/27-62-94-990-801 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Outboard Ground Spoiler Actuator Installation
Figure 401/27-62-94-990-801 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2413302 S00061532304_V2

Outboard Ground Spoiler Actuator Installation
Figure 401/27-62-94-990-801 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-62-94-400-801

3. Outboard Ground Spoiler Actuator Installation

(Figure 401)

A. References

Reference	Title
27-51-00-860-804	Retract the Trailing Edge Flaps (P/B 201)
27-62-00-800-801	Apply Pressure to the Speed Brake Hydraulic Systems A and B (P/B 201)
27-62-00-820-802	Ground Spoiler Adjustment (P/B 501)
27-62-00-840-801	Shut Down the Speed Brake Hydraulic Systems A and B (P/B 201)
27-62-00-840-802	Pressurization of the Speed Brake Hydraulic Systems A and B to its Usual Condition (P/B 201)
27-62-91-000-802	Outboard Ground Spoiler Test (P/B 401)

B. Consumable Materials

Reference	Description	Specification
D00054	Fluid - Hydraulic, Assembly Lubricant, MCS 352B	
D00153	Fluid - Hydraulic Fluid, Fire Resistant (Interchangeable And Intermixable With BMS 3-11 Type V)	BMS3-11 Type IV
D00633	Grease - Aircraft General Purpose	BMS3-33
G50237	Compound - Corrosion Inhibiting, Non-drying - Cor-Ban 27L	BMS3-38

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
8	Pin	27-62-94-01-125	PTF ALL
10	Actuator	27-62-94-01-155	PTF ALL
16	Packing	27-62-94-01-180	PTF ALL
17	Packing	27-62-94-01-180	PTF ALL

D. Location Zones

Zone	Area
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50
566	Left Wing - Spoiler No. 1
666	Right Wing - Spoiler No. 12

E. Actuator Installation

SUBTASK 27-62-94-420-001

(1) Remove the caps from the hydraulic ports.

SUBTASK 27-62-94-020-013

(2) Disconnect the union [15] and restrictor [18].

SUBTASK 27-62-94-020-014

(3) Remove and discard the packing [16] and packing [17].

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SUBTASK 27-62-94-640-001

(4) Apply MCS 352B fluid, D00054, or hydraulic fluid, D00153, to the union [15], restrictor [18], new packing [16], and new packing [17].

SUBTASK 27-62-94-420-004

(5) Install the packing [16], packing [17], union [15], and restrictor [18] in the hydraulic ports at the actuator [10].

SUBTASK 27-62-94-420-002

(6) Install the actuator [10]:

(a) Put the actuator [10] in its position and make sure that the hydraulic lines [9] align with the fittings.

(b) Install the pillow blocks [11] on the actuator [10]:

1) Apply grease, D00633, to the split ball bearings [12].

2) Assemble the split ball bearings [12].

3) Install the split ball bearings [12] in the pillow blocks [11].

4) Make sure that the split ball bearings [12] move freely.

WARNING USE NITRILE GLOVES FOR SKIN PROTECTION WHEN YOU USE COR-BAN 27L. IF IT GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER. GET MEDICAL AID. THIS MATERIAL CONTAINS FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

5) Apply Cor-Ban 27L Compound, G50237, to the bolts [13].

6) Install the bolts [13] and washers [14].

7) Tighten the bolts [13] to 141 in-lb (15.9 N·m) - 172 in-lb (19.4 N·m).

(c) Connect the piston rod to the outboard ground spoiler [1]:

1) Apply grease, D00633, to all surfaces of the bolt [2], bushing [4], and bushing [5].

2) Install the bolt [2], washer [3], washer [19], bushing [4], bushing [5], washer [6], and nut [7] in the piston rod.

3) Tighten the nut [7] to 660 in-lb (74.6 N·m) - 780 in-lb (88.1 N·m).

4) Install the new pin [8].

SUBTASK 27-62-94-420-003

(7) Connect the hydraulic lines [9] to the actuator [10]:

(a) Remove the plugs from the hydraulic lines [9].

(b) Apply hydraulic fluid, D00153, or MCS 352B fluid, D00054, to the hydraulic fittings.

(c) Connect the hydraulic lines [9] to the union [15] and restrictor [18] on the actuator [10].

(d) Make sure that the actuator [10] is installed with the actuator ports connected to the hydraulic lines [9] that face towards the spoiler panel.

(e) Tighten the B-nuts on the hydraulic lines [9] to 140 ±7 in-lb (16 ±1 N·m).

SUBTASK 27-62-94-860-007

(8) Do this task: Pressurization of the Speed Brake Hydraulic Systems A and B to its Usual Condition, TASK 27-62-00-840-802.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-62-94-860-008

(9) Do this task: Apply Pressure to the Speed Brake Hydraulic Systems A and B, TASK 27-62-00-800-801.

SUBTASK 27-62-94-870-001

(10) Move the speed brake control lever to bleed the spoiler actuators.

SUBTASK 27-62-94-700-001

(11) Make sure that there is no leakage of hydraulic fluid at the actuator [10].

SUBTASK 27-62-94-820-001

(12) Do this task: Ground Spoiler Adjustment, TASK 27-62-00-820-802.

SUBTASK 27-62-94-820-002

(13) Do this task: Outboard Ground Spoiler Test, TASK 27-62-91-000-802.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 27-62-94-860-009

(1) Do this task: Retract the Trailing Edge Flaps, TASK 27-51-00-860-804.

SUBTASK 27-62-94-860-010

(2) Move the speed brake lever to the DOWN position to lower the ground spoilers.

SUBTASK 27-62-94-860-011

(3) Do this task: Shut Down the Speed Brake Hydraulic Systems A and B, TASK 27-62-00-840-801.

SUBTASK 27-62-94-860-012

(4) Put the SPOILER A and B switches in the ON position.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OUTBOARD GROUND SPOILER ACTUATOR - ADJUSTMENT/TEST

1. General

A. This Procedure has this task:

(1) A functional test of the outboard ground spoiler actuator.

TASK 27-62-94-720-801

2. Outboard Ground Spoiler Actuator Test

A. References

Reference	Title
27-61-00-840-801	Shut Down the Spoiler Hydraulic Systems A and B (P/B 201)
27-62-00-800-801	Apply Pressure to the Speed Brake Hydraulic Systems A and B (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1691	Protractor - Digital, Vertical or Horizontal Part #: 21465A82 Supplier: 3A054 Part #: 31-038-3 (Model Pro 360) Supplier: 1X822 Part #: 31-040-9 (Model Pro 3600) Supplier: 1X822 Part #: 950-317 (Model Pro 360) Supplier: 01EP7 Part #: 950-318 (Model Pro 3600) Supplier: 01EP7 Part #: KS5549 Supplier: 75245 Part #: KS6005 Supplier: 75245

C. Location Zones

Zone	Area
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50
566	Left Wing - Spoiler No. 1
666	Right Wing - Spoiler No. 12

D. Prepare for the Test

SUBTASK 27-62-94-860-014

WARNING

KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Do this task: Apply Pressure to the Speed Brake Hydraulic Systems A and B, TASK 27-62-00-800-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-62-94-480-002

CAUTION

INSTALL THE PROTRACTOR ON THE SPOILER TOP SURFACE. IF THE PROTRACTOR INSTALLED ON THE BOTTOM OF THE SPOILER, DAMAGE TO AIRCRAFT CAN OCCUR.

(2) Install the protractor, COM-1691, on the upper surface of outboard ground spoilers 1 and 12.

SUBTASK 27-62-94-820-003

(3) Adjust the protractor, COM-1691 to zero.

E. Outboard Ground Spoiler Actuator Functional Test

SUBTASK 27-62-94-020-005

(1) Move the speed brake lever to the UP position to lift the spoilers.

SUBTASK 27-62-94-020-006

(2) Make sure the spoilers No. 1 and 12 are 60 +/-2 degrees up from the retracted position.

SUBTASK 27-62-94-020-007

(3) Make sure the ground spoilers No. 1 and 12 align in the limit of 2 degrees.

SUBTASK 27-62-94-020-008

(4) Move the speed brake lever to the DOWN position to lower the spoilers.

SUBTASK 27-62-94-820-004

(5) Make sure the spoilers move to the fully retracted position.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 27-62-94-410-001

(1) Do this task: Shut Down the Spoiler Hydraulic Systems A and B, TASK 27-61-00-840-801.

SUBTASK 27-62-94-080-001

(2) Remove the protractor, COM-1691, from the upper surface of the spoilers No. 1 and 12.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OUTBOARD GROUND SPOILER ACTUATOR - INSPECTION/CHECK

1. General

A. This procedure contains a task to do a check of the outboard ground spoiler actuator.

B. This procedure does not remove or install the outboard ground spoiler actuator, refer to the Outboard Ground Spoiler Actuator - Removal/Installation for procedures to remove and install the outboard ground spoiler actuator.

TASK 27-62-94-200-801

2. Outboard Ground Spoiler Actuator Inspection

(Figure 601)

A. General

(1) This procedure gives wear limits and repair data for the outboard ground spoiler actuator.

B. References

Reference	Title
27-62-94-000-801	Outboard Ground Spoiler Actuator Removal (P/B 401)
27-62-94-400-801	Outboard Ground Spoiler Actuator Installation (P/B 401)

C. Tools/Equipment

Reference	Description
STD-1096	Micrometer - Depth, 0-1 inch, Readable to 1/1000 inch
STD-1097	Caliper - Vernier, 0-6 inch, Readable to 1/1000 inch

D. Location Zones

Zone	Area
566	Left Wing - Spoiler No. 1
666	Right Wing - Spoiler No. 12

E. Procedure

SUBTASK 27-62-94-020-009

(1) Remove the outboard ground spoiler actuator. To remove it, do this task: Outboard Ground Spoiler Actuator Removal, TASK 27-62-94-000-801.

SUBTASK 27-62-94-020-010

(2) Use these tools to check for wear on the parts of the outboard ground spoiler actuator:

(a) micrometer (0-1 inch, readable to 1/1000 inch), STD-1096

(b) vernier caliper (0 - 6 inch, readable to 1/1000 inch), STD-1097

SUBTASK 27-62-94-020-011

(3) Examine the parts of the outboard ground spoiler actuator for worn areas:

(a) Remove the parts to be checked.

(b) Measure the bolts, bearing, and bushings as necessary with a micrometer (0-1 inch, readable to 1/1000 inch), STD-1096 or a vernier caliper (0 - 6 inch, readable to 1/1000 inch), STD-1097.

(c) Compare the dimensions you measured with the permitted wear dimensions that follow.

(d) If the parts are not in the tolerance, replace the part as required.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Table 601/27-62-94-993-801 WEAR LIMITS TABLE

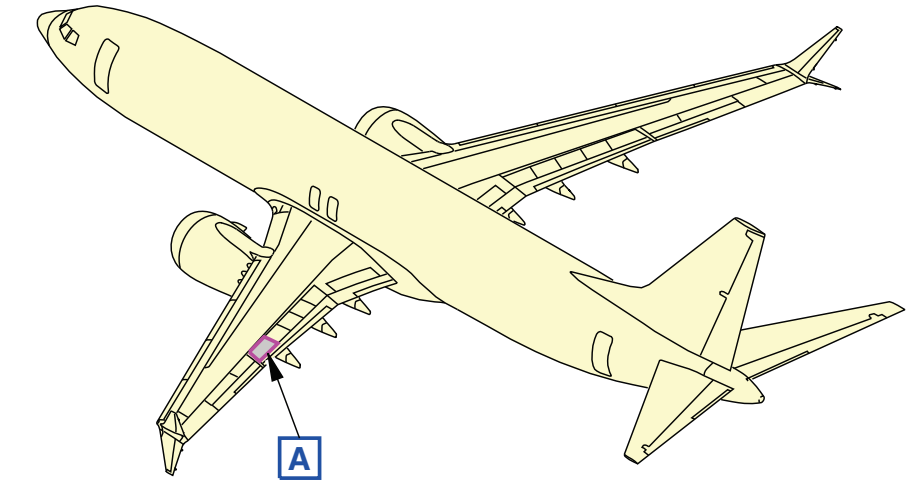
NAME OF THE PARTS THAT ARE IN CONTACT	DIM.	DIAMETER DESIGN LIMITS		WEAR LIMITS	
		MINIMUM inches/mm	MAXIMUM inches/mm	PERMITTED MAXIMUM I.D./ MINIMUM O.D. OF WORN PART	MAXIMUM SERVICE DIAMETRIC CLEARANCE
#1 BUSHING	I.D.	0.7500 in 19.05 mm	0.7515 in 19.09 mm	0.7535 in 19.14 mm	0.0050 in 0.13 mm
#1 BUSHING	O.D.	0.7485 in 19.01 mm	0.7490 in 19.02 mm	0.7470 in 18.97 mm	
#2 BUSHING	I.D.	0.6245 in 15.86 mm	0.6250 in 15.88 mm	0.6255 in 15.89 mm	0.0025 in 0.06 mm
#2 BOLT	O.D.	0.6230 in 15.82 mm	0.6240 in 15.85 mm	0.6226 in 15.81 mm	
#3 BEARING	I.D.	0.6245 in 15.86 mm	0.6250 in 15.88 mm	0.6255 in 15.89 mm	0.0025 in 0.06 mm
#3 BOLT	O.D.	0.6230 in 15.82 mm	0.6240 in 15.85 mm	0.6226 in 15.81 mm	

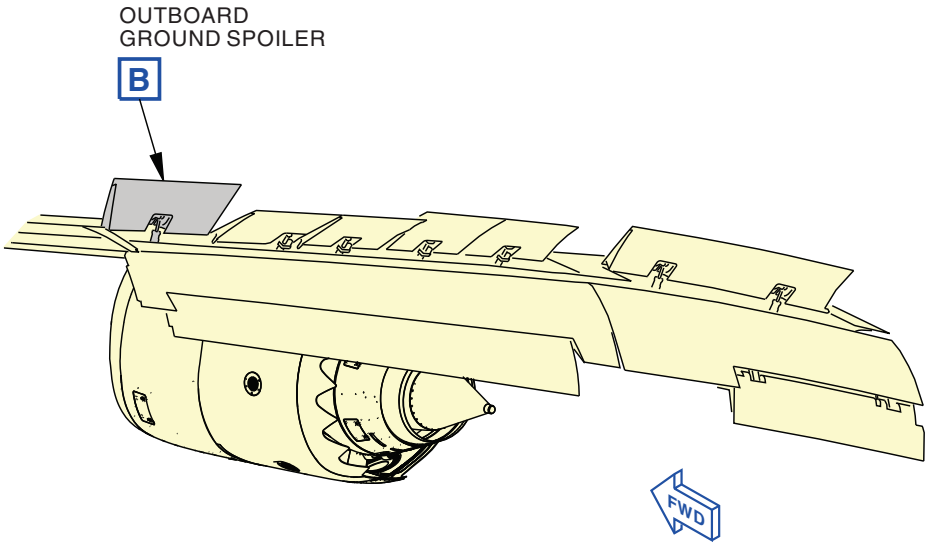
SUBTASK 27-62-94-020-012

(4) Install the outboard ground spoiler actuator. To install it, do this task: Outboard Ground Spoiler Actuator Installation, TASK 27-62-94-400-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details





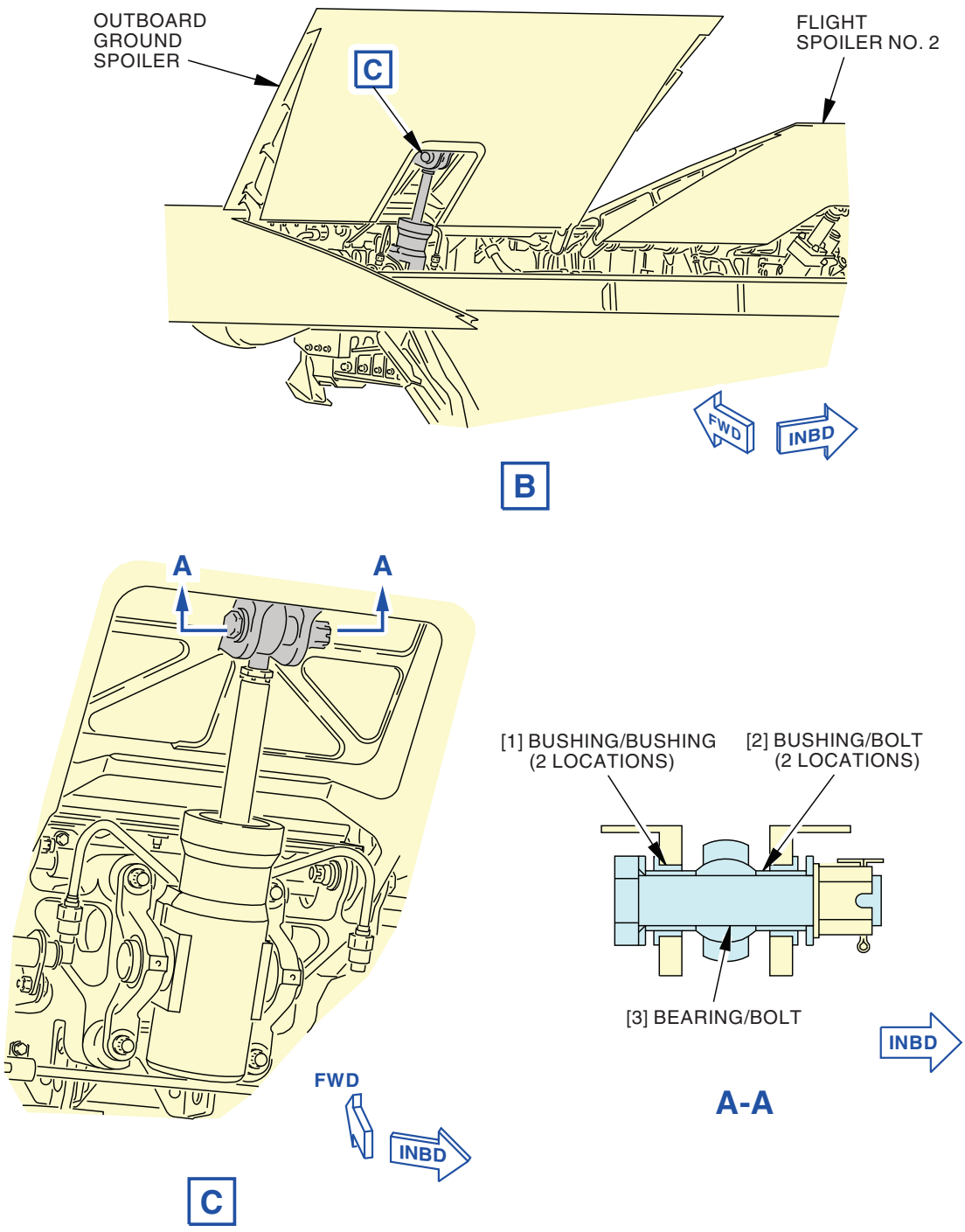
LEFT WING
(RIGHT WING IS EQUIVALENT)

A

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Outboard Ground Spoiler Actuator Wear Limits
Figure 601/27-62-94-990-802 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Outboard Ground Spoiler Actuator Wear Limits
Figure 601/27-62-94-990-802 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

LEADING EDGE FLAP AND SLAT CONTROL SYSTEM - MAINTENANCE PRACTICES

1. General

A. This procedure contains these tasks for the leading edge (LE) flap and slat control system:

(1) A task to deactivate the leading edge flaps and slats

(2) A task to reactivate the leading edge flaps and slats

(3) A task to operate the leading edge flaps and slats with primary control

(4) A task to operate the leading edge flaps and slats with alternate control

(5) A task to extend the leading edge flaps and slats

(6) A task to retract the leading edge flaps and slats

(7) A task to manually extend the leading edge flaps and slats

(8) A task to manually retract the leading edge flaps and slats

(9) A task to install the leading edge flap and slat locks

(10) A task to remove the leading edge flap and slat locks

TASK 27-81-00-040-801

2. Leading Edge Flaps and Slats - Deactivation

(Figure 201, Figure 202, Figure 203)

A. General

WARNING	MAKE SURE THAT FOR AIRPLANES WITH THE -1001 CRUISE DEPRESSURIZATION VALVE, YOU REVIEW THIS WARNING. MAKE SURE THAT YOU CLOSE THE FAN COWLS AND THRUST REVERSERS IF YOU MOVE THE TRAILING EDGE FLAPS WHILE THE LEADING EDGE FLAPS AND SLATS ARE DISCONNECTED. THE -1001 CRUISE DEPRESSURIZATION CREEP CAN CAUSE MOVEMENT OF THE LEADING EDGE FLAPS AND SLATS WHEN YOU MOVE THE FLAP CONTROL LEVER. IF YOU DO NOT OBEY, INJURY TO PERSONNEL OR DAMAGE TO THE EQUIPMENT CAN OCCUR.

WARNING	MAKE SURE ALL PERSONS AND EQUIPMENT ARE CLEAR OF THE FLIGHT CONTROLS SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

CAUTION	MAKE SURE THAT THE INBOARD FAN COWL PANEL AND THE INBOARD AND OUTBOARD THRUST REVERSERS ARE CLOSED AND IN THE STOWED POSITION OR REMOVED BEFORE YOU EXTEND THE LEADING EDGE FLAPS AND SLATS. THERE IS NOT SUFFICIENT CLEARANCE FOR THE FLAPS AND SLATS TO EXTEND IF THE INBOARD FAN COWL PANEL AND THE INBOARD AND OUTBOARD THRUST REVERSERS ARE IN THE OPEN POSITION. THIS CAN CAUSE DAMAGE TO EQUIPMENT.

(1) Before you do maintenance on the Leading Edge (LE) flaps and slats, you must deactivate the LE control surfaces. This is to prevent injury to persons and damage to equipment because of accidental operation of the LE control surfaces.

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(2) A LE device creep condition has been associated with the -1001 cruise depressurization valve. It can cause hydraulic pressure leakage through the valve when manually closed.

(a) This cruise depress creep condition can permit movement of the LE flaps and slats when the Trailing Edge (TE) flaps are moved.

(b) The redesigned -1003 cruise depressurization valve is not susceptible to this creep condition.

B. References

Reference	Title
20-60-07-910-802	Inspection, Cleaning and Sealing of Electrical Connectors (P/B 201)
27-51-00-860-804	Retract the Trailing Edge Flaps (P/B 201)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
29-21-00-000-802	Standby Hydraulic System Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-09-00-860-802	Return the Airplane to the Ground Mode (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
SWPM 20-60-03	Special Protection of Electrical Connectors

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1675	Lock Assembly - Flight Controls (used on valve p/n 10-60581-2 and 10-60581-3) Part #: F80049-83 Supplier: 81205 Opt Part #: F80049-56 Supplier: 81205 Opt Part #: F80049-65 Supplier: 81205
SPL-1734	Lockout Set - Leading Edge Slats and Krueger Flaps Actuators Part #: C27051-29 Supplier: 81205
SPL-1746	Pin - Lockout, L.E. Cruise Depressurization Valve Part #: C27038-6 Supplier: 81205
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right
512	Left Wing - Krueger Flap No. 2
513	Left Wing - Krueger Flap No. 1
522	Left Wing - Slat No. 4

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Zone	Area
523	Left Wing - Slat No. 3
524	Left Wing - Slat No. 2
525	Left Wing - Slat No. 1
612	Right Wing - Krueger Flap No. 3
613	Right Wing - Krueger Flap No. 4
622	Right Wing - Slat No. 5
623	Right Wing - Slat No. 6
624	Right Wing - Slat No. 7
625	Right Wing - Slat No. 8

E. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
192BR	ECS Ram Air Inlet Mixing Duct Panel - Forward

F. Prepare for the Deactivation

SUBTASK 27-81-00-480-001

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 27-81-00-860-001

(2) Make sure that the TE flaps and the LE flaps and slats are in the retracted position.

(a) If they are not in the retracted position, do this task: Retract the Trailing Edge Flaps, TASK 27-51-00-860-804.

SUBTASK 27-81-00-860-002

(3) If hydraulic systems A and B are pressurized, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 27-81-00-860-003

(4) If the standby hydraulic system is pressurized, do this task: Standby Hydraulic System Power Removal, TASK 29-21-00-000-802.

SUBTASK 27-81-00-860-107

(5) Make sure that the Proximity Sensor Electronics Unit (PSEU) is in GROUND MODE (TASK 32-09-00-860-802).

G. Deactivate the Leading Edge Flaps and Slats in the Retract (UP) Position

SUBTASK 27-81-00-860-004

(1) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	7	C00210	FLIGHT CONTROL FLAP SHUTOFF VALVES

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-81-00-860-005

(2) Make sure that the ALTERNATE FLAPS ARM switch is in the OFF position.

SUBTASK 27-81-00-210-001

(3) Attach a DO NOT OPERATE tag, STD-858, on the flap control lever and the ALTERNATE FLAPS switches.

SUBTASK 27-81-00-860-006

WARNING

MAKE SURE THAT FOR AIRPLANES WITH THE -1001 CRUISE DEPRESSURIZATION VALVE, YOU REVIEW THIS WARNING. MAKE SURE THAT YOU CLOSE THE FAN COWLS AND THRUST REVERSERS IF YOU MOVE THE TRAILING EDGE FLAPS WHILE THE LEADING EDGE FLAPS AND SLATS ARE DISCONNECTED. THE -1001 CRUISE DEPRESSURIZATION CREEP CAN CAUSE MOVEMENT OF THE LEADING EDGE FLAPS AND SLATS WHEN YOU MOVE THE FLAP CONTROL LEVER. IF YOU DO NOT OBEY, INJURY TO PERSONNEL OR DAMAGE TO THE EQUIPMENT CAN OCCUR.

(4) Do these steps to move the manual override lever (handle) on the LE cruise depressurization valve to the OFF position (Figure 201):

WARNING

MAKE SURE THAT THE TAB HAS BEEN CUT FROM THE VALVE. THE DEPRESSURIZATION VALVE HANDLE CAN MOVE QUICKLY. IF YOU DO NOT OBEY, INJURY TO PERSONNEL CAN OCCUR.

(a) If the valve has a tab, cut the tab from the valve.

NOTE: For more information, please refer to Service Bulletin 737–27–1242.

(b) Remove the lockwire from the manual override lever on the LE cruise depressurization valve.

(c) Move the manual override lever on the LE cruise depressurization valve to the OFF position.

NOTE: Do not force movement of the lever with tools.

1) If you cannot move the manual override lever, use the Flap/Slat Electronics Unit (FSEU) to operate the LE cruise depressurization valve:

NOTE: Two persons will be necessary when the FSEU is used to operate the LE cruise depressurization valve.

a) To get access to the main equipment center, do this step:

<1> Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

b) Close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	7	C00210	FLIGHT CONTROL FLAP SHUTOFF VALVES

c) Push the ON/OFF button on the front panel of the FSEU to turn on the display.

NOTE: The display will show EXISTING FAULTS?

d) Push the NO button until OTHER FUNCTNS? shows on the display.

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e) Push the YES button to select OTHER FUNCTNS? on the FSEU.

f) Push the YES button to select SET OUTPUTS? from the OTHER FUNCTNS? menu.

g) Push the YES button to select LE DEPR VALVE? from the SET OUTPUTS? menu.

h) Push the YES button at the SET ON? display.

NOTE: This will drive the valve to the closed position.

WARNING

MAKE SURE THAT YOU MOVE THE MANUAL OVERRIDE HANDLE TO THE OFF POSITION AND INSTALL THE RIG PIN IN LESS THAN 75 SECONDS. THE FSEU WILL AUTOMATICALLY CHANGE THE POSITION OF THE VALVE AND THE OVERRIDE HANDLE AFTER 75 SECONDS. THE MANUAL OVERRIDE HANDLE CAN MOVE SUDDENLY AND CAUSE INJURY TO PERSONS.

i) Move the manual override lever (handle) on the LE cruise depressurization valve to the OFF position.

NOTE: Do not force movement of the lever with tools.

j) If you still cannot move the manual override lever (handle) on the LE cruise depressurization valve, do these steps:

<1> Move the flaps and slats to the fully retracted position, if necessary.

WARNING

KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

<2> Supply system B hydraulic power, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

<3> Move the flap control lever to the UP position to retract the flaps and slats (if not retracted).

<4> Remove system B hydraulic power, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

<5> Use the FSEU to operate the valve again.

<6> Move the manual override lever (handle) on the LE cruise depressurization valve to the OFF position.

NOTE: Do not force movement of the lever with tools.

k) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	7	C00210	FLIGHT CONTROL FLAP SHUTOFF VALVES

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(d) Install the lockout pin, SPL-1746, in the manual override lever (handle) of the cruise depressurization valve to keep the handle in the OFF position.

NOTE: If the newer -1003 cruise depressurization valve is installed on the airplane, then the lockout pin, SPL-1746, will stop the movement of the LE flaps and slats when hydraulic power is supplied.

NOTE: If the older -1001 cruise depressurization valve is installed on the airplane, then the LE flaps and slats are subject to the cruise depress creep condition when hydraulic power is supplied and the TE flaps are moved.

1) If the lockout pin, SPL-1746, cannot be easily installed then it is necessary to supply system B hydraulic power and then use the FSEU to relieve the pressure on the LE depressurization valve.

2) Attach the DO NOT OPERATE tag, STD-858, on the lockout pin, SPL-1746, after installed in the LE cruise depressurization valve.

SUBTASK 27-81-00-480-002

(5) To install the flight controls lock assembly (lock, SPL-1675), to prevent LE devices extension by the standby hydraulic system, do these steps:

(a) Disconnect the electrical connector from the LE standby drive shutoff valve.

(b) After you disconnect the electrical connector, examine the connector for corrosion (TASK 20-60-07-910-802).

NOTE: The inspection of the electrical connector can be done after the removal step or before the installation step of the electrical connector. This is an operator option.

(c) If there was corrosion, correct the problem (SWPM 20-60-03).

WARNING

MAKE SURE THAT THE MANUAL OVERRIDE LEVER IS IN POSITION 1 TO PREVENT THE MOVEMENT OF THE LEADING EDGE SLATS. IF THE LEADING EDGE SLATS MOVE, DAMAGE TO AN OPEN THRUST REVERSER AND INJURIES TO PERSONNEL CAN OCCUR.

(d) Make sure that the manual override lever (handle) at the LE standby drive shutoff valve is in the Position 1 (standby hydraulic power OFF).

NOTE: The LE slats will extend fully if the manual override lever is in the Position 2 and the standby hydraulic system is pressurized.

WARNING

MAKE SURE TO INSTALL THE GROUND LOCK ON THE STANDBY-DRIVE SHUTOFF-VALVE FOR THE LEADING EDGE. THIS PREVENTS ACCIDENTAL STANDBY SYSTEM OPERATION OF THE LEADING EDGE FLAPS AND SLATS. THE LEADING EDGE FLAPS AND SLATS CAN MOVE QUICKLY. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(e) Install the flight controls lock assembly (lock, SPL-1675), on the LE standby drive shutoff valve.

(f) Install the detent pin.

SUBTASK 27-81-00-700-001

(6) Do the steps in 'Leading Edge Flaps and Slats - Tryout'.

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H. Leading Edge Flaps and Slats Deactivation in the Extend (Intermediate) Position

SUBTASK 27-81-00-860-125

> **CAUTION**
>
> MAKE SURE THAT THE INBOARD FAN COWL PANEL AND THE INBOARD AND OUTBOARD THRUST REVERSERS ARE CLOSED, AND IN THE STOWED POSITION, OR REMOVED BEFORE YOU EXTEND THE LEADING EDGE FLAPS AND SLATS. THERE IS NOT SUFFICIENT CLEARANCE FOR THE FLAPS AND SLATS TO EXTEND IF THE INBOARD FAN COWL PANEL AND THE INBOARD AND OUTBOARD THRUST REVERSERS ARE IN THE OPEN POSITION. THIS CAN CAUSE DAMAGE TO EQUIPMENT.

(1) Make sure that the thrust reversers and fan cowls are closed.

(a) If necessary, do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

(b) If necessary, do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

SUBTASK 27-81-00-860-007

(2) If the LE flaps and slats are not in the extend position, do this task: Leading Edge Flaps and Slats Extension, TASK 27-81-00-860-803.

SUBTASK 27-81-00-860-008

(3) Remove hydraulic system power, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 27-81-00-860-104

(4) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	7	C00210	FLIGHT CONTROL FLAP SHUTOFF VALVES

SUBTASK 27-81-00-210-002

(5) Attach the DO NOT OPERATE tags, STD-858, on the flap control lever and the ALTERNATE FLAPS switches.

SUBTASK 27-81-00-860-009

> **WARNING**
>
> MAKE SURE THAT FOR AIRPLANES WITH THE -1001 CRUISE DEPRESSURIZATION VALVE, YOU REVIEW THIS WARNING. MAKE SURE THAT YOU CLOSE THE FAN COWLS AND THRUST REVERSERS IF YOU MOVE THE TRAILING EDGE FLAPS WHILE THE LEADING EDGE FLAPS AND SLATS ARE DISCONNECTED. THE -1001 CRUISE DEPRESSURIZATION CREEP CAN CAUSE MOVEMENT OF THE LEADING EDGE FLAPS AND SLATS WHEN YOU MOVE THE FLAP CONTROL LEVER. IF YOU DO NOT OBEY, INJURY TO PERSONNEL OR DAMAGE TO THE EQUIPMENT CAN OCCUR.

(6) Do these steps to move the manual override lever (handle) on the LE cruise depressurization valve to the OFF position (Figure 201):

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING MAKE SURE THAT THE TAB HAS BEEN CUT FROM THE VALVE. THE DEPRESSURIZATION VALVE HANDLE CAN MOVE QUICKLY. IF YOU DO NOT OBEY, INJURY TO PERSONNEL CAN OCCUR.

(a) If the valve has a tab, cut the tab from the valve.

NOTE: For more information, please refer to Service Bulletin 737–27–1242.

(b) Remove the lockwire from the manual override lever on the LE cruise depressurization valve.

(c) Move the manual override lever on the LE cruise depressurization valve to the OFF position.

NOTE: Do not force movement of the lever with tools.

1) If you cannot move the manual override lever, use the FSEU to operate the LE cruise depressurization valve:

NOTE: Two persons will be necessary when the FSEU is used to operate the LE cruise depressurization valve.

a) To get access to the main equipment center, do this step:

<1> Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

b) Close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	7	C00210	FLIGHT CONTROL FLAP SHUTOFF VALVES

c) Push the ON/OFF button on the front panel of the FSEU to turn on the display.

NOTE: The display will show EXISTING FAULTS?

d) Push the NO button until OTHER FUNCTNS? shows on the display.

e) Push the YES button to select OTHER FUNCTNS? on the FSEU.

f) Push the YES button to select SET OUTPUTS? from the OTHER FUNCTNS? menu.

g) Push the YES button to select LE DEPR VALVE? from the SET OUTPUTS? menu.

h) Push the YES button at the SET ON? display.

NOTE: This will drive the valve to the closed position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING MAKE SURE THAT YOU MOVE THE MANUAL OVERRIDE HANDLE TO THE OFF POSITION AND INSTALL THE RIG PIN IN LESS THAN 75 SECONDS. THE FSEU WILL AUTOMATICALLY CHANGE THE POSITION OF THE VALVE AND THE OVERRIDE HANDLE AFTER 75 SECONDS. THE MANUAL OVERRIDE HANDLE CAN MOVE SUDDENLY AND CAUSE INJURY TO PERSONS.

i) Move the manual override lever (handle) on the LE cruise depressurization valve to the OFF position.

NOTE: Do not force movement of the lever with tools.

j) If you still cannot move the manual override lever (handle) on the LE cruise depressurization valve, do these steps:

<1> Move the flaps and slats to the fully retracted position, if necessary.

WARNING KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

<2> Supply system B hydraulic power, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

<3> Move the flap control lever to the UP position to retract the flaps and slats (if not retracted).

<4> Remove system B hydraulic power, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

<5> Use the FSEU to operate the valve again.

<6> Move the manual override lever (handle) on the LE cruise depressurization valve to the OFF position.

NOTE: Do not force movement of the lever with tools.

k) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	7	C00210	FLIGHT CONTROL FLAP SHUTOFF VALVES

(d) Install the lockout pin, SPL-1746, in the manual override lever (handle) of the LE cruise depressurization valve to keep the lever in the OFF position.

NOTE: If the newer -1003 cruise depressurization valve is installed on the airplane, then the lockout pin, SPL-1746, will stop the movement of the LE flaps and slats when hydraulic power is supplied.

NOTE: If the older -1001 cruise depressurization valve is installed on the airplane, then the LE flaps and slats are subject to the cruise depress creep condition when hydraulic power is supplied and the TE flaps are moved.

1) If the lockout pin, SPL-1746, cannot be easily installed then it is necessary to supply system B hydraulic power and then use the FSEU to relieve the pressure on the LE depressurization valve.

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a) Use the FSEU procedure below but replace the selection of the LE Uncommanded Motion (UCM) shutoff valve for the LE DEPR valve.

2) Attach the DO NOT OPERATE tag, STD-858, on the lockout pin, SPL-1746, after it is installed in the LE cruise depressurization valve.

SUBTASK 27-81-00-010-001

(7) Open this access panel to get access to the LE UCM shutoff valve:

Number	Name/Location
192BR	ECS Ram Air Inlet Mixing Duct Panel - Forward

SUBTASK 27-81-00-480-005

(8) Remove the lockwire from the manual override lever (handle) on the LE UCM shutoff valve (Figure 203).

SUBTASK 27-81-00-480-011

(9) Move the manual override lever on the LE UCM shutoff valve to the UP (OFF) position.

NOTE: The lever should move with very little force. If the lever is tight, you may need hydraulic pressure and the FSEU to release the pressure on the LE UCM SHUTOFF VALVE.

(a) If necessary, do the steps that follow:

1) Supply system B hydraulic power (TASK 29-11-00-860-801).

2) Use the FSEU to release the pressure on the LE UCM shutoff valve, as follows:

NOTE: Two persons will be necessary when the FSEU is used to release the pressure on the LE UCM shutoff valve.

a) To get access to the main equipment center, do this step:

<1> Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

b) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	7	C00210	FLIGHT CONTROL FLAP SHUTOFF VALVES

c) Push the ON/OFF button.

d) EXISTING FAULTS.....press NO.

e) Press NO until OTHER FUNCTIONS.

f) OTHER FUNCTIONS....press YES.

g) SET OUTPUTS.....press YES.

h) Press NO until LE UCM SHUTOFF VALVE.

i) LE UCM SHUTOFF VALVE....press YES.

j) SET ON.....press YES.

k) Remove system B hydraulic power (TASK 29-11-00-860-805).

l) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	7	C00210	FLIGHT CONTROL FLAP SHUTOFF VALVES

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING MAKE SURE THAT YOU MOVE THE MANUAL OVERRIDE HANDLE TO THE OFF POSITION AND INSTALL THE RIG PIN IN LESS THAN 75 SECONDS. THE FSEU WILL AUTOMATICALLY CHANGE THE POSITION OF THE VALVE AND THE OVERRIDE HANDLE AFTER 75 SECONDS. THE MANUAL OVERRIDE HANDLE CAN MOVE SUDDENLY AND CAUSE INJURY TO PERSONS.

3) Move the manual override lever on the LE UCM shutoff valve to the UP (OFF) position.

SUBTASK 27-81-00-860-136

(10) Install the lockout pin, SPL-1746, through the lever (handle).

NOTE: The lockout pin, SPL-1746, will keep the handle in the UP (OFF) position.

SUBTASK 27-81-00-480-022

(11) Attach a DO-NOT-REMOVE tag on the lockout pin, SPL-1746.

SUBTASK 27-81-00-480-018

(12) To install the flight controls lock assembly (lock, SPL-1675), to prevent LE devices extension by the standby hydraulic system, do these steps:

(a) Disconnect the electrical connector from the LE standby drive shutoff valve.

(b) After you disconnect the electrical connector, examine the connector for corrosion (TASK 20-60-07-910-802).

NOTE: The inspection of the electrical connector can be done after the removal step or before the installation step of the electrical connector. This is an operator option.

(c) If there was corrosion, correct the problem (SWPM 20-60-03).

WARNING MAKE SURE THAT THE MANUAL OVERRIDE LEVER IS IN POSITION 1 TO PREVENT THE MOVEMENT OF THE LEADING EDGE SLATS. IF THE LEADING EDGE SLATS MOVE, DAMAGE TO AN OPEN THRUST REVERSER AND INJURIES TO PERSONNEL CAN OCCUR.

(d) Make sure that the manual override lever (handle) at the LE standby drive shutoff valve is in the Position 1 (standby hydraulic power OFF).

NOTE: The LE slats will extend fully if the manual override lever is in the Position 2 and the standby hydraulic system is pressurized.

WARNING MAKE SURE TO INSTALL THE GROUND LOCK ON THE STANDBY-DRIVE SHUTOFF-VALVE FOR THE LEADING EDGE. THIS PREVENTS ACCIDENTAL STANDBY SYSTEM OPERATION OF THE LEADING EDGE FLAPS AND SLATS. THE LEADING EDGE FLAPS AND SLATS CAN MOVE QUICKLY. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(e) Install the flight controls lock assembly (lock, SPL-1675), on the LE standby drive shutoff valve.

(f) Install the detent pin.

SUBTASK 27-81-00-700-002

(13) Do the steps in 'Leading Edge Flaps and Slats - Tryout'.

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I. Deactivate the Leading Edge Flaps and Slats in the Full Extend Position

SUBTASK 27-81-00-860-126

CAUTION

MAKE SURE THAT THE INBOARD FAN COWL PANEL AND THE INBOARD AND OUTBOARD THRUST REVERSERS ARE CLOSED AND IN THE STOWED POSITION OR REMOVED BEFORE YOU EXTEND THE LEADING EDGE FLAPS AND SLATS. THERE IS NOT SUFFICIENT CLEARANCE FOR THE FLAPS AND SLATS TO EXTEND IF THE INBOARD FAN COWL PANEL AND THE INBOARD AND OUTBOARD THRUST REVERSERS ARE IN THE OPEN POSITION. THIS CAN CAUSE DAMAGE TO EQUIPMENT.

(1) Make sure that the thrust reversers and fan cowls are closed.

(a) If necessary, do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

(b) If necessary, do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

SUBTASK 27-81-00-860-127

WARNING

MAKE SURE ALL PERSONS AND EQUIPMENT ARE CLEAR OF THE FLIGHT CONTROLS SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(2) If the LE flaps and slats are not in the full extend position, do this task: Leading Edge Flaps and Slats Extension, TASK 27-81-00-860-803.

SUBTASK 27-81-00-860-128

(3) Remove the hydraulic power from the airplane. Do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 27-81-00-860-129

(4) Make sure that the ALTERNATE FLAPS ARM switch is in the OFF position.

SUBTASK 27-81-00-480-019

(5) Attach DO NOT OPERATE tags, STD-858, on the flap control lever and the ALTERNATE FLAPS switches.

SUBTASK 27-81-00-480-020

WARNING

MAKE SURE TO INSTALL THE LEADING EDGE FLAP AND SLAT ACTUATORS LOCKOUT SET TO PREVENT ACCIDENTAL OPERATION OF THE LEADING EDGE FLAPS AND SLATS. THE LEADING EDGE FLAPS AND SLATS CAN MOVE QUICKLY. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(6) Install the lockout set, SPL-1734, on the extended actuators of the LE flaps and slats. Do this task: Leading Edge Flap and Slat Locks Installation, TASK 27-81-00-480-801.

NOTE: This will prevent retraction of the LE devices.

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SUBTASK 27-81-00-860-130

WARNING

MAKE SURE THAT FOR AIRPLANES WITH THE -1001 CRUISE DEPRESSURIZATION VALVE, YOU REVIEW THIS WARNING. MAKE SURE THAT YOU CLOSE THE FAN COWLS AND THRUST REVERSERS IF YOU MOVE THE TRAILING EDGE FLAPS WHILE THE LEADING EDGE FLAPS AND SLATS ARE DISCONNECTED. THE -1001 CRUISE DEPRESSURIZATION CREEP CAN CAUSE MOVEMENT OF THE LEADING EDGE FLAPS AND SLATS WHEN YOU MOVE THE FLAP CONTROL LEVER. IF YOU DO NOT OBEY, INJURY TO PERSONNEL OR DAMAGE TO THE EQUIPMENT CAN OCCUR.

(7) Do these steps to move the manual override lever (handle) on the LE cruise depressurization valve to the OFF position (Figure 201):

WARNING

MAKE SURE THAT THE TAB HAS BEEN CUT FROM THE VALVE. THE DEPRESSURIZATION VALVE HANDLE CAN MOVE QUICKLY. IF YOU DO NOT OBEY, INJURY TO PERSONNEL CAN OCCUR.

(a) If the valve has a tab, cut the tab from the valve.

NOTE: For more information, please refer to Service Bulletin 737–27–1242.

(b) Remove the lockwire from the manual override lever on the LE cruise depressurization valve.

(c) Move the manual override lever on the LE cruise depressurization valve to the OFF position.

NOTE: Do not force movement of the lever with tools.

1) If you cannot move the manual override lever, use the FSEU to operate the LE cruise depressurization valve:

NOTE: Two persons will be necessary when the FSEU is used to operate the LE cruise depressurization valve.

a) To get access to the main equipment center, do this step:

<1> Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

b) Close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	7	C00210	FLIGHT CONTROL FLAP SHUTOFF VALVES

c) Push the ON/OFF button on the front panel of the FSEU to turn on the display.

NOTE: The display will show EXISTING FAULTS?

d) Push the NO button until OTHER FUNCTNS? shows on the display.

e) Push the YES button to select OTHER FUNCTNS? on the FSEU.

f) Push the YES button to select SET OUTPUTS? from the OTHER FUNCTNS? menu.

g) Push the YES button to select LE DEPR VALVE? from the SET OUTPUTS? menu.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

h) Push the YES button at the SET ON? display.

NOTE: This will drive the valve to the closed position.

WARNING

MAKE SURE THAT YOU MOVE THE MANUAL OVERRIDE HANDLE TO THE OFF POSITION AND INSTALL THE RIG PIN IN LESS THAN 75 SECONDS. THE FSEU WILL AUTOMATICALLY CHANGE THE POSITION OF THE VALVE AND THE OVERRIDE HANDLE AFTER 75 SECONDS. THE MANUAL OVERRIDE HANDLE CAN MOVE SUDDENLY AND CAUSE INJURY TO PERSONS.

i) Move the manual override lever (handle) on the LE cruise depressurization valve to the OFF position.

NOTE: Do not force movement of the lever with tools.

j) If you still cannot move the manual override lever (handle) on the LE cruise depressurization valve, do these steps:

<1> Move the flaps and slats to the fully retracted position, if necessary.

WARNING

KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

<2> Supply system B hydraulic power, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

<3> Move the flap control lever to the UP position to retract the flaps and slats (if not retracted).

<4> Remove system B hydraulic power, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

<5> Use the FSEU to operate the valve again.

<6> Move the manual override lever (handle) on the LE cruise depressurization valve to the OFF position.

NOTE: Do not force movement of the lever with tools.

k) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	7	C00210	FLIGHT CONTROL FLAP SHUTOFF VALVES

(d) Install the lockout pin, SPL-1746, in the manual override lever (handle) of the LE cruise depressurization valve to keep the lever in the OFF position.

NOTE: If the newer -1003 cruise depressurization valve is installed on the airplane, then the lockout pin, SPL-1746, will stop the movement of the LE flaps and slats when hydraulic power is supplied.

NOTE: If the older -1001 cruise depressurization valve is installed on the airplane, then the LE flaps and slats are subject to the cruise depress creep condition when hydraulic power is supplied and the TE flaps are moved.

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1) If the lockout pin, SPL-1746, cannot be easily installed then it is necessary to supply system B hydraulic power and then use the FSEU to relieve the pressure on the LE depressurization valve.
   a) Use the FSEU procedure below but replace the selection of the LE UCM shutoff valve for the LE DEPR valve.
2) Attach the DO NOT OPERATE tag, STD-858, on the lockout pin, SPL-1746, after it is installed in the LE cruise depressurization valve.

SUBTASK 27-81-00-010-015

(8) Open this access panel to get access to the LE UCM shutoff valve:

Number	Name/Location
192BR	ECS Ram Air Inlet Mixing Duct Panel - Forward

SUBTASK 27-81-00-000-001

(9) Remove the lockwire from the manual override lever (handle) on the LE UCM shutoff valve (Figure 203).

SUBTASK 27-81-00-860-131

(10) Move the manual override lever on the LE UCM shutoff valve to the UP (OFF) position.

NOTE: The lever should move with very little force. If the lever is tight, you may need hydraulic pressure and the FSEU to release the pressure on the LE UCM SHUTOFF VALVE.

(a) If necessary, do the steps that follow:
   1) Supply system B hydraulic power (TASK 29-11-00-860-801).
   2) Use the FSEU to release the pressure on the LE UCM shutoff valve, as follows:

      NOTE: Two persons will be necessary when the FSEU is used to release the pressure on the LE UCM shutoff valve.

      a) To get access to the main equipment center, do this step:
         <1> Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

      b) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	7	C00210	FLIGHT CONTROL FLAP SHUTOFF VALVES

      c) Push the ON/OFF button.
      d) EXISTING FAULTS.....press NO.
      e) Press NO until OTHER FUNCTIONS.
      f) OTHER FUNCTIONS....press YES.
      g) SET OUTPUTS.....press YES.
      h) Press NO until LE UCM SHUTOFF VALVE.
      i) LE UCM SHUTOFF VALVE....press YES.
      j) SET ON.....press YES.
      k) Remove system B hydraulic power (TASK 29-11-00-860-805).

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l) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	7	C00210	FLIGHT CONTROL FLAP SHUTOFF VALVES

WARNING

MAKE SURE THAT YOU MOVE THE MANUAL OVERRIDE HANDLE TO THE OFF POSITION AND INSTALL THE RIG PIN IN LESS THAN 75 SECONDS. THE FSEU WILL AUTOMATICALLY CHANGE THE POSITION OF THE VALVE AND THE OVERRIDE HANDLE AFTER 75 SECONDS. THE MANUAL OVERRIDE HANDLE CAN MOVE SUDDENLY AND CAUSE INJURY TO PERSONS.

3) Move the manual override lever on the LE UCM shutoff valve to the UP (OFF) position.

SUBTASK 27-81-00-480-023

(11) Install the lockout pin, SPL-1746, through the lever (handle).

NOTE: The lockout pin, SPL-1746, will keep the handle in the UP (OFF) position.

SUBTASK 27-81-00-480-024

(12) Attach a DO-NOT-REMOVE tag on the lockout pin, SPL-1746.

SUBTASK 27-81-00-700-003

(13) Do the steps in 'Leading Edge Flaps and Slats - Tryout'.

J. Leading Edge Flaps and Slats - Tryout

NOTE: This tryout is to make sure that the LE flaps and slats are in a zero energy state.

SUBTASK 27-81-00-860-011

(1) Move the flap control lever to the 40 detent position.

(a) Make sure that the LE flaps or slats do not move.

SUBTASK 27-81-00-860-133

(2) Move the flap control lever to the FLAP UP position.

(a) Make sure that the LE flaps or slats do not move.

SUBTASK 27-81-00-860-012

(3) Move the flap control lever back to its initial flap position.

SUBTASK 27-81-00-860-103

(4) Set the ALTERNATE FLAPS ARM switch in the ARM position.

SUBTASK 27-81-00-860-013

(5) Move the ALTERNATE FLAPS CONTROL switch to the UP position for approximately one second and then to the DOWN position for approximately one second.

(a) Make sure that the LE flap or slat does not move.

SUBTASK 27-81-00-860-123

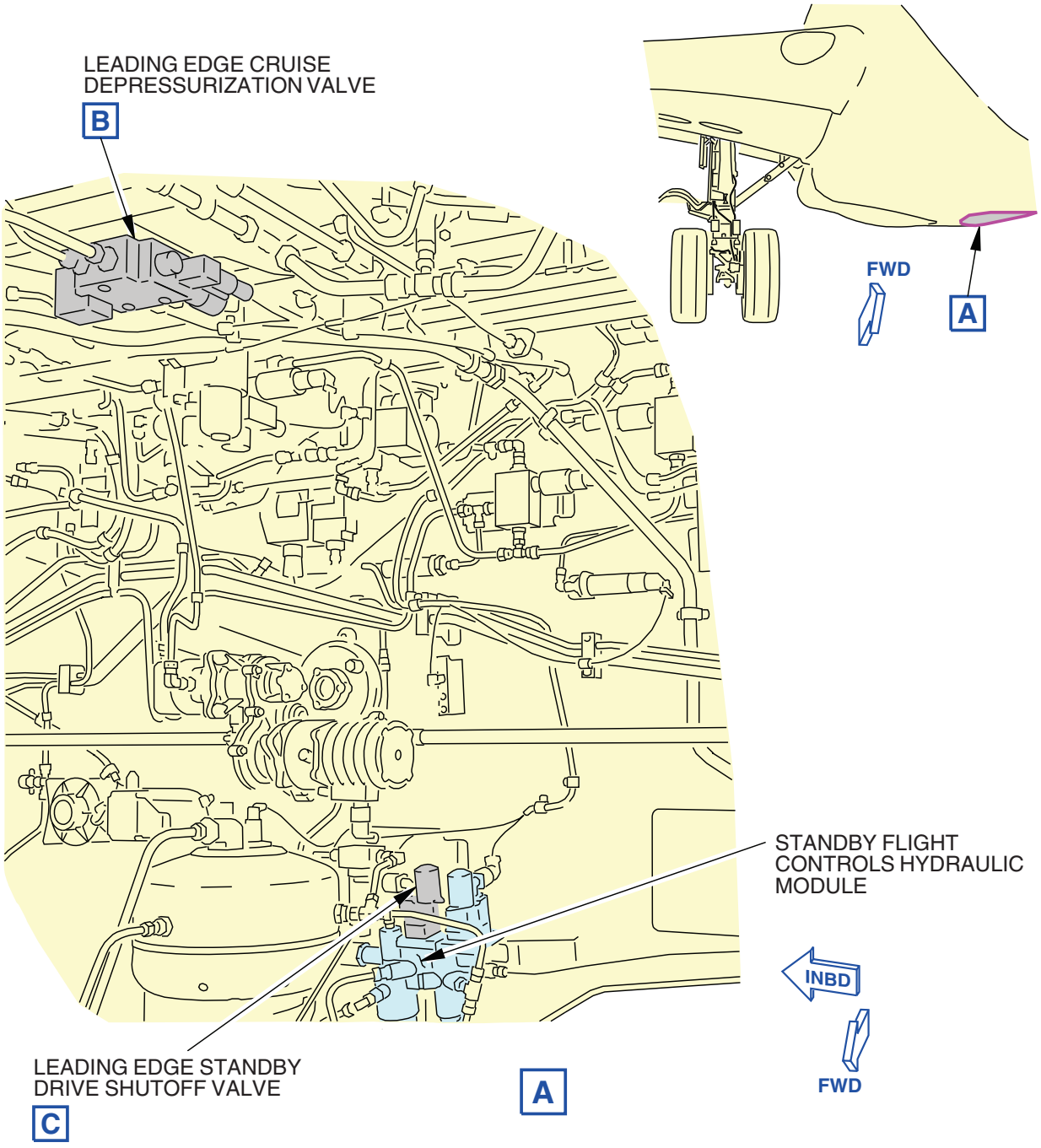
(6) Set the ALTERNATE FLAPS ARM switch to the OFF position.

SUBTASK 27-81-00-420-011

(7) Install the DO NOT OPERATE tag, STD-858, to the flap control lever and the ALTERNATE FLAPS control switch.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

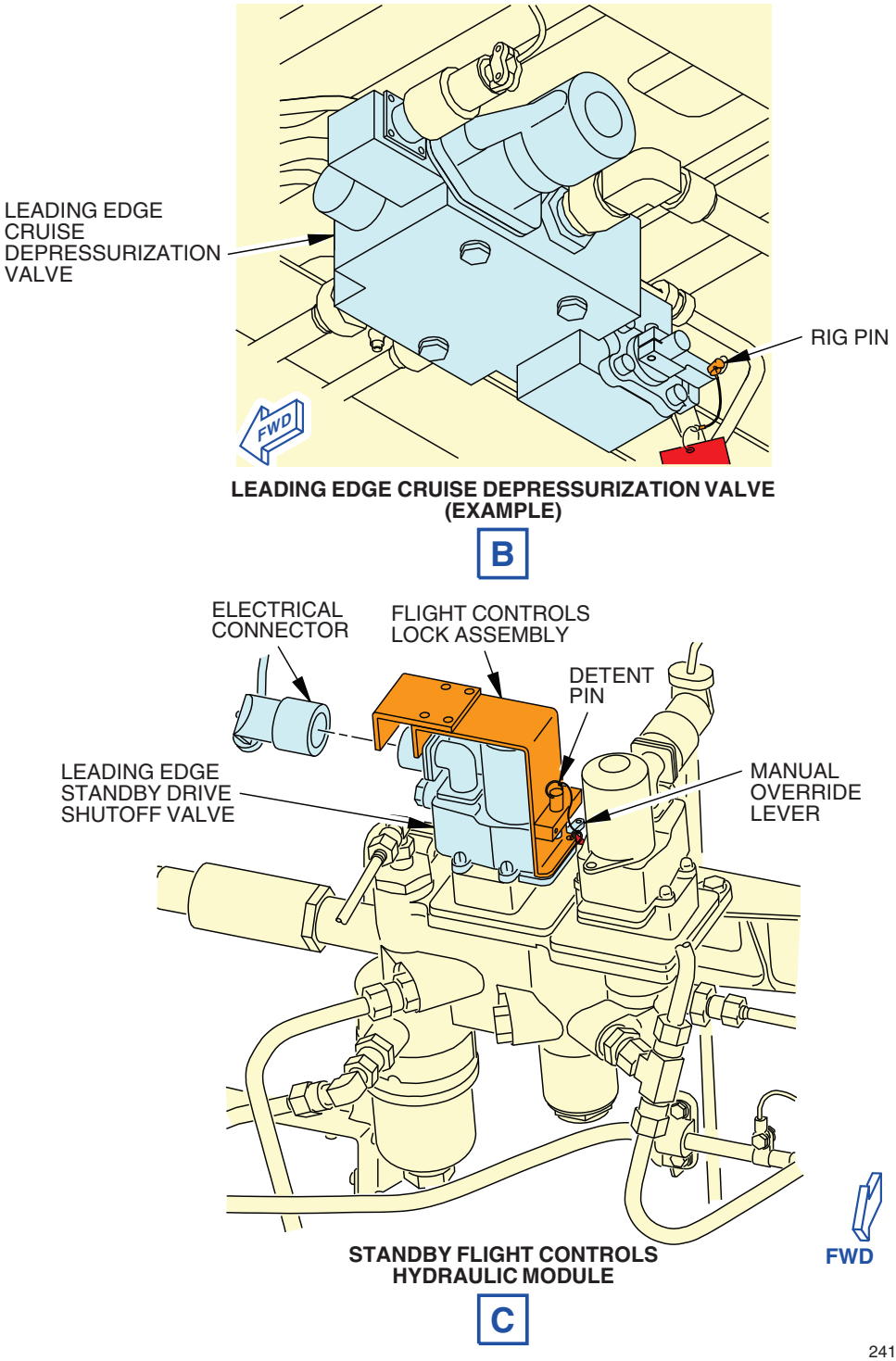


**LEADING EDGE CRUISE DEPRESSURIZATION VALVE
(MAIN LANDING GEAR WHEEL WELL)**

2413305 S00061532315_V3

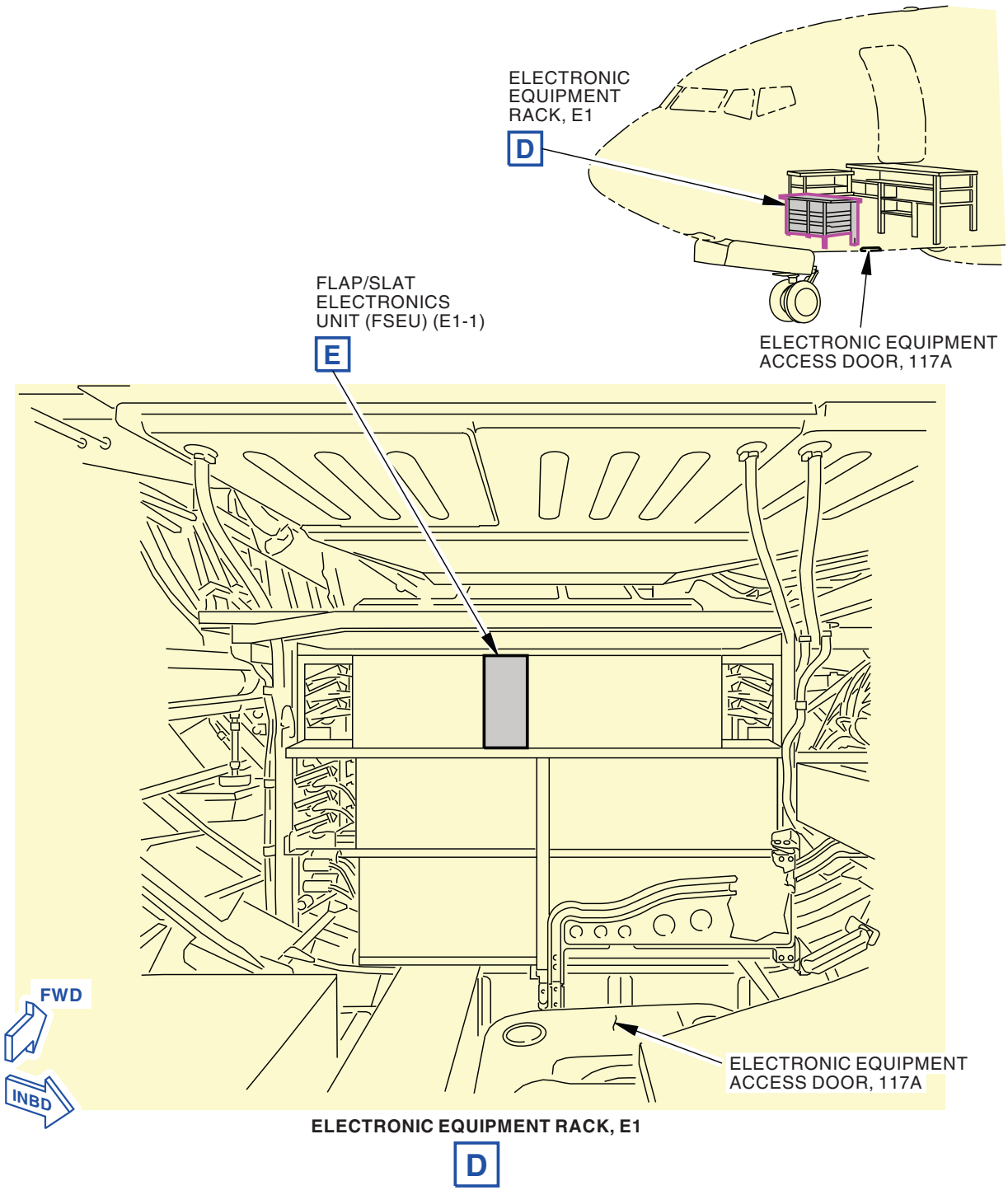
**Leading Edge Cruise Depressurization Valve and FSEU
Figure 201/27-81-00-990-801 (Sheet 1 of 4)**

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Leading Edge Cruise Depressurization Valve and FSEU
Figure 201/27-81-00-990-801 (Sheet 2 of 4)

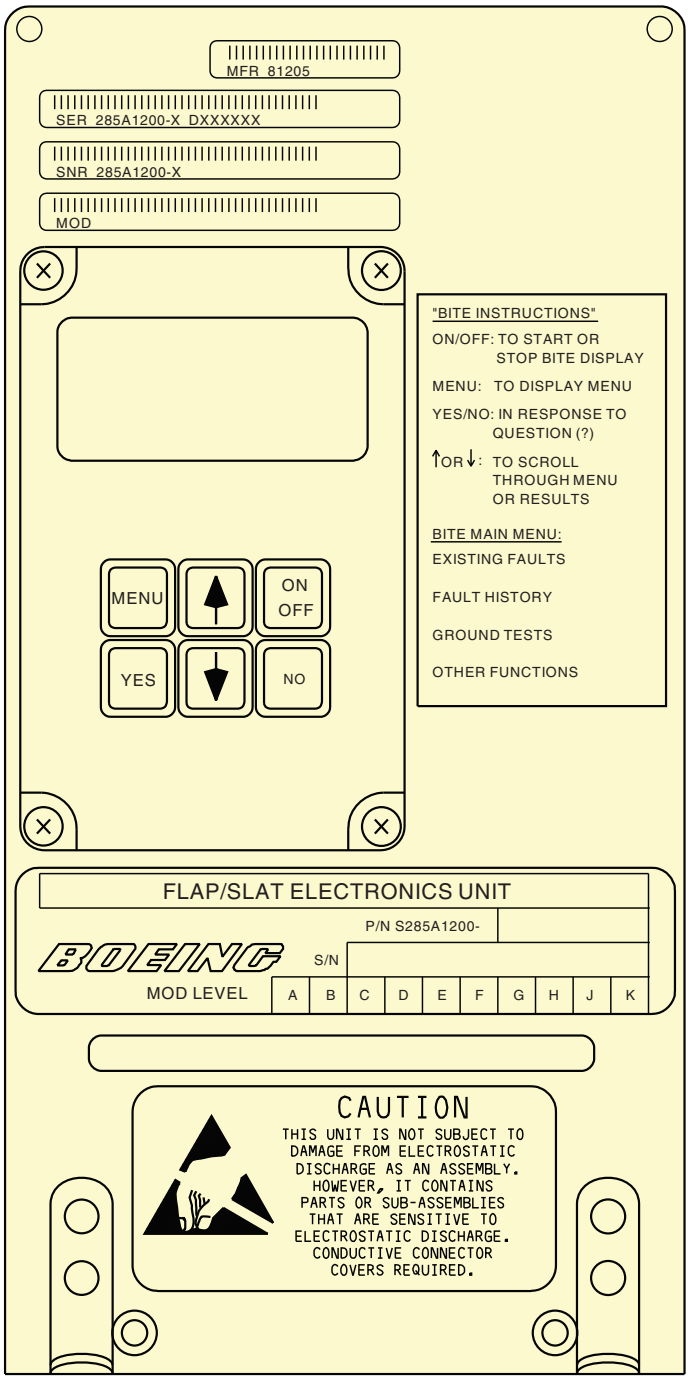
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2413307 S00061532317_V2

Leading Edge Cruise Depressurization Valve and FSEU
Figure 201/27-81-00-990-801 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



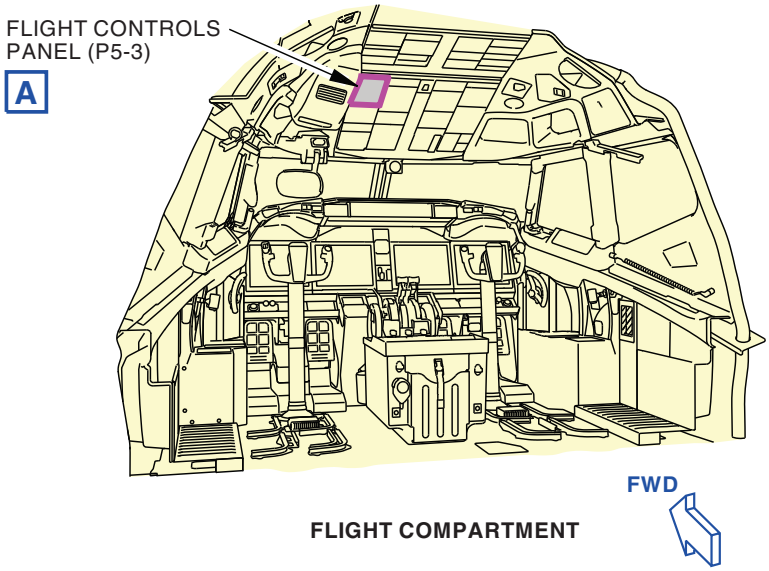
FLAP/SLAT ELECTRONICS UNIT (FSEU)

E

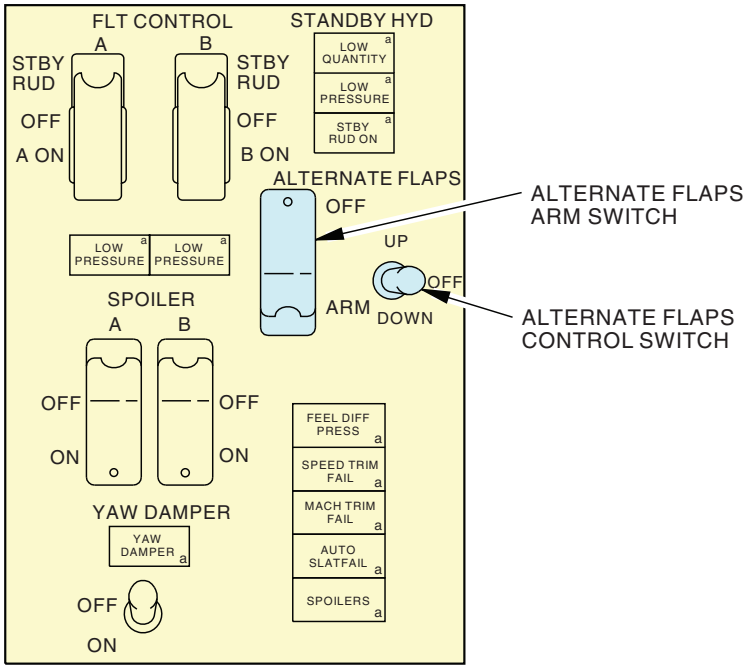
2413308 S00061532318_V2

Leading Edge Cruise Depressurization Valve and FSEU
Figure 201/27-81-00-990-801 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



FLIGHT COMPARTMENT



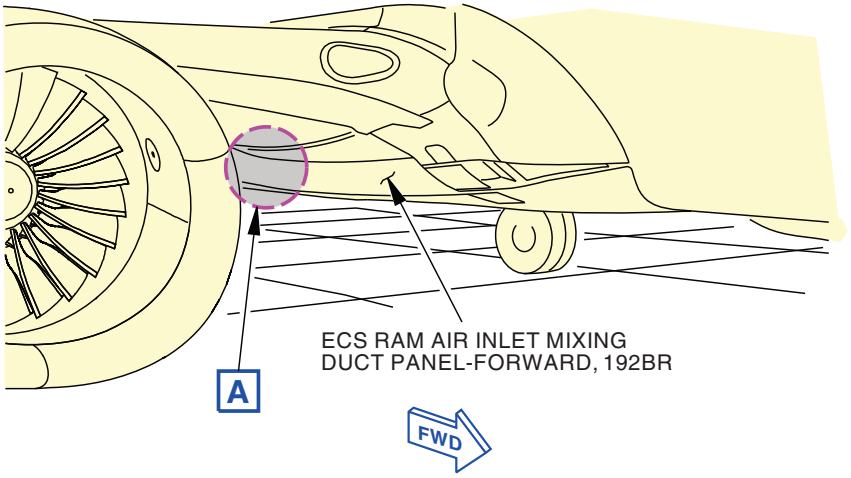
FLIGHT CONTROLS PANEL (P5-3)

A

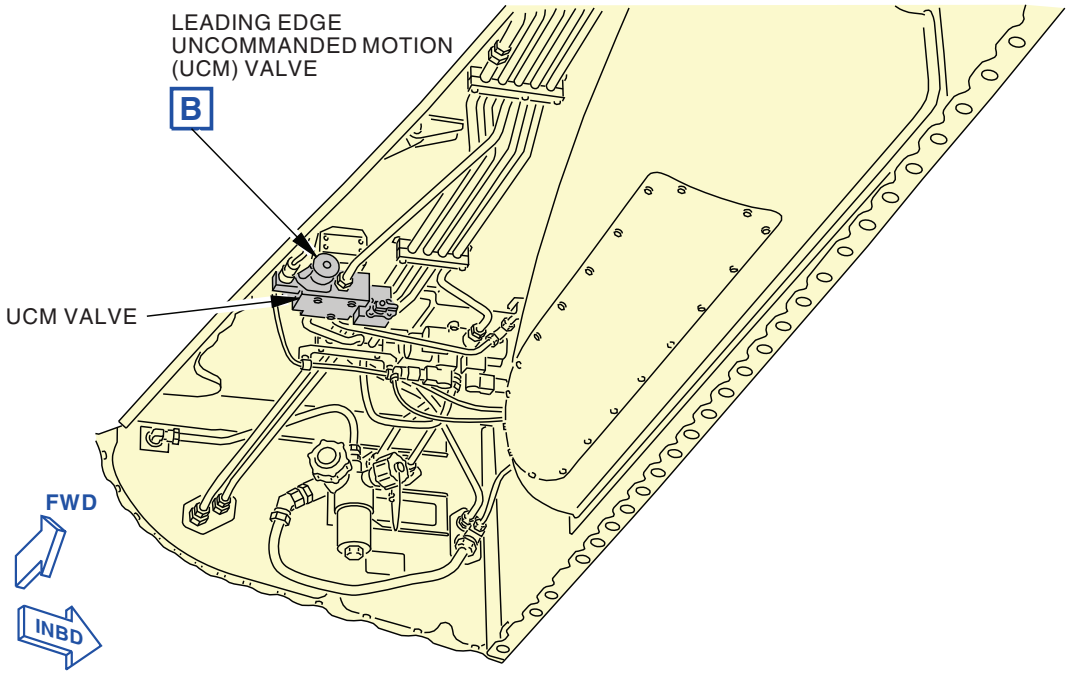
2412724 S00061531225_V1

Leading Edge Flap and Slat Controls
Figure 202/27-81-00-990-802

ECCN 9E991 BOEING PROPRIETARY - See title page for details



(EXAMPLE)



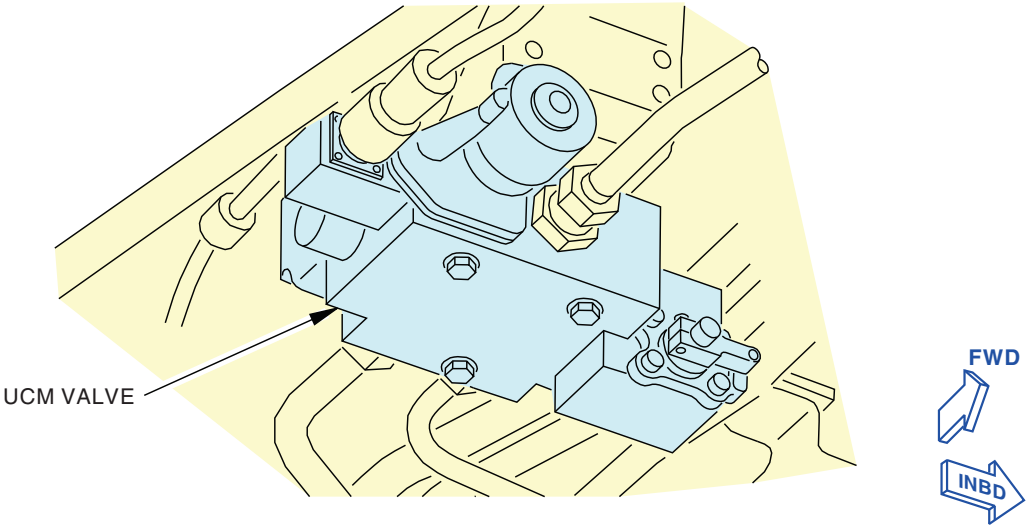
VIEW THROUGH THE ECS RAM AIR INLET
MIXING DUCT PANEL - FORWARD, 192BR
(EXAMPLE)

A

2500192 S0000586198_V2

Leading Edge Uncommanded Motion (UCM) Valve
Figure 203/27-81-00-990-803 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LEADING EDGE UNCOMMANDED MOTION (UCM) VALVE

B

2500197 S0000586199_V1

Leading Edge Uncommanded Motion (UCM) Valve
Figure 203/27-81-00-990-803 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-81-00-440-801

3. Leading Edge Flaps and Slats - Activation

(Figure 201)

A. General

WARNING	MAKE SURE THAT PERSONNEL AND EQUIPMENT ARE CLEAR OF ALL FLIGHT CONTROL SURFACES, THE THRUST REVERSERS AND THE LANDING GEAR. THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS AND THE LANDING GEAR CAN MOVE QUICKLY. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

CAUTION	MAKE SURE THAT THE INBOARD FAN COWL PANEL AND THE INBOARD AND OUTBOARD THRUST REVERSERS ARE CLOSED AND IN THE STOWED POSITION OR REMOVED BEFORE YOU EXTEND THE LEADING EDGE FLAPS AND SLATS. THERE IS NOT SUFFICIENT CLEARANCE FOR THE FLAPS AND SLATS TO EXTEND IF THE INBOARD FAN COWL PANEL AND THE INBOARD AND OUTBOARD THRUST REVERSERS ARE IN THE OPEN POSITION. THIS CAN CAUSE DAMAGE TO EQUIPMENT.

(1) If you do the deactivation task with the Leading Edge (LE) flaps and slats in the retract position, then do the steps to reactivate the LE flaps and slats in the retract position.

(2) If you do the deactivation task with the LE flaps and slats in the extend (intermediate) position, then do the steps to reactivate the LE flaps and slats in the extend (intermediate) position.

(3) If you do the deactivation task with the LE flaps and slats in the full extend position, then do the steps to reactivate the LE flaps and slats in the full extend position.

B. References

Reference	Title
20-10-44-400-801	Lockwire, Cotter Pins, and Lockrings - Installation (P/B 401)
20-60-07-910-802	Inspection, Cleaning and Sealing of Electrical Connectors (P/B 201)
27-81-00-710-801	Leading Edge Flap and Slat System Standby Actuation Operational Test (P/B 501)
27-81-91-720-801	Leading Edge Standby Shutoff Valve Test (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1675	Lock Assembly - Flight Controls (used on valve p/n 10-60581-2 and 10-60581-3) Part #: F80049-83 Supplier: 81205 Opt Part #: F80049-56 Supplier: 81205 Opt Part #: F80049-65 Supplier: 81205
SPL-1734	Lockout Set - Leading Edge Slats and Krueger Flaps Actuators Part #: C27051-29 Supplier: 81205

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Reference	Description
SPL-1746	Pin - Lockout, L.E. Cruise Depressurization Valve Part #: C27038-6 Supplier: 81205
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
G01048	Lockwire - MS20995C32, Corrosion Resistant Steel - 0.032 Inch (0.8128 mm) Diameter	NASM20995
G50171	Compound - Corrosion Inhibiting Compound, Interior Application - D5026NS or ZC-026	

E. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

F. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

G. Reactivate the Leading Edge Flaps and Slats In the Retract (UP) Position

SUBTASK 27-81-00-080-003

(1) To remove the flight controls lock assembly (lock, SPL-1675), from the LE standby drive shutoff valve, do these steps (Figure 201):

(a) Remove the detent pin.

(b) Remove the flight controls lock assembly (lock, SPL-1675), from the LE standby drive shutoff valve.

(c) Make sure that the override lever at the LE standby drive shutoff valve is in the position 1.

(d) Do this task: Inspection, Cleaning and Sealing of Electrical Connectors, TASK 20-60-07-910-802.

NOTE: The inspection of the electrical connector(s) can be done after the removal step or before the installation step of the electrical connector(s). This is an operator option.

(e) Apply D5026NS or ZC-026 compound, G50171, to the connector (SWPM 20-60-03).

(f) Connect the electrical connector to the LE standby drive shutoff valve.

SUBTASK 27-81-00-020-001

(2) Remove the DO-NOT-REMOVE tag with the lockout pin, SPL-1746, from the manual override lever on the LE cruise depressurization valve (Figure 201).

(a) Make sure that the manual override lever moves from the OFF position to the ON position.

(b) Install MS20995C32 lockwire, G01048, on the manual override lever (TASK 20-10-44-400-801).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-81-00-020-003

(3) Remove the DO NOT OPERATE tags, STD-858, from the flap control lever and ALTERNATE FLAPS switches.

SUBTASK 27-81-00-860-105

(4) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	7	C00210	FLIGHT CONTROL FLAP SHUTOFF VALVES

SUBTASK 27-81-00-410-001

(5) If it is open, close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 27-81-00-710-001

(6) Do this task: Leading Edge Flap and Slat System Standby Actuation Operational Test, TASK 27-81-00-710-801.

NOTE: The LE flap and slat system standby actuation operational test will make sure that the alternate control system is working correctly.

SUBTASK 27-81-00-710-006

(7) Do this task: Leading Edge Standby Shutoff Valve Test, TASK 27-81-91-720-801.

H. Reactivate the Leading Edge Flaps and Slats in the Extend (intermediate) Position

SUBTASK 27-81-00-080-016

(1) To remove the flight controls lock assembly (lock, SPL-1675), from the LE standby drive shutoff valve, do these steps (Figure 201):

(a) Remove the detent pin.

(b) Remove the flight controls lock assembly (lock, SPL-1675), from the LE standby drive shutoff valve.

(c) Make sure that the override lever at the LE standby drive shutoff valve is in the position 1.

(d) Do this task: Inspection, Cleaning and Sealing of Electrical Connectors, TASK 20-60-07-910-802.

NOTE: The inspection of the electrical connector(s) can be done after the removal step or before the installation step of the electrical connector(s). This is an operator option.

(e) Apply D5026NS or ZC-026 compound, G50171, to the connector (SWPM 20-60-03).

(f) Connect the electrical connector to the LE standby drive shutoff valve.

SUBTASK 27-81-00-020-004

(2) Remove the DO-NOT-REMOVE tag with the lockout pin, SPL-1746, from the manual override lever (handle) on the LE cruise depressurization valve (Figure 201).

NOTE: Do not force movement of the lever (handle) with tools. The lever should move with very little force. If the lever is tight, you could need hydraulic pressure and Flap/Slat Electronics Unit (FSEU) to relieve the pressure on the LE CRUISE DEPRESSURIZATION VALVE.

(a) If the lockout pin, SPL-1746, will not come out easily, then do the steps that follow:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) If it is necessary, supply system B hydraulic power and then use the FSEU to relieve the pressure on the LE depressurization valve.
   a) Use the FSEU steps below but replace the selection of the LE Uncommanded Motion (UCM) SHUTOFF VALVE for the LE DEPR VALVE.

SUBTASK 27-81-00-860-139

(3) Make sure that the (LE cruise depressurization valve) manual override lever moves from the OFF position to the ON position.

SUBTASK 27-81-00-420-017

(4) Install MS20995C32 lockwire, G01048, on the manual override lever on the LE cruise depressurization valve (TASK 20-10-44-400-801).

SUBTASK 27-81-00-020-005

(5) Remove the DO-NOT-REMOVE tag with the lockout pin, SPL-1746, from the manual override lever (handle) on the LE UCM shutoff valve.
   (a) If the lockout pin, SPL-1746, will not come out easily, then do the steps that follow:
      1) Supply system B hydraulic power, do this task: Leading Edge Flap and Slat System Operation With Primary Control, TASK 27-81-00-860-801.
      2) Use the FSEU to relieve the pressure on the LE UCM SHUTOFF VALVE as follows:
         a) Push the ON/OFF button.
         b) EXISTING FAULTS.....press NO.
         c) Press NO until OTHER FUNCTIONS.
         d) OTHER FUNCTIONS....press YES.
         e) SET OUTPUTS.....press YES.
         f) Press NO until LE UCM SHUTOFF VALVE.
         g) LE UCM SHUTOFF VALVE....press YES.
         h) SET ON.....press YES.

WARNING MAKE SURE THAT YOU MOVE THE MANUAL OVERRIDE HANDLE TO THE OFF POSITION AND INSTALL THE RIG PIN IN LESS THAN 75 SECONDS. THE FSEU WILL AUTOMATICALLY CHANGE THE POSITION OF THE VALVE AND THE OVERRIDE HANDLE AFTER 75 SECONDS. THE MANUAL OVERRIDE HANDLE CAN MOVE SUDDENLY AND CAUSE INJURY TO PERSONS.

      3) Remove the lockout pin, SPL-1746, from the LE UCM SHUTOFF VALVE within 75 seconds of pushing YES on the FSEU.
         a) If the pin will not come out easily, then repeat the steps above pressurizing the hydraulic system B before operating the FSEU.
         b) Remove the B pressure after the valve has moved.

SUBTASK 27-81-00-860-137

(6) Make sure that the manual override lever on the UCM shutoff valve moves to the down (ON) position.

NOTE: The lever (handle) should move with very little force. If the handle is tight, you could need hydraulic pressure and FSEU to relieve the pressure on the LE UCM SHUTOFF VALVE.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-81-00-420-015

(7) Install MS20995C32 lockwire, G01048, when the LE UCM shutoff lever is in the down position (TASK 20-10-44-400-801).

SUBTASK 27-81-00-860-106

(8) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	7	C00210	FLIGHT CONTROL FLAP SHUTOFF VALVES

I. Reactivate the Leading Edge Flaps and Slats In the Full Extend Position

SUBTASK 27-81-00-080-005

(1) Remove the lockout set, SPL-1734, from the LE devices:

(a) If the hydraulic system B is pressurized, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

WARNING REMOVE THE LOCK FROM LEADING EDGE FLAP ACTUATOR CAREFULLY TO PREVENT ACCIDENTAL OPERATION OF THE LEADING EDGE FLAPS. THE LEADING EDGE FLAPS CAN MOVE QUICKLY. THIS CAN CAUSE INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT.

(b) Remove the lockout set, SPL-1734, from the extended actuators of the LE flaps and slats, do this task: Leading Edge Flap and Slat Locks Removal, TASK 27-81-00-080-801.

SUBTASK 27-81-00-080-017

(2) Remove the DO-NOT-REMOVE tag with the lockout pin, SPL-1746, from the manual override lever (handle) on the LE cruise depressurization valve (Figure 201).

NOTE: Do not force movement of the lever (handle) with tools. The lever should move with very little force. If the lever is tight, you could need hydraulic pressure and FSEU to relieve the pressure on the LE CRUISE DEPRESSURIZATION VALVE.

(a) If the lockout pin, SPL-1746, will not come out easily, then do the steps that follow:

1) If it is necessary, supply system B hydraulic power and then use the FSEU to relieve the pressure on the LE depressurization valve.

a) Use the FSEU steps below but replace the selection of the LE UCM SHUTOFF VALVE for the LE DEPR VALVE.

(b) Move the (LE cruise depressurization valve) manual override lever from the OFF position to the ON position.

(c) Install MS20995C32 lockwire, G01048, on the manual override lever on the LE cruise depressurization valve (TASK 20-10-44-400-801).

SUBTASK 27-81-00-080-018

(3) Remove the DO-NOT-REMOVE tag with the lockout pin, SPL-1746, from the manual override lever (handle) on the LE UCM shutoff valve.

(a) If the lockout pin, SPL-1746, will not come out easily, then do the steps that follow:

1) Supply system B hydraulic power, do this task: Leading Edge Flap and Slat System Operation With Primary Control, TASK 27-81-00-860-801.

2) Use the FSEU to relieve the pressure on the LE UCM SHUTOFF VALVE as follows:

a) Push the ON/OFF button.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

b) EXISTING FAULTS.....press NO.

c) Press NO until OTHER FUNCTIONS.

d) OTHER FUNCTIONS....press YES.

e) SET OUTPUTS.....press YES.

f) Press NO until LE UCM SHUTOFF VALVE.

g) LE UCM SHUTOFF VALVE....press YES.

h) SET ON.....press YES.

WARNING

MAKE SURE THAT YOU MOVE THE MANUAL OVERRIDE HANDLE TO THE OFF POSITION AND INSTALL THE RIG PIN IN LESS THAN 75 SECONDS. THE FSEU WILL AUTOMATICALLY CHANGE THE POSITION OF THE VALVE AND THE OVERRIDE HANDLE AFTER 75 SECONDS. THE MANUAL OVERRIDE HANDLE CAN MOVE SUDDENLY AND CAUSE INJURY TO PERSONS.

3) Remove the lockout pin, SPL-1746, from the LE UCM SHUTOFF VALVE within 75 seconds of pushing YES on the FSEU.

a) If the pin will not come out easily, then repeat the steps above pressurizing the hydraulic system B before operating the FSEU.

b) Remove the B pressure after the valve has moved.

SUBTASK 27-81-00-860-138

(4) Make sure that the manual override lever on the UCM shutoff valve moves to the down (ON) position.

NOTE: The lever (handle) should move with very little force. If the handle is tight, you could need hydraulic pressure and FSEU to relieve the pressure on the LE UCM SHUTOFF VALVE.

SUBTASK 27-81-00-420-016

(5) Install MS20995C32 lockwire, G01048, when the LE UCM shutoff lever is in the down position (TASK 20-10-44-400-801).

SUBTASK 27-81-00-860-135

(6) Do this task: Leading Edge Flaps and Slats Retraction, TASK 27-81-00-860-804.

J. Put the Airplane Back to Its Usual Condition

SUBTASK 27-81-00-860-014

(1) Make sure that the safety tag is removed and this circuit breaker is closed:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	7	C00210	FLIGHT CONTROL FLAP SHUTOFF VALVES

——— END OF TASK ———

TASK 27-81-00-860-801

4. Leading Edge Flap and Slat System Operation With Primary Control

A. General

(1) The primary control system uses the hydraulic system B to move the flaps and slats.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

B. References

Reference	Title
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Procedure

SUBTASK 27-81-00-860-015

WARNING KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) To pressurize hydraulic system B, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 27-81-00-860-017

WARNING MAKE SURE ALL PERSONS AND EQUIPMENT ARE MOVED AWAY FROM THE CONTROL SURFACES BEFORE YOU START MOVEMENT. THIS WILL PREVENT INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

CAUTION MAKE SURE THAT THE INBOARD FAN COWL PANEL AND THE INBOARD AND OUTBOARD THRUST REVERSERS ARE CLOSED AND IN THE STOWED POSITION OR REMOVED BEFORE YOU EXTEND THE LEADING EDGE FLAPS AND SLATS. THERE IS NOT SUFFICIENT CLEARANCE FOR THE FLAPS AND SLATS TO EXTEND IF THE INBOARD FAN COWL PANEL AND THE INBOARD AND OUTBOARD THRUST REVERSERS ARE IN THE OPEN POSITION. THIS CAN CAUSE DAMAGE TO EQUIPMENT.

(2) Move the flap control lever to the applicable detent position (Table 201).

Table 201/27-81-00-993-801 Leading Edge (LE) Flap and Slat Position

Flap Control Lever Detent Position	LE Flap Position	LE Slat Position
UP	Retract	Retract
1 thru 25	Full Extend	Extend
30 thru 40	Full Extend	Full Extend

E. Put the Airplane Back to Its Usual Condition

SUBTASK 27-81-00-860-018

(1) To remove power from hydraulic system B, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-81-00-860-802

5. Leading Edge Flap and Slat System Operation With Alternate Control

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) The alternate control system uses electrical power to power the standby hydraulic system which moves the leading edge slats and flaps.

B. References

Reference	Title
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Procedure

SUBTASK 27-81-00-860-020

WARNING

KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

CAUTION

MAKE SURE THAT THE INBOARD FAN COWL PANEL AND THE INBOARD AND OUTBOARD THRUST REVERSERS ARE CLOSED AND IN THE STOWED POSITION OR REMOVED BEFORE YOU EXTEND THE LEADING EDGE FLAPS AND SLATS. THERE IS NOT SUFFICIENT CLEARANCE FOR THE FLAPS AND SLATS TO EXTEND IF THE INBOARD FAN COWL PANEL AND THE INBOARD AND OUTBOARD THRUST REVERSERS ARE IN THE OPEN POSITION. THIS CAN CAUSE DAMAGE TO EQUIPMENT.

(1) Move the ALTERNATE FLAPS ARM switch on the P5 panel to the ARM position.

SUBTASK 27-81-00-860-021

(2) Move the ALTERNATE FLAPS control switch to the DOWN position to extend the flaps and slats.

NOTE: The leading edge flaps and slats will not retract with alternate control.

SUBTASK 27-81-00-860-022

(3) Move the ALTERNATE FLAPS control switch to the OFF position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Put the Airplane Back to Its Usual Condition

SUBTASK 27-81-00-860-023

WARNING MAKE SURE THE POSITION OF THE FLAPS AGREES WITH THE POSITION OF THE FLAP CONTROL LEVER. WHEN YOU SUPPLY HYDRAULIC POWER, THE FLAPS AND SLATS WILL MOVE AUTOMATICALLY TO THE POSITION OF THE FLAP CONTROL LEVER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Move the ALTERNATE FLAPS control switch until the flaps and slats are in the same position as the flap control lever or move the flap control lever to the position of the flaps and slats.

SUBTASK 27-81-00-860-024

(2) Move the ALTERNATE FLAPS control switch to the OFF position.

SUBTASK 27-81-00-860-025

(3) Move the ALTERNATE FLAPS ARM switch to the OFF position to disarm the alternate control system.

SUBTASK 27-81-00-860-026

(4) If it is necessary to retract the leading edge flaps and slats, do these steps:

WARNING KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(a) Pressurize hydraulic system B (TASK 29-11-00-860-801).

(b) Move the flap control lever to the UP position to retract the leading edge flaps and slats.

(c) Remove the power from hydraulic system B (TASK 29-11-00-860-805).

——— END OF TASK ———

TASK 27-81-00-860-803

6. Leading Edge Flaps and Slats Extension

A. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

B. Procedure

SUBTASK 27-81-00-860-027

(1) Do one of these tasks to extend the leading edge flaps and slats:

WARNING MAKE SURE THAT PERSONNEL AND EQUIPMENT STAY AWAY FROM THE LEADING EDGE SLATS, TRAILING EDGE FLAPS, AND DRIVE MECHANISMS. THE FLAPS, SLATS, AND DRIVE MECHANISMS MOVE QUICKLY. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

CAUTION	MAKE SURE THAT THE INBOARD FAN COWL PANEL AND THE INBOARD AND OUTBOARD THRUST REVERSERS ARE CLOSED AND IN THE STOWED POSITION OR REMOVED BEFORE YOU EXTEND THE LEADING EDGE FLAPS AND SLATS. THERE IS NOT SUFFICIENT CLEARANCE FOR THE FLAPS AND SLATS TO EXTEND IF THE INBOARD FAN COWL PANEL AND THE INBOARD AND OUTBOARD THRUST REVERSERS ARE IN THE OPEN POSITION. THIS CAN CAUSE DAMAGE TO EQUIPMENT.

(a) To extend the flaps and slats with primary control, do this task: Leading Edge Flap and Slat System Operation With Primary Control, TASK 27-81-00-860-801.

WARNING	MAKE SURE THAT PERSONNEL AND EQUIPMENT STAY AWAY FROM THE LEADING EDGE SLATS, TRAILING EDGE FLAPS, AND DRIVE MECHANISMS. THE FLAPS, SLATS, AND DRIVE MECHANISMS MOVE QUICKLY. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

CAUTION	MAKE SURE THAT THE INBOARD FAN COWL PANEL AND THE INBOARD AND OUTBOARD THRUST REVERSERS ARE CLOSED AND IN THE STOWED POSITION OR REMOVED BEFORE YOU EXTEND THE LEADING EDGE FLAPS AND SLATS. THERE IS NOT SUFFICIENT CLEARANCE FOR THE FLAPS AND SLATS TO EXTEND IF THE INBOARD FAN COWL PANEL AND THE INBOARD AND OUTBOARD THRUST REVERSERS ARE IN THE OPEN POSITION. THIS CAN CAUSE DAMAGE TO EQUIPMENT.

(b) To extend the flaps and slats with alternate control, do this task: Leading Edge Flap and Slat System Operation With Alternate Control, TASK 27-81-00-860-802.

——— END OF TASK ———

TASK 27-81-00-860-804

7. Leading Edge Flaps and Slats Retraction

A. General

(1) The leading edge flaps and slats will not retract with alternate control.

B. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Procedure

SUBTASK 27-81-00-860-028

(1) To retract the flaps and slats with primary control, do this task: Leading Edge Flap and Slat System Operation With Primary Control, TASK 27-81-00-860-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-81-00-800-801

8. Leading Edge Flaps and Slats Manual Extension

(Figure 204)

A. General

(1) Use this task to manually extend the leading edge flaps and slats using a portable hydraulic cart, COM-163.

B. References

Reference	Title
27-81-00-700-803	Leading Edge Flap and Slat Timing Test in the Standby Mode (P/B 501)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
29-21-00-000-802	Standby Hydraulic System Power Removal (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-163	Cart - Hydraulic, Portable 3000 PSI, 30 GPM Part #: 5833 Supplier: 59603 Part #: 673A5FCBBCLMNY4 Supplier: 59603 Part #: 9780-0327 Supplier: 05172 Part #: HGPU50-30-1SX Supplier: 0022N Part #: HPU2 Supplier: 0T652 Part #: HPU3.3 Supplier: 0T652 Part #: HPU4 Supplier: 0T652 Part #: HT2000-1-E/1-S Supplier: H6394 Part #: HT2000-1-E/1-S2 Supplier: H6394 Part #: HT2000-1-F-S Supplier: H6394 Part #: HT2000-1-F-S2 Supplier: H6394 Part #: PH50D Supplier: 10000 Part #: PH50E Supplier: 10000 Part #: TMH3000 Supplier: L1238 Opt Part #: 57301FC3BBCILMNY Supplier: 59603 Opt Part #: 57305KV3BBCI-A1 Supplier: 59603
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
118	Electrical and Electronics Compartment - Right
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
211	Flight Compartment - Left
212	Flight Compartment - Right

E. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Prepare for the Leading Edge Flaps and Slats Extension

SUBTASK 27-81-00-480-013

WARNING

MAKE SURE THAT PERSONNEL AND EQUIPMENT STAY AWAY FROM THE LEADING EDGE SLATS, TRAILING EDGE FLAPS, AND DRIVE MECHANISMS. THE FLAPS, SLATS, AND DRIVE MECHANISMS MOVE QUICKLY. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(1) Install a DO NOT OPERATE tag, STD-858, on the flap control lever.

SUBTASK 27-81-00-860-108

(2) Do this task to depressurize Hydraulic System B: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

(a) Attach a DO NOT OPERATE tag, STD-858, on the HYDRAULIC PUMP B ELEC 1 switch from the overhead HYDRAULIC PUMPS PANEL (P5-8).

SUBTASK 27-81-00-860-109

(3) Do this task: Standby Hydraulic System Power Removal, TASK 29-21-00-000-802.

SUBTASK 27-81-00-480-014

(4) Attach a DO NOT OPERATE tags, STD-858, on the ALTERNATE FLAPS switches.

SUBTASK 27-81-00-860-110

(5) Make sure that these circuit breakers are open and have safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	15	C01081	HYDRAULIC SYSTEM PTU VALVE CONT 1
A	16	C01085	HYDRAULIC SYSTEM PTU VALVE CONT 2

SUBTASK 27-81-00-010-012

(6) To get access to the P92 panel, open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 27-81-00-860-111

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(7) Make sure that this circuit breaker is open and has safety tag:

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
F	2	C01449	STANDBY HYDRAULIC PUMP

G. Manual Leading Edge Flaps and Slats Extension

CAUTION

MAKE SURE THAT THE INBOARD FAN COWL PANEL AND THE INBOARD AND OUTBOARD THRUST REVERSERS ARE CLOSED AND IN THE STOWED POSITION OR REMOVED BEFORE YOU EXTEND THE LEADING EDGE FLAPS AND SLATS. THERE IS NOT SUFFICIENT CLEARANCE FOR THE FLAPS AND SLATS TO EXTEND IF THE INBOARD FAN COWL PANEL AND THE INBOARD AND OUTBOARD THRUST REVERSERS ARE IN THE OPEN POSITION. THIS CAN CAUSE DAMAGE TO EQUIPMENT.

SUBTASK 27-81-00-860-112

(1) Do these steps to manually extend the leading edge flaps and slats (Figure 204):

(a) Disconnect the standby hydraulic line at the flow limiting valve.

(b) Install a cap on the flow limiting valve.

(c) If it is necessary, disconnect the block clamp that holds the standby hydraulic line.

(d) Connect the portable hydraulic cart, COM-163, to the standby hydraulic line.

(e) Pressurize the standby hydraulic line with the portable hydraulic cart, COM-163, to extend the leading edge flaps and slats.

NOTE: The leading edge surfaces may move slightly when hydraulic pressure is removed.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 27-81-00-420-012

(1) Disconnect the standby hydraulic line from the portable hydraulic cart, COM-163.

(a) Connect the standby hydraulic line to the flow limiting valve.

(b) If it is necessary, connect the block clamp that holds the standby hydraulic line.

SUBTASK 27-81-00-080-010

(2) Remove the DO NOT OPERATE tag, STD-858, from the flap control lever.

SUBTASK 27-81-00-080-011

(3) Remove the DO NOT OPERATE tags, STD-858, from the ALTERNATE FLAPS switches.

SUBTASK 27-81-00-080-012

(4) Remove the DO NOT OPERATE tag, STD-858, from the he HYDRAULIC PUMP B ELEC 1 switch.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-81-00-860-113

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(5) Remove the safety tag and close this circuit breaker:

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
F	2	C01449	STANDBY HYDRAULIC PUMP

SUBTASK 27-81-00-410-010

(6) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 27-81-00-860-114

(7) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	15	C01081	HYDRAULIC SYSTEM PTU VALVE CONT 1
A	16	C01085	HYDRAULIC SYSTEM PTU VALVE CONT 2

SUBTASK 27-81-00-730-002

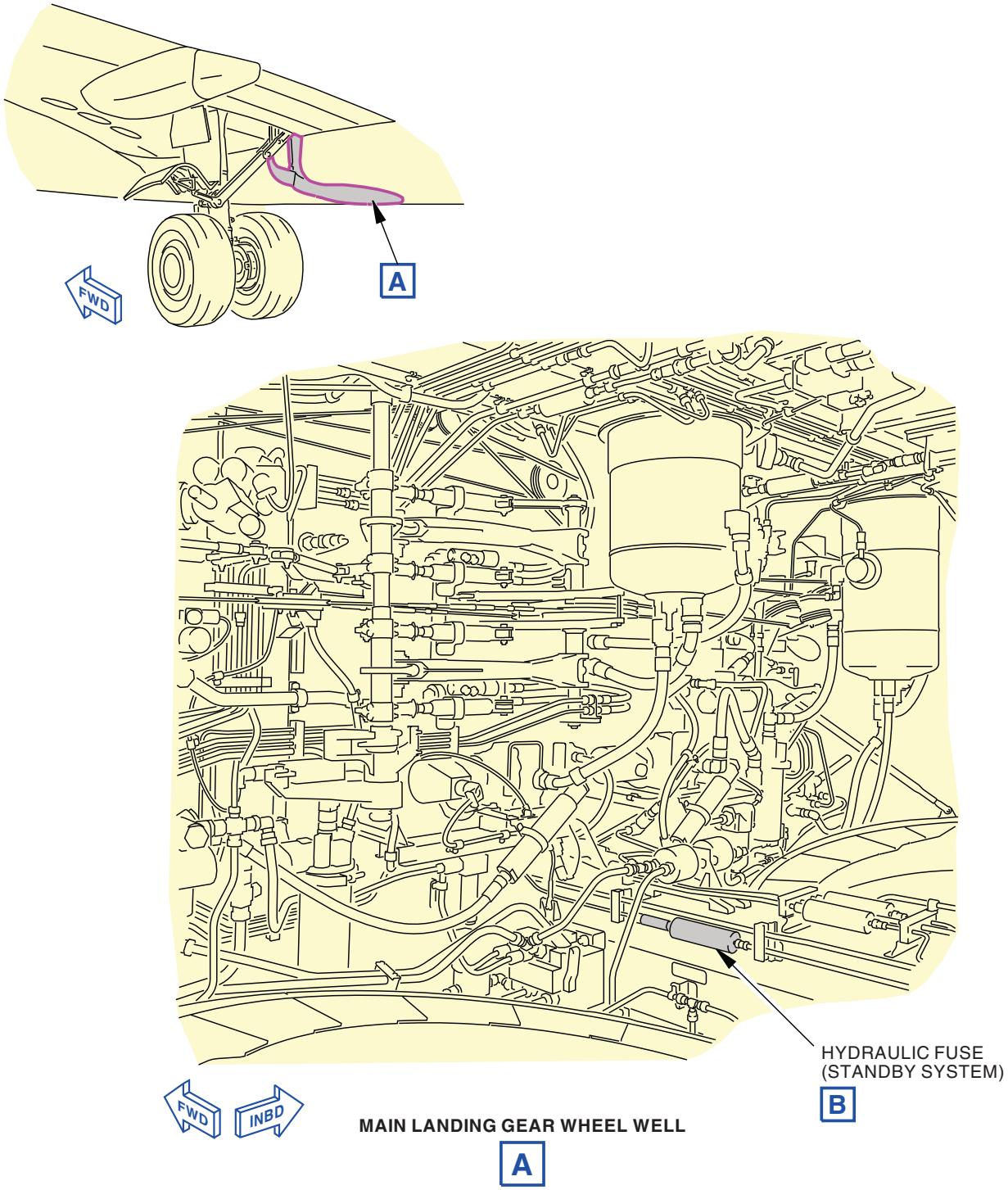
WARNING

MAKE SURE THAT PERSONNEL AND EQUIPMENT ARE CLEAR OF THE FLIGHT CONTROLS SURFACES. MAKE SURE THAT THE INBOARD FAN DUCT COWL AND THE INBOARD AND OUTBOARD THRUST REVERSERS ARE CLOSED. THE FLIGHT CONTROL SURFACES CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(8) Do this task: Leading Edge Flap and Slat Timing Test in the Standby Mode, TASK 27-81-00-700-803.

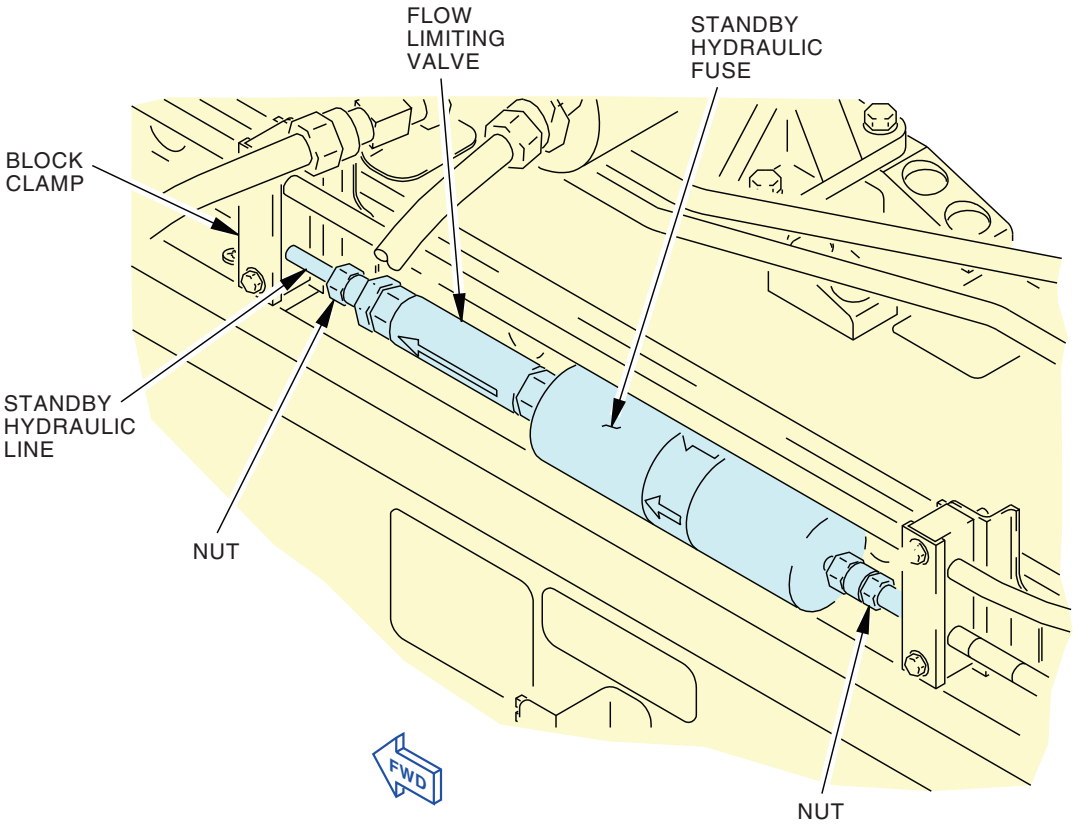
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Leading Edge Flaps and Slats Manual Extension
Figure 204/27-81-00-990-813 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Leading Edge Flaps and Slats Manual Extension
Figure 204/27-81-00-990-813 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-81-00-800-802

9. Leading Edge Flaps and Slats Manual Retraction

(Figure 205)

A. General

(1) Use this task to manually retract the leading edge flaps and slats using a portable hydraulic cart, COM-163.

B. References

Reference	Title
27-81-00-700-802	Leading Edge Flap and Slat Timing Test in Primary Mode (P/B 501)
27-81-00-700-804	Leading Edge Uncommanded Motion Protection Test (P/B 501)
27-83-00-710-801	Leading Edge Autoslat System - Functional Test (P/B 501)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
29-21-00-000-802	Standby Hydraulic System Power Removal (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-163	Cart - Hydraulic, Portable 3000 PSI, 30 GPM Part #: 5833 Supplier: 59603 Part #: 673A5FCBBCLMNY4 Supplier: 59603 Part #: 9780-0327 Supplier: 05172 Part #: HGPU50-30-1SX Supplier: 0022N Part #: HPU2 Supplier: 0T652 Part #: HPU3.3 Supplier: 0T652 Part #: HPU4 Supplier: 0T652 Part #: HT2000-1-E/1-S Supplier: H6394 Part #: HT2000-1-E/1-S2 Supplier: H6394 Part #: HT2000-1-F-S Supplier: H6394 Part #: HT2000-1-F-S2 Supplier: H6394 Part #: PH50D Supplier: 10000 Part #: PH50E Supplier: 10000 Part #: TMH3000 Supplier: L1238 Opt Part #: 57301FC3BBCILMNY Supplier: 59603 Opt Part #: 57305KV3BBCI-A1 Supplier: 59603
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
118	Electrical and Electronics Compartment - Right
132	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Right
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right

E. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
192BR	ECS Ram Air Inlet Mixing Duct Panel - Forward

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Prepare for the Leading Edge Flaps and Slats Retraction

SUBTASK 27-81-00-480-015

WARNING

MAKE SURE THAT PERSONNEL AND EQUIPMENT STAY AWAY FROM THE LEADING EDGE SLATS, TRAILING EDGE FLAPS, AND DRIVE MECHANISMS. THE FLAPS, SLATS, AND DRIVE MECHANISMS MOVE QUICKLY. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(1) Attach a DO NOT OPERATE tag, STD-858, on the flap control lever.

SUBTASK 27-81-00-860-115

(2) Do this task to depressurize Hydraulic System B: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

(a) Attach a DO NOT OPERATE tag, STD-858, on the HYDRAULIC PUMP B ELEC 1 switch from the overhead HYDRAULIC PUMPS PANEL (P5-8).

SUBTASK 27-81-00-860-116

(3) Do this task: Standby Hydraulic System Power Removal, TASK 29-21-00-000-802.

SUBTASK 27-81-00-480-016

(4) Attach a DO NOT OPERATE tags, STD-858, on the ALTERNATE FLAPS switches.

SUBTASK 27-81-00-860-117

(5) Make sure that these circuit breakers are open and have safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	15	C01081	HYDRAULIC SYSTEM PTU VALVE CONT 1
A	16	C01085	HYDRAULIC SYSTEM PTU VALVE CONT 2

SUBTASK 27-81-00-010-013

(6) To get access to the P92 panel, open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 27-81-00-860-118

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(7) Make sure that this circuit breaker is open and has safety tag:

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
F	2	C01449	STANDBY HYDRAULIC PUMP

SUBTASK 27-81-00-010-014

(8) Open this access panel:

Number	Name/Location
192BR	ECS Ram Air Inlet Mixing Duct Panel - Forward

G. Manual Leading Edge Flaps and Slats Retraction

SUBTASK 27-81-00-860-119

(1) If it is necessary, do these steps (Figure 205):

NOTE: Pressurizing the hydraulic line to retract the flaps and slats will add hydraulic fluid to Hydraulic Reservoir B. If the additional hydraulic fluid does not cause an overfill condition then the following steps are not necessary. An overfill condition in the reservoir can result in hydraulic fumes in the passenger cabin.

NOTE: Hydraulic fluid will be pushed back through the C1 and C2 lines during retraction.

(a) Disconnect the C1 line from the TEE fitting that is connected to the autoslat control valve.

1) Install a cap on the TEE fitting.

2) Put a container on the C1 line to catch the hydraulic fluid.

(b) Disconnect the C2 line from the autoslat control valve.

1) Install a cap on the open port of the autoslat control valve.

2) Put a container on the C1 line to catch the hydraulic fluid.

SUBTASK 27-81-00-860-120

(2) Do these steps to manually retract the leading edge flaps and slats (Figure 205):

(a) Disconnect the hydraulic line from the CYL port on the leading edge Uncommanded Motion (UCM) valve.

(b) Install a cap on the CYL port.

(c) Connect the portable hydraulic cart, COM-163, to the hydraulic line.

(d) Pressurize the hydraulic line with the portable hydraulic cart, COM-163, to retract the leading edge flaps and slats.

NOTE: The leading edge surfaces may move slightly when hydraulic pressure is removed.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 27-81-00-420-013

(1) Disconnect the hydraulic line from the portable hydraulic cart, COM-163.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Connect the hydraulic line to the CYL port on the UCM.

SUBTASK 27-81-00-420-014

(2) If the C1 line and the C2 line were disconnected, do these steps:

(a) Connect the C1 line to the TEE fitting that is connected to the autoslat control valve.

(b) Connect the C2 line to the autoslat control valve.

SUBTASK 27-81-00-080-013

(3) Remove the DO NOT OPERATE tag, STD-858, from the flap control lever.

SUBTASK 27-81-00-080-014

(4) Remove the DO NOT OPERATE tags, STD-858, from the ALTERNATE FLAPS switches.

SUBTASK 27-81-00-080-015

(5) Remove the DO NOT OPERATE tag, STD-858, from the HYDRAULIC PUMP B ELEC 1 switch.

SUBTASK 27-81-00-860-121

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(6) Remove the safety tag and close this circuit breaker:

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
F	2	C01449	STANDBY HYDRAULIC PUMP

SUBTASK 27-81-00-410-011

(7) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 27-81-00-860-122

(8) Close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	15	C01081	HYDRAULIC SYSTEM PTU VALVE CONT 1
A	16	C01085	HYDRAULIC SYSTEM PTU VALVE CONT 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-81-00-730-003

WARNING

MAKE SURE THAT PERSONNEL AND EQUIPMENT ARE CLEAR OF THE FLIGHT CONTROLS SURFACES. MAKE SURE THAT THE INBOARD FAN DUCT COWL AND THE INBOARD AND OUTBOARD THRUST REVERSERS ARE CLOSED. THE FLIGHT CONTROL SURFACES CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(9) Do this task: Leading Edge Flap and Slat Timing Test in Primary Mode, TASK 27-81-00-700-802.

SUBTASK 27-81-00-730-004

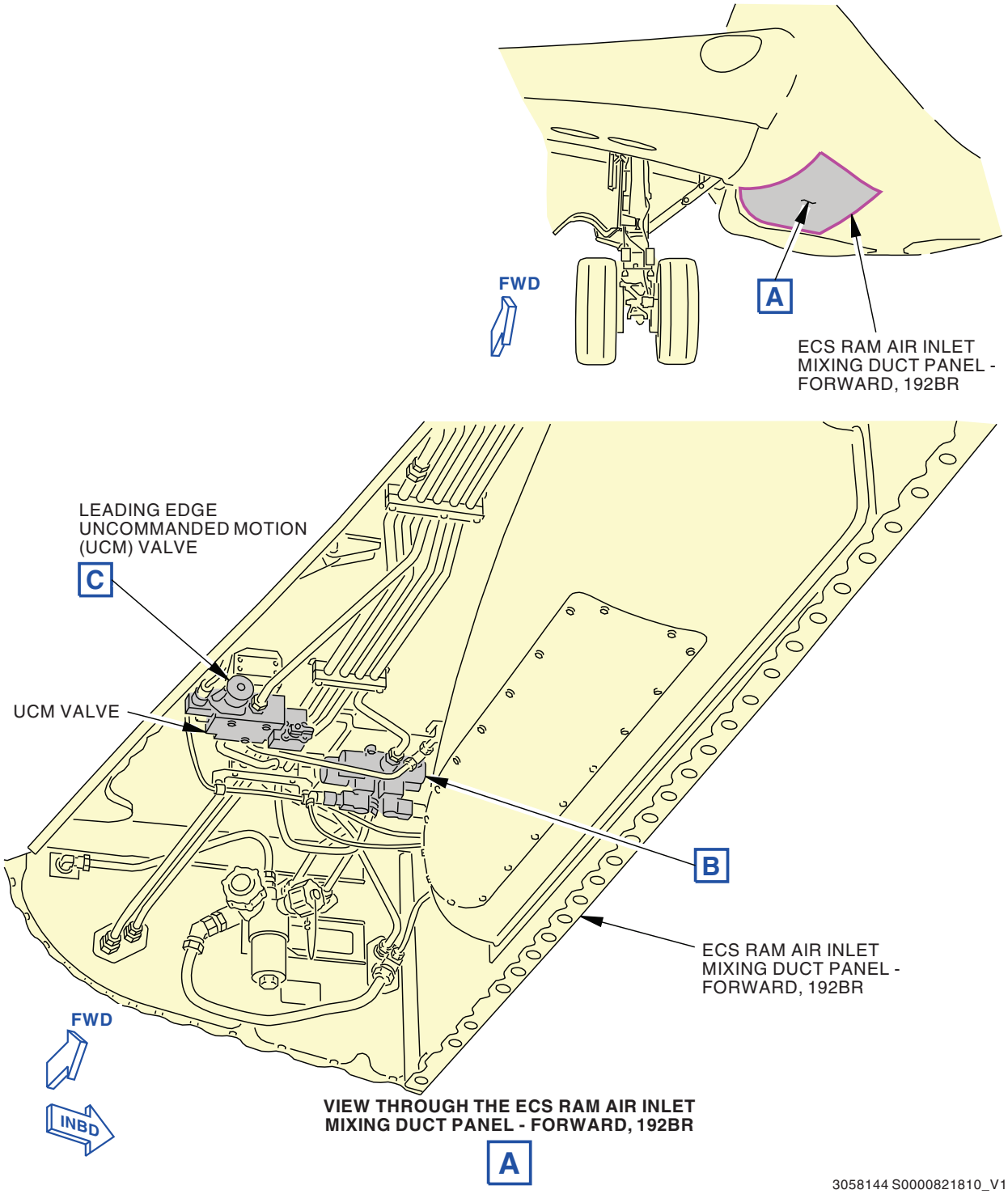
(10) Do this task: Leading Edge Autoslat System - Functional Test, TASK 27-83-00-710-801.

SUBTASK 27-81-00-730-005

(11) Do this task: Leading Edge Uncommanded Motion Protection Test, TASK 27-81-00-700-804.

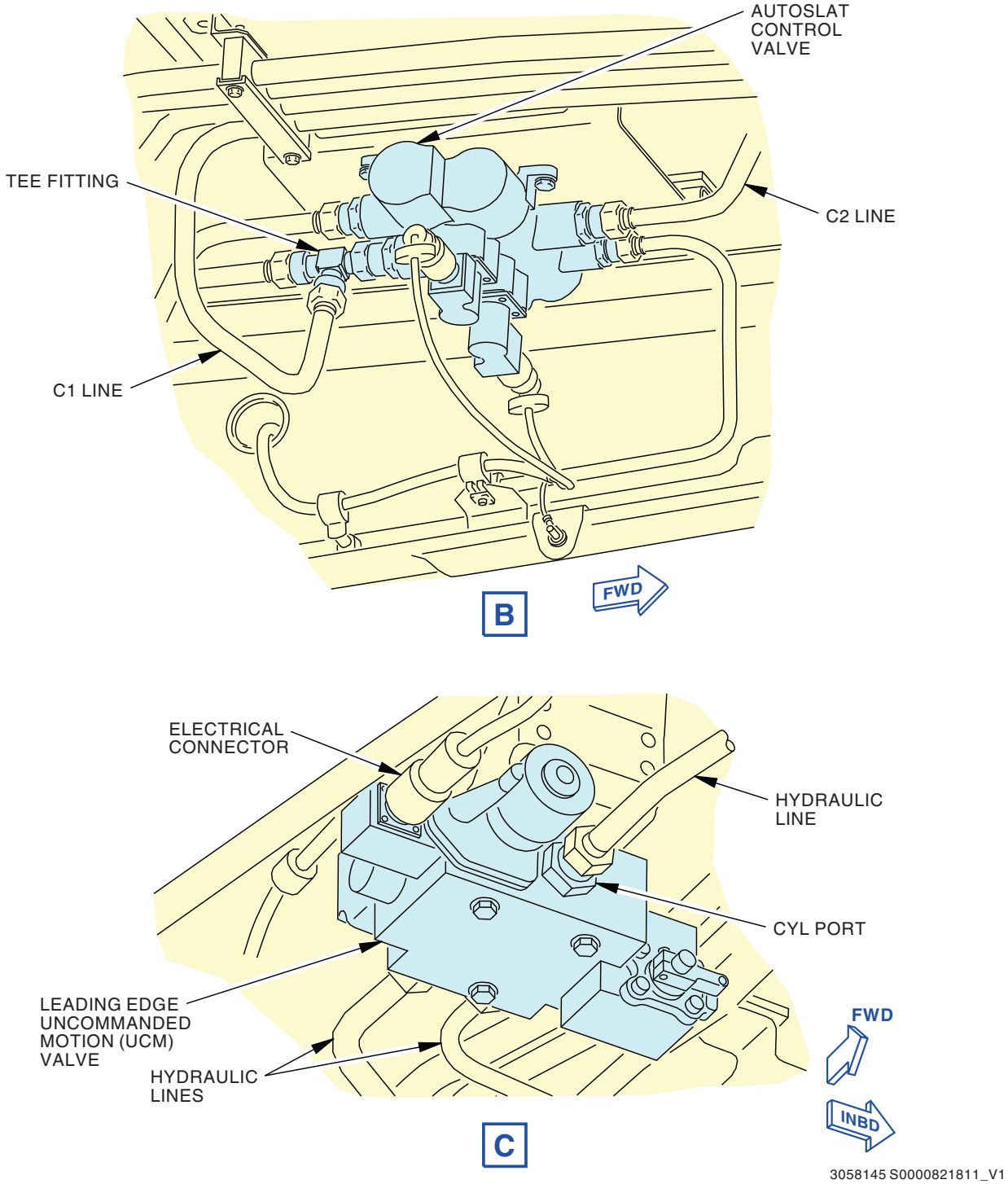
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ECCN 9E991 BOEING PROPRIETARY - See title page for details



Leading Edge Flaps and Slats Manual Retraction
Figure 205/27-81-00-990-814 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Leading Edge Flaps and Slats Manual Retraction
Figure 205/27-81-00-990-814 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-81-00-480-801

10. Leading Edge Flap and Slat Locks Installation

A. References

Reference	Title
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1734	Lockout Set - Leading Edge Slats and Krueger Flaps Actuators Part #: C27051-29 Supplier: 81205

C. Location Zones

Zone	Area
512	Left Wing - Krueger Flap No. 2
513	Left Wing - Krueger Flap No. 1
522	Left Wing - Slat No. 4
523	Left Wing - Slat No. 3
524	Left Wing - Slat No. 2
525	Left Wing - Slat No. 1
612	Right Wing - Krueger Flap No. 3
613	Right Wing - Krueger Flap No. 4
622	Right Wing - Slat No. 5
623	Right Wing - Slat No. 6
624	Right Wing - Slat No. 7
625	Right Wing - Slat No. 8

D. Leading Edge Flap and Slat Locks Installation

SUBTASK 27-81-00-860-100

WARNING MAKE SURE ALL PERSONS AND EQUIPMENT ARE CLEAR OF THE FLIGHT CONTROLS SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

CAUTION MAKE SURE THAT THE INBOARD FAN COWL PANEL AND THE INBOARD AND OUTBOARD THRUST REVERSERS ARE CLOSED AND IN THE STOWED POSITION OR REMOVED BEFORE YOU EXTEND THE LEADING EDGE FLAPS AND SLATS. THERE IS NOT SUFFICIENT CLEARANCE FOR THE FLAPS AND SLATS TO EXTEND IF THE INBOARD FAN COWL PANEL AND THE INBOARD AND OUTBOARD THRUST REVERSERS ARE IN THE OPEN POSITION. THIS CAN CAUSE DAMAGE TO EQUIPMENT.

(1) If the leading edge flaps and slats are not in the full extend (30-unit) position, do this task: Leading Edge Flaps and Slats Extension, TASK 27-81-00-860-803.

SUBTASK 27-81-00-860-101

(2) Make sure that the hydraulic power is removed (TASK 29-11-00-860-805).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-81-00-480-007

(3) Install the lockout set, SPL-1734, to prevent retraction of the leading edge devices.

WARNING MAKE SURE TO INSTALL THE LEADING EDGE FLAP AND SLAT ACTUATORS LOCKOUT SET TO PREVENT ACCIDENTAL OPERATION OF THE LEADING EDGE FLAPS AND SLATS. THE LEADING EDGE FLAPS AND SLATS CAN MOVE QUICKLY. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(a) Install the lockout set, SPL-1734, on the extended actuators of the leading edge flaps and slats.

——— END OF TASK ———

TASK 27-81-00-080-801

11. Leading Edge Flap and Slat Locks Removal

A. References

Reference	Title
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1734	Lockout Set - Leading Edge Slats and Krueger Flaps Actuators Part #: C27051-29 Supplier: 81205

C. Location Zones

Zone	Area
512	Left Wing - Krueger Flap No. 2
513	Left Wing - Krueger Flap No. 1
522	Left Wing - Slat No. 4
523	Left Wing - Slat No. 3
524	Left Wing - Slat No. 2
525	Left Wing - Slat No. 1
612	Right Wing - Krueger Flap No. 3
613	Right Wing - Krueger Flap No. 4
622	Right Wing - Slat No. 5
623	Right Wing - Slat No. 6
624	Right Wing - Slat No. 7
625	Right Wing - Slat No. 8

D. Leading Edge Flap and Slat Locks Removal

SUBTASK 27-81-00-080-006

(1) Remove the lockout set, SPL-1734 from the leading edge devices:

(a) If the hydraulic system B is pressurized, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING

CAREFULLY REMOVE THE LEADING EDGE FLAP AND SLAT ACTUATORS LOCKOUT SET. THE LEADING EDGE FLAPS AND SLATS CAN MOVE QUICKLY IF YOU DO NOT REMOVE THE LOCKS CORRECTLY. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(b) Remove the lockout set, SPL-1734 from the extended actuators of the leading edge flaps and slats.

(c) Do this task: Leading Edge Flaps and Slats Retraction, TASK 27-81-00-860-804.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

LEADING EDGE FLAP AND SLAT - ADJUSTMENT/TEST

1. General

A. This procedure has these tasks:

(1) An adjustment of the leading edge flaps

(2) An adjustment of the leading edge actuators

(3) An adjustment of the leading edge slats

(4) An operational test of the leading edge flaps and slats in primary mode

(5) A timing test of the leading edge flaps and slats in primary mode

(6) An operational test of the leading edge flaps and slats in standby mode

(7) A timing test of the leading edge flaps and slats in standby mode

(8) A test of the leading edge cruise depressurization system

(9) A test of the leading edge uncommanded motion protection system

(10) A test of the standby hydraulic system leading edge uncommanded motion

TASK 27-81-00-820-801

2. Leading Edge Flaps Adjustment

(Figure 501)

A. References

Reference	Title
27-81-00-080-801	Leading Edge Flap and Slat Locks Removal (P/B 201)
27-81-00-480-801	Leading Edge Flap and Slat Locks Installation (P/B 201)
27-81-00-860-803	Leading Edge Flaps and Slats Extension (P/B 201)
27-81-00-860-804	Leading Edge Flaps and Slats Retraction (P/B 201)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1691	Protractor - Digital, Vertical or Horizontal Part #: 21465A82 Supplier: 3A054 Part #: 31-038-3 (Model Pro 360) Supplier: 1X822 Part #: 31-040-9 (Model Pro 3600) Supplier: 1X822 Part #: 950-317 (Model Pro 360) Supplier: 01EP7 Part #: 950-318 (Model Pro 3600) Supplier: 01EP7 Part #: KS5549 Supplier: 75245 Part #: KS6005 Supplier: 75245
SPL-1734	Lockout Set - Leading Edge Slats and Krueger Flaps Actuators Part #: C27051-29 Supplier: 81205

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. Consumable Materials

Reference	Description	Specification
G01912	Lockwire - MS20995NC32, Monel - 0.032 Inch (0.8128 mm) Diameter	NASM20995

D. Location Zones

Zone	Area
512	Left Wing - Krueger Flap No. 2
513	Left Wing - Krueger Flap No. 1
612	Right Wing - Krueger Flap No. 3
613	Right Wing - Krueger Flap No. 4

E. Prepare for the Adjustment

SUBTASK 27-81-00-860-029

WARNING

KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) To pressurize hydraulic system B, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 27-81-00-860-030

WARNING

MAKE SURE THAT PERSONNEL AND EQUIPMENT STAY AWAY FROM THE LEADING EDGE SLATS, TRAILING EDGE FLAPS, AND DRIVE MECHANISMS. THE FLAPS, SLATS, AND DRIVE MECHANISMS MOVE QUICKLY. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

CAUTION

MAKE SURE THAT THE INBOARD FAN COWL PANEL AND THE INBOARD AND OUTBOARD THRUST REVERSERS ARE CLOSED AND IN THE STOWED POSITION OR REMOVED BEFORE YOU EXTEND THE LEADING EDGE FLAPS AND SLATS. THERE IS NOT SUFFICIENT CLEARANCE FOR THE FLAPS AND SLATS TO EXTEND IF THE INBOARD FAN COWL PANEL AND THE INBOARD AND OUTBOARD THRUST REVERSERS ARE IN THE OPEN POSITION. THIS CAN CAUSE DAMAGE TO EQUIPMENT.

(2) Move the flap control lever to the 1-unit detent position to extend the leading edge flaps.

SUBTASK 27-81-00-860-031

(3) To remove power from hydraulic system B, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 27-81-00-480-009

WARNING

INSTALL THE LOCK ON LEADING EDGE FLAP ACTUATOR CAREFULLY TO PREVENT THE ACCIDENTAL OPERATION OF THE LEADING EDGE FLAPS. THE LEADING EDGE FLAPS CAN MOVE QUICKLY. THIS CAN CAUSE INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT.

(4) Install the lockout set, SPL-1734 on all the adjacent flaps that you will not adjust. To install the lockout set, do this task: Leading Edge Flap and Slat Locks Installation, TASK 27-81-00-480-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Leading Edge Flaps Adjustment

SUBTASK 27-81-00-860-032

WARNING MAKE SURE THAT PERSONNEL AND EQUIPMENT STAY AWAY FROM THE LEADING EDGE SLATS, TRAILING EDGE FLAPS, AND DRIVE MECHANISMS. THE FLAPS, SLATS, AND DRIVE MECHANISMS MOVE QUICKLY. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(1) Move the flap control lever to the UP detent to retract the leading edge flaps.

SUBTASK 27-81-00-220-001

(2) Do a check of the full extend angle of the leading edge flap 1 [1] and leading edge flap 2 [2].

(a) Attach a protractor, COM-1691 to the inboard edge of the leading edge flap 1 [1] or outboard edge of the leading edge flap 2 [2].

(b) Set the protractor, COM-1691 to zero with the leading edge flaps retracted.

WARNING MAKE SURE THAT PERSONNEL AND EQUIPMENT STAY AWAY FROM THE LEADING EDGE SLATS, TRAILING EDGE FLAPS, AND DRIVE MECHANISMS. THE FLAPS, SLATS, AND DRIVE MECHANISMS MOVE QUICKLY. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

CAUTION MAKE SURE THAT THE INBOARD FAN COWL PANEL AND THE INBOARD AND OUTBOARD THRUST REVERSERS ARE CLOSED AND IN THE STOWED POSITION OR REMOVED BEFORE YOU EXTEND THE LEADING EDGE FLAPS AND SLATS. THERE IS NOT SUFFICIENT CLEARANCE FOR THE FLAPS AND SLATS TO EXTEND IF THE INBOARD FAN COWL PANEL AND THE INBOARD AND OUTBOARD THRUST REVERSERS ARE IN THE OPEN POSITION. THIS CAN CAUSE DAMAGE TO EQUIPMENT.

(c) Move the flap control lever to the 1-unit detent to extend the leading edge flap.

(d) Make sure the full extend angle of the leading edge flap 1 [1] or leading edge flap 2 [2] is 98 +/- 0.5 degrees.

1) If the full extend angle is out of tolerance, do these steps to adjust the rod end [8] of the flap actuator [7]:

a) Remove the pin [15], nut [14], washer [13], bushing [12], washer [11], and bolt [10] to disconnect the flap actuator [7] from the leading edge flap 1 [1] or leading edge flap 2 [2].

b) Loosen the nut [9] on the flap actuator [7].

c) Adjust the rod end [8] until you get the correct full extend angle.

NOTE: The limit for maximum adjustment is two full turns.

(e) Remove the protractor, COM-1691 from the inboard edge of the leading edge flap.

SUBTASK 27-81-00-220-002

(3) Make sure the mismatch between the inboardleading edge flap 2 [2] and the outboardleading edge flap 1 [1] is 0.000 ±0.040 in. (0.0 ±1.0 mm).

SUBTASK 27-81-00-860-033

(4) Apply preload at the flap bullnose to the leading edge flap 1 [1] or leading edge flap 2 [2].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Make sure the flap bullnose touches the leading edge flap 1 [1] or leading edge flap 2 [2].

NOTE: To apply the correct preload, the flap bullnose must first touch the leading edge flap.

1) If the flap bullnose does not touch the leading edge flap 1 [1] or leading edge flap 2 [2], adjust the length of the flap rod.

(b) Adjust the flap rod an additional 1/2 turn to apply the correct preload at the flap bullnose to the leading edge flap 1 [1] or leading edge flap 2 [2].

(c) Install MS20995NC32 lockwire, G01912 on the flap rod.

SUBTASK 27-81-00-860-034

WARNING

KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(5) Do this task: Leading Edge Flaps and Slats Retraction, TASK 27-81-00-860-804.

SUBTASK 27-81-00-220-003

(6) Do a check of the mismatch at the forward edge of the leading edge flap 1 [1] or leading edge flap 2 [2].

(a) Make sure the mismatch is 0.000 +0.062 / -0.071 in. (0.0 +1.6 / -1.8 mm).

1) If it is necessary, adjust the mismatch dimension.

a) Loosen the nuts [3] to adjust the seal [4].

b) Adjust the seal [4] until you get the correct mismatch requirement.

c) Tighten the nuts [3].

SUBTASK 27-81-00-220-004

(7) Do a check of the gap and mismatch at the aft edge of the leading edge flap 1 [1] or leading edge flap 2 [2].

(a) Make sure the gap is 0.050 in. (1.3 mm) to 0.240 in. (6.1 mm).

(b) Make sure the mismatch is 0.000 +0.040 / -0.072 in. (0.0 +1.0 / -1.8 mm).

1) If the mismatch is out of tolerance, adjust the rod end [8] of the flap actuator [7] again.

SUBTASK 27-81-00-420-001

(8) If you made adjustment to the rod end [8] of the flap actuator [7], do these steps:

(a) Install the bolt [10], washer [11], bushing [12], washer [13] and nut [14] to connect the flap actuator [7] to the leading edge flap 1 [1] or leading edge flap 2 [2].

(b) Tighten the nut [14] to 660 in-lb (74.6 N·m) - 980 in-lb (110.7 N·m).

(c) Install the pin [15].

(d) Tighten the nut [9] to 550 in-lb (62.1 N·m) - 650 in-lb (73.4 N·m).

(e) Install the MS20995NC32 lockwire, G01912 on the nut [9].

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SUBTASK 27-81-00-860-035

WARNING

MAKE SURE THAT PERSONNEL AND EQUIPMENT STAY AWAY FROM THE LEADING EDGE SLATS, TRAILING EDGE FLAPS, AND DRIVE MECHANISMS. THE FLAPS, SLATS, AND DRIVE MECHANISMS MOVE QUICKLY. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

CAUTION

MAKE SURE THAT THE INBOARD FAN COWL PANEL AND THE INBOARD AND OUTBOARD THRUST REVERSERS ARE CLOSED AND IN THE STOWED POSITION OR REMOVED BEFORE YOU EXTEND THE LEADING EDGE FLAPS AND SLATS. THERE IS NOT SUFFICIENT CLEARANCE FOR THE FLAPS AND SLATS TO EXTEND IF THE INBOARD FAN COWL PANEL AND THE INBOARD AND OUTBOARD THRUST REVERSERS ARE IN THE OPEN POSITION. THIS CAN CAUSE DAMAGE TO EQUIPMENT.

(9) Extend the leading edge flaps to the 1-unit detent position with hydraulic power. To extend the leading edge flaps, do this task: Leading Edge Flaps and Slats Extension, TASK 27-81-00-860-803.

SUBTASK 27-81-00-860-036

(10) Do a check of the flap panel clearances.

(a) For all but the aft 5 in. (127 mm) of the flap, make sure the gap between the outboard leading edge flap 1 [1] and the inboard leading edge flap 2 [2] is 0.060 in. (1.52 mm) to 0.150 in. (3.81 mm).

(b) For the aft 5 in. (127 mm) of the flap, make sure the gap between the outboard leading edge flap 1 [1] and the inboard leading edge flap 2 [2] is 0.06 in. (1.5 mm) to 0.25 in. (6.4 mm).

SUBTASK 27-81-00-210-003

(11) If the flap panel clearances are out of tolerance, do a check of the wing and flap panel for damage or deformation.

NOTE: This check is to make sure that the flap and the wing align correctly. There is no adjustment that you can do.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 27-81-00-080-008

WARNING

CAREFULLY REMOVE THE LEADING EDGE FLAP AND SLAT ACTUATORS LOCKOUT SET. THE LEADING EDGE FLAPS AND SLATS CAN MOVE QUICKLY IF YOU DO NOT REMOVE THE LOCKS CORRECTLY. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Remove the lockout set, SPL-1734 from the leading flaps. To remove the lockout set, do this task: Leading Edge Flap and Slat Locks Removal, TASK 27-81-00-080-801.

SUBTASK 27-81-00-860-037

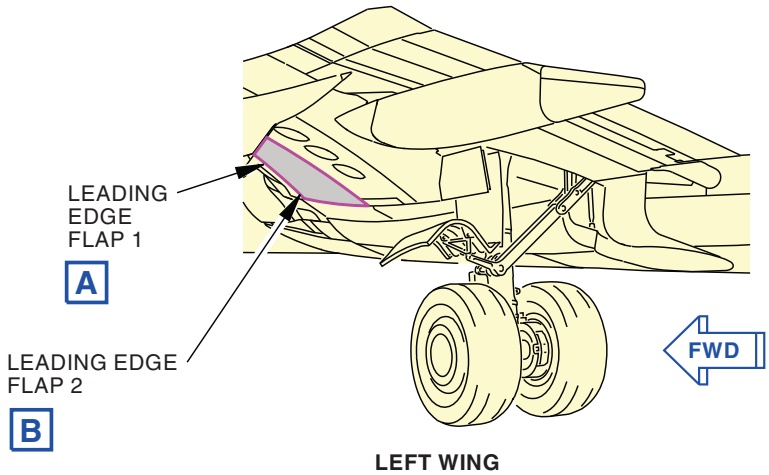
WARNING

MAKE SURE THAT PERSONNEL AND EQUIPMENT STAY AWAY FROM THE LEADING EDGE SLATS, TRAILING EDGE FLAPS, AND DRIVE MECHANISMS. THE FLAPS, SLATS, AND DRIVE MECHANISMS MOVE QUICKLY. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

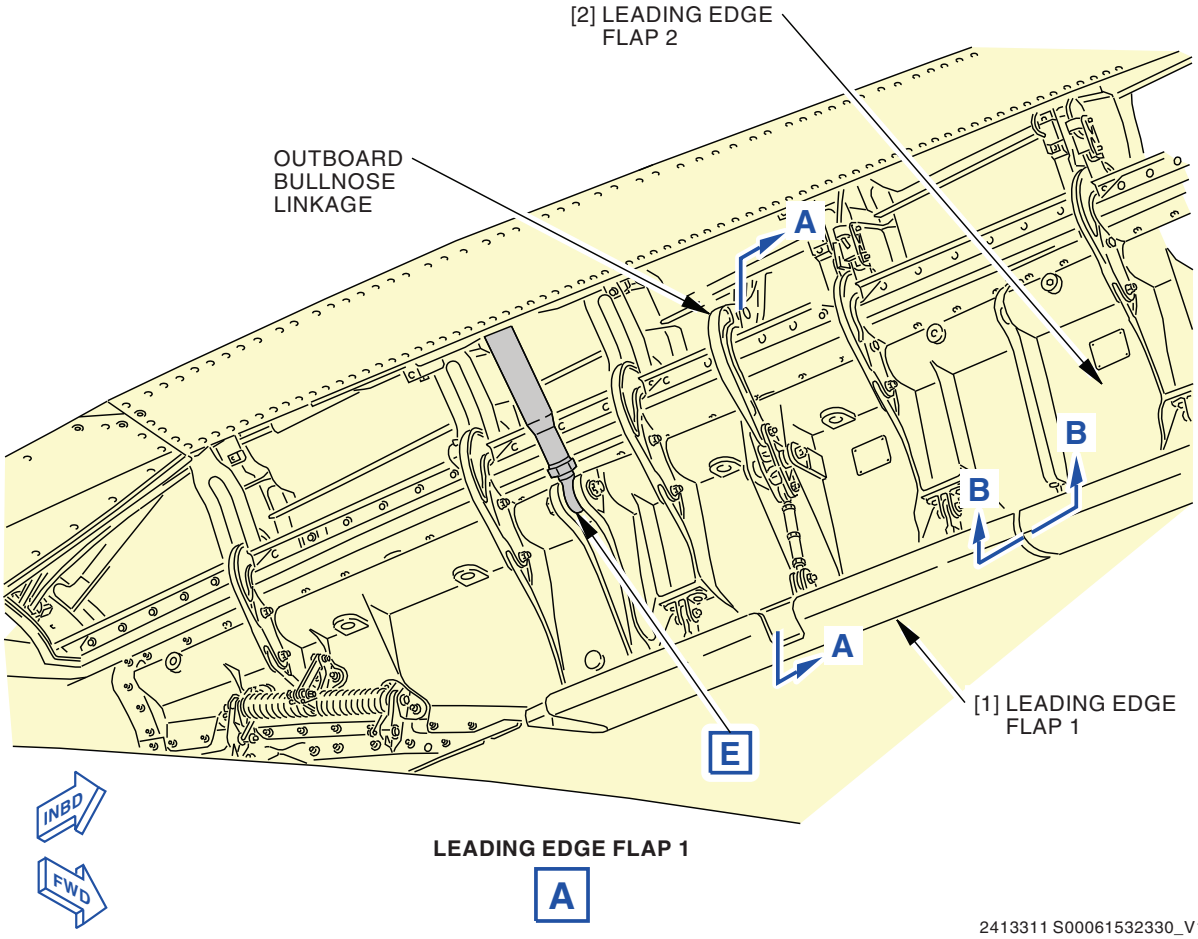
(2) Do this task: Leading Edge Flaps and Slats Retraction, TASK 27-81-00-860-804.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LEFT WING
(RIGHT WING IS OPPOSITE)



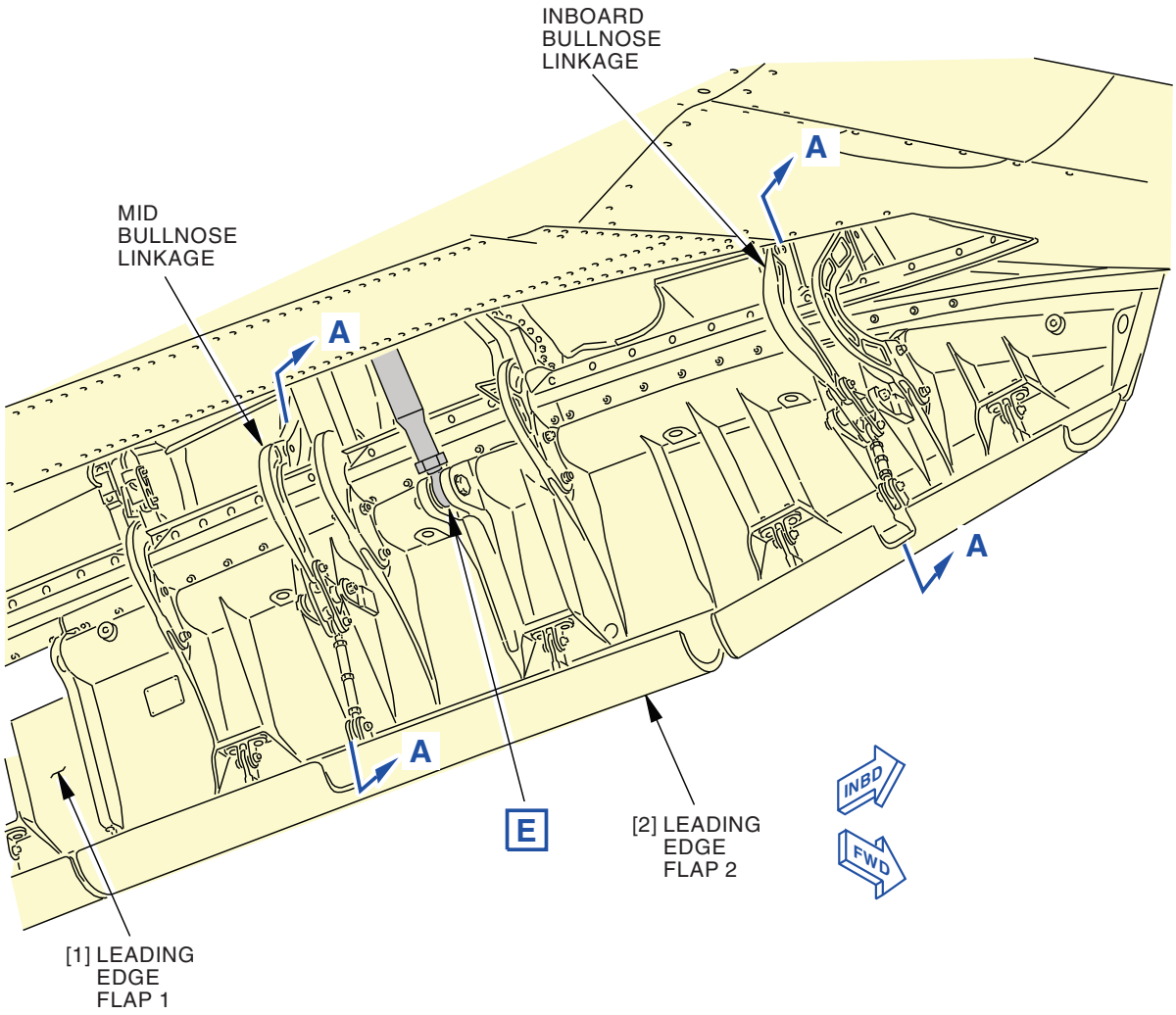
LEADING EDGE FLAP 1

A

2413311 S00061532330_V1

Leading Edge Flap Adjustment
Figure 501/27-81-00-990-804 (Sheet 1 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



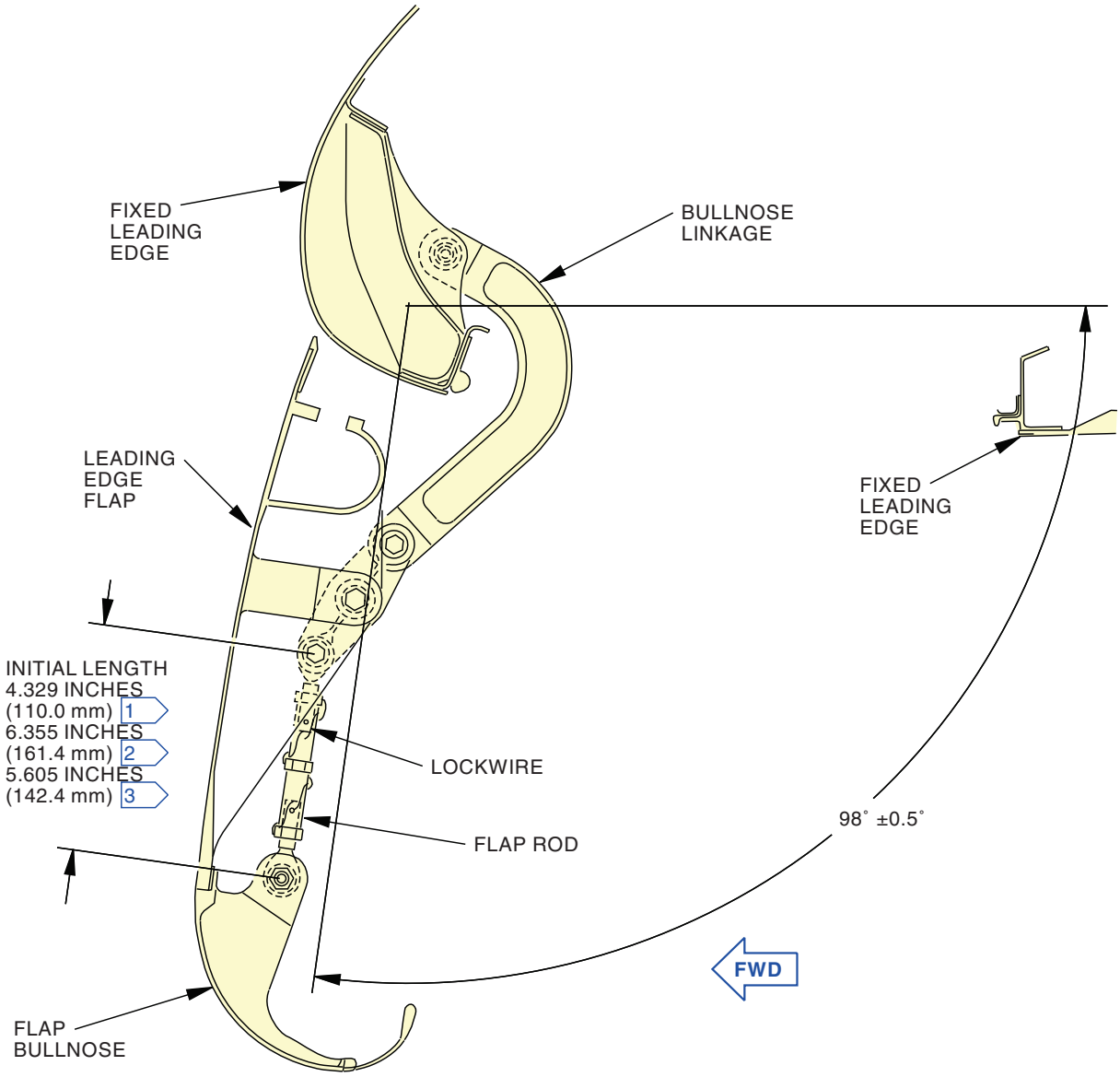
LEADING EDGE FLAP 2

B

2413312 S00061532331_V1

Leading Edge Flap Adjustment
Figure 501/27-81-00-990-804 (Sheet 2 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



OUTBOARD BULLNOSE LINKAGE
(MID AND INBOARD BULLNOSE LINKAGES ARE EQUIVALENT)
(FLAP IS SHOWN EXTENDED)

A-A

1 INBOARD BULLNOSE LINKAGE

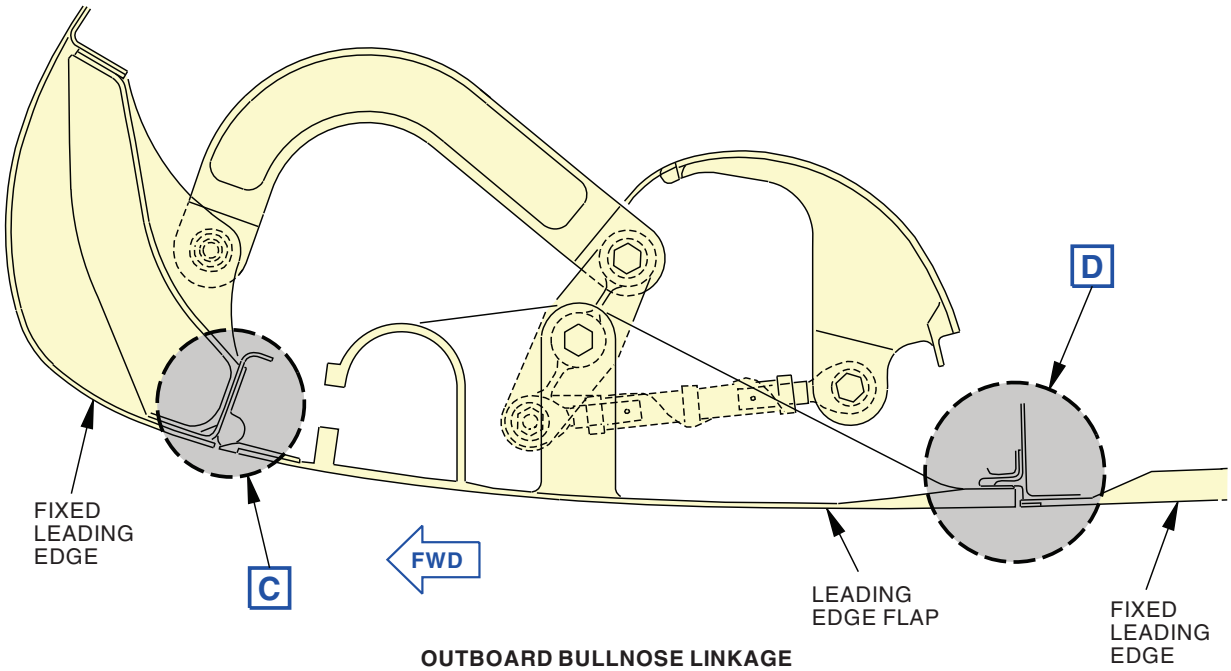
2 MID BULLNOSE LINKAGE

3 OUTBOARD BULLNOSE LINKAGE

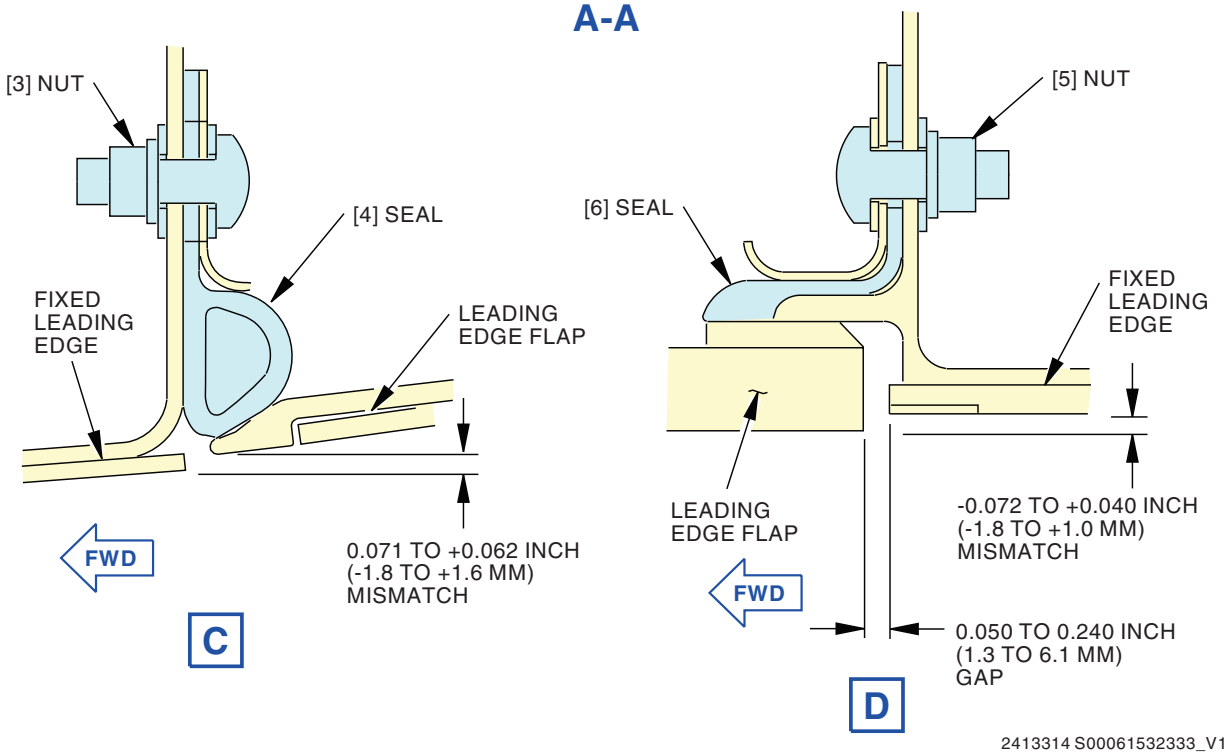
2413313 S00061532332_V1

Leading Edge Flap Adjustment
Figure 501/27-81-00-990-804 (Sheet 3 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

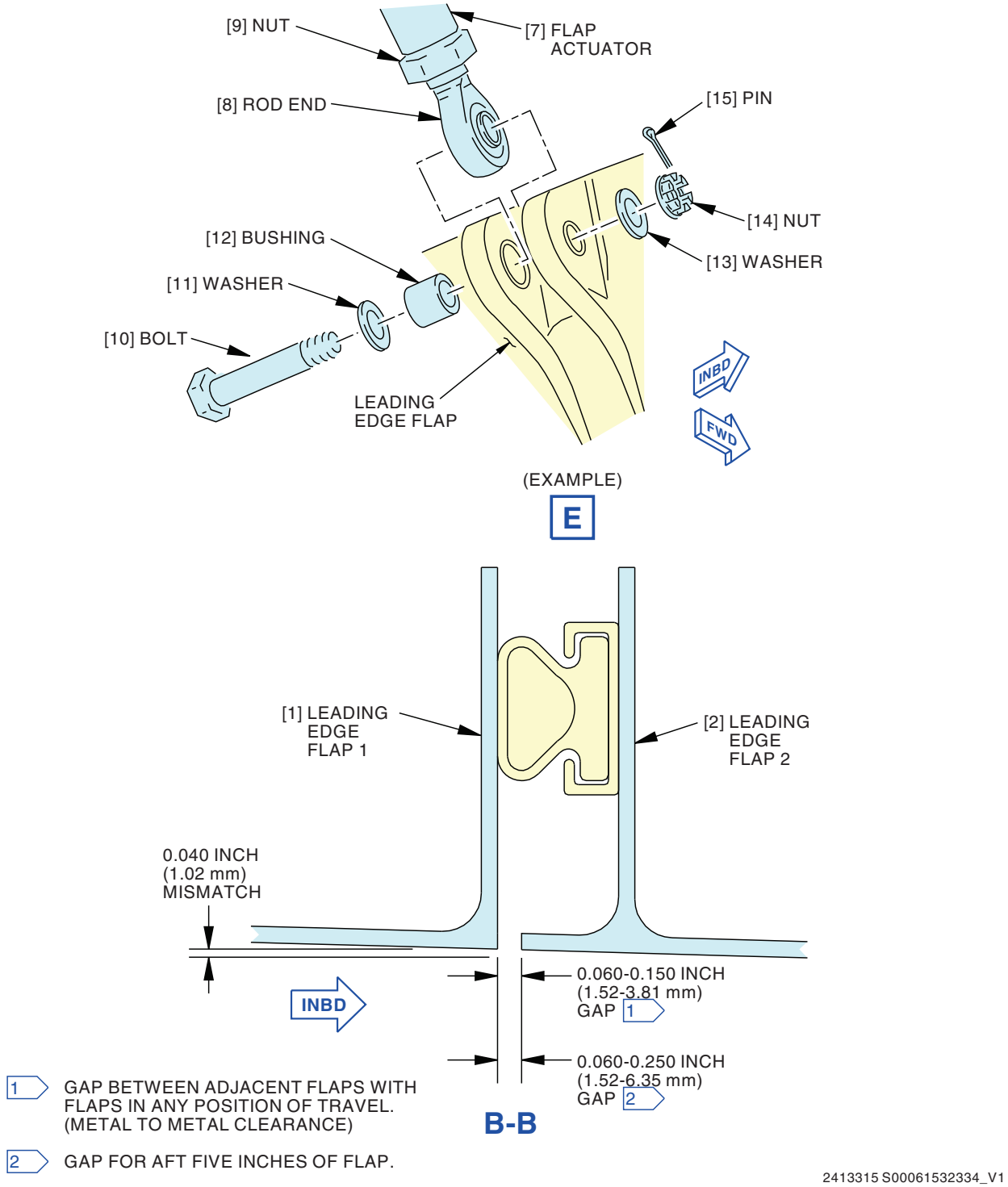


OUTBOARD BULLNOSE LINKAGE
(MID AND INBOARD BULLNOSE LINKAGES ARE EQUIVALENT)
(FLAP IS SHOWN RETRACTED)



Leading Edge Flap Adjustment
Figure 501/27-81-00-990-804 (Sheet 4 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Leading Edge Flap Adjustment
Figure 501/27-81-00-990-804 (Sheet 5 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-81-00-820-802

3. Leading Edge Slat Actuator Adjustment

(Figure 502)

A. General

(1) You must adjust the leading edge slats with the wing unsupported outboard of the engine nacelle. You can adjust the leading edge slats with the airplane on the gear or on jacks inboard of the engine nacelle.

B. References

Reference	Title
27-81-00-080-801	Leading Edge Flap and Slat Locks Removal (P/B 201)
27-81-00-480-801	Leading Edge Flap and Slat Locks Installation (P/B 201)
27-81-00-860-804	Leading Edge Flaps and Slats Retraction (P/B 201)
27-81-21-400-801	Leading Edge Slat Installation (P/B 401)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1734	Lockout Set - Leading Edge Slats and Krueger Flaps Actuators Part #: C27051-29 Supplier: 81205

D. Consumable Materials

Reference	Description	Specification
C00528	Compound - Corrosion Preventive, Petroleum Hot Application (Soft Film)	MIL-C-11796 Class III

E. Location Zones

Zone	Area
522	Left Wing - Slat No. 4
523	Left Wing - Slat No. 3
524	Left Wing - Slat No. 2
525	Left Wing - Slat No. 1
622	Right Wing - Slat No. 5
623	Right Wing - Slat No. 6
624	Right Wing - Slat No. 7
625	Right Wing - Slat No. 8

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Slat Actuator Adjustment

SUBTASK 27-81-00-860-038

WARNING KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) To pressurize hydraulic system B, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 27-81-00-860-039

CAUTION MAKE SURE THAT THE INBOARD FAN COWL PANEL AND THE INBOARD AND OUTBOARD THRUST REVERSERS ARE CLOSED AND IN THE STOWED POSITION OR REMOVED BEFORE YOU EXTEND THE LEADING EDGE FLAPS AND SLATS. THERE IS NOT SUFFICIENT CLEARANCE FOR THE FLAPS AND SLATS TO EXTEND IF THE INBOARD FAN COWL PANEL AND THE INBOARD AND OUTBOARD THRUST REVERSERS ARE IN THE OPEN POSITION. THIS CAN CAUSE DAMAGE TO EQUIPMENT.

(2) Move the flap control lever to the 30-unit position, to extend the leading edge slat [31] to the full extend position.

SUBTASK 27-81-00-480-010

WARNING INSTALL LEADING EDGE SLAT ACTUATOR LOCKS. LEADING EDGE SLATS CAN CAUSE INJURY TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(3) Install the lockout set, SPL-1734, on the adjacent leading edge slat [31]. Do this task: Leading Edge Flap and Slat Locks Installation, TASK 27-81-00-480-801.

SUBTASK 27-81-00-860-040

(4) Move the flap control lever to the 1-unit position, to retract the leading edge slat [31] to the extend position.

SUBTASK 27-81-00-860-041

(5) Remove power from hydraulic system B. Do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 27-81-00-200-001

WARNING DO NOT LET OBJECTS GET IN THE HOUSING ASSEMBLY OF THE SLAT TRACK. THIS WILL HELP PREVENT A PUNCTURE OF THE HOUSING ASSEMBLY THAT COULD CAUSE A FUEL LEAK. THE FUEL LEAK COULD CAUSE A FIRE AND POSSIBLE DEATH OR INJURY TO PERSONNEL.

(6) Keep the housing (can) assemblies of the slat main tracks clean and free of all unwanted objects Foreign Object Damage (FOD) at all times.

SUBTASK 27-81-00-020-006

(7) Remove the slat actuator door [39] from the slat actuator [40]:

(a) Remove the bolts [35], washers [36], washers [37], and nuts [38] that attach the slat actuator door [39] to the slat actuator [40].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Remove the bolts [32], washers [33] and nuts [34] that attach the slat actuator door [39] and the bracket to the slat actuator [40].

(c) Remove the slat actuator door [39].

(d) Temporarily install the bolts [32], washers [33] and nuts [34] to attach the bracket to the slat actuator [40].

SUBTASK 27-81-00-020-007

(8) Disconnect the slat actuator [40] from the leading edge slat [31]:

(a) Remove the bolts [46] and washers [47] that attach the bolt retainer [48] to the leading edge slat [31].

(b) Remove the bolt retainer [48] from the leading edge slat [31].

WARNING MAKE SURE YOU HOLD THE LEADING EDGE SLAT IN ITS POSITION. THE LEADING EDGE SLAT CAN FALL TO THE FULL EXTEND POSITION. THIS CAN CAUSE INJURY TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(c) Hold the leading edge slat [31] while you remove the nut [43], washer [42], bushing [41], washer [49] and bolt [50].

SUBTASK 27-81-00-860-042

(9) Carefully let the leading edge slat [31] move down, to the full extend position.

SUBTASK 27-81-00-200-002

(10) Make sure that the downstop assembly is installed on the main tracks (TASK 27-81-21-400-801).

SUBTASK 27-81-00-420-002

(11) Connect the slat actuator [40] to the leading edge slat [31]:

WARNING MAKE SURE YOU HOLD THE LEADING EDGE SLAT IN ITS POSITION. THE LEADING EDGE SLAT CAN FALL TO THE FULL EXTEND POSITION. THIS CAN CAUSE INJURY TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(a) Move the leading edge slat [31], until it aligns with the slat actuator [40].

1) Hold the leading edge slat [31], while you install the fasteners.

(b) Install the bushing [41], bolt [50], washer [49].

1) The head of the bolt [50] must be on the outboard side of the slat actuator [40].

(c) Install the washer [42] and nut [43].

1) Tighten the nut [43] to 660 in-lb (74.6 N·m) - 980 in-lb (110.7 N·m).

a) Optional method: Tighten the head of the bolt [50] to 882 in-lb (99.7 N·m) - 1078 in-lb (121.8 N·m).

(d) Install the bolt retainer [48]:

1) Apply compound, C00528, to the shank and threads of the bolts [46].

2) Install the bolts [46] and washers [47] to attach the bolt retainer [48].

(e) Install the lockwire at the slat actuator [40] rod end:

1) Bend the lockwire tail ends parallel to the actuator rod.

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2) Make sure that the lockwire tail ends do not touch the leading edge panel assembly.

NOTE: If the lockwire tail ends touch the leading edge assembly, it can cause chafing and damage to the leading edge panel.

SUBTASK 27-81-00-860-124

(12) Use hydraulic power to extend the slat actuator [40] to the full extend position:

WARNING

KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(a) Pressurize hydraulic system B. Do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

CAUTION

MAKE SURE THAT THE INBOARD FAN COWL PANEL AND THE INBOARD AND OUTBOARD THRUST REVERSERS ARE CLOSED AND IN THE STOWED POSITION OR REMOVED BEFORE YOU EXTEND THE LEADING EDGE FLAPS AND SLATS. THERE IS NOT SUFFICIENT CLEARANCE FOR THE FLAPS AND SLATS TO EXTEND IF THE INBOARD FAN COWL PANEL AND THE INBOARD AND OUTBOARD THRUST REVERSERS ARE IN THE OPEN POSITION. THIS CAN CAUSE DAMAGE TO EQUIPMENT.

(b) Move the flap control lever to the 30-unit detent position, to extend the slat actuator [40] to the full extend position.

(c) Remove power from hydraulic system B. Do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 27-81-00-220-005

(13) With the slats correctly adjusted (rigged), do the step that follow:

NOTE: If the leading edge slat is in the full extend position, the slat actuator will be preloaded against the leading edge slat.

(a) Apply hand load to the end of each slat assembly (as necessary), to make sure that preload exists.

(b) If no preload exists, do as follow:

1) Make sure that the possible under-travel dimension of the slat actuator [40] does not exceed 0.0200 in. (0.5080 mm) maximum, between the attach point at the rod end of the slat actuator [40] and the clevis bores of the slat attach fittings.

NOTE: The under-travel dimension of the slat actuator indicates that there is no preload of the slat actuator in relationship to the slat actuator attach fitting.

a) If the dimension is greater than 0.0200 in. (0.5080 mm), make sure that the part numbers of the main track assemblies are correct.

NOTE: When the slat main tracks are incorrectly installed visual cues (excessive mismatches, misalignment, skew, etc.) will sometimes indicate a problem in advance depending on which slat is affected.

SUBTASK 27-81-00-420-003

(14) Install the slat actuator door [39] on the slat actuator [40]:

(a) Put the slat actuator door [39] in its position on the slat actuator [40].

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(b) Remove the bolts [32], washers [33] and nuts [34] that attach the bracket to the slat actuator [40].

(c) Put the slat actuator door [39] in its position on the slat actuator [40].

(d) Install the bolt [32], washers [33] and nuts [34] that attach the slat actuator door [39] and bracket to the slat actuator [40].

(e) Install the bolts [35], washers [36], washers [37] and nuts [38] that attach the slat actuator door [39] to the slat actuator [40].

SUBTASK 27-81-00-200-003

WARNING DO NOT LET OBJECTS GET IN THE HOUSING ASSEMBLY OF THE SLAT TRACK. THIS WILL HELP PREVENT A PUNCTURE OF THE HOUSING ASSEMBLY THAT COULD CAUSE A FUEL LEAK. THE FUEL LEAK COULD CAUSE A FIRE AND POSSIBLE DEATH OR INJURY TO PERSONNEL.

(15) Keep the housing (can) assemblies of the slat main tracks clean and free of all unwanted objects FOD at all time.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 27-81-00-080-009

WARNING REMOVE LEADING EDGE ACTUATOR LOCKS. LEADING EDGE SLATS CAN CAUSE INJURY TO PERSONNEL AND DAMAGE TO EQUIPMENT.

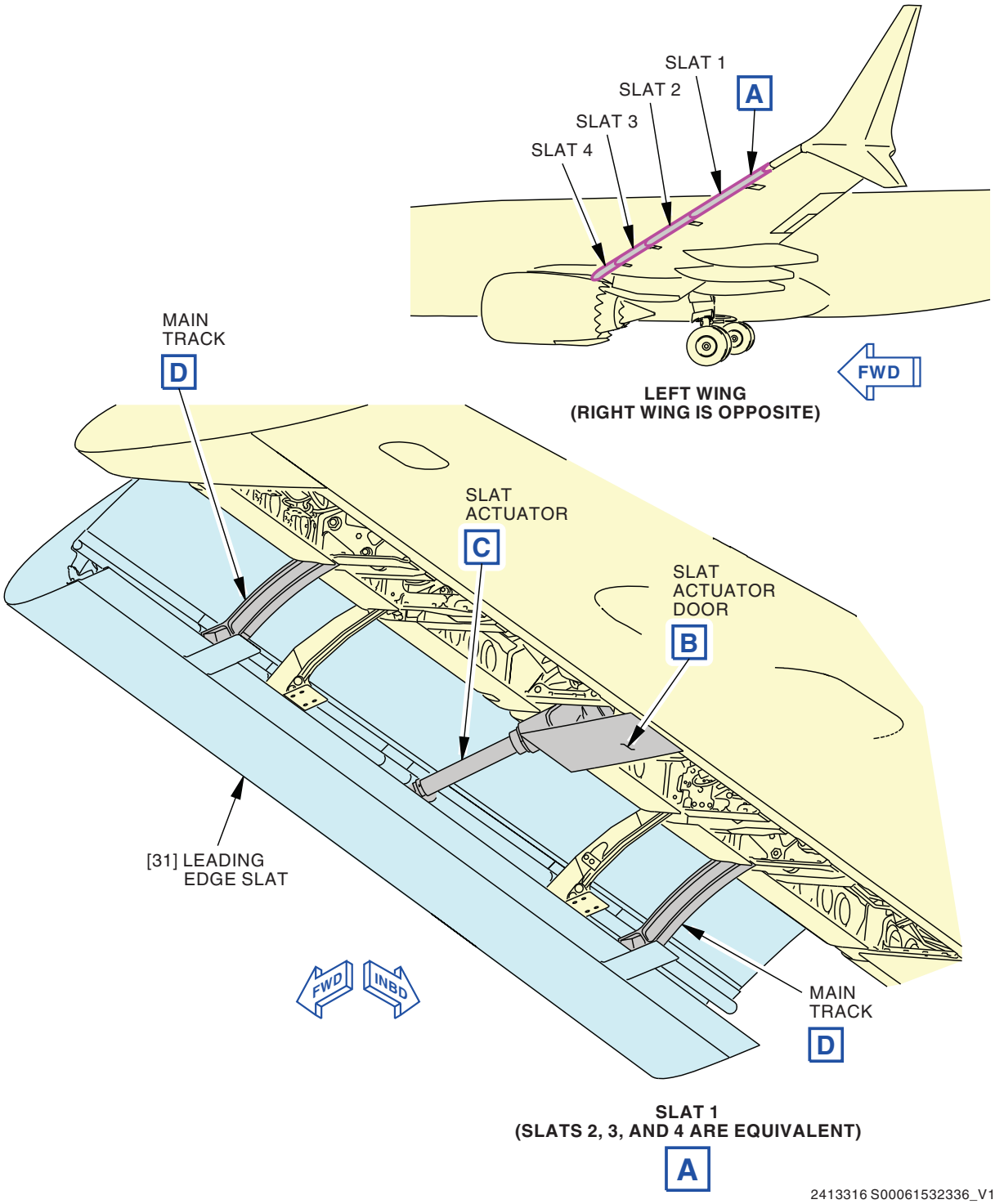
(1) Do this task: Leading Edge Flap and Slat Locks Removal, TASK 27-81-00-080-801.

SUBTASK 27-81-00-860-044

(2) Do this task: Leading Edge Flaps and Slats Retraction, TASK 27-81-00-860-804.

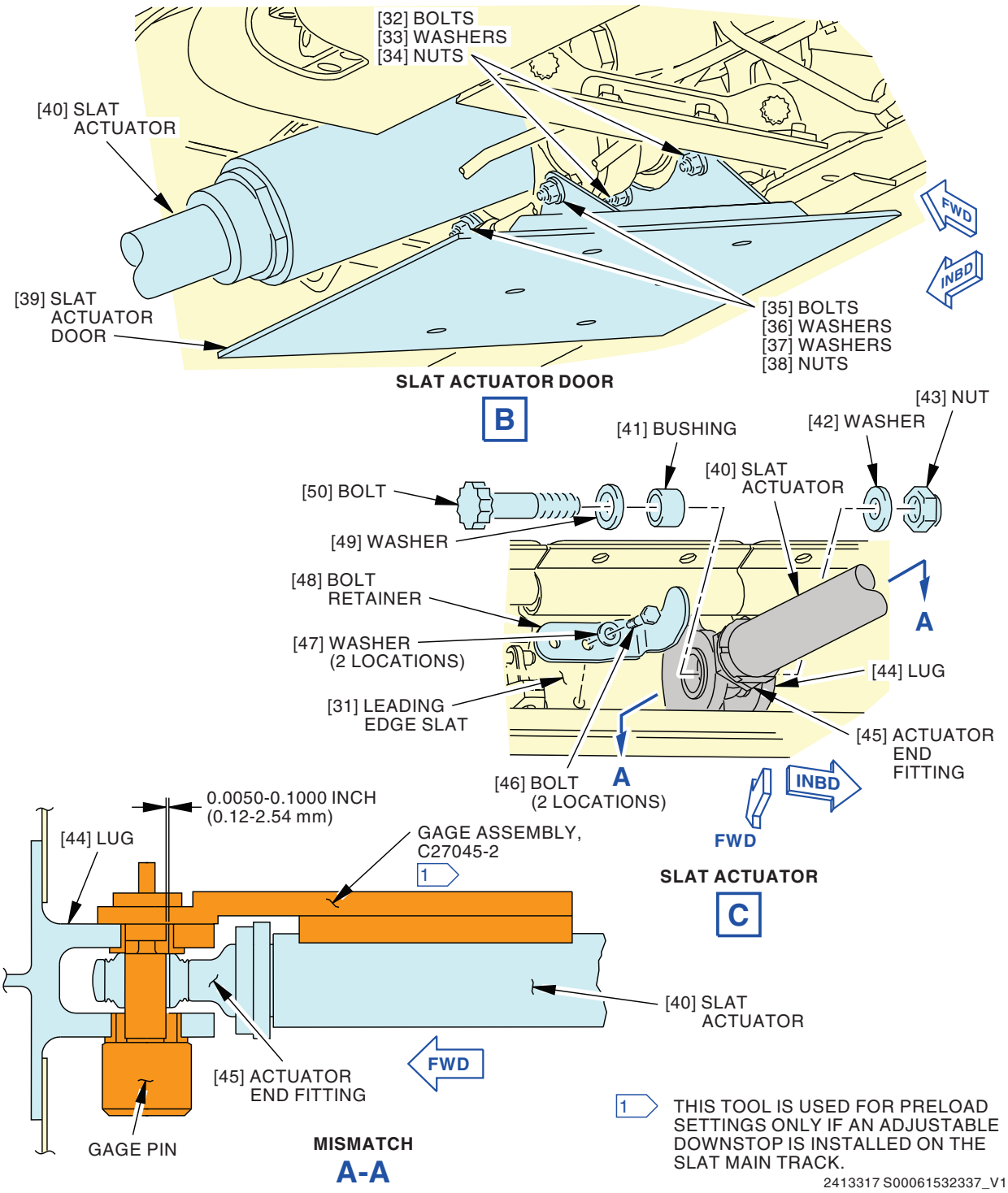
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



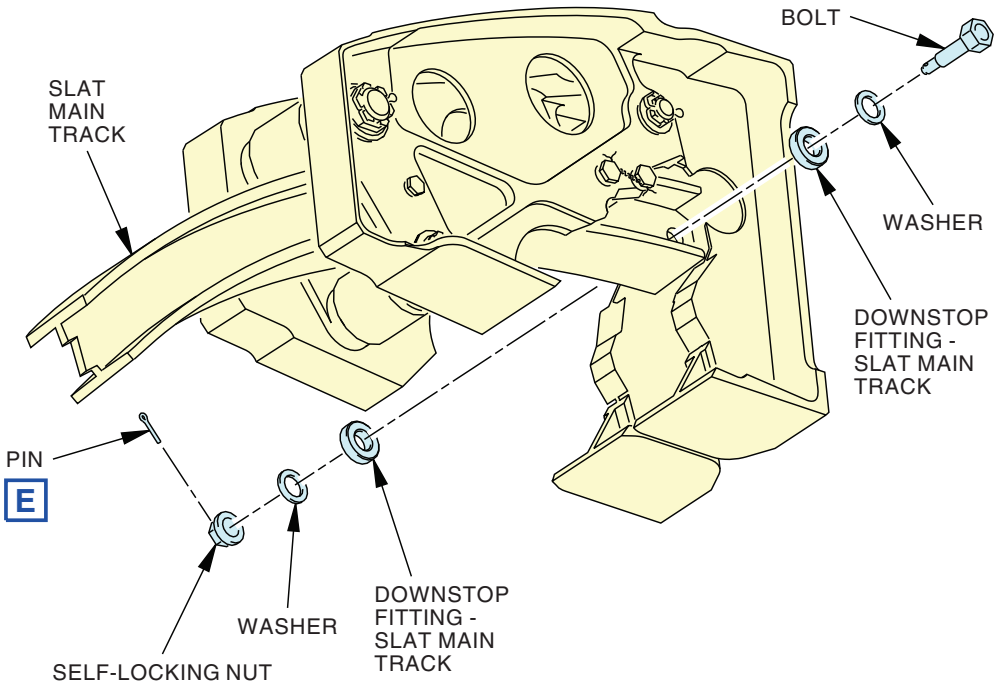
Leading Edge Slat Actuator Adjustment
Figure 502/27-81-00-990-805 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



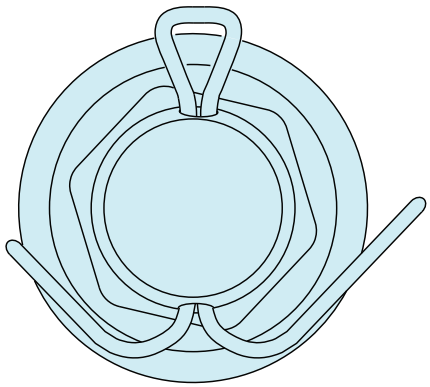
Leading Edge Slat Actuator Adjustment
Figure 502/27-81-00-990-805 (Sheet 2 of 3)

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TYPICAL NON-ADJUSTABLE DOWNSTOP INSTALLATION

D



E

2413318 S00061532338_V1

Leading Edge Slat Actuator Adjustment
Figure 502/27-81-00-990-805 (Sheet 3 of 3)

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TASK 27-81-00-820-803

4. Leading Edge Slats Adjustment

(Figure 503, Figure 504, and Figure 505)

A. General

(1) You must adjust the leading edge slats with the wing unsupported outboard of the engine nacelle. You can adjust the leading edge slats with the airplane on the gear or on jacks inboard of the engine nacelle. The airplane must not have fuel on the outboard wing.

(2) It is recommended that you do all the steps in this task, the full slat adjustment, for the most consistent and lowest drag adjustment.

(3) Where time and/or airplane condition do not support the use of the full slat adjustment, a minimum slat adjustment is permitted. Use of this minimum slat adjustment can result in increased drag compared with the full slat adjustment. In this task, all the steps that are not necessary for a minimum slat adjustment will be identified.

B. References

Reference	Title
27-81-00-860-803	Leading Edge Flaps and Slats Extension (P/B 201)
27-81-00-860-804	Leading Edge Flaps and Slats Retraction (P/B 201)
27-88-11 P/B 401	LEADING EDGE FLAP AND SLAT PROXIMITY SENSOR - REMOVAL/INSTALLATION
57-41-02-000-801	Leading Edge Access Panel Removal (P/B 201)
57-41-02-400-801	Leading Edge Access Panel Installation (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1739	Rigging Equipment - Slat Part #: C27045-1 Supplier: 81205
STD-755	Scale - Spring, Push/Pull Type, 0 - 150lbs (0 - 68kg) Capacity

D. Consumable Materials

Reference	Description	Specification
G01048	Lockwire - MS20995C32, Corrosion Resistant Steel - 0.032 Inch (0.8128 mm) Diameter	NASM20995
G50136	Compound - Corrosion Inhibiting, Non-drying Paste	BMS3-38
G50237	Compound - Corrosion Inhibiting, Non-drying - Cor-Ban 27L	BMS3-38

E. Location Zones

Zone	Area
522	Left Wing - Slat No. 4
523	Left Wing - Slat No. 3
524	Left Wing - Slat No. 2
525	Left Wing - Slat No. 1
622	Right Wing - Slat No. 5

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(Continued)

Zone	Area
623	Right Wing - Slat No. 6
624	Right Wing - Slat No. 7
625	Right Wing - Slat No. 8

F. Access Panels

Number	Name/Location
521CB	Lower Leading Edge Access Panel
521DB	Lower Leading Edge Access Panel
521EB	Lower Leading Edge Access Panel
521FB	Lower Leading Edge Access Panel
521JB	Lower Leading Edge Access Panel
521KB	Lower Leading Edge Access Panel
521LB	Lower Leading Edge Access Panel
521MB	Lower Leading Edge Access Panel
521QB	Lower Leading Edge Access Panel
521RB	Lower Leading Edge Access Panel
521SB	Lower Leading Edge Access Panel
521TB	Lower Leading Edge Access Panel
521WB	Lower Leading Edge Access Panel
521XB	Lower Leading Edge Access Panel
521YB	Lower Leading Edge Access Panel
521ZB	Lower Leading Edge Access Panel
621CB	Lower Leading Edge Access Panel
621DB	Lower Leading Edge Access Panel
621EB	Lower Leading Edge Access Panel
621FB	Lower Leading Edge Access Panel
621JB	Lower Leading Edge Access Panel
621KB	Lower Leading Edge Access Panel
621LB	Lower Leading Edge Access Panel
621MB	Lower Leading Edge Access Panel
621QB	Lower Leading Edge Access Panel
621RB	Lower Leading Edge Access Panel
621SB	Lower Leading Edge Access Panel
621TB	Lower Leading Edge Access Panel
621VB	Lower Leading Edge Access Panel
621WB	Lower Leading Edge Access Panel
621XB	Lower Leading Edge Access Panel
621YB	Lower Leading Edge Access Panel
621ZB	Lower Leading Edge Access Panel

G. Prepare for the Adjustment

SUBTASK 27-81-00-010-002

(1) Remove the applicable access panels (TASK 57-41-02-000-801):

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(a) For the left wing, remove these access panels:

Number	Name/Location
521CB	Lower Leading Edge Access Panel
521DB	Lower Leading Edge Access Panel
521EB	Lower Leading Edge Access Panel
521FB	Lower Leading Edge Access Panel
521JB	Lower Leading Edge Access Panel
521KB	Lower Leading Edge Access Panel
521LB	Lower Leading Edge Access Panel
521MB	Lower Leading Edge Access Panel
521QB	Lower Leading Edge Access Panel
521RB	Lower Leading Edge Access Panel
521SB	Lower Leading Edge Access Panel
521TB	Lower Leading Edge Access Panel
521WB	Lower Leading Edge Access Panel
521XB	Lower Leading Edge Access Panel
521YB	Lower Leading Edge Access Panel
521ZB	Lower Leading Edge Access Panel

(b) For the right wing, remove these access panels:

Number	Name/Location
621CB	Lower Leading Edge Access Panel
621DB	Lower Leading Edge Access Panel
621EB	Lower Leading Edge Access Panel
621FB	Lower Leading Edge Access Panel
621JB	Lower Leading Edge Access Panel
621KB	Lower Leading Edge Access Panel
621LB	Lower Leading Edge Access Panel
621QB	Lower Leading Edge Access Panel
621RB	Lower Leading Edge Access Panel
621SB	Lower Leading Edge Access Panel
621TB	Lower Leading Edge Access Panel
621VB	Lower Leading Edge Access Panel
621WB	Lower Leading Edge Access Panel
621XB	Lower Leading Edge Access Panel
621YB	Lower Leading Edge Access Panel

SUBTASK 27-81-00-200-004

WARNING DO NOT LET OBJECTS GET IN THE HOUSING ASSEMBLY OF THE SLAT TRACK. THIS WILL HELP PREVENT A PUNCTURE OF THE HOUSING ASSEMBLY THAT COULD CAUSE A FUEL LEAK. THE FUEL LEAK COULD CAUSE A FIRE AND POSSIBLE DEATH OR INJURY TO PERSONNEL.

(2) Keep the housing (can) assemblies of the slat main tracks clean and free of all unwanted objects (FOD) at all time.

SUBTASK 27-81-00-820-001

(3) Do a preliminary adjustment of the auxiliary track forward support:

(a) Install the preload shim assembly, that is part of the rigging equipment, SPL-1739, at the forward supports for the auxiliary tracks.

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1) Remove the nuts [74] and washers [75] from the clevis [76].
2) Lift the auxiliary track [72] with the clevis [76] through the auxiliary track support.
3) Install the preload shim assembly, which is part of the rigging equipment, SPL-1739, and shim [77] on the clevis [76].
4) Put the clevis [76] back in its position on the auxiliary track support.
5) Install the washers [75] and nuts [74] on the clevis [76].
6) Make sure that two BACW10P390TF, PTFE (Polytetraflouroethylene (Teflon)) washers are installed in the clevis/auxiliary track/detent arm joint, if you disconnected it. Use the applicable Illustrated Part Catalog for the required washers at this location.

NOTE: If the washers are damaged or missing, stabilization of the detent arm is reduced and it is possible for the detent arm to be deflected laterally during slat extension or retraction and the possibility of misalignment between the detent arm and the detent roller increases.

NOTE: If the washers are damaged or missing, it may be possible for the roller to come out of the detent arm track, thus affecting the operation of the detent system. Also extreme misalignment of the slat roller attachment could move the slat down sensor target, causing fault indications or physical damage to the sensor itself.

SUBTASK 27-81-00-820-002

(4) Do a preliminary adjustment of the auxiliary track aft support:

NOTE: If a minimum slat adjustment is done, preliminary adjustment of the auxiliary track aft support is not necessary.

(a) Loosen the nut [90] on the locking bolt [91] at the upper aft auxiliary track.
(b) Loosen the jamnut [82].
(c) Loosen the nut [80] on the locking bolt [79].
(d) Turn the adjustment nut [81] of the auxiliary track aft support to get the required trailing edge clearance until the dimension A is as specified (Table 501).

NOTE: Optional. In order to adjust the auxiliary track aft support to the dimensions given in the table below, adjust the auxiliary track up until the slot in the aft upper corner of the track touches the locking bolt [91] at the upper aft auxiliary track

Table 501/27-81-00-993-802

Location	Dimension A	Dimension B (reference)
Slat 1 and 8 (Outbd)	0.830 in. (21.08 mm)	2.380 in. (60.45 mm)
Slat 1 and 8 (Inbd)	0.950 in. (24.13 mm)	2.500 in. (63.50 mm)
Slat 2 and 7 (Outbd)	0.950 in. (24.13 mm)	2.500 in. (63.50 mm)
Slat 2 and 7 (Inbd)	1.450 in. (36.83 mm)	3.000 in. (76.20 mm)
Slat 3 and 6 (Outbd)	1.450 in. (36.83 mm)	3.000 in. (76.20 mm)
Slat 3 and 6 (Inbd)	1.450 in. (36.83 mm)	3.000 in. (76.20 mm)
Slat 4 and 5 (Outbd)	1.500 in. (38.10 mm)	3.050 in. (77.47 mm)
Slat 4 and 5 (Inbd)	2.100 in. (53.34 mm)	3.650 in. (92.71 mm)

(e) Tighten the nut [80] by hand on the locking bolt [79].

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(f) Temporarily tighten the jamnut [82].

NOTE: Do not install lockwire on the jam nut.

(g) Temporarily tighten the upper aft auxiliary track locking bolt [91].

SUBTASK 27-81-00-820-003

(5) Do a preliminary adjustment of the upstop pad (View C, Figure 503):

(a) Remove the retainers [85], nuts [86], and bolts [89].

(b) Add shims [87] under each of the stop pads [88].

NOTE: The total shim thickness will be 0.060 +0.005 / -0.002 in. (1.52 +0.13 / -0.05 mm).

(c) Install the bolts [89], nuts [86], and retainers [85].

SUBTASK 27-81-00-820-004

(6) Do a preliminary adjustment of the endstop pad (View C, Figure 503):

NOTE: If you do a minimum slat adjustment, preliminary adjustment of the endstop pad is not necessary.

(a) Remove the retainers [85], nuts [86], and bolts [89].

(b) Add shims [87] under each of the stop pads [88].

NOTE: The total shim thickness will be 0.060 +0.005 / -0.002 in. (1.52 +0.13 / -0.05 mm).

(c) Apply Cor-Ban 27L Compound, G50237, or corrosion inhibiting material, G50136, to the bolts [89].

(d) Install the bolts [89], nuts [86], and retainers [85].

1) Tighten the bolts [89] to 16 in-lb (1.8 N·m) - 20 in-lb (2.3 N·m).

SUBTASK 27-81-00-820-005

(7) Do a preliminary adjustment of the detent arm preload (Views C-C and D-D, Figure 503):

NOTE: If you do a minimum slat adjustment, preliminary adjustment of the detent arm preload is not necessary.

NOTE: When measuring the 0.100 in. (2.54 mm) minimum clearance between the torsion rod [109] and shoulder bolt [105], the detent arm [78] must be flush and in contact with the shoulder bolt [105]. This dimension only applies during the initial measurement of the torsion rod [109]. This dimension will not apply after the torsion rod [109] is preloaded during the slat adjustment.

(a) Make sure that three (CRES material, 0.063 in. (1.60 mm) nominal thickness) washers are installed in the detent roller/track/arm/sensor target joint (one washer under the bolt head and two washers installed on both sides of the auxiliary arm) per the applicable Illustrated Part Catalog.

NOTE: If the washers are thinner than 0.063 in. (1.60 mm), the lateral location of the roller bearing will be changed and it will no longer align with the centerline of the detent arm.

NOTE: The thinner washers will not support the torque preload and operating load imposed on the joint. As the thinner washers deform, the fastener torque preload is lost, and the joint may move under normal operating loads, inducing additional misalignment of the bearing. This joint holds the slat sensor target, so a loosened bolt will permit the sensor target to move. Target movement could lead to sensor indication problems, and the loose target could physically contact the sensor as the slat moves and cause additional damage.

(b) Adjust all preload on the detent arm to 0 lb (0 kg).

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1) Turn the detent arm preload adjustment bolt [73] clockwise to increase preload or counterclockwise to reduce preload (View D-D, Figure 503).

2) Adjust the preload on all the detent arms [78] to 0 lb (0 kg).

SUBTASK 27-81-00-860-046

(8) Make sure that the bolts [102] on the skin tab [92] are loose on the slat auxiliary arms (View F, Figure 503).

NOTE: This is to prevent damage to the fixed leading edge when the slat is retracted during the adjustment.

H. Leading Edge Slats Adjustment

NOTE: The following operations may require slat position change to complete the adjustment requirements and may result in dimensional changes when moving the slats.

SUBTASK 27-81-00-200-005

WARNING

DO NOT LET OBJECTS GET IN THE HOUSING ASSEMBLY OF THE SLAT TRACK. THIS WILL HELP PREVENT A PUNCTURE OF THE HOUSING ASSEMBLY THAT COULD CAUSE A FUEL LEAK. THE FUEL LEAK COULD CAUSE A FIRE AND POSSIBLE DEATH OR INJURY TO PERSONNEL.

(1) Keep the housing (can) assemblies of the slat main tracks clean and free of all unwanted objects (FOD) at all times.

SUBTASK 27-81-00-820-006

(2) Do the initial adjustment of the leading edge slat in the (intermediate) extend position:

NOTE: If you do a minimum slat adjustment, initial adjustment of the slat in the (intermediate) extend position is not necessary.

WARNING

MAKE SURE THAT PERSONNEL AND EQUIPMENT STAY AWAY FROM THE LEADING EDGE SLATS, TRAILING EDGE FLAPS, AND DRIVE MECHANISMS. THE FLAPS, SLATS, AND DRIVE MECHANISMS MOVE QUICKLY. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

CAUTION

MAKE SURE THAT THE INBOARD FAN COWL PANEL AND THE INBOARD AND OUTBOARD THRUST REVERSERS ARE CLOSED AND IN THE STOWED POSITION OR REMOVED BEFORE YOU EXTEND THE LEADING EDGE FLAPS AND SLATS. THERE IS NOT SUFFICIENT CLEARANCE FOR THE FLAPS AND SLATS TO EXTEND IF THE INBOARD FAN COWL PANEL AND THE INBOARD AND OUTBOARD THRUST REVERSERS ARE IN THE OPEN POSITION. THIS CAN CAUSE DAMAGE TO EQUIPMENT.

(a) Extend the leading edge slat to the (intermediate) extend (1-unit detent) position with hydraulic power, do this task: Leading Edge Flaps and Slats Extension, TASK 27-81-00-860-803.

(b) Align the noses of the slats to get a mismatch of 0.000 in. (0.00 mm) to 0.060 in. (1.52 mm) between the adjacent slats.

1) Dimension may apply at any one point within the 1.0000 in. (25.40 mm) band width (Figure 505).

NOTE: This dimension is normal to the skin surface where the measurement is taken. Vertical mismatch is permissible.

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(c) Adjust the forward support of the auxiliary track [72] (View A-A, Figure 503).

1) Make sure that two BACW10P390TF, PTFE washers are installed in the clevis/auxiliary track/detent arm joint, if you disconnected it. Use the applicable Illustrated Part Catalog for the required washers at this location.

NOTE: If the washers are damaged or missing, stabilization of the detent arm is reduced and it is possible for the detent arm to be deflected laterally during slat extension or retraction and the possibility of misalignment between the detent arm and the detent roller increases.

NOTE: If the washers are damaged or missing, it may be possible for the roller to come out of the detent arm track, thus affecting the operation of the detent system. Also extreme misalignment of the slat roller attachment could move the slat down sensor target, causing fault indications or physical damage to the sensor itself.

2) Tighten the nut [110] to 130 in-lb (14.7 N·m) - 200 in-lb (22.6 N·m) after the PTFE washers / detent arm are re-installed.

3) Remove the nut [74] and washer [75] from the clevis [76].

4) Lift the auxiliary track [72] with the clevis [76] through the auxiliary track support.

5) Add or remove the shims [77] until the gap between the slat and fixed leading edge is 0.002 +0.006 / -0.000 in. (0.05 +0.15 / -0.00 mm).

a) Measure the clearance at the tightest point, within 2.00 in. (50.8 mm) on either side of the each auxiliary track centerline.

b) Measure the clearance to a depth of 0.40 in. (10.16 mm) to 0.60 in. (15.24 mm).

6) Move the slat, if required, to remove shims [77].

7) Make sure that the total shim thickness is 0.030 in. (0.76 mm) to 0.270 in. (6.86 mm).

8) Re-align the noses of the slats to get a mismatch of 0.000 in. (0.00 mm) to 0.060 in. (1.52 mm) between the adjacent slats, if/as required.

9) Put the clevis [76] back in its positions on the auxiliary track support.

10) Install the washer [75] and nut [74] on the clevis [76].

(d) Adjust the cam plate [83].

NOTE: The cam plates are located on the forward end of the detent arm and are used to align and hold the slats in the extend position.

1) Loosen the two nuts [84] that attach the cam plate [83] to the detent arm [78].

a) Adjust the cam plates to the dimension (Views B-B and C-C, Figure 503).

<1> If the cam plate serrations do not mate with corresponding serrations on the detent arm when located per the required dimension then move the cam plate aft into the next available serration.

2) Move the cam plate [83] forward or aft, as required, until you get these conditions:

a) Engage the serrations in the required position and tighten the nuts [84] to a setpoint of 85 in-lb (9.6 N·m) (preferred method) or bolts [103] to a setpoint of 100 in-lb (11.3 N·m) (optional method).

NOTE: The recommended method is to tighten the nuts.

NOTE: The optional method is to tighten the bolts.

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b) The detent roller touch the root of each detent arm [78].

c) Make sure that the clearance between the detent roller and aft surface of the cam plate is 0.000 in. (0.00 mm) to 0.030 in. (0.76 mm).

3) Cycle the slats and make sure that the 0.002 +0.006 / -0.000 in. (0.05 +0.15 / -0.00 mm) clearance is kept.

4) Adjust if/as required the alignment of the slat noses and detent arm cam plates.

(e) Move the slat from the extend position until the detent roller is completely clear of the detent arm [78] and cam plate [83].

(f) On slats 1, 4, 5 and 8, adjust the detent arm preload to 80 ±5 lbf (355.9 ±22.3 N), measured with a spring scale, STD-755, at the throat of the cam plate [83], and 0.100 +0.060 / -0.040 in. (2.54 +1.52 / -1.02 mm) dimension at the lower portion of the detent arm [78].

NOTE: The detent arm preload adjustment bolt [73] is located next to the auxiliary tracks at the far end of the torsion rod [109] and is used to change the preload tension of the torsion rod [109] and detent arm [78]. Adjust the preload by rotating the detent arm preload adjustment bolt [73] clockwise for increased preload and counterclockwise for reduced preload.

1) If it is necessary, remove the nut at the lower auxiliary track attach bolt to see this clearance.

(g) On slats 2, 3, 6 and 7, adjust the detent arm preload to 50 ±5 lbf (222.4 ±22.3 N), measured with a spring scale, STD-755, at the throat of the cam plate [83], and 0.100 +0.060 / -0.040 in. (2.54 +1.52 / -1.02 mm) dimension at the lower portion of the detent arm [78].

(h) Make sure that there is 0.010 in. (0.2540 mm) minimum clearance between the torsion rod [109] and shoulder bolt [104], if not, do the steps that follow (View D-D, Figure 503):

NOTE: The shoulder bolt [104] is installed through both of the flanges of the auxiliary track support clevis.

NOTE: The clearance measurement is taken where the torsion rod [109] exits the shoulder bolt [104] on the detent preload side of the mechanism.

1) Remove the preload from the detent arm [78].

2) Adjust the detent arm preload adjustment bolt [73] with the steps that follow:

a) Re-align the torsion rod [109] by backing off the nut [108] and adjusting the shoulder bolt [105] up or down in the slotted hole in the torsion rod support bracket assembly.

b) Tighten the nut [108] to 450 in-lb (51 N·m) - 600 in-lb (68 N·m).

3) Adjust the detent arm [78] preload.

a) Adjust the preload by rotating the detent arm preload adjustment bolt [73] clockwise for increased preload and counterclockwise for reduced preload.

b) After final adjustment fasten the detent arm preload adjustment bolt [73] to its lower trunnion with safety wire (single wire method permissible).

4) Make sure that it is a correct clearance between the torsion rod [109] and shoulder bolt [104].

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SUBTASK 27-81-00-820-007

(3) Do the initial adjustment of the leading edge slat in the retract position:

NOTE: If a minimum slat adjustment is done, initial adjustment of the slat in the retract position is not necessary.

(a) Do this task: Leading Edge Flaps and Slats Retraction, TASK 27-81-00-860-804.

(b) Position the ends of the slats using the end stop pad adjustments to get a 0.000 in. (0.00 mm) to 0.050 in. (1.27 mm) fair between all adjacent slat ends; between the outboard end of slats 1 and 8 and on the adjacent fixed leading edge; and between the inboard end of slat 4 and 5 and adjacent fixed leading edge.

NOTE: The end stops are located near both ends of each slat and can be used to adjust the trailing edge clearance near the slat center, clearance and misfair of the flex skin to the lower fixed leading edge panels, and misfair at the slat ends. Each stop location has a horizontal and vertical pad. To adjust each stop pad, add or subtract in any combination of 0.010 in. (0.25 mm), 0.020 in. (0.51 mm), 0.040 in. (1.02 mm) and 0.063 in. (1.60 mm) thick solid shims. The total permitted shim thickness is 0.000 in. (0.00 mm) to 0.128 in. (3.25 mm) and must be such that the slat flex skin keeps solid contact with the fixed leading edge.

(c) Make sure that the slat upstops touch the horizontal and vertical stop pads.

NOTE: The upstop pads are located 5.3 in. (135 mm) on either side of each actuator and can be used to adjust the trailing edge clearance near the slat center, clearance and misfair of the flex skin to the lower fixed leading edge panels, and misfair at the slat ends. Each stop location has a horizontal and vertical pad.

(d) If the slat upstops do not touch the horizontal and vertical stop pads, adjust the stop pad:

NOTE: Use these stops to adjust the slat nose to match the adjacent slats.

1) Remove the retainers [85], nuts [86], and bolts [89].

2) Add or remove shims [87] as it is necessary until the slat fairs with the adjacent slat.

NOTE: Add or remove only a minimum amount of shim necessary for the slat upstop to touch the stop pad. To adjust each stop pad, add or subtract in any combination of 0.010 in. (0.25 mm), 0.020 in. (0.51 mm), 0.040 in. (1.02 mm) and 0.063 in. (1.60 mm) thick solid shims. The total permitted shim thickness is 0.000 in. (0.00 mm) to 0.128 in. (3.25 mm) and must be such that the slat flex skin keeps solid contact with the fixed leading edge.

3) Install the bolts [89], nuts [86], and retainers [85].

(e) Adjust the aft support of the auxiliary track [72] (Views C-C and D-D, Figure 503):

1) Loosen the nut [90] on the locking bolt [91] at the upper aft auxiliary track.

2) Loosen the jamnut [82].

3) Loosen the nut [80] on the locking bolt [79].

4) Turn the adjustment nut [81] until the clearance between the slat and fixed leading edge is 0.002 +0.006 / -0.000 in. (0.05 +0.15 / -0.00 mm) (View D, Figure 503).

a) Measure the clearance at the tightest point, within 2.00 in. (50.8 mm) on either side of the each auxiliary track centerline.

b) Measure the clearance to a depth of 0.40 in. (10.2 mm) to 0.60 in. (15.2 mm).

5) Temporarily tighten the locking bolt [79].

6) Temporarily tighten the jamnut [82].

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7) Temporarily tighten the locking bolt [91] at the upper aft auxiliary track .

(f) Measure and calculate the weighted average misfair.

1) Measure the misfair at six locations for each specified gap (Figure 504, Table 502):

NOTE: There will be a total of 30 locations on each wing that you will need to measure.

a) The gap between all adjacent slat ends.

b) The clearance between the outboard end of slats 1 and 8 and adjacent fixed leading edge.

c) The clearance between the inboard end of slats 4 and 5 and adjacent fixed leading edge.

Table 502/27-81-00-993-803

Point	Location	Value
F1	Outbd slat 1/8	
F2	Outbd slat 1/8	
F3	Outbd slat 1/8	
F4	Outbd slat 1/8	
F5	Outbd slat 1/8	
F6	Outbd slat 1/8	
F11	Inbd slat 1/8	
F12	Inbd slat 1/8	
F13	Inbd slat 1/8	
F14	Inbd slat 1/8	
F15	Inbd slat 1/8	
F16	Inbd slat 1/8	
F21	Inbd slat 2/7	
F22	Inbd slat 2/7	
F23	Inbd slat 2/7	
F24	Inbd slat 2/7	
F25	Inbd slat 2/7	
F26	Inbd slat 2/7	
F31	Inbd slat 3/6	
F32	Inbd slat 3/6	
F33	Inbd slat 3/6	
F34	Inbd slat 3/6	
F35	Inbd slat 3/6	
F36	Inbd slat 3/6	
F41	Inbd slat 4/5	

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Table 502/27-81-00-993-803 (Continued)

Point	Location	Value
F42	Inbd slat 4/5	
F43	Inbd slat 4/5	
F44	Inbd slat 4/5	
F45	Inbd slat 4/5	
F46	Inbd slat 4/5	

2) Multiply all negative fair measurements by -0.5.

3) Add all the misfair values (positive and negative misfairs).

4) Divide the total misfair value by 6.

5) Make sure that the result is 0.000 in. (0.00 mm) to 0.050 in. (1.27 mm).

(g) Make sure that the slat endstops touch the horizontal and vertical stop pads.

(h) If the slat endstops do not touch the horizontal and vertical stop pads, adjust the stop pad:

1) Remove the retainers [85], nuts [86], and bolts [89].

2) Add or remove shims [87] as it is necessary until the slat fairs with the adjacent slat.

a) Add or remove only a minimum amount of shim necessary for the slat upstop to touch the stop pad.

NOTE: Add or remove only a minimum amount of shim necessary for the slat upstop to touch the stop pad. To adjust each stop pad, add or subtract in any combination of 0.010 in. (0.25 mm), 0.020 in. (0.51 mm), 0.040 in. (1.02 mm) and 0.063 in. (1.60 mm) thick solid shims. The total permitted shim thickness is 0.000 in. (0.00 mm) to 0.128 in. (3.25 mm) and must be such that the slat flex skin keeps solid contact with the fixed leading edge.

3) Apply Cor-Ban 27L Compound, G50237, or corrosion inhibiting material, G50136, to the bolts [89].

4) Install the bolts [89], nuts [86], and retainers [85].

a) Tighten the bolts [89] to 16 in-lb (1.8 N·m) - 20 in-lb (2.3 N·m).

SUBTASK 27-81-00-820-008

(4) Do the final adjustment of the leading edge slat in the (intermediate) extend position:

WARNING

MAKE SURE THAT PERSONNEL AND EQUIPMENT STAY AWAY FROM THE LEADING EDGE SLATS, TRAILING EDGE FLAPS, AND DRIVE MECHANISMS. THE FLAPS, SLATS, AND DRIVE MECHANISMS MOVE QUICKLY. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

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(WARNING PRECEDES)

CAUTION

MAKE SURE THAT THE INBOARD FAN COWL PANEL AND THE INBOARD AND OUTBOARD THRUST REVERSERS ARE CLOSED AND IN THE STOWED POSITION OR REMOVED BEFORE YOU EXTEND THE LEADING EDGE FLAPS AND SLATS. THERE IS NOT SUFFICIENT CLEARANCE FOR THE FLAPS AND SLATS TO EXTEND IF THE INBOARD FAN COWL PANEL AND THE INBOARD AND OUTBOARD THRUST REVERSERS ARE IN THE OPEN POSITION. THIS CAN CAUSE DAMAGE TO EQUIPMENT.

(a) To extend the leading edge slat to the (intermediate) extend (1-unit detent) position, do this task: Leading Edge Flaps and Slats Extension, TASK 27-81-00-860-803.

(b) Align the noses of the slats to get a mismatch of 0.000 in. (0.00 mm) to 0.060 in. (1.52 mm) between the adjacent slats.

1) Dimension may apply at any one point within the 1.0000 in. (25.40 mm) band width (Figure 505).

NOTE: This dimension is normal to the skin surface where the measurement is taken. Vertical mismatch is permissible.

(c) Make sure that the detent roller touch the root of each detent arm.

(d) Make sure that the clearance between the detent roller and aft surface of the cam plate is 0.000 in. (0.00 mm) to 0.030 in. (0.76 mm).

(e) If the detent roller does not touch the root of the detent arm and the gap between the detent roller and cam plate [83] is out of tolerance, adjust the cam plate [83]:

1) Loosen the two nuts [84] that attach the cam plate [83] to the detent arm [78].

2) Move the cam plate [83] forward or aft as it is necessary.

3) Temporarily tighten the two nuts [84].

(f) Make sure that the clearance between the slat and fixed leading edge is 0.002 +0.006 / -0.000 in. (0.05 +0.15 / -0.00 mm) (View D, Figure 503).

1) Measure the clearance at the tightest point, within 2.00 in. (50.8 mm) on either side of each auxiliary track centerline.

2) Measure the clearance to a depth of 0.40 in. (10.2 mm) to 0.60 in. (15.2 mm).

(g) If the clearance between the slat and fixed leading edge is not 0.002 +0.006 / -0.000 in. (0.05 +0.15 / -0.00 mm), adjust the forward support of the auxiliary track:

1) Remove the nut [74] and washer [75] from the clevis [76].

2) Lift the auxiliary track [72] with the clevis [76] through the auxiliary track support.

3) Add or remove the shims [77] to a total thickness of 0.030 in. (0.76 mm) to 0.270 in. (6.86 mm).

4) Put the clevis [76] back in its position on the auxiliary track support.

5) Install the washer [75] and nut [74] on the clevis [76].

(h) Remove the preload shim assembly, which is part of the rigging equipment, SPL-1739, at the forward supports for the auxiliary tracks:

1) Remove the nut [74] and washer [75] from the clevis [76].

2) Lift the auxiliary track [72] with the clevis [76] through the auxiliary track support.

3) Remove the preload shim assembly.

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4) Put the clevis [76] back in its position on the auxiliary track support.

5) Install the washer [75] and nut [74] on the clevis [76].

(i) Make sure that the minimum clearance at these locations is 0.20 in. (5.1 mm):

1) The clearance between all adjacent slat ends.

2) The clearance between the outboard end of slats 1 and 8 and adjacent fixed leading edge.

3) The clearance between the inboard end of slats 4 and 5 and adjacent fixed leading edge.

(j) Extend the slat until the detent roller is completely clear of the detent arm [78] and cam plate [83].

(k) On slats 1, 4, 5 and 8, make sure that the detent arm preload is 80 ±5 lbf (355.9 ±22.3 N), measured with a spring scale, STD-755, at the throat of the cam plate [83], and 0.100 +0.060 / -0.040 in. (2.54 +1.52 / -1.02 mm) dimension at the lower portion of the detent arm [78].

(l) On slats 2, 3, 6 and 7, make sure that the detent arm preload is 50 ±5 lbf (222.4 ±22.3 N), measured with a spring scale, STD-755, at the throat of the cam plate [83], and 0.100 +0.060 / -0.040 in. (2.54 +1.52 / -1.02 mm) dimension at the lower portion of the detent arm [78].

1) If it is necessary, turn the detent arm preload adjustment bolt [73] clockwise to increase the preload or counterclockwise to reduce the preload.

2) Install MS20995C32 lockwire, G01048, on the detent arm preload adjustment bolt [73].

(m) Make sure that the gap at the lower portion of the detent arm slot is 0.100 +0.060 / -0.040 in. (2.54 +1.52 / -1.02 mm).

SUBTASK 27-81-00-820-009

(5) Do the final adjustment of the leading edge slat in the retract position:

WARNING MAKE SURE THAT PERSONNEL AND EQUIPMENT STAY AWAY FROM THE LEADING EDGE SLATS, TRAILING EDGE FLAPS, AND DRIVE MECHANISMS. THE FLAPS, SLATS, AND DRIVE MECHANISMS MOVE QUICKLY. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(a) Do this task: Leading Edge Flaps and Slats Retraction, TASK 27-81-00-860-804.

(b) Make sure that the slat upstops touch the horizontal and vertical stop pads.

(c) If the slat upstops do not touch the horizontal and vertical stop pads, adjust the stop pad:

1) Remove the retainers [85], nuts [86], and bolts [89].

2) Add or remove shims [87] as it is necessary until the slat fairs with the adjacent slat.

NOTE: Add or remove only a minimum amount of shim necessary for the slat upstop to touch the stop pad. To adjust each stop pad, add or subtract in any combination of 0.010 in. (0.25 mm), 0.020 in. (0.51 mm), 0.040 in. (1.02 mm) and 0.063 in. (1.60 mm) thick solid shims. The total permitted shim thickness is 0.000 in. (0.00 mm) to 0.128 in. (3.25 mm) and must be such that the slat flex skin keeps solid contact with the fixed leading edge.

3) Install the bolts [89], nuts [86], and retainers [85].

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(d) Make sure that the clearance between the slat and fixed leading edge is 0.002 +0.006 / -0.000 in. (0.05 +0.15 / -0.00 mm) (View D, Figure 503).

1) Measure the clearance at the tightest point, within 2.00 in. (50.8 mm) on either side of each auxiliary track centerline.

2) Measure the clearance to a depth of 0.40 in. (10.2 mm) to 0.60 in. (15.2 mm).

(e) If the clearance between the slat and fixed leading edge is not in tolerance, adjust the aft support of the auxiliary track:

1) Loosen the nut [90] on the locking bolt [91] at the upper aft auxiliary track.

2) Loosen the jamnut [82].

3) Loosen the nut [80] on the locking bolt [79].

4) Turn the adjustment nut [81] until the clearance between the slat and fixed leading edge is 0.002 +0.006 / -0.000 in. (0.05 +0.15 / -0.00 mm) (View D, Figure 503).

5) Temporarily tighten the nut [80] on the locking bolt [79].

6) Temporarily tighten the jamnut [82].

(f) Reduce the dimension A by 0.021 ±0.003 in. (0.53 ±0.08 mm) to preload the slat trailing edge:

1) Loosen the jamnut [82].

2) Loosen the nut [80] to loosen the locking bolt [79].

3) Turn the adjustment nut [81] 180 ±30 degrees clockwise, viewed from below.

(g) Temporarily tighten the locking bolt [79].

(h) Temporarily tighten the jamnut [82].

(i) Make sure that the weighted average misfair is 0.000 in. (0.00 mm) to 0.050 in. (1.27 mm).

(j) Make sure that the slat endstops touch the horizontal and vertical stop pads.

(k) If the slat endstops do not touch the horizontal and vertical stop pads, adjust the stop pad:

1) Remove the retainers [85], nuts [86], and bolts [89].

2) Add or remove shims [87] as it is necessary until the slat fairs with the adjacent slat.

NOTE: Add or remove only a minimum amount of shim necessary for the slat upstop to touch the stop pad. To adjust each stop pad, add or subtract in any combination of 0.010 in. (0.25 mm), 0.020 in. (0.51 mm), 0.040 in. (1.02 mm) and 0.063 in. (1.60 mm) thick solid shims. The total permitted shim thickness is 0.000 in. (0.00 mm) to 0.128 in. (3.25 mm) and must be such that the slat flex skin keeps solid contact with the fixed leading edge.

3) Apply Cor-Ban 27L Compound, G50237, or corrosion inhibiting material, G50136, to the bolts [89].

4) Install the bolts [89], nuts [86], and retainers [85].

a) Tighten the bolts [89] to 16 in-lb (1.8 N·m) - 20 in-lb (2.3 N·m).

(l) Measure and write down the misfairs at six locations for each of these gaps (Figure 504, Table 502):

1) The clearance between all adjacent slat ends

2) The clearance between the outboard end of slats 1 and 8 and adjacent fixed leading edge

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3) The clearance between the inboard end of slats 4 and 5 and adjacent fixed leading edge.

(m) Measure the slat end clearance at three locations for each slat (Figure 504, Table 503).

Table 503/27-81-00-993-804

Point	Location	Value
G1	Outbd slat 1/8	
G2	Outbd slat 1/8	
G3	Outbd slat 1/8	
G11	Inbd slat 1/8	
G12	Inbd slat 1/8	
G13	Inbd slat 1/8	
G21	Inbd slat 2/7	
G22	Inbd slat 2/7	
G23	Inbd slat 2/7	
G31	Inbd slat 3/6	
G32	Inbd slat 3/6	
G33	Inbd slat 3/6	
G41	Inbd slat 4/5	
G42	Inbd slat 4/5	
G43	Inbd slat 4/5	

1) Make sure that the clearance between all adjacent slat ends is 0.260 +0.150 / -0.060 in. (6.60 +3.81 / -1.53 mm).

2) Make sure that the clearance between the slat ends and the adjacent fixed leading edge is 0.250 +0.150 / -0.060 in. (6.35 +3.81 / -1.52 mm).

SUBTASK 27-81-00-860-047

WARNING MAKE SURE THAT PERSONNEL AND EQUIPMENT STAY AWAY FROM THE LEADING EDGE SLATS, TRAILING EDGE FLAPS, AND DRIVE MECHANISMS. THE FLAPS, SLATS, AND DRIVE MECHANISMS MOVE QUICKLY. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

CAUTION MAKE SURE THAT THE INBOARD FAN COWL PANEL AND THE INBOARD AND OUTBOARD THRUST REVERSERS ARE CLOSED AND IN THE STOWED POSITION OR REMOVED BEFORE YOU EXTEND THE LEADING EDGE FLAPS AND SLATS. THERE IS NOT SUFFICIENT CLEARANCE FOR THE FLAPS AND SLATS TO EXTEND IF THE INBOARD FAN COWL PANEL AND THE INBOARD AND OUTBOARD THRUST REVERSERS ARE IN THE OPEN POSITION. THIS CAN CAUSE DAMAGE TO EQUIPMENT.

(6) To extend the leading edge slats to the full extend (30-unit detent) position, do this task: Leading Edge Flaps and Slats Extension, TASK 27-81-00-860-803.

SUBTASK 27-81-00-220-006

(7) Make sure that the minimum clearance at these locations is 0.20 in. (5.1 mm):

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(a) The clearance between all adjacent slat ends.

(b) The clearance between the outboard end of slats 1 and 8 and adjacent fixed leading edge.

(c) The clearance between the inboard end of slats 4 and 5 and adjacent fixed leading edge.

SUBTASK 27-81-00-860-048

WARNING MAKE SURE THAT PERSONNEL AND EQUIPMENT STAY AWAY FROM THE LEADING EDGE SLATS, TRAILING EDGE FLAPS, AND DRIVE MECHANISMS. THE FLAPS, SLATS, AND DRIVE MECHANISMS MOVE QUICKLY. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

CAUTION MAKE SURE THAT THE INBOARD FAN COWL PANEL AND THE INBOARD AND OUTBOARD THRUST REVERSERS ARE CLOSED AND IN THE STOWED POSITION OR REMOVED BEFORE YOU EXTEND THE LEADING EDGE FLAPS AND SLATS. THERE IS NOT SUFFICIENT CLEARANCE FOR THE FLAPS AND SLATS TO EXTEND IF THE INBOARD FAN COWL PANEL AND THE INBOARD AND OUTBOARD THRUST REVERSERS ARE IN THE OPEN POSITION. THIS CAN CAUSE DAMAGE TO EQUIPMENT.

(8) To extend the leading edge slats to the (intermediate) extend (1-unit detent) position, do this task: Leading Edge Flaps and Slats Extension, TASK 27-81-00-860-803.

SUBTASK 27-81-00-820-010

(9) Make sure that the detent roller touch the root of each detent arm.

NOTE: It is possible to manually apply load to the slats to help the slats meet the requirement.

SUBTASK 27-81-00-820-011

(10) Align the noses of the slats to get a mismatch of 0.000 in. (0.00 mm) to 0.060 in. (1.52 mm) between the adjacent slats.

(a) Dimension may apply at any one point within the 1.0000 in. (25.40 mm) band width (Figure 505).

NOTE: This dimension is normal to the skin surface where the measurement is taken. Vertical mismatch is permissible.

(b) Measure the clearance between the detent roller and aft surface of the cam plate.

(c) Make sure that the sum of all of the clearances is 0.000 in. (0.00 mm) to 0.060 in. (1.52 mm).

SUBTASK 27-81-00-420-004

(11) Tighten the nut [80] on the locking bolt [79] for the auxiliary track aft support to 65 in-lb (7.3 N·m) - 100 in-lb (11.3 N·m).

SUBTASK 27-81-00-420-005

(12) Tighten the jamnut [82] on the auxiliary track aft support.

(a) Finger tight until the jamnut [82] touches the support shaft and tighten an additional quarter turn.

1) As an option, it is possible to tighten the jamnut [82] to a setpoint of 65 in-lb (7.3 N·m).

NOTE: There is no lockwire required on the jamnut [82].

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SUBTASK 27-81-00-420-006

(13) Tighten the nut [90] on the locking bolt [91] at the upper aft auxiliary track to 130 in-lb (14.7 N·m) - 200 in-lb (22.6 N·m).

SUBTASK 27-81-00-420-007

(14) Tighten the nut [74] on the auxiliary track forward support to a setpoint of 530 in-lb (59.9 N·m).

SUBTASK 27-81-00-420-008

(15) Tighten the nuts [84] on the cam plate to a setpoint of 85 in-lb (9.6 N·m) (preferred method) or the bolts [103] to a setpoint of 100 in-lb (11.3 N·m) (optional method).

NOTE: The recommended method is to tighten the nuts.

NOTE: The optional method is to tighten the bolts.

SUBTASK 27-81-00-420-009

(16) Install MS20995C32 lockwire, G01048, between the holes in the shims [77] of the auxiliary track forward support.

SUBTASK 27-81-00-420-010

(17) Install MS20995C32 lockwire, G01048, on the detent arm preload adjustment bolt [73].

SUBTASK 27-81-00-220-007

(18) Make sure that the clearances between the slat proximity sensors and slat auxiliary track targets are correct and adjust any out of adjustment clearance if necessary (LEADING EDGE FLAP AND SLAT PROXIMITY SENSOR - REMOVAL/INSTALLATION, PAGEBLOCK 27-88-11/401).

SUBTASK 27-81-00-200-006

WARNING DO NOT LET OBJECTS GET IN THE HOUSING ASSEMBLY OF THE SLAT TRACK. THIS WILL HELP PREVENT A PUNCTURE OF THE HOUSING ASSEMBLY THAT COULD CAUSE A FUEL LEAK. THE FUEL LEAK COULD CAUSE A FIRE AND POSSIBLE DEATH OR INJURY TO PERSONNEL.

(19) Keep the housing (can) assemblies of the slat main tracks clean and free of all unwanted objects (FOD) at all times.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 27-81-00-860-049

WARNING MAKE SURE THAT PERSONNEL AND EQUIPMENT STAY AWAY FROM THE LEADING EDGE SLATS, TRAILING EDGE FLAPS, AND DRIVE MECHANISMS. THE FLAPS, SLATS, AND DRIVE MECHANISMS MOVE QUICKLY. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(1) Do this task: Leading Edge Flaps and Slats Retraction, TASK 27-81-00-860-804.

SUBTASK 27-81-00-410-002

(2) Install the applicable access panels (TASK 57-41-02-400-801):

NOTE: At the main track locations, the clearance between the access panel and flex tab must be 0.100 +0.150 / -0.040 in. (2.54 +3.81 / -1.02 mm).

(a) For the left wing, install these access panels:

Number	Name/Location
521CB	Lower Leading Edge Access Panel
521DB	Lower Leading Edge Access Panel
521EB	Lower Leading Edge Access Panel

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(Continued)

Number	Name/Location
521FB	Lower Leading Edge Access Panel
521JB	Lower Leading Edge Access Panel
521KB	Lower Leading Edge Access Panel
521LB	Lower Leading Edge Access Panel
521MB	Lower Leading Edge Access Panel
521QB	Lower Leading Edge Access Panel
521RB	Lower Leading Edge Access Panel
521SB	Lower Leading Edge Access Panel
521TB	Lower Leading Edge Access Panel
521WB	Lower Leading Edge Access Panel
521XB	Lower Leading Edge Access Panel
521YB	Lower Leading Edge Access Panel
521ZB	Lower Leading Edge Access Panel

(b) For the right wing, install these access panels:

Number	Name/Location
621CB	Lower Leading Edge Access Panel
621DB	Lower Leading Edge Access Panel
621EB	Lower Leading Edge Access Panel
621FB	Lower Leading Edge Access Panel
621JB	Lower Leading Edge Access Panel
621KB	Lower Leading Edge Access Panel
621LB	Lower Leading Edge Access Panel
621MB	Lower Leading Edge Access Panel
621QB	Lower Leading Edge Access Panel
621RB	Lower Leading Edge Access Panel
621SB	Lower Leading Edge Access Panel
621TB	Lower Leading Edge Access Panel
621VB	Lower Leading Edge Access Panel
621WB	Lower Leading Edge Access Panel
621XB	Lower Leading Edge Access Panel
621YB	Lower Leading Edge Access Panel
621ZB	Lower Leading Edge Access Panel

SUBTASK 27-81-00-820-012

(3) Adjust the auxiliary arm skin tabs [92] at the auxiliary track locations:

(a) Adjust the skin tab [92] to get the correct gap to the flex skin and access panels [93]:

1) Loosen the nuts [95] on the bolts [102].

2) Adjust the position of the skin tab [92] to get a gap of 0.100 in. (2.54 mm) between the skin tab [92] and flex skin.

3) Adjust the position of the skin tab [92] to get a gap of 0.100 +0.080 / -0.040 in. (2.54 +2.03 / -1.02 mm) between the skin tab [92] and access panels [93].

4) Tighten the nuts [95].

(b) Adjust the position of the skin tab [92] on the skin tab support [96] and skin tab support [97] to get the correct misfair:

1) Loosen the nuts [101] on the bolts [98].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) Adjust the position of the skin tab [92] until these misfairs are between the slat and skin tab [92]:
   a) -0.010 ±0.020 in. (-0.25 ±0.51 mm) for a minimum of 75 percent of the slat span.
   b) -0.010 ±0.040 in. (-0.25 ±1.02 mm) for a maximum of 25 percent of the slat span.
3) Tighten the nuts [101].

SUBTASK 27-81-00-220-008

(4) With the slats completely adjusted (rigged), do the step that follow:
   (a) Apply hand load to the end of each slat assembly (as necessary) to make sure that preload exists.
   (b) If no preload exists, do these steps:
      1) Make sure that the possible under-travel dimension of the slat actuator does not exceed 0.020 in. (0.51 mm) maximum between the attach point at the rod end of the slat actuator and clevis bores of the slat attach fittings.

         NOTE: The under-travel dimension of the slat actuator indicates that there is no preload of the slat actuator in relationship to the slat actuator attach fitting.

         a) If the dimension is greater than 0.020 in. (0.51 mm), make sure that the part numbers of the main track assemblies are correct.

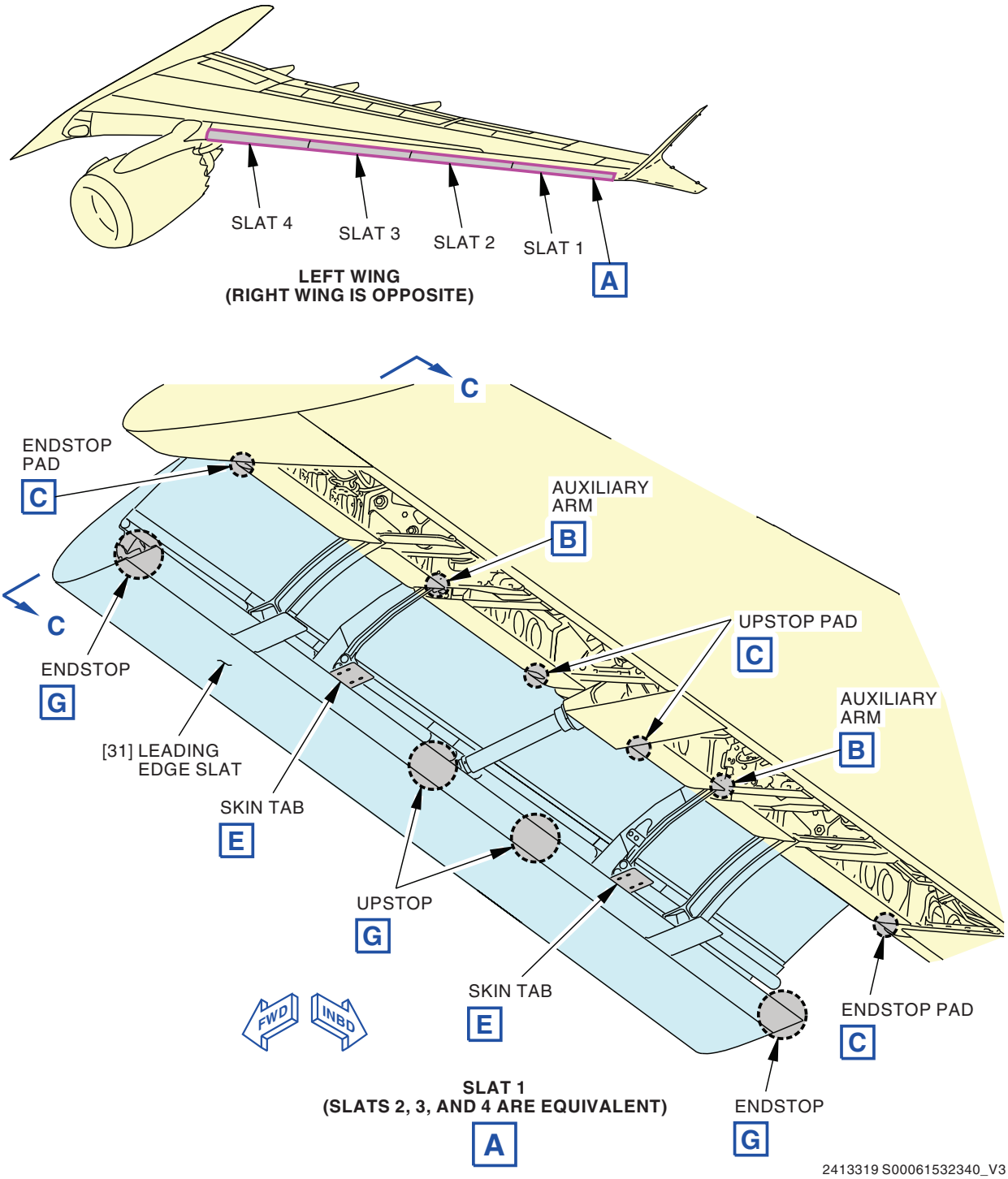
            NOTE: When the slat main tracks are incorrectly installed visual cues (excessive mismatches, misalignment, skew, etc.) will sometimes indicate a problem in advance depending on which slat is affected.

SUBTASK 27-81-00-710-005

(5) Do this task: Leading Edge Flap and Slat Operational Test in Primary Mode, TASK 27-81-00-700-801.

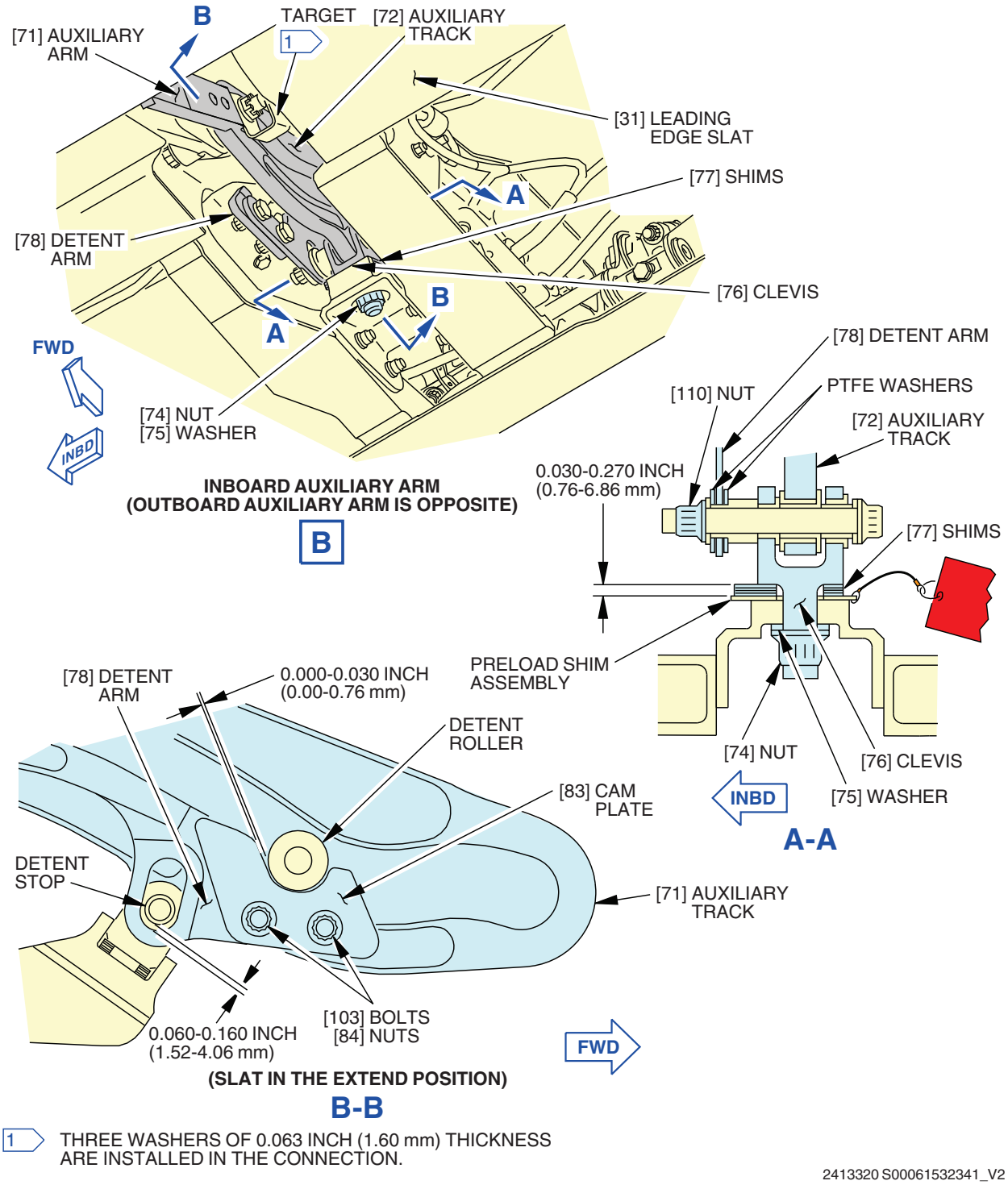
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Leading Edge Slat Adjustment
Figure 503/27-81-00-990-806 (Sheet 1 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

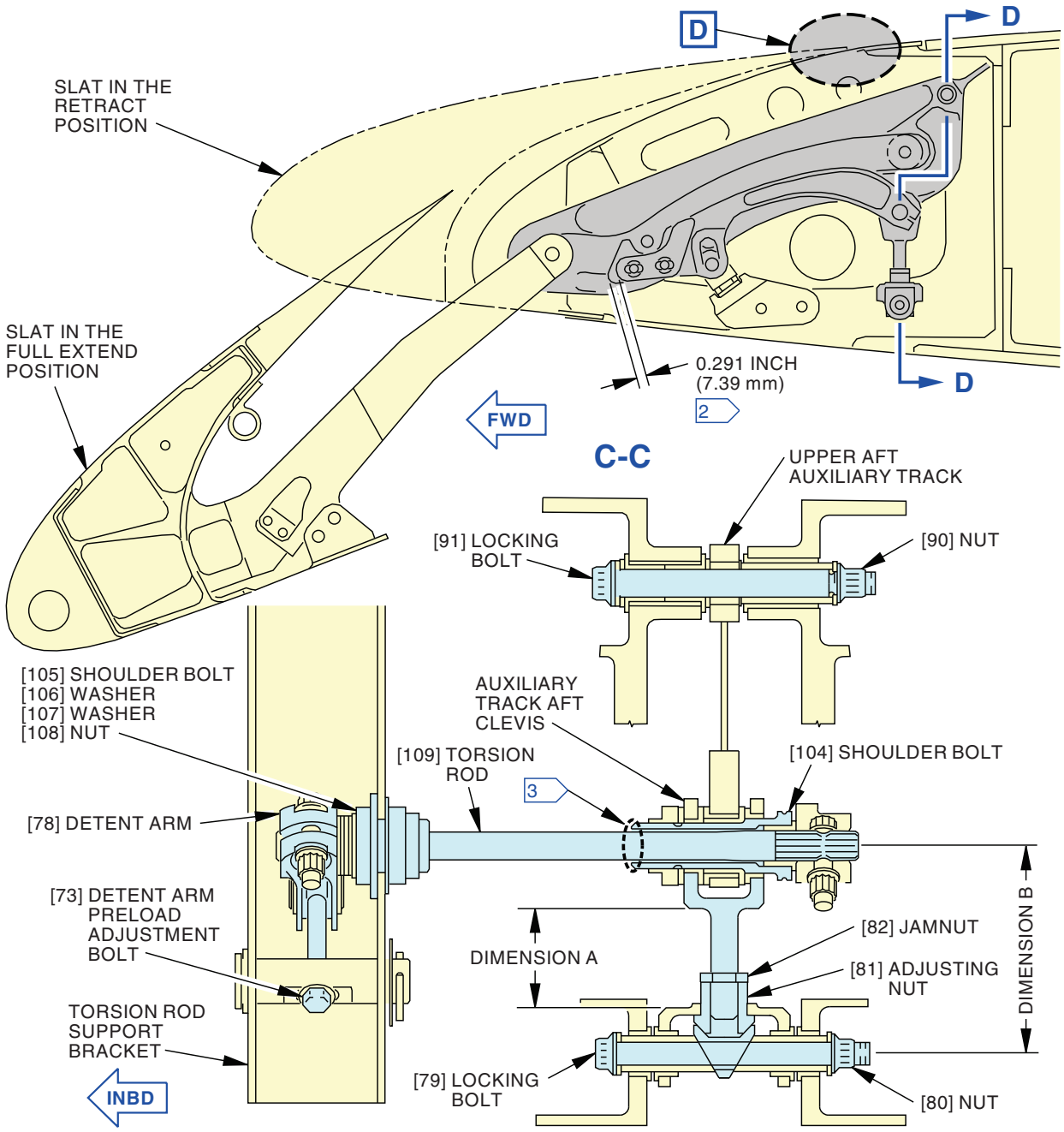


1 THREE WASHERS OF 0.063 INCH (1.60 mm) THICKNESS ARE INSTALLED IN THE CONNECTION.

2413320 S00061532341_V2

Leading Edge Slat Adjustment
Figure 503/27-81-00-990-806 (Sheet 2 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



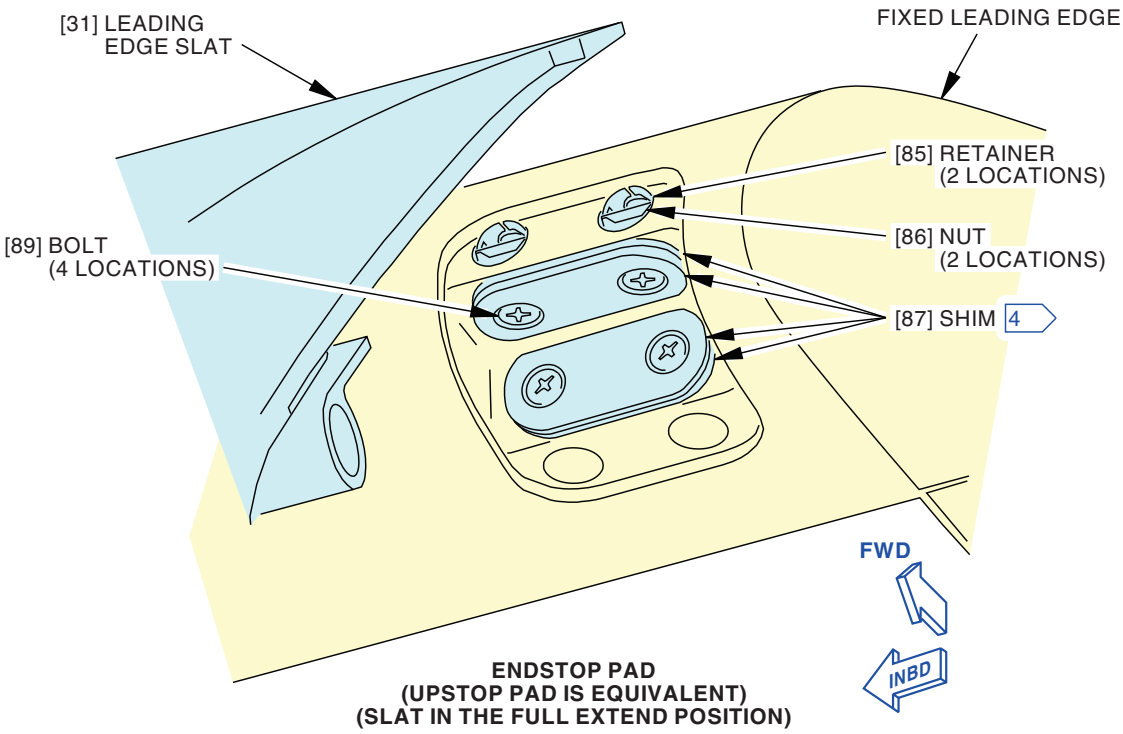
2 PRELIMINARY CAM PLATE SETTING DIMENSIONS ON SLATS 4 AND 5 AT 2 LOCATIONS PER SLAT.

3 0.010 INCH (0.25 mm) CLEARANCE REQUIRED IN THIS AREA. THIS DIMENSION ONLY APPLIES DURING INITIAL TORSION ROD INSTALLATION.

2413321 S00061532342_V3

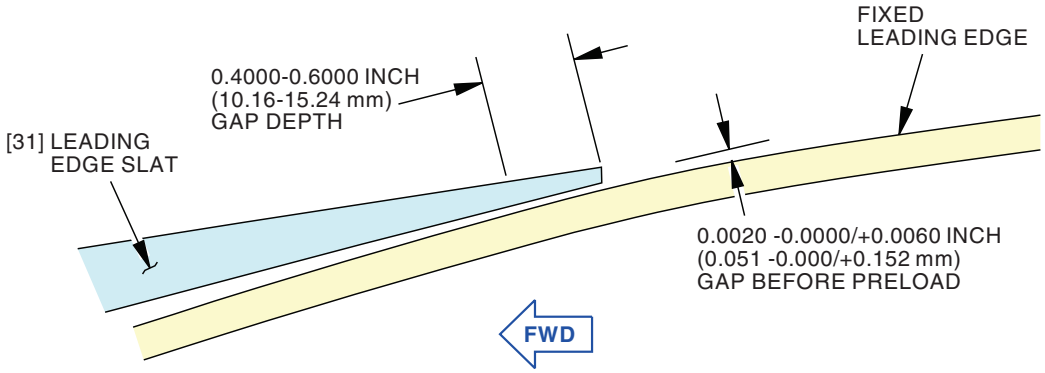
Leading Edge Slat Adjustment
Figure 503/27-81-00-990-806 (Sheet 3 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



ENDSTOP PAD
(UPSTOP PAD IS EQUIVALENT)
(SLAT IN THE FULL EXTEND POSITION)

C



(SLAT SHOWN IN RETRACT POSITION,
THE INTERMEDIATE POSITION IS DIMENSIONALLY EQUIVALENT)

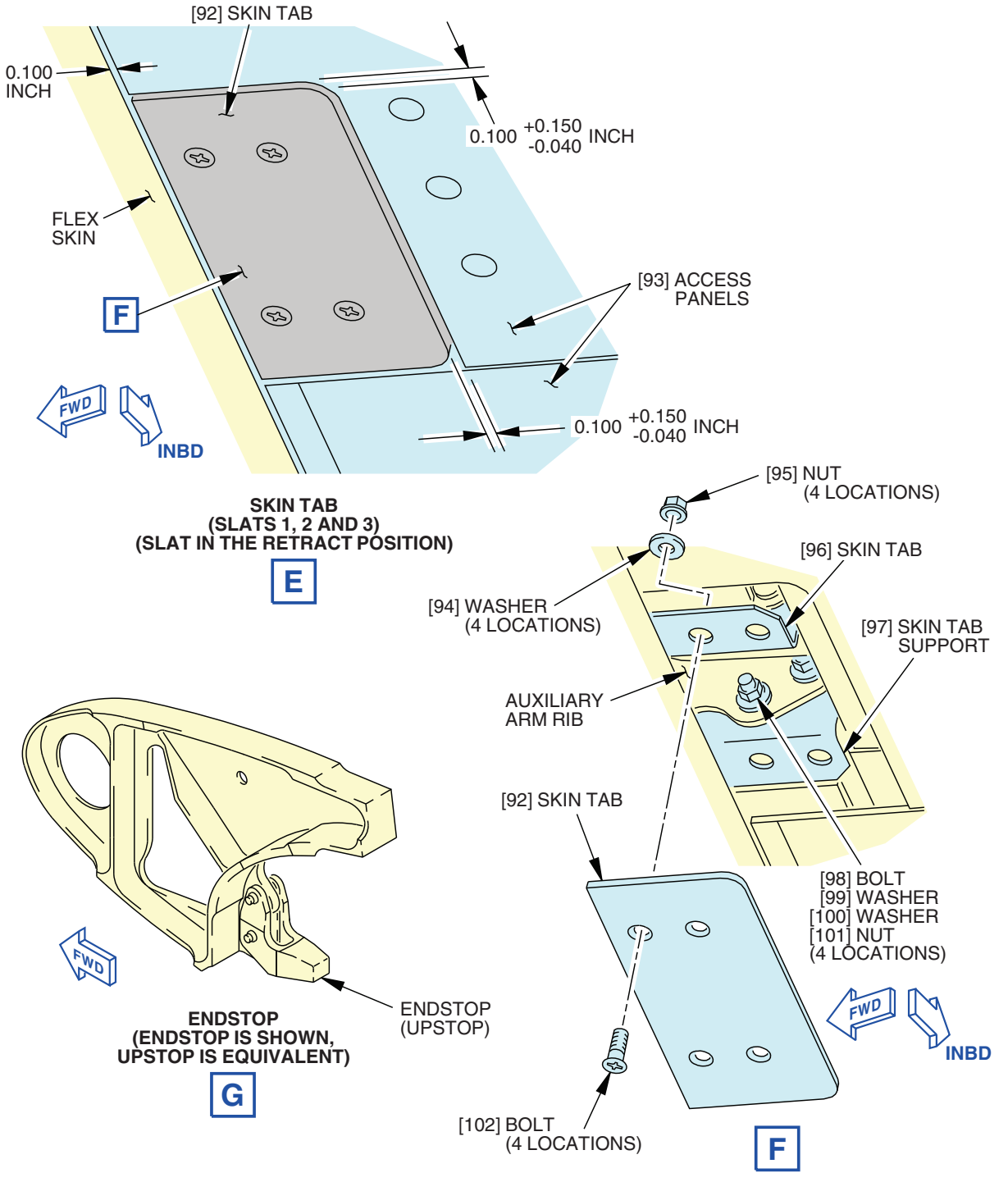
D

4 THE TOTAL SHIM THICKNESS AFTER THE FINAL ADJUSTMENT SHALL BE 0.0000-0.1280 INCH (0.000 - 3.251 mm) AND SHALL BE SUCH THAT THE SLAT FLEX SKIN MAINTAINS SOLID CONTACT WITH THE FORWARD LEADING EDGE.

2413322 S00061532343_V3

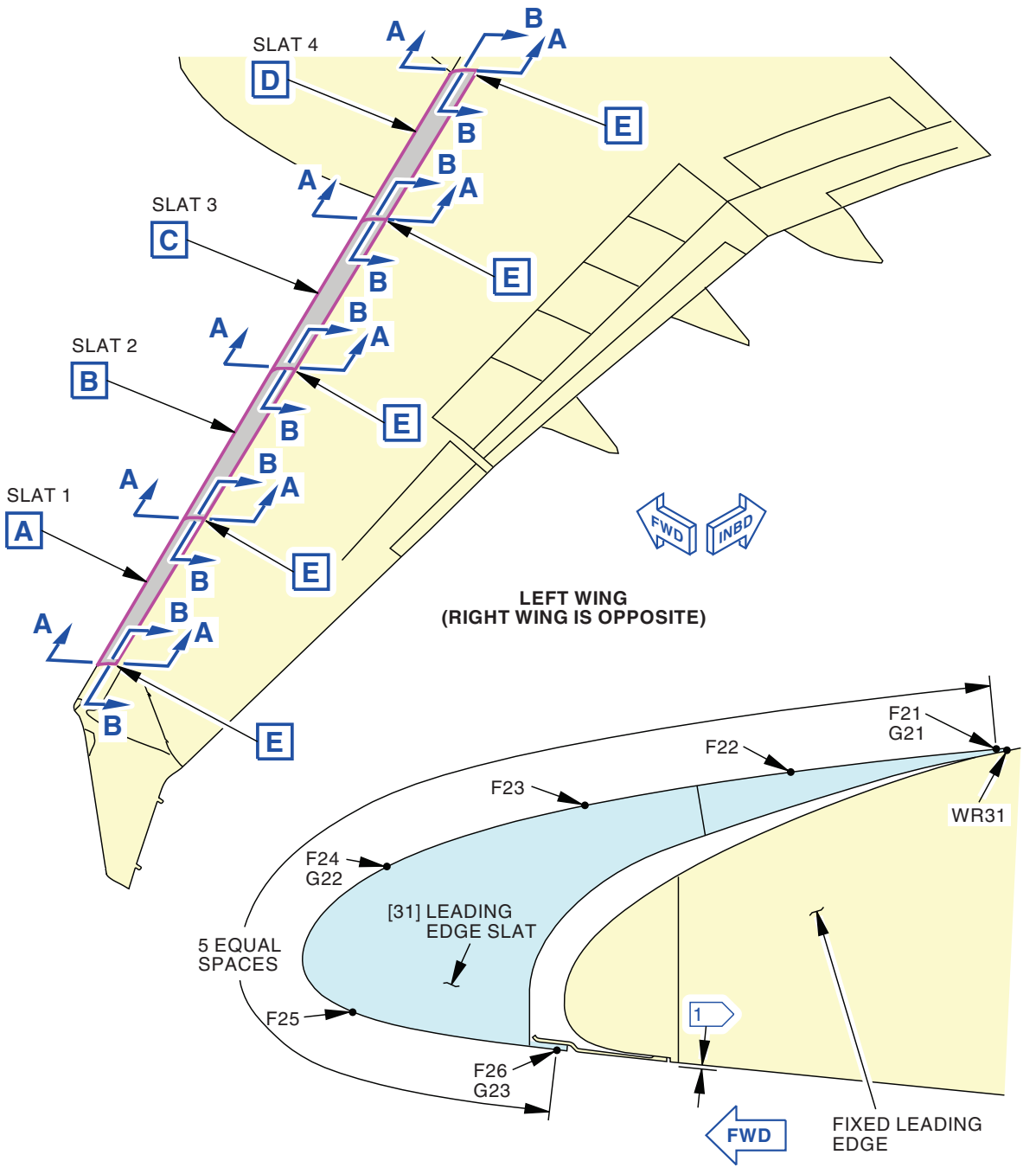
Leading Edge Slat Adjustment
Figure 503/27-81-00-990-806 (Sheet 4 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Leading Edge Slat Adjustment
Figure 503/27-81-00-990-806 (Sheet 5 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

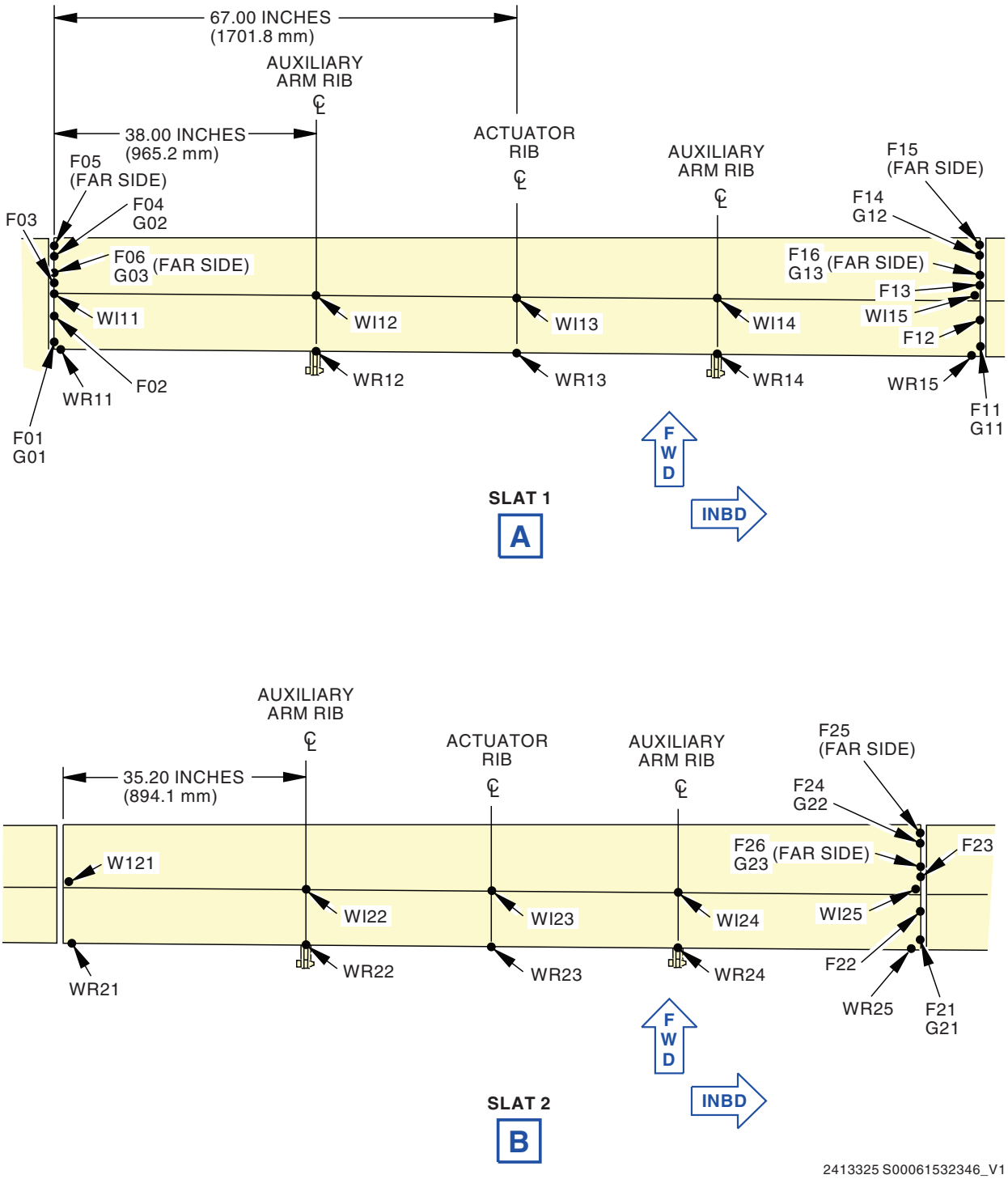


1 PERMITTED MISMATCH IS:
0.010 ±0.020 INCH FOR MINIMUM OF 75% SPAN.
0.010 ±0.040 INCH FOR MAXIMUM OF 25% SPAN.

2413324 S00061532345_V2

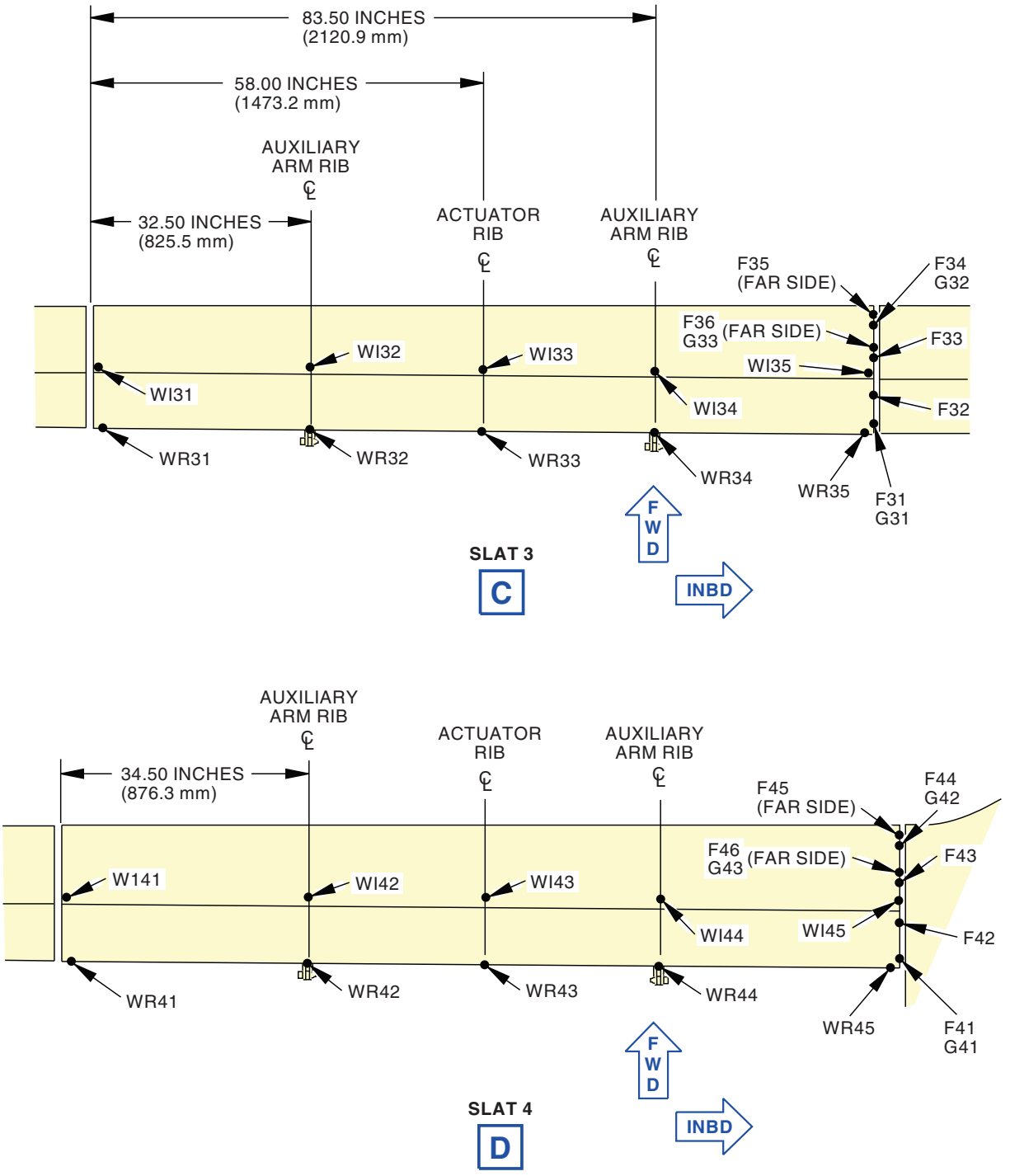
Leading Edge Slat Adjustment Points
Figure 504/27-81-00-990-811 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Leading Edge Slat Adjustment Points
Figure 504/27-81-00-990-811 (Sheet 2 of 4)

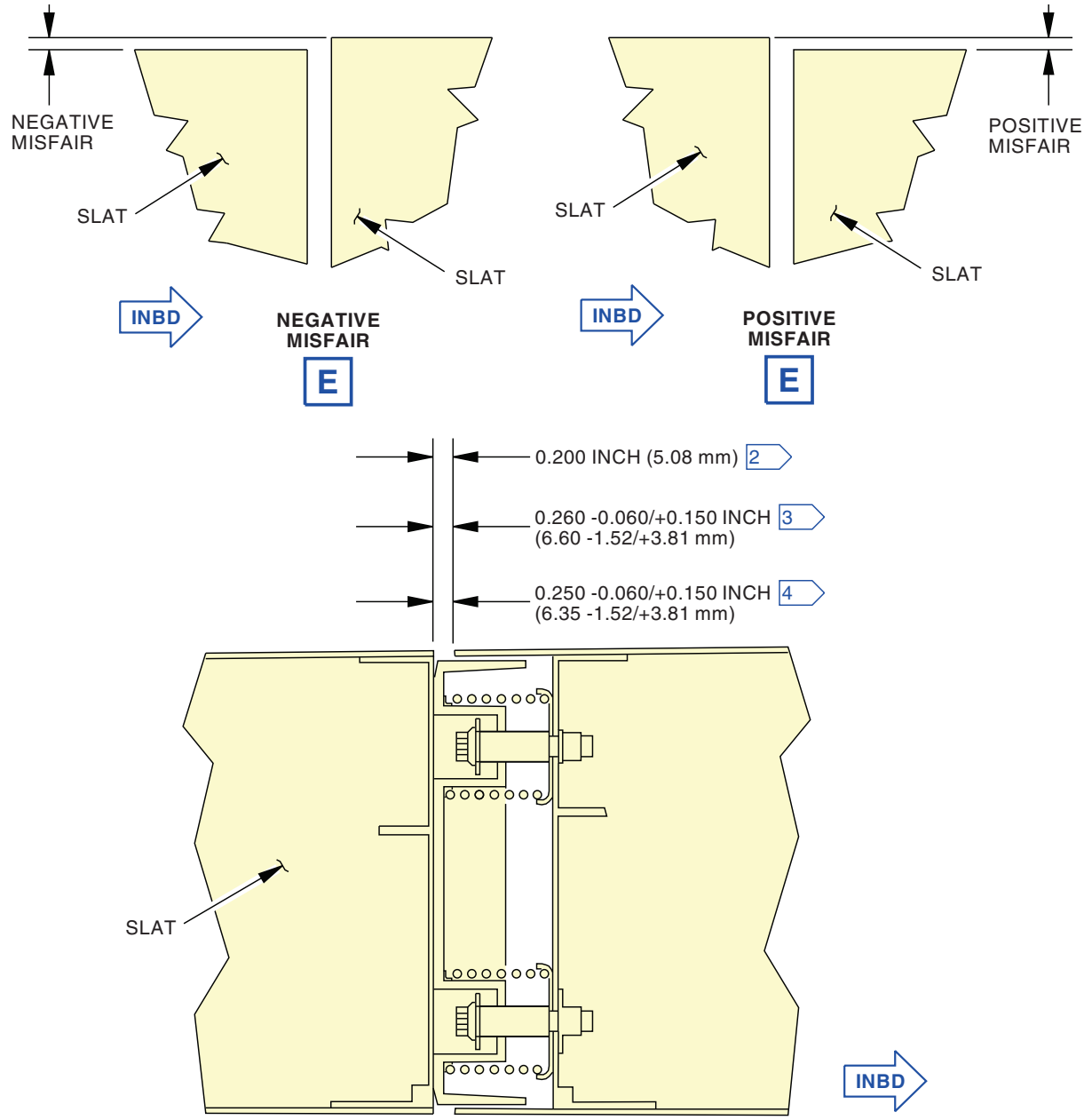
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2413326 S00061532347_V1

Leading Edge Slat Adjustment Points
Figure 504/27-81-00-990-811 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



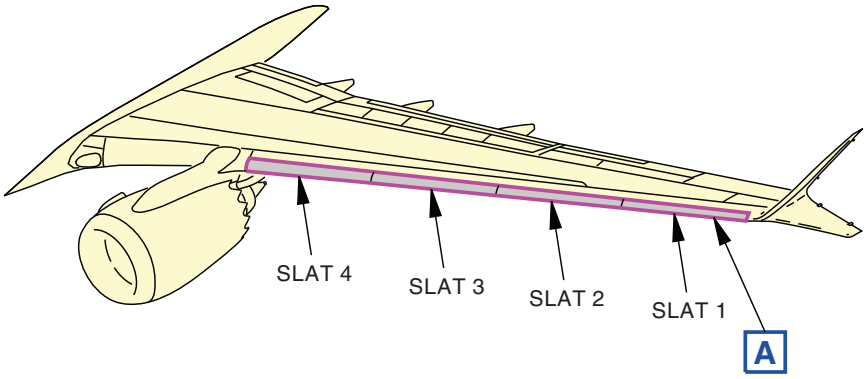
B-B

2 MIN FOR SLAT EXTEND AND FULL EXTEND POSITIONS.
3 SLAT TO SLAT
4 SLAT TO FIXED LEADING EDGE.

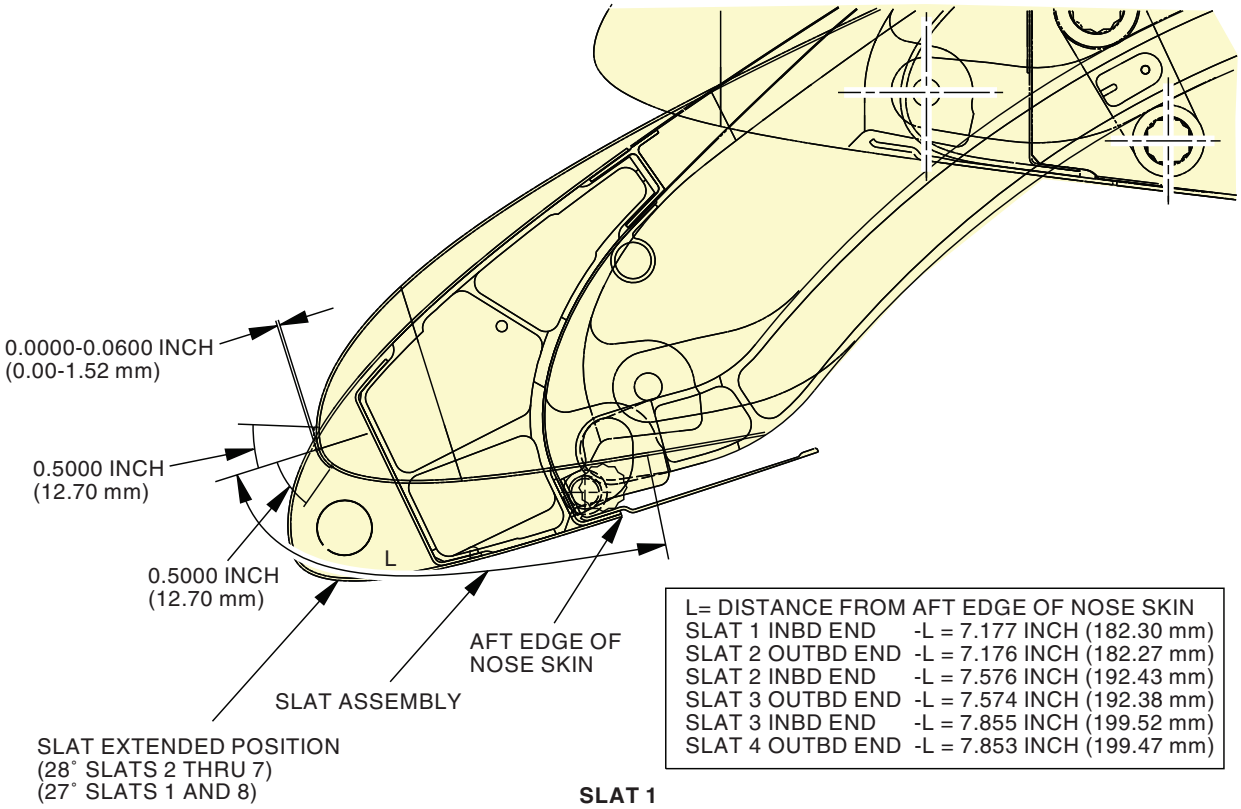
2413327 S00061532348_V1

Leading Edge Slat Adjustment Points
Figure 504/27-81-00-990-811 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LEFT WING
(RIGHT WING IS OPPOSITE)



SLAT 1
(SLAT 2, 3 AND 4 ARE EQUIVALENT)

A

2789765 S0000634505_V2

Leading Edge Slat Adjustment within the Band Width
Figure 505/27-81-00-990-812

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-81-00-700-801

5. Leading Edge Flap and Slat Operational Test in Primary Mode

(Figure 506, Figure 507)

A. General

(1) The leading edge slats move to the extend position for flap lever positions 1-unit detent through 25-unit detent. The leading edge slats move to the full extend position at the 30-unit detent flap lever position.

B. References

Reference	Title
27-51-00-740-801	FSEU Display and Existing Faults Test (P/B 501)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)

C. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
211	Flight Compartment - Left
212	Flight Compartment - Right
500	Left Wing
600	Right Wing

D. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

E. Prepare for the Test

SUBTASK 27-81-00-860-051

WARNING KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) To pressurize hydraulic system B, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 27-81-00-860-052

(2) If the flap control lever is not in the UP position, do these steps:

WARNING MAKE SURE THAT PERSONNEL AND EQUIPMENT STAY AWAY FROM THE LEADING EDGE SLATS, TRAILING EDGE FLAPS, AND DRIVE MECHANISMS. THE FLAPS, SLATS, AND DRIVE MECHANISMS MOVE QUICKLY. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(a) Move the flap control lever to the UP position.

(b) Let the leading edge flaps and slats and the trailing edge flaps move to the retract position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-81-00-010-003

(3) Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

F. Leading Edge Flap and Slat Operational Test in Primary Mode

SUBTASK 27-81-00-740-001

(1) Do this task: FSEU Display and Existing Faults Test, TASK 27-51-00-740-801.

SUBTASK 27-81-00-730-001

WARNING

KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

CAUTION

MAKE SURE THAT THE INBOARD FAN COWL PANEL AND THE INBOARD AND OUTBOARD THRUST REVERSERS ARE CLOSED AND IN THE STOWED POSITION OR REMOVED BEFORE YOU EXTEND THE LEADING EDGE FLAPS AND SLATS. THERE IS NOT SUFFICIENT CLEARANCE FOR THE FLAPS AND SLATS TO EXTEND IF THE INBOARD FAN COWL PANEL AND THE INBOARD AND OUTBOARD THRUST REVERSERS ARE IN THE OPEN POSITION. THIS CAN CAUSE DAMAGE TO EQUIPMENT.

(2) Do a test of the leading edge flap and slat operational test in primary mode:

(a) Use the Flap/Slat Electronics Unit (FSEU) front panel to do a check of the position of the leading edge flaps and slats (Figure 506):

1) Push the ON/OFF button on the front panel of the FSEU to turn on the display.

NOTE: The display will show EXISTING FAULTS?.

2) Push the down arrow until OTHER FUNCTNS? shows on the display.

3) Push the YES button to select OTHER FUNCTNS? on the FSEU.

4) Push the down arrow until I/O MONITOR? shows on the display.

5) Push the YES button to select I/O MONITOR? from the menu.

6) Push the YES button to select PROX SENSORS? from the I/O MONITOR? menu.

(b) Push the down arrow to step through the proximity sensor states at the leading edge flaps and slats UP position.

(c) Make sure that these displays and the proximity sensor states show on the FSEU (FSEU DISPLAY AND PROXIMITY SENSOR STATE - LE FLAPS AND SLATS RETRACTED (FLAP UP)/Table 504):

Table 504/27-81-00-993-805 FSEU DISPLAY AND PROXIMITY SENSOR STATE - LE FLAPS AND SLATS RETRACTED (FLAP UP)

Display	State
S1 OUTBD	GAP NEAR
S1 INBD	GAP NEAR
S2 OUTBD	GAP FAR

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Table 504/27-81-00-993-805 FSEU DISPLAY AND PROXIMITY SENSOR STATE - LE FLAPS AND SLATS RETRACTED (FLAP UP) (Continued)

Display	State
S2 INBD	GAP FAR
S2 RETR	GAP NEAR
S3 OUTBD	GAP FAR
S3 INBD	GAP FAR
S3 RETR	GAP NEAR
S4 OUTBD	GAP FAR
S4 INBD	GAP FAR
S4 RETR	GAP NEAR
S5 OUTBD	GAP FAR
S5 INBD	GAP FAR
S5 RETR	GAP NEAR
S6 OUTBD	GAP FAR
S6 INBD	GAP FAR
S6 RETR	GAP NEAR
S7 OUTBD	GAP FAR
S7 INBD	GAP FAR
S7 RETR	GAP NEAR
S8 OUTBD	GAP NEAR
S8 INBD	GAP NEAR
F1 RETR	GAP NEAR
F1 EXT	GAP FAR
F2 RETR	GAP NEAR
F2 EXT	GAP FAR
F3 RETR	GAP NEAR
F3 EXT	GAP FAR
F4 RETR	GAP NEAR
F4 EXT	GAP FAR

(d) Move the flap control lever to the 1-unit detent position.

(e) Make sure that the leading edge flaps 1 to 4 move to the extend position.

(f) Make sure that the leading edge slats 1 to 8 move to the extend position.

(g) When the leading edge flaps and slats start to move to the commanded 1-unit detent position, look for these conditions(Figure 507):

1) Make sure that the amber LE FLAPS TRANSIT light on the engine indication display of the captain's inboard Display Unit (DU) on panel P1, comes on.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) Make sure that these lights come on in the leading edge flap/slat position annunciator, P5-12, on the aft overhead panel.
   a) Make sure that the annunciator LE DEVICES FLAPS 1 through 4, amber TRANSIT lights come on.
   b) Make sure that the annunciator LE DEVICES SLATS 1 through 8, amber TRANSIT lights come on.

(h) When the leading edge flaps and slats stop at the commanded 1-unit detent position, look for these conditions:
1) Make sure that the amber LE FLAPS TRANSIT light, on the captain's inboard DU engine indication display, goes off.
2) Make sure that the green LE FLAPS EXT light, on the captain's inboard DU engine indication display, comes on.
3) Make sure that the annunciator LE DEVICES FLAPS 1 through 4, amber TRANSIT lights go off.
4) Make sure that the annunciator LE DEVICES SLATS 1 through 8, amber TRANSIT lights go off.
5) Make sure that the annunciator LE DEVICES FLAPS 1 through 4, green EXT lights come on.
6) Make sure that the annunciator LE DEVICES SLATS 1 through 8, green EXT lights come on.
7) Make sure that the leading edge flaps 1 through 4 are at the extend position.
8) Make sure that the leading edge slats 1 through 8 are at the extend position.

(i) Push the down arrow on the FSEU to step through the proximity sensor states.
1) Make sure these displays and the proximity sensor states show on the FSEU (FSEU DISPLAY AND PROXIMITY SENSOR STATE - LE FLAPS AND SLATS EXTEND (1-UNIT DETENT)/Table 505):

Table 505/27-81-00-993-806 FSEU DISPLAY AND PROXIMITY SENSOR STATE - LE FLAPS AND SLATS EXTEND (1-UNIT DETENT)

Display	State
S1 OUTBD	GAP FAR
S1 INBD	GAP NEAR
S2 OUTBD	GAP FAR
S2 INBD	GAP NEAR
S2 RETR	GAP FAR
S3 OUTBD	GAP FAR
S3 INBD	GAP NEAR
S3 RETR	GAP FAR
S4 OUTBD	GAP FAR
S4 INBD	GAP NEAR
S4 RETR	GAP FAR
S5 OUTBD	GAP FAR

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Table 505/27-81-00-993-806 FSEU DISPLAY AND PROXIMITY SENSOR STATE - LE FLAPS AND SLATS EXTEND (1-UNIT DETENT) (Continued)

Display	State
S5 INBD	GAP NEAR
S5 RETR	GAP FAR
S6 OUTBD	GAP FAR
S6 INBD	GAP NEAR
S6 RETR	GAP FAR
S7 OUTBD	GAP FAR
S7 INBD	GAP NEAR
S7 RETR	GAP FAR
S8 OUTBD	GAP FAR
S8 INBD	GAP NEAR
F1 RETR	GAP FAR
F1 EXT	GAP NEAR
F2 RETR	GAP FAR
F2 EXT	GAP NEAR
F3 RETR	GAP FAR
F3 EXT	GAP NEAR
F4 RETR	GAP FAR
F4 EXT	GAP NEAR

(j) Move the flap control lever to the 30-unit detent position.

1) Make sure that the leading edge flaps 1 to 4 remain at the extend position.

2) Make sure that the leading edge slats 1 to 8 move to the full extend position.

(k) Do these checks while the leading edge slats move to the commanded 30-unit detent position (Figure 507):

1) Make sure that the green LE FLAPS EXT light, on the captain's inboard DU engine indication display, goes off.

2) Make sure that the amber LE FLAPS TRANSIT light, on the captain's inboard DU engine indication display, comes on.

3) Make sure that the annunciator LE DEVICES SLATS 1 through 8, green EXT lights, go off.

4) Make sure that the annunciator LE DEVICES SLATS 1 through 8, amber TRANSIT lights, come on.

(l) Do these checks when the leading edge slats stop at the commanded 30-unit detent position:

1) Make sure that the amber LE FLAPS TRANSIT light, on the captain's inboard DU engine indication display, goes off.

2) Make sure that the green LE FLAPS EXT light, on the captain's inboard DU engine indication display, comes on.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

3) Make sure that the annunciator LE DEVICES FLAPS 1 through 4, green EXT lights, are on.

4) Make sure that the annunciator LE DEVICES SLATS 1 through 8, amber TRANSIT lights, go off.

5) Make sure that the annunciator LE DEVICES SLATS 1 through 8, green FULL EXT lights, come on.

(m) Push the down arrow on the FSEU to step through the proximity sensor states.

1) Make sure these displays and the proximity sensor states show on the FSEU (FSEU DISPLAY AND PROXIMITY SENSOR STATE - LE FLAPS AND SLATS FULL EXTEND (30-UNIT DETENT)/Table 506):

Table 506/27-81-00-993-807 FSEU DISPLAY AND PROXIMITY SENSOR STATE - LE FLAPS AND SLATS FULL EXTEND (30-UNIT DETENT)

Display	State
S1 OUTBD	GAP NEAR
S1 INBD	GAP FAR
S2 OUTBD	GAP NEAR
S2 INBD	GAP FAR
S2 RETR	GAP FAR
S3 OUTBD	GAP NEAR
S3 INBD	GAP FAR
S3 RETR	GAP FAR
S4 OUTBD	GAP NEAR
S4 INBD	GAP FAR
S4 RETR	GAP FAR
S5 OUTBD	GAP NEAR
S5 INBD	GAP FAR
S5 RETR	GAP FAR
S6 OUTBD	GAP NEAR
S6 INBD	GAP FAR
S6 RETR	GAP FAR
S7 OUTBD	GAP NEAR
S7 INBD	GAP FAR
S7 RETR	GAP FAR
S8 OUTBD	GAP NEAR
S8 INBD	GAP FAR
F1 RETR	GAP FAR
F1 EXT	GAP NEAR
F2 RETR	GAP FAR

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Table 506/27-81-00-993-807 FSEU DISPLAY AND PROXIMITY SENSOR STATE - LE FLAPS AND SLATS FULL EXTEND (30-UNIT DETENT) (Continued)

Display	State
F2 EXT	GAP NEAR
F3 RETR	GAP FAR
F3 EXT	GAP NEAR
F4 RETR	GAP FAR
F4 EXT	GAP NEAR

(n) Push the ON/OFF button on the front panel of the FSEU to turn the display off.

NOTE: The FSEU will turn itself off after 75 seconds.

SUBTASK 27-81-00-860-053

(3) Move the flap control lever to the UP position.

(a) Make sure that the leading edge flaps 1 to 4 retract to the up position.

(b) Make sure that the leading edge slats 1 to 8 retract to the up position.

(c) Look for these conditions as the leading edge flaps and slats move to the retract position.

1) Make sure that the green LE FLAPS EXT light, on the captain's inboard DU engine indication display, goes off.

2) Make sure that the amber LE FLAPS TRANSIT light on the engine indication display of the captain's inboard DU on panel P1, comes on.

3) Make sure that the annunciator LE DEVICES FLAPS 1 through 4, green EXT lights go off.

4) Make sure that the annunciator LE DEVICES SLATS 1 through 8, green EXT lights go off.

5) Make sure that the annunciator LE DEVICES FLAPS 1 through 4, amber TRANSIT lights come on.

6) Make sure that the annunciator LE DEVICES SLATS 1 through 8, amber TRANSIT lights come on.

(d) Look for these conditions when the leading edge flaps and slats stop at the retract position:

1) Make sure that the annunciator LE DEVICES FLAPS 1 through 4, amber TRANSIT lights go off.

2) Make sure that the annunciator LE DEVICES SLATS 1 through 8, amber TRANSIT lights go off.

3) Make sure that all annunciator lights are off.

4) Make sure that the amber LE FLAPS TRANSIT light on the engine indication display of the captain's inboard DU on panel P1, goes off.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 27-81-00-410-003

(1) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

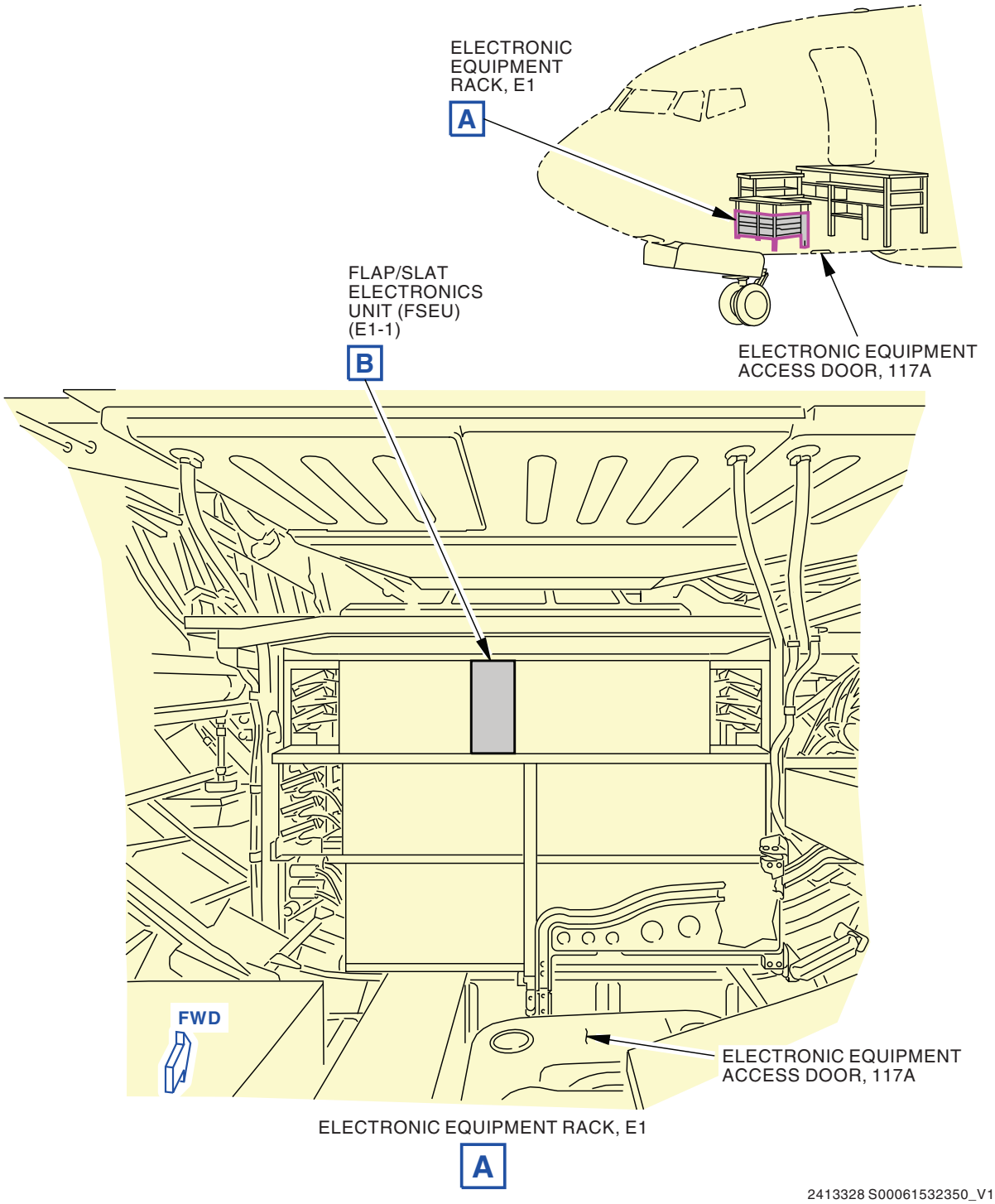
ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-81-00-860-054

(2) Remove hydraulic power, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

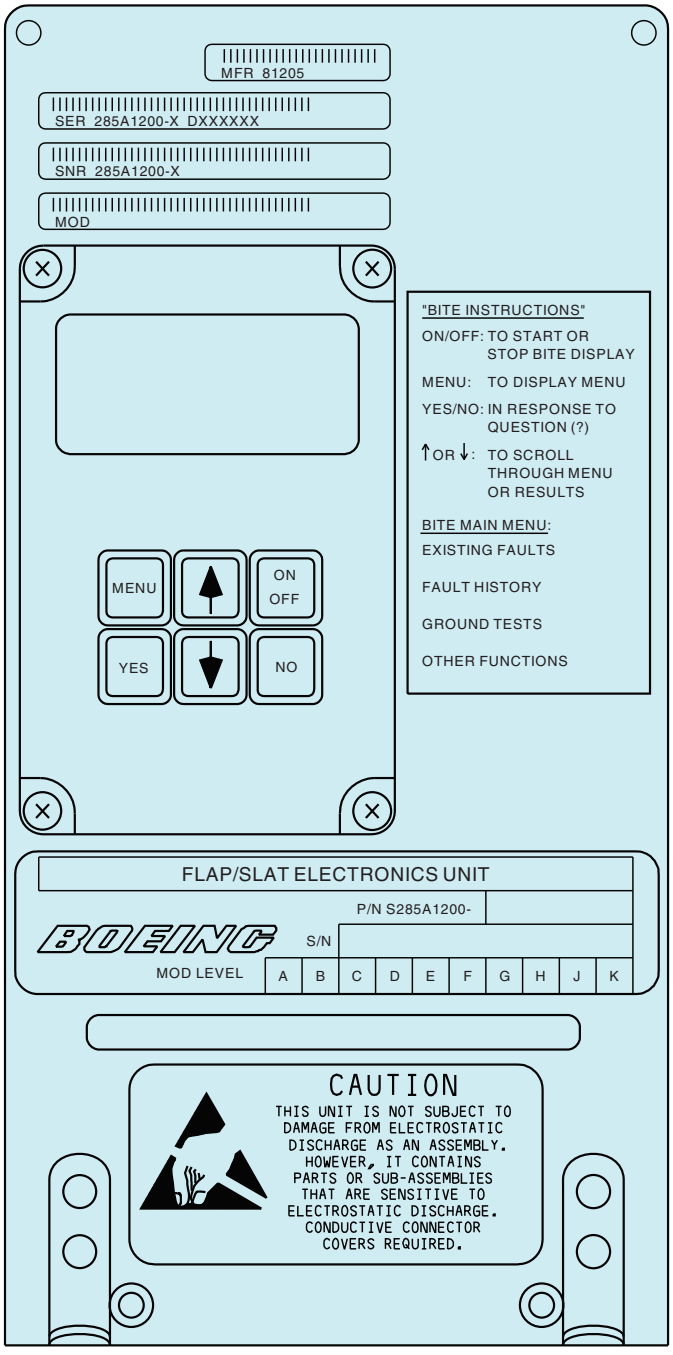
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Flap/Slat Electronics Unit
Figure 506/27-81-00-990-807 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



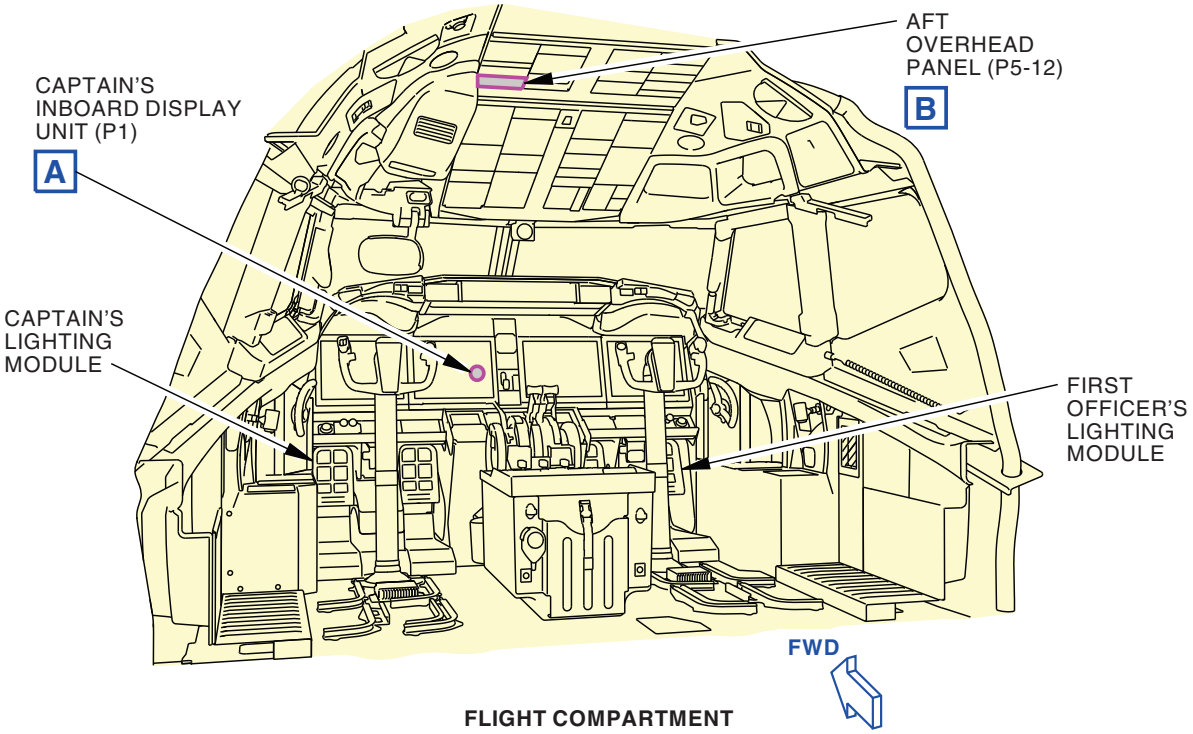
FLAP/SLAT ELECTRONICS UNIT (FSEU)

B

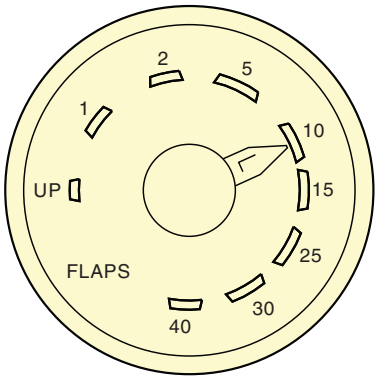
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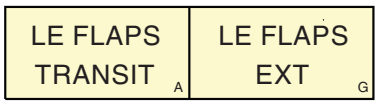
Flap/Slat Electronics Unit
Figure 506/27-81-00-990-807 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



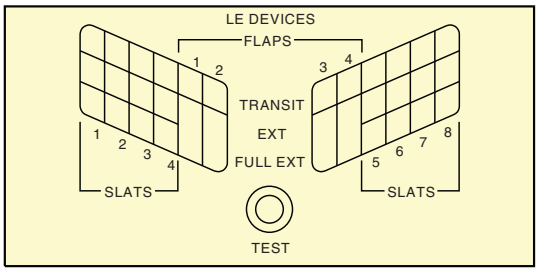
FLIGHT COMPARTMENT





CAPTAIN'S INBOARD DISPLAY UNIT (P1)

A



AFT OVERHEAD PANEL (P5-12)

B

2413330 S00061532352_V2

Leading Edge Flap and Slat Position Indicator
Figure 507/27-81-00-990-808

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-81-00-700-802

6. Leading Edge Flap and Slat Timing Test in Primary Mode

(Figure 507)

A. General

(1) The leading edge slats move to the extend position at the 25-unit detent. flap lever position. The leading edge slats move to the full extend position at the 30-unit detent flap lever position.

B. References

Reference	Title
29-11-00-860-802	Hydraulic System A or B Pressurization with a Portable Hydraulic Cart (P/B 201)
29-11-00-860-803	Hydraulic System Pressurization with an Electric Motor-Driven Pump (EMDP) (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)

C. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Prepare for the Test

SUBTASK 27-81-00-860-056

WARNING KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) To pressurize hydraulic system B, do this task: Hydraulic System Pressurization with an Electric Motor-Driven Pump (EMDP), TASK 29-11-00-860-803 or Hydraulic System A or B Pressurization with a Portable Hydraulic Cart, TASK 29-11-00-860-802.

E. Leading Edge Flap and Slat Timing Test in Primary Mode

SUBTASK 27-81-00-720-001

WARNING MAKE SURE THAT PERSONNEL AND EQUIPMENT STAY AWAY FROM THE LEADING EDGE SLATS, TRAILING EDGE FLAPS, AND DRIVE MECHANISMS. THE FLAPS, SLATS, AND DRIVE MECHANISMS MOVE QUICKLY. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(1) Do a timing test of the leading edge flap and slat control system in primary mode:

NOTE: Measure the time intervals between the time the surfaces start to move to the time the surfaces stop moving.

(a) If the flap control lever is not in the UP position, do these steps:

1) Move the flap control lever to the UP position.

2) Let the leading edge flaps and slats and the trailing edge flaps move to the retract position.

(b) Move the flap control lever to the 5-unit detent position.

1) Using hydraulic system B pressurization with a portable hydraulic cart:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

a) Make sure that the leading edge flaps extend fully in 5.5 to 7.5 seconds.

b) Make sure that the leading edge slats move to the extend position in 7.5 to 9.5 seconds.

2) Using hydraulic system B pressurization with an Electric Motor-Driven Pump (EMDP):

a) Make sure that the leading edge flaps extend fully in 7.0 to 9.0 seconds.

b) Make sure that the leading edge slats move to the extend position in 9.0 to 11.0 seconds.

(c) Move the flap control lever to the 30-unit detent. Do one of the following steps:

1) Using hydraulic system B pressurization with a portable hydraulic cart:

a) Make sure that the leading edge slats move to the full extend position in 2.0 to 3.5 seconds.

2) Using hydraulic system B pressurization with an EMDP:

a) Make sure that the leading edge slats move to the full extend position in 3.5 to 5.5 seconds.

(d) Make sure that the LE indication flaps EXTEND and slats FULL EXTEND lights on the aft overhead panel are on (Figure 507).

(e) Move the flap control lever to the 1-unit detent position. Do one of the following steps:

1) Using hydraulic system B pressurization with a portable hydraulic cart:

a) Make sure that the leading edge slats move to the extend position in 3.5 to 5.5 seconds.

2) Using hydraulic system B pressurization with an EMDP:

a) Make sure that the leading edge slats move to the extend position in 4.5 to 6.5 seconds.

3) Let the trailing edge flaps move to the correct position.

(f) Move the flap control lever to the UP detent position. Do one of the following steps:

1) Using hydraulic system B pressurization with a portable hydraulic cart:

a) Make sure that the leading edge flaps move to the retract position in 7.0 to 9.0 seconds.

b) Make sure that the leading edge slats move to the retract position in 6.5 to 8.5 seconds.

2) Using hydraulic system B pressurization with an EMDP:

a) Make sure that the leading edge flaps move to the retract position in 8.0 to 10.0 seconds.

b) Make sure that the leading edge slats move to the retract position in 7.0 to 9.0 seconds.

(g) Make sure that the LE indication flaps and slats TRANSIT, EXTEND and FULL EXTEND lights on the aft overhead panel are off.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 27-81-00-860-057

(1) Do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-81-00-710-801

7. Leading Edge Flap and Slat System Standby Actuation Operational Test

(Figure 507, Figure 508)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) The alternate control system uses electrical power to power the standby hydraulic system which moves the leading edge slats and flaps.

B. References

Reference	Title
27-81-00-860-804	Leading Edge Flaps and Slats Retraction (P/B 201)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Prepare for the Operational Test

SUBTASK 27-81-00-860-059

WARNING MAKE SURE THAT PERSONNEL AND EQUIPMENT STAY AWAY FROM THE LEADING EDGE SLATS, TRAILING EDGE FLAPS, AND DRIVE MECHANISMS. THE FLAPS, SLATS, AND DRIVE MECHANISMS MOVE QUICKLY. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(1) If the leading edge flaps and slats are not in the retract position, do this task: Leading Edge Flaps and Slats Retraction, TASK 27-81-00-860-804.

E. Leading Edge Standby System Operational Test

SUBTASK 27-81-00-860-060

CAUTION MAKE SURE THAT THE INBOARD FAN COWL PANEL AND THE INBOARD AND OUTBOARD THRUST REVERSERS ARE CLOSED AND IN THE STOWED POSITION OR REMOVED BEFORE YOU EXTEND THE LEADING EDGE FLAPS AND SLATS. THERE IS NOT SUFFICIENT CLEARANCE FOR THE FLAPS AND SLATS TO EXTEND IF THE INBOARD FAN COWL PANEL AND THE INBOARD AND OUTBOARD THRUST REVERSERS ARE IN THE OPEN POSITION. THIS CAN CAUSE DAMAGE TO EQUIPMENT.

(1) Move the ALTERNATE FLAPS ARM switch on the P5 panel to the ARM position (Figure 508).

NOTE: The standby hydraulic pump motor will operate when the ALTERNATE FLAPS ARM switch is in the ARM position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-81-00-860-061

WARNING

MAKE SURE THAT PERSONNEL AND EQUIPMENT STAY AWAY FROM THE LEADING EDGE SLATS, TRAILING EDGE FLAPS, AND DRIVE MECHANISMS. THE FLAPS, SLATS, AND DRIVE MECHANISMS MOVE QUICKLY. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(2) Move the flap control lever to the 30-unit position.

NOTE: This will decrease the load on the flap electric motor. The leading edge and trailing edge surfaces will not move when you move the flap control lever.

SUBTASK 27-81-00-710-002

(3) Move the ALTERNATE FLAPS control switch to the DOWN position to extend the leading edge flaps and slats to the full extend position.

NOTE: It is not necessary to extend the trailing edge flaps beyond the 30-unit position.

(a) When the flaps and slats start to extend, make sure all the LE indication flaps and slats TRANSIT lights, on the aft overhead panel, P5, come on (Figure 507).

(b) Do these checks of the aft overhead lights when the leading edge flaps and slats are in the full extend position:

1) Make sure all the TRANSIT lights go off.

2) Make sure the flaps EXT lights come on.

3) Make sure the slats FULL EXT lights come on.

SUBTASK 27-81-00-860-062

(4) Move the ALTERNATE FLAPS control switch to the OFF position.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 27-81-00-860-063

WARNING

MAKE SURE THE POSITION OF THE FLAPS AGREES WITH THE POSITION OF THE FLAP CONTROL LEVER. WHEN YOU SUPPLY HYDRAULIC POWER, THE FLAPS AND SLATS WILL MOVE AUTOMATICALLY TO THE POSITION OF THE FLAP CONTROL LEVER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Make sure the flap control lever is in the same position as the trailing edge flaps and leading edge flaps and slats.

SUBTASK 27-81-00-860-064

(2) Move the ALTERNATE FLAPS ARM switch to the OFF position to disarm the alternate control system.

SUBTASK 27-81-00-860-065

WARNING

KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(3) Supply system B hydraulic power, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-81-00-860-066

(4) Move the flap control lever to the UP position to retract the leading edge flaps and slats and the trailing edge flaps.

NOTE: The leading edge flaps and slats will not retract with alternate control.

SUBTASK 27-81-00-860-067

(5) Remove system B hydraulic power, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

——— END OF TASK ———

TASK 27-81-00-700-803

8. Leading Edge Flap and Slat Timing Test in the Standby Mode

(Figure 508)

A. References

Reference	Title
27-81-00-860-804	Leading Edge Flaps and Slats Retraction (P/B 201)

B. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Prepare for the Test

SUBTASK 27-81-00-860-069

WARNING MAKE SURE THAT PERSONNEL AND EQUIPMENT STAY AWAY FROM THE LEADING EDGE SLATS, TRAILING EDGE FLAPS, AND DRIVE MECHANISMS. THE FLAPS, SLATS, AND DRIVE MECHANISMS MOVE QUICKLY. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(1) If the leading edge flaps and slats are not in the retract position, do this task: Leading Edge Flaps and Slats Retraction, TASK 27-81-00-860-804.

D. Leading Edge Flap and Slat Timing in the Standby Mode

SUBTASK 27-81-00-710-003

(1) Do the leading edge flap and slat timing test in the standby mode:

NOTE: The leading edge flaps and slats do not retract in the standby mode.

(a) Put the ALTERNATE FLAPS arm switch, on the overhead panel, P5, in the ARM position (Figure 508).

(b) Move the ALTERNATE FLAPS control switch to the DOWN position for 5 seconds, then release.

(c) When the flaps and slats start to extend, make sure all the LE indication flaps and slats TRANSIT lights, on the aft overhead panel, P5, come on.

(d) Make sure that the leading edge flaps extend in 29 to 37 seconds.

(e) Make sure that the leading edge slats extend in 43 to 55 seconds.

(f) Do these checks of the lights on the Aft Overhead Panel, P5-12, when the leading edge flaps and slats are in the full extended position(Figure 507);

1) Make sure that all the TRANSIT lights go off.

2) Make sure that the flaps EXT lights come on.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

3) Make sure that the slats FULL EXT lights come on.

(g) Put the ALTERNATE FLAPS arm switch, on the overhead panel, P5, in the OFF position (Figure 508).

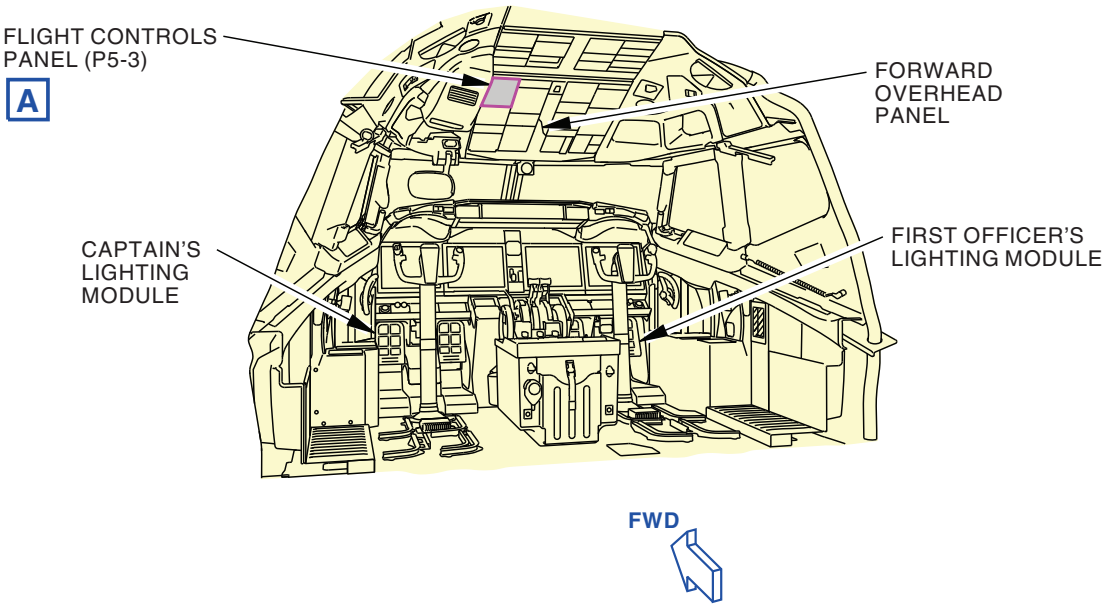
E. Put the Airplane Back to Its Usual Condition

SUBTASK 27-81-00-860-070

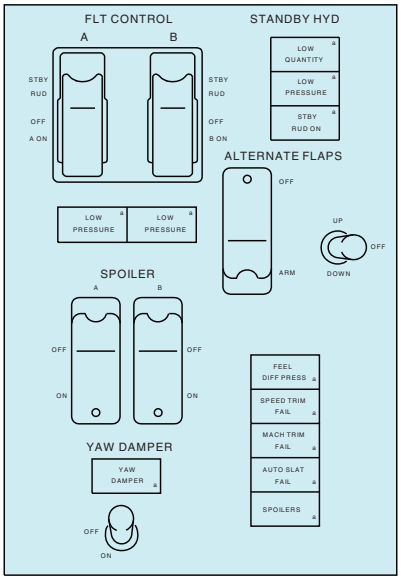
(1) Do this task: Leading Edge Flaps and Slats Retraction, TASK 27-81-00-860-804.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



FLIGHT COMPARTMENT



FLIGHT CONTROLS PANEL (P5-3)

A

2413331 S00061532356_V1

Flight Control Switches
Figure 508/27-81-00-990-809

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-81-00-730-801

9. Leading Edge Cruise Depressurization Test

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)

B. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

D. Prepare for the Test

SUBTASK 27-81-00-010-004

(1) Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 27-81-00-860-072

WARNING

KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(2) To pressurize the hydraulic system B, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

E. Leading Edge Cruise Depressurization Test

SUBTASK 27-81-00-860-073

WARNING

MAKE SURE THAT PERSONNEL AND EQUIPMENT STAY AWAY FROM THE LEADING EDGE SLATS, TRAILING EDGE FLAPS, AND DRIVE MECHANISMS. THE FLAPS, SLATS, AND DRIVE MECHANISMS MOVE QUICKLY. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(1) Move the flap control lever to the 5-unit position to extend the leading edge flaps, slats, and trailing edge flaps.

SUBTASK 27-81-00-740-002

(2) Do this test of the leading edge cruise depressurization valve:

(a) Make sure that the FLT CONTROL and ALTERNATE FLAPS switches, on the P5 forward overhead panel, are in the OFF position.

(b) Move the ALTERNATE FLAPS ARM switch, on the P5-3 panel, to the ARM position.

(c) Move the flap control lever to the UP position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(d) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	7	C00210	FLIGHT CONTROL FLAP SHUTOFF VALVES

(e) Use the FSEU to operate the leading edge cruise depressurization valve:

1) Push the ON/OFF button on the front panel of the FSEU to turn on the display.

NOTE: The display will show EXISTING FAULTS?.

2) Push the NO button until OTHER FUNCTNS? shows on the display.

3) Push the YES button to select OTHER FUNCTNS? on the FSEU.

4) Push the YES button to select SET OUTPUTS? from the OTHER FUNCTNS? menu.

5) Push the YES button to select LE DEPR VALVE? from the menu.

6) Push the YES button to select SET ON? on the display.

7) Press the YES switch on the FSEU.

(f) Make sure that the leading edge flaps and slats do not move.

(g) Wait until the FSEU display shows LE DEPR VALVE?.

NOTE: This will take approximately 75 seconds from pressing YES button to SET ON?.

(h) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	7	C00210	FLIGHT CONTROL FLAP SHUTOFF VALVES

(i) Push the YES button on the FSEU to display SET ON?.

1) Make sure that the trailing edge flaps remain stationary.

WARNING MAKE SURE THAT PERSONNEL AND EQUIPMENT STAY AWAY FROM THE LEADING EDGE SLATS, TRAILING EDGE FLAPS, AND DRIVE MECHANISMS. THE FLAPS, SLATS, AND DRIVE MECHANISMS MOVE QUICKLY. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(j) Push the YES button to select SET ON? on the display.

(k) Make sure that the leading edge slats retract and that the trailing edge flaps stay extended.

NOTE: The LE FLAPS TRANSIT lights will stay on.

(l) Wait until the FSEU display shows LE DEPR VALVE? and the leading edge surfaces extend to the extend position.

NOTE: This will take approximately 75 seconds.

(m) Move the flap control lever to the 1-unit position.

NOTE: The flap position indicator stays at the 5-unit position when the flaps and slats move to the 1-unit position.

(n) Push the YES button on the FSEU to display SET ON?.

(o) Push the YES button to select SET ON? on the display.

(p) Make sure that the flaps and slats don't move.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(q) Move the ALTERNATE FLAPS ARM switch to the OFF position.

NOTE: The leading edge and trailing edge flaps will move to the 1-unit position.

F. Put the Airplane Back to Its Usual Condition.

SUBTASK 27-81-00-860-074

(1) Push the ON/OFF button, on the front panel, of the FSEU to turn the display off.

SUBTASK 27-81-00-860-140

(2) Put the FLT CONTROL A and B switches, on the P5 forward overhead panel, to the ON position

SUBTASK 27-81-00-860-075

(3) Move the flap control lever to the UP position to retract the flaps and slats.

SUBTASK 27-81-00-410-004

(4) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 27-81-00-860-076

(5) To remove pressure from hydraulic system B, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

——— END OF TASK ———

TASK 27-81-00-700-804

10. Leading Edge Uncommanded Motion Protection Test

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)

B. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

D. Prepare for the Test

SUBTASK 27-81-00-010-005

(1) Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-81-00-860-078

WARNING KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(2) Pressurize hydraulic system B, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

E. Leading Edge Uncommanded Motion Protection Test

SUBTASK 27-81-00-860-079

WARNING MAKE SURE THAT PERSONNEL AND EQUIPMENT STAY AWAY FROM THE LEADING EDGE SLATS, TRAILING EDGE FLAPS, AND DRIVE MECHANISMS. THE FLAPS, SLATS, AND DRIVE MECHANISMS MOVE QUICKLY. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(1) Move the flap control lever to the UP position.

SUBTASK 27-81-00-720-002

(2) Do this test of the leading edge UCM shutoff valve:

(a) Make sure that the flap control lever is in the UP detent position.

(b) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	7	C00210	FLIGHT CONTROL FLAP SHUTOFF VALVES

(c) Push the ON/OFF button on the front panel of the FSEU to turn on the display.

NOTE: The display will show EXISTING FAULTS?.

(d) Push the down arrow until OTHER FUNCTNS? shows on the display.

(e) When OTHER FUNCTNS? shows, press YES.

(f) When SET OUTPUTS? shows, press YES.

(g) When LE DEPR VALVE? shows, press NO.

(h) When LE UCM SHUTOFF? shows, press YES.

(i) When SET ON? shows, press YES and move the flap control lever to the 1-unit detent position within 10 seconds.

NOTE: If you do not move the lever within 10 seconds, the test will not work.

(j) Make sure that the leading edge flaps and slats extend before the TE flap indicator needles are inside the 1-unit detent indicator (View A, Figure 507).

(k) Make sure that the leading edge indication flaps EXTEND light on the aft overhead panel is on (View B, Figure 507).

NOTE: The Amber LE FLAPS TRANSIT indication stays ON and then turns OFF when the Green LE FLAPS EXT indication shows.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(l) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	7	C00210	FLIGHT CONTROL FLAP SHUTOFF VALVES

(m) Wait until the FSEU display shows LE UCM SHUTOFF?.

NOTE: This will take approximately 75 seconds.

(n) Push the YES button on the FSEU to display SET ON?.

(o) When SET ON? shows, press YES and move the flap control lever to the UP detent position within 10 seconds.

NOTE: If you do not move the lever within 10 seconds, the test will not work.

(p) Make sure that the leading edge indication flaps and slats EXTEND lights on the aft overhead panel are on (View B, Figure 507).

(q) Make sure that the leading edge flaps and slats move to the UP position.

NOTE: This will take approximately 75 seconds from pressing YES button to SET ON?.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 27-81-00-860-080

(1) Push the ON/OFF button on the front panel of the FSEU to turn the display off.

SUBTASK 27-81-00-860-081

(2) Move the flap control lever to the UP position to retract the flaps and slats.

SUBTASK 27-81-00-410-005

(3) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 27-81-00-860-082

(4) To remove pressure from hydraulic system B, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

——— END OF TASK ———

TASK 27-81-00-710-802

11. Standby Hydraulic System Leading Edge Uncommanded Motion Test

(Figure 508)

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)

B. Location Zones

Zone	Area
118	Electrical and Electronics Compartment - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

D. Prepare for the Test

SUBTASK 27-81-00-860-084

(1) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	8	C00211	FLIGHT CONTROL FSEU DC 1
A	9	C01468	FLIGHT CONTROL FSEU DC 2
A	11	C01443	FC TE FLAP POS/SKEW SNSR & IND RIGHT
A	12	C00213	FC TE FLAP POS/SKEW SNSR & IND LEFT
C	11	C00362	FLIGHT CONT SHUTOFF VALVES STBY RUD

SUBTASK 27-81-00-010-011

(2) Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 27-81-00-860-102

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

MAKE SURE THAT YOU DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(3) Make sure that this circuit breaker is closed:

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
F	2	C01449	STANDBY HYDRAULIC PUMP

SUBTASK 27-81-00-410-009

(4) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 27-81-00-860-085

(5) Make sure that the leading edge flaps and slats, and trailing edge flaps are in the retracted position:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(a) Supply system B hydraulic power, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

(b) Move the flap control lever to the UP position to move the leading edge flaps and slats, and trailing edge flaps to the retracted position.

(c) Remove system B hydraulic power, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

E. Standby Hydraulic System Leading Edge Uncommanded Motion Test

SUBTASK 27-81-00-720-003

(1) Pressurize the standby hydraulic system:

(a) Set the ALTERNATE FLAPS, on the P5 forward overhead panel, to the ARM position.

NOTE: This will start the standby hydraulic pump.

(b) Make sure that the STANDBY HYD LOW PRESSURE light, on the P5 panel, comes on and quickly goes out.

NOTE: This makes sure that the standby system is pressurized.

SUBTASK 27-81-00-210-004

(2) Observe the leading edges on both the left and right wings:

(a) Make sure that the leading edges do not move when you pressurize the standby hydraulic system.

NOTE: The leading edge flaps can slowly extend because of gravity.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 27-81-00-860-086

(1) Move the ALTERNATE FLAPS ARM switch on the forward overhead panel to the OFF position.

NOTE: This will turn off the standby hydraulic system.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

LEADING EDGE FLAP AND SLAT CONTROL SYSTEM - INSPECTION/CHECK

1. General

A. This procedure has these tasks:

(1) Inspection of the Wing Leading Edge Flap and Slat Actuators and Leading Edge Flap and Slat Actuation Mechanism.

(2) Internal Hydraulic Leakage Check of the Wing Leading Edge Flap and Slat System.

TASK 27-81-00-210-801

2. Leading Edge Flap and Slat Actuators and Actuation Mechanism - Inspection

(Figure 601)

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
27-81-00-040-801	Leading Edge Flaps and Slats - Deactivation (P/B 201)
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)
27-81-00-860-803	Leading Edge Flaps and Slats Extension (P/B 201)
27-81-00-860-804	Leading Edge Flaps and Slats Retraction (P/B 201)
57-41-02-000-801	Leading Edge Access Panel Removal (P/B 201)
57-41-02-400-801	Leading Edge Access Panel Installation (P/B 201)

B. Location Zones

Zone	Area
512	Left Wing - Krueger Flap No. 2
513	Left Wing - Krueger Flap No. 1
522	Left Wing - Slat No. 4
523	Left Wing - Slat No. 3
524	Left Wing - Slat No. 2
525	Left Wing - Slat No. 1
612	Right Wing - Krueger Flap No. 3
613	Right Wing - Krueger Flap No. 4
622	Right Wing - Slat No. 5
623	Right Wing - Slat No. 6
624	Right Wing - Slat No. 7
625	Right Wing - Slat No. 8

C. Access Panels

Number	Name/Location
521CB	Lower Leading Edge Access Panel
521FB	Lower Leading Edge Access Panel
521JB	Lower Leading Edge Access Panel
521MB	Lower Leading Edge Access Panel
521QB	Lower Leading Edge Access Panel
521TB	Lower Leading Edge Access Panel
521WB	Lower Leading Edge Access Panel
521ZB	Lower Leading Edge Access Panel

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
621CB	Lower Leading Edge Access Panel
621FB	Lower Leading Edge Access Panel
621JB	Lower Leading Edge Access Panel
621MB	Lower Leading Edge Access Panel
621QB	Lower Leading Edge Access Panel
621TB	Lower Leading Edge Access Panel
621WB	Lower Leading Edge Access Panel
621ZB	Lower Leading Edge Access Panel

D. Prepare for the Inspection

SUBTASK 27-81-00-010-006

(1) Extend the leading edge flaps and slats (TASK 27-81-00-860-803).

SUBTASK 27-81-00-040-001

(2) If it is necessary, do this task: Leading Edge Flaps and Slats - Deactivation, TASK 27-81-00-040-801.

SUBTASK 27-81-00-010-009

(3) If it is necessary, remove the applicable lower leading edge access panels for the left wing (TASK 57-41-02-000-801).

(a) Open these access panels:

Number	Name/Location
521CB	Lower Leading Edge Access Panel
521FB	Lower Leading Edge Access Panel
521JB	Lower Leading Edge Access Panel
521MB	Lower Leading Edge Access Panel
521QB	Lower Leading Edge Access Panel
521TB	Lower Leading Edge Access Panel
521WB	Lower Leading Edge Access Panel
521ZB	Lower Leading Edge Access Panel

SUBTASK 27-81-00-010-010

(4) If it is necessary, remove the applicable lower leading edge access panels for the right wing (TASK 57-41-02-000-801).

(a) Open these access panels:

Number	Name/Location
621CB	Lower Leading Edge Access Panel
621FB	Lower Leading Edge Access Panel
621JB	Lower Leading Edge Access Panel
621MB	Lower Leading Edge Access Panel
621QB	Lower Leading Edge Access Panel
621TB	Lower Leading Edge Access Panel
621WB	Lower Leading Edge Access Panel
621ZB	Lower Leading Edge Access Panel

E. Leading Edge Flap and Slat Actuators and Actuation Mechanism Inspection

SUBTASK 27-81-00-210-005

(1) For the left wing inspection:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Do a general visual inspection of the left wing leading edge flap and slat actuators and the leading edge flap and slat actuation mechanism (main and auxiliary tracks and rollers).

SUBTASK 27-81-00-210-006

(2) For the right wing inspection:

(a) Do a general visual inspection of the right wing leading edge flap and slat actuators and the leading edge flap and slat actuation mechanism (main and auxiliary tracks and rollers).

F. Put the Airplane Back to Its Usual Condition

SUBTASK 27-81-00-410-007

(1) If it is necessary, install the applicable lower leading edge access panels on the left wing (TASK 57-41-02-400-801).

(a) Close these access panels:

Number	Name/Location
521CB	Lower Leading Edge Access Panel
521FB	Lower Leading Edge Access Panel
521JB	Lower Leading Edge Access Panel
521MB	Lower Leading Edge Access Panel
521QB	Lower Leading Edge Access Panel
521TB	Lower Leading Edge Access Panel
521WB	Lower Leading Edge Access Panel
521ZB	Lower Leading Edge Access Panel

SUBTASK 27-81-00-410-008

(2) If it is necessary, install the applicable lower leading edge access panels on the right wing (TASK 57-41-02-400-801).

(a) Close these access panels:

Number	Name/Location
621CB	Lower Leading Edge Access Panel
621FB	Lower Leading Edge Access Panel
621JB	Lower Leading Edge Access Panel
621MB	Lower Leading Edge Access Panel
621QB	Lower Leading Edge Access Panel
621TB	Lower Leading Edge Access Panel
621WB	Lower Leading Edge Access Panel
621ZB	Lower Leading Edge Access Panel

SUBTASK 27-81-00-440-001

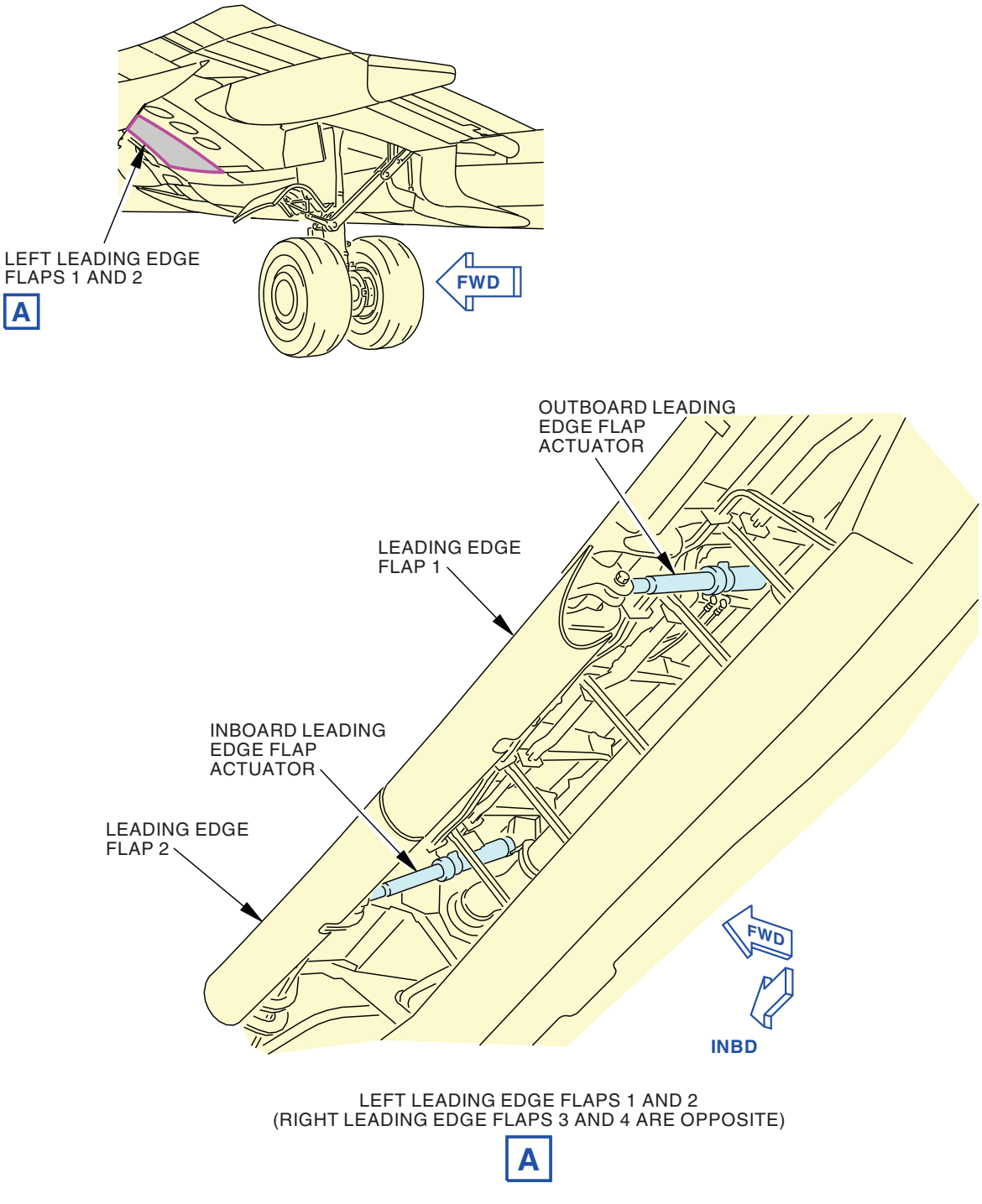
(3) If it is necessary, do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801.

SUBTASK 27-81-00-010-007

(4) Retract the leading edge flaps and slats (TASK 27-81-00-860-804).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



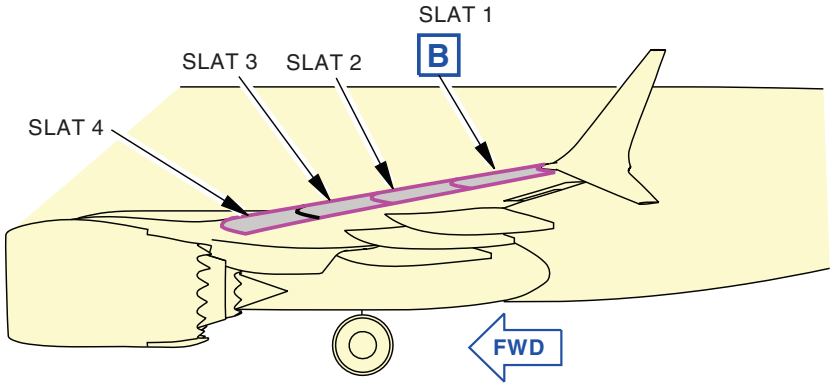
LEFT LEADING EDGE FLAPS 1 AND 2
(RIGHT LEADING EDGE FLAPS 3 AND 4 ARE OPPOSITE)

A

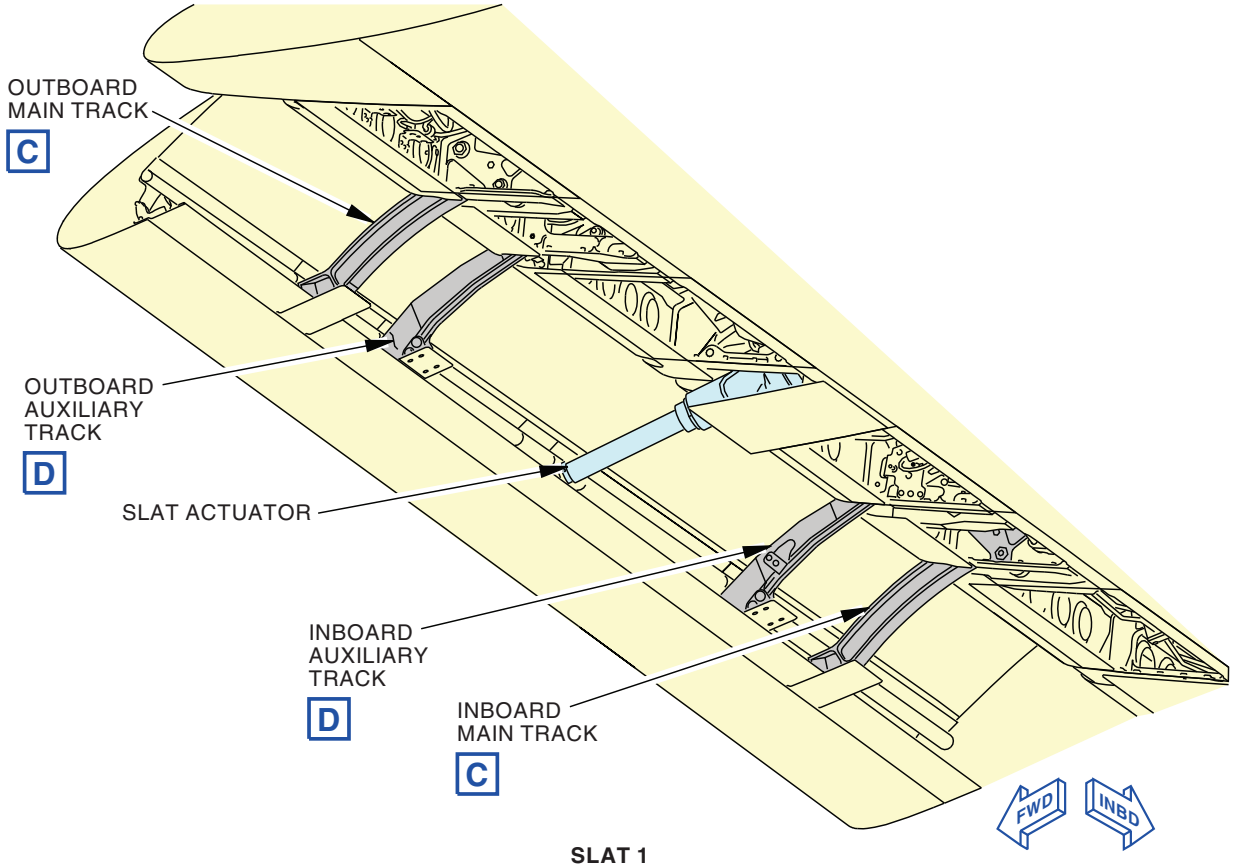
2413332 S00061532362_V1

Leading Edge Flap and Slat Actuators and Actuation Mechanism Inspection
Figure 601/27-81-00-990-810 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LEFT WING
(RIGHT WING IS EQUIVALENT)



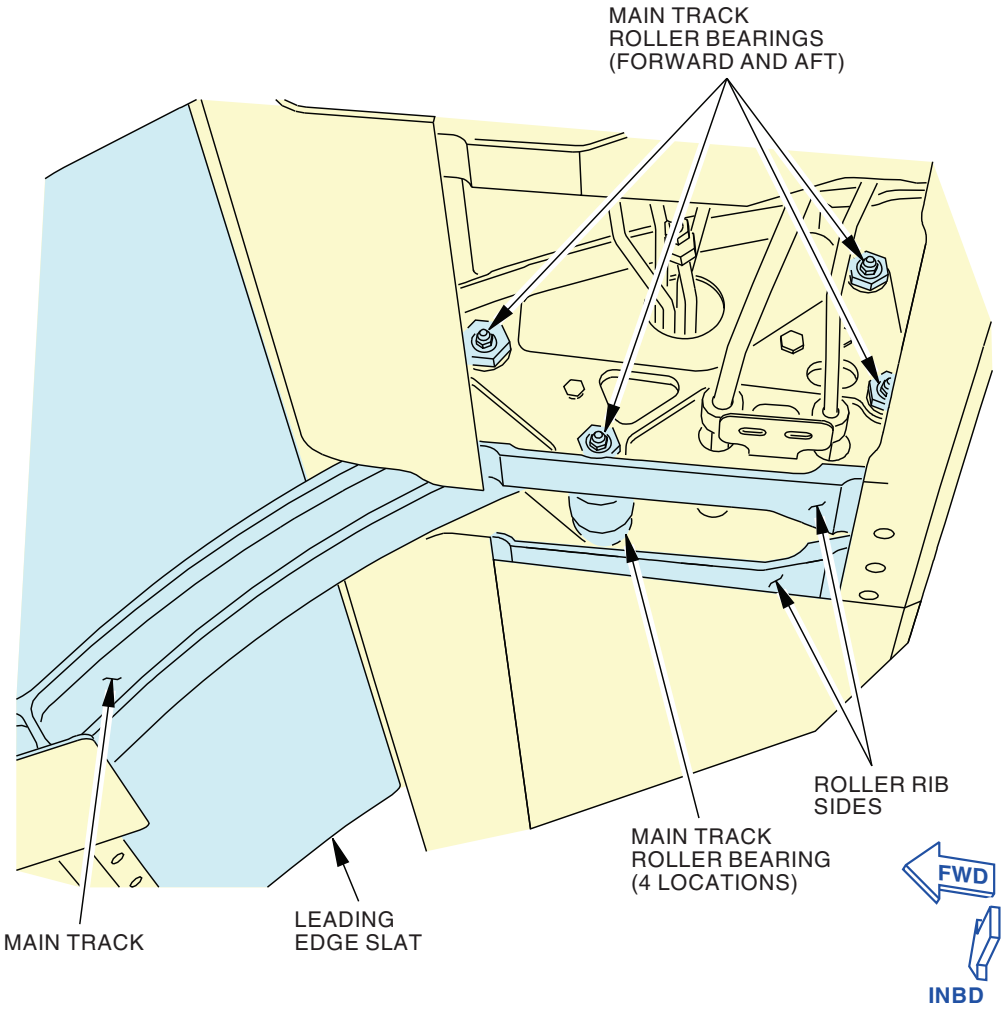
SLAT 1
(SLATS 2, 3, AND 4 ARE EQUIVALENT)

B

2413333 S00061532363_V2

Leading Edge Flap and Slat Actuators and Actuation Mechanism Inspection
Figure 601/27-81-00-990-810 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



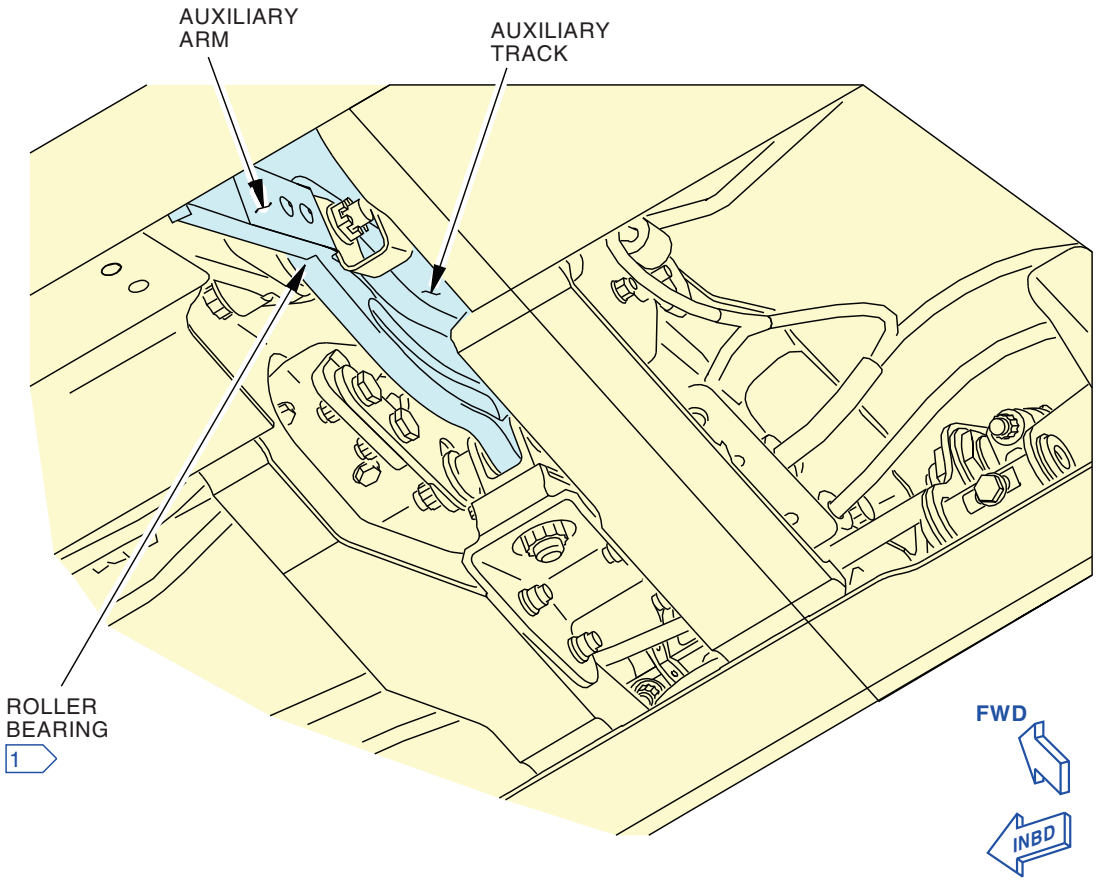
INBOARD MAIN TRACK
(OUTBOARD MAIN TRACK IS OPPOSITE)
(SLAT 1 IS SHOWN, SLATS 2, 3, AND 4 ARE EQUIVALENT)

C

2413334 S00061532364_V2

Leading Edge Flap and Slat Actuators and Actuation Mechanism Inspection
Figure 601/27-81-00-990-810 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



INBOARD AUXILIARY TRACK
(OUTBOARD AUXILIARY TRACK IS OPPOSITE)

D

1 INSTALLED BETWEEN THE ARM AND TRACK CONNECTION.

2413335 S00061532365_V2

Leading Edge Flap and Slat Actuators and Actuation Mechanism Inspection
Figure 601/27-81-00-990-810 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-81-00-790-801

3. Leading Edge Flap and Slat System Internal Hydraulic Leakage Check

A. General

(1) This procedure does an internal hydraulic leakage check for the leading edge flap and slat valves. The purpose of this leakage test is to identify the total leakage value at the retracted and full extended position.

(2) You must measure the amount of leakage of hydraulic fluid during different conditions of operation to measure the fluid flow rate. It will be necessary to disconnect the return hydraulic lines.

(3) Hydraulic pressure that pushes the hydraulic fluid to blow past the internal O-ring seals in the valves and actuators is internal hydraulic leakage. This is an indication of an internal seal problem. Large internal leakage can cause slow actuation times.

(4) Normal system operation uses the hydraulic system B to move the Leading Edge (LE) flaps and slats.

(5) Hydraulic pressure goes through the LE cruise depressurization valve to the LE flap and slat control valve. The LE flap and slat control valve sends extend pressure to the LE flap and slat actuators. This causes the LE flaps and slats to move to the extend position. The hydraulic fluid goes through hydraulic fuses in the lines.

(6) The LE flap and slat system connects to the hydraulic system return at these three locations: the leading edge standby line, the return line on the LE flap and slat control valve, and the return line for the autoslat module and LE Uncommanded Motion (UCM) shutoff valve.

B. References

Reference	Title
12-12-00-610-801	Hydraulic Reservoir Servicing (P/B 301)
24-22-00-860-801	Supply Electrical Power (P/B 201)
27-81-00-700-801	Leading Edge Flap and Slat Operational Test in Primary Mode (P/B 501)
27-81-00-860-801	Leading Edge Flap and Slat System Operation With Primary Control (P/B 201)
27-81-52-000-801	LE Uncommanded Motion (UCM) or LE Cruise Depressurization Valve Removal (P/B 401)
27-81-52-400-801	LE Uncommanded Motion (UCM) or LE Cruise Depressurization Valve Installation (P/B 401)
27-81-81-000-801	Leading Edge Flap and Slat Control Valve Removal (P/B 401)
27-81-81-400-801	Leading Edge Flap and Slat Control Valve Installation (P/B 401)
27-81-91-000-801	Leading Edge Standby Shutoff Valve Removal (P/B 201)
27-81-91-400-801	Leading Edge Standby Shutoff Valve Installation (P/B 201)
29-09-00-860-801	Hydraulic Reservoirs Pressurization (P/B 201)
29-09-00-860-802	Hydraulic Reservoirs Depressurization (P/B 201)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)

C. Tools/Equipment

Reference	Description
STD-6338	Container - Graduated - 1000 ml measuring flask
STD-14686	Fittings - Hydraulic, Caps/Plugs (ref. AMM 20-10-XX)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. Location Zones

Zone	Area
132	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Right
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
192	Lower Wing-To-Body Fairing - Under Wing Box
211	Flight Compartment - Left
212	Flight Compartment - Right

E. Access Panels

Number	Name/Location
192BR	ECS Ram Air Inlet Mixing Duct Panel - Forward

F. Prepare for the Internal Leakage Check

SUBTASK 27-81-00-860-087

(1) Supply electrical power (TASK 24-22-00-860-801).

SUBTASK 27-81-00-010-008

(2) To get access to the autoslat control valve, do this step:

(a) Remove this access panel:

Number	Name/Location
192BR	ECS Ram Air Inlet Mixing Duct Panel - Forward

G. Retracted LE Flap and Slat System Internal Hydraulic Leakage Check

SUBTASK 27-81-00-860-088

WARNING

KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Supply system B hydraulic power (TASK 29-11-00-860-801).

SUBTASK 27-81-00-860-089

(2) Pressurize the system B hydraulic reservoir (TASK 29-09-00-860-801).

SUBTASK 27-81-00-860-090

(3) Make sure the flap control lever is at the FLAP UP position to do the leakage check at the retracted position.

SUBTASK 27-81-00-860-091

(4) Remove the pressure from the system B reservoir (TASK 29-09-00-860-802).

NOTE: It is necessary to remove the pressure from the reservoir. When you disconnect a hydraulic line below the level of the reservoir, all the hydraulic fluid will flow and drain out of the reservoir if pressure is applied.

SUBTASK 27-81-00-860-092

(5) Remove the system B hydraulic power (TASK 29-11-00-860-805).

SUBTASK 27-81-00-790-001

(6) Do a check of the LE flap and slat control valve for hydraulic fluid leakage:

(a) Put a graduated container, STD-6338, under the LE flap and slat control valve.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Disconnect the return line from the LE flap and slat control valve (TASK 27-81-81-000-801).

NOTE: Only do the applicable steps in the task.

(c) Install a cap/plug, STD-14686, on the hydraulic line that goes to the hydraulic reservoir.

(d) Supply system B hydraulic power (TASK 29-11-00-860-801).

(e) Pressurize the system B hydraulic reservoir (TASK 29-09-00-860-801).

(f) Measure and make a record of the leakage rate of the hydraulic fluid from the LE flap and slat control valve.

(g) Remove the pressure from the system B reservoir (TASK 29-09-00-860-802).

(h) Remove the system B hydraulic power (TASK 29-11-00-860-805).

(i) Remove the cap/plug, STD-14686, from the hydraulic line.

(j) Reconnect the return line to the LE flap and slat control valve (TASK 27-81-81-400-801).

NOTE: Only do the applicable steps in the task.

SUBTASK 27-81-00-790-002

(7) Do a check of the LE standby shutoff valve for hydraulic fluid leakage:

(a) Put a graduated container, STD-6338, under the leading edge standby shutoff valve.

(b) Disconnect the return line from the leading edge standby shutoff valve (TASK 27-81-91-000-801).

NOTE: Only do the applicable steps in the task.

(c) Install a cap/plug, STD-14686, on the hydraulic line that goes to the hydraulic reservoir.

(d) Supply system B hydraulic power (TASK 29-11-00-860-801).

(e) Pressurize the system B hydraulic reservoir (TASK 29-09-00-860-801).

(f) Measure and make a record of the leakage rate of the hydraulic fluid from the leading edge standby shutoff valve.

(g) Remove the pressure from the system B reservoir (TASK 29-09-00-860-802).

(h) Remove the system B hydraulic power (TASK 29-11-00-860-805).

(i) Remove the cap/plug, STD-14686, from the hydraulic line.

(j) Re-connect the return line to the leading edge standby shutoff valve (TASK 27-81-91-400-801).

NOTE: Only do the applicable steps in the task.

SUBTASK 27-81-00-790-003

(8) Do a check of the LE UCM shutoff valve and the autoslat valve for hydraulic fluid leakage:

(a) Put a graduated container, STD-6338, under the tee for the return line on the LE UCM shutoff valve and the autoslat valve.

(b) Disconnect the return line from the tee (TASK 27-81-52-000-801).

NOTE: Only do the applicable steps in the task.

(c) Install a cap/plug, STD-14686, on the hydraulic line that goes to the hydraulic reservoir.

(d) Supply system B hydraulic power (TASK 29-11-00-860-801).

(e) Pressurize the system B hydraulic reservoir (TASK 29-09-00-860-801).

(f) Measure and make a record of the leakage rate of the hydraulic fluid from the tee.

(g) Remove the pressure from the system B reservoir (TASK 29-09-00-860-802).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(h) Remove the system B hydraulic power (TASK 29-11-00-860-805).

(i) Remove the cap/plug, STD-14686, from the hydraulic line.

(j) Re-connect the return line to the tee (TASK 27-81-52-400-801).

NOTE: Only do the applicable steps in the task.

SUBTASK 27-81-00-790-004

(9) Calculate the internal hydraulic leakage for the retracted LE flap and slats:

(a) Add the three leakage rates together to find the total leakage rate with the LE flaps and slats in the retracted position.

(b) If the total leakage rate is more than 0.2 gal (0.8 l) per minute, then there is excessive leakage in the LE system.

NOTE: The hydraulic fuse will set when the leakage rate is more than 0.6 gal (2.3 l) per minute. When the leakage rate is between 0.2 gal (0.8 l) per minute and 0.6 gal (2.3 l) per minute, the fuse could set.

(c) If the total leakage rate is 0.2 gal (0.8 l) per minute or less, then continue to find the total leakage rate for the LE flaps and slats in the full extended position.

NOTE: A hydraulic fuse that sets at a leakage rate less than 0.2 gal (0.8 l) per minute should be replaced.

H. Extended LE Flap and Slat System Internal Hydraulic Leakage Check

SUBTASK 27-81-00-860-093

WARNING

KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Supply system B hydraulic power (TASK 29-11-00-860-801).

SUBTASK 27-81-00-860-094

(2) Pressurize the system B hydraulic reservoir (TASK 29-09-00-860-801).

SUBTASK 27-81-00-860-095

(3) Extend the LE flaps and slats (TASK 27-81-00-860-801).

(a) Move the flap control lever to the 1-unit position to move the LE flaps to the full extend position and LE slats to the extend position.

SUBTASK 27-81-00-790-005

(4) Do these procedures to make a record of the leakage rate with the LE flaps at the full extend position and LE slats at the extend position.

(a) Do a check of the LE flap and slat control valve for hydraulic fluid leakage.

(b) Do a check of the LE standby shutoff valve for hydraulic fluid leakage.

(c) Do a check of the LE UCM shutoff valve and the autoslat valve for hydraulic fluid leakage.

SUBTASK 27-81-00-860-096

(5) Extend the LE flaps and slats (TASK 27-81-00-860-801).

(a) Move the flap control lever to the 30-unit position to move the LE flaps and LE slats to the full extend position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-81-00-790-006

(6) Do these steps to make a record of the leakage rate with the LE flaps and LE slats at the full extend position.

(a) Do a check of the LE flap and slat control valve for hydraulic fluid leakage.

(b) Do a check of the LE standby shutoff valve for hydraulic fluid leakage.

(c) Do a check of the LE UCM shutoff valve and the autoslat valve for hydraulic fluid leakage.

SUBTASK 27-81-00-860-097

(7) Remove the pressure from the system B reservoir (TASK 29-09-00-860-802).

SUBTASK 27-81-00-860-098

(8) Remove the system B hydraulic power (TASK 29-11-00-860-805).

SUBTASK 27-81-00-790-007

(9) Calculate the internal hydraulic leakage for the full extended flap and slats.

(a) Add the three leakage rates together to find the total leakage rate with the LE in the full extended position.

(b) If the total leakage rate is more than 0.2 gal (0.8 l) per minute, then there is excessive leakage in the leading edge system.

NOTE: The hydraulic fuse will set when the leakage rate is more than 0.6 gal (2.3 l) per minute. When the leakage rate is between 0.2 gal (0.8 l) per minute and 0.6 gal (2.3 l) per minute, the fuse could set.

SUBTASK 27-81-00-790-008

(10) If the total leakage rate is 0.2 gal (0.8 l) per minute or less, then there can be a problem with the hydraulic fuses.

NOTE: A hydraulic fuse that sets at a leakage rate less than 0.2 gal (0.8 l) per minute should be replaced.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 27-81-00-610-001

(1) If it is necessary to service the hydraulic reservoir, do this task: Hydraulic Reservoir Servicing, TASK 12-12-00-610-801.

SUBTASK 27-81-00-710-004

(2) Do an operational test of the LE flaps and slats (TASK 27-81-00-700-801).

SUBTASK 27-81-00-790-009

(3) Make sure there are no hydraulic fluid leaks at the LE flap and slat control valve, LE standby shutoff valve, LE UCM shutoff valve and autoslat control valve.

SUBTASK 27-81-00-410-006

(4) Install this access panel:

Number	Name/Location
192BR	ECS Ram Air Inlet Mixing Duct Panel - Forward

SUBTASK 27-81-00-860-099

(5) Remove the hydraulic power (TASK 29-11-00-860-805).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

LEADING EDGE FLAPS - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the leading edge flaps.

(2) An installation of the leading edge flaps.

B. This procedure applies to the inboard and outboard leading edge flaps.

TASK 27-81-11-000-801

2. Leading Edge Flap Removal

(Figure 401)

A. References

Reference	Title
27-81-00-480-801	Leading Edge Flap and Slat Locks Installation (P/B 201)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1734	Lockout Set - Leading Edge Slats and Krueger Flaps Actuators Part #: C27051-29 Supplier: 81205

C. Location Zones

Zone	Area
512	Left Wing - Krueger Flap No. 2
513	Left Wing - Krueger Flap No. 1
612	Right Wing - Krueger Flap No. 3
613	Right Wing - Krueger Flap No. 4

D. Prepare for the Removal

SUBTASK 27-81-11-860-001

WARNING

KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Pressurize hydraulic system B (TASK 29-11-00-860-801).

SUBTASK 27-81-11-860-002

WARNING

KEEP PERSONNEL AND EQUIPMENT AWAY FROM THE FLAPS AND SLATS. THE FLAPS AND SLATS CAN ACCIDENTALLY MOVE AND CAUSE INJURIES TO PERSONNEL OR DAMAGE TO EQUIPMENT.

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(WARNING PRECEDES)

CAUTION

DO NOT EXTEND THE LEADING EDGE FLAPS AND SLATS UNTIL YOU CLOSE THE THRUST REVERSER, AND FAN DUCT COWL. THE LEADING EDGE FLAPS AND SLATS WILL HIT THE THRUST REVERSER, OR FAN DUCT COWL IF THEY ARE OPEN. IF THEY HIT, DAMAGE TO EQUIPMENT WILL OCCUR.

(2) Move the flap control lever to the 1-unit detent to extend the leading edge flaps.

SUBTASK 27-81-11-860-003

(3) Remove power from hydraulic system B (TASK 29-11-00-860-805).

SUBTASK 27-81-11-480-001

WARNING

INSTALL THE LOCK ON LEADING EDGE FLAP ACTUATOR CAREFULLY TO PREVENT THE ACCIDENTAL OPERATION OF THE LEADING EDGE FLAPS. THE LEADING EDGE FLAPS CAN MOVE QUICKLY. THIS CAN CAUSE INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT.

(4) Install the lockout set, SPL-1734, on the actuator of the adjacent leading edge flap (TASK 27-81-00-480-801).

E. Leading Edge Flap Removal

SUBTASK 27-81-11-020-001

(1) Disconnect the flap actuator [4] from the leading edge flap [2] or leading edge flap [1]:

(a) Remove the pin [6], nut [7], washer [8], washer [5], bushing [11], washer [10], and bolt [9] to disconnect the flap actuator [4] from the outboard leading edge flap [1].

(b) Remove the pin [15], nut [16], washer [17], washer [18], bushing [12], washer [13], and bolt [14] to disconnect the flap actuator [4] from the inboard leading edge flap [2].

SUBTASK 27-81-11-020-002

(2) Do these steps to remove the outboard leading edge flap [1] from the fixed leading edge:

(a) Disconnect the outboard bullnose linkage [53] from the fixed leading edge flap.

1) Remove the pin [55], nut [54], washer [56], anti-rotation washers [41], bushing [44], washer [43], and bolt [42].

(b) Remove the nut [32], washers [31], and bolt [30] to disconnect the bonding jumper [29] from the fixed leading edge.

(c) Disconnect hinges 1 through 4 on the leading edge flap [1] from the fixed leading edge.

1) At hinges 1 and 4, remove the pins [21], nuts [22], washers [20], bushings [19], washers [24], and bolts [23].

2) At hinges 2 and 3, remove the pins [34], nuts [33], washers [25], bushings [26], washers [28], and bolts [27].

(d) Remove the leading edge flap [1] from the wing.

SUBTASK 27-81-11-020-003

(3) Do these steps to remove the inboard leading edge flap [2] from the fixed leading edge:

(a) Disconnect the mid and inboard bullnose linkages [53] from the fixed leading edge flap.

1) Remove the pin [55], nut [54], washer [56], anti-rotation washers [41], bushing [44], washer [43], and bolt [42].

(b) Remove the nut [32], washers [31], and bolt [30] to disconnect the bonding jumper [29] from the leading edge flap [2].

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(c) Disconnect hinges 5 through 8 on the leading edge flap [2] from the fixed leading edge flap.

1) At hinge 5, remove the pin [21], nut [22], washer [20], bushing [19], washer [24], and bolt [23].

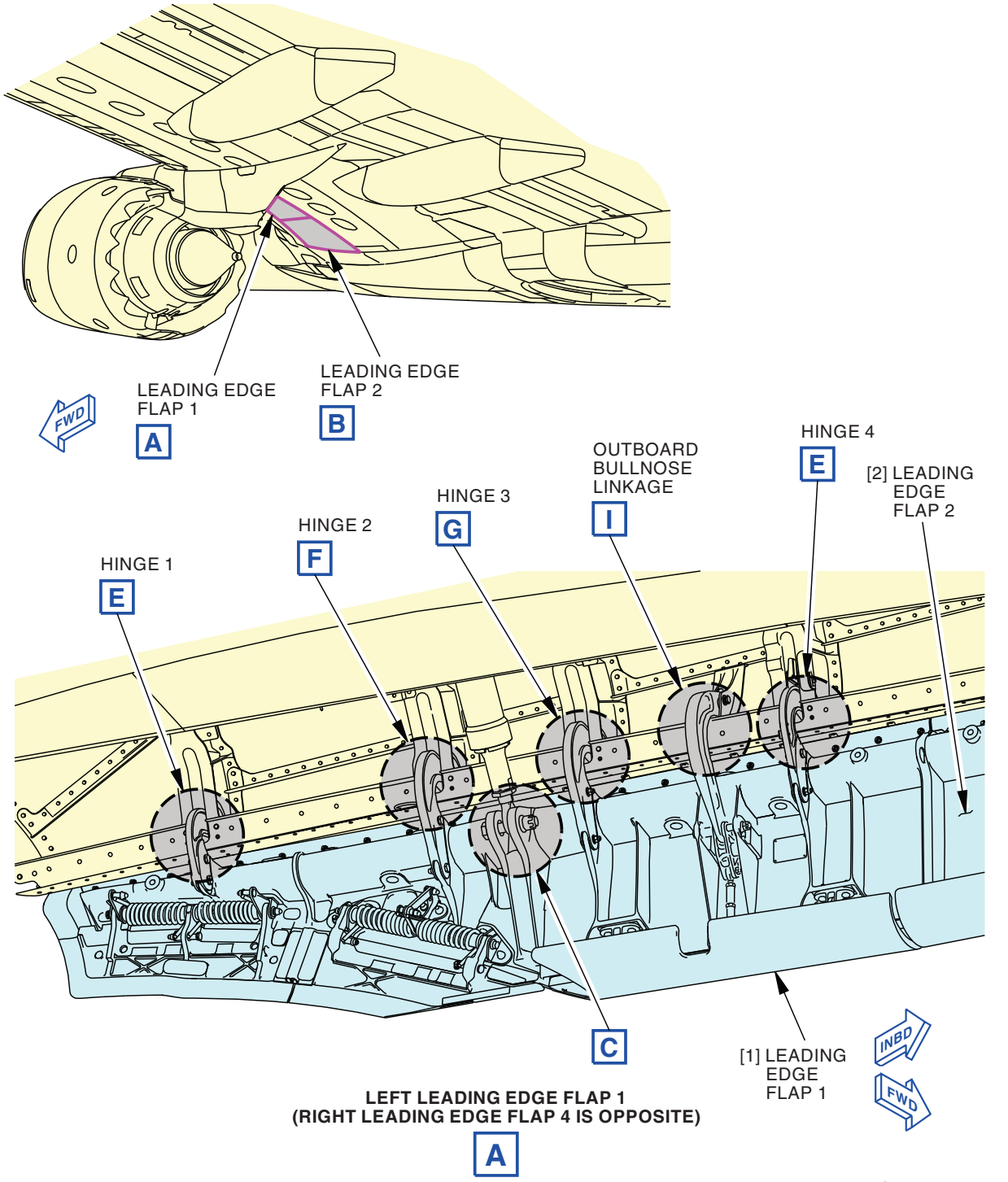
2) At hinges 6 and 7, remove the pins [34], nuts [33], washers [25], bushings [26], washers [28], and bolts [27].

3) At hinge 8, remove the pin [40], nut [39], washer [35], bushing [38], washer [36], and bolt [37].

(d) Remove the leading edge flap [2] from the wing.

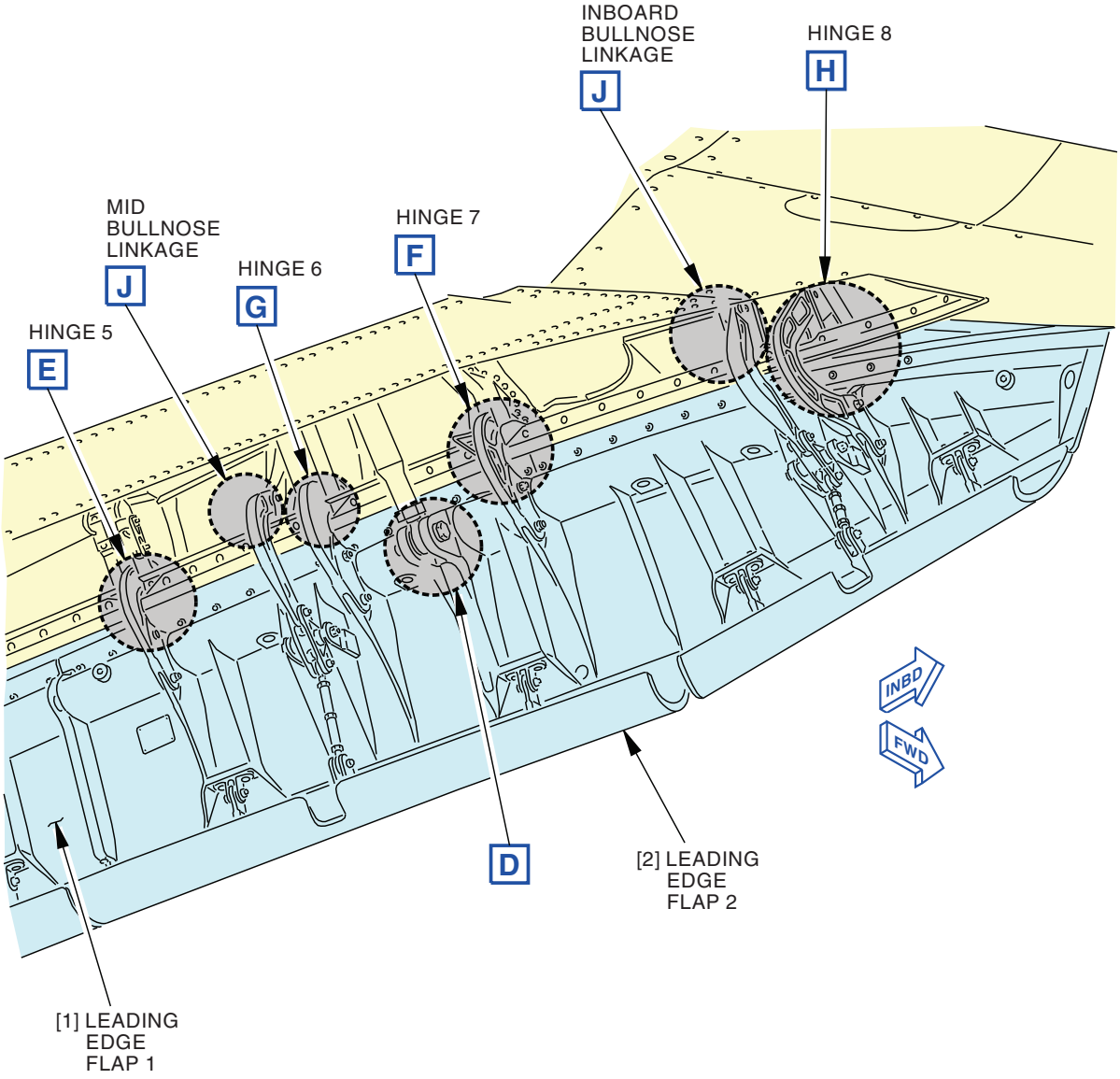
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Leading Edge Flap Installation
Figure 401/27-81-11-990-801 (Sheet 1 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



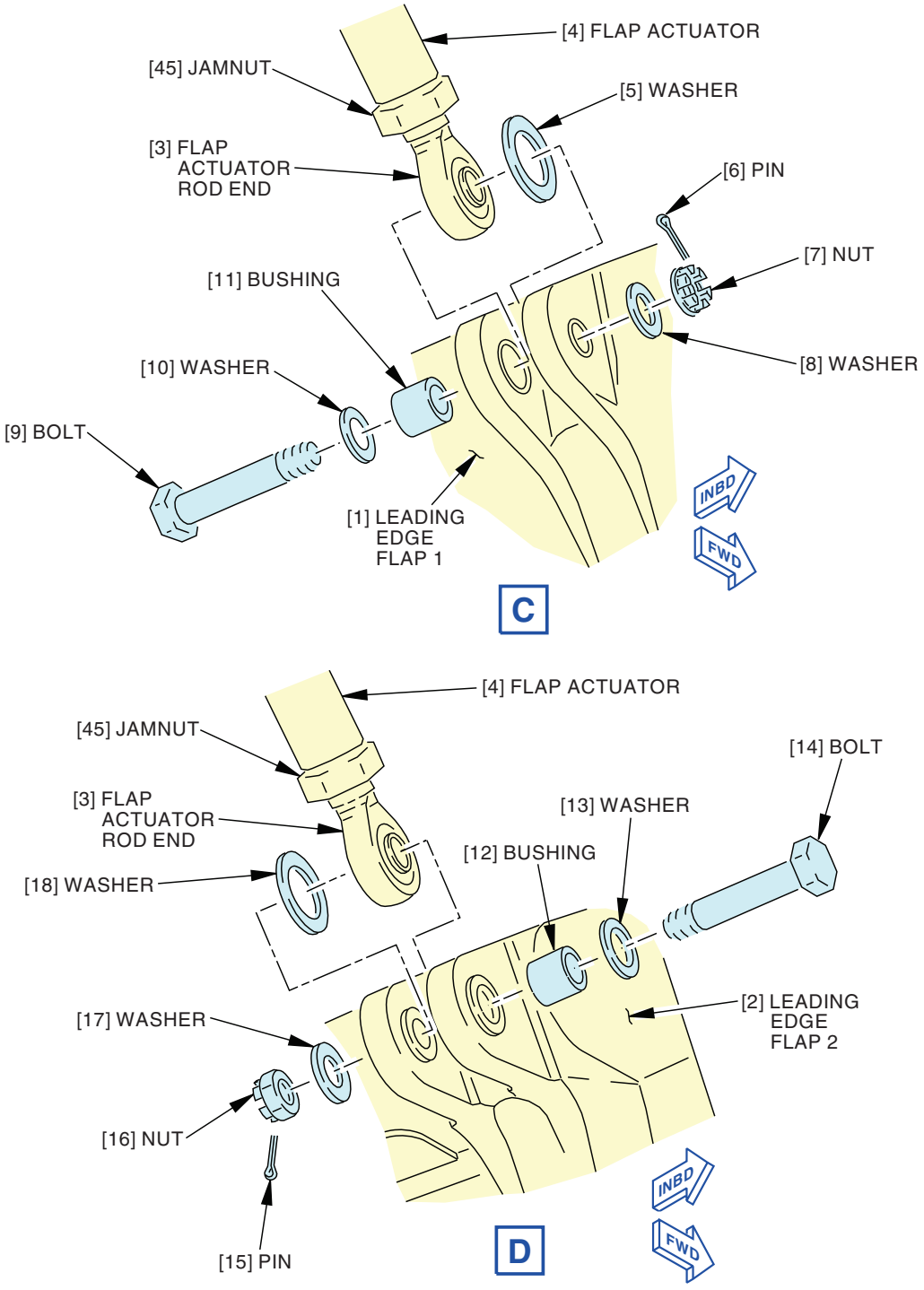
LEFT LEADING EDGE FLAP 2
(RIGHT LEADING EDGE FLAP 3 IS OPPOSITE)

B

2413337 S00061532370_V2

Leading Edge Flap Installation
Figure 401/27-81-11-990-801 (Sheet 2 of 9)

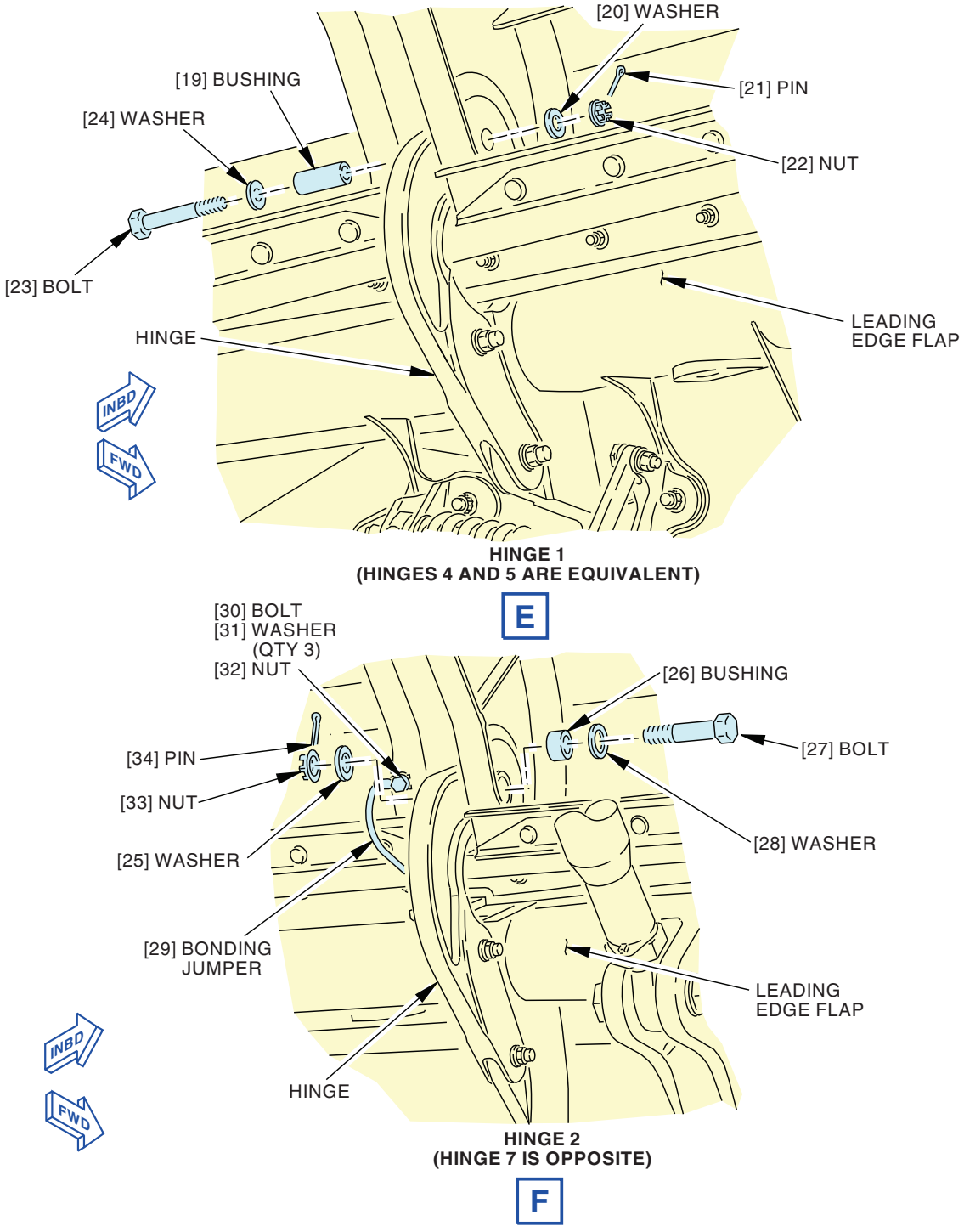
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2413338 S00061532371_V1

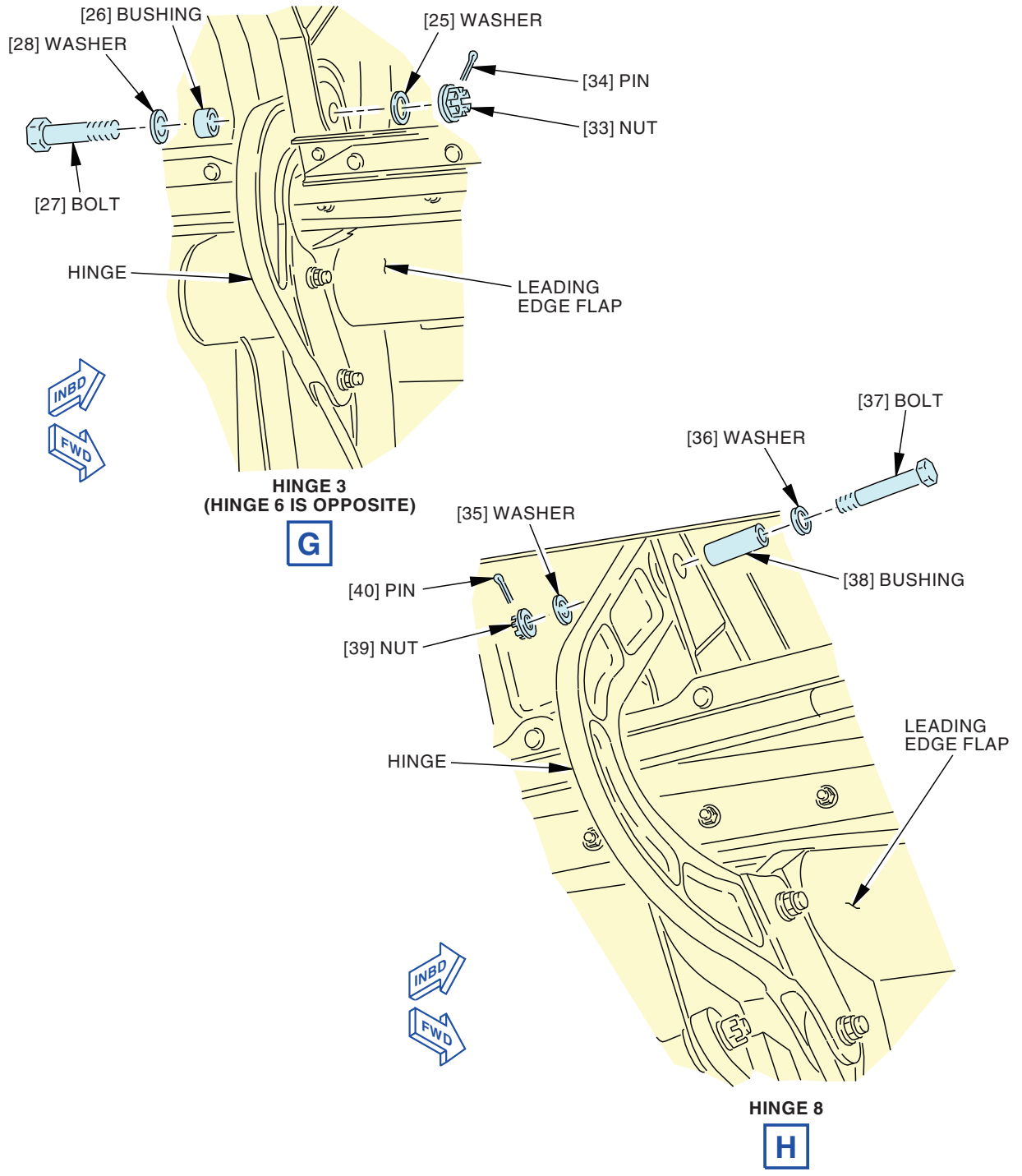
Leading Edge Flap Installation
Figure 401/27-81-11-990-801 (Sheet 3 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



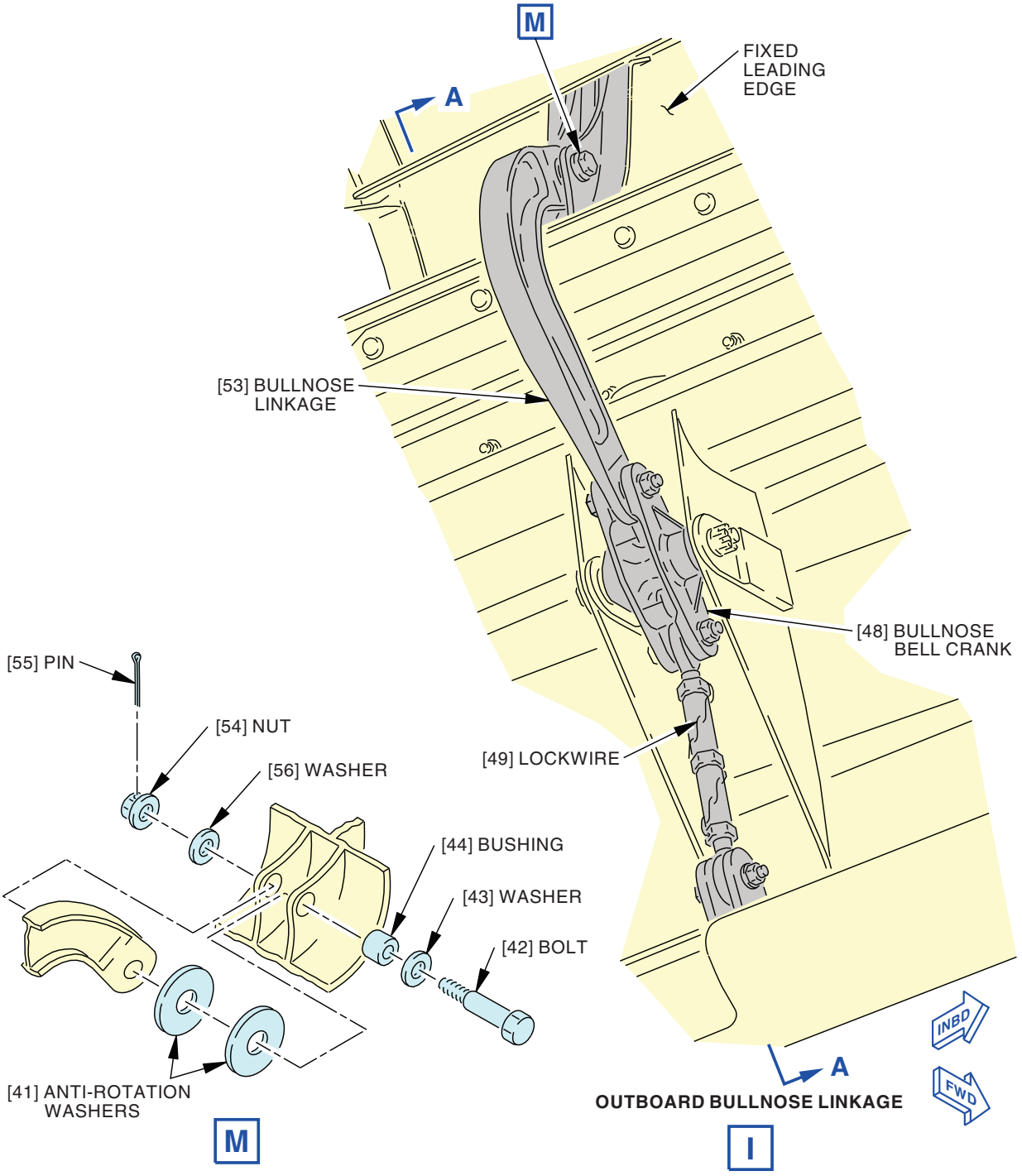
Leading Edge Flap Installation
Figure 401/27-81-11-990-801 (Sheet 4 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



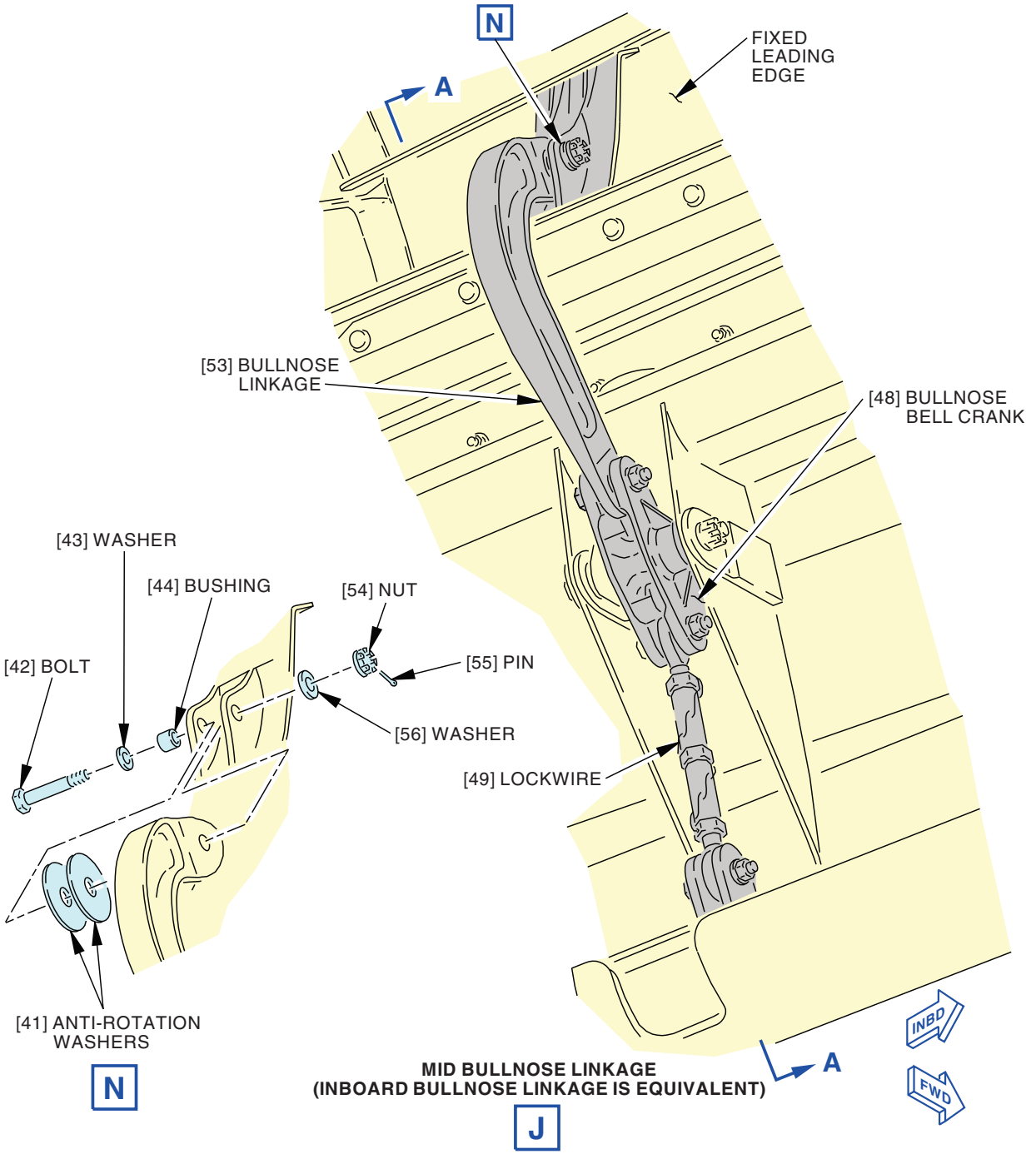
Leading Edge Flap Installation
Figure 401/27-81-11-990-801 (Sheet 5 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



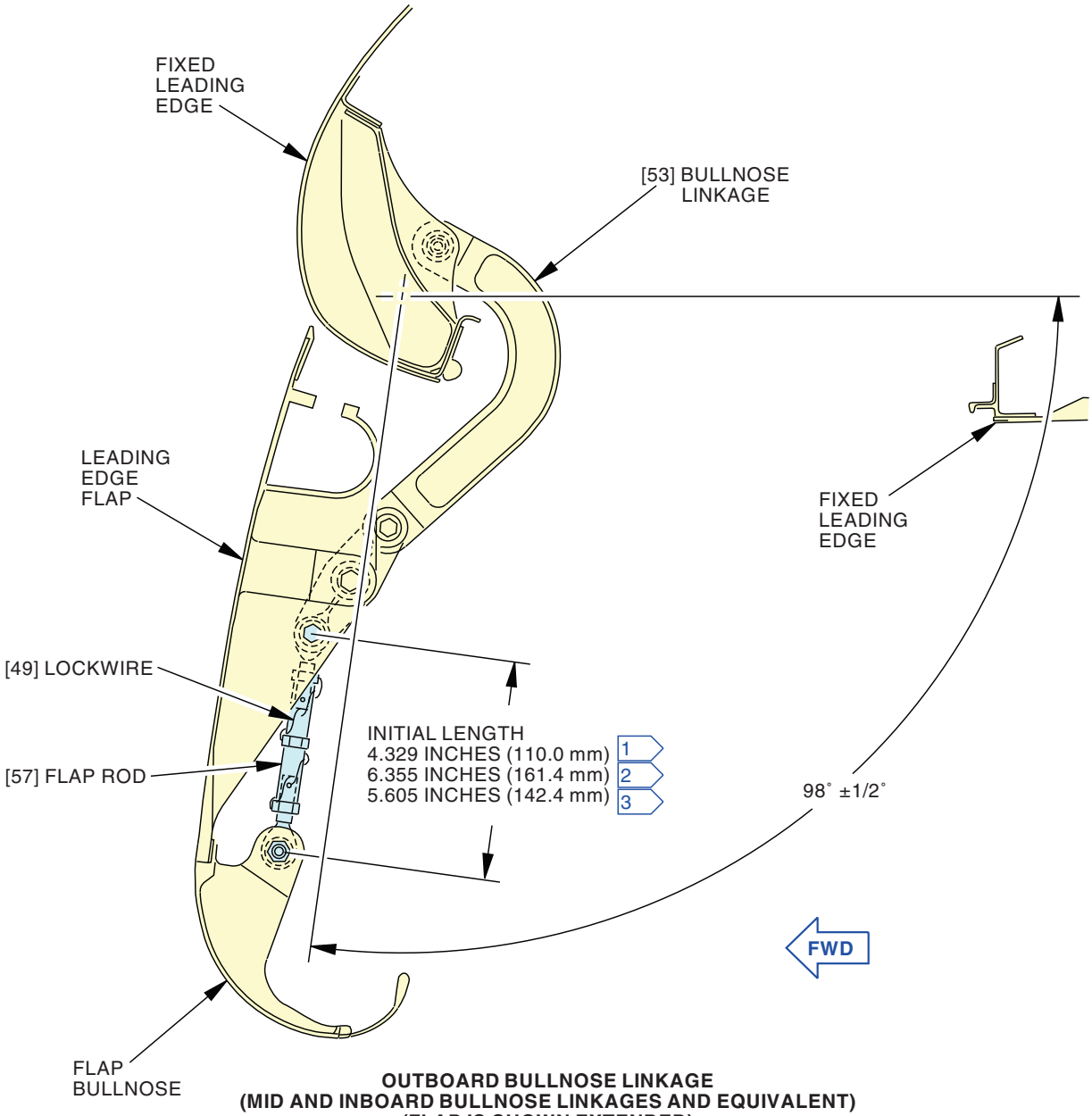
Leading Edge Flap Installation
Figure 401/27-81-11-990-801 (Sheet 6 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Leading Edge Flap Installation
Figure 401/27-81-11-990-801 (Sheet 7 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



OUTBOARD BULLNOSE LINKAGE
(MID AND INBOARD BULLNOSE LINKAGES AND EQUIVALENT)
(FLAP IS SHOWN EXTENDED)

A-A

1 INBOARD BULLNOSE LINKAGE

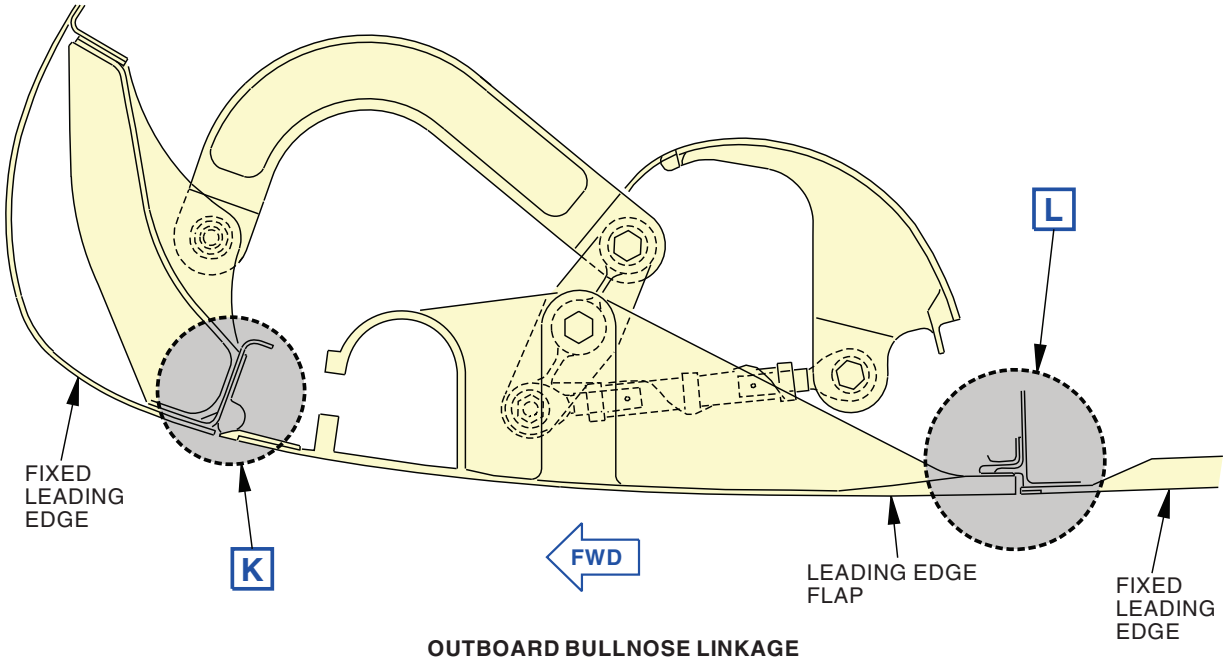
2 MID BULLNOSE LINKAGE

3 OUTBOARD BULLNOSE LINKAGE

2413343 S00061532376_V1

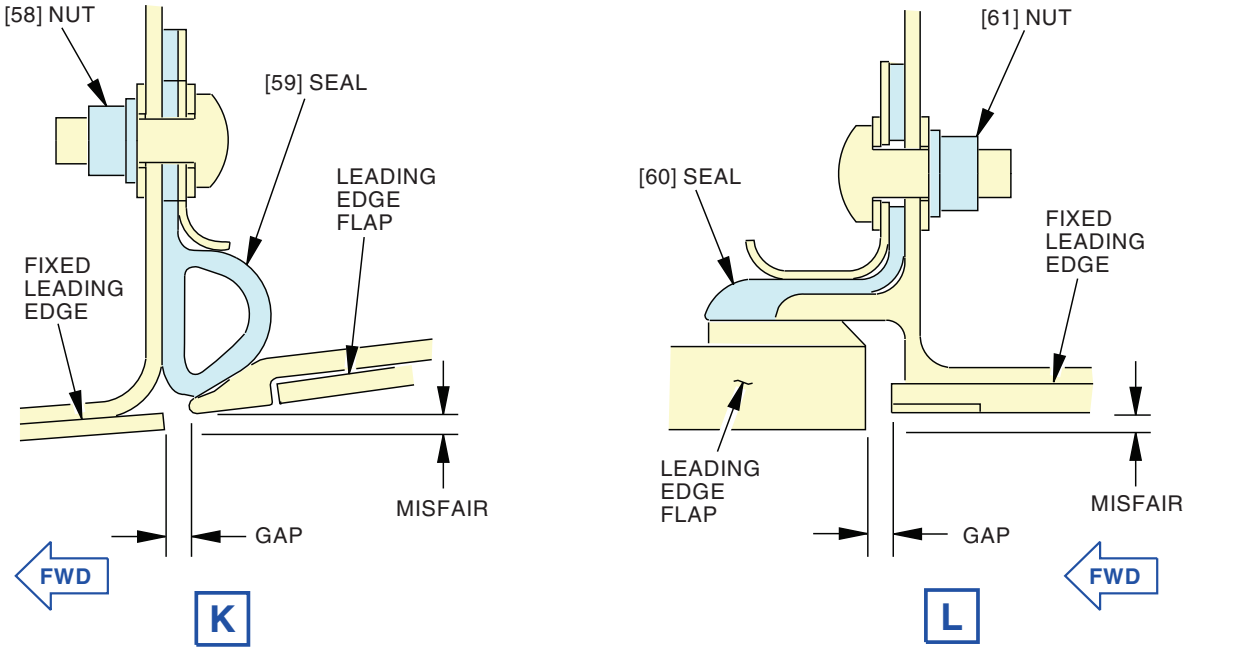
Leading Edge Flap Installation
Figure 401/27-81-11-990-801 (Sheet 8 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



OUTBOARD BULLNOSE LINKAGE
(MID AND INBOARD BULLNOSE LINKAGES AND EQUIVALENT)
(FLAP IS SHOWN RETRACTED)

A-A



2413344 S00061532377_V2

Leading Edge Flap Installation
Figure 401/27-81-11-990-801 (Sheet 9 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-81-11-400-801

3. Leading Edge Flap Installation

(Figure 401, Figure 402)

A. References

Reference	Title
27-81-00-080-801	Leading Edge Flap and Slat Locks Removal (P/B 201)
27-81-11-200-801	Leading Edge Flap Inspection (P/B 601)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
51-22-00-160-801	Prepare For Sealing (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659
COM-1691	Protractor - Digital, Vertical or Horizontal Part #: 21465A82 Supplier: 3A054 Part #: 31-038-3 (Model Pro 360) Supplier: 1X822 Part #: 31-040-9 (Model Pro 3600) Supplier: 1X822 Part #: 950-317 (Model Pro 360) Supplier: 01EP7 Part #: 950-318 (Model Pro 3600) Supplier: 01EP7 Part #: KS5549 Supplier: 75245 Part #: KS6005 Supplier: 75245
SPL-1734	Lockout Set - Leading Edge Slats and Krueger Flaps Actuators Part #: C27051-29 Supplier: 81205

C. Consumable Materials

Reference	Description	Specification
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95
A00767	Sealant - Fuel Tank	BMS5-45
A50231	Sealant - Pressure And Environmental - Chromate Type	BMS5-95 Class B
D00633	Grease - Aircraft General Purpose	BMS3-33
G01912	Lockwire - MS20995NC32, Monel - 0.032 Inch (0.8128 mm) Diameter	NASM20995

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Leading edge flap	27-81-11-01-185	PTF ALL

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(Continued)

AMM Item	Description	AIPC Reference	AIPC Effectivity
2	Leading edge flap	27-81-11-01-195	PTF ALL
6	Pin	27-62-94-01-125	PTF ALL
15	Pin	27-81-11-01-135	PTF ALL
21	Pin	27-81-11-01-020	PTF ALL
34	Pin	27-81-11-01-025	PTF ALL
40	Pin	27-81-11-01-020	PTF ALL
55	Pin	27-81-11-01-085	PTF ALL

E. Location Zones

Zone	Area
512	Left Wing - Krueger Flap No. 2
513	Left Wing - Krueger Flap No. 1
612	Right Wing - Krueger Flap No. 3
613	Right Wing - Krueger Flap No. 4

F. Leading Edge Flap Installation

SUBTASK 27-81-11-220-001

(1) If you install used bearings, bushings, and bolts, do this task: Leading Edge Flap Inspection, TASK 27-81-11-200-801.

SUBTASK 27-81-11-420-001

(2) If you removed the outboard leading edge flap [1], connect the outboard leading edge flap [1] to the fixed leading edge:

(a) Connect hinges 1 through 4 on the leading edge flap [1] to the fixed leading edge.

1) Apply grease, D00633, on the mating surfaces of the bolts [23] and bolts [27].

2) At hinges 1 and 4, install the bolts [23], washer [24], bushings [19], washers [20], and nuts [22].

a) Install the bolts [23] with the head outboard.

3) Tighten the nuts [22] to 60 in-lb (6.8 N·m) - 95 in-lb (10.7 N·m).

4) Install the new pins [21].

5) At hinges 2 and 3, install the bolts [27], washers [28], bushings [26], washers [25], and nuts [33].

a) At hinge 2, install the bolt [27] with the head inboard.

b) At hinge 3, install the bolt [27] with the head outboard.

6) Tighten the nuts [33] to 290 in-lb (32.8 N·m) - 510 in-lb (57.6 N·m).

7) Install the new pins [34].

(b) Connect the bonding jumper [29] to the fixed leading edge.

1) Install the bolt [30], washers [31], and nut [32] to attach the bonding jumper [29].

2) Tighten the nut [32] to 28 in-lb (3 N·m) - 35 in-lb (4 N·m).

3) Measure the electrical bonding resistance between the bonding jumper [29] and the fixed leading edge.

a) Use an intrinsically safe approved bonding meter, COM-1550.

b) Make sure that the resistance is less than 0.001 ohms.

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4) Do these steps to make a fay seal (SWPM 20-20-00):
   a) Clean the area before you apply the sealant (TASK 51-22-00-160-801).
   b) Apply sealant, A00767, or sealant, A50231, around the bolt [30], washers [31], nut [32] and end of the bonding jumper [29].

(c) Connect the outboard bullnose linkage [53] to the fixed leading edge.
   1) Install the bolt [42], washer [43], bushing [44], anti-rotation washers [41], washer [56], and nut [54].
      a) Install the bolt [42] with the head inboard.
   2) Tighten the nut [54] to 30 in-lb (3.4 N·m) - 50 in-lb (5.6 N·m).
   3) Install the new pin [55].

SUBTASK 27-81-11-420-002

(3) If you removed the inboard leading edge flap [2], connect the inboard leading edge flap [2] to the fixed leading edge:
   (a) Connect hinges 5 through 8 on the leading edge flap [2] to the fixed leading edge.
      1) Apply grease, D00633, on the mating surfaces of the bolts [23] and bolts [27].
      2) At hinge 5, install the bolt [23], washer [24], bushing [19], washer [20], and nut [22].
         a) Install the bolt [23] with the head outboard.
         b) Tighten the nut [22] to 60 in-lb (6.8 N·m) - 95 in-lb (10.7 N·m).
         c) Install the new pin [21].
      3) At hinges 6 and 7, install the bolts [27], washers [28], bushings [26], washers [25], and nuts [33].
         a) At hinge 6, install the bolt [27] with the head inboard.
         b) At hinge 7, install the bolt [27] with the head outboard.
         c) Tighten the nuts [33] to 290 in-lb (32.8 N·m) - 510 in-lb (57.6 N·m).
         d) Install the new pins [34].
      4) At hinge 8, install the bolt [37], washer [36], bushing [38], washer [35], and nut [39].
         a) Install the bolt [37] with the head inboard.
         b) Tighten the nut [39] to 60 in-lb (6.8 N·m) - 95 in-lb (10.7 N·m).
         c) Install the new pin [40].
   (b) Connect the bonding jumper [29] to the fixed leading edge.
      1) Install the bolt [30], washers [31], and nut [32] to attach the bonding jumper [29].
      2) Tighten the nut [32] to 28 in-lb (3 N·m) - 35 in-lb (4 N·m).
      3) Measure the electrical bonding resistance between the bonding jumper [29] and the fixed leading edge.
         a) Use an intrinsically safe approved bonding meter, COM-1550.
         b) Make sure that the resistance is less than 0.001 ohms.
      4) Do these steps to make a fay seal (SWPM 20-20-00):
         a) Clean the area before you apply the sealant (TASK 51-22-00-160-801).
         b) Apply sealant, A00767, or sealant, A50231, around the bolt [30], washers [31], nut [32] and end of the bonding jumper [29].
   (c) Connect the mid and inboard bullnose linkages [53] to the fixed leading edge.

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1) Install the bolt [42], washer [43], bushing [44], anti-rotation washers [41], washer [56], and nut [54].
   a) Install the bolt [42] with the heads outboard.
2) Tighten the nut [54] to 30 in-lb (3.4 N·m) - 50 in-lb (5.6 N·m).
3) Install the new pin [55].

SUBTASK 27-81-11-210-001

WARNING KEEP PERSONNEL AND EQUIPMENT AWAY FROM THE FLAPS AND SLATS. THE FLAPS AND SLATS CAN ACCIDENTALLY MOVE AND CAUSE INJURIES TO PERSONNEL OR DAMAGE TO EQUIPMENT.

WARNING MOVE ALL PERSONS AND EQUIPMENT AWAY FROM THE TRAILING EDGE FLAPS. THE FLAPS WILL MOVE AND CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(4) Do a check of the clearances between the leading edge flap [1] or leading edge flap [2] and the fixed leading edge:
   (a) Temporarily install the bolt [9] or the bolt [14].
      1) Make sure that the flap bullnose can move freely.
   (b) Adjust the flap rod [57] until the flap bullnose touches the leading edge flap [1] or leading edge flap [2].
   (c) Remove the bolt [9] or the bolt [14] from the rod end [3].
   (d) To pressurize hydraulic system B, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.
   (e) Move the flap control lever to the UP detent to retract the leading edge flaps.
      1) Make sure that the flap actuator is retracted.
   (f) To remove power from hydraulic system B, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.
   (g) Do this check while you retract the leading edge flap [1] or leading edge flap [2] manually:
      1) Make sure that there is sufficient clearance between the leading edge flap [2] or leading edge flap [1] and the fixed leading edge.

SUBTASK 27-81-11-860-017

(5) Manually extend the leading edge flap [1] or leading edge flap [2].

SUBTASK 27-81-11-860-012

WARNING KEEP PERSONNEL AND EQUIPMENT AWAY FROM THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(6) To pressurize hydraulic system B, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

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SUBTASK 27-81-11-860-015

WARNING KEEP PERSONNEL AND EQUIPMENT AWAY FROM THE FLAPS AND SLATS. THE FLAPS AND SLATS CAN ACCIDENTALLY MOVE AND CAUSE INJURIES TO PERSONNEL OR DAMAGE TO EQUIPMENT.

(7) Move the flap control lever to the 1-unit detent to extend the leading edge flaps.

SUBTASK 27-81-11-860-016

(8) Make sure that the flap actuator [4] is extended.

SUBTASK 27-81-11-860-013

(9) To remove power from hydraulic system B, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 27-81-11-420-003

(10) Connect the flap actuator [4] to the leading edge flap [1] or leading edge flap [2].

(a) Apply grease, D00633, on the mating surfaces of the bolt [14] or bolt [9].

(b) For the outboard leading edge flap [1], install the bolt [9], washer [10], bushing [11], washer [5], washer [8], and nut [7].

NOTE: You can bond the washer [5] to the leading edge flap [1] with sealant, A00247, for easier installation.

(c) For the inboard leading edge flap [2], install the bolt [14], washer [13], bushing [12], washer [18], washer [17], and nut [16].

NOTE: You can bond the washer [18] to the leading edge flap [2] with sealant, A00247, for easier installation.

(d) Tighten the nut [7] or nut [16] to 660 in-lb (74.6 N·m) - 980 in-lb (110.7 N·m).

(e) Install the new pin [6] or new pin [15].

SUBTASK 27-81-11-860-004

WARNING KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(11) Pressurize hydraulic system B (TASK 29-11-00-860-801).

SUBTASK 27-81-11-860-014

WARNING KEEP PERSONNEL AND EQUIPMENT AWAY FROM THE FLAPS AND SLATS. THE FLAPS AND SLATS CAN ACCIDENTALLY MOVE AND CAUSE INJURIES TO PERSONNEL OR DAMAGE TO EQUIPMENT.

(12) Move the flap control lever to the UP detent to retract the leading edge flaps.

SUBTASK 27-81-11-220-002

(13) Measure the full extend angle of the leading edge flap [1] or leading edge flap [2].

(a) Attach a protractor, COM-1691, to the inboard edge of the outboard leading edge flap [1] or the outboard edge of the inboard leading edge flap [2].

(b) Set the protractor, COM-1691, to zero with the leading edge flaps retracted.

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WARNING KEEP PERSONNEL AND EQUIPMENT AWAY FROM THE FLAPS AND SLATS. THE FLAPS AND SLATS CAN ACCIDENTALLY MOVE AND CAUSE INJURIES TO PERSONNEL OR DAMAGE TO EQUIPMENT.

(c) Move the flap control lever to the 1-unit detent to extend the leading edge flaps.

(d) Make sure that the full extend angle of the leading edge flap [1] or leading edge flap [2] is 98 ±0.5 degrees.

1) If the extend angle is out of tolerance, adjust the length of the flap actuator [4].

a) Loosen the jamnut [45] on the rod end [3].

b) Turn the rod end [3] until you get the correct full extend angle.

NOTE: The limit for maximum adjustment is two full turns.

c) Tighten the jamnut [45] to 550 in-lb (62.1 N·m) - 650 in-lb (73.4 N·m).

d) Install MS20995NC32 lockwire, G01912, on the jamnut [45].

(e) Remove the protractor, COM-1691.

SUBTASK 27-81-11-220-003

(14) Make sure that the mismatch between the inboard leading edge flap [2] and the outboard leading edge flap [1] is 0.000 ±0.040 in. (0.0 ±1.0 mm).

SUBTASK 27-81-11-860-006

(15) Apply preload at the flap bullnose to the leading edge flap [1] or leading edge flap [2].

(a) Make sure the flap bullnose touches the leading edge flap [1] or leading edge flap [2].

NOTE: To apply the correct preload, the flap bullnose must first touch the leading edge flap [1] or leading edge flap [2].

1) If the flap bullnose does not touch the leading edge flap [1] or leading edge flap [2], adjust the flap rod [57].

(b) Adjust the flap rod [57] an additional 1/2 turn to apply the correct preload at the flap bullnose to the leading edge flap [1] or leading edge flap [2].

(c) Install MS20995NC32 lockwire, G01912, on the flap rod [57].

SUBTASK 27-81-11-860-007

WARNING KEEP PERSONNEL AND EQUIPMENT AWAY FROM THE FLAPS AND SLATS. THE FLAPS AND SLATS CAN ACCIDENTALLY MOVE AND CAUSE INJURIES TO PERSONNEL OR DAMAGE TO EQUIPMENT.

(16) Move the flap control lever to the UP detent to retract the leading edge flaps.

SUBTASK 27-81-11-220-004

(17) Do a check of the gap at the edge of the leading edge flap [1] or leading edge flap [2].

(a) Do a check of the gap at location 1-4, 6-10, and 13-18 (Figure 402).

1) If all the measurements taken at locations 1-4, 6-10, 13-18 are less than 0.240 in. (6.1 mm), then the gap is satisfactory.

2) If one or more of the gap measurements exceeds 0.240 in. (6.1 mm), convert each of the measurements into a Net Effect Value (NEV) (Figure 402).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

3) Add the NEV measurements together and divide by the total number of measurements.

NOTE: This gives the total NEV.

NOTE: For 85% of the Krueger flap trailing edge length, the gap cannot exceed 0.300 in. (7.620 mm). For the remaining 15% of the Krueger flap trailing edge interface, the gap cannot exceed 0.450 in. (11.430 mm).

4) Make sure that the calculated total NEV is less than 1.0 (the Net Effect Limit (NEL) for this measurement).

SUBTASK 27-81-11-220-005

(18) Do a check of the mismatch at the forward edge of the leading edge flap [1] and the leading edge flap [2] (Figure 401).

(a) Do a check of the mismatch at the forward edge of the leading edge flap [1] and leading edge flap [2] at locations 1-4 and 6-10 (Figure 402).

1) If all the measurements taken at locations 1-4 and 6-10 are within the range 0.000 +0.062 / -0.071 in. (0.00 +1.57 / -1.80 mm), the misfair at the forward edge of the Krueger flap is acceptable as is without any additional adjustments.

NOTE: See Figure 402 for definition of negative and positive misfair measurements.

2) If one or more of the misfair measurements is outside the range 0.000 +0.062 / -0.071 in. (0.00 +1.57 / -1.80 mm), convert each of the measurements into a NEV (Figure 402).

3) Add the NEV measurements together and divide by the total number of measurements.

NOTE: This gives the total NEV.

NOTE: For 75% of the Krueger flap leading edge length, the fair cannot exceed ±0.100 in. (2.540 mm). For the remaining 25% of the Krueger flap leading edge interface, the fair cannot exceed ±0.130 in. (3.302 mm).

4) Make sure that the calculated total NEV is less than 1.0 (the NEL for this measurement).

5) If it is necessary, adjust the mismatch dimension.

a) Loosen the jamnut [45] on the rod end [3].

b) Turn the rod end [3] to adjust the length of the flap actuator [4] until you get the correct mismatch.

NOTE: The limit for maximum adjustment is two full turns.

c) Tighten the jamnut [45] to 550 in-lb (62.1 N·m) - 650 in-lb (73.4 N·m).

d) Install MS20995NC32 lockwire, G01912, on the jamnut [45].

SUBTASK 27-81-11-220-006

(19) Do a check of the mismatch at the aft edge of the leading edge flap [1] and leading edge flap [2] (Figure 401).

(a) Do a check of the mismatch at the aft edge of the leading edge flap [1] and leading edge flap [2] at locations 13-18 (Figure 402).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) If all the measurements taken at locations 13-18 are within the range 0.000 +0.040 / -0.072 in. (0.00 +1.02 / -1.83 mm), the misfair at the aft edge of the Krueger flap is acceptable as is without any additional adjustments.

   NOTE: See Figure 402 for definition of negative and positive misfair measurements.

2) If one or more of the misfair measurements is outside the range 0.000 +0.040 / -0.072 in. (0.00 +1.02 / -1.83 mm), convert each of the measurements into a NEV (Figure 402).

3) Add the NEV measurements together and divide by the total number of measurements.

   NOTE: This gives the total NEV.

   NOTE: For 70% of the Krueger flap trailing edge length, the fair cannot exceed ±0.080 in. (2.032 mm). For 25% of the Krueger flap trailing edge length, the fair cannot exceed ±0.100 in. (2.540 mm). For the remaining 5% of the Krueger flap trailing edge interface, the fair cannot exceed ±0.160 in. (4.064 mm).

4) Make sure that the calculated total NEV is less than 1.0 (the NEL for this measurement).

5) If the mismatch dimension is out of tolerance for the outboard leading edge flap [1], do these steps:

   a) Remove the nut [7], washer [8], washer [5], bushing [11], washer [10], and bolt [9] to disconnect the flap actuator [4] from the leading edge flap [1].

   b) Loosen the jamnut [45] on the rod end [3].

   c) Turn the rod end [3] to adjust the length of the flap actuator [4] until you get the specified mismatch dimension.

      NOTE: The limit for maximum adjustment is two full turns. The full extend angle of the leading edge flap [1] must still be 98 ±0.5 degrees while you make this adjustment.

   d) Tighten the jamnut [45] to 550 in-lb (62.1 N·m) - 650 in-lb (73.4 N·m).

   e) Install the MS20995NC32 lockwire, G01912, on the jamnut [45].

   f) Connect the flap actuator [4] to the leading edge flap [1] with bolt [9], washer [10], bushing [11], washer [5], washer [8], and nut [7].

6) If the mismatch dimension is out of tolerance for the inboard leading edge flap [2], do these steps:

   a) Remove the nut [16], washer [17], washer [18], bushing [12], washer [13], and bolt [14] to disconnect the flap actuator [4] from the leading edge flap [2].

   b) Loosen the jamnut [45] on the rod end [3].

   c) Turn the rod end [3] to adjust the length of the flap actuator [4] until you get the specified mismatch dimension.

      NOTE: The limit for maximum adjustment is two full turns. The full extend angle of the leading edge flap [2] must still be 98 ±0.5 degrees while you make this adjustment.

   d) Tighten the jamnut [45] to 550 in-lb (62.1 N·m) - 650 in-lb (73.4 N·m).

   e) Install the MS20995NC32 lockwire, G01912, on the jamnut [45].

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f) Connect the flap actuator [4] to the leading edge flap [2] with bolt [14], washer [13], bushing [12], washer [18], washer [17], and nut [16].

SUBTASK 27-81-11-420-004

(20) Tighten the nut [7] or nut [16] to 660 in-lb (74.6 N·m) - 980 in-lb (110.7 N·m).

SUBTASK 27-81-11-420-005

(21) Install the new pin [6] or new pin [15].

SUBTASK 27-81-11-860-008

(22) Move the flap control lever to the 1-unit detent position to extend the leading edge flaps.

G. Leading Edge Flap Installation Test

SUBTASK 27-81-11-080-001

WARNING REMOVE THE LOCK FROM LEADING EDGE FLAP ACTUATOR CAREFULLY TO PREVENT ACCIDENTAL OPERATION OF THE LEADING EDGE FLAPS. THE LEADING EDGE FLAPS CAN MOVE QUICKLY. THIS CAN CAUSE INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT.

(1) Remove the lockout set, SPL-1734, from the actuator of the leading edge flap (TASK 27-81-00-080-801).

SUBTASK 27-81-11-860-009

WARNING KEEP PERSONNEL AND EQUIPMENT AWAY FROM THE FLAPS AND SLATS. THE FLAPS AND SLATS CAN ACCIDENTALLY MOVE AND CAUSE INJURIES TO PERSONNEL OR DAMAGE TO EQUIPMENT.

(2) Move the flap control lever to the FLAP UP detent to retract the leading edge flaps.

SUBTASK 27-81-11-860-010

(3) Do a check of the flap panel clearances.

(a) For all but the aft 5 in. (127 mm) of the flap, make sure the gap between the inboard leading edge flap [2] and the outboard leading edge flap [1] is 0.060 in. (1.52 mm) - 0.150 in. (3.81 mm).

(b) For the aft 5 in. (127 mm) of the flap, make sure the gap between the inboard leading edge flap [2] and the outboard leading edge flap [1] is 0.060 in. (1.52 mm) - 0.250 in. (6.35 mm).

SUBTASK 27-81-11-210-002

(4) If the flap panel clearances are out of tolerance, do a check of the wing and flap panel for damage or warping.

NOTE: This check is to make sure that the flap and the wing align correctly. There is no adjustment that you can do.

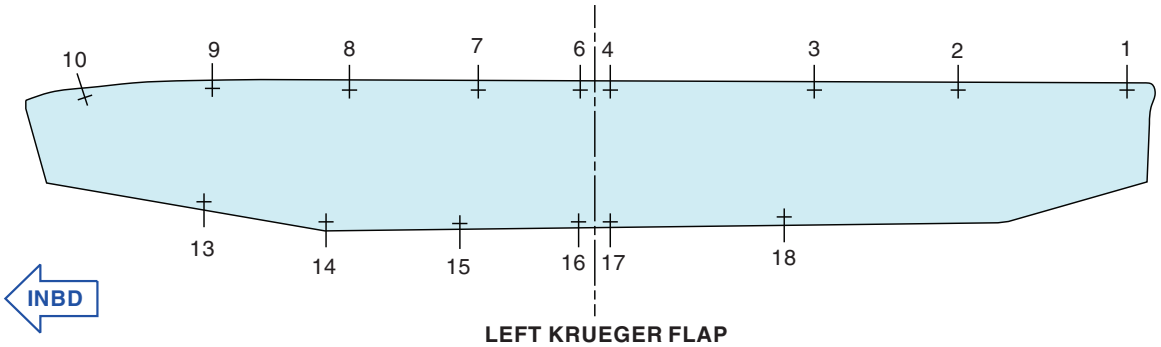
H. Put the Airplane Back to Its Usual Condition

SUBTASK 27-81-11-860-011

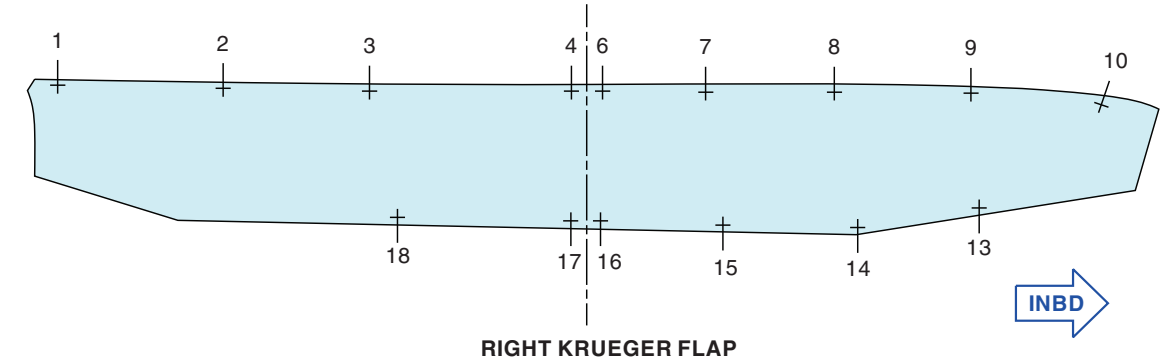
(1) Remove power from hydraulic system B (TASK 29-11-00-860-805).

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LEFT KRUEGER FLAP
(VIEW IN THE AFT DIRECTION)



RIGHT KRUEGER FLAP
(VIEW IN THE AFT DIRECTION)

POINT	DISTANCE FROM K-FLAP CENTERLINE, INCH (mm) [1]
1	56.59 (1437.38)
2	32.77 (832.35)
3	20.78 (527.81)
4	1.31 (33.27)
6	-1.06 (-26.92)
7	-19.98 (-507.49)
8	-33.89 (-860.80)
9	-51.17 (-1299.71)
10	-63.04 (-1601.21)

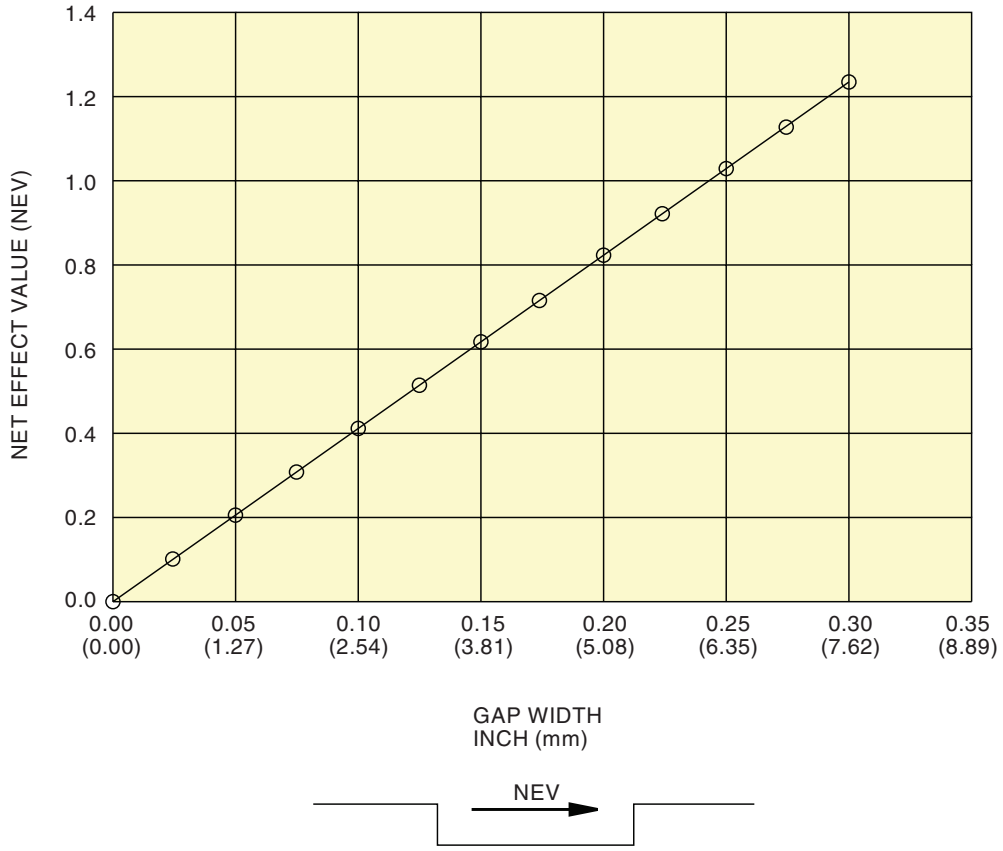
POINT	DISTANCE FROM K-FLAP CENTERLINE, INCH (mm) [1]
13	-52.02 (-1321.30)
14	-37.76 (-959.10)
15	-23.76 (-603.50)
16	-1.06 (-26.92)
17	1.18 (29.97)
18	16.66 (423.16)

[1] OUTBD DISTANCE ARE POSITIVE (+), INBOARD DISTANCES ARE NEGATIVE (-)

2413345 S00061532379_V2

Leading Edge Gap/Misfair Measurement
Figure 402/27-81-11-990-802 (Sheet 1 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NET EFFECT VALUE CHART
KRUEGER FLAP - TRAILING EDGE GAPS

NOTE:

THE NET EFFECT LIMIT (N.E.L.) FOR THIS ITEM IS 1.0

IF ALL MEASURED GAP VALUES FOR THIS ITEM ARE LESS THAN 0.240 INCH (6.09 mm)
THEN PASSAGE OF N.E.L. IS ASSURED AND AN ACTUAL N.E.L. CALCULATION IS NOT REQUIRED.

2413346 S00061532380_V3

Leading Edge Gap/Misfair Measurement
Figure 402/27-81-11-990-802 (Sheet 2 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

GAP (IN)	GAP (mm)	NET EFFECT VALUE (NEV)
0.000	0.000	0.000
0.025	0.635	0.105
0.050	1.270	0.209
0.075	1.905	0.314
0.100	2.540	0.418
0.125	3.175	0.523
0.150	3.810	0.621
0.175	4.445	0.725
0.200	5.080	0.830
0.225	5.715	0.935
0.250	6.350	1.039
0.275	6.985	1.144
0.300	7.620	1.248

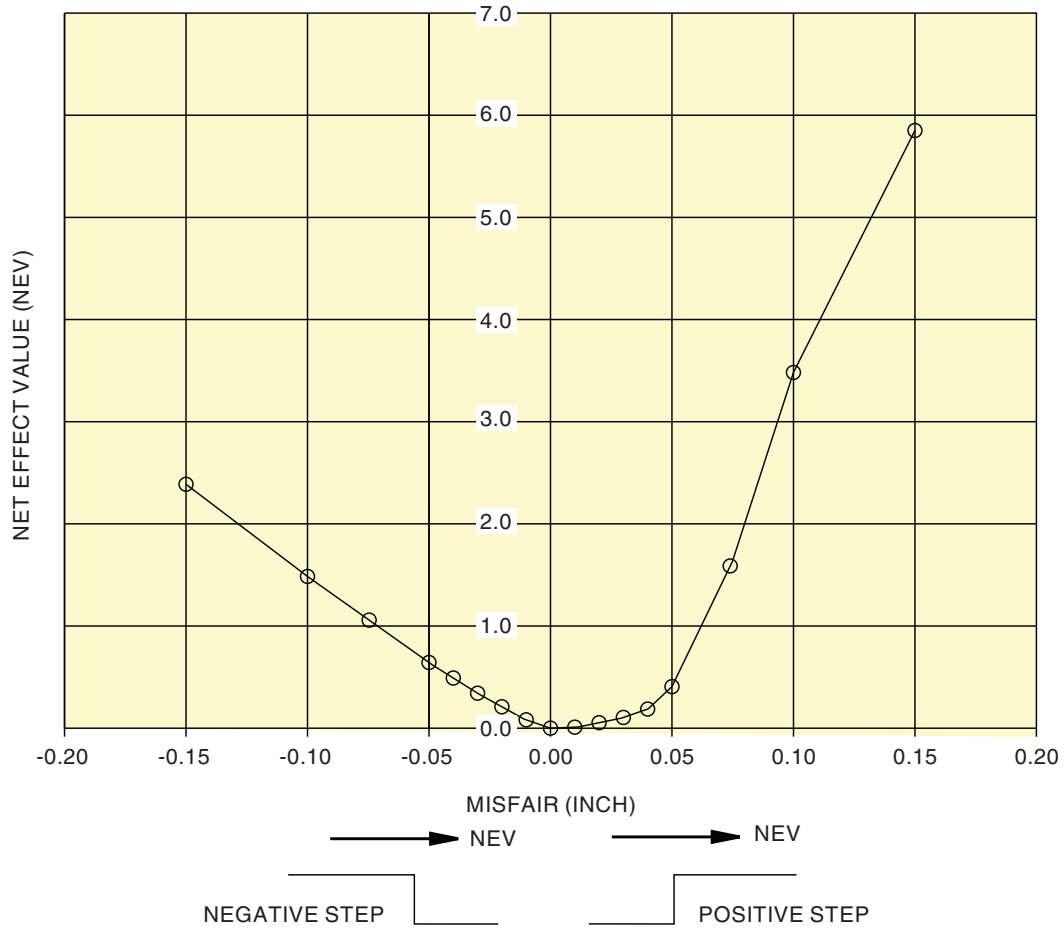
NET EFFECT VALUE TABLE
KRUEGER FLAP - TRAILING EDGE GAPS

NOTE:
N.E.V. OF 1.0 CORRESPONDS TO GAP OF 0.240 INCH (6.09 mm)

2413347 S00061532381_V2

Leading Edge Gap/Misfair Measurement
Figure 402/27-81-11-990-802 (Sheet 3 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NET EFFECT VALUE CHART
KRUEGER FLAP - LEADING EDGE MISFAIR ONLY

NOTE:

THE NET EFFECT LIMIT (N.E.L.) FOR THIS ITEM IS 1.0

IF ALL MEASURED MISFAIR VALUES FOR THIS ITEM ARE BETWEEN -0.07 INCH (-1.80 mm) AND + 0.062 INCH (+1.57 mm) THEN PASSAGE OF N.E.L. IS ASSURED AND AN ACTUAL N.E.L. CALCULATION IS NOT REQUIRED.

2413348 S00061532382_V3

Leading Edge Gap/Misfair Measurement
Figure 402/27-81-11-990-802 (Sheet 4 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

STEP HEIGHT (IN)	STEP HEIGHT (mm)	NET EFFECT VALUE (NEV)
-0.150	-3.810	2.401
-0.100	-2.540	1.491
-0.075	-1.905	1.066
-0.050	-1.270	0.653
-0.040	-1.016	0.497
-0.030	-0.762	0.347
-0.020	-0.508	0.216
-0.010	-0.254	0.090
0.000	0.000	0.000
0.010	0.254	0.012
0.020	0.508	0.054
0.030	0.762	0.108
0.040	1.016	0.192
0.050	1.270	0.407
0.075	1.905	1.641
0.100	2.540	3.491
0.150	3.810	5.844

NET EFFECT VALUE TABLE
KRUEGER FLAP - LEADING EDGE MISFAIR ONLY

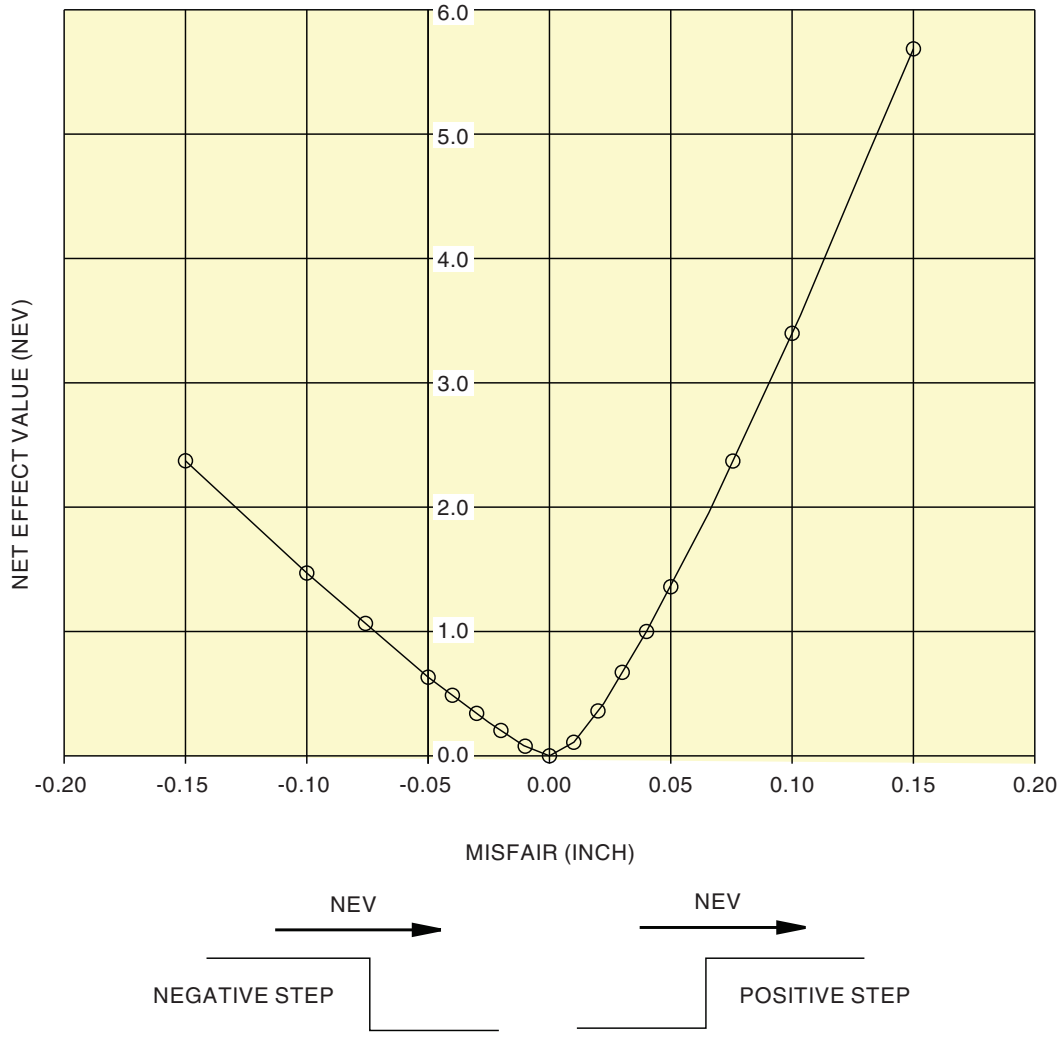
NOTE:

A NEGATIVE STEP HEIGHT INDICATES AN AFT FACING STEP.
N.E.V. OF 1.0 CORRESPONDS TO MISFAIR OF -0.071 INCH (-1.80 mm)
AND +0.062 INCH (+1.57 mm)

2413349 S00061532383_V2

Leading Edge Gap/Misfair Measurement
Figure 402/27-81-11-990-802 (Sheet 5 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NET EFFECT VALUE CHART
KRUEGER FLAP - ALL EDGES EXCEPT LEADING (FORWARD) EDGE MISFAIR

NOTE:

THE NET EFFECT LIMIT (N.E.L.) FOR THIS ITEM IS 1.0

IF ALL MEASURED STEP VALUES FOR THIS ITEM ARE BETWEEN -0.072 INCH (-1.83 mm) AND +0.040 INCH (+1.02 mm) THEN PASSAGE OF N.E.L. IS ASSURED AND AN ACTUAL N.E.L. CALCULATION IS NOT REQUIRED.

2413350 S00061532384_V3

Leading Edge Gap/Misfair Measurement
Figure 402/27-81-11-990-802 (Sheet 6 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

STEP HEIGHT (IN)	STEP HEIGHT (mm)	NET EFFECT VALUE (NEV)
-0.150	-3.810	2.387
-0.100	-2.540	1.482
-0.075	-1.905	1.051
-0.050	-1.270	0.646
-0.040	-1.016	0.493
-0.030	-0.762	0.347
-0.020	-0.508	0.208
-0.010	-0.254	0.088
0.000	0.000	0.000
0.010	0.254	0.113
0.020	0.508	0.365
0.030	0.762	0.668
0.040	1.016	1.000
0.050	1.270	1.361
0.075	1.905	2.336
0.100	2.540	3.394
0.150	3.810	5.686

NET EFFECT VALUE TABLE
KRUEGER FLAP - ALL EDGES EXCEPT LEADING (FORWARD) EDGE MISFAIR

NOTE:

A NEGATIVE STEP HEIGHT INDICATES AN AFT FACING STEP.
N.E.V. OF 1.0 CORRESPONDS TO MISFAIR OF -0.072 INCH (-1.83 mm) AND +0.040 INCH (+1.02 mm)

2413351 S00061532385_V2

Leading Edge Gap/Misfair Measurement
Figure 402/27-81-11-990-802 (Sheet 7 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

LEADING EDGE FLAP - INSPECTION/CHECK

1. General

A. This procedure contains the task to do an inspection of the leading edge flap.

TASK 27-81-11-200-801

2. Leading Edge Flap Inspection

(Figure 601, Table 601)

A. General

(1) This task gives wear limits and repair data for the leading edge flap inspection.

B. References

Reference	Title
27-81-11-000-801	Leading Edge Flap Removal (P/B 401)
27-81-11-400-801	Leading Edge Flap Installation (P/B 401)

C. Tools/Equipment

Reference	Description
STD-1096	Micrometer - Depth, 0-1 inch, Readable to 1/1000 inch
STD-1097	Caliper - Vernier, 0-6 inch, Readable to 1/1000 inch

D. Location Zones

Zone	Area
512	Left Wing - Krueger Flap No. 2
513	Left Wing - Krueger Flap No. 1
612	Right Wing - Krueger Flap No. 3
613	Right Wing - Krueger Flap No. 4

E. Leading Edge Flap Inspection

SUBTASK 27-81-11-020-004

(1) Do this task: Leading Edge Flap Removal, TASK 27-81-11-000-801.

SUBTASK 27-81-11-020-005

(2) Use these tools to do a check for wear on the parts of the leading edge flap:

(a) micrometer (0-1 inch, readable to 1/1000 inch), STD-1096

(b) vernier caliper (0 - 6 inch, readable to 1/1000 inch), STD-1097.

SUBTASK 27-81-11-020-006

(3) Examine the parts of the leading edge flap for worn areas:

NOTE: The Nomex cloth common to the leading edge (krueger) flap seals can be frayed provided there is no damage to the silicon rubber portion of the seal. If the silicon rubber portion is damaged then the seal should be repaired or replaced. Contact Boeing for repair instructions for the seal.

NOTE: Damage to the wear pads is acceptable for continued service, however they must be replaced before through-thickness wear.

(a) Remove the parts you want to do a check.

(b) Measure the parts with a micrometer (0-1 inch, readable to 1/1000 inch), STD-1096, or a vernier caliper (0 - 6 inch, readable to 1/1000 inch), STD-1097.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Compare the dimensions you measured with the permitted wear dimensions in Table 601.

(d) If the parts are out of tolerance, replace or repair the parts.

Table 601/27-81-11-993-801 Diameter Design and Wear Limits

NAME OF THE PARTS THAT ARE IN CONTACT	DIM.	DIAMETER DESIGN LIMITS		WEAR LIMITS	
		MINIMUM inches/mm	MAXIMUM inches/mm	PERMITTED MAXIMUM I.D./ MINIMUM O.D. OF WORN PART	MAXIMUM SERVICE DIAMETRIC CLEARANCE
#1 BUSHING	I.D.	0.3125 in 7.938 mm	0.3130 in 7.950 mm	0.3150 in 8.000 mm	0.0030 in 0.076 mm
#1 BOLT	O.D.	0.3110 in 7.899 mm	0.3120 in 7.924 mm	0.3100 in 7.874 mm	
#2 BUSHING	I.D.	0.3125 in 7.938 mm	0.3136 in 7.965 mm	0.3159 in 8.023 mm	0.0039 in 0.099 mm
#2 BOLT	O.D.	0.3110 in 7.899 mm	0.3120 in 7.924 mm	0.3097 in 7.866 mm	
#3 BEARING	I.D.	0.4375 in 11.113 mm	0.4385 in 11.137 mm	0.4400 in 11.175 mm	0.0030 in 0.076 mm
#3 BUSHING	O.D.	0.4365 in 11.087 mm	0.4370 in 11.099 mm	0.4350 in 11.061 mm	
#4 BUSHING	I.D.	0.4375 in 11.113 mm	0.4386 in 11.140 mm	0.4402 in 11.180 mm	0.0032 in 0.081 mm
#4 BUSHING	O.D.	0.4365 in 11.087 mm	0.4370 in 11.099 mm	0.4354 in 11.059 mm	
#5 BUSHING	I.D.	0.5000 in 12.700 mm	0.5007 in 12.717 mm	0.5004 in 12.710 mm	0.0033 in 0.083 mm
#5 BOLT	O.D.	0.4985 in 12.662 mm	0.4995 in 12.687 mm	0.4974 in 12.634 mm	
#6 BUSHING	I.D.	0.6870 in 17.450 mm	0.6877 in 17.467 mm	0.6891 in 17.503 mm	0.0026 in 0.066 mm
#6 BUSHING	O.D.	0.6860 in 17.424 mm	0.6865 in 17.437 mm	0.6851 in 17.401 mm	
#7 BUSHING	I.D.	0.5000 in 12.700 mm	0.5005 in 12.712 mm	0.5025 in 12.763 mm	0.0030 in

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Table 601/27-81-11-993-801 Diameter Design and Wear Limits (Continued)

NAME OF THE PARTS THAT ARE IN CONTACT	DIM.	DIAMETER DESIGN LIMITS		WEAR LIMITS	
		MINIMUM inches/mm	MAXIMUM inches/mm	PERMITTED MAXIMUM I.D./ MINIMUM O.D. OF WORN PART	MAXIMUM SERVICE DIAMETRIC CLEARANCE
#7 BOLT	O.D.	0.4985 in 12.662 mm	0.4995 in 12.687 mm	0.4975 in 12.636 mm	0.076 mm
#8 BEARING	I.D.	0.5000 in 12.700 mm	0.5005 in 12.712 mm	0.5025 in 12.763 mm	0.0030 in
#8 BOLT	O.D.	0.4985 in 12.662 mm	0.4995 in 12.687 mm	0.4975 in 12.636 mm	0.076 mm
#9 BUSHING	I.D.	0.3125 in 7.938 mm	0.3136 in 7.965 mm	0.3159 in 8.023 mm	0.0039 in
#9 BOLT	O.D.	0.3110 in 7.899 mm	0.3120 in 7.924 mm	0.3097 in 7.866 mm	0.099 mm
#10 BUSHING	I.D.	0.3125 in 7.938 mm	0.3130 in 7.950 mm	0.3150 in 8.000 mm	0.0030 in
#10 BOLT	O.D.	0.3110 in 7.899	0.3120 in 7.924 mm	0.3100 in 7.874 mm	0.076 mm
#11 BUSHING	I.D.	0.4375 in 11.113 mm	0.4386 in 11.140 mm	0.4402 in 11.180 mm	0.0032 in
#11 BUSHING	O.D.	0.4365 in 11.087 mm	0.4370 in 11.099 mm	0.4354 in 11.059 mm	0.081 mm
#12 BEARING	I.D.	0.4375 in 1.113 mm	0.4385 in 11.137 mm	0.4400 in 11.175 mm	0.0030 in
#12 BUSHING	O.D.	0.4365 in 11.087 mm	0.4370 in 11.099 mm	0.4355 in 11.061 mm	0.076 mm
#13 BUSHING	I.D.	0.2500 in 6.350 mm	0.2505 in 6.362 mm	0.2525 in 6.340 mm	0.0030 in
#13 BOLT	O.D.	0.2485 in 6.312 mm	0.2495 in 6.337 mm	0.2475 in 6.286 mm	0.076 mm
#14 BUSHING	I.D.	0.2500 in 6.350 mm	0.2505 in 6.362 mm	0.2525 in 6.413 mm	0.0030 in

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Table 601/27-81-11-993-801 Diameter Design and Wear Limits (Continued)

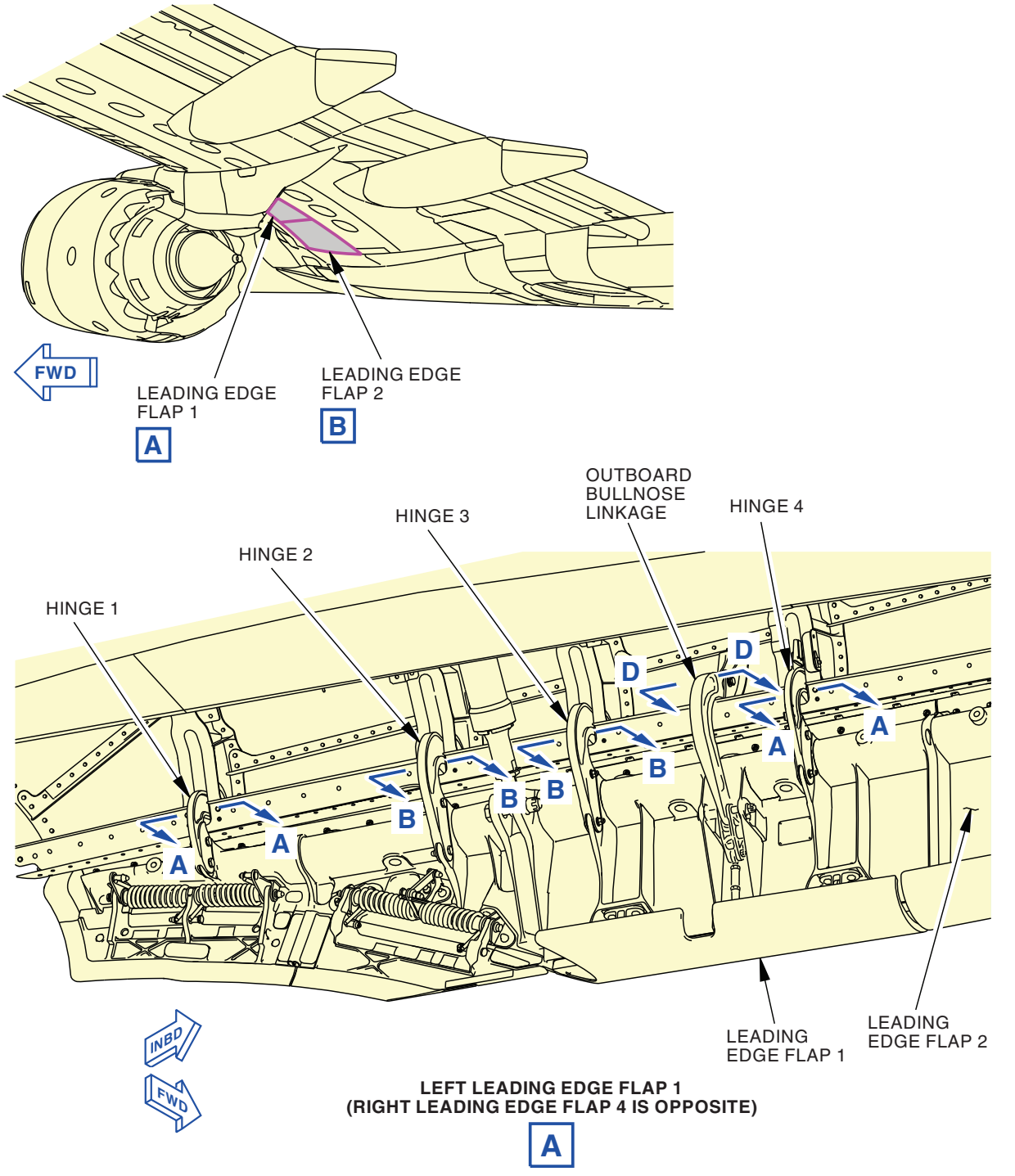
NAME OF THE PARTS THAT ARE IN CONTACT	DIM.	DIAMETER DESIGN LIMITS		WEAR LIMITS	
		MINIMUM inches/mm	MAXIMUM inches/mm	PERMITTED MAXIMUM I.D./ MINIMUM O.D. OF WORN PART	MAXIMUM SERVICE DIAMETRIC CLEARANCE
#14 BOLT	O.D.	0.2485 in 6.312 mm	0.2495 in 6.337 mm	0.2475 in 6.286 mm	0.076 mm
#15 BUSHING	I.D.	0.3750 in 9.525 mm	0.3756 in 9.540 mm	0.3769 in 9.572 mm	0.0024 in
#15 BUSHING	O.D.	0.3740 in 9.500 mm	0.3745 in 9.512 mm	0.3732 in 9.480 mm	0.060 mm
#16 BEARING	I.D.	0.2500 in 6.350 mm	0.2505 in 6.362 mm	0.2525 in 6.413 mm	0.0030 in
#16 BOLT	O.D.	0.2485 in 6.311 mm	0.2495 in 6.337 mm	0.2475 in 6.286 mm	0.076 mm

SUBTASK 27-81-11-420-006

(4) Do this task: Leading Edge Flap Installation, TASK 27-81-11-400-801.

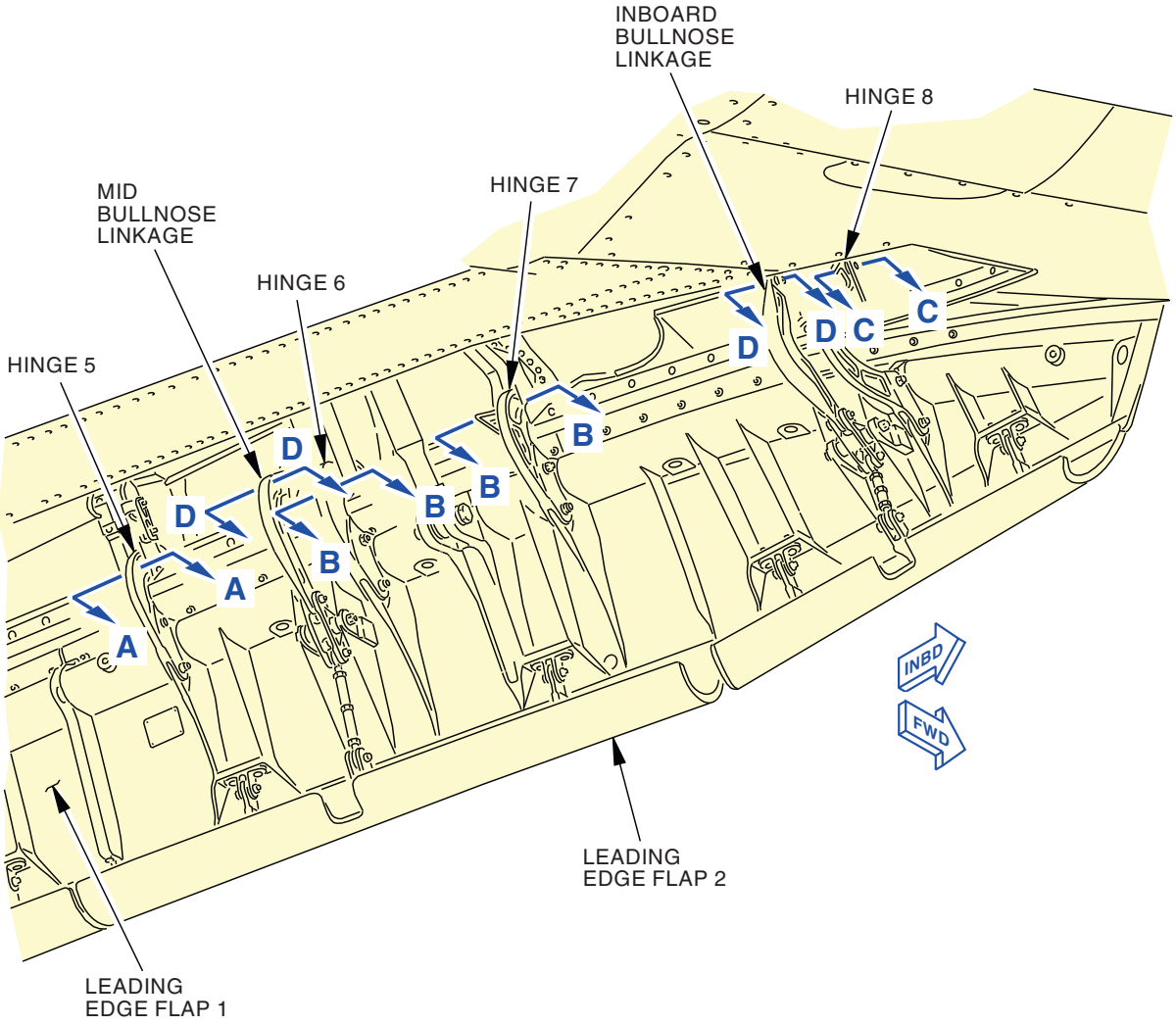
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Leading Edge Flap Wear Limits
Figure 601/27-81-11-990-803 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



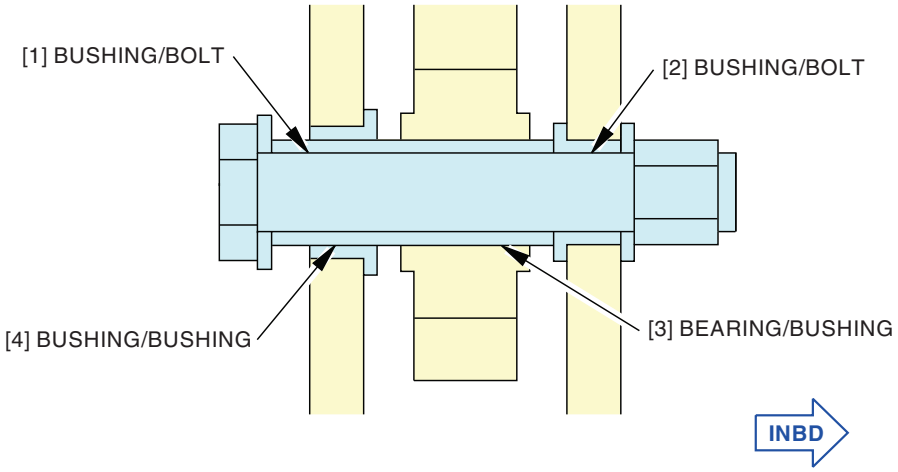
LEFT LEADING EDGE FLAP 2
(RIGHT LEADING EDGE FLAP 3 IS OPPOSITE)

B

2413353 S00061532389_V2

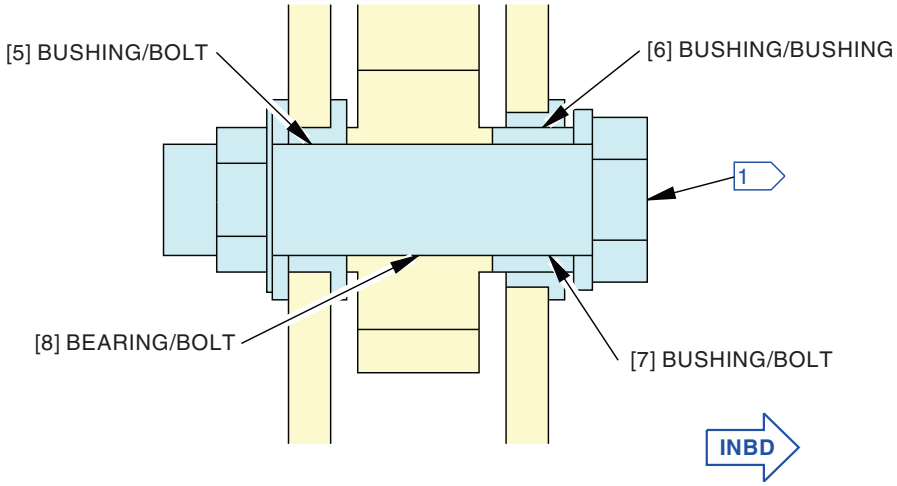
Leading Edge Flap Wear Limits
Figure 601/27-81-11-990-803 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



HINGE 1
(HINGES 4 AND 5 ARE EQUIVALENT)

A-A



HINGE 2
(HINGES 3, 6 AND 7 ARE EQUIVALENT)

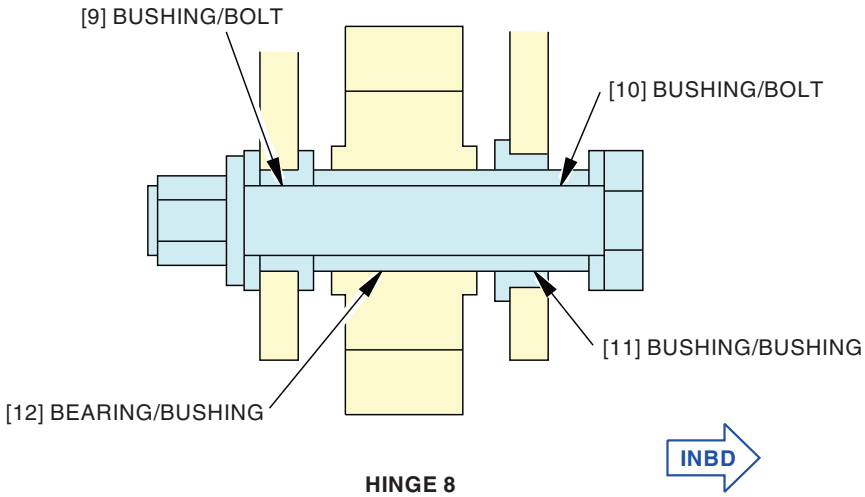
B-B

1 THE BOLT HEAD IS INBOARD AT HINGES 2 AND 6, AND OUTBOARD AT HINGES 3 AND 7.

2413354 S00061532390_V2

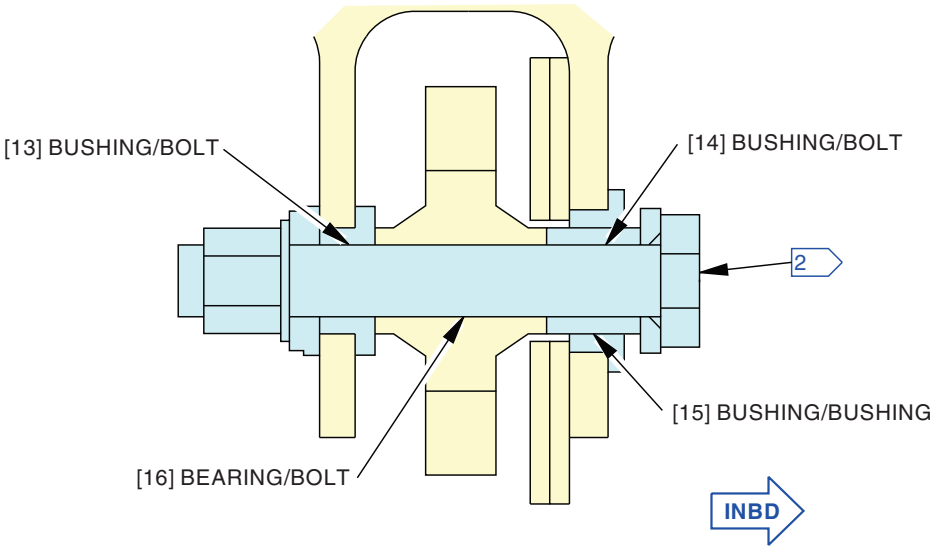
Leading Edge Flap Wear Limits
Figure 601/27-81-11-990-803 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



HINGE 8

C-C



OUTBOARD BULLNOSE LINKAGE
(MID AND INBOARD BULLNOSE LINKAGES ARE EQUIVALENT)

D-D

2 THE BOLT HEAD IS OUTBOARD AT THE MID AND INBOARD BULLNOSE LIKAGES.

2413355 S00061532391_V2

Leading Edge Flap Wear Limits
Figure 601/27-81-11-990-803 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

LEADING EDGE SLAT - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Leading Edge Slat Removal

(2) Leading Edge Slat Installation

(3) Leading Edge Slat Main Track Removal

(4) Leading Edge Slat Main Track Installation.

B. There are four slats on each wing. This procedure is applicable for the leading edge slats 1 thru 8.

C. You can do this procedure with the airplane on jacks or off jacks, but the wing should not be supported outboard of the engines.

TASK 27-81-21-000-801

2. Leading Edge Slat Removal

(Figure 401, Figure 402)

A. References

Reference	Title
27-81-00-040-801	Leading Edge Flaps and Slats - Deactivation (P/B 201)
27-81-00-480-801	Leading Edge Flap and Slat Locks Installation (P/B 201)
27-81-00-860-804	Leading Edge Flaps and Slats Retraction (P/B 201)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
30-11-31-000-801	Wing Anti-Ice Telescoping Duct Seal Removal (P/B 401)
57-41-02-000-801	Leading Edge Access Panel Removal (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1734	Lockout Set - Leading Edge Slats and Krueger Flaps Actuators Part #: C27051-29 Supplier: 81205
SPL-2041	Sling - Leading Edge Slat Removal/Installation Part #: C27056-35 Supplier: 81205 Opt Part #: C27056-19 Supplier: 81205

C. Location Zones

Zone	Area
521	Left Wing - Leading Edge to Front Spar
522	Left Wing - Slat No. 4
523	Left Wing - Slat No. 3
524	Left Wing - Slat No. 2
525	Left Wing - Slat No. 1
621	Right Wing - Leading Edge to Front Spar
622	Right Wing - Slat No. 5
623	Right Wing - Slat No. 6

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Zone	Area
624	Right Wing - Slat No. 7
625	Right Wing - Slat No. 8

D. Access Panels

Number	Name/Location
521AAB	Lower Leading Edge Access Panel
521BB	Outboard Lower Fixed Leading Edge Access Panel
521CB	Lower Leading Edge Access Panel
521DB	Lower Leading Edge Access Panel
521EB	Lower Leading Edge Access Panel
521FB	Lower Leading Edge Access Panel
521GB	Lower Leading Edge Access Panel
521HB	Lower Leading Edge Access Panel
521JB	Lower Leading Edge Access Panel
521KB	Lower Leading Edge Access Panel
521LB	Lower Leading Edge Access Panel
521MB	Lower Leading Edge Access Panel
521NB	Lower Leading Edge Access Panel
521PB	Lower Leading Edge Access Panel
521QB	Lower Leading Edge Access Panel
521RB	Lower Leading Edge Access Panel
521SB	Lower Leading Edge Access Panel
521TB	Lower Leading Edge Access Panel
521UB	Lower Leading Edge Access Panel
521VB	Lower Leading Edge Access Panel
521WB	Lower Leading Edge Access Panel
521XB	Lower Leading Edge Access Panel
521YB	Lower Leading Edge Access Panel
521ZB	Lower Leading Edge Access Panel
621AAB	Lower Leading Edge Access Panel
621BB	Outboard Lower Fixed Leading Edge Access Panel
621CB	Lower Leading Edge Access Panel
621DB	Lower Leading Edge Access Panel
621EB	Lower Leading Edge Access Panel
621FB	Lower Leading Edge Access Panel
621GB	Refuel Valve Door
621JB	Lower Leading Edge Access Panel
621KB	Lower Leading Edge Access Panel
621LB	Lower Leading Edge Access Panel
621MB	Lower Leading Edge Access Panel
621NB	Lower Leading Edge Access Panel
621PB	Lower Leading Edge Access Panel
621QB	Lower Leading Edge Access Panel
621RB	Lower Leading Edge Access Panel

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
621SB	Lower Leading Edge Access Panel
621TB	Lower Leading Edge Access Panel
621UB	Lower Leading Edge Access Panel
621VB	Lower Leading Edge Access Panel
621WB	Lower Leading Edge Access Panel
621XB	Lower Leading Edge Access Panel
621YB	Lower Leading Edge Access Panel
621ZB	Lower Leading Edge Access Panel

E. Prepare for the Removal

SUBTASK 27-81-21-010-002

(1) Remove these access panels from the lower surface of the fixed leading edge (TASK 57-41-02-000-801):

(a) For the left wing, open these access panels:

Number	Name/Location
521AAB	Lower Leading Edge Access Panel
521BB	Outboard Lower Fixed Leading Edge Access Panel
521CB	Lower Leading Edge Access Panel
521DB	Lower Leading Edge Access Panel
521EB	Lower Leading Edge Access Panel
521FB	Lower Leading Edge Access Panel
521GB	Lower Leading Edge Access Panel
521HB	Lower Leading Edge Access Panel
521JB	Lower Leading Edge Access Panel
521KB	Lower Leading Edge Access Panel
521LB	Lower Leading Edge Access Panel
521MB	Lower Leading Edge Access Panel
521NB	Lower Leading Edge Access Panel
521PB	Lower Leading Edge Access Panel
521QB	Lower Leading Edge Access Panel
521RB	Lower Leading Edge Access Panel
521SB	Lower Leading Edge Access Panel
521TB	Lower Leading Edge Access Panel
521UB	Lower Leading Edge Access Panel
521VB	Lower Leading Edge Access Panel
521WB	Lower Leading Edge Access Panel
521XB	Lower Leading Edge Access Panel
521YB	Lower Leading Edge Access Panel
521ZB	Lower Leading Edge Access Panel

(b) For the right wing, open these access panels:

Number	Name/Location
621AAB	Lower Leading Edge Access Panel
621BB	Outboard Lower Fixed Leading Edge Access Panel
621CB	Lower Leading Edge Access Panel
621DB	Lower Leading Edge Access Panel

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
621EB	Lower Leading Edge Access Panel
621FB	Lower Leading Edge Access Panel
621GB	Refuel Valve Door
621JB	Lower Leading Edge Access Panel
621KB	Lower Leading Edge Access Panel
621LB	Lower Leading Edge Access Panel
621MB	Lower Leading Edge Access Panel
621NB	Lower Leading Edge Access Panel
621PB	Lower Leading Edge Access Panel
621QB	Lower Leading Edge Access Panel
621RB	Lower Leading Edge Access Panel
621SB	Lower Leading Edge Access Panel
621TB	Lower Leading Edge Access Panel
621UB	Lower Leading Edge Access Panel
621VB	Lower Leading Edge Access Panel
621WB	Lower Leading Edge Access Panel
621XB	Lower Leading Edge Access Panel
621YB	Lower Leading Edge Access Panel
621ZB	Lower Leading Edge Access Panel

SUBTASK 27-81-21-860-001

WARNING KEEP PERSONNEL AND EQUIPMENT AWAY FROM THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(2) Supply hydraulic power to system B (TASK 29-11-00-860-801).

SUBTASK 27-81-21-860-002

WARNING MAKE SURE THAT PERSONNEL AND EQUIPMENT STAY AWAY FROM THE LEADING EDGE SLATS, TRAILING EDGE FLAPS, AND DRIVE MECHANISMS. THE FLAPS, SLATS, AND DRIVE MECHANISMS MOVE QUICKLY. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

CAUTION MAKE SURE THAT THE INBOARD FAN COWL PANEL AND THE INBOARD AND OUTBOARD THRUST REVERSERS ARE CLOSED AND IN THE STOWED POSITION OR REMOVED BEFORE YOU EXTEND THE LEADING EDGE FLAPS AND SLATS. THERE IS NOT SUFFICIENT CLEARANCE FOR THE FLAPS AND SLATS TO EXTEND IF THE INBOARD FAN COWL PANEL AND THE INBOARD AND OUTBOARD THRUST REVERSERS ARE IN THE OPEN POSITION. THIS CAN CAUSE DAMAGE TO EQUIPMENT.

(3) Move the flap control lever to the 30-unit detent to extend the leading edge slats to the full extended position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-81-21-480-001

WARNING: INSTALL THE LOCK ON LEADING EDGE FLAP ACTUATOR CAREFULLY TO PREVENT THE ACCIDENTAL OPERATION OF THE LEADING EDGE FLAPS. THE LEADING EDGE FLAPS CAN MOVE QUICKLY. THIS CAN CAUSE INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT.

(4) Install the lockout set, SPL-1734, on the actuators of the adjacent leading edge slats (TASK 27-81-00-480-801).

SUBTASK 27-81-21-860-003

(5) Move the flap control lever to the 1-unit detent to retract the leading edge slat.

NOTE: If the slat is in the full extend position, the slat actuators will be preloaded against the leading edge slat. When you put the flap control lever to the 1-unit detent it removes the preload, which makes it easier removal of the slat actuator.

SUBTASK 27-81-21-860-004

(6) Remove power from hydraulic system B (TASK 29-11-00-860-805).

SUBTASK 27-81-21-200-001

WARNING: DO NOT LET OBJECTS GET IN THE HOUSING ASSEMBLY OF THE SLAT TRACK. THIS WILL HELP PREVENT A PUNCTURE OF THE HOUSING ASSEMBLY THAT COULD CAUSE A FUEL LEAK. THE FUEL LEAK COULD CAUSE A FIRE AND POSSIBLE DEATH OR INJURY TO PERSONNEL.

(7) Keep clean and free of all Foreign Object Damage (FOD), the housing (can) assemblies of the slat main tracks, at all time.

F. Leading Edge Slat Removal

SUBTASK 27-81-21-020-001

(1) Disconnect the slat actuator [9] from the leading edge slat [17] (Figure 401).

(a) Remove the bolt retainer [18] from the leading edge slat [17].

1) Remove the bolts [15] and washers [16] that attach the bolt retainer [18] to the leading edge slat [17].

2) Remove the bolt retainer [18] from the leading edge slat [17].

WARNING: MAKE SURE YOU HOLD THE LEADING EDGE SLAT IN ITS POSITION. THE LEADING EDGE SLAT CAN FALL TO THE FULL EXTEND POSITION. THIS CAN CAUSE INJURY TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(b) Hold the leading edge slat [17] while you remove the nut [12], washer [11], bushing [10], washer [19], and bolt [20].

SUBTASK 27-81-21-980-001

(2) Carefully let the leading edge slat [17] move down to the full extend position.

SUBTASK 27-81-21-860-005

(3) Use hydraulic power to retract the slat actuator [9] (TASK 27-81-00-860-804).

NOTE: The slat actuator can hang from the upper fitting.

SUBTASK 27-81-21-040-001

(4) Do this task: Leading Edge Flaps and Slats - Deactivation, TASK 27-81-00-040-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-81-21-480-002

(5) If you use the sling, SPL-2041, to help with the removal of the leading edge slat [17], do these steps to install the sling, SPL-2041, on the leading edge slat [17] (Figure 402):

(a) Remove the stowage A/P fastener (shown in the placard of the sling tool box) from the holes of the slat structure that you will use to install the screws [64] and screws [67] of the sling, SPL-2041, kit.

(b) Attach the wedges [69] on the trailing edge of the leading edge slat [17] where they will line up with the main tracks [21].

(c) Attach the strap assembly [61] to the bottom of the leading edge slat [17] with screws [64], washers [65] and nuts [66].

(d) Attach the end fittings [63] to the leading edge slat [17] with screws [67] and washers [68].

(e) Attach the strap assembly [61] to the end fittings [63], wedges [69] and spreader bar [62].

SUBTASK 27-81-21-480-003

(6) Connect the overhead hoist to the sling, SPL-2041.

SUBTASK 27-81-21-020-002

(7) Remove the leading edge slat [17] (Figure 401).

(a) Disconnect the auxiliary arms [28] from the auxiliary tracks [29].

1) Remove the pins [40], nuts [41], washers [42], targets [43], spacers [48], detent arm roller bearings [45], washers [47], and bolts [46].

2) Move the auxiliary arms [28] away from the auxiliary tracks [29].

NOTE: The auxiliary arms will remain attached to the leading edge slat structure.

3) Remove the auxiliary arm roller bearings [44] from the auxiliary tracks [29].

4) Disconnect the leading edge slat [17] from the main tracks [21].

NOTE: The main track will remain attached to the fixed leading edge structure.

a) Remove the bolt retainer [70] (attached to the leading edge slat [17] structure) by removing two adjustable bolts [76] and two washers [77] at each of the main track [21] attach locations.

b) Remove the nut [75], washer [74], plain bushing [73], washer [72] from the bolt [71] at each of the main track attach locations.

(b) Remove the main tracks [21] from the fixed leading edge structure, as necessary.

1) Manually move the main track [21] until the downstops [22] on the main tracks [21] align with the access holes in the track ribs.

2) Remove the cotter pins [53], nuts [52], washers [51], stop fittings [49], and bolts [50].

SUBTASK 27-81-21-020-003

(8) If you remove slats No. 2, 3, 4, 5, 6 or 7, disconnect the thermal anti-icing duct from the leading edge slat [17] (TASK 30-11-31-000-801).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

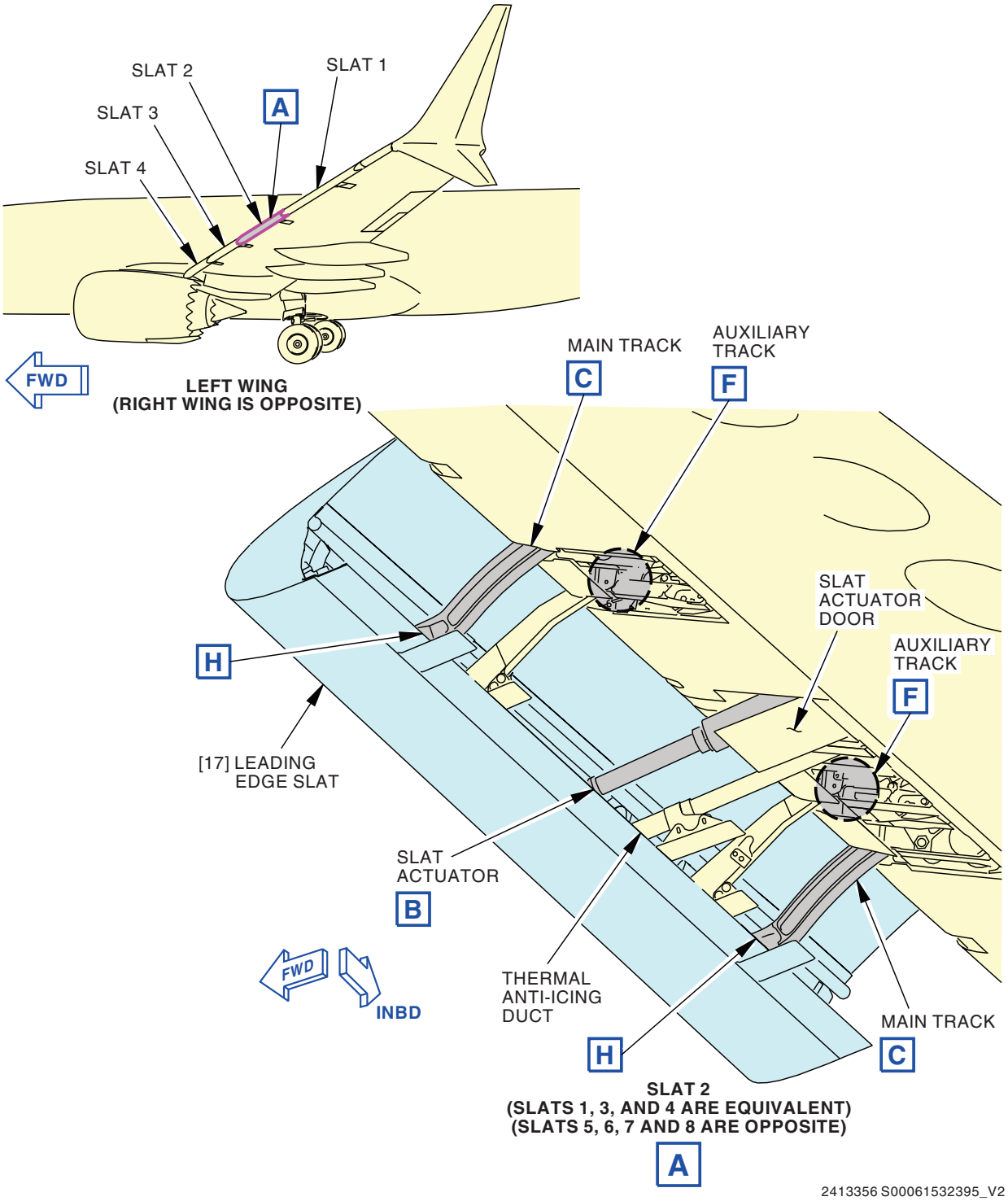
SUBTASK 27-81-21-200-002

WARNING DO NOT LET OBJECTS GET IN THE HOUSING ASSEMBLY OF THE SLAT TRACK. THIS WILL HELP PREVENT A PUNCTURE OF THE HOUSING ASSEMBLY THAT COULD CAUSE A FUEL LEAK. THE FUEL LEAK COULD CAUSE A FIRE AND POSSIBLE DEATH OR INJURY TO PERSONNEL.

(9) Keep clean and free of all FOD, the housing (can) assemblies of the slat main tracks, at all time.

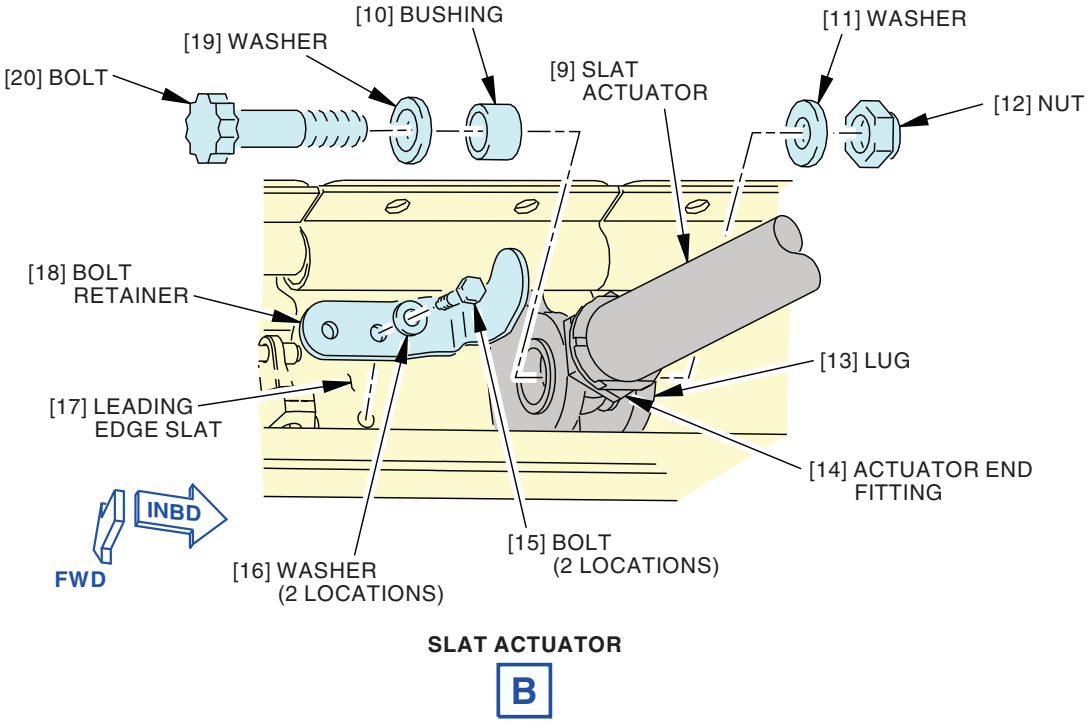
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Leading Edge Slat Installation
Figure 401/27-81-21-990-801 (Sheet 1 of 7)

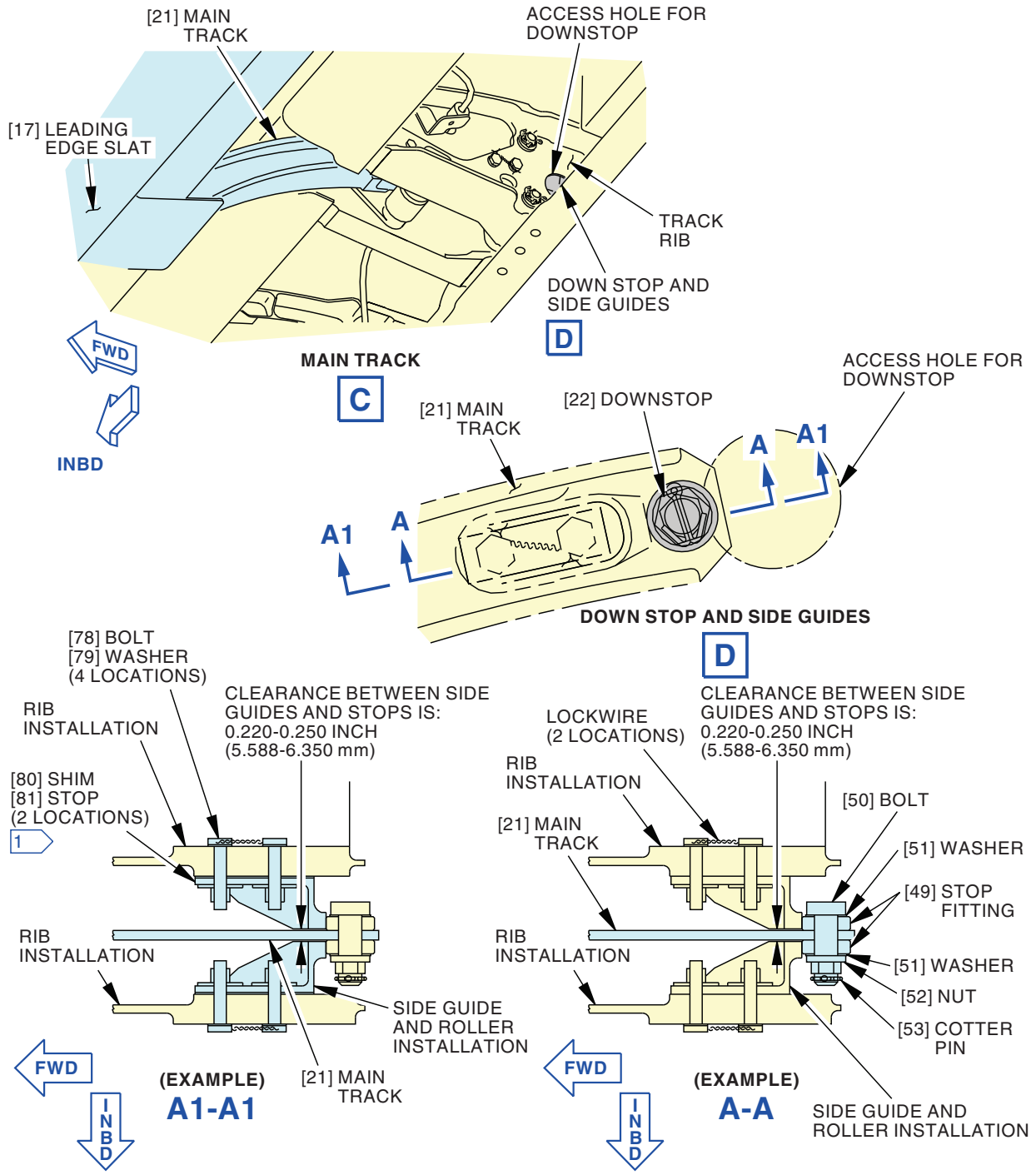
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2413357 S00061532396_V2

Leading Edge Slat Installation
Figure 401/27-81-21-990-801 (Sheet 2 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

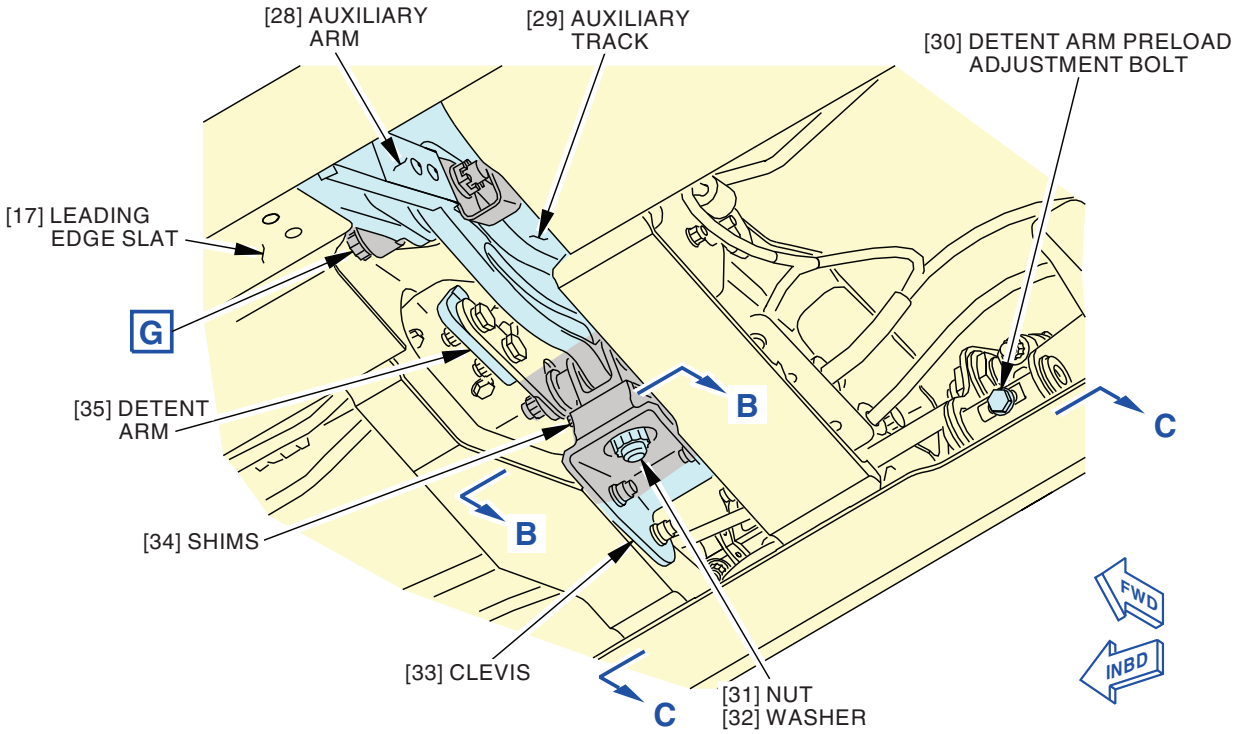


1 SHIM BOTH SIDES EQUALLY IF AND AS REQUIRED TO MAINTAIN THE CLEARANCE.

2413358 S00061532397_V3

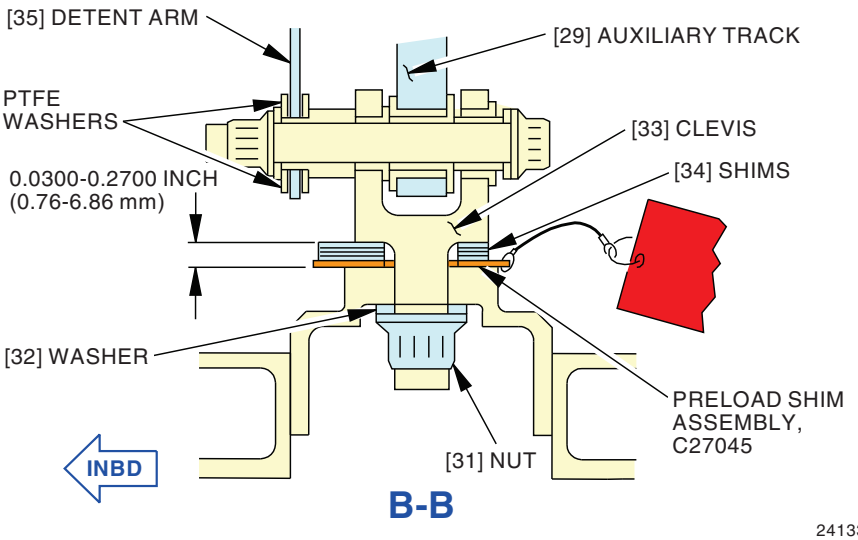
Leading Edge Slat Installation
Figure 401/27-81-21-990-801 (Sheet 3 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



INBOARD AUXILIARY TRACK
(OUTBOARD AUXILIARY TRACK IS OPPOSITE)

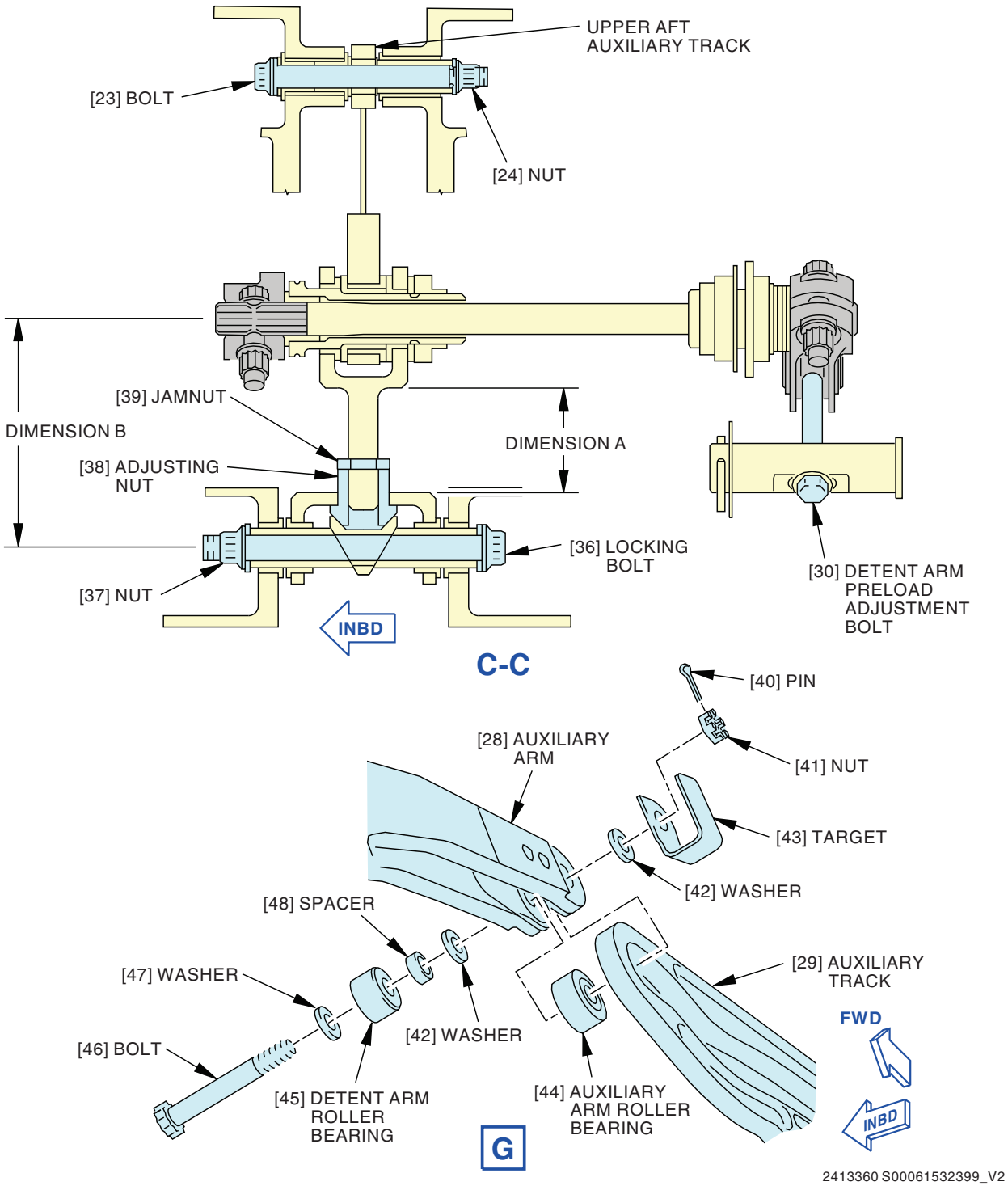
F



2413359 S00061532398_V3

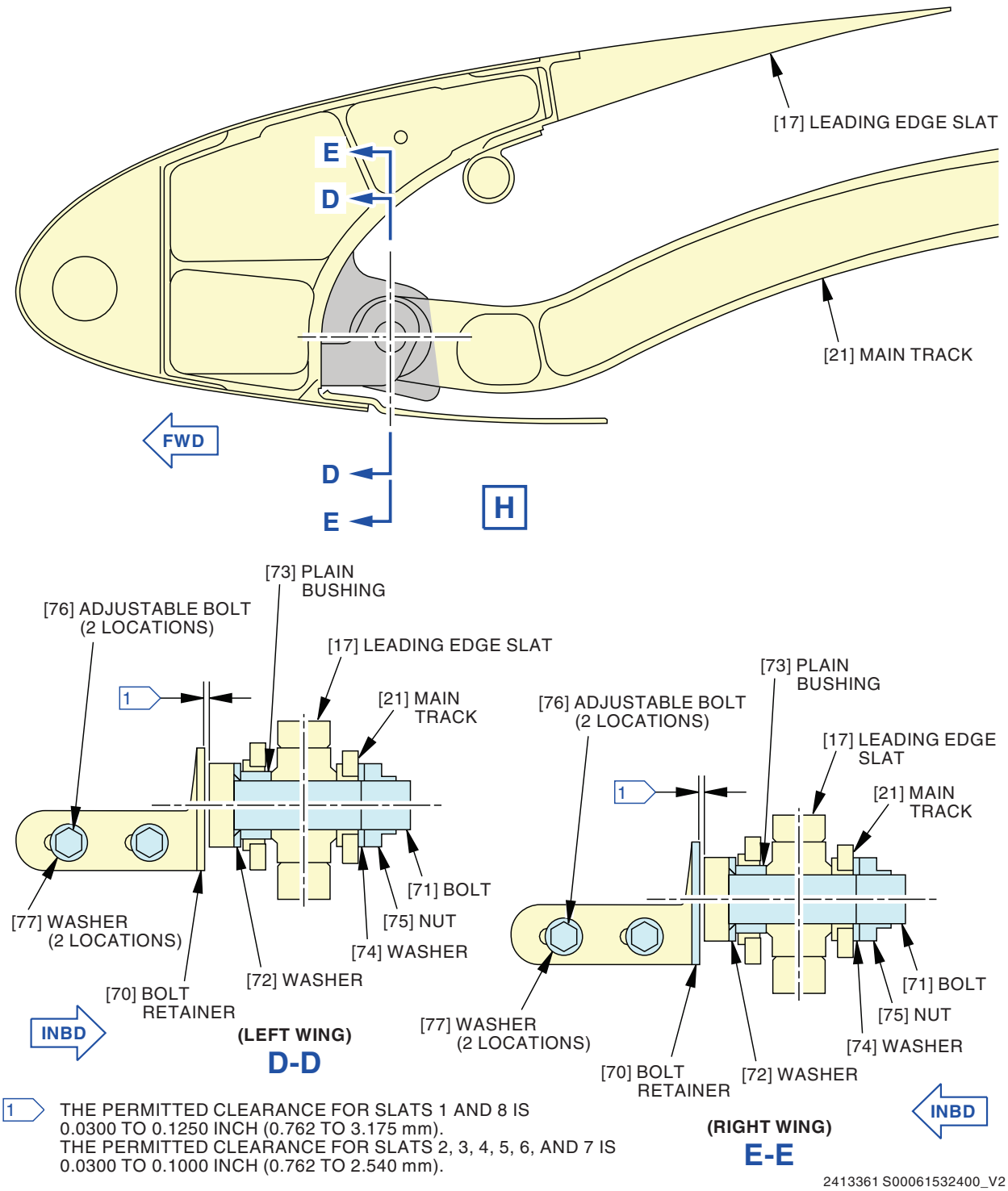
Leading Edge Slat Installation
Figure 401/27-81-21-990-801 (Sheet 4 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



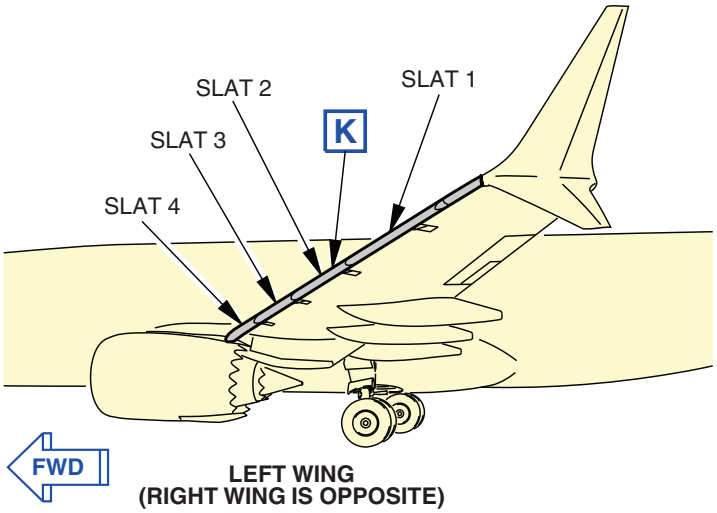
Leading Edge Slat Installation
Figure 401/27-81-21-990-801 (Sheet 5 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

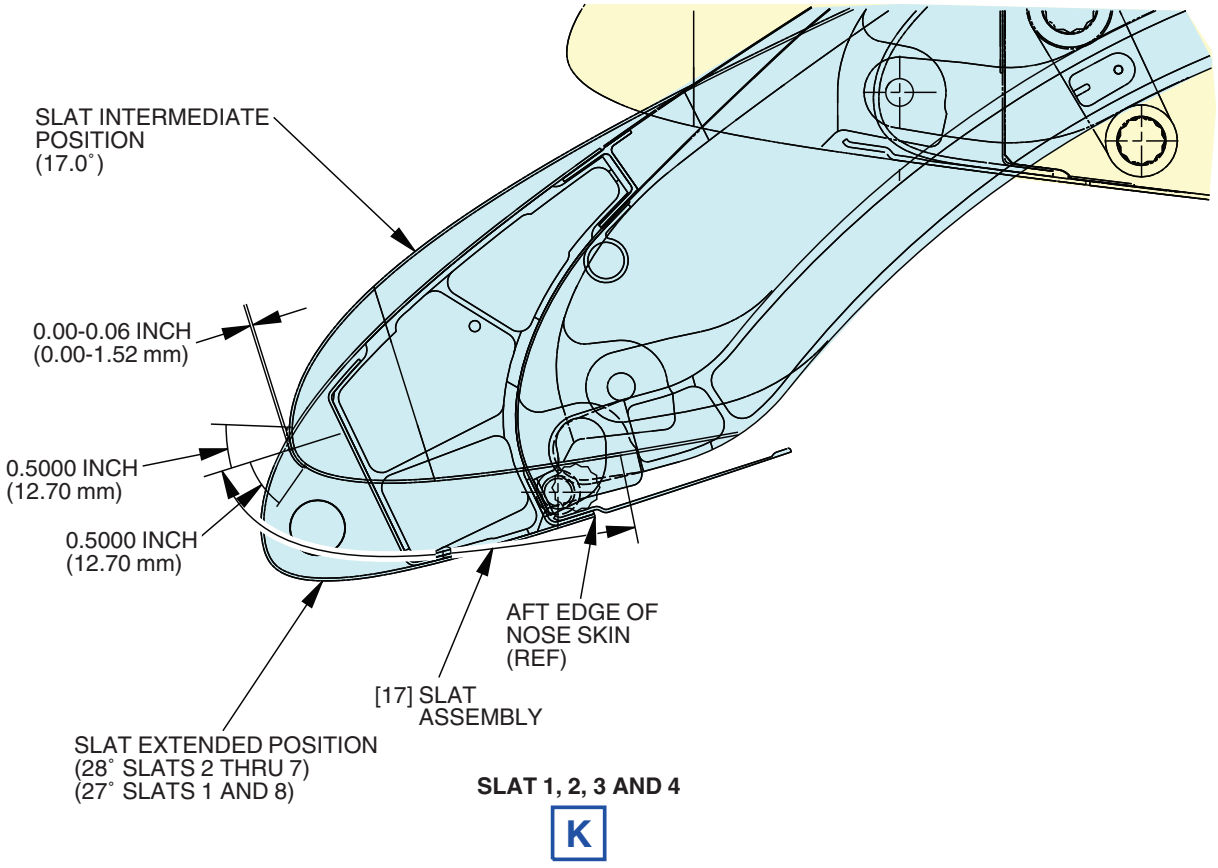


Leading Edge Slat Installation
Figure 401/27-81-21-990-801 (Sheet 6 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LEFT WING
(RIGHT WING IS OPOSITE)

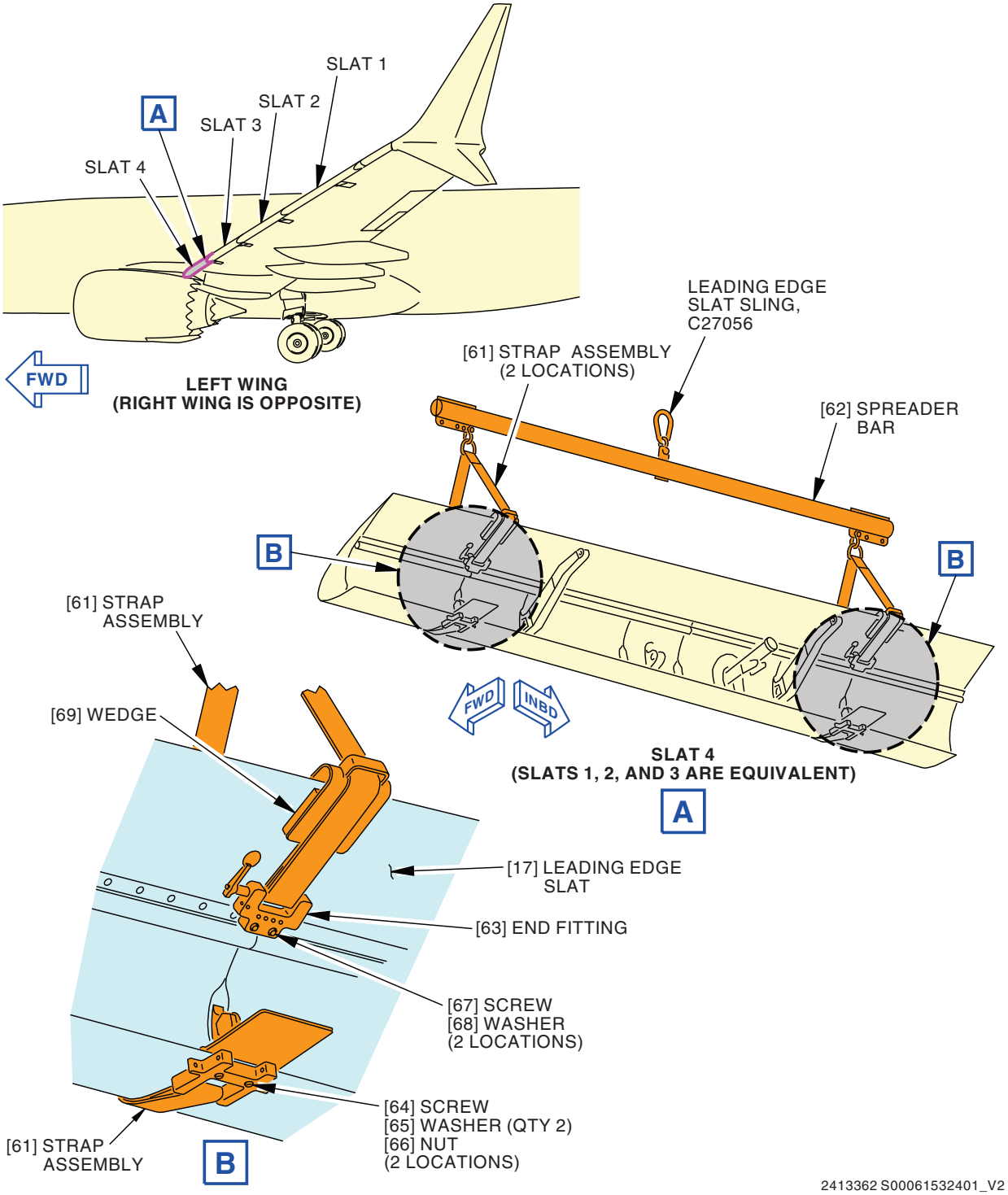


SLAT 1, 2, 3 AND 4

2796995 S0000639073_V2

Leading Edge Slat Installation
Figure 401/27-81-21-990-801 (Sheet 7 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Leading Edge Slat Sling Installation
Figure 402/27-81-21-990-802

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-81-21-400-801

3. Leading Edge Slat Installation

(Figure 401, Figure 402)

A. References

Reference	Title
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)
27-81-00-480-801	Leading Edge Flap and Slat Locks Installation (P/B 201)
27-81-00-820-803	Leading Edge Slats Adjustment (P/B 501)
27-81-00-860-804	Leading Edge Flaps and Slats Retraction (P/B 201)
30-11-31-400-801	Wing Anti-Ice Telescoping Duct Seal Installation (P/B 401)
57-41-02-400-801	Leading Edge Access Panel Installation (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1734	Lockout Set - Leading Edge Slats and Krueger Flaps Actuators Part #: C27051-29 Supplier: 81205
SPL-1739	Rigging Equipment - Slat Part #: C27045-1 Supplier: 81205
SPL-2041	Sling - Leading Edge Slat Removal/Installation Part #: C27056-35 Supplier: 81205 Opt Part #: C27056-19 Supplier: 81205

C. Consumable Materials

Reference	Description	Specification
C00528	Compound - Corrosion Preventive, Petroleum Hot Application (Soft Film)	MIL-C-11796 Class III
D00633	Grease - Aircraft General Purpose	BMS3-33
G50237	Compound - Corrosion Inhibiting, Non-drying - Cor-Ban 27L	BMS3-38

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
17	Leading edge slat	27-81-21-20-190	PTF ALL
53	Cotter pin	27-81-21-05-010	PTF ALL

E. Location Zones

Zone	Area
521	Left Wing - Leading Edge to Front Spar
522	Left Wing - Slat No. 4
523	Left Wing - Slat No. 3
524	Left Wing - Slat No. 2
525	Left Wing - Slat No. 1
621	Right Wing - Leading Edge to Front Spar
622	Right Wing - Slat No. 5
623	Right Wing - Slat No. 6

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Zone	Area
624	Right Wing - Slat No. 7
625	Right Wing - Slat No. 8

F. Access Panels

Number	Name/Location
521AAB	Lower Leading Edge Access Panel
521BB	Outboard Lower Fixed Leading Edge Access Panel
521GB	Lower Leading Edge Access Panel
521HB	Lower Leading Edge Access Panel
521NB	Lower Leading Edge Access Panel
521PB	Lower Leading Edge Access Panel
521UB	Lower Leading Edge Access Panel
521VB	Lower Leading Edge Access Panel
621AAB	Lower Leading Edge Access Panel
621BB	Outboard Lower Fixed Leading Edge Access Panel
621GB	Refuel Valve Door
621NB	Lower Leading Edge Access Panel
621PB	Lower Leading Edge Access Panel
621UB	Lower Leading Edge Access Panel

G. Leading Edge Slat Installation

SUBTASK 27-81-21-200-003

WARNING DO NOT LET OBJECTS GET IN THE HOUSING ASSEMBLY OF THE SLAT TRACK. THIS WILL HELP PREVENT A PUNCTURE OF THE HOUSING ASSEMBLY THAT COULD CAUSE A FUEL LEAK. THE FUEL LEAK COULD CAUSE A FIRE AND POSSIBLE DEATH OR INJURY TO PERSONNEL.

(1) Keep clean and free of all unwanted objects (FOD), the housing (can) assemblies of the slat main tracks, at all time.

SUBTASK 27-81-21-860-006

(2) Prepare the auxiliary tracks [29] and the detent arm [35] for the installation:

NOTE: If you will install the same slat that you removed, it is recommended that you do these steps, but is not necessary.

(a) Adjust the aft support for the auxiliary tracks [29] (View C-C, Figure 401).

1) Loosen nut [24] on the bolt [23] at the upper aft auxiliary track.

2) Loosen the jamnut [39].

3) Loosen the nut [37] to loosen the locking bolt [36].

4) Turn the adjusting nut [38] of the auxiliary track aft support to get the necessary trailing edge clearance until the dimension A is as specified in Table 401.

a) Optional. In order to adjust the auxiliary track aft support to the dimensions given in the table below, adjust the auxiliary track [29] up until the slot in the aft upper corner of the track touches the bolt [23] at the upper aft auxiliary track.

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Table 401/27-81-21-993-802

Location	Dimension A	Dimension B (reference)
Slat 1 and 8 (Outbd)	0.830 in. (21.08 mm)	2.38 in. (60.5 mm)
Slat 1 and 8 (Inbd)	0.950 in. (24.13 mm)	2.50 in. (63.5 mm)
Slat 2 and 7 (Outbd)	0.950 in. (24.13 mm)	2.50 in. (63.5 mm)
Slat 2 and 7 (Inbd)	1.450 in. (36.83 mm)	3.00 in. (76.2 mm)
Slat 3 and 6 (Outbd)	1.450 in. (36.83 mm)	3.00 in. (76.2 mm)
Slat 3 and 6 (Inbd)	1.450 in. (36.83 mm)	3.00 in. (76.2 mm)
Slat 4 and 5 (Outbd)	1.500 in. (38.10 mm)	3.05 in. (77.5 mm)
Slat 4 and 5 (Inbd)	2.100 in. (53.34 mm)	3.65 in. (92.7 mm)

5) Temporarily tighten the nut [37] on the locking bolt [36].

NOTE: You will apply final torque on the locking bolt during the completion of the slat adjustment (TASK 27-81-00-820-803).

6) Finger tighten the jamnut [39] until in contact with the adjusting nut [38] of the aft support.

7) Tighten the jamnut [39] an additional quarter turn.

8) Temporarily tighten the nut [24] on the bolt [23] at the upper aft auxiliary track.

NOTE: You will apply final torque on the nut during the completion of the slat adjustment (TASK 27-81-00-820-803).

There is no lockwire required on the jamnut.

(b) Adjust the preload on the detent arm [35] to zero:

NOTE: It is recommended that you do these steps, but it is not necessary.

1) Turn the preload adjustment bolt [30] in the clockwise direction to increase preload or in the counterclockwise direction to reduce preload.

2) Adjust the preload on all the detent arm [35] to zero pound.

SUBTASK 27-81-21-480-004

(3) If you disconnected the clevis [33]/auxiliary track [29]/detent arm [35] then install the preload shim assembly, that is part of the rigging equipment, SPL-1739, at the forward supports for the auxiliary tracks [29] (View B-B, Figure 401):

NOTE: This step is necessary to make sure that the leading edge slat is not preloaded against the fixed leading edge.

(a) Loosen the nut [31] on the clevis [33].

(b) Lift the auxiliary track [29] with the clevis [33] through the auxiliary track support.

(c) Install the preload shim assembly, that is part of the rigging equipment, SPL-1739, with the shims [34] on the clevis [33].

(d) Put the clevis [33] back in its position on the auxiliary track support.

(e) Tighten the nut [31] on the clevis [33].

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(f) Make sure that two polytetrafluoroethylene (PTFE) washers are installed in the clevis/auxiliary track/ detent arm joint, if they were disconnected.

NOTE: If the washers are damaged or missing, stabilization of the detent arm is reduced and it is possible for the detent arm to be deflected laterally during slat extension or retraction and the possibility of misalignment between the detent arm and the detent roller increases.

NOTE: If the washers are damaged or missing, it may be possible for the roller to come out of the detent arm track, thus affecting the operation of the detent system. Also extreme misalignment of the slat roller attachment could move the slat down sensor target, causing fault indications or physical damage to the sensor itself.

SUBTASK 27-81-21-020-004

(4) If you removed slats No. 2, 3, 4, 5, 6, or 7, connect the thermal anti-icing duct to the leading edge slat [17] (TASK 30-11-31-400-801).

SUBTASK 27-81-21-420-001

CAUTION

MAKE SURE THAT THE LOWER SURFACE OF THE SLAT TRAILING EDGE AND THE UPPER SKIN PANELS OF THE FIXED LEADING EDGE ARE SMOOTH BEFORE YOU INSTALL THE LEADING EDGE SLATS. MAKE SURE THAT THE SURFACE AND PANELS ARE ALSO CLEAN AND FREE OF CONTAMINATIONS, NICKS, OR BURRS. DAMAGE TO EQUIPMENT CAN OCCUR IF THE SURFACE AND PANELS ARE NOT SMOOTH AND THERE ARE CONTAMINATIONS.

(5) Install the leading edge slat [17] with the steps that follow:

WARNING

DO NOT LET OBJECTS GET IN THE HOUSING ASSEMBLY OF THE SLAT TRACK. THIS WILL HELP PREVENT A PUNCTURE OF THE HOUSING ASSEMBLY THAT COULD CAUSE A FUEL LEAK. THE FUEL LEAK COULD CAUSE A FIRE AND POSSIBLE DEATH OR INJURY TO PERSONNEL.

(a) Keep clean and free of all unwanted objects (FOD), the housing (can) assemblies of the slat main tracks, at all time.

(b) If you removed them, install the main tracks [21] on the fixed leading edge structure (View A-A, Figure 401):

1) Make sure that the clearance between the side guides and stops is 0.220 in. (5.59 mm) - 0.250 in. (6.35 mm).

a) If the clearance is not within the specified limit, adjust the shims [80], as necessary (TASK 27-81-21-400-802).

2) Apply a layer of grease, D00633, to the vertical web surfaces of the main tracks [21] that contact the side guides.

NOTE: It is not necessary to grease the rolling surface, only the side guide contact surface.

3) Install the stop fittings [49] to the main tracks [21]:

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WARNING USE NITRILE GLOVES FOR SKIN PROTECTION WHEN YOU USE COR-BAN 27L. IF IT GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER. GET MEDICAL AID. THIS MATERIAL CONTAINS FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

CAUTION INSTALL THE WASHERS CORRECTLY. IF YOU DO NOT, THE MAIN TRACK/STRUCTURE JOINT CAN BECOME LOOSE AND THE DOWNSTOP ASSEMBLY CAN FALL. THIS CAN CAUSE DAMAGE TO THE SLAT TRACK HOUSING.

a) Get access to the stop fittings [49] through the access holes in the track ribs.

b) Do a check on the fit of the down stop assembly bolt in the hole in the slat main track.

<1> If the bolt will not enter the hole in the main track [21] using hand pressure (because the hole diameter is less than the bolt's shank diameter) , then do the steps that follow:

<a> Ream the hole up to a maximum of 0.4375 in. (11.1125 mm). If necessary, remove the slat.

<b> Clean the area as necessary.

<c> Apply Cor-Ban 27L Compound, G50237, corrosion inhibiting material to the reamed hole.

<2> If the bolt has side-to-side play in the hole, then do the steps that follow:

<a> Measure the hole and if the hole diameter is 0.4421 in. (11.2293 mm) - 0.5005 in. (12.7127 mm), then install a bushing.

<b> If the hole diameter is greater than 0.5005 in. (12.7127 mm), then replace the slat main track [21].

c) Coat the shank of the bolt [50] and the face of the first stop fitting [49] with Cor-Ban 27L Compound, G50237, and all the surface of the hole in the main track [21].

d) Install one washer [51] under the bolt head and the first stop fitting [49] on the bolt [50].

NOTE: If it is necessary, an additional washer is permitted under the bolt head.

e) Install the bolt [50] attached to the washer [51] and the first stop fitting [49] to the main track [21].

NOTE: The down stop assembly can be installed with the bolt head on either the inboard side or the outboard side of the slat main track.

f) Install the second stop fitting [49], one washer [51], and the nut [52] on the shank of the bolt [50] that you have it installed on the main track [21].

NOTE: No grip length increase is permitted on the bolt.

<1> Tighten the nut [52] (recommended) or the bolt [50] (optional) to 180 in-lb (20.3 N·m) - 200 in-lb (22.6 N·m).

NOTE: The recommended method is to tighten the nuts.

The optional method is to tighten the bolts.

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g) Install the new cotter pin [53].

<1> Wrap the cotter pin [53] around the shank of the bolt [50].

h) Make sure by the visual inspection that the bolts [50], washers [51], stop fittings [49], nuts [52], and cotter pins [53] are all installed correctly.

i) Make sure that the slat main track housing (can) is free of unwanted material.

(c) Lift the leading edge slat [17] into its position.

NOTE: On slats No. 2, 3, 4, 5, 6, or 7, engage the thermal anti-ice duct with the leading edge slat.

(d) Attach the main tracks [21] to the leading edge slat [17] at each of the main track [21] locations with the steps that follow (View D-D, Figure 401):

1) Apply grease, D00633, to the shank and threads before the installation of the bolts [71].

2) Install the plain bushing [73] in the main track [21].

3) Install the bolt [71], washer [72], washer [74], and the nut [75].

NOTE: The slats 1 and 8 are attached with 0.500 in. (12.700 mm) nominal diameter bolts. The slats 2 through 7 are attached with 0.625 in. (15.875 mm) nominal diameter bolts.

a) Tighten the nut [75] to 388 in-lb (43.8 N·m) - 412 in-lb (46.5 N·m) for slats 1 and 8.

b) Tighten the nut [75] to 795 in-lb (89.8 N·m) - 845 in-lb (95.5 N·m) for slats 2 through 7.

4) Install the bolt retainer [70] with two adjustable bolts [76] and two washers [77] on the leading edge slat [17] structure.

a) Make sure that there is a permitted clearance dimension between the bolt retainer [70] and the head of the bolt [71].

b) If it is necessary, adjust the clearance by loosening the two adjustable bolts [76], moving the bolt retainer [70] along the slotted holes and tightening the two adjustable bolts [76].

(e) Attach the auxiliary arms [28] to the auxiliary tracks [29] (View G, Figure 401):

1) Apply a layer of grease, D00633, to the auxiliary tracks [29] and the side guards on the auxiliary arms [28].

2) Lift the leading edge slat [17] partially for better access to the roller attachment location.

3) Install the auxiliary arm roller bearings [44] to the auxiliary tracks [29].

4) Install the bolts [46], washers [47], detent arm roller bearings [45], spacers [48], washers [42], targets [43], and nuts [41].

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a) Make sure that three washers (CRES material, 0.063 in. (1.60 mm) nominal thickness) are installed in the detent roller/track/arm/sensor target joint (one washer [47] under the bolt head and two washers [42] installed on both sides of the auxiliary arm).

NOTE: If the washers are thinner than 0.063 in. (1.6 mm), the lateral location of the roller bearing will be changed and it will no longer align with the centerline of the detent arm.

NOTE: The thinner washers will not support the torque preload and operating load imposed on the joint. As the thinner washers deform, the fastener torque preload is lost, and the joint may move under normal operating loads, inducing additional misalignment of the bearing. This joint holds the slat sensor target, so a loosened bolt will permit the sensor target to move. Target movement could lead to sensor indication problems, and the loose target could physically contact the sensor as the slat moves and cause additional damage.

b) Make sure that the flat edges of the target detail aligns with the corresponding face on the auxiliary arm, otherwise the target detail can be damaged during the movement of the slat.

NOTE: The target are “U” shaped and are oriented as such with the base facing down.

c) Add the washer [42] under the nut [41] if and as it is necessary to align the cotter pin hole in the bolt [46] with the slot in the nut [41].

d) Make sure that the bolt [46] head clears the fixed LE structure (auxiliary ribs) on slats 1 and 8.

<1> If clearance is limited, make sure that the auxiliary track support is fully adjusted up.

5) Tighten the nuts [41] (recommended) to 95 in-lb (10.7 N·m) - 160 in-lb (18.1 N·m), or the bolts [46] (optional) to 144 in-lb (16.3 N·m) - 192 in-lb (21.7 N·m).

NOTE: The recommended method is to tighten the nuts.

The optional method is to tighten the bolts.

NOTE: It is recommended to orient the bolt for better access to the hole for the pin.

6) Install the pins [40].

(f) Manually extend and retract the leading edge slat [17] to make sure that it moves freely on the tracks.

(g) Make sure that the main tracks [21] move parallel with each other and non-binding between main tracks [21] and side guides.

1) Make sure that the bolt head clears the fixed LE structure (auxiliary ribs) on slats 1 and 8.

a) If clearance is limited, make sure that the auxiliary track aft support is fully adjusted up.

SUBTASK 27-81-21-440-001

(6) Do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801.

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SUBTASK 27-81-21-860-007

WARNING

MAKE SURE THAT PERSONNEL AND EQUIPMENT STAY AWAY FROM THE LEADING EDGE SLATS, TRAILING EDGE FLAPS, AND DRIVE MECHANISMS. THE FLAPS, SLATS, AND DRIVE MECHANISMS MOVE QUICKLY. IF YOU DO NOT OBEY, THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(7) Use hydraulic power to retract the slat actuator [9] to the 1-unit position (TASK 27-81-00-860-804).

NOTE: This will permit you to install the bolt.

SUBTASK 27-81-21-710-001

(8) Manually retract the leading edge slat.

SUBTASK 27-81-21-420-002

(9) Connect the slat actuator [9] to the leading edge slat [17] (View B, Figure 401):

(a) Move the leading edge slat [17] until it aligns with the slat actuator [9].

(b) Install the bushing [10], bolt [20], washer [19], washer [11], and nut [12] to attach the slat actuator [9] to the leading edge slat [17].

1) Make sure that the head of the bolt is on the outboard side of the slat actuator.

2) Tighten the nut [12] (recommended) to 660 in-lb (74.6 N·m) - 980 in-lb (110.7 N·m) or the bolt [20] (optional) to 882 in-lb (99.7 N·m) - 1078 in-lb (121.8 N·m).

NOTE: The recommended method is to tighten the nuts.

The optional method is to tighten the bolts.

(c) Install the bolt retainer [18]:

1) Apply compound, C00528, to the shank and threads of the bolts [15].

2) Install the bolts [15] and washers [16] to attach the bolt retainer [18].

SUBTASK 27-81-21-080-001

(10) If you installed the sling, SPL-2041, remove the sling, SPL-2041, from the leading edge slat [17] (Figure 402):

(a) Hold and remove the spreader bar [62] from the overhead hoist.

(b) Remove the strap assembly [61] from end fittings [63] and the wedges [69].

(c) Remove the screws [67] and washers [68] to remove the end fittings [63].

(d) Remove the nuts [66], washers [65], and screws [64] to remove the strap assembly [61] from bottom of the leading edge slat [17].

(e) Remove the wedges [69] from the leading edge slat [17].

(f) Install the stowage A/P fastener (shown in the placard of the tool box) in the holes of the slat structure that you use to install the screws [64] and screws [67] of the sling, SPL-2041, kit.

SUBTASK 27-81-21-211-001

(11) Visually check all leading edge slat bull-noses to make sure that they align within 0 in. (0 mm) - 0.060 in. (1.52 mm) when in the extended intermediate position (View K, Figure 401).

(a) Dimension may apply at any one point within the 1.0000 in. (25.40 mm) band width.

NOTE: This dimension is normal to the skin surface where the measurement is taken. Vertical mismatch is permissible.

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(b) Make sure that the sum of all of the clearances is 0.0000 in. (0.00 mm) to 0.0600 in. (1.52 mm).

(c) If they do not, make sure that all main tracks [21] are correctly installed.

NOTE: The part number of the main track is located on the forward lower non-wear surface.

SUBTASK 27-81-21-820-001

(12) Do this task: Leading Edge Slats Adjustment, TASK 27-81-00-820-803.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 27-81-21-080-002

WARNING

CAREFULLY REMOVE THE LEADING EDGE FLAP AND SLAT ACTUATORS LOCKOUT SET. THE LEADING EDGE FLAPS AND SLATS CAN MOVE QUICKLY IF YOU DO NOT REMOVE THE LOCKS CORRECTLY. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Remove the lockout set, SPL-1734, from the actuators of the adjacent leading edge slats (TASK 27-81-00-480-801).

SUBTASK 27-81-21-200-004

WARNING

DO NOT LET OBJECTS GET IN THE HOUSING ASSEMBLY OF THE SLAT TRACK. THIS WILL HELP PREVENT A PUNCTURE OF THE HOUSING ASSEMBLY THAT COULD CAUSE A FUEL LEAK. THE FUEL LEAK COULD CAUSE A FIRE AND POSSIBLE DEATH OR INJURY TO PERSONNEL.

(2) Keep clean and free of all unwanted objects (FOD), the housing (can) assemblies of the slat main tracks, at all time.

SUBTASK 27-81-21-940-001

(3) Install these access panels on the lower surface of the fixed leading edge (TASK 57-41-02-400-801):

(a) For the left wing, close these access panels:

Number	Name/Location
521AAB	Lower Leading Edge Access Panel
521BB	Outboard Lower Fixed Leading Edge Access Panel
521GB	Lower Leading Edge Access Panel
521HB	Lower Leading Edge Access Panel
521NB	Lower Leading Edge Access Panel
521PB	Lower Leading Edge Access Panel
521UB	Lower Leading Edge Access Panel
521VB	Lower Leading Edge Access Panel

(b) For the right wing, close these access panels:

Number	Name/Location
621AAB	Lower Leading Edge Access Panel
621BB	Outboard Lower Fixed Leading Edge Access Panel
621GB	Refuel Valve Door
621NB	Lower Leading Edge Access Panel
621PB	Lower Leading Edge Access Panel
621UB	Lower Leading Edge Access Panel

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SUBTASK 27-81-21-860-008

(4) Do this task: Leading Edge Flaps and Slats Retraction, TASK 27-81-00-860-804.

END OF TASK

TASK 27-81-21-000-802

4. Leading Edge Slat Main Track Removal

(Figure 401)

A. General

(1) This procedure contains steps to remove one main track.

(2) You can remove only one main track per slat at a time.

B. References

Reference	Title
27-81-00-040-801	Leading Edge Flaps and Slats - Deactivation (P/B 201)
27-81-00-860-801	Leading Edge Flap and Slat System Operation With Primary Control (P/B 201)
57-41-02-000-801	Leading Edge Access Panel Removal (P/B 201)

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
522	Left Wing - Slat No. 4
523	Left Wing - Slat No. 3
524	Left Wing - Slat No. 2
525	Left Wing - Slat No. 1
622	Right Wing - Slat No. 5
623	Right Wing - Slat No. 6
624	Right Wing - Slat No. 7
625	Right Wing - Slat No. 8

D. Access Panels

Number	Name/Location
521AAB	Lower Leading Edge Access Panel
521BB	Outboard Lower Fixed Leading Edge Access Panel
521CB	Lower Leading Edge Access Panel
521FB	Lower Leading Edge Access Panel
521GB	Lower Leading Edge Access Panel
521HB	Lower Leading Edge Access Panel
521JB	Lower Leading Edge Access Panel
521MB	Lower Leading Edge Access Panel
521NB	Lower Leading Edge Access Panel
521PB	Lower Leading Edge Access Panel
521QB	Lower Leading Edge Access Panel
521TB	Lower Leading Edge Access Panel
521UB	Lower Leading Edge Access Panel
521VB	Lower Leading Edge Access Panel

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(Continued)

Number	Name/Location
521WB	Lower Leading Edge Access Panel
521ZB	Lower Leading Edge Access Panel
621ABB	Lower Leading Edge Access Panel
621BB	Outboard Lower Fixed Leading Edge Access Panel
621CB	Lower Leading Edge Access Panel
621FB	Lower Leading Edge Access Panel
621GB	Refuel Valve Door
621JB	Lower Leading Edge Access Panel
621MB	Lower Leading Edge Access Panel
621NB	Lower Leading Edge Access Panel
621PB	Lower Leading Edge Access Panel
621QB	Lower Leading Edge Access Panel
621TB	Lower Leading Edge Access Panel
621UB	Lower Leading Edge Access Panel
621VB	Lower Leading Edge Access Panel
621WB	Lower Leading Edge Access Panel
621ZB	Lower Leading Edge Access Panel

E. Leading Edge Slat Main Track Removal

SUBTASK 27-81-21-860-010

(1) Extend the slats to the full extend position (TASK 27-81-00-860-801).

SUBTASK 27-81-21-040-003

(2) Do this task: Leading Edge Flaps and Slats - Deactivation, TASK 27-81-00-040-801.

SUBTASK 27-81-21-010-004

(3) Remove the applicable access panels (TASK 57-41-02-000-801):

(a) Open these access panels:

Number	Name/Location
521AAB	Lower Leading Edge Access Panel
521BB	Outboard Lower Fixed Leading Edge Access Panel
521CB	Lower Leading Edge Access Panel
521FB	Lower Leading Edge Access Panel
521GB	Lower Leading Edge Access Panel
521HB	Lower Leading Edge Access Panel
521JB	Lower Leading Edge Access Panel
521MB	Lower Leading Edge Access Panel
521NB	Lower Leading Edge Access Panel
521PB	Lower Leading Edge Access Panel
521QB	Lower Leading Edge Access Panel
521TB	Lower Leading Edge Access Panel
521UB	Lower Leading Edge Access Panel
521VB	Lower Leading Edge Access Panel
521WB	Lower Leading Edge Access Panel
521ZB	Lower Leading Edge Access Panel

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(b) Open these access panels:

Number	Name/Location
621ABB	Lower Leading Edge Access Panel
621BB	Outboard Lower Fixed Leading Edge Access Panel
621CB	Lower Leading Edge Access Panel
621FB	Lower Leading Edge Access Panel
621GB	Refuel Valve Door
621JB	Lower Leading Edge Access Panel
621MB	Lower Leading Edge Access Panel
621NB	Lower Leading Edge Access Panel
621PB	Lower Leading Edge Access Panel
621QB	Lower Leading Edge Access Panel
621TB	Lower Leading Edge Access Panel
621UB	Lower Leading Edge Access Panel
621VB	Lower Leading Edge Access Panel
621WB	Lower Leading Edge Access Panel
621ZB	Lower Leading Edge Access Panel

SUBTASK 27-81-21-020-009

(4) Disconnect the leading edge slat [17] from the main tracks [21].

(a) Remove the bolt retainer [70] attached to the structure of the leading edge slat [17].

1) Remove two adjustable bolts [76] and two washers [77] at each of the main track [21] attach location.

(b) Remove the nut [75], washer [74], bolt [71], washer [72], and plain bushing [73] at each of the main track [21] attach location.

SUBTASK 27-81-21-020-010

(5) Remove the cotter pin [53], nut [52], washer [51], stop fittings [49], and bolt [50] from the aft end of the main track [21].

SUBTASK 27-81-21-020-011

(6) Remove the main track [21] from the housing assembly.

(a) Carefully lift the slat manually to give clearance to remove the main track.

NOTE: The bolt [20] can be removed to give additional clearance for the main track removal.

(b) Remove the main track [21].

NOTE: The weight of the slat on the auxiliary tracks can cause an inadvertent movement in the assembly. This can cause the slat to be out of rig.

——— END OF TASK ———

TASK 27-81-21-400-802

5. Leading Edge Slat Main Track Installation

(Figure 401)

A. General

(1) This procedure contains steps to install one main track.

(2) You can remove only one main track per slat at a time.

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B. References

Reference	Title
20-10-44-400-801	Lockwire, Cotter Pins, and Lockrings - Installation (P/B 401)
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)
27-81-00-820-803	Leading Edge Slats Adjustment (P/B 501)
27-81-00-860-801	Leading Edge Flap and Slat System Operation With Primary Control (P/B 201)
27-81-00-860-803	Leading Edge Flaps and Slats Extension (P/B 201)
57-41-02-400-801	Leading Edge Access Panel Installation (P/B 201)

C. Consumable Materials

Reference	Description	Specification
C00259	Coating - Chemical And Solvent Resistant Finish, Corrosion Inhibiting Primer	BMS10-11 Type I
C00528	Compound - Corrosion Preventive, Petroleum Hot Application (Soft Film)	MIL-C-11796 Class III
D00633	Grease - Aircraft General Purpose	BMS3-33
D50180	Grease - Aircraft General Purpose (AeroShell Grease 33)	BMS3-33, MIL-PRF-23827
D50306	Grease - Synthetic Aircraft Grease (Mobilgrease 33)	BMS3-33 Type 1, MIL-PRF-23827 Type 1
G01912	Lockwire - MS20995NC32, Monel - 0.032 Inch (0.8128 mm) Diameter	NASM20995
G50136	Compound - Corrosion Inhibiting, Non-drying Paste	BMS3-38
G50237	Compound - Corrosion Inhibiting, Non-drying - Cor-Ban 27L	BMS3-38

D. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
522	Left Wing - Slat No. 4
523	Left Wing - Slat No. 3
524	Left Wing - Slat No. 2
525	Left Wing - Slat No. 1
622	Right Wing - Slat No. 5
623	Right Wing - Slat No. 6
624	Right Wing - Slat No. 7
625	Right Wing - Slat No. 8

E. Access Panels

Number	Name/Location
521AAB	Lower Leading Edge Access Panel
521BB	Outboard Lower Fixed Leading Edge Access Panel
521CB	Lower Leading Edge Access Panel
521FB	Lower Leading Edge Access Panel
521GB	Lower Leading Edge Access Panel

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(Continued)

Number	Name/Location
521HB	Lower Leading Edge Access Panel
521JB	Lower Leading Edge Access Panel
521MB	Lower Leading Edge Access Panel
521NB	Lower Leading Edge Access Panel
521PB	Lower Leading Edge Access Panel
521QB	Lower Leading Edge Access Panel
521TB	Lower Leading Edge Access Panel
521UB	Lower Leading Edge Access Panel
521VB	Lower Leading Edge Access Panel
521WB	Lower Leading Edge Access Panel
521ZB	Lower Leading Edge Access Panel
621ABB	Lower Leading Edge Access Panel
621BB	Outboard Lower Fixed Leading Edge Access Panel
621CB	Lower Leading Edge Access Panel
621FB	Lower Leading Edge Access Panel
621GB	Refuel Valve Door
621JB	Lower Leading Edge Access Panel
621MB	Lower Leading Edge Access Panel
621NB	Lower Leading Edge Access Panel
621PB	Lower Leading Edge Access Panel
621QB	Lower Leading Edge Access Panel
621TB	Lower Leading Edge Access Panel
621UB	Lower Leading Edge Access Panel
621VB	Lower Leading Edge Access Panel
621WB	Lower Leading Edge Access Panel
621ZB	Lower Leading Edge Access Panel

F. Leading Edge Slat Main Track Installation

SUBTASK 27-81-21-160-001

WARNING DO NOT LET OBJECTS GET IN THE HOUSING ASSEMBLY OF THE SLAT TRACK. THIS WILL HELP PREVENT A PUNCTURE OF THE HOUSING ASSEMBLY THAT COULD CAUSE A FUEL LEAK. THE FUEL LEAK COULD CAUSE A FIRE AND POSSIBLE DEATH OR INJURY TO PERSONNEL.

(1) Make sure that the housing assemblies and parts of the slat main tracks are clean and free of unwanted materials.

SUBTASK 27-81-21-210-001

(2) Do a check on the fit of the bolt [50] in the aft hole in the main track [21].

(a) If the bolt will not enter the hole in the main track [21] using hand pressure, then do the steps that follow:

1) Ream the hole up to a maximum of 0.4375 in. (11.1125 mm).

2) Clean the area as necessary.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING USE NITRILE GLOVES FOR SKIN PROTECTION WHEN YOU USE COR-BAN 27L. IF IT GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER. GET MEDICAL AID. THIS MATERIAL CONTAINS FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

3) Apply Cor-Ban 27L Compound, G50237, (preferred) or corrosion inhibiting material, G50136, (alternate) to the reamed hole.

(b) If the bolt has side-to-side play in the hole, then do the steps that follow:

1) Measure the hole and if the hole diameter is 0.4421 in. (11.2293 mm) - 0.5005 in. (12.7127 mm), then install a bushing.

2) If the hole diameter is greater than 0.5005 in. (12.7127 mm), then replace the slat main track.

SUBTASK 27-81-21-420-003

(3) Install the new main track [21] in the housing assembly.

(a) Carefully lift the slat manually to give clearance to install the main track.

(b) Install the main track in the housing assembly.

(c) Make sure that the clearance between the side guides and stops is 0.220 in. (5.588 mm) - 0.250 in. (6.350 mm) (View A-A, Figure 401).

1) If the clearance is not within the specified limit, adjust the shims [80], as necessary (View A1-A1, Figure 401).

a) Remove and discard the lockwire from the bolts [78].

b) Remove the bolts [78], washers [79], and stops [81] as necessary to adjust the shims [80].

c) Apply primer, C00259, to all areas of the holes where the bolts [78] will be installed.

d) Apply compound, C00528, to all areas of the holes where the bolts [78] will be installed.

e) Install the stops [81], washers [79], and bolts [78].

f) Install MS20995NC32 lockwire, G01912, on the bolts [78] (TASK 20-10-44-400-801).

(d) Apply a layer of grease, D00633, to the vertical web surfaces of the main track [21] that contact the side guides.

NOTE: It is not required to grease the rolling surface, only the side guide contact surface.

(e) Lower the slat to align the main track [21] and the leading edge slat [17].

SUBTASK 27-81-21-420-004

(4) Install the stop fittings [49] to the main track [21]:

(a) Get access to the aft hole of the main track [21] through the access holes in the track ribs.

(b) Coat the shank of the bolt [50], face of the first stop fitting [49], and all the surface of the hole in the main track [21] with Cor-Ban 27L Compound, G50237, (preferred) or corrosion inhibiting material, G50136, (alternate).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CAUTION INSTALL THE WASHERS CORRECTLY. IF YOU DO NOT, THE MAIN TRACK/STRUCTURE JOINT CAN BECOME LOOSE AND THE DOWNSTOP ASSEMBLY CAN FALL. THIS CAN CAUSE DAMAGE TO THE SLAT TRACK HOUSING.

(c) Install one washer [51] under the bolt head and first stop fitting [49] on the bolt [50].

NOTE: An additional washer is permitted under the bolt head, if required.

(d) Install the bolt [50] attached to the washer [51] and first stop fitting [49] to the main track [21].

NOTE: The down stop assembly can be installed with the bolt head on either the inboard side or the outboard side of the slat main track.

(e) Install the second stop fitting [49], one washer [51], and nut [52] on the shank of the bolt [50] that is installed on the main track [21].

NOTE: No grip length increase is permitted on the bolt [50].

(f) Tighten the nut [52] (preferred) or the bolt head to the torque range of 180 in-lb (20.3 N·m) - 200 in-lb (22.6 N·m).

NOTE: The recommended method is to tighten the nuts.

The optional method is to tighten the bolts.

(g) Install the new cotter pin [53].

1) Wrap the cotter pin [53] around the shank of the bolt [50].

(h) Make sure by the visual inspection that the bolt [50], washers [51], stop fittings [49], nut [52], and cotter pin [53] are all installed correctly.

(i) Make sure that the slat main track housing (can) is free of unwanted material.

SUBTASK 27-81-21-420-005

(5) Attach the main track [21] to the leading edge slat [17] with the steps that follow:

(a) Apply Mobilgrease 33, D50306, or AeroShell Grease 33, D50180, to the shank and threads of the bolt [71] and outer diameter of plain bushing [73].

(b) Install the plain bushing [73] in the main track [21].

(c) Install the bolt [71], washer [72], washer [74], and nut [75].

NOTE: The slats 1 and 8 are attached with 0.500 in. (12.700 mm) nominal diameter bolts. The slats 2 through 7 are attached with 0.625 in. (15.875 mm) nominal diameter bolts.

1) Tighten the nut [75] to 388 in-lb (43.8 N·m) - 412 in-lb (46.5 N·m) for slats 1 and 8.

2) Tighten the nut [75] to 795 in-lb (89.8 N·m) - 845 in-lb (95.5 N·m) for slats 2 through 7.

(d) Install the bolt retainer [70] with two bolts [76] and two washers [77] on the leading edge slat [17] structure.

1) Make sure that there is a permitted clearance dimension between the bolt retainer [70] and the head of the bolt [71] (View H, flagnote 1, Figure 401).

2) If necessary, adjust the clearance by loosening the two bolts [76], moving the bolt retainer [70] along the slotted holes and tightening the two bolts [76].

SUBTASK 27-81-21-420-006

(6) If removed, install the bolt [20] to connect the slat actuator [9] to the leading edge slat [17].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Install the bushing [10], bolt [20], washer [19], washer [11], and nut [12] to attach the slat actuator [9] to the leading edge slat [17].

1) Make sure that the head of the bolt [20] is on the outboard side of the slat actuator [9].

(b) Tighten the nut [12] to 660 in-lb (74.6 N·m) - 980 in-lb (110.7 N·m) or the bolt [20] to 882 in-lb (99.7 N·m) - 1078 in-lb (121.8 N·m).

NOTE: The recommended method is to tighten the nuts.

The optional method is to tighten the bolts.

(c) Install the bolt retainer [18]:

1) Apply compound, C00528, to the shank and threads of the bolts [15].

2) Install the bolts [15] and washers [16] to attach the bolt retainer [18].

SUBTASK 27-81-21-210-002

(7) Visually check that all leading edge slat [17] bull-noses align within 0.00 in. (0.00 mm) - 0.06 in. (1.52 mm) when in the extended intermediate position.

(a) If not, make sure that all main tracks [21] are correctly installed.

NOTE: The part number of the main track [21] is located on the forward lower non-wear surface.

(b) If necessary, adjust the slat (TASK 27-81-00-820-803).

SUBTASK 27-81-21-860-011

WARNING

MAKE SURE THAT PERSONNEL AND EQUIPMENT STAY AWAY FROM THE LEADING EDGE SLATS, TRAILING EDGE FLAPS, AND DRIVE MECHANISMS. THE FLAPS, SLATS, AND DRIVE MECHANISMS MOVE QUICKLY. IF YOU DO NOT OBEY, THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

CAUTION

MAKE SURE THAT THE INBOARD FAN COWL PANEL AND THE INBOARD AND OUTBOARD THRUST REVERSERS ARE CLOSED, AND IN THE STOWED POSITION, OR REMOVED BEFORE YOU EXTEND THE LEADING EDGE FLAPS AND SLATS. THERE IS NOT SUFFICIENT CLEARANCE FOR THE FLAPS AND SLATS TO EXTEND IF THE INBOARD FAN COWL PANEL AND THE INBOARD AND OUTBOARD THRUST REVERSERS ARE IN THE OPEN POSITION. THIS CAN CAUSE DAMAGE TO EQUIPMENT.

(8) To move the leading edge slats to the full extend position, do this task: Leading Edge Flaps and Slats Extension, TASK 27-81-00-860-803.

SUBTASK 27-81-21-210-003

(9) With the slats extended, do these steps:

(a) Apply hand load to the end of the slat assembly to make sure that preload exists.

(b) If no preload exists, do as follow:

1) Make sure that the possible under-travel dimension of the slat actuator does not exceed 0.02 in. (0.508 mm) maximum between the attach point at the rod end of the slat actuator and the clevis bores of the slat attach fittings.

NOTE: The under-travel dimension of the slat actuator indicates that there is no preload of the slat actuator in relationship to the slat actuator attach fitting.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

a) If the dimension is greater than 0.02 in. (0.508 mm), make sure that the part numbers of the main track assemblies are correct.

NOTE: When the slat main tracks are incorrectly installed visual cues (excessive mismatches, misalignment, skew, etc.) will sometimes indicate a problem in advance depending on that slat is affected.

(c) If necessary, adjust the slat (TASK 27-81-00-820-803).

SUBTASK 27-81-21-440-002

(10) Do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801.

SUBTASK 27-81-21-710-002

(11) Do this task: Leading Edge Flap and Slat System Operation With Primary Control, TASK 27-81-00-860-801.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 27-81-21-410-002

(1) Install the applicable access panels (TASK 57-41-02-400-801):

(a) Close these access panels:

Number	Name/Location
521AAB	Lower Leading Edge Access Panel
521BB	Outboard Lower Fixed Leading Edge Access Panel
521CB	Lower Leading Edge Access Panel
521FB	Lower Leading Edge Access Panel
521GB	Lower Leading Edge Access Panel
521HB	Lower Leading Edge Access Panel
521JB	Lower Leading Edge Access Panel
521MB	Lower Leading Edge Access Panel
521NB	Lower Leading Edge Access Panel
521PB	Lower Leading Edge Access Panel
521QB	Lower Leading Edge Access Panel
521TB	Lower Leading Edge Access Panel
521UB	Lower Leading Edge Access Panel
521VB	Lower Leading Edge Access Panel
521WB	Lower Leading Edge Access Panel
521ZB	Lower Leading Edge Access Panel

(b) Close these access panels:

Number	Name/Location
621ABB	Lower Leading Edge Access Panel
621BB	Outboard Lower Fixed Leading Edge Access Panel
621CB	Lower Leading Edge Access Panel
621FB	Lower Leading Edge Access Panel
621GB	Refuel Valve Door
621JB	Lower Leading Edge Access Panel
621MB	Lower Leading Edge Access Panel
621NB	Lower Leading Edge Access Panel
621PB	Lower Leading Edge Access Panel
621QB	Lower Leading Edge Access Panel
621TB	Lower Leading Edge Access Panel
621UB	Lower Leading Edge Access Panel

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(Continued)

Number	Name/Location
621VB	Lower Leading Edge Access Panel
621WB	Lower Leading Edge Access Panel
621ZB	Lower Leading Edge Access Panel

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

LEADING EDGE FLAP ACTUATOR - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the leading edge flap actuator.

(2) An installation of the leading flap actuator.

B. The leading edge flap actuator is referred to as the actuator in this procedure.

TASK 27-81-31-000-801

2. Leading Edge Flap Actuator Removal

(Figure 401)

A. References

Reference	Title
27-51-00-860-804	Retract the Trailing Edge Flaps (P/B 201)
27-81-00-480-801	Leading Edge Flap and Slat Locks Installation (P/B 201)
29-09-00-860-802	Hydraulic Reservoirs Depressurization (P/B 201)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1734	Lockout Set - Leading Edge Slats and Krueger Flaps Actuators Part #: C27051-29 Supplier: 81205

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
512	Left Wing - Krueger Flap No. 2
513	Left Wing - Krueger Flap No. 1
612	Right Wing - Krueger Flap No. 3
613	Right Wing - Krueger Flap No. 4

D. Prepare for the Removal

SUBTASK 27-81-31-040-001

WARNING DO THE DEACTIVATION PROCEDURE TO PREVENT THE OPERATION OF THE THRUST REVERSER. THE ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the thrust reversers are activated, then deactivate them. Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-81-31-860-010

CAUTION

MAKE SURE THAT THE INBOARD FAN COWL PANEL AND THE INBOARD AND OUTBOARD THRUST REVERSERS ARE CLOSED AND IN THE STOWED POSITION OR REMOVED BEFORE YOU EXTEND THE LEADING EDGE FLAPS AND SLATS. THERE IS NOT SUFFICIENT CLEARANCE FOR THE FLAPS AND SLATS TO EXTEND IF THE INBOARD FAN COWL PANEL AND THE INBOARD AND OUTBOARD THRUST REVERSERS ARE IN THE OPEN POSITION. THIS CAN CAUSE DAMAGE TO EQUIPMENT.

(2) Make sure that the inboard fan duct cowl and the inboard and outboard thrust reversers are closed in the stowed position or are removed.

SUBTASK 27-81-31-860-011

CAUTION

MAKE SURE THAT THE TRAILING EDGE FLAPS ARE FULLY RETRACTED. IF THEY ARE NOT FULLY RETRACTED, THE ACTUATOR WILL EXTEND. THE ACTUATOR CAN CAUSE DAMAGE TO THE WING RIB.

(3) If the trailing edge flaps are not fully retracted, then retract them. Do this task: Retract the Trailing Edge Flaps, TASK 27-51-00-860-804.

SUBTASK 27-81-31-860-002

(4) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	15	C01081	HYDRAULIC SYSTEM PTU VALVE CONT 1
A	16	C01085	HYDRAULIC SYSTEM PTU VALVE CONT 2

SUBTASK 27-81-31-860-012

WARNING

MAKE SURE ALL PERSONS AND EQUIPMENT ARE CLEAR OF THE FLIGHT CONTROLS SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(5) Pressurize the hydraulic system B. Do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 27-81-31-860-003

WARNING

MAKE SURE PERSONS AND EQUIPMENT ARE CLEAR OF THE LEADING EDGE AND TRAILING EDGE CONTROL SURFACES. THE LEADING EDGE AND TRAILING EDGE CONTROL SURFACES CAN EXTEND AND RETRACT QUICKLY. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(6) Move the flap control lever to the 1-unit position to extend the leading edge flaps.

SUBTASK 27-81-31-860-013

(7) Remove power from the hydraulic system B. Do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

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SUBTASK 27-81-31-480-001

WARNING MAKE SURE TO INSTALL THE LEADING EDGE FLAP AND SLAT ACTUATORS LOCKOUT SET TO PREVENT ACCIDENTAL OPERATION OF THE LEADING EDGE FLAPS AND SLATS. THE LEADING EDGE FLAPS AND SLATS CAN MOVE QUICKLY. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(8) Install the lockout set, SPL-1734, on the actuator for the adjacent leading edge flap. Do this task: Leading Edge Flap and Slat Locks Installation, TASK 27-81-00-480-801.

E. Leading Edge Flap Actuator Removal

SUBTASK 27-81-31-020-001

(1) Disconnect the leading edge flap actuator from the leading edge flap. Do the steps for the applicable actuator:

(a) If you are removing the outboard leading edge flap actuator, then do these steps (View E, Figure 401):

1) Remove the cotter pin [19].

a) Discard the cotter pin [19].

2) Remove the nut [24] and the washer [23].

3) Remove the bolt [18], the washer [20], and the bushing [21].

4) Remove the actuator [11] and the washer [22] from the leading edge flap [1].

(b) If you are removing the inboard leading edge flap actuator, then do these steps (View F, Figure 401):

1) Remove the cotter pin [26].

a) Discard the cotter pin [26].

2) Remove the nut [30] and the washer [29].

3) Remove the bolt [25], the washer [27], and the bushing [28].

4) Remove the actuator [12] and the washer [22] from the leading edge flap [2].

SUBTASK 27-81-31-860-014

WARNING MAKE SURE ALL PERSONS AND EQUIPMENT ARE CLEAR OF THE FLIGHT CONTROLS SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(2) Pressurize the hydraulic system B. Do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 27-81-31-860-004

(3) Move the flap control lever to the UP position to retract the actuator [11] or the actuator [12].

SUBTASK 27-81-31-860-017

(4) For the hydraulic system B, do this task: Hydraulic Reservoirs Depressurization, TASK 29-09-00-860-802.

SUBTASK 27-81-31-020-002

(5) Remove the leading edge flap actuator from the airplane. Do these steps (Views B, C, and D, Figure 401):

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Disconnect the three hydraulic lines from the hydraulic fittings. Do these steps:

1) Attach identification tags or labels to the hydraulic lines and to their related hydraulic fittings for correct connections during the installation procedure.

2) Remove the four screws [6], the four nuts [7], and two clamps [5] that attach the hydraulic line [4] to the bracket [8].

3) Disconnect the hydraulic line [4], the hydraulic line [9], and the hydraulic line [10] from the three hydraulic fittings [3].

4) Install plugs in the open ends of the hydraulic lines.

5) Install caps in the three hydraulic fittings [3].

(b) Remove the nut [17] and the washer [16].

(c) Remove the bolt [13], the washer [14], and the bushing [15].

(d) Remove the actuator [11] or the actuator [12] from the structure.

1) Remove the hydraulic line [4], the hydraulic line [9], and the hydraulic line [10] with the actuator [11] or the actuator [12].

SUBTASK 27-81-31-020-003

(6) If you will install a new leading edge flap actuator, then remove the hydraulic lines from the old actuator. Do these steps for the applicable actuator:

(a) Attach identification tags or labels to the hydraulic lines and to their related hydraulic ports for correct connections during the installation procedure.

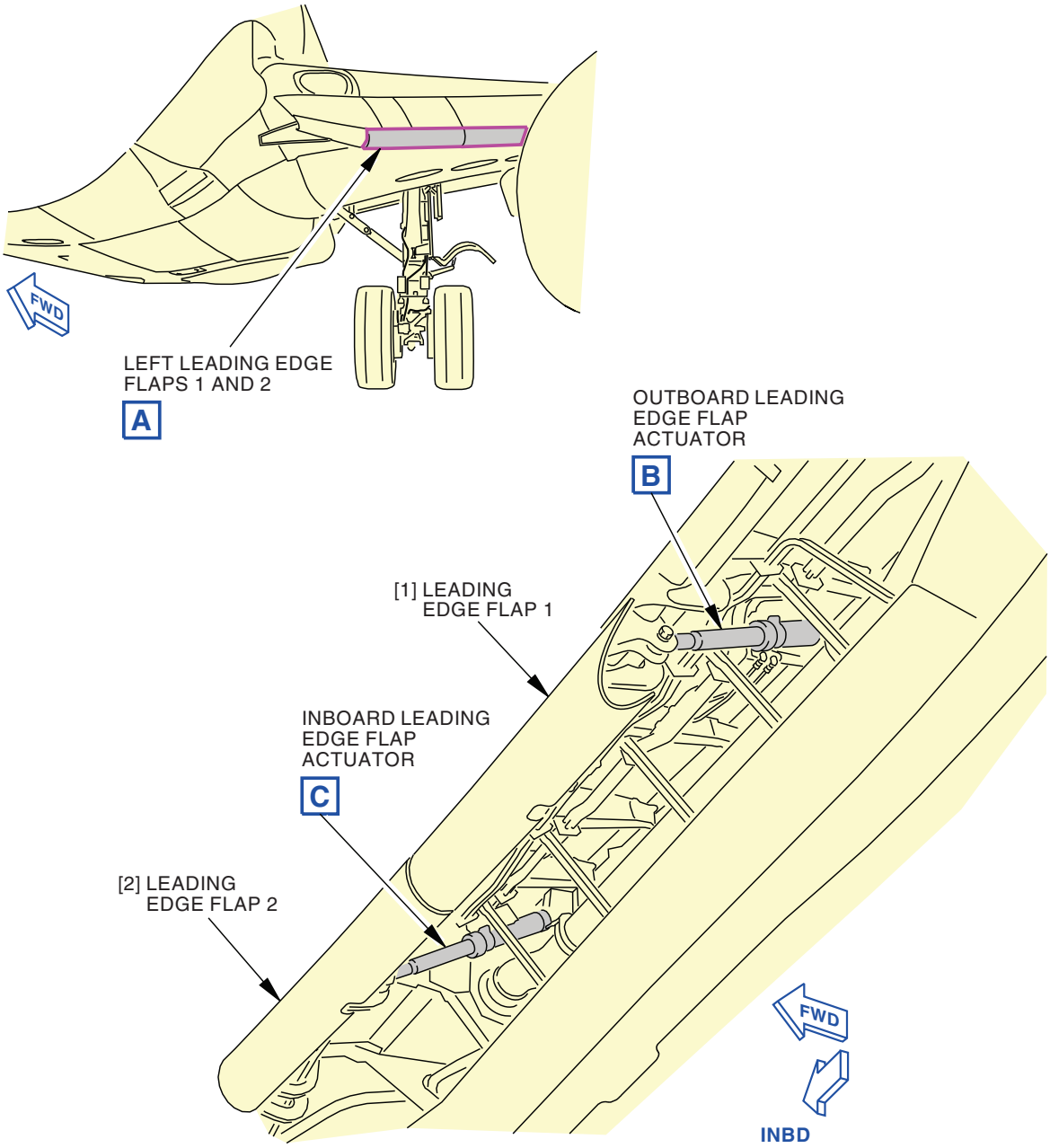
(b) Remove the hydraulic line [4], the hydraulic line [9], and the hydraulic line [10] from the actuator [11] or the actuator [12].

(c) Install plugs in the open ends of the hydraulic lines.

(d) Install caps in the hydraulic ports that are on the actuator [11] or the actuator [12].

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

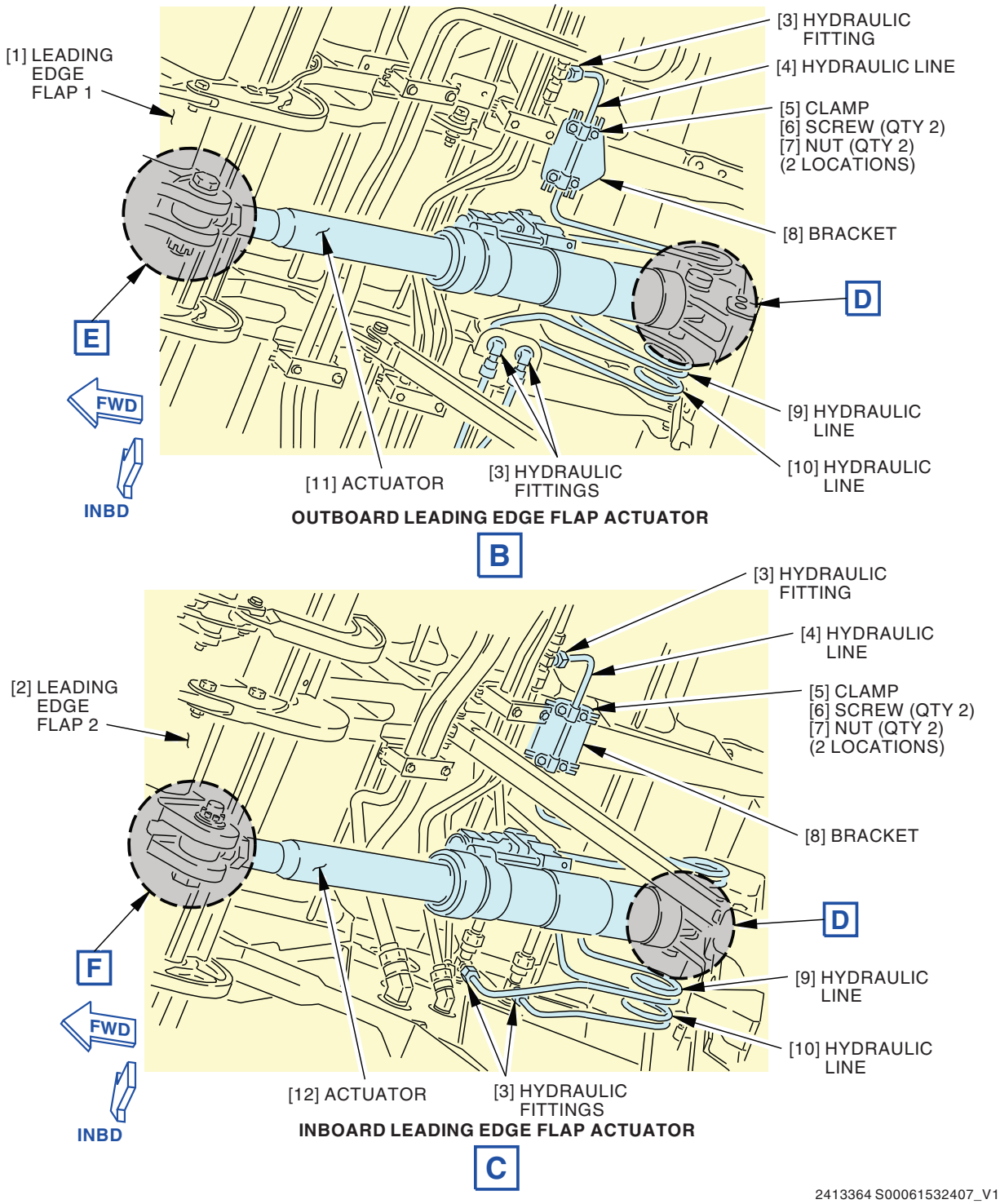


LEFT LEADING EDGE FLAPS 1 AND 2
(RIGHT LEADING EDGE FLAPS 3 AND 4 ARE OPPOSITE)
A

2413363 S00061532406_V2

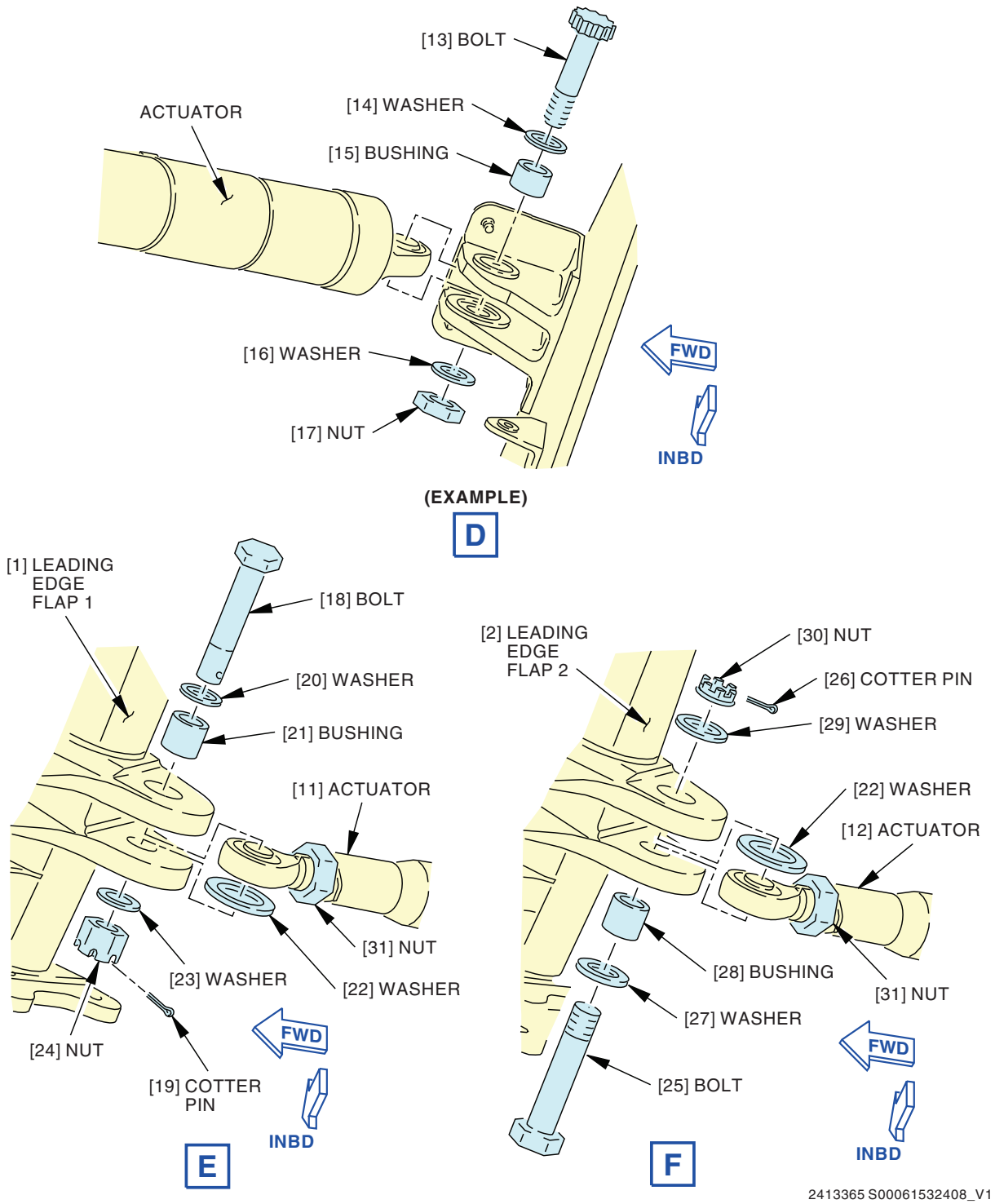
Leading Edge Flap Actuator Installation
Figure 401/27-81-31-990-801 (Sheet 1 of 3)

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Leading Edge Flap Actuator Installation
Figure 401/27-81-31-990-801 (Sheet 2 of 3)

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Leading Edge Flap Actuator Installation
Figure 401/27-81-31-990-801 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-81-31-400-801

3. Leading Edge Flap Actuator Installation

(Figure 401)

A. References

Reference	Title
20-10-44-400-801	Lockwire, Cotter Pins, and Lockrings - Installation (P/B 401)
27-51-00-860-804	Retract the Trailing Edge Flaps (P/B 201)
27-81-00-080-801	Leading Edge Flap and Slat Locks Removal (P/B 201)
27-81-00-700-802	Leading Edge Flap and Slat Timing Test in Primary Mode (P/B 501)
27-81-00-710-801	Leading Edge Flap and Slat System Standby Actuation Operational Test (P/B 501)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1734	Lockout Set - Leading Edge Slats and Krueger Flaps Actuators Part #: C27051-29 Supplier: 81205

C. Consumable Materials

Reference	Description	Specification
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95
D00633	Grease - Aircraft General Purpose	BMS3-33
G50347	Lockwire - MS20995NC32, Monel - 0.032 Inch (0.8128 mm) Diameter	NASM20995

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
19	Cotter pin	27-81-11-01-135	PTF ALL
26	Cotter pin	27-81-11-01-135	PTF ALL

E. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
512	Left Wing - Krueger Flap No. 2
513	Left Wing - Krueger Flap No. 1
612	Right Wing - Krueger Flap No. 3
613	Right Wing - Krueger Flap No. 4

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Prepare for the Installation

SUBTASK 27-81-31-040-002

WARNING DO THE DEACTIVATION PROCEDURE TO PREVENT THE OPERATION OF THE THRUST REVERSER. THE ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the thrust reversers are activated, then deactivate them. Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

SUBTASK 27-81-31-860-015

CAUTION MAKE SURE THAT THE INBOARD FAN COWL PANEL AND THE INBOARD AND OUTBOARD THRUST REVERSERS ARE CLOSED AND IN THE STOWED POSITION OR REMOVED BEFORE YOU EXTEND THE LEADING EDGE FLAPS AND SLATS. THERE IS NOT SUFFICIENT CLEARANCE FOR THE FLAPS AND SLATS TO EXTEND IF THE INBOARD FAN COWL PANEL AND THE INBOARD AND OUTBOARD THRUST REVERSERS ARE IN THE OPEN POSITION. THIS CAN CAUSE DAMAGE TO EQUIPMENT.

(2) Make sure that the inboard fan duct cowl and the inboard and outboard thrust reversers are closed in the stowed position or are removed.

SUBTASK 27-81-31-860-016

CAUTION MAKE SURE THAT THE TRAILING EDGE FLAPS ARE FULLY RETRACTED. IF THEY ARE NOT FULLY RETRACTED, THE ACTUATOR WILL EXTEND. THE ACTUATOR CAN CAUSE DAMAGE TO THE WING RIB.

(3) If the trailing edge flaps are not fully retracted, then retract them. Do this task: Retract the Trailing Edge Flaps, TASK 27-51-00-860-804.

G. Leading Edge Flap Actuator Installation

SUBTASK 27-81-31-820-002

(1) Make sure that the rod end of the leading edge flap actuator that you will install is the same length as the rod end of the actuator that you removed.

(a) If the lengths are not the same, then adjust the rod end length of the actuator that you will install. Do these steps:

1) Remove the lockwire from the nut [31].

a) Discard the lockwire.

2) Loosen the nut [31].

3) Adjust the rod end of the actuator [11] or the actuator [12] to the correct length.

4) Tighten the nut [31] from 550 in-lb (62 N·m) to 650 in-lb (73 N·m).

5) Safety the nut [31] with a new MS20995NC32 lockwire, G50347. Do this task: Lockwire, Cotter Pins, and Lockrings - Installation, TASK 20-10-44-400-801.

SUBTASK 27-81-31-420-001

(2) If the hydraulic lines are not installed on the leading edge flap actuator, then connect the hydraulic lines to the actuator. Do these steps:

(a) Remove the caps from the hydraulic ports that are on the actuator [11] or the actuator [12].

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(b) Remove the plugs from the end of each hydraulic line that will attach to the actuator [11] or the actuator [12].

(c) Connect the hydraulic line [4], the hydraulic line [9], and the hydraulic line [10] to the hydraulic ports that are on the actuator [11] or the actuator [12].

(d) Tighten the B-nuts that are on the hydraulic lines to 140 ±7 in-lb (16 ±1 N·m).

(e) Safety each B-nut with a new MS20995NC32 lockwire, G50347. Do this task: Lockwire, Cotter Pins, and Lockrings - Installation, TASK 20-10-44-400-801.

(f) Remove the identification tags or labels from the hydraulic lines and from their related hydraulic ports that are on the actuator.

SUBTASK 27-81-31-420-005

(3) Install the leading edge flap actuator. Do these steps (Views B, C, and D, Figure 401):

(a) Put the actuator [11] or the actuator [12] in its position on the fitting that is on the structure.

(b) If you are installing a leading edge flap actuator in the left wing, then do this step:

1) Install the bushing [15], the washer [14], and the bolt [13] on the outboard side of the structure fitting.

(c) If you are installing a leading edge flap actuator in the right wing, then do these steps:

1) Install the bushing [15] on the inboard side of the structure fitting.

2) Install the washer [14] and the bolt [13] on the inboard side or the outboard side of the structure fitting.

NOTE: The washer [14] must be installed next to the head of the bolt [13].

(d) Install the washer [16] and the nut [17].

1) Tighten the nut [17] from 660 in-lb (75 N·m) to 980 in-lb (111 N·m).

SUBTASK 27-81-31-420-006

(4) Connect the hydraulic lines to the hydraulic fittings. Do these steps (Views B and C, Figure 401):

(a) Remove the plugs from the hydraulic lines.

(b) Remove the caps from the three hydraulic fittings [3].

(c) Connect the hydraulic line [4], the hydraulic line [9], and the hydraulic line [10] to the three hydraulic fittings [3].

(d) Tighten the B-nuts that are on the hydraulic lines to 140 ±7 in-lb (16 ±1 N·m).

(e) Safety each B-nut with a new MS20995NC32 lockwire, G50347. Do this task: Lockwire, Cotter Pins, and Lockrings - Installation, TASK 20-10-44-400-801.

(f) Remove the identification tags or labels from the hydraulic lines and from their related hydraulic fittings.

SUBTASK 27-81-31-420-007

(5) Attach the hydraulic line [4] to the bracket [8]. Do these steps:

(a) Put the hydraulic line [4] in its position on the bracket [8].

(b) Put the two clamps [5] in their positions on the hydraulic line [4] and the bracket [8].

(c) Install the four screws [6] and the four nuts [7].

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SUBTASK 27-81-31-860-006

WARNING MAKE SURE ALL PERSONS AND EQUIPMENT ARE CLEAR OF THE FLIGHT CONTROLS SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(6) Pressurize the hydraulic system B. Do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 27-81-31-860-007

WARNING MAKE SURE PERSONS AND EQUIPMENT ARE CLEAR OF THE LEADING EDGE AND TRAILING EDGE CONTROL SURFACES. THE LEADING EDGE AND TRAILING EDGE CONTROL SURFACES CAN EXTEND AND RETRACT QUICKLY. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(7) Move the flap control lever to the 1-unit position to extend the actuator [11] or the actuator [12].

SUBTASK 27-81-31-860-008

(8) Remove power from the hydraulic system B. Do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 27-81-31-420-008

(9) Connect the leading edge flap actuator to the leading edge flap. Do the steps for the applicable actuator:

(a) If you are installing the outboard leading edge flap actuator, then do these steps (View E, Figure 401):

1) Apply the grease, D00633, on the mating surfaces of the bolt [18].
2) Put the actuator [11] and the washer [22] in their positions on the fitting of the leading edge flap [1].
a) You can bond the washer [22] to the leading edge flap [1] with the sealant, A00247, for easier installation.
3) Install the bushing [21], the washer [20], and the bolt [18].
4) Install the washer [23] and the nut [24].
a) Tighten the nut [24] from 660 in-lb (75 N·m) to 980 in-lb (111 N·m).
<1> Align a slot that is on the nut [24] with the cotter pin hole that is in the bolt [18].
5) Install a new cotter pin [19] in the bolt [18] hole.

(b) If you are installing the inboard leading edge flap actuator, then do these steps (View F, Figure 401):

1) Apply the grease, D00633, on the mating surfaces of the bolt [25].
2) Put the actuator [12] and the washer [22] in their positions on the fitting of the leading edge flap [2].
a) You can bond the washer [22] to the leading edge flap [2] with the sealant, A00247, for easier installation.
3) Install the bushing [28], the washer [27], and the bolt [25].
4) Install the washer [29] and the nut [30].

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a) Tighten the nut [30] from 660 in-lb (75 N·m) to 980 in-lb (111 N·m).

<1> Align a slot that is on the nut [30] with the cotter pin hole that is in the bolt [25].

5) Install a new cotter pin [26] in the bolt [25] hole.

SUBTASK 27-81-31-080-001

WARNING CAREFULLY REMOVE THE LEADING EDGE FLAP AND SLAT ACTUATORS LOCKOUT SET. THE LEADING EDGE FLAPS AND SLATS CAN MOVE QUICKLY IF YOU DO NOT REMOVE THE LOCKS CORRECTLY. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(10) Carefully remove each lockout set, SPL-1734, that is installed. Do this task: Leading Edge Flap and Slat Locks Removal, TASK 27-81-00-080-801.

SUBTASK 27-81-31-790-001

(11) Do a check for hydraulic leaks. Do these steps:

WARNING MAKE SURE ALL PERSONS AND EQUIPMENT ARE CLEAR OF THE FLIGHT CONTROLS SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(a) Pressurize the hydraulic system B. Do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

WARNING MAKE SURE PERSONS AND EQUIPMENT ARE CLEAR OF THE LEADING EDGE AND TRAILING EDGE CONTROL SURFACES. THE LEADING EDGE AND TRAILING EDGE CONTROL SURFACES CAN EXTEND AND RETRACT QUICKLY. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

CAUTION CAREFULLY MONITOR THE FLAP-NOSE CLEARANCE AND THE ACTUATOR CLEARANCES DURING THE FLAP EXTENSION AND RETRACTION. IF THERE IS NOT SUFFICIENT CLEARANCE, YOU CAN CAUSE DAMAGE TO THE STRUCTURE.

(b) Operate the leading edge flaps for a minimum of two cycles to bleed air from the system.

1) Leave the flaps in the extended position.

(c) Make sure that there are no hydraulic leaks from the actuator [11] or the actuator [12].

(d) Make sure that there are no hydraulic leaks from the hydraulic lines.

WARNING MAKE SURE PERSONS AND EQUIPMENT ARE CLEAR OF THE LEADING EDGE AND TRAILING EDGE CONTROL SURFACES. THE LEADING EDGE AND TRAILING EDGE CONTROL SURFACES CAN EXTEND AND RETRACT QUICKLY. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(e) Move the flap control lever to the UP position to retract the leading edge flaps.

(f) Remove power from the hydraulic system B. Do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

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SUBTASK 27-81-31-720-001

(12) Do this task: Leading Edge Flap and Slat Timing Test in Primary Mode, TASK 27-81-00-700-802.

SUBTASK 27-81-31-710-002

(13) Do this task: Leading Edge Flap and Slat System Standby Actuation Operational Test, TASK 27-81-00-710-801.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 27-81-31-860-009

(1) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	15	C01081	HYDRAULIC SYSTEM PTU VALVE CONT 1
A	16	C01085	HYDRAULIC SYSTEM PTU VALVE CONT 2

SUBTASK 27-81-31-440-001

(2) Activate the thrust reversers. Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

LEADING EDGE FLAP ACTUATOR - INSPECTION/CHECK

1. General

A. This procedure contains the task to do a check of the leading edge flap actuator.

TASK 27-81-31-200-801

2. Leading Edge Flap Actuator Inspection

(Figure 601)

A. General

(1) This task gives the wear limits and the repair data for the leading edge flap actuator inspection.

B. References

Reference	Title
27-81-31-000-801	Leading Edge Flap Actuator Removal (P/B 401)
27-81-31-400-801	Leading Edge Flap Actuator Installation (P/B 401)

C. Tools/Equipment

Reference	Description
STD-1096	Micrometer - Depth, 0-1 inch, Readable to 1/1000 inch
STD-1097	Caliper - Vernier, 0-6 inch, Readable to 1/1000 inch

D. Location Zones

Zone	Area
512	Left Wing - Krueger Flap No. 2
513	Left Wing - Krueger Flap No. 1
612	Right Wing - Krueger Flap No. 3
613	Right Wing - Krueger Flap No. 4

E. Procedure

SUBTASK 27-81-31-020-004

(1) Do this task: Leading Edge Flap Actuator Removal, TASK 27-81-31-000-801.

SUBTASK 27-81-31-220-002

(2) Examine the parts of the leading edge flap actuator for worn areas:

(a) Remove the parts that you want to do a check.

(b) Measure the parts with a micrometer (0-1 inch, readable to 1/1000 inch), STD-1096, or a vernier caliper (0 - 6 inch, readable to 1/1000 inch), STD-1097.

(c) Compare the dimensions that you measured with the permitted wear dimensions that are shown in Table 601.

(d) If the parts are out of tolerance, replace or repair the parts if it is necessary.

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Table 601/27-81-31-993-801

NAME OF THE PARTS THAT ARE IN CONTACT	DIM.	DIAMETER DESIGN LIMITS		WEAR LIMITS	
		MINIMUM inches/mm	MAXIMUM inches/mm	PERMITTED MAXIMUM I.D./ MINIMUM O.D. OF WORN PART	MAXIMUM SERVICE DIAMETRIC CLEARANCE
#1 BUSHING	I.D.	0.6245 in. (15.862 mm)	0.6252 in. (15.880 mm)	0.6258 in. (15.895 mm)	0.0033 in. (0.084 mm)
#1 BOLT	O.D.	0.6230 in. (15.824 mm)	0.6240 in. (15.850 mm)	0.6225 in. (15.811 mm)	
#2 BUSHING	I.D.	0.8745 in. (22.212 mm)	0.8753 in. (22.233 mm)	0.8757 in. (22.243 mm)	0.0027 in. (0.069 mm)
#2 BUSHING	O.D.	0.8735 in. (22.187 mm)	0.8740 in. (22.200 mm)	0.8730 in. (22.174 mm)	
#3 BUSHING	I.D.	0.6245 in. (15.862 mm)	0.6250 in. (15.875 mm)	0.6255 in. (15.888 mm)	0.0030 in. (0.076 mm)
#3 BOLT	O.D.	0.6230 in. (15.824 mm)	0.6240 in. (15.850 mm)	0.6225 in. (15.811 mm)	
#4 BEARING	I.D.	0.6245 in. (15.862 mm)	0.6250 in. (15.875 mm)	0.6255 in. (15.888 mm)	0.0030 in. (0.076 mm)
#4 BOLT	O.D.	0.6230 in. (15.824 mm)	0.6240 in. (15.850 mm)	0.6225 in. (15.811 mm)	
#5 BUSHING	I.D.	0.6245 in. (15.862 mm)	0.6250 in. (15.875 mm)	0.6255 in. (15.888 mm)	0.0030 in. (0.076 mm)
#5 BOLT	O.D.	0.6230 in. (15.824 mm)	0.6240 in. (15.850 mm)	0.6225 in. (15.811 mm)	
#6 BEARING	I.D.	0.6245 in. (15.862 mm)	0.6250 in. (15.875 mm)	0.6255 in. (15.888 mm)	0.0030 in. (0.076 mm)
#6 BOLT	O.D.	0.6230 in. (15.824 mm)	0.6240 in. (15.850 mm)	0.6225 in. (15.811 mm)	
#7 BUSHING	I.D.	0.6245 in. (15.862 mm)	0.6252 in. (15.880 mm)	0.6258 in. (15.895 mm)	0.0033 in. (0.084 mm)
#7 BOLT	O.D.	0.6230 in. (15.824 mm)	0.6240 in. (15.850 mm)	0.6225 in. (15.811 mm)	

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Table 601/27-81-31-993-801 (Continued)

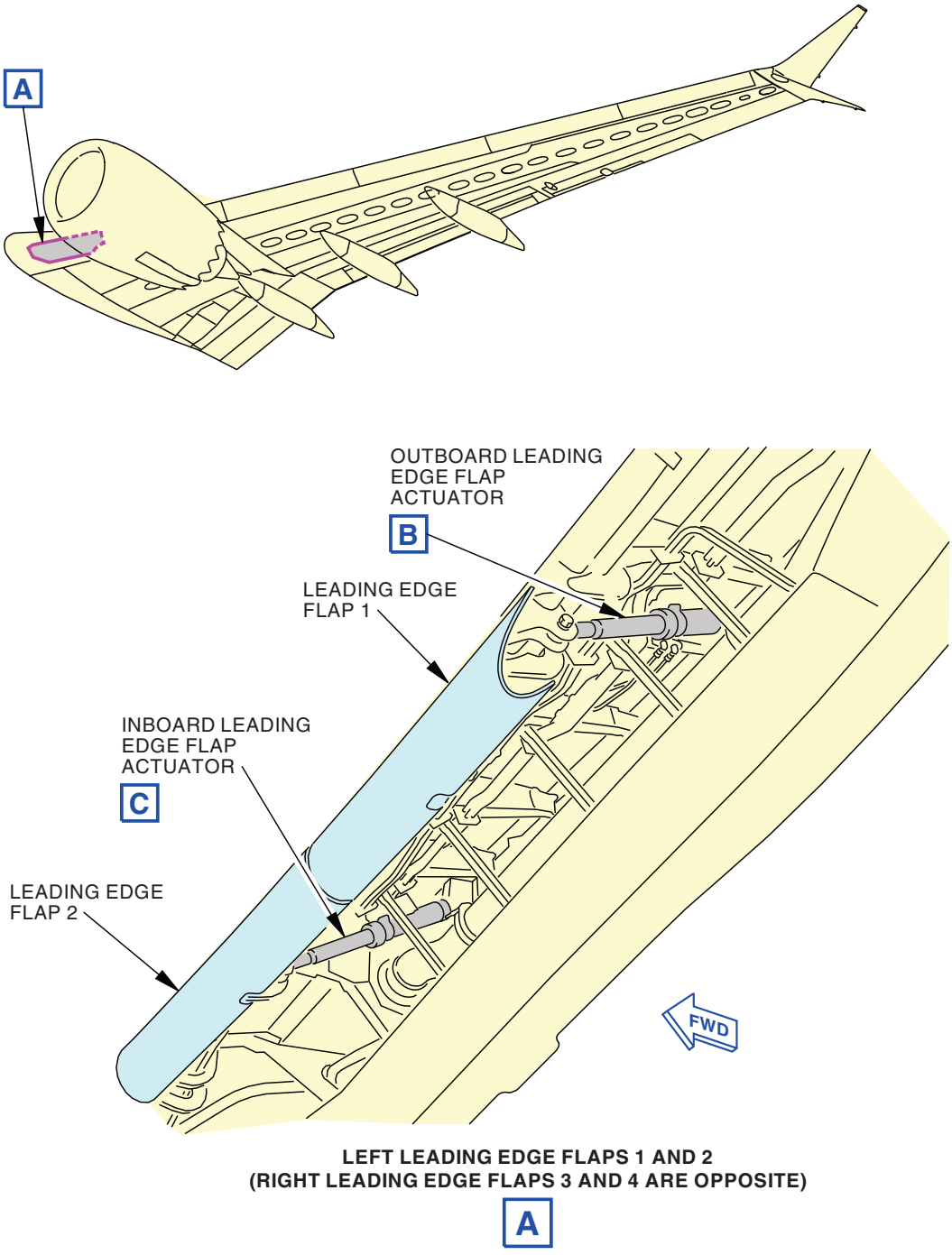
NAME OF THE PARTS THAT ARE IN CONTACT	DIM.	DIAMETER DESIGN LIMITS		WEAR LIMITS	
		MINIMUM inches/mm	MAXIMUM inches/mm	PERMITTED MAXIMUM I.D./ MINIMUM O.D. OF WORN PART	MAXIMUM SERVICE DIAMETRIC CLEARANCE
#8 BUSHING	I.D.	0.8745 in. (22.212 mm)	0.8753 in. (22.233 mm)	0.8757 in. (22.243 mm)	0.0027 in. (0.069 mm)
#8 BUSHING	O.D.	0.8735 in. (22.187 mm)	0.8740 in. (22.200 mm)	0.8730 in. (22.174 mm)	
#9 BUSHING	I.D.	0.6245 in. (15.862 mm)	0.6252 in. (15.880 mm)	0.6258 in. (15.895 mm)	0.0033 in. (0.084 mm)
#9 BOLT	O.D.	0.6230 in. (15.824 mm)	0.6240 in. (15.850 mm)	0.6225 in. (15.811 mm)	
#10 BUSHING	I.D.	0.8745 in. (22.212 mm)	0.8753 in. (22.233 mm)	0.8757 in. (22.243 mm)	0.0027 in. (0.069 mm)
#10 BUSHING	O.D.	0.8735 in. (22.187 mm)	0.8740 in. (22.200 mm)	0.8730 in. (22.174 mm)	
#11 BUSHING	I.D.	0.6245 in. (15.862 mm)	0.6250 in. (15.875 mm)	0.6255 in. (15.888 mm)	0.0030 in. (0.076 mm)
#11 BOLT	O.D.	0.6230 in. (15.824 mm)	0.6240 in. (15.850 mm)	0.6225 in. (15.811 mm)	
#12 BEARING	I.D.	0.6245 in. (15.862 mm)	0.6250 in. (15.875 mm)	0.6255 in. (15.888 mm)	0.0030 in. (0.076 mm)
#12 BOLT	O.D.	0.6230 in. (15.824 mm)	0.6240 in. (15.850 mm)	0.6225 in. (15.811 mm)	

SUBTASK 27-81-31-420-004

(3) Do this task: Leading Edge Flap Actuator Installation, TASK 27-81-31-400-801.

——— END OF TASK ———

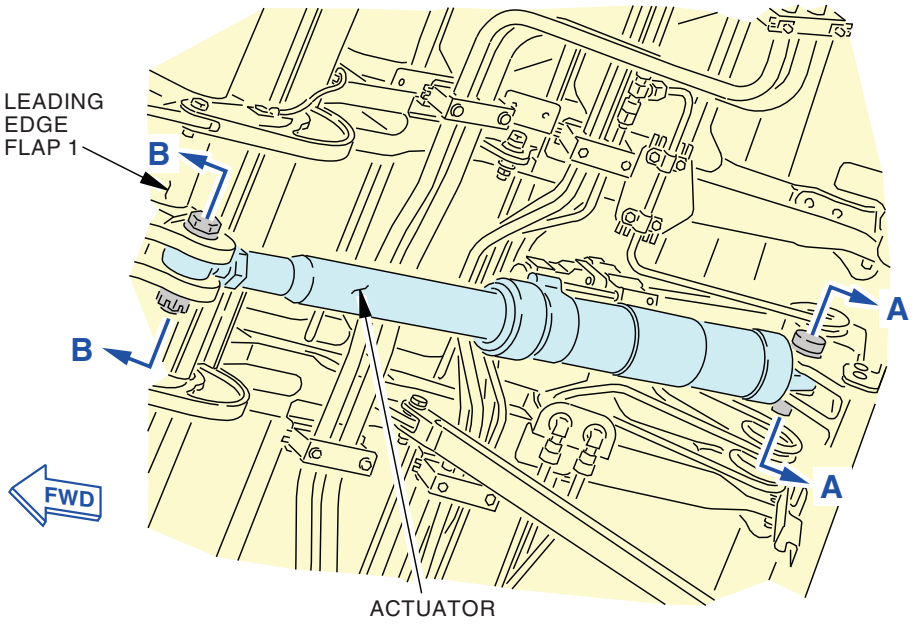
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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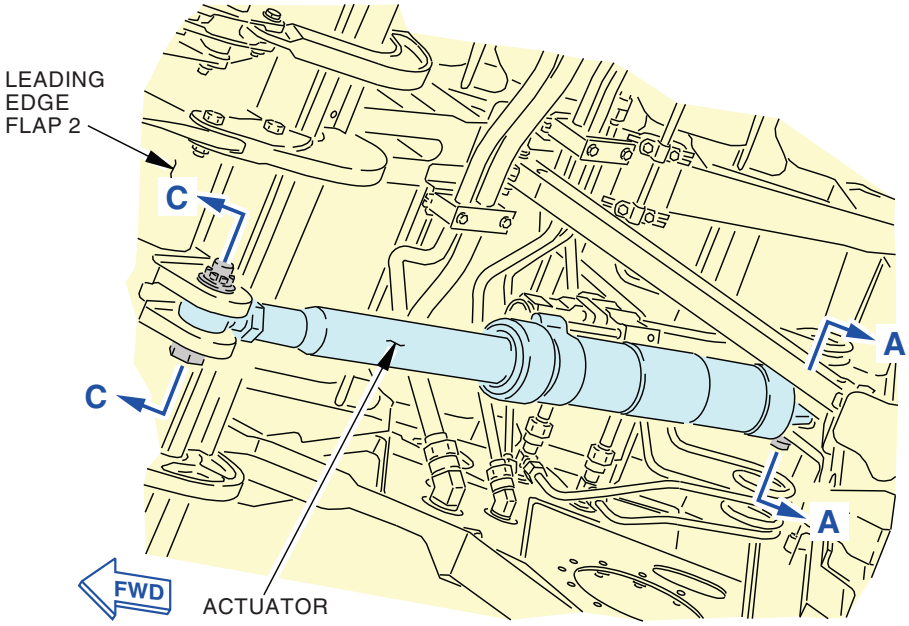
Leading Edge Flap Actuator Wear Limits
Figure 601/27-81-31-990-802 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



OUTBOARD LEADING EDGE FLAP ACTUATOR

B



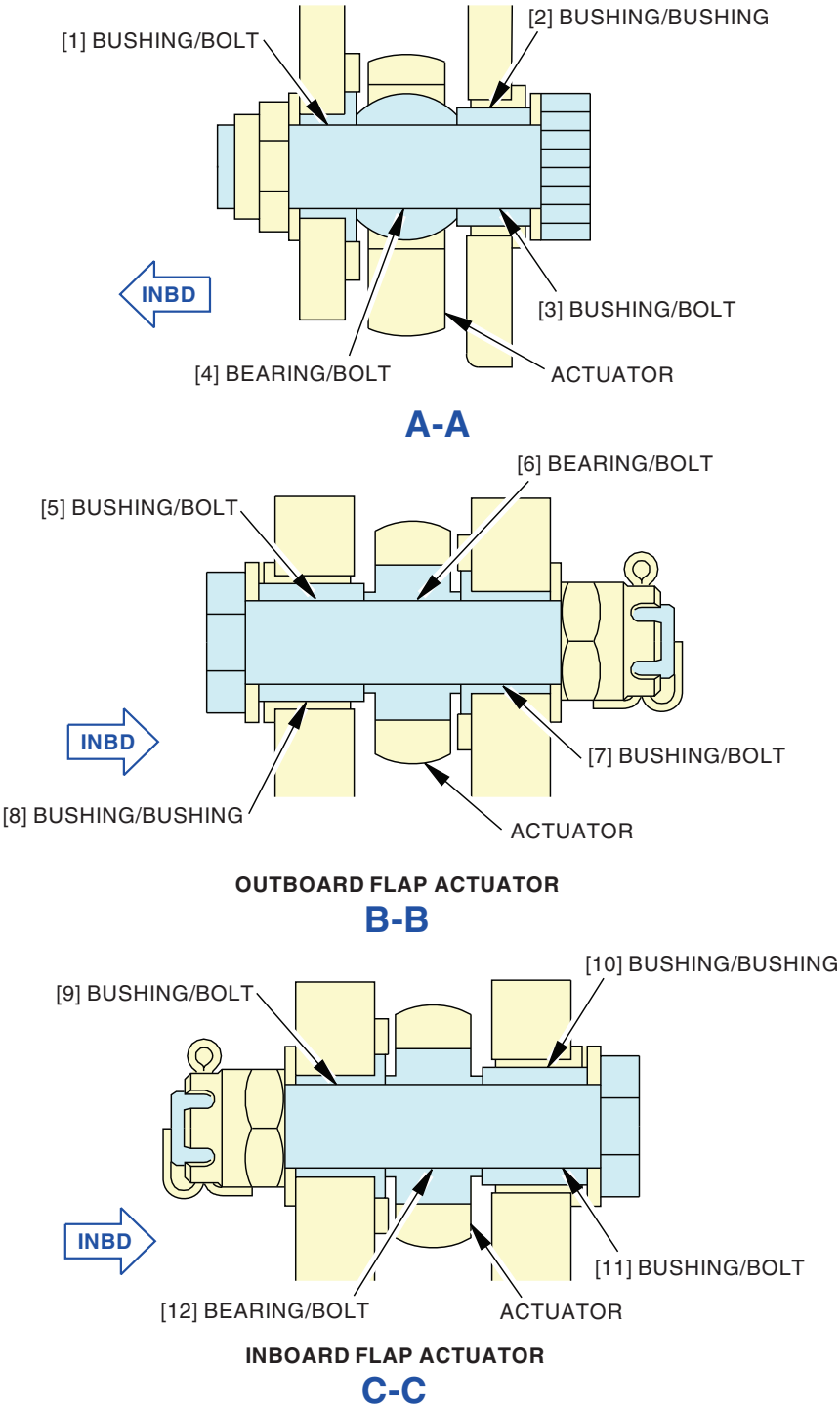
INBOARD LEADING EDGE FLAP ACTUATOR

C

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Leading Edge Flap Actuator Wear Limits
Figure 601/27-81-31-990-802 (Sheet 2 of 3)

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Leading Edge Flap Actuator Wear Limits
Figure 601/27-81-31-990-802 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

LEADING EDGE SLAT ACTUATOR - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the actuator of the leading edge slat.

(2) An installation of the actuator of the leading edge slat.

TASK 27-81-41-000-801

2. Leading Edge Slat Actuator Removal

(Figure 401)

A. References

Reference	Title
27-51-00-860-804	Retract the Trailing Edge Flaps (P/B 201)
27-81-00-480-801	Leading Edge Flap and Slat Locks Installation (P/B 201)
27-81-00-820-803	Leading Edge Slats Adjustment (P/B 501)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
57-41-02-000-801	Leading Edge Access Panel Removal (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1734	Lockout Set - Leading Edge Slats and Krueger Flaps Actuators Part #: C27051-29 Supplier: 81205

C. Location Zones

Zone	Area
521	Left Wing - Leading Edge to Front Spar
522	Left Wing - Slat No. 4
523	Left Wing - Slat No. 3
524	Left Wing - Slat No. 2
525	Left Wing - Slat No. 1
621	Right Wing - Leading Edge to Front Spar
622	Right Wing - Slat No. 5
623	Right Wing - Slat No. 6
624	Right Wing - Slat No. 7
625	Right Wing - Slat No. 8

D. Access Panels

Number	Name/Location
521CB	Lower Leading Edge Access Panel
521DB	Lower Leading Edge Access Panel
521EB	Lower Leading Edge Access Panel

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
521FB	Lower Leading Edge Access Panel
521JB	Lower Leading Edge Access Panel
521KB	Lower Leading Edge Access Panel
521LB	Lower Leading Edge Access Panel
521MB	Lower Leading Edge Access Panel
521QB	Lower Leading Edge Access Panel
521RB	Lower Leading Edge Access Panel
521SB	Lower Leading Edge Access Panel
521TB	Lower Leading Edge Access Panel
521WB	Lower Leading Edge Access Panel
521XB	Lower Leading Edge Access Panel
521YB	Lower Leading Edge Access Panel
521ZB	Lower Leading Edge Access Panel
621CB	Lower Leading Edge Access Panel
621DB	Lower Leading Edge Access Panel
621EB	Lower Leading Edge Access Panel
621FB	Lower Leading Edge Access Panel
621JB	Lower Leading Edge Access Panel
621KB	Lower Leading Edge Access Panel
621MB	Lower Leading Edge Access Panel
621QB	Lower Leading Edge Access Panel
621RB	Lower Leading Edge Access Panel
621TB	Lower Leading Edge Access Panel
621WB	Lower Leading Edge Access Panel
621XB	Lower Leading Edge Access Panel
621ZB	Lower Leading Edge Access Panel

E. Prepare for the Removal

SUBTASK 27-81-41-040-001

WARNING DO THE DEACTIVATION PROCEDURE TO PREVENT THE OPERATION OF THE THRUST REVERSER. THE ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the thrust reversers are activated, do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

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SUBTASK 27-81-41-860-013

CAUTION

MAKE SURE THAT THE INBOARD FAN COWL PANEL AND THE INBOARD AND OUTBOARD THRUST REVERSERS ARE CLOSED AND IN THE STOWED POSITION OR REMOVED BEFORE YOU EXTEND THE LEADING EDGE FLAPS AND SLATS. THERE IS NOT SUFFICIENT CLEARANCE FOR THE FLAPS AND SLATS TO EXTEND IF THE INBOARD FAN COWL PANEL AND THE INBOARD AND OUTBOARD THRUST REVERSERS ARE IN THE OPEN POSITION. THIS CAN CAUSE DAMAGE TO EQUIPMENT.

(2) Make sure that the inboard fan duct cowl and the inboard and outboard thrust reversers are closed in the stowed position or are removed.

SUBTASK 27-81-41-860-014

CAUTION

MAKE SURE THAT THE TRAILING EDGE FLAPS ARE FULLY RETRACTED. IF THEY ARE NOT FULLY RETRACTED, THE ACTUATOR WILL EXTEND. THE ACTUATOR CAN CAUSE DAMAGE TO THE WING RIB.

(3) If the trailing edge flaps are not fully retracted, do this task: Retract the Trailing Edge Flaps, TASK 27-51-00-860-804.

SUBTASK 27-81-41-860-001

(4) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	15	C01081	HYDRAULIC SYSTEM PTU VALVE CONT 1
A	16	C01085	HYDRAULIC SYSTEM PTU VALVE CONT 2

SUBTASK 27-81-41-860-002

WARNING

MAKE SURE ALL PERSONS AND EQUIPMENT ARE CLEAR OF THE FLIGHT CONTROLS SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(5) Pressurize the hydraulic system B, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 27-81-41-860-003

WARNING

MAKE SURE PERSONS AND EQUIPMENT ARE CLEAR OF THE LEADING EDGE AND TRAILING EDGE CONTROL SURFACES. THE LEADING EDGE AND TRAILING EDGE CONTROL SURFACES CAN EXTEND AND RETRACT QUICKLY. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(6) Move the flap control lever to the 30-unit position to move the leading edge slats to the full extend position.

SUBTASK 27-81-41-040-004

(7) Remove power from the hydraulic system B, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-81-41-480-001

WARNING MAKE SURE TO INSTALL THE LEADING EDGE FLAP AND SLAT ACTUATORS LOCKOUT SET TO PREVENT ACCIDENTAL OPERATION OF THE LEADING EDGE FLAPS AND SLATS. THE LEADING EDGE FLAPS AND SLATS CAN MOVE QUICKLY. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(8) Install the lockout set, SPL-1734, on the leading edge slat actuators that you will not remove, do this task: Leading Edge Flap and Slat Locks Installation, TASK 27-81-00-480-801.

SUBTASK 27-81-41-860-018

WARNING MAKE SURE ALL PERSONS AND EQUIPMENT ARE CLEAR OF THE FLIGHT CONTROLS SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(9) Pressurize the hydraulic system B, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 27-81-41-860-004

WARNING MAKE SURE PERSONS AND EQUIPMENT ARE CLEAR OF THE LEADING EDGE AND TRAILING EDGE CONTROL SURFACES. THE LEADING EDGE AND TRAILING EDGE CONTROL SURFACES CAN EXTEND AND RETRACT QUICKLY. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(10) Move the flap control lever to the 1-unit position to retract the applicable leading edge slats to the intermediate position.

NOTE: A slat actuator is preloaded against a fully extended slat. You will remove a slat actuator when it is not preloaded.

SUBTASK 27-81-41-040-002

(11) Remove power from the hydraulic system B, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 27-81-41-010-002

(12) Open the applicable access panels that are adjacent to the actuator (TASK 57-41-02-000-801).

(a) For the left wing, open these access panels:

Number	Name/Location
521DB	Lower Leading Edge Access Panel
521EB	Lower Leading Edge Access Panel
521KB	Lower Leading Edge Access Panel
521LB	Lower Leading Edge Access Panel
521RB	Lower Leading Edge Access Panel
521SB	Lower Leading Edge Access Panel
521XB	Lower Leading Edge Access Panel
521YB	Lower Leading Edge Access Panel

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(b) For the right wing, open these access panels:

Number	Name/Location
621DB	Lower Leading Edge Access Panel
621EB	Lower Leading Edge Access Panel
621JB	Lower Leading Edge Access Panel
621KB	Lower Leading Edge Access Panel
621QB	Lower Leading Edge Access Panel
621RB	Lower Leading Edge Access Panel
621WB	Lower Leading Edge Access Panel
621XB	Lower Leading Edge Access Panel

SUBTASK 27-81-41-200-001

WARNING DO NOT LET OBJECTS GET IN THE HOUSING ASSEMBLY OF THE SLAT TRACK. THIS WILL HELP PREVENT A PUNCTURE OF THE HOUSING ASSEMBLY THAT COULD CAUSE A FUEL LEAK. THE FUEL LEAK COULD CAUSE A FIRE AND POSSIBLE DEATH OR INJURY TO PERSONNEL.

(13) Keep the housing (can) assemblies of the slat main tracks clean and free of unwanted objects (FOD) at all times.

F. Leading Edge Slat Actuator Removal

SUBTASK 27-81-41-020-001

(1) Remove the slat actuator door [3] from the slat actuator [2], do these steps (Views B and D, Figure 401):

(a) Do this step for slat 1 or slat 8:

1) Remove the bolts [14], washers [15], washers [16], and nuts [17] that attach the slat actuator door [3] to the slat actuator [2].

(b) Do this step for slat 2, 3, 4, 5, 6, or 7:

1) Remove the bolts [14], washers [15], washers [16], and nuts [17] that attach the slat actuator door [3] and clamp [52] to the slat actuator [2].

(c) Remove the bolts [4], washers [5], and nuts [6] that attach the slat actuator door [3] and bracket [32] to the slat actuator [2].

(d) Remove the slat actuator door [3].

(e) Install the bolts [4], washers [5], and nuts [6] to temporarily attach the bracket [32] to the slat actuator [2].

SUBTASK 27-81-41-020-002

(2) Remove the dropout links [7], do these steps (View B, Figure 401):

(a) Do this step for slat 4 or slat 5:

1) Remove the cotter pins [13].

a) Discard the cotter pins [13].

(b) Remove the nuts [12] and washers [11].

(c) Remove the bolts [8], washers [9], and bushings [10].

(d) Remove the dropout links [7].

SUBTASK 27-81-41-020-003

(3) Disconnect the slat actuator [2] from the leading edge slat [1], do these steps (View C, Figure 401):

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(a) Remove the bolt retainer [23] from the leading edge slat [1], do these steps:

1) Remove the bolts [21] and washers [22].

2) Remove the bolt retainer [23] from the leading edge slat [1].

WARNING MAKE SURE YOU HOLD THE LEADING EDGE SLAT IN ITS POSITION. THE LEADING EDGE SLAT CAN FALL TO THE FULL EXTEND POSITION. THIS CAN CAUSE INJURY TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(b) Remove the slat actuator [2] from the leading edge slat [1] and hold the leading edge slat [1] while you do these steps:

1) Remove the nut [19] and washer [20].

2) Remove the bolt [25], washer [24], and bushing [18].

SUBTASK 27-81-41-020-006

(4) Manually move the applicable leading edge slat [1] to the 30 (full extend) position while you hold the slat actuator [2].

(a) Remove the applicable access panels:

1) For slat 1, remove these access panels:

Number	Name/Location
521WB	Lower Leading Edge Access Panel
521ZB	Lower Leading Edge Access Panel

2) For slat 2, remove these access panels:

Number	Name/Location
521QB	Lower Leading Edge Access Panel
521TB	Lower Leading Edge Access Panel

3) For slat 3, remove these access panels:

Number	Name/Location
521JB	Lower Leading Edge Access Panel
521MB	Lower Leading Edge Access Panel

4) For slat 4, remove these access panels:

Number	Name/Location
521CB	Lower Leading Edge Access Panel
521FB	Lower Leading Edge Access Panel

5) For slat 5, remove these access panels:

Number	Name/Location
621CB	Lower Leading Edge Access Panel
621FB	Lower Leading Edge Access Panel

6) For slat 6, remove these access panels:

Number	Name/Location
621JB	Lower Leading Edge Access Panel
621MB	Lower Leading Edge Access Panel

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7) For slat 7, remove these access panels:

Number	Name/Location
621QB	Lower Leading Edge Access Panel
621TB	Lower Leading Edge Access Panel

8) For slat 8, remove these access panels:

Number	Name/Location
621WB	Lower Leading Edge Access Panel
621ZB	Lower Leading Edge Access Panel

(b) Disconnect the actuator rod from the slat.

(c) Pull down on the two detent arms that are installed on each auxiliary track of the slats 1, 4, 5 or 8 with a hand pressure of 75 lbf (334 N) to 85 lbf (378 N) or with a hand pressure of 45 lbf (200 N) to 55 lbf (245 N) on the slats 2, 3, 6 or 7 Leading Edge Slats Adjustment, TASK 27-81-00-820-803.

1) Measure the applied force with a spring force scale at the throat of the cam plate.

NOTE: It is acceptable to wrap rags around the detent arms for leverage to pull down.

2) Pull the slat down to the 30 (full extend) position when the roller bearings on the auxiliary track are out of detent.

SUBTASK 27-81-41-980-002

(5) Let the slat actuator [2] hang from the support fitting [46].

SUBTASK 27-81-41-020-004

(6) Remove the slat actuator [2], do these steps (View D, Figure 401):

(a) Attach identification tags or labels to the hydraulic lines and to their related ports that are on the swivels [31] for correct connections during the installation task.

(b) Do this step for slat 1 or slat 8:

1) Disconnect the hydraulic line [29], hydraulic line [30], hydraulic line [40], and hydraulic line [41] from the swivels [31].

NOTE: The hydraulic lines, bracket, and clamps are removed from the airplane with the slat actuator.

(c) Do this step for slat 2, 3, 4, 5, 6, or 7:

1) Disconnect the hydraulic line [53], hydraulic line [54], hydraulic line [40], and hydraulic line [41] from the swivels [31].

NOTE: The hydraulic lines, bracket, and clamps are removed from the airplane with the slat actuator.

(d) Install caps on the hydraulic ports that are on the swivels [31].

(e) Install plugs in the open ends of the hydraulic lines.

(f) Remove the nut [47] and washer [48] from the bolt [44].

1) Discard the nut [47].

(g) Hold the slat actuator [2] while you remove the bolt [44], washer [45], and bushing [43].

(h) Remove the slat actuator [2] from the airplane.

SUBTASK 27-81-41-020-005

(7) If it is necessary, do the following steps to remove the hydraulic lines and bracket [32] from the slat actuator [2]:

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(a) Attach identification tags or labels to the hydraulic lines and to their related ports that are on the slat actuator [2] for correct connections during the installation task.

(b) Do this step for slat 1 or slat 8:

1) Disconnect the hydraulic line [29], hydraulic line [30], hydraulic line [40], and hydraulic line [41] from the slat actuator [2].

(c) Do this step for slat 2, 3, 4, 5, 6, or 7:

1) Disconnect the hydraulic line [53], hydraulic line [54], hydraulic line [40], and hydraulic line [41] from the slat actuator [2].

(d) Install caps on the hydraulic ports that are on the slat actuator [2].

(e) Install plugs in the open ends of the hydraulic lines.

(f) Remove the nut [28], bolt [27] and clamps [26] from the hydraulic lines.

(g) Do this step for slat 2, 3, 4, 5, 6, or 7:

1) Remove the nut [51], washer [50] and the clamp [52].

(h) Remove the nuts [39], washers [38], bolts [37], and clamps [36] that attach the hydraulic lines to the bracket [32].

(i) Remove the nuts [6], washers [5], and bolts [4] that attach the bracket [32] to the slat actuator [2].

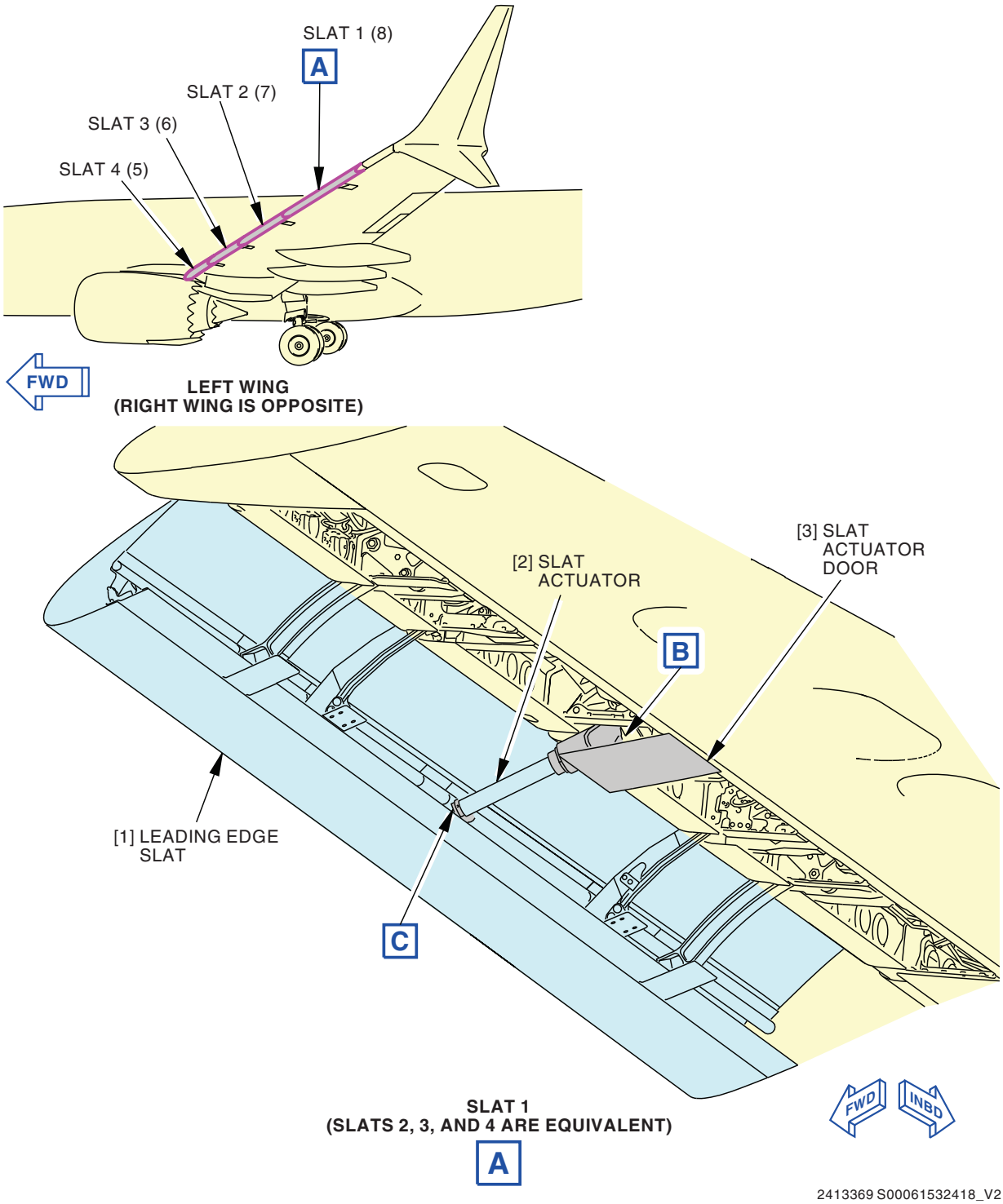
SUBTASK 27-81-41-200-002

WARNING DO NOT LET OBJECTS GET IN THE HOUSING ASSEMBLY OF THE SLAT TRACK. THIS WILL HELP PREVENT A PUNCTURE OF THE HOUSING ASSEMBLY THAT COULD CAUSE A FUEL LEAK. THE FUEL LEAK COULD CAUSE A FIRE AND POSSIBLE DEATH OR INJURY TO PERSONNEL.

(8) Keep the housing (can) assemblies of the slat main tracks clean and free of unwanted objects (FOD) at all times.

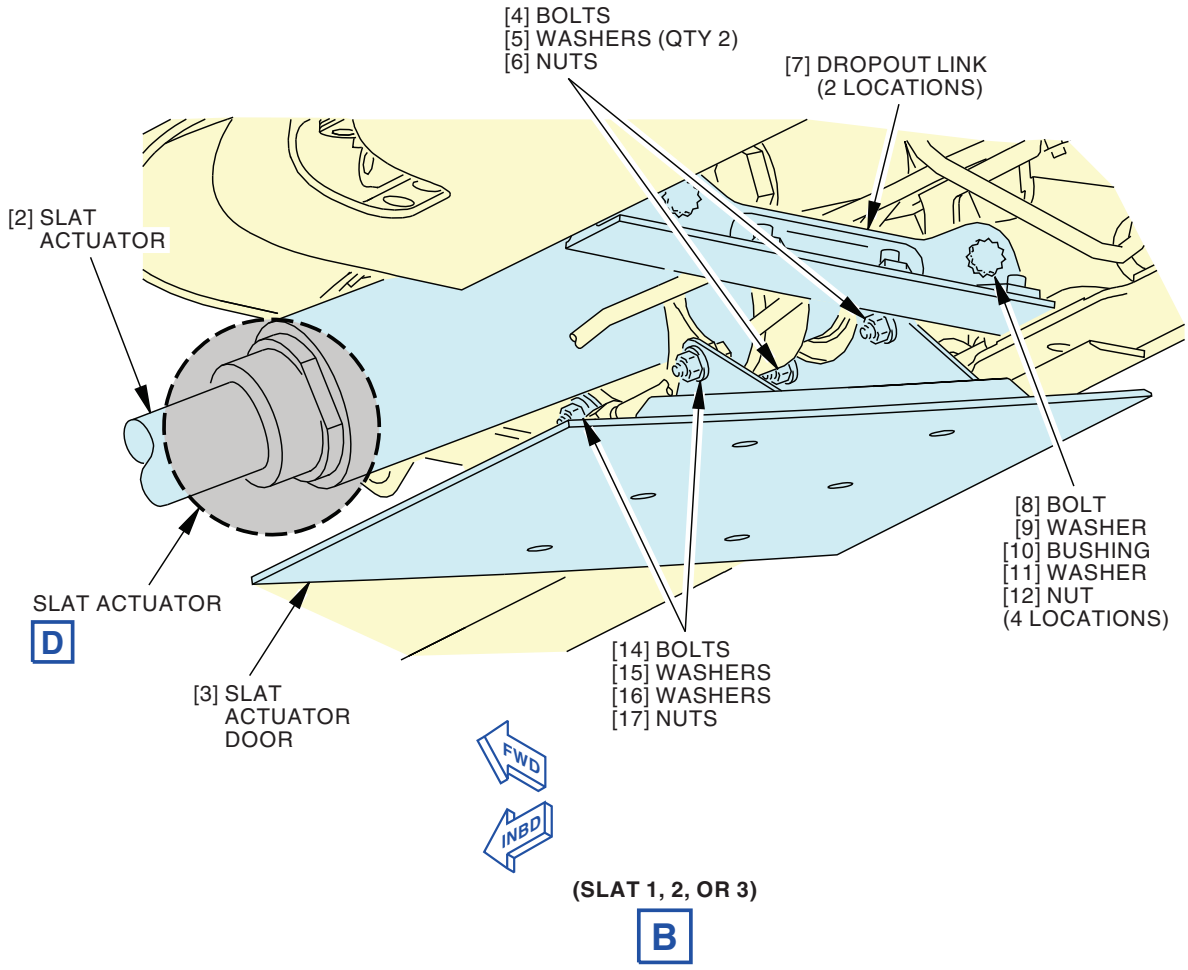
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Leading Edge Slat Actuator Installation
Figure 401/27-81-41-990-801 (Sheet 1 of 5)

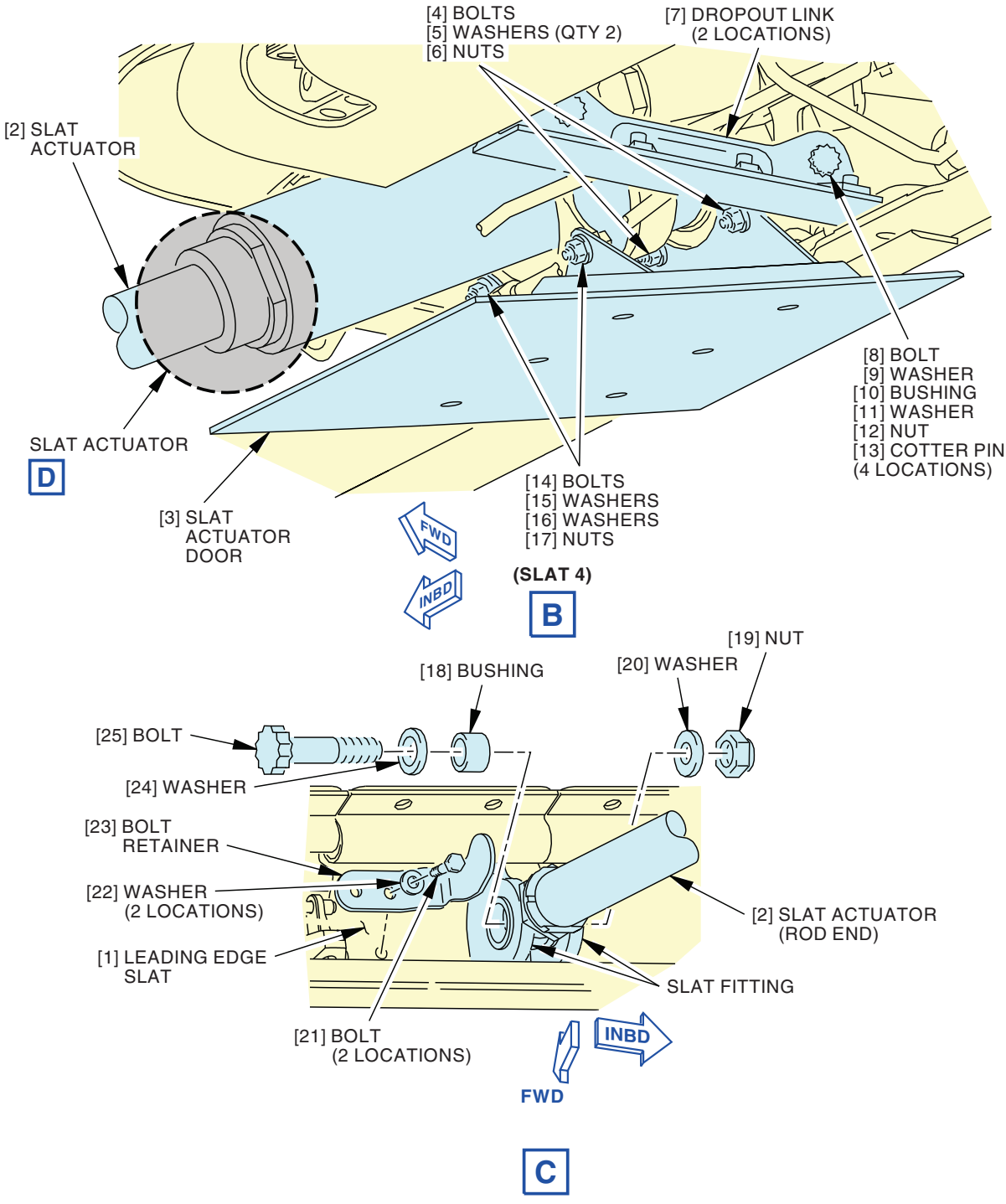
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Leading Edge Slat Actuator Installation
Figure 401/27-81-41-990-801 (Sheet 2 of 5)

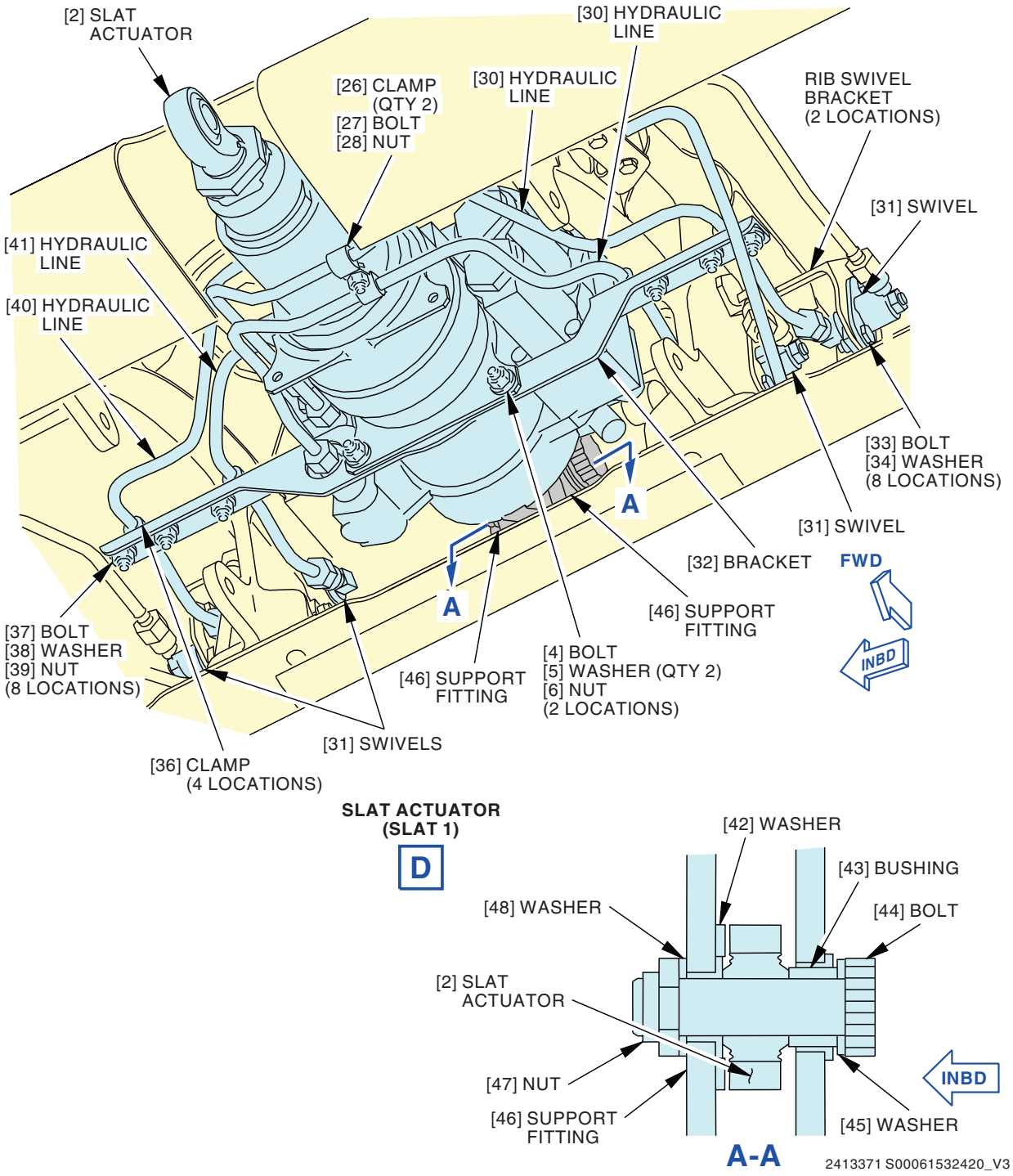
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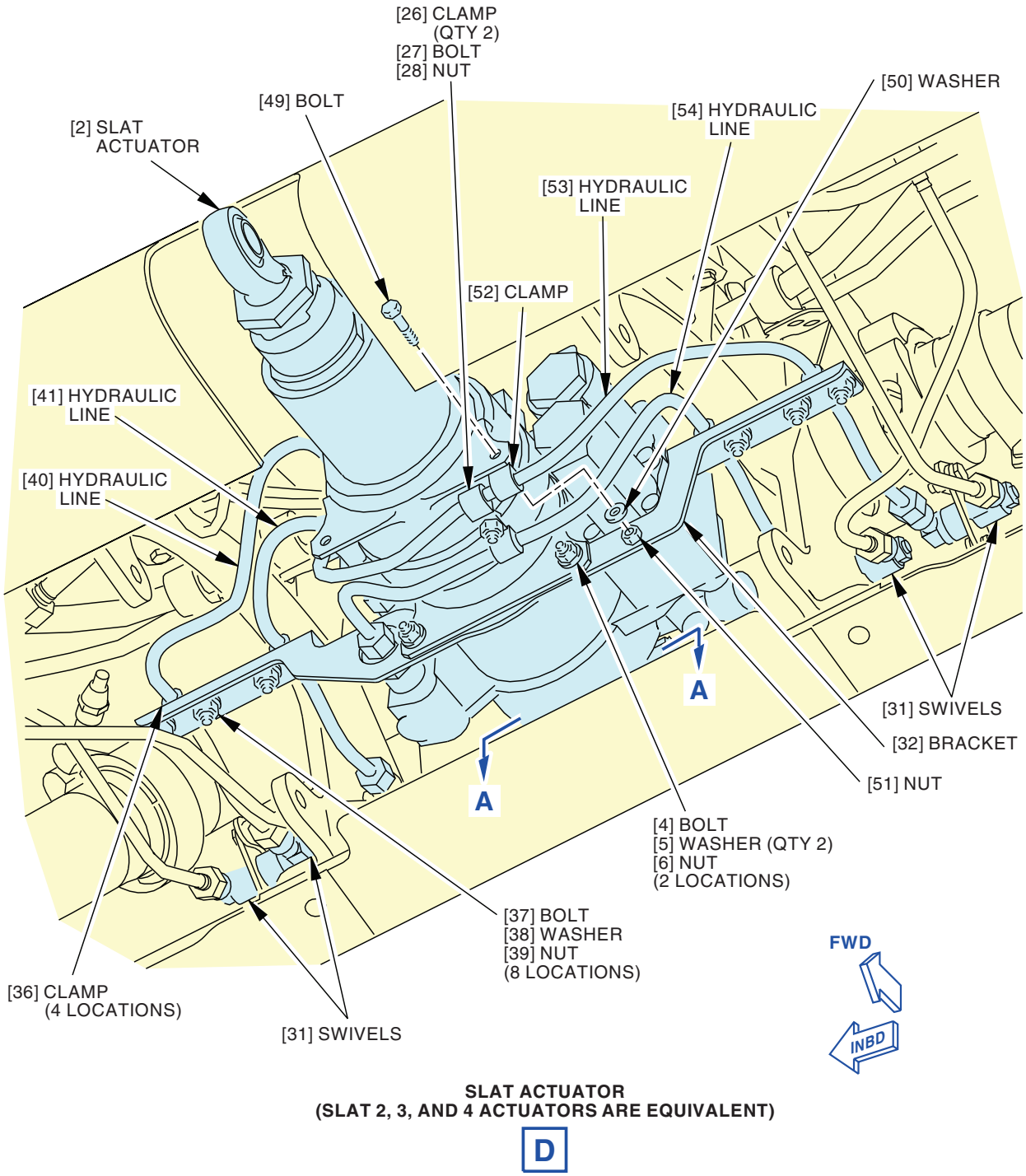
Leading Edge Slat Actuator Installation
Figure 401/27-81-41-990-801 (Sheet 3 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



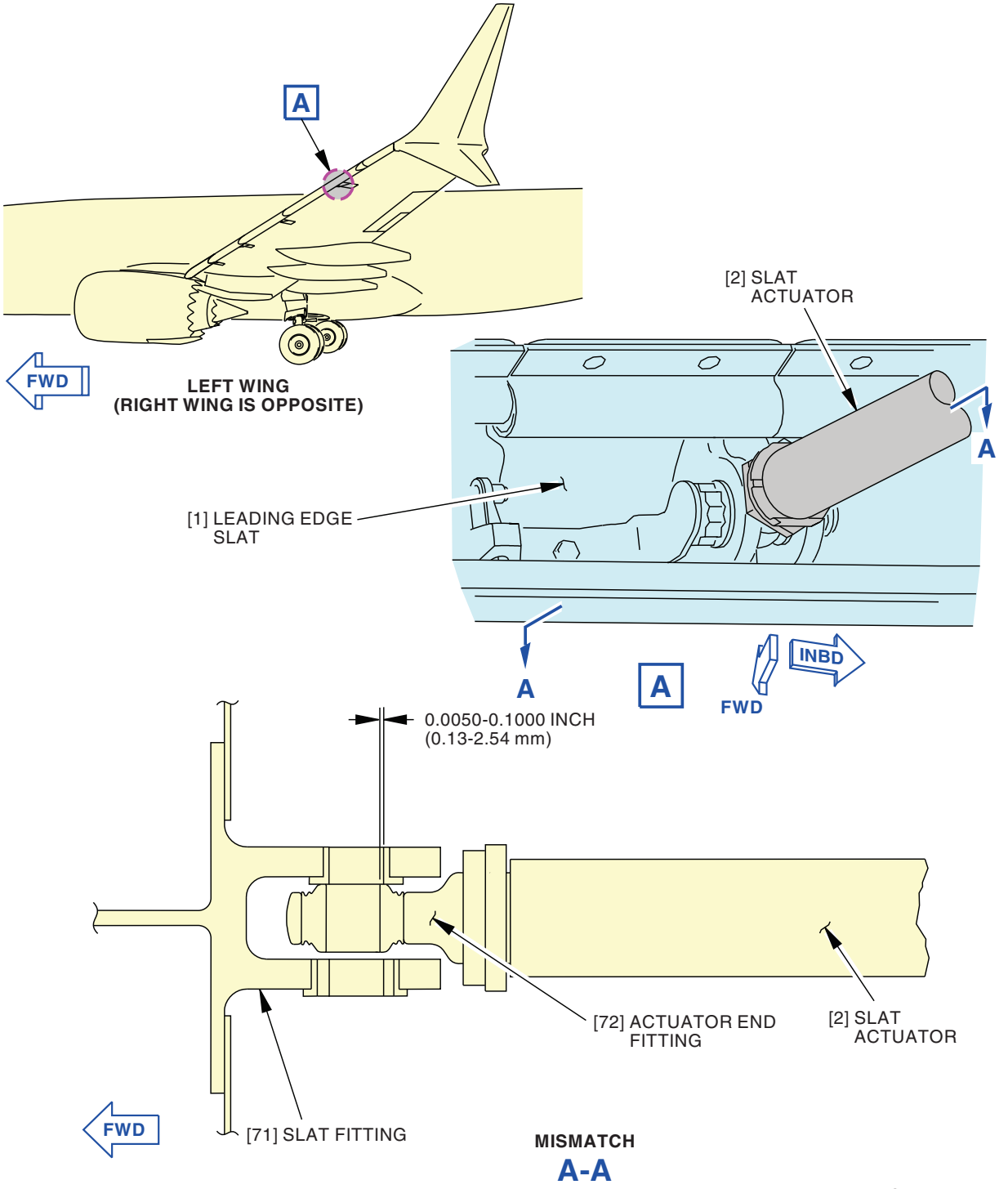
Leading Edge Slat Actuator Installation
Figure 401/27-81-41-990-801 (Sheet 4 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Leading Edge Slat Actuator Installation
Figure 401/27-81-41-990-801 (Sheet 5 of 5)

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Leading Edge Slat Actuator Preload Adjustment
Figure 402/27-81-41-990-804

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TASK 27-81-41-400-801

3. Leading Edge Slat Actuator Installation

(Figure 401, Figure 402)

A. References

Reference	Title
20-50-11-910-801	Standard Torque Values (P/B 201)
27-51-00-860-804	Retract the Trailing Edge Flaps (P/B 201)
27-81-00-080-801	Leading Edge Flap and Slat Locks Removal (P/B 201)
27-81-00-700-802	Leading Edge Flap and Slat Timing Test in Primary Mode (P/B 501)
27-81-00-710-801	Leading Edge Flap and Slat System Standby Actuation Operational Test (P/B 501)
27-81-00-820-802	Leading Edge Slat Actuator Adjustment (P/B 501)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
57-41-02-400-801	Leading Edge Access Panel Installation (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1734	Lockout Set - Leading Edge Slats and Krueger Flaps Actuators Part #: C27051-29 Supplier: 81205

C. Consumable Materials

Reference	Description	Specification
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
2	Slat actuator	27-81-41-10-095	PTF ALL
		27-81-41-10-100	PTF ALL
13	Cotter pin	57-44-51-27-090	PTF ALL
		57-44-51-27-235	PTF ALL
47	Nut	27-81-41-10-085	PTF ALL

E. Location Zones

Zone	Area
521	Left Wing - Leading Edge to Front Spar
522	Left Wing - Slat No. 4
523	Left Wing - Slat No. 3
524	Left Wing - Slat No. 2
525	Left Wing - Slat No. 1

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(Continued)

Zone	Area
621	Right Wing - Leading Edge to Front Spar
622	Right Wing - Slat No. 5
623	Right Wing - Slat No. 6
624	Right Wing - Slat No. 7
625	Right Wing - Slat No. 8

F. Access Panels

Number	Name/Location
521CB	Lower Leading Edge Access Panel
521DB	Lower Leading Edge Access Panel
521EB	Lower Leading Edge Access Panel
521FB	Lower Leading Edge Access Panel
521JB	Lower Leading Edge Access Panel
521KB	Lower Leading Edge Access Panel
521LB	Lower Leading Edge Access Panel
521MB	Lower Leading Edge Access Panel
521QB	Lower Leading Edge Access Panel
521RB	Lower Leading Edge Access Panel
521SB	Lower Leading Edge Access Panel
521TB	Lower Leading Edge Access Panel
521WB	Lower Leading Edge Access Panel
521XB	Lower Leading Edge Access Panel
521YB	Lower Leading Edge Access Panel
521ZB	Lower Leading Edge Access Panel
621CB	Lower Leading Edge Access Panel
621DB	Lower Leading Edge Access Panel
621EB	Lower Leading Edge Access Panel
621FB	Lower Leading Edge Access Panel
621JB	Lower Leading Edge Access Panel
621KB	Lower Leading Edge Access Panel
621MB	Lower Leading Edge Access Panel
621QB	Lower Leading Edge Access Panel
621RB	Lower Leading Edge Access Panel
621TB	Lower Leading Edge Access Panel
621WB	Lower Leading Edge Access Panel
621XB	Lower Leading Edge Access Panel
621ZB	Lower Leading Edge Access Panel

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G. Prepare for the Installation

SUBTASK 27-81-41-040-003

WARNING DO THE DEACTIVATION PROCEDURE TO PREVENT THE OPERATION OF THE THRUST REVERSER. THE ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the thrust reversers are activated, do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

SUBTASK 27-81-41-860-015

CAUTION MAKE SURE THAT THE INBOARD FAN COWL PANEL AND THE INBOARD AND OUTBOARD THRUST REVERSERS ARE CLOSED AND IN THE STOWED POSITION OR REMOVED BEFORE YOU EXTEND THE LEADING EDGE FLAPS AND SLATS. THERE IS NOT SUFFICIENT CLEARANCE FOR THE FLAPS AND SLATS TO EXTEND IF THE INBOARD FAN COWL PANEL AND THE INBOARD AND OUTBOARD THRUST REVERSERS ARE IN THE OPEN POSITION. THIS CAN CAUSE DAMAGE TO EQUIPMENT.

(2) Make sure that the inboard fan duct cowl and the inboard and outboard thrust reversers are closed in the stowed position or are removed.

SUBTASK 27-81-41-860-016

CAUTION MAKE SURE THAT THE TRAILING EDGE FLAPS ARE FULLY RETRACTED. IF THEY ARE NOT FULLY RETRACTED, THE ACTUATOR WILL EXTEND. THE ACTUATOR CAN CAUSE DAMAGE TO THE WING RIB.

(3) If the trailing edge flaps are not fully retracted, do this task: Retract the Trailing Edge Flaps, TASK 27-51-00-860-804.

H. Leading Edge Slat Actuator Installation

SUBTASK 27-81-41-200-003

WARNING DO NOT LET OBJECTS GET IN THE HOUSING ASSEMBLY OF THE SLAT TRACK. THIS WILL HELP PREVENT A PUNCTURE OF THE HOUSING ASSEMBLY THAT COULD CAUSE A FUEL LEAK. THE FUEL LEAK COULD CAUSE A FIRE AND POSSIBLE DEATH OR INJURY TO PERSONNEL.

(1) Keep the housing (can) assemblies of the slat main tracks clean and free of unwanted objects (FOD) at all times.

SUBTASK 27-81-41-420-001

(2) If the hydraulic lines and bracket [32] are not installed on the slat actuator [2], do these steps to install them (View D, Figure 401):

(a) Install the bracket [32] on the slat actuator [2], do these steps:

1) Put the bracket [32] in its position on the slat actuator [2].

2) Install the bolts [4], washers [5], and nuts [6].

(b) Attach the hydraulic lines to the bracket [32], do these steps:

1) Put the hydraulic lines in their positions on the bracket [32].

2) Put the clamps [36] in their positions on the hydraulic lines.

3) Install the bolts [37], washers [38], and nuts [39].

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(c) Remove the plug from the end of each hydraulic line that will attach to the slat actuator [2].

(d) Remove the caps from the hydraulic ports that are on the slat actuator [2].

(e) Do these steps for slat 1 or slat 8:

1) Connect the hydraulic line [29], hydraulic line [30], hydraulic line [40], and hydraulic line [41] to the hydraulic ports that are on the slat actuator [2].

2) Tighten the B-nuts that are on the hydraulic line [29], hydraulic line [30], hydraulic line [40], and hydraulic line [41] to 140 ±7 in-lb (15.8 ±0.8 N·m).

3) Put the clamps [26] on the hydraulic line [29] and hydraulic line [30].

4) Install the bolt [27] and nut [28].

(f) Do these steps for slat 2, 3, 4, 5, 6, or 7:

1) Connect the hydraulic line [53], hydraulic line [54], hydraulic line [40], and hydraulic line [41] to the hydraulic ports that are on the slat actuator [2].

2) Tighten the B-nuts that are on the hydraulic line [53], hydraulic line [54], hydraulic line [40], and hydraulic line [41] to 140 ±7 in-lb (15.8 ±0.8 N·m).

3) Put the clamps [26] on the hydraulic line [53] and hydraulic line [54].

4) Install the bolt [27] and nut [28].

5) Put the clamp [52] on hydraulic line [53].

(g) Remove the identification tags or labels from the hydraulic lines and from their related hydraulic ports that are on the slat actuator [2].

SUBTASK 27-81-41-980-003

(3) Make sure that the leading edge slat [1] is in the 30-unit (full extend) position.

SUBTASK 27-81-41-420-002

(4) Connect the aft end of the slat actuator [2] to the support fitting [46], do these steps (View D, Figure 401):

(a) If the washer [42] is not bonded to the support fitting [46], do these steps:

1) Make sure that the mating surface of the support fitting [46] is clean.

2) Make sure that the mating surface of the washer [42] is clean.

3) Apply the sealant, A00247, to the mating surface of the washer [42] and to the mating surface of the support fitting [46].

4) Put the washer [42] in its position on the support fitting [46].

(b) Connect the slat actuator [2] to the support fitting [46], do these steps:

1) Put the slat actuator [2] in its position on the support fitting [46].

2) Install the bushing [43], washer [45], and bolt [44].

a) Make sure that the bushing [43] and head of the bolt [44] are on the outboard side of the slat actuator [2].

3) Install the washer [48] and new nut [47].

a) Tighten the nut [47] to 660 in-lb (74.6 N·m) - 980 in-lb (110.7 N·m).

<1> Optional method: Tighten the bolt head to 882 in-lb (99.7 N·m) - 1078 in-lb (121.8 N·m).

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SUBTASK 27-81-41-420-007

(5) If the installation of a repaired or new swivel [31] is necessary, do these steps for each applicable swivel [31] (View D, Figure 401):

CAUTION

MAKE SURE THAT YOU POSITION THE SWIVEL CORRECTLY TO THE STRUCTURE. MAKE SURE THAT THE MOVEMENT SIDE IS CONNECTED TO THE HYDRAULIC TUBES THAT MOVE WITH THE ACTUATOR. AN INCORRECT REVERSE INSTALLATION OF THE SWIVEL CAN CAUSE TUBE FRACTURE AND HYDRAULIC SYSTEM FLUID LOSS. THIS CAN CAUSE DAMAGE.

(a) Install the repaired or new swivel [31] on the rib swivel bracket, do these steps:

1) Put the swivel [31] in its position on the rib swivel bracket.

CAUTION

DO NOT FULLY TIGHTEN THE BOLTS TO INSTALL THE SWIVEL. LOOSELY TIGHTEN THE BOLTS TO LET THE SWIVEL ALIGN CORRECTLY DURING THE MOVEMENT OF THE SLATS. DAMAGE TO THE HYDRAULIC LINES CAN OCCUR IF THE BOLTS FULLY TIGHTENED.

2) Loosely install the washers [34] and bolts [33].

a) Do not fully tighten the bolts [33] at this time.

NOTE: The bolts must be tightened with the slats in the fully retracted position at the end of the actuator installation task.

SUBTASK 27-81-41-420-008

CAUTION

MAKE SURE THAT YOU SAFELY CONNECT THE HYDRAULIC LINES TO THE CORRECT PORTS AND FITTINGS. IF YOU CONNECT THE HYDRAULIC LINES TO THE INCORRECT PORTS AND FITTINGS, DAMAGE CAN OCCUR. IF YOU DO NOT CONNECT ALL OF THE HYDRAULIC LINES, THEN DAMAGE TO THE ACTUATOR CAN OCCUR.

(6) Connect the hydraulic lines to the swivels [31], do these steps:

(a) Remove the caps from the swivels [31].

(b) Remove the plugs from the hydraulic lines.

(c) Do these steps for slat 1 or slat 8:

1) Connect the hydraulic line [29], hydraulic line [30], hydraulic line [40], and hydraulic line [41] to their related hydraulic ports that are on the swivels [31].

2) Tighten the B-nuts that are on the hydraulic line [29], hydraulic line [30], hydraulic line [40], and hydraulic line [41] to 140 ±7 in-lb (15.8 ±0.8 N·m).

(d) Do these steps for slat 2, 3, 4, 5, 6, or 7:

1) Connect the hydraulic line [53], hydraulic line [54], hydraulic line [40], and hydraulic line [41] to their related hydraulic ports that are on the swivels [31].

2) Tighten the B-nuts that are on the hydraulic line [53], hydraulic line [54], hydraulic line [40], and hydraulic line [41] to 140 ±7 in-lb (15.8 ±0.8 N·m).

(e) Remove the identification tags or labels from the hydraulic lines and from their related hydraulic ports that are on the swivels [31].

SUBTASK 27-81-41-860-006

(7) Use hydraulic power to extend the slat actuator [2] to the full extend position, do these steps:

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WARNING MAKE SURE ALL PERSONS AND EQUIPMENT ARE CLEAR OF THE FLIGHT CONTROLS SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(a) Pressurize the hydraulic system B, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

WARNING MAKE SURE PERSONS AND EQUIPMENT ARE CLEAR OF THE LEADING EDGE AND TRAILING EDGE CONTROL SURFACES. THE LEADING EDGE AND TRAILING EDGE CONTROL SURFACES CAN EXTEND AND RETRACT QUICKLY. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(b) Move the flap control lever to the 30-unit position to extend the slat actuator [2] to the full extend position.

(c) Remove power from the hydraulic system B, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 27-81-41-820-001

(8) Make sure that the leading edge slat [1] is in the correct position, do these steps (Figure 402):

NOTE: The slat actuator must be extended farther than the leading edge slat. This permits the slat actuator to be pre-loaded against the leading edge slat when it is fully extended (View A-A, Figure 402).

(a) Put the actuator end fitting [72] in its position on the slat fitting [71].

(b) Make sure that the mismatch between the actuator end fitting [72] and the slat fitting [71] is from 0.0050 in. (0.1270 mm) to 0.1000 in. (2.5400 mm).

SUBTASK 27-81-41-820-002

(9) Adjust the leading edge slat actuator (TASK 27-81-00-820-802).

SUBTASK 27-81-41-420-004

(10) Install the dropout links [7], do these steps (View B, Figure 401):

(a) Put the dropout links [7] in their positions.

(b) Install the bushings [10], washers [9], and bolts [8].

(c) Install the washers [11] and nuts [12].

(d) Do these steps for each nut [12] and its related bolt [8]:

1) Tighten the nut [12] to 50 in-lb (5.6 N·m).

2) Make sure that you align the nut [12] castellations with the bolt [8] hole.

a) If it is necessary, continue to tighten the nut [12] until the castellations align with the bolt [8] hole.

<1> Do not tighten the nut [12] more than 80 in-lb (9.0 N·m).

3) Do this step for slat 4 or slat 5:

a) Install new cotter pins [13] in the nuts [12] hole.

SUBTASK 27-81-41-860-017

(11) Use hydraulic power to extend the leading edge slat [1] to the full extend position, do these steps:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING MAKE SURE ALL PERSONS AND EQUIPMENT ARE CLEAR OF THE FLIGHT CONTROLS SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(a) Pressurize the hydraulic system B, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

WARNING MAKE SURE PERSONS AND EQUIPMENT ARE CLEAR OF THE LEADING EDGE AND TRAILING EDGE CONTROL SURFACES. THE LEADING EDGE AND TRAILING EDGE CONTROL SURFACES CAN EXTEND AND RETRACT QUICKLY. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(b) Move the flap control lever to the 30-unit position to extend the leading edge slat [1] to the full extend position.

(c) Remove power from the hydraulic system B, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 27-81-41-420-005

(12) Install the slat actuator door [3] on the slat actuator [2], do these steps (Views B and D, Figure 401):

(a) Remove the bolts [4], washers [5], and nuts [6] that attach the bracket [32] to the slat actuator [2].

1) If there are three washers installed on each bolt [4], then discard the thicker aluminum washers.

(b) Put the slat actuator door [3] and bracket [32] in their positions on the slat actuator [2].

(c) Install the bolts [4], washers [5], and nuts [6] that attach the slat actuator door [3] and bracket [32] to the slat actuator [2].

(d) Do this step for slat 1 or slat 8:

1) Install the bolts [14], washers [15], washers [16], and nuts [17] that attach the slat actuator door [3] to the slat actuator [2].

(e) Do this step for slat 2, 3, 4, 5, 6, or 7:

1) Install the bolts [14], washers [15], washers [16], and nuts [17] that attach the slat actuator door [3] and clamp [52] to the slat actuator [2].

SUBTASK 27-81-41-080-001

WARNING CAREFULLY REMOVE THE LEADING EDGE FLAP AND SLAT ACTUATORS LOCKOUT SET. THE LEADING EDGE FLAPS AND SLATS CAN MOVE QUICKLY IF YOU DO NOT REMOVE THE LOCKS CORRECTLY. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(13) Carefully remove each lockout set, SPL-1734, that is installed, do this task: Leading Edge Flap and Slat Locks Removal, TASK 27-81-00-080-801.

SUBTASK 27-81-41-720-001

(14) Do this task: Leading Edge Flap and Slat Timing Test in Primary Mode, TASK 27-81-00-700-802.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-81-41-710-001

(15) Do this task: Leading Edge Flap and Slat System Standby Actuation Operational Test, TASK 27-81-00-710-801.

SUBTASK 27-81-41-790-001

(16) Do the check for hydraulic leaks, do these steps:

WARNING MAKE SURE ALL PERSONS AND EQUIPMENT ARE CLEAR OF THE FLIGHT CONTROLS SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(a) Pressurize the hydraulic system B, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

WARNING MAKE SURE PERSONS AND EQUIPMENT ARE CLEAR OF THE LEADING EDGE AND TRAILING EDGE CONTROL SURFACES. THE LEADING EDGE AND TRAILING EDGE CONTROL SURFACES CAN EXTEND AND RETRACT QUICKLY. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(b) Operate the leading edge slats for a minimum of two cycles to bleed air from the system.

1) Leave the slats in the extended position.

(c) Make sure that there are no hydraulic leaks from the slat actuator [2] or the hydraulic lines.

WARNING MAKE SURE PERSONS AND EQUIPMENT ARE CLEAR OF THE LEADING EDGE AND TRAILING EDGE CONTROL SURFACES. THE LEADING EDGE AND TRAILING EDGE CONTROL SURFACES CAN EXTEND AND RETRACT QUICKLY. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(d) Move the flap control lever to the UP position to retract the leading edge slats.

(e) Remove power from the hydraulic system B, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 27-81-41-420-006

(17) If you installed one or more repaired or new swivels [31], do these steps:

(a) Apply the final torque to the bolts [33] that attach each applicable swivel [31] to the rib swivel brackets (TASK 20-50-11-910-801).

(b) Apply the final torque to all of the B-Nuts and fittings that are on the hydraulic tubes.

1) Do not tighten the B-Nuts that are common to the swivels [31] until all of the hydraulic tube assemblies have been tightened.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 27-81-41-860-008

(1) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	15	C01081	HYDRAULIC SYSTEM PTU VALVE CONT 1

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	16	C01085	HYDRAULIC SYSTEM PTU VALVE CONT 2

SUBTASK 27-81-41-200-005

WARNING DO NOT LET OBJECTS GET IN THE HOUSING ASSEMBLY OF THE SLAT TRACK. THIS WILL HELP PREVENT A PUNCTURE OF THE HOUSING ASSEMBLY THAT COULD CAUSE A FUEL LEAK. THE FUEL LEAK COULD CAUSE A FIRE AND POSSIBLE DEATH OR INJURY TO PERSONNEL.

(2) Keep the housing (can) assemblies of the slat main tracks clean and free of unwanted objects (FOD) at all times.

SUBTASK 27-81-41-410-003

(3) Close the applicable access panels (TASK 57-41-02-400-801).

(a) For the left wing, close these access panels:

Number	Name/Location
521CB	Lower Leading Edge Access Panel
521DB	Lower Leading Edge Access Panel
521EB	Lower Leading Edge Access Panel
521FB	Lower Leading Edge Access Panel
521JB	Lower Leading Edge Access Panel
521KB	Lower Leading Edge Access Panel
521LB	Lower Leading Edge Access Panel
521MB	Lower Leading Edge Access Panel
521QB	Lower Leading Edge Access Panel
521RB	Lower Leading Edge Access Panel
521SB	Lower Leading Edge Access Panel
521TB	Lower Leading Edge Access Panel
521WB	Lower Leading Edge Access Panel
521XB	Lower Leading Edge Access Panel
521YB	Lower Leading Edge Access Panel
521ZB	Lower Leading Edge Access Panel

(b) For the right wing, close these access panels:

Number	Name/Location
621CB	Lower Leading Edge Access Panel
621DB	Lower Leading Edge Access Panel
621EB	Lower Leading Edge Access Panel
621FB	Lower Leading Edge Access Panel
621JB	Lower Leading Edge Access Panel
621KB	Lower Leading Edge Access Panel
621MB	Lower Leading Edge Access Panel
621QB	Lower Leading Edge Access Panel
621RB	Lower Leading Edge Access Panel
621TB	Lower Leading Edge Access Panel
621WB	Lower Leading Edge Access Panel
621XB	Lower Leading Edge Access Panel

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
621ZB	Lower Leading Edge Access Panel

SUBTASK 27-81-41-440-001

(4) Activate the thrust reversers, do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

LEADING EDGE SLAT ACTUATOR - INSPECTION/CHECK

1. General

A. There is one task in this procedure. The procedure has this task:

(1) A visual check of the leading edge slat actuator lock indicator.

TASK 27-81-41-200-801

2. Leading Edge Slat Actuator Lock Indicator Visual Check

(Figure 601)

A. References

Reference	Title
27-81-00-080-801	Leading Edge Flap and Slat Locks Removal (P/B 201)
27-81-00-480-801	Leading Edge Flap and Slat Locks Installation (P/B 201)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1734	Lockout Set - Leading Edge Slats and Krueger Flaps Actuators Part #: C27051-29 Supplier: 81205

C. Location Zones

Zone	Area
522	Left Wing - Slat No. 4
523	Left Wing - Slat No. 3
524	Left Wing - Slat No. 2
525	Left Wing - Slat No. 1
622	Right Wing - Slat No. 5
623	Right Wing - Slat No. 6
624	Right Wing - Slat No. 7
625	Right Wing - Slat No. 8

D. Prepare for the Visual Check

SUBTASK 27-81-41-860-009

WARNING

MAKE SURE THAT PERSONNEL AND EQUIPMENT ARE CLEAR OF THE FLIGHT CONTROLS SURFACES. MAKE SURE THAT THE INBOARD FAN DUCT COWL AND THE INBOARD AND OUTBOARD THRUST REVERSERS ARE CLOSED. THE FLIGHT CONTROL SURFACES CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) To pressurize hydraulic system B, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-81-41-860-010

WARNING MAKE SURE THAT PERSONNEL AND EQUIPMENT STAY AWAY FROM THE LEADING EDGE SLATS, TRAILING EDGE FLAPS, AND DRIVE MECHANISMS. THE FLAPS, SLATS, AND DRIVE MECHANISMS MOVE QUICKLY. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

CAUTION MAKE SURE THAT THE INBOARD FAN COWL PANEL AND THE INBOARD AND OUTBOARD THRUST REVERSERS ARE CLOSED AND IN THE STOWED POSITION OR REMOVED BEFORE YOU EXTEND THE LEADING EDGE FLAPS AND SLATS. THERE IS NOT SUFFICIENT CLEARANCE FOR THE FLAPS AND SLATS TO EXTEND IF THE INBOARD FAN COWL PANEL AND THE INBOARD AND OUTBOARD THRUST REVERSERS ARE IN THE OPEN POSITION. THIS CAN CAUSE DAMAGE TO EQUIPMENT.

(2) Move the flap control lever to the 30-unit position.

(a) Make sure that the leading edge slats move to the full extend position.

SUBTASK 27-81-41-480-002

(3) Install the lockout set, SPL-1734 on the adjacent leading edge slats. To install it, do this task: Leading Edge Flap and Slat Locks Installation, TASK 27-81-00-480-801.

SUBTASK 27-81-41-200-006

WARNING DO NOT LET OBJECTS GET IN THE HOUSING ASSEMBLY OF THE SLAT TRACK. THIS WILL HELP PREVENT A PUNCTURE OF THE HOUSING ASSEMBLY THAT COULD CAUSE A FUEL LEAK. THE FUEL LEAK COULD CAUSE A FIRE AND POSSIBLE DEATH OR INJURY TO PERSONNEL.

(4) Keep clean and free of all unwanted objects (FOD), the housing (can) assemblies of the slat main tracks, at all time.

E. Leading Edge Slat Actuator Lock Indicator Visual Check

SUBTASK 27-81-41-010-003

(1) Remove the slat actuator door [8] from the slat actuator [9]:

(a) Remove the bolts [4], washers [5], washers [6] and nuts [7] that attach the slat actuator door [8] to the slat actuator [9].

(b) Remove the bolts [1], washers [2] and nuts [3] that attach the slat actuator door [8] and the bracket to the slat actuator [9].

(c) Remove the slat actuator door [8].

(d) Temporarily install the bolts [1], washers [2] and nuts [3] to attach the bracket to the slat actuator [9].

SUBTASK 27-81-41-280-001

(2) Do a visual check of the leading edge slat actuator lock indicator:

(a) Move the flap control lever to the UP detent position to retract the slats.

(b) Make sure that the lock indicator hole is open.

(c) Put a 0.115-0.125 inch (2.92-3.18 millimeters) diameter steel rod in the lock indicator hole.

(d) Move the flap control lever to the 1-unit detent position.

(e) Make sure that the actuator is not locked and extends to the extend position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(f) Remove the steel rod from the lock indicator hole.

(g) Make sure that the lock indicator hole is not open.

SUBTASK 27-81-41-410-002

(3) Install the slat actuator door [8] on the slat actuator [9].

(a) Remove the bolts [1], washers [2] and nuts [3] that attach the bracket to the slat actuator [9].

1) If there are three washers installed on the bolts [1], discard the thicker aluminum washer.

(b) Put the slat actuator door [8] in its position on the slat actuator [9].

(c) Install the bolts [1], washers [2] and nuts [3] that attach the slat actuator door [8] and the bracket to the slat actuator [9].

(d) Install the bolts [4], washers [5], washers [6] and nuts [7] that attach the slat actuator door [8] to the slat actuator [9].

F. Put the Airplane Back to Its Usual Condition

SUBTASK 27-81-41-080-002

(1) To remove the lockout set, SPL-1734 from the adjacent leading edge slats, do this task: Leading Edge Flap and Slat Locks Removal, TASK 27-81-00-080-801.

SUBTASK 27-81-41-200-007

WARNING DO NOT LET OBJECTS GET IN THE HOUSING ASSEMBLY OF THE SLAT TRACK. THIS WILL HELP PREVENT A PUNCTURE OF THE HOUSING ASSEMBLY THAT COULD CAUSE A FUEL LEAK. THE FUEL LEAK COULD CAUSE A FIRE AND POSSIBLE DEATH OR INJURY TO PERSONNEL.

(2) Keep clean and free of all unwanted objects (FOD), the housing (can) assemblies of the slat main tracks, at all time.

SUBTASK 27-81-41-860-011

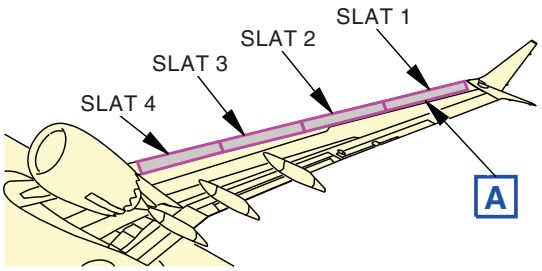
(3) Move the flap control lever to the UP detent position to retract the slats.

SUBTASK 27-81-41-860-012

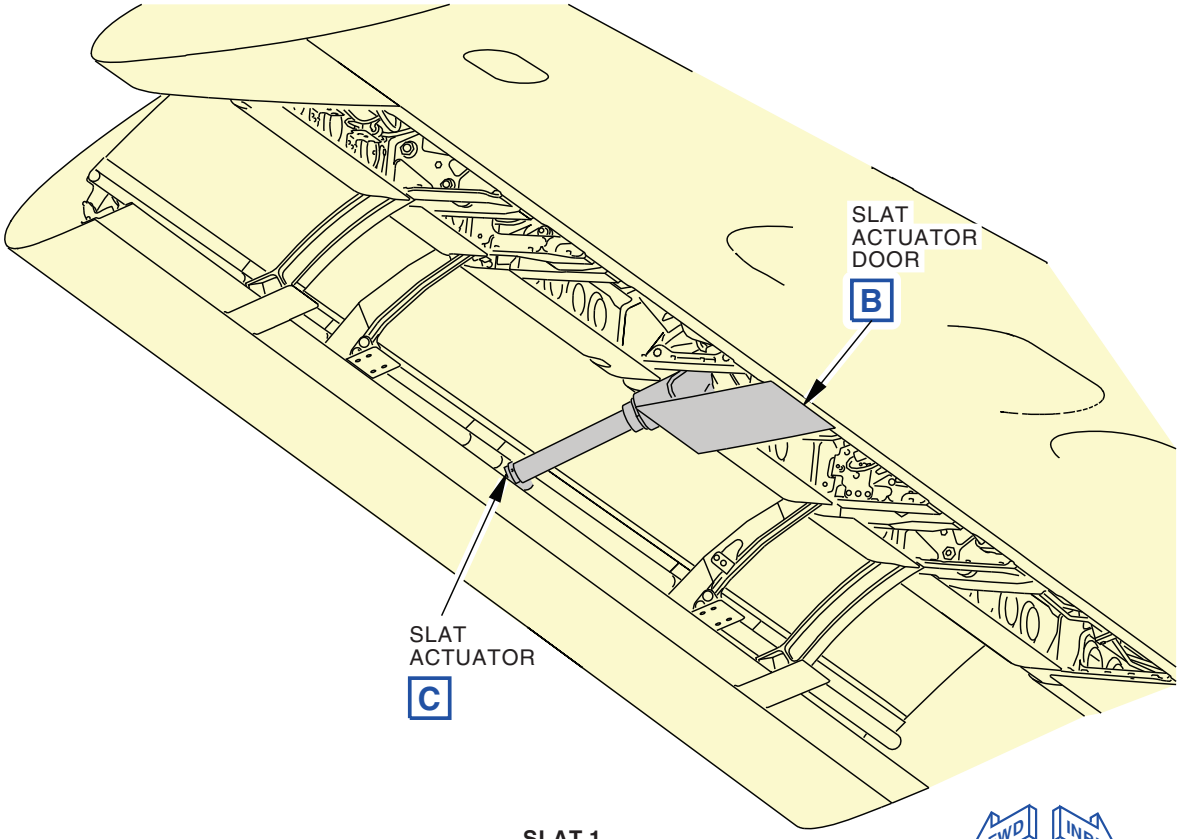
(4) To remove power from hydraulic system B, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LEFT WING
(RIGHT WING IS OPPOSITE)



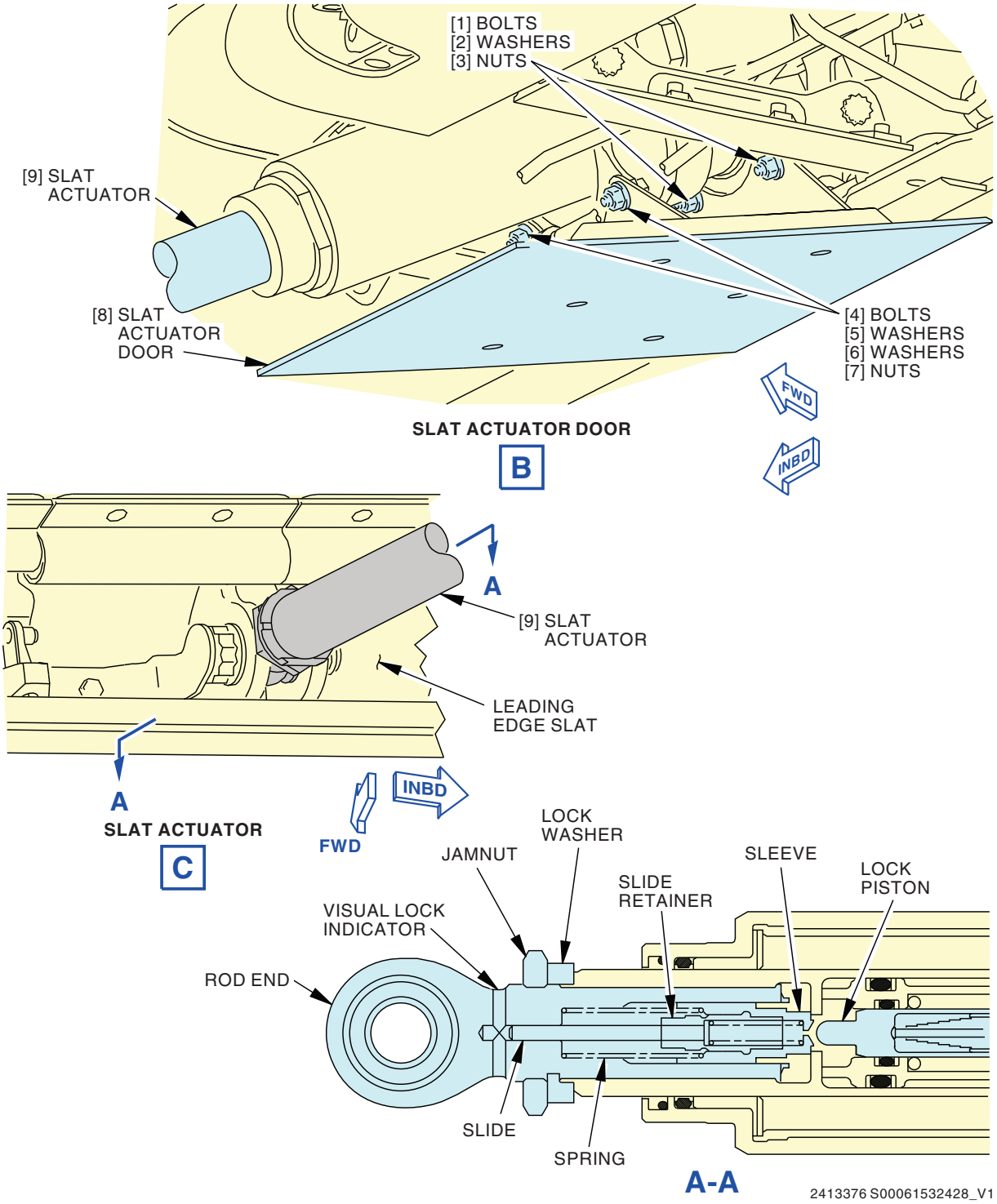
SLAT 1
(SLATS 2, 3, AND 4 ARE EQUIVALENT)

A

2413375 S00061532427_V1

Leading Edge Slat Actuator Lock Indicator Visual Inspection
Figure 601/27-81-41-990-803 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Leading Edge Slat Actuator Lock Indicator Visual Inspection
Figure 601/27-81-41-990-803 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

LEADING EDGE UNCOMMANDED MOTION (UCM) VALVE AND LEADING EDGE CRUISE DEPRESSURIZATION VALVE - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) LE Uncommanded Motion (UCM) or LE Cruise Depressurization Valve Removal

(2) LE Uncommanded Motion (UCM) or LE Cruise Depressurization Valve Installation.

B. There are two valves installed on the airplane. The Leading Edge (LE) cruise depressurization valve and the LE UCM shutoff valve. This procedure is applicable for the two valves.

C. The leading edge flap and slat uncommanded motion (UCM) or cruise depressurization valve is referred as the valve [3] in this procedure.

D. The LE cruise depressurization valve will be operated each flight to depressurize the leading edge extend at cruise. The LE UCM shutoff valve will be operated only when there is uncommanded motion of the leading edge.

TASK 27-81-52-000-801

2. LE Uncommanded Motion (UCM) or LE Cruise Depressurization Valve Removal

(Figure 401)

A. References

Reference	Title
20-60-07-910-802	Inspection, Cleaning and Sealing of Electrical Connectors (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Tools/Equipment

Reference	Description
STD-1054	Container - Fuel Resistant, 5-Gallon (19-Liter)

C. Location Zones

Zone	Area
132	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Right
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right

D. Access Panels

Number	Name/Location
192BR	ECS Ram Air Inlet Mixing Duct Panel - Forward

E. Prepare for the Removal

SUBTASK 27-81-52-480-001

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed on all the landing gears, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-81-52-860-001

(2) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	7	C00210	FLIGHT CONTROL FLAP SHUTOFF VALVES

SUBTASK 27-81-52-860-002

(3) If there is hydraulic power on hydraulic system B, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 27-81-52-010-001

(4) To gain access to the UCM valve, do this step:

Remove this access panel:

Number	Name/Location
192BR	ECS Ram Air Inlet Mixing Duct Panel - Forward

F. Leading Edge Flap and Slat Valve Removal

SUBTASK 27-81-52-020-001

(1) Remove the valve [3]:

CAUTION DO NOT LET HYDRAULIC FLUID GET ON EQUIPMENT. IF HYDRAULIC FLUID GETS ON EQUIPMENT, IMMEDIATELY CLEAN IT FROM THE EQUIPMENT. HYDRAULIC FLUID CAN CAUSE DAMAGE TO EQUIPMENT.

(a) Put a 5-gallon (19-liter) fuel resistant container, STD-1054 below the valve [3] to catch hydraulic fluid.

(b) Disconnect the hydraulic line [4], hydraulic line [5], and hydraulic line [6] from the valve [3].

1) Install caps on the hydraulic lines.

2) Install plugs in the hydraulic ports of the valve [3].

(c) Disconnect the electrical connectors from the valve [3].

(d) After you disconnect the electrical connector(s), examine the connector(s) for corrosion (TASK 20-60-07-910-802).

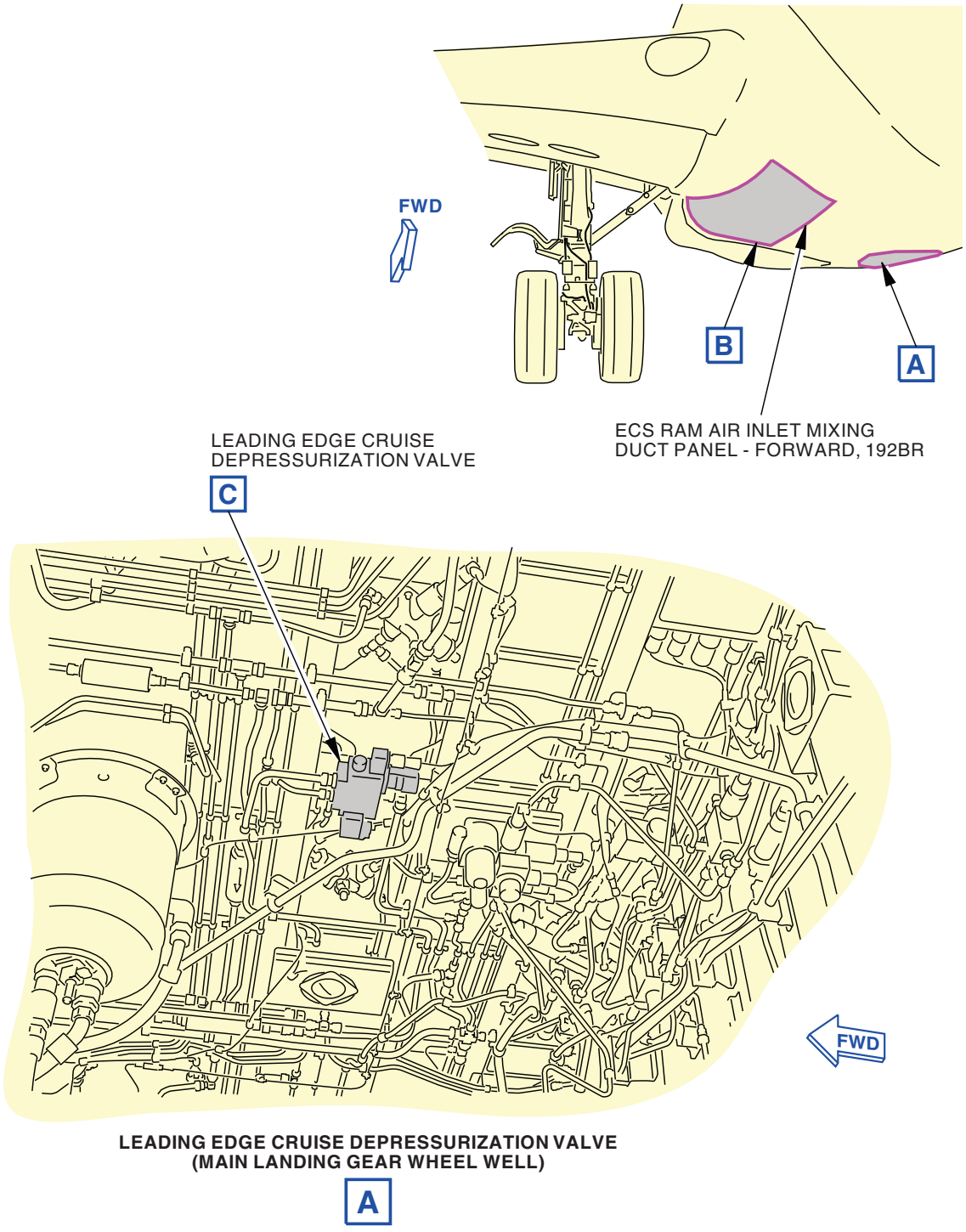
NOTE: The inspection of the electrical connector(s) can be done after the removal step or before the installation step of the electrical connector(s). This is an operator option.

(e) Remove the bolts [1] and washers [2].

(f) Remove the valve [3].

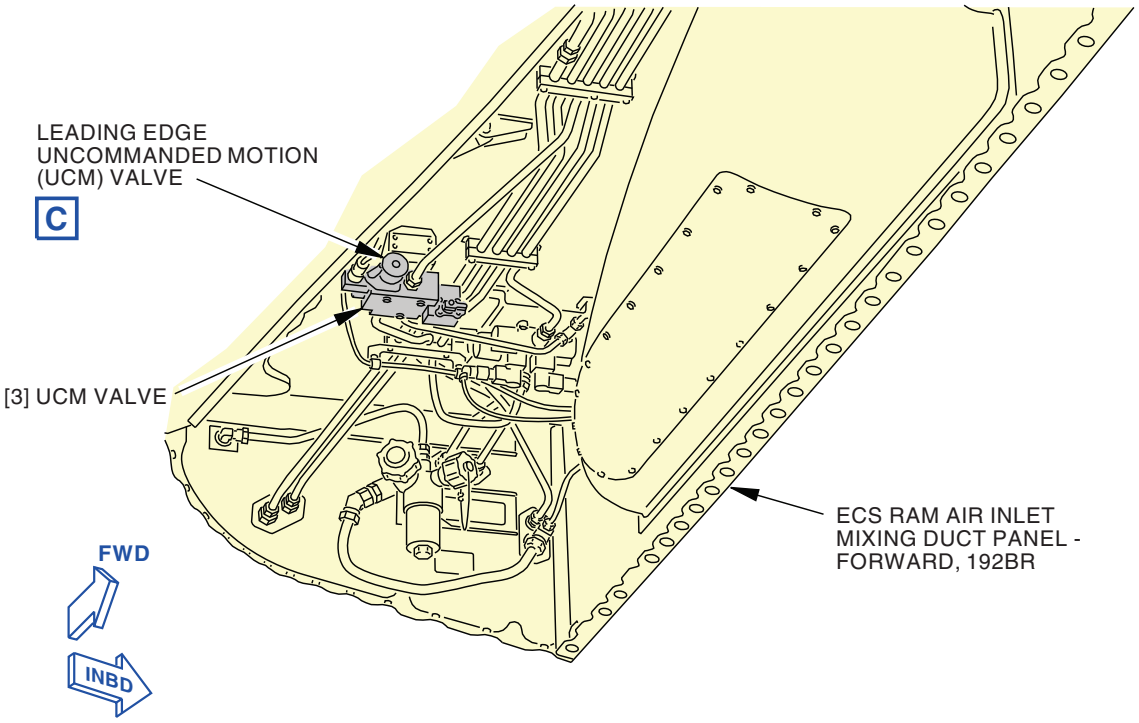
END OF TASK

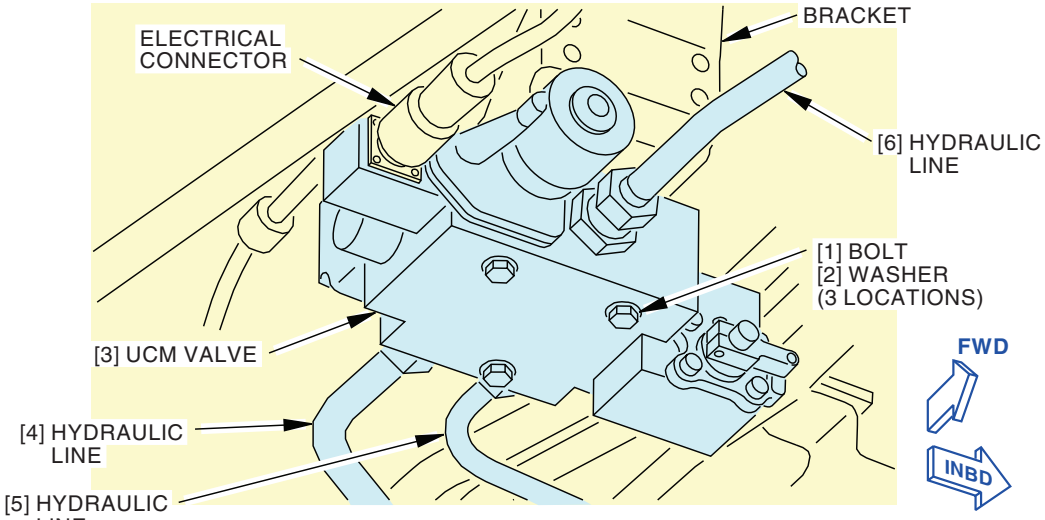
ECCN 9E991 BOEING PROPRIETARY - See title page for details



Leading Edge Uncommanded Motion (UCM) Valve Installation
Figure 401/27-81-52-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details





**LEADING EDGE UNCOMMANDED MOTION (UCM) VALVE
(LEADING EDGE VALVE IS SHOWN, LEADING EDGE
CRUISE DEPRESSURIZATION VALVE IS EQUIVALENT)**

C

2413378 S00061532433_V2

**Leading Edge Uncommanded Motion (UCM) Valve Installation
Figure 401/27-81-52-990-801 (Sheet 2 of 2)**

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-81-52-400-801

3. LE Uncommanded Motion (UCM) or LE Cruise Depressurization Valve Installation

(Figure 401)

A. References

Reference	Title
20-60-07-910-802	Inspection, Cleaning and Sealing of Electrical Connectors (P/B 201)
27-81-00-700-804	Leading Edge Uncommanded Motion Protection Test (P/B 501)
27-81-00-730-801	Leading Edge Cruise Depressurization Test (P/B 501)
SWPM 20-20-00, Paragraph 2	CLEANING OF FAY SURFACES
SWPM 20-20-00, Section 14-D	Standard Wiring Practices Manual
SWPM 20-20-00, Section 21-A	Resistance of Faying Surface Bonds
SWPM 20-60-03	Special Protection of Electrical Connectors

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659

C. Consumable Materials

Reference	Description	Specification
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95
G50171	Compound - Corrosion Inhibiting Compound, Interior Application - D5026NS or ZC-026	

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
3	Valve	27-81-52-01-015	PTF ALL
		27-81-52-02-015	PTF ALL

E. Location Zones

Zone	Area
132	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Right
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Access Panels

Number	Name/Location
192BR	ECS Ram Air Inlet Mixing Duct Panel - Forward

G. Leading Edge Flap and Slat Valve Installation

SUBTASK 27-81-52-420-001

(1) Install the valve [3]:

(a) Clean the fay surfaces of the aluminum bracket and valve [3] (CLEANING OF FAY SURFACES, SWPM 20-20-00, Paragraph 2)

(b) Put the valve [3] in its position.

(c) Install the bolts [1] and washers [2].

(d) Apply thin layer of sealant, A00247 along the edges of the joint between the mating surfaces (SWPM 20-20-00, Section 14-D).

(e) Connect the hydraulic lines to the valve [3].

1) Remove the caps from the hydraulic lines.

2) Remove the plugs from the hydraulic ports on the valve [3].

3) Connect the hydraulic line [4] and hydraulic line [6] to the valve [3].

4) Tighten the B-nuts on hydraulic line [4] and hydraulic line [6] to 500 ±25 in-lb (56 ±3 N·m).

5) Connect hydraulic line [5] to the valve [3].

6) Tighten the B-nut on hydraulic line [5] to 270 ±14 in-lb (31 ±2 N·m).

(f) If you did not examine the electrical connector(s) at the removal step, do this task: Inspection, Cleaning and Sealing of Electrical Connectors, TASK 20-60-07-910-802.

(g) Apply the D5026NS or ZC-026 compound, G50171 to the connectors (SWPM 20-60-03).

(h) If you did not examine the electrical connector(s) at the removal step, do this task: Inspection, Cleaning and Sealing of Electrical Connectors, TASK 20-60-07-910-802.

NOTE: The inspection of the electrical connector(s) can be done after the removal step or before the installation step of the electrical connector(s). This is an operator option.

SUBTASK 27-81-52-860-003

(2) Use an intrinsically safe approved bonding meter, COM-1550 to measure the resistance between the valve [3] and the bracket (SWPM 20-20-00, Section 21-A).

(a) Make sure the maximum resistance between the valve [3] and the bracket is 0.0025 ohms (2.5 milliohms).

SUBTASK 27-81-52-860-004

(3) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	7	C00210	FLIGHT CONTROL FLAP SHUTOFF VALVES

SUBTASK 27-81-52-720-001

(4) If you replace the Leading Edge (LE) cruise depressurization valve, do this task: Leading Edge Cruise Depressurization Test, TASK 27-81-00-730-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-81-52-720-002

(5) If you replace the LE UCM shutoff valve, do this task: Leading Edge Uncommanded Motion Protection Test, TASK 27-81-00-700-804.

H. Put the plane back to its usual condition.

SUBTASK 27-81-52-940-001

(1) Install this access panel:

Number	Name/Location
192BR	ECS Ram Air Inlet Mixing Duct Panel - Forward

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

LEADING EDGE CRUISE DEPRESSURIZATION VALVE - INSPECTION/CHECK

1. General

A. This task does a check for internal hydraulic leakage in the leading edge cruise depressurization valve.

TASK 27-81-52-790-801

2. Leading Edge Cruise Depressurization Valve Hydraulic Internal Leakage Check

A. General

(1) This procedure does an internal hydraulic leakage check for the leading edge cruise depressurization valves. The purpose of this leakage test is to identify a high leakage valve.

(2) You must measure the amount of leakage of hydraulic fluid during different conditions of operation to measure the fluid flow rate. It will be necessary to disconnect the return hydraulic line.

B. References

Reference	Title
12-12-00-610-801	Hydraulic Reservoir Servicing (P/B 301)
27-81-00-730-801	Leading Edge Cruise Depressurization Test (P/B 501)
27-81-52-000-801	LE Uncommanded Motion (UCM) or LE Cruise Depressurization Valve Removal (P/B 401)
27-81-52-400-801	LE Uncommanded Motion (UCM) or LE Cruise Depressurization Valve Installation (P/B 401)
29-09-00-860-801	Hydraulic Reservoirs Pressurization (P/B 201)
29-09-00-860-802	Hydraulic Reservoirs Depressurization (P/B 201)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-09-00-860-801	Put the Airplane in the Air Mode (P/B 201)
32-09-00-860-802	Return the Airplane to the Ground Mode (P/B 201)

C. Tools/Equipment

Reference	Description
STD-6505	Container - Measuring, 500 cc, Calibrated in cc (ml)

D. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right

E. Prepare for the Internal Leakage Check

SUBTASK 27-81-52-860-006

(1) If the trailing edge flaps are not in the retracted position, then do these steps:

(a) Supply system B hydraulic power. To supply hydraulic power, do this task:
Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

(b) Move the flap control lever to the FLAP UP position to retract the trailing edge flaps.

(c) Remove system B hydraulic power. To remove hydraulic power, do this task:
Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-81-52-860-007

(2) Make sure the flap control lever is at the FLAP UP position.

F. Leading Edge Cruise Depressurization Valve Internal Leakage Check

SUBTASK 27-81-52-790-001

(1) Put a measuring container, STD-6505 under the return line on the cruise depressurization valve.

SUBTASK 27-81-52-790-002

(2) Disconnect the return line from the cruise depressurization valve. Do the applicable steps for this task:

LE Uncommanded Motion (UCM) or LE Cruise Depressurization Valve Removal, TASK 27-81-52-000-801

NOTE: The return line is the smaller of two hydraulic lines that go forward from the cruise depressurization valve.

SUBTASK 27-81-52-790-003

(3) Install a cap or plug on the hydraulic line that goes to the hydraulic reservoir.

SUBTASK 27-81-52-860-008

(4) Supply system B hydraulic power. To supply hydraulic power, do this task:

Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 27-81-52-860-009

(5) Pressurize the system B hydraulic reservoir. To pressurize the reservoir, do this task:

Hydraulic Reservoirs Pressurization, TASK 29-09-00-860-801.

SUBTASK 27-81-52-790-004

(6) Measure the leakage from the cruise depressurization valve with the solenoids not energized:

(a) Measure and record the leakage rate from the return port on the cruise depressurization valve.

NOTE: The leakage rate must not be more than 100 cc/minute.

SUBTASK 27-81-52-860-010

(7) Do this task:Put the Airplane in the Air Mode, TASK 32-09-00-860-801.

NOTE: The solenoids on the cruise depressurization valve will be energized three seconds after the PSEU is in air mode. A small amount of hydraulic fluid can leak while the valve moves.

SUBTASK 27-81-52-790-005

(8) Measure the leakage from the cruise depressurization valve with the solenoids energized:

(a) Measure and record the leakage rate from the return port on the cruise depressurization valve.

NOTE: The leakage rate must not be more than 100 cc/minute.

SUBTASK 27-81-52-860-011

(9) Do this task: Return the Airplane to the Ground Mode, TASK 32-09-00-860-802.

NOTE: The solenoids on the cruise depressurization valve will be de-energized three seconds after the PSEU is in ground mode. A small amount of hydraulic fluid can leak while the valve moves.

SUBTASK 27-81-52-860-012

(10) Remove the pressure from the system B reservoir. To remove the pressure, do this task:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Hydraulic Reservoirs Depressurization, TASK 29-09-00-860-802.

SUBTASK 27-81-52-860-013

(11) Remove system B hydraulic power. To remove hydraulic power, do this task:

Hydraulic System A or B Power Removal, TASK 29-11-00-860-805

G. Put the Airplane Back to the Usual Condition

SUBTASK 27-81-52-790-006

(1) Remove the cap or plug from the hydraulic line that goes to the hydraulic reservoir.

SUBTASK 27-81-52-790-007

(2) Re-connect the return line to the leading edge cruise depressurization valve. Do the applicable steps from this task:

LE Uncommanded Motion (UCM) or LE Cruise Depressurization Valve Installation, TASK 27-81-52-400-801.

SUBTASK 27-81-52-610-001

(3) If it is necessary for the hydraulic reservoir, do this task:

Hydraulic Reservoir Servicing, TASK 12-12-00-610-801.

SUBTASK 27-81-52-720-003

(4) Do this task:

Leading Edge Cruise Depressurization Test, TASK 27-81-00-730-801.

SUBTASK 27-81-52-210-001

(5) Make sure there are no hydraulic fluid leaks at the leading edge cruise depressurization valve.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

LEADING EDGE FLAP AND SLAT HYDRAULIC FUSE - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the hydraulic fuses of the leading edge flap and slat.

(2) An installation of the hydraulic fuses of the leading edge flap and slat.

B. There are two hydraulic fuses (extend and retract) in the system B and one hydraulic fuse in the standby system. The hydraulic fuses are all located in the airplane main landing gear wheel well.

C. This procedure is applicable for all three hydraulic fuses.

TASK 27-81-61-000-801

2. Leading Edge Flap and Slat Hydraulic Fuse Removal (LE System B or Standby System Fuses)

(Figure 401)

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
29-21-00-000-802	Standby Hydraulic System Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left

C. Leading Edge Flap and Slat Hydraulic Fuse Removal (LE System B or Standby System Fuses)

SUBTASK 27-81-61-480-001

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed on all the landing gears, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 27-81-61-860-001

(2) To remove power from hydraulic system B, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 27-81-61-860-006

(3) To remove power from the standby hydraulic system, do this task: Standby Hydraulic System Power Removal, TASK 29-21-00-000-802.

SUBTASK 27-81-61-020-001

(4) Do these steps to remove the two hydraulic fuses [2] in system B:

(a) Loosen nut [1] and nut [3] to disconnect the hydraulic tubes from the hydraulic fuse [2].

(b) Remove the hydraulic fuse [2].

SUBTASK 27-81-61-020-002

(5) Do these steps to remove the standby hydraulic fuse [5] in the standby system:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Loosen nut [6] and nut [7] to disconnect the hydraulic tubes from flow limiting valve [4] and standby hydraulic fuse [5].

(b) Remove the standby hydraulic fuse [5] with the flow limiting valve [4].

(c) Remove the standby hydraulic fuse [5] from the flow limiting valve [4].

1) Remove and discard the packing [8] installed between the standby hydraulic fuse [5] and flow limiting valve [4].

D. Test the LE Hydraulic Fuses

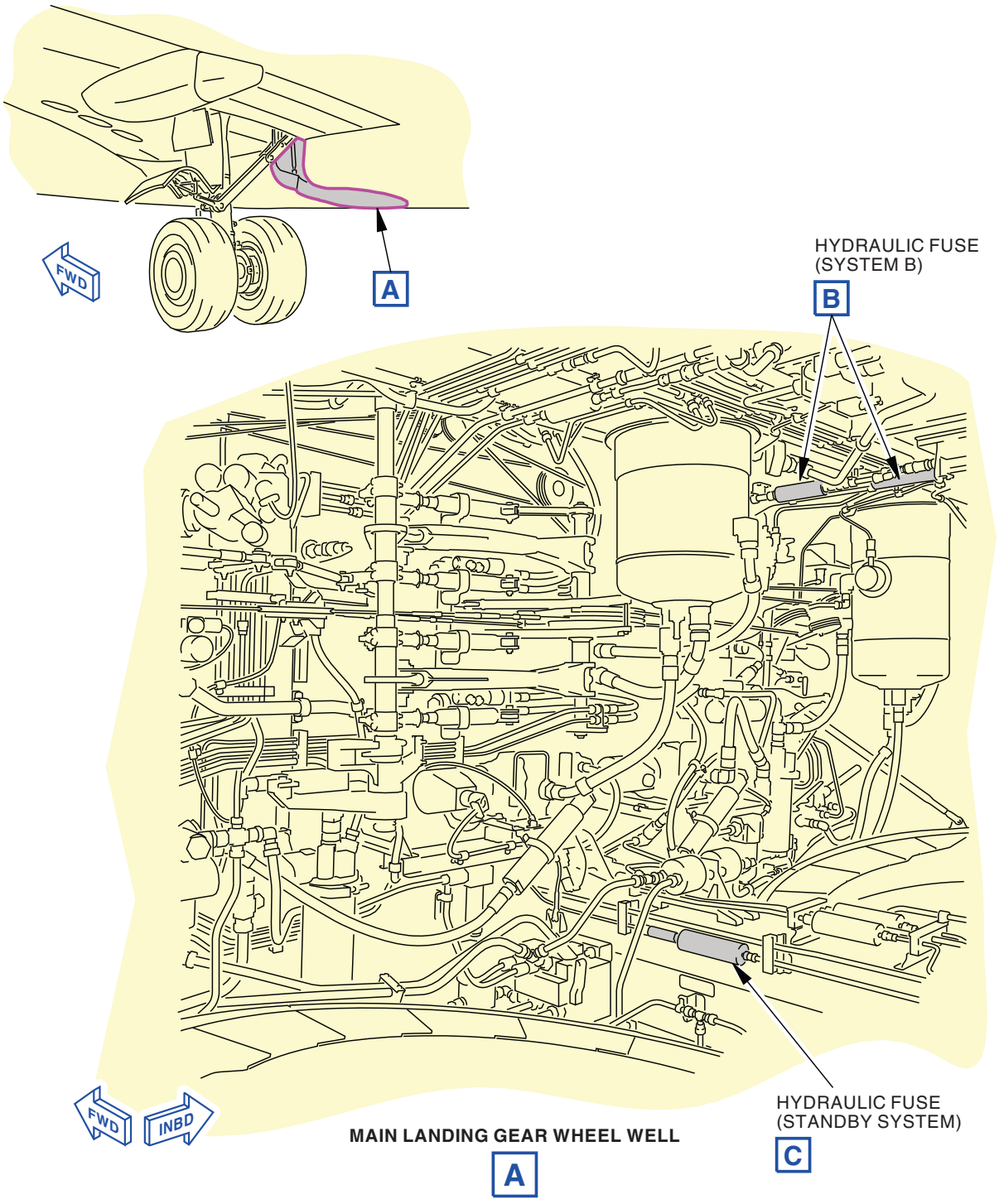
SUBTASK 27-81-61-720-001

(1) Do the functional test of the Leading Edge (LE) hydraulic fuses with the suppliers recommended component maintenance test instructions and test equipment.

NOTE: The suppliers instructions give a reverse flow test, internal leakage test, pressure drop test, volumetric capacity test and reset test for the fuses. This is an off-airplane bench test.

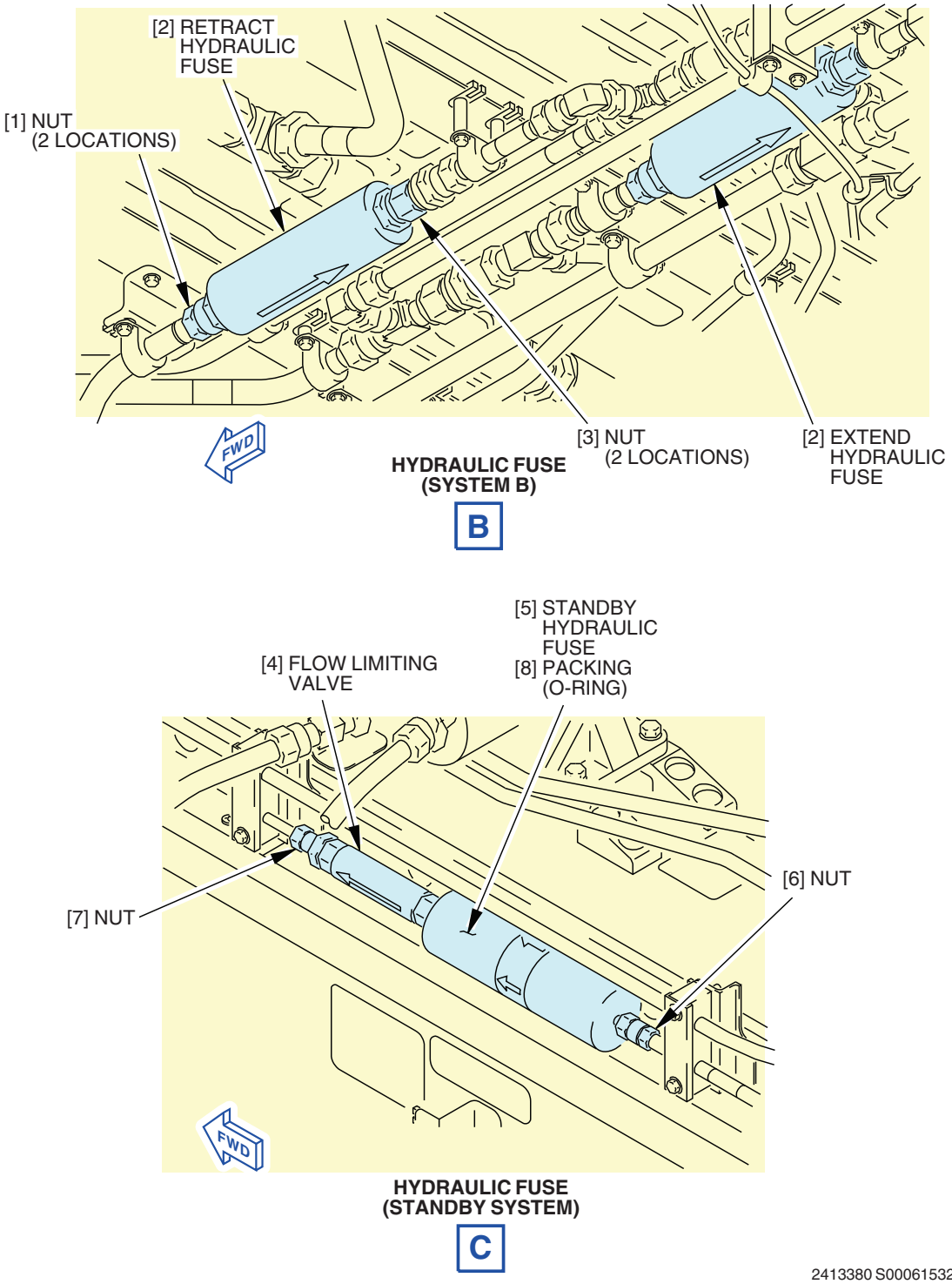
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Leading Edge and Slat Hydraulic Fuse Installation
Figure 401/27-81-61-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Leading Edge and Slat Hydraulic Fuse Installation
Figure 401/27-81-61-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-81-61-400-801

3. Leading Edge Flap and Slat Hydraulic Fuse Installation (LE System B or Standby System Fuses)

(Figure 401)

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
12-12-00-610-801	Hydraulic Reservoir Servicing (P/B 301)
27-81-00-700-802	Leading Edge Flap and Slat Timing Test in Primary Mode (P/B 501)
27-81-00-700-803	Leading Edge Flap and Slat Timing Test in the Standby Mode (P/B 501)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
29-21-00-000-801	Standby Hydraulic System Pressurization (P/B 201)
29-21-00-000-802	Standby Hydraulic System Power Removal (P/B 201)

B. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
2	Hydraulic fuse	27-81-61-05-010	PTF ALL
5	Standby hydraulic fuse	27-81-61-01-010	PTF ALL
8	Packing	27-81-61-01-005	PTF ALL

C. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left

D. Leading Edge Flap and Slat Hydraulic Fuse Installation (LE System B or Standby System Fuses)

SUBTASK 27-81-61-420-001

(1) Do these steps to install the two hydraulic fuses [2] in system B:

(a) Hold the hydraulic fuse [2] in its position.

NOTE: The flow direction arrow on the hydraulic fuse must agree with the arrow on the pressure bulkhead.

(b) Connect the hydraulic tubes to the hydraulic fuse [2].

(c) Tighten the nut [1] to a value of 700.0 ±35.0 in-lb (79.1 ±4.0 N·m).

(d) Tighten the nut [3] to a value of 500.0 ±25.0 in-lb (56.5 ±2.8 N·m).

SUBTASK 27-81-61-420-002

(2) Do these steps to install the standby hydraulic fuse [5] in the standby system:

(a) Install the standby hydraulic fuse [5] to the flow limiting valve [4].

1) Install a new packing [8] between the standby hydraulic fuse [5] and flow limiting valve [4].

2) Tighten the flow limiting valve [4] to the fuse assembly end fitting to 170.0 ±8.5 in-lb (19.2 ±1.0 N·m).

(b) Hold the standby hydraulic fuse [5] and the flow limiting valve [4] in their position.

NOTE: The flow direction arrow on the fuse assembly and the flow limiting valve must agree with the arrow on the pressure bulkhead.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Connect the hydraulic tubes to the standby hydraulic fuse [5] and the flow limiting valve [4].

(d) Tighten the nut [6] to a value of 270.0 ±14.0 in-lb (30.5 ±1.6 N·m).

(e) Tighten the nut [7] to a value of 140.0 ±7.0 in-lb (15.8 ±0.8 N·m).

SUBTASK 27-81-61-860-002

(3) To pressurize hydraulic system B, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

(a) If you replaced the hydraulic fuse [2], you will need to pressurize the hydraulic system B.

SUBTASK 27-81-61-860-007

(4) To pressurize the standby hydraulic system, do this task: Standby Hydraulic System Pressurization, TASK 29-21-00-000-801.

(a) If you replaced the standby hydraulic fuse [5], you will need to pressurize the standby hydraulic system.

SUBTASK 27-81-61-210-001

(5) Do a visual check of the hydraulic fuse [2] or standby hydraulic fuse [5] for hydraulic leaks.

SUBTASK 27-81-61-710-001

(6) Do the Leading Edge Flap and Slat Timing Test in Primary Mode, TASK 27-81-00-700-802 for the extend or retract fuse (hydraulic system B fuses).

SUBTASK 27-81-61-710-002

(7) Do the Leading Edge Flap and Slat Timing Test in the Standby Mode, TASK 27-81-00-700-803 for the standby fuse.

SUBTASK 27-81-61-860-003

(8) To remove power from hydraulic system B, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 27-81-61-860-008

(9) To remove power from the standby hydraulic system, do this task: Standby Hydraulic System Power Removal, TASK 29-21-00-000-802.

SUBTASK 27-81-61-610-001

(10) If it is necessary to service the hydraulic reservoirs, do this task: Hydraulic Reservoir Servicing, TASK 12-12-00-610-801.

——— END OF TASK ———

TASK 27-81-61-860-801

4. Flap and Slat Hydraulic Fuses (System B) Reset

Figure 401

A. References

Reference	Title
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)

B. Tools/Equipment

Reference	Description
STD-1154	Container - 5 Gallon (19 Liter)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Procedure

SUBTASK 27-81-61-860-004

(1) Do these steps to reset the hydraulic fuses [2] for system B:

(a) Make sure that the flap control lever is in the UP position.

(b) Make sure that there is no power on hydraulic system B. If it is necessary to remove power from hydraulic system B, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

(c) Between the retract or extend hydraulic fuse [2] and the check valve upstream, loosen the nut [1] sufficiently to bleed off the hydraulic fluid into a 5 gallon (19 liter) container, STD-1154.

NOTE: The pressure difference across the hydraulic fuse [2] must be less than 5 psi (34.5 kPa) for a minimum of 5 seconds to put the hydraulic fuse [2] back to the open position.

(d) Tighten the nut [1] after you put the hydraulic fuse [2] back to the open position.



WARNING KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(e) To pressurize the hydraulic system B, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

(f) Do a check of the hydraulic fuse [2] for hydraulic leaks.

(g) To remove power from hydraulic system B, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

END OF TASK

TASK 27-81-61-860-802

5. Flap and Slat Hydraulic Fuse (Standby System) Reset

A. References

Reference	Title
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)

B. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Procedure

SUBTASK 27-81-61-860-005

(1) Do these steps to reset the hydraulic fuse for the standby system:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Make sure that the flap control lever is in the UP position.

(b) Remove power from hydraulic system B. To remove power from hydraulic system B, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

LEADING EDGE FLAP FLOW LIMITING VALVE - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Leading Edge Flap Flow Limiting Valve Removal

(2) Leading Edge Flap Flow Limiting Valve Installation.

B. The Leading Edge Flap Flow Limiting Valve is referred to as the flow limiting valve in these procedures.

TASK 27-81-71-000-801

2. Leading Edge Flap Flow Limiting Valve Removal

(Figure 401)

A. References

Reference	Title
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left

C. Prepare for the Removal

SUBTASK 27-81-71-480-001

WARNING MAKE SURE THAT YOU INSTALL THE DOWNLOCK PINS ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed on all the landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 27-81-71-860-001

(2) Do this task to remove power from hydraulic system B: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

D. Leading Edge Flap Flow Limiting Valve Removal

SUBTASK 27-81-71-020-001

(1) Do these steps to remove the flow limiting valve [1]:

(a) Loosen nut [3] and nut [4] to disconnect the hydraulic tubes from the flow limiting valve [1] and the hydraulic fuse [2].

(b) Remove the hydraulic fuse [2] with the flow limiting valve [1].

(c) Remove the flow limiting valve [1] from the hydraulic fuse [2].

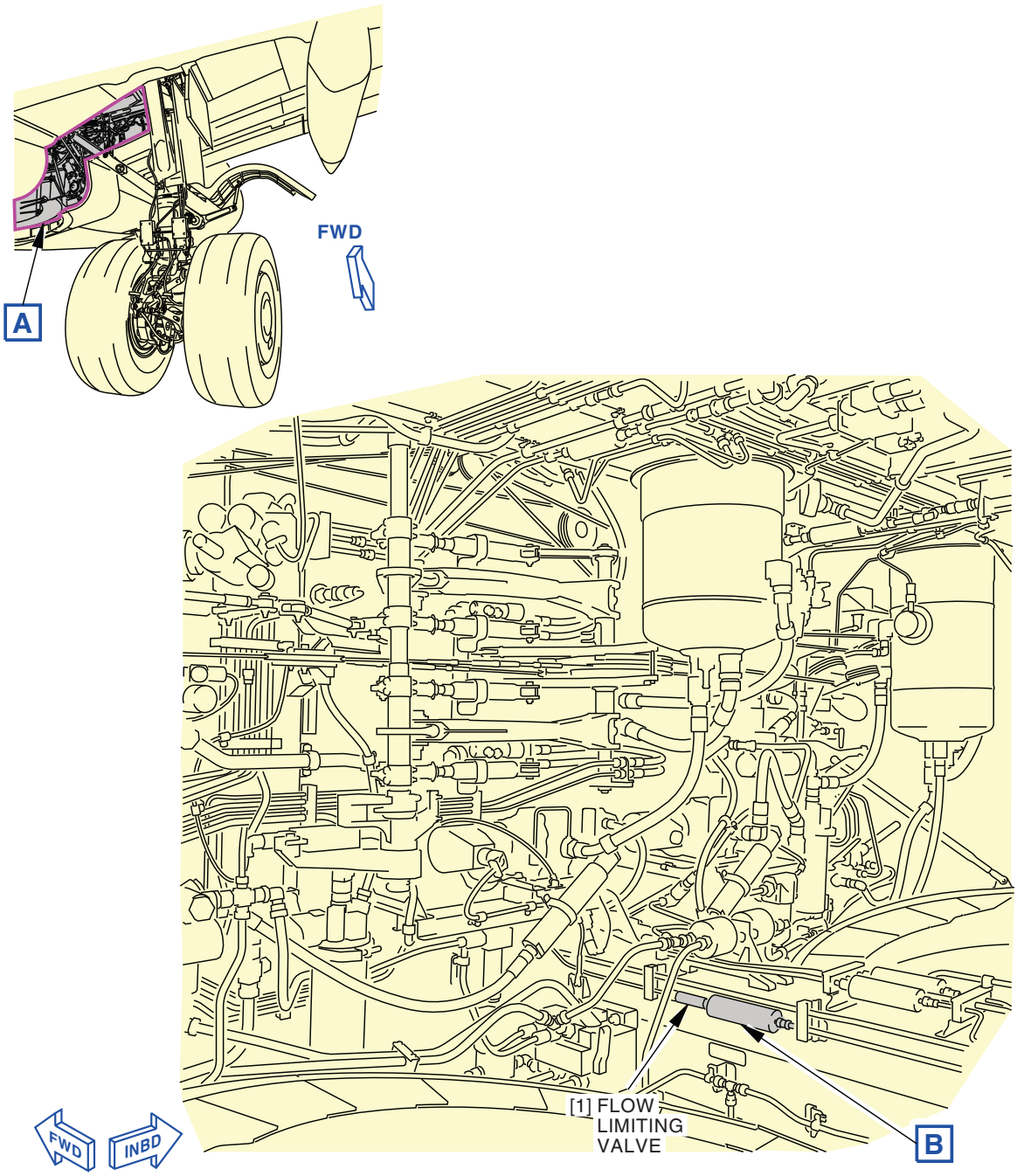
1) Remove and discard the packing [5] installed between the hydraulic fuse [2] and flow limiting valve [1].

(d) Remove the union [6] from the flow limiting valve [1].

1) Remove and discard the packing [5] installed between the union [6] and flow limiting valve [1].

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



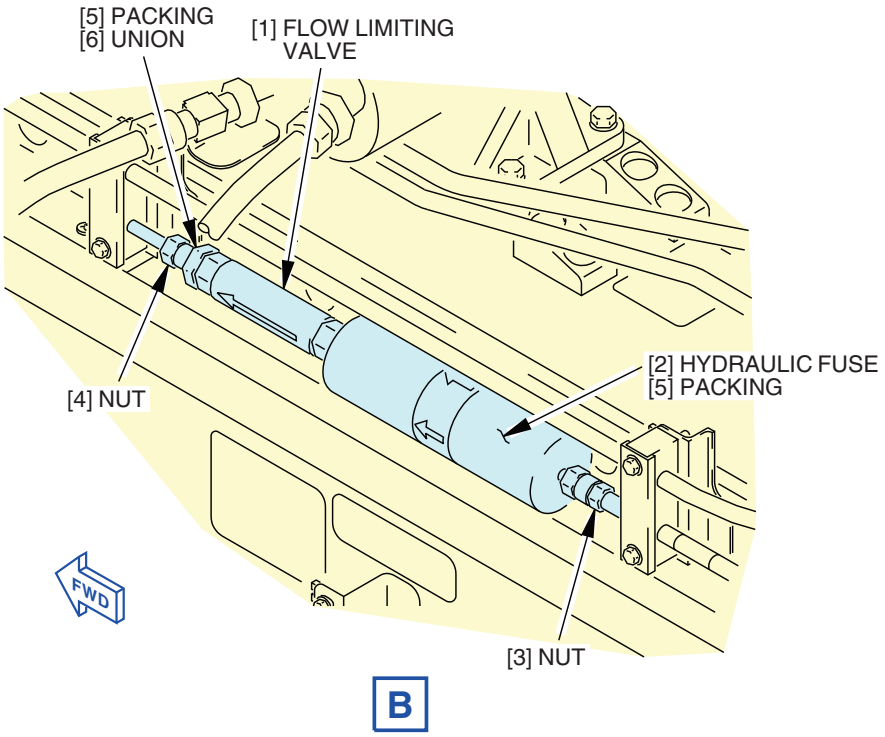
MAIN LANDING GEAR WHEEL WELL
(LEFT SIDE IS SHOWN, RIGHT SIDE IS EQUIVALENT)

A

2413381 S00061532446_V1

Leading Edge Flap Flow Limiting Valve Installation
Figure 401/27-81-71-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2413382 S00061532447_V3

Leading Edge Flap Flow Limiting Valve Installation
Figure 401/27-81-71-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-81-71-400-801

3. Leading Edge Flap Flow Limiting Valve Installation

(Figure 401)

A. References

Reference	Title
12-12-00-610-801	Hydraulic Reservoir Servicing (P/B 301)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)

B. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Flow limiting valve	27-81-71-01-015	PTF ALL
5	Packing	27-81-61-01-005	PTF ALL
		27-81-71-01-010	PTF ALL

C. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left

D. Leading Edge Flap Flow Limiting Valve Installation

SUBTASK 27-81-71-420-001

(1) Do these steps to install the flow limiting valve [1]:

(a) Install the union [6] on the flow limiting valve [1]:

1) Install a new packing [5] between the union [6] and flow limiting valve [1].

(b) Connect the flow limiting valve [1] to the hydraulic fuse [2].

1) Install a new packing [5] between the hydraulic fuse [2] and flow limiting valve [1].

2) Tighten the flow limiting valve [1] to the hydraulic fuse end fitting to 170.0 ±8.5 in-lb (19.2 ±1.0 N·m).

(c) Hold the flow limiting valve [1] and hydraulic fuse [2] in their position.

1) Make sure that the flow direction arrow on the flow limiting valve [1] and hydraulic fuse [2] agree with the arrow on the pressure bulkhead.

(d) Connect the hydraulic tubes to the flow limiting valve [1] and hydraulic fuse [2].

(e) Tighten nut [3] to 270.0 ±14.0 in-lb (30.5 ±1.6 N·m).

(f) Tighten nut [4] to 140.0 ±7.0 in-lb (15.8 ±0.8 N·m).

SUBTASK 27-81-71-610-001

(2) If it is necessary to service the hydraulic reservoirs, do this task: Hydraulic Reservoir Servicing, TASK 12-12-00-610-801.

SUBTASK 27-81-71-710-001

(3) Do these steps to test the flap flow limiting valve [1]:

(a) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	13	C00841	FLIGHT CONTROL ALTN T.E. FLAP DRIVE

(b) Do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) If it is necessary, move the flap control lever to the FLAP UP detent position to retract the leading edge flaps and slats.

WARNING MOVE ALL PERSONS AND EQUIPMENT AWAY FROM THE TRAILING EDGE FLAPS. THE FLAPS WILL MOVE AND CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

CAUTION MAKE SURE THAT YOU CLOSE OR REMOVE THE INBOARD FAN COWL AND THRUST REVERSERS BEFORE YOU RETRACT THE LEADING EDGE FLAPS. THE CLEARANCE IS NOT SUFFICIENT FOR THE FLAPS TO RETRACT WITH THE INBOARD FAN DUCT COWL AND THE THRUST REVERSERS IN THE OPEN POSITION. THIS CAN CAUSE DAMAGE TO EQUIPMENT.

(d) Put the alternate flap arm switch in the ARM position.

(e) Put the alternate flap control switch in the DOWN position.

1) Make sure that the leading edge flaps and slats fully extend.

(f) Put the alternate flap control switch in the OFF position.

(g) Put the alternate flap arm switch in the OFF position.

1) Make sure that the leading edge flaps and slats fully retract.

(h) Examine the hydraulic fittings of the flow limiting valve [1] for leakage.

E. Put the Airplane Back to Its Usual Condition

SUBTASK 27-81-71-860-002

(1) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	13	C00841	FLIGHT CONTROL ALTN T.E. FLAP DRIVE

SUBTASK 27-81-71-860-003

(2) Do this task to remove pressure from hydraulic system B: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 27-81-71-610-002

(3) If it is necessary to service the hydraulic reservoirs, do this task: Hydraulic Reservoir Servicing, TASK 12-12-00-610-801.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

LEADING EDGE FLAP AND SLAT CONTROL VALVE - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the leading edge flap and slat control valve.

(2) An installation of the leading edge flap and slat control valve.

B. The leading edge flap and slat control valve is attached to the flap control unit forward of the trailing edge flap control valve.

C. The leading edge flap and slat control valve is referred to as the "control valve" in this procedure.

TASK 27-81-81-000-801

2. Leading Edge Flap and Slat Control Valve Removal

(Figure 401)

A. References

Reference	Title
27-81-00-860-804	Leading Edge Flaps and Slats Retraction (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205
SPL-12421	Rig Pin, F70207-116 (included in F70207 Kit) Part #: F70207-151 Supplier: 81205
STD-1154	Container - 5 Gallon (19 Liter)

C. Location Zones

Zone	Area
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right

D. Prepare for the Removal

SUBTASK 27-81-81-480-001

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed on all the landing gears, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 27-81-81-860-001

(2) If the leading edge flaps and slats are not in the retracted position, do this task: Leading Edge Flaps and Slats Retraction, TASK 27-81-00-860-804.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-81-81-860-002

(3) To remove power from hydraulic system B, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

E. Leading Edge Flap and Slat Control Valve Removal

SUBTASK 27-81-81-020-001

(1) Remove the lower pan [8] from the flap control unit [6].

(a) Remove the screw [7] that attaches the lower pan [8] to the flap control unit [6].

(b) Remove the lower pan [8] from the flap control unit [6].

SUBTASK 27-81-81-480-002

(2) Install the rig pin [16] through the cam follower arm [15] and trailing edge flap control valve cam [14].

(a) Use rig pin, F70207-116, SPL-12421, from the rigging pin kit, SPL-1585.

SUBTASK 27-81-81-020-002

(3) Remove the leading edge flap and slat control valve assembly [1].

CAUTION

DO NOT LET HYDRAULIC FLUID GET ON EQUIPMENT. IF HYDRAULIC FLUID GETS ON EQUIPMENT, IMMEDIATELY CLEAN IT FROM THE EQUIPMENT. HYDRAULIC FLUID CAN CAUSE DAMAGE TO EQUIPMENT.

(a) Put a 5 gallon (19 liter) container, STD-1154, below the leading edge flap and slat control valve assembly [1] to catch hydraulic fluid.

(b) Disconnect the hydraulic lines [3] and hydraulic line [18] from the leading edge flap and slat control valve assembly [1].

1) Install caps on the hydraulic lines [3] and hydraulic line [18].

2) Install plugs in the hydraulic ports of the leading edge flap and slat control valve assembly [1].

(c) Remove the nut [13], washer [12], and bolt [11] that attach the leading edge flap and slat control valve assembly [1] to the leading edge cam link [17].

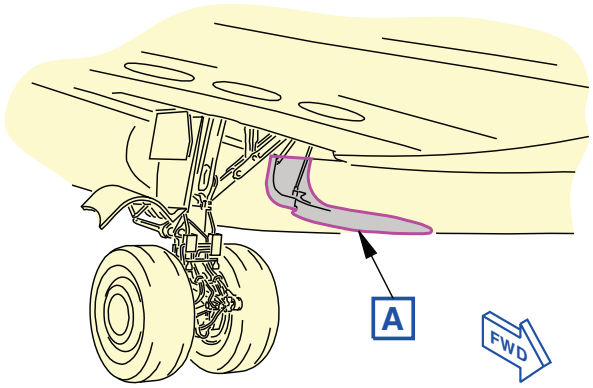
(d) Remove the lockwire or safety cable from the bolts [9].

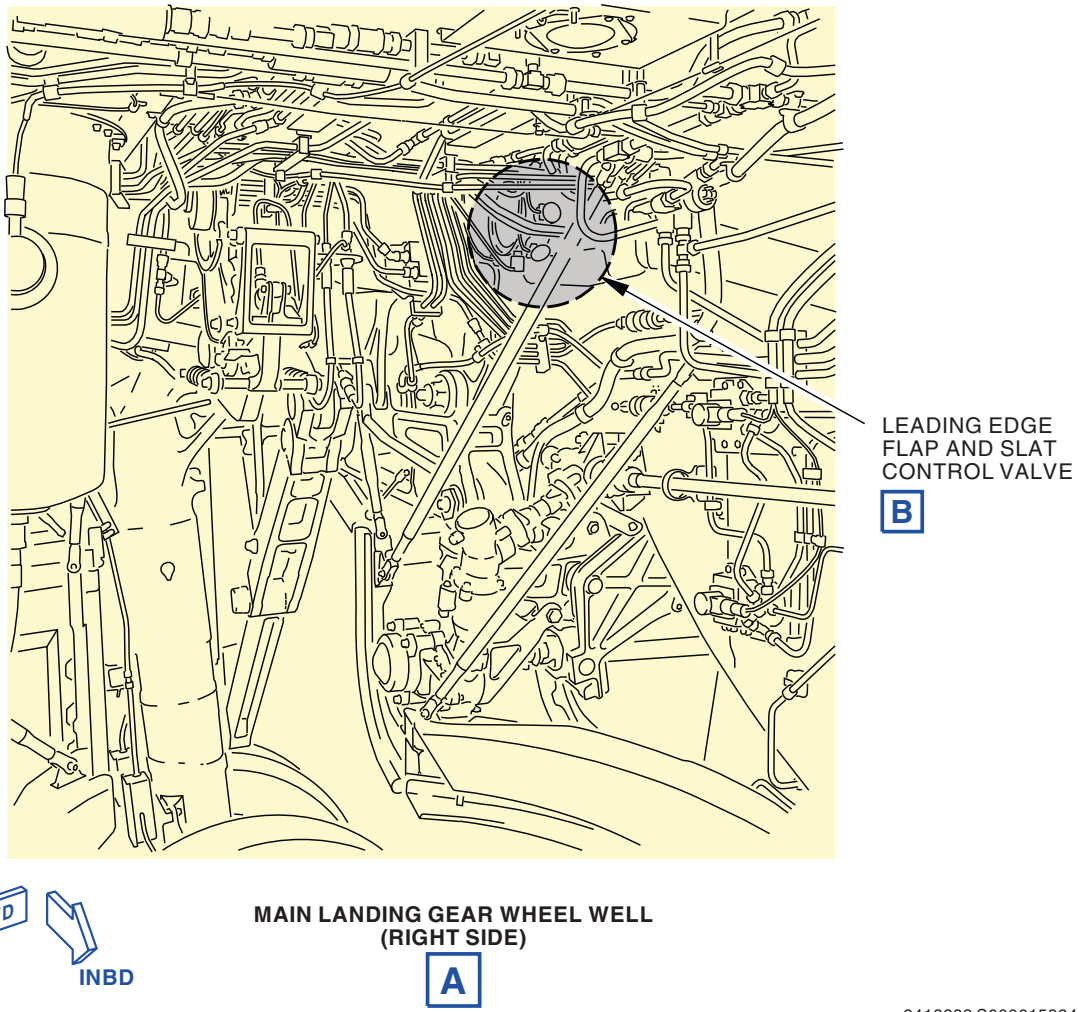
(e) Remove the bolts [9] and washers [10] that attach the leading edge flap and slat control valve assembly [1] to the flap control unit [6].

(f) Carefully remove the leading edge flap and slat control valve assembly [1] from the flap control unit [6].

END OF TASK

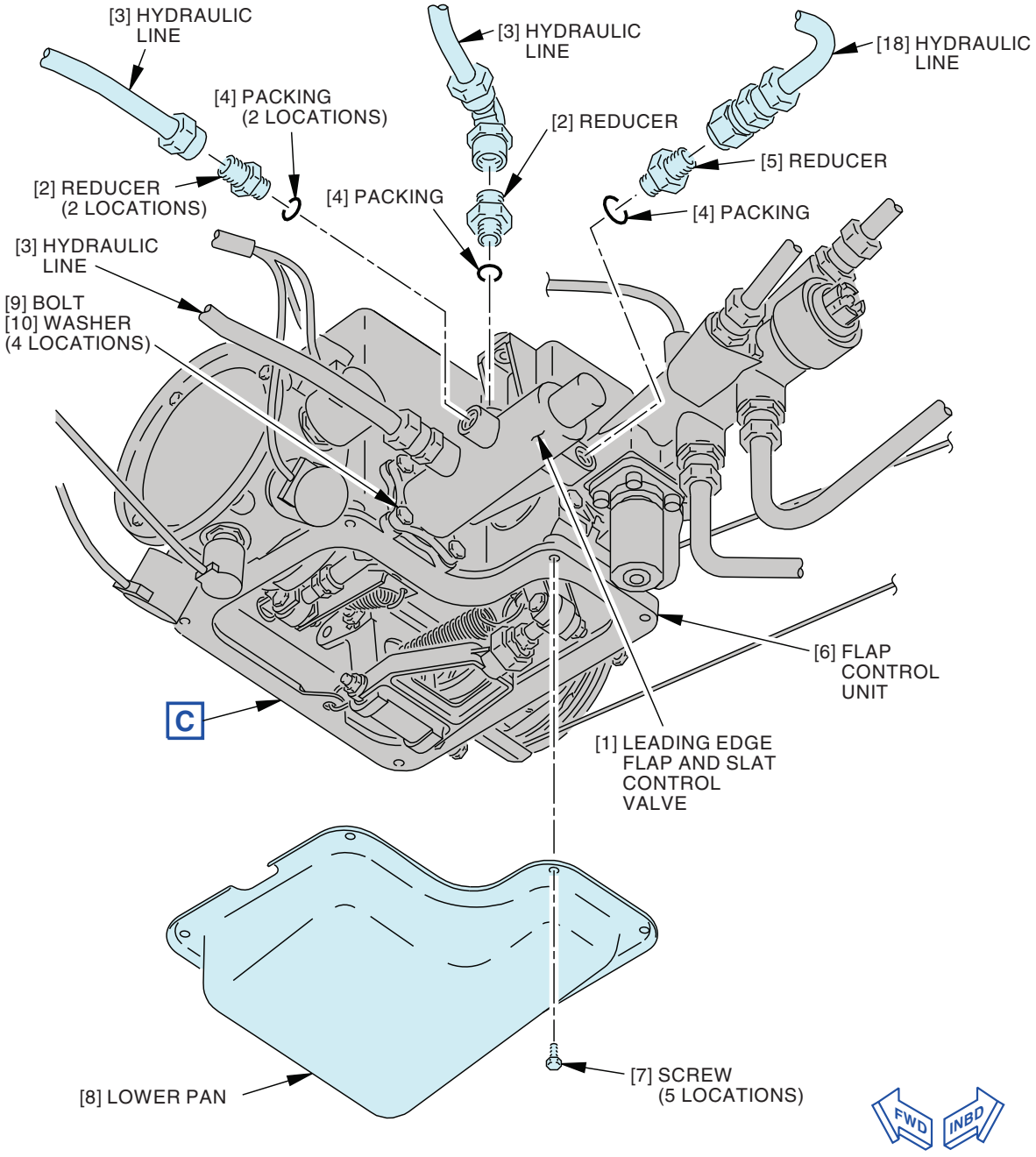
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Leading Edge Flap and Slat Control Valve Installation
Figure 401/27-81-81-990-801 (Sheet 1 of 3)

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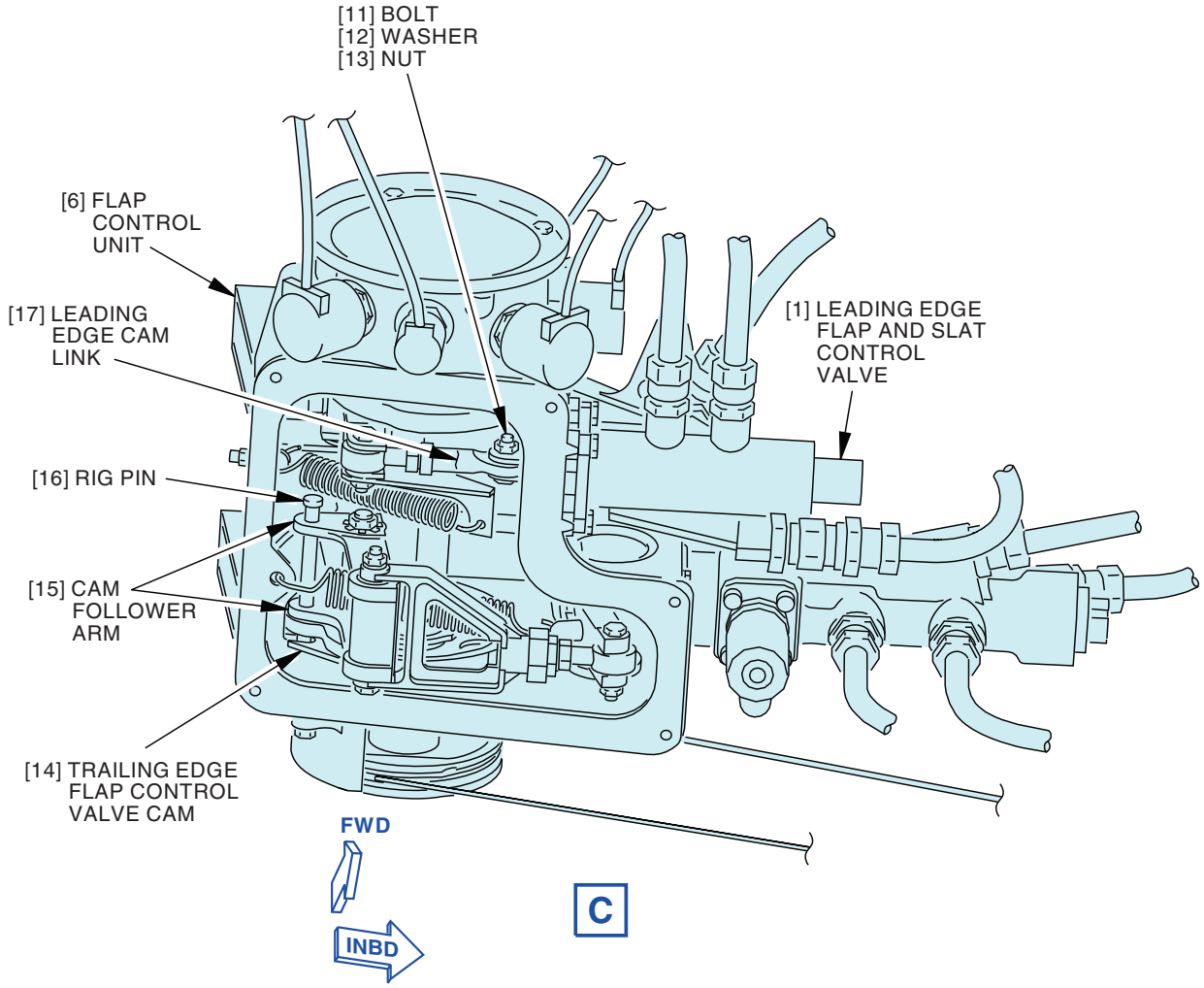
LEADING EDGE FLAP AND SLAT CONTROL VALVE

B

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Leading Edge Flap and Slat Control Valve Installation
Figure 401/27-81-81-990-801 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Leading Edge Flap and Slat Control Valve Installation
Figure 401/27-81-81-990-801 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-81-81-400-801

3. Leading Edge Flap and Slat Control Valve Installation

(Figure 401)

A. References

Reference	Title
12-12-00-610-801	Hydraulic Reservoir Servicing (P/B 301)
27-81-00-860-801	Leading Edge Flap and Slat System Operation With Primary Control (P/B 201)
27-81-81-820-801	Leading Edge Flap and Slat Control Valve Adjustment (P/B 501)

B. Consumable Materials

Reference	Description	Specification
D00054	Fluid - Hydraulic, Assembly Lubricant, MCS 352B	
D00153	Fluid - Hydraulic Fluid, Fire Resistant (Interchangeable And Intermixable With BMS 3-11 Type V)	BMS3-11 Type IV
G50237	Compound - Corrosion Inhibiting, Non-drying - Cor-Ban 27L	BMS3-38

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Leading edge flap and slat control valve assembly	27-81-81-01A-050	PTF ALL
4	Packing	27-81-81-01A-065	PTF ALL

D. Location Zones

Zone	Area
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right

E. Leading Edge Flap and Slat Control Valve Installation

SUBTASK 27-81-81-420-001

(1) If they are not installed, install the packings [4], reducers [2], and reducer [5] in the ports of the leading edge flap and slat control valve assembly [1].

(a) Apply hydraulic fluid, D00153, or MCS 352B fluid, D00054, to the new packings [4].

(b) Put the packings [4], reducers [2], and reducer [5] in their location on the leading edge flap and slat control valve assembly [1].

SUBTASK 27-81-81-420-002

(2) Install the leading edge flap and slat control valve assembly [1] in the flap control unit [6].

(a) Carefully put the leading edge flap and slat control valve assembly [1] on the flap control unit [6].

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION WHEN YOU USE COR-BAN 27L. IF IT GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER. GET MEDICAL AID. THIS MATERIAL CONTAINS FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(b) Apply Cor-Ban 27L Compound, G50237, to the shank and threads of the bolts [9].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Install the bolts [9] and washers [10] to attach the leading edge flap and slat control valve assembly [1] to the flap control unit [6].

(d) Install lockwire or safety cable on the bolts [9].

(e) Connect the leading edge flap and slat control valve assembly [1] to the leading edge cam link [17] with the bolt [11], washer [12], and nut [13].

(f) Connect the hydraulic lines [3] and hydraulic line [18] to the leading edge flap and slat control valve assembly [1].

1) Remove the caps from the hydraulic lines [3] and hydraulic line [18].

2) Remove the plugs from the hydraulic ports on the leading edge flap and slat control valve assembly [1].

3) Connect the hydraulic lines [3] to the leading edge flap and slat control valve assembly [1].

a) Tighten the B-nuts on hydraulic lines [3] to 500 ±25 in-lb (56 ±3 N·m).

b) Tighten the B-nut on hydraulic line [18] to 280 ±14 in-lb (32 ±2 N·m).

SUBTASK 27-81-81-080-001

(3) Remove the rig pin [16] from the flap control unit [6].

SUBTASK 27-81-81-860-003

WARNING

MAKE SURE PERSONS AND EQUIPMENT ARE CLEAR OF THE LEADING EDGE AND TRAILING EDGE CONTROL SURFACES. THE LEADING EDGE AND TRAILING EDGE CONTROL SURFACES CAN EXTEND AND RETRACT QUICKLY. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(4) Use the primary system to extend and retract the flaps and slats.

(a) To extend and retract the flaps and slats, do this task: Leading Edge Flap and Slat System Operation With Primary Control, TASK 27-81-00-860-801.

SUBTASK 27-81-81-210-001

(5) Examine the hydraulic lines at the control valve for leakage.

SUBTASK 27-81-81-820-001

(6) Do this task: Leading Edge Flap and Slat Control Valve Adjustment, TASK 27-81-81-820-801.

SUBTASK 27-81-81-420-003

(7) Install the lower pan [8] on the flap control unit [6].

(a) Put the lower pan [8] in its location on the flap control unit [6].

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION WHEN YOU USE COR-BAN 27L. IF IT GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER. GET MEDICAL AID. THIS MATERIAL CONTAINS FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(b) Apply Cor-Ban 27L Compound, G50237, to the shank and threads of the screws [7].

(c) Install the screws [7] to attach the lower pan [8] to the flap control unit [6].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Put the Airplane Back to Its Usual Condition

SUBTASK 27-81-81-610-001

(1) If it is necessary to fill the hydraulic reservoirs, do this task: Hydraulic Reservoir Servicing, TASK 12-12-00-610-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

LEADING EDGE FLAP AND SLAT CONTROL VALVE - ADJUSTMENT/TEST

1. General

A. This procedure has a task to adjust the leading edge flap and slat control valve.

TASK 27-81-81-820-801

2. Leading Edge Flap and Slat Control Valve Adjustment

(Figure 501)

A. References

Reference	Title
27-81-00-700-802	Leading Edge Flap and Slat Timing Test in Primary Mode (P/B 501)
27-81-00-860-804	Leading Edge Flaps and Slats Retraction (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205
SPL-12421	Rig Pin, F70207-116 (included in F70207 Kit) Part #: F70207-151 Supplier: 81205

C. Consumable Materials

Reference	Description	Specification
G01048	Lockwire - MS20995C32, Corrosion Resistant Steel - 0.032 Inch (0.8128 mm) Diameter	NASM20995
G50237	Compound - Corrosion Inhibiting, Non-drying - Cor-Ban 27L	BMS3-38

D. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
211	Flight Compartment - Left
212	Flight Compartment - Right

E. Prepare for the Adjustment

SUBTASK 27-81-81-480-003

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed on all the landing gears, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-81-81-860-004

(2) If the leading edge flaps and slats are not in the retract position, do this task: Leading Edge Flaps and Slats Retraction, TASK 27-81-00-860-804.

SUBTASK 27-81-81-010-001

(3) Remove the lower pan [3] from the flap control unit [1].

(a) Remove the screws [4] that attach the lower pan [3] to the flap control unit [1].

(b) Remove the lower pan [3].

SUBTASK 27-81-81-420-004

(4) Install the rig pin [7] through the cam follower arm [6] and the valve cam [5].

(a) Use rig pin, F70207-116, SPL-12421, from the rigging pin kit, SPL-1585.

F. Leading Edge Flap and Slat Control Valve Adjustment

SUBTASK 27-81-81-820-002

(1) Do the adjustment of the leading edge flap and slat control valve [2].

(a) Remove the MS20995C32 lockwire, G01048.

(b) Loosen the jam nut [14].

(c) Remove the nut [10], washer [9], washer [15], and bolt [8].

1) If there is no enough clearance to remove the bolt [8], remove the bolt [16], washer [17], washer [18], and nut [19].

(d) Turn the rod end [13] in 180 degree increments until you can get the valve slide [11] and the index surface [12] to align.

(e) Install the bolt [8] or bolt [16].

(f) Make sure that the valve slide [11] aligns with the index surface [12].

(g) If you installed the bolt [8], install the washer [9], washer [15], and nut [10].

(h) If you installed the bolt [16], install the washer [17], washer [18], and new nut [19].

1) Make sure that the bolt [16] is installed in proper direction.

(i) Make sure that the dimension from the center of the rod end [13] to the jam nut [14] is not more than 1 in. (25.4 mm).

(j) Tighten the jam nut [14] to 55 in-lb (6 N·m).

(k) Install the MS20995C32 lockwire, G01048, with the double twist method.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 27-81-81-020-003

(1) Remove the rig pin [7] from the cam follower arm [6] and the valve cam [5].

SUBTASK 27-81-81-410-001

(2) Install the lower pan [3] on the flap control unit [1].

(a) Put the lower pan [3] in its location on the flap control unit [1].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION WHEN YOU USE COR-BAN 27L. IF IT GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER. GET MEDICAL AID. THIS MATERIAL CONTAINS FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(b) Apply Cor-Ban 27L Compound, G50237, to the shank and threads of the screws [4].

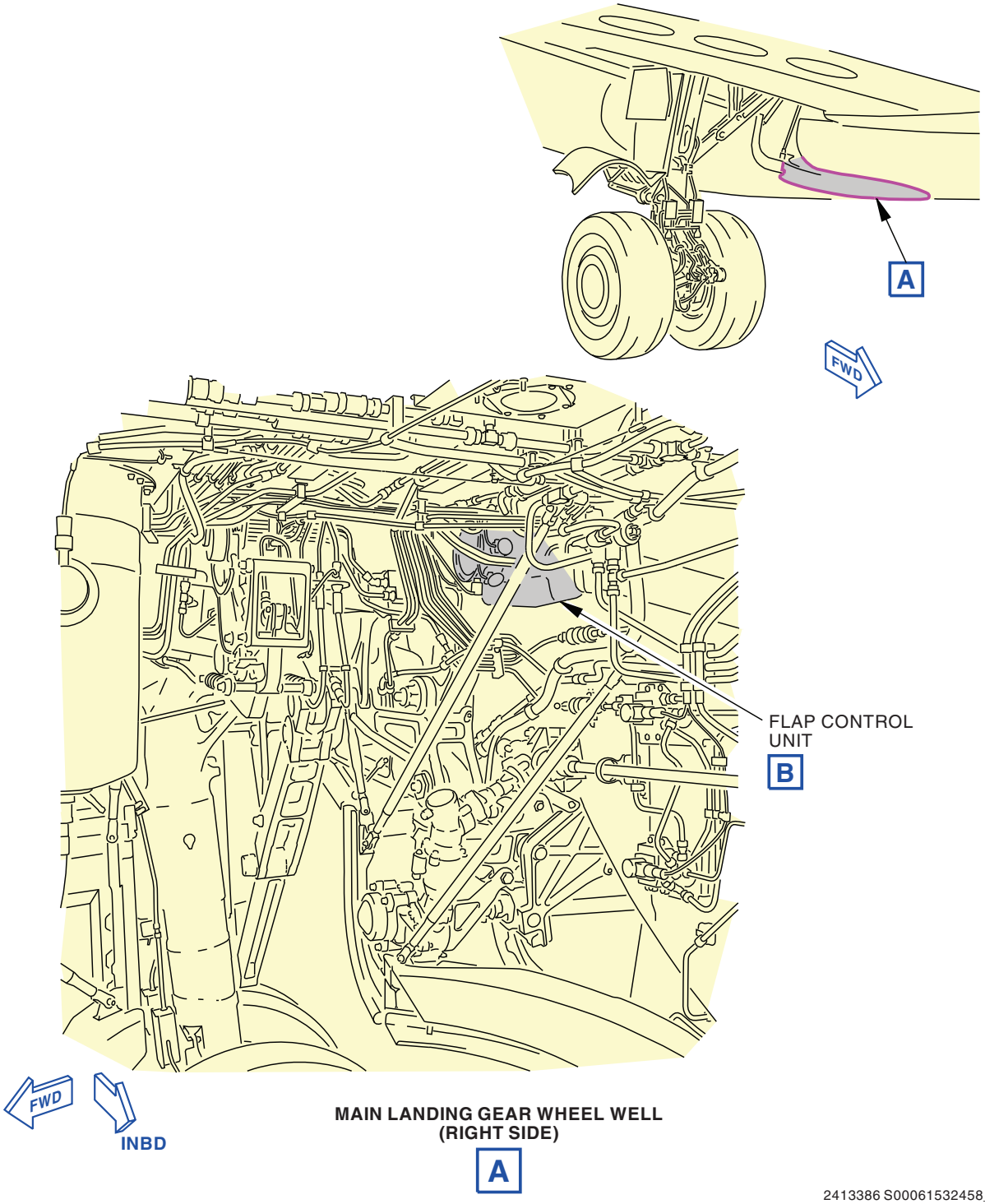
(c) Install the screws [4] that attach the lower pan [3] to the flap control unit [1].

SUBTASK 27-81-81-720-001

(3) Do this task: Leading Edge Flap and Slat Timing Test in Primary Mode, TASK 27-81-00-700-802.

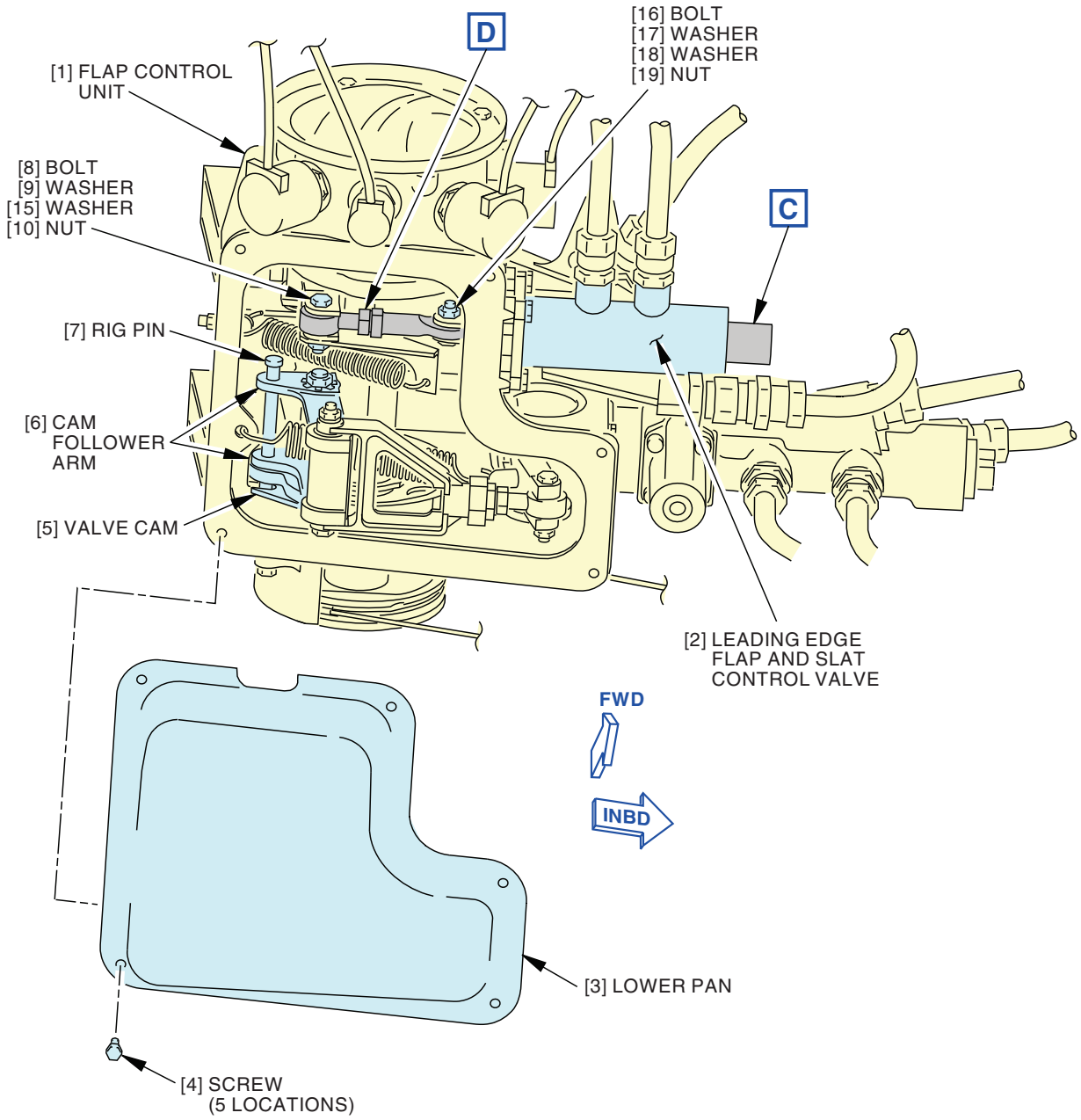
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Leading Edge Flap and Slat Control Valve Adjustment
Figure 501/27-81-81-990-802 (Sheet 1 of 3)

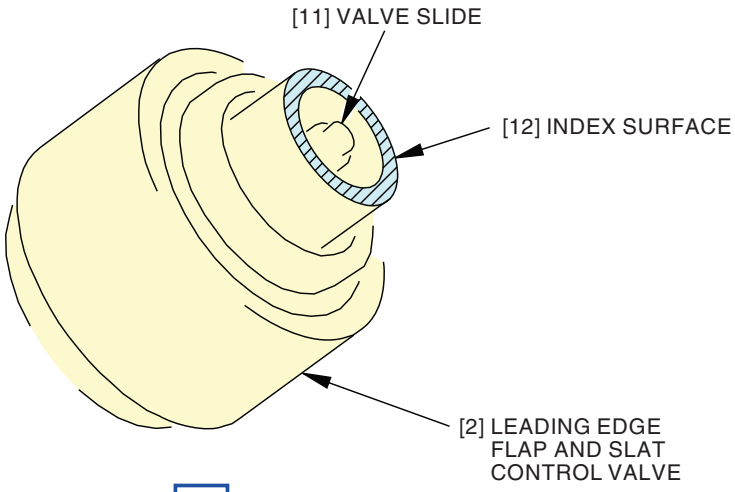
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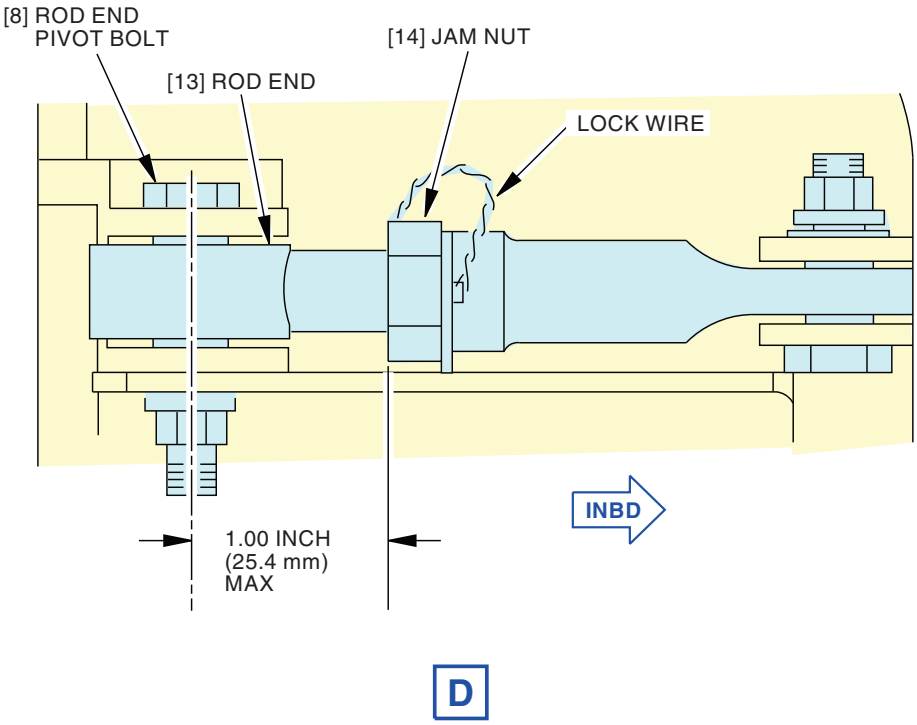


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Leading Edge Flap and Slat Control Valve Adjustment
Figure 501/27-81-81-990-802 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details





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Leading Edge Flap and Slat Control Valve Adjustment
Figure 501/27-81-81-990-802 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

LEADING EDGE STANDBY SHUTOFF VALVE - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) Leading edge standby shutoff valve removal

(2) Leading edge standby shutoff valve installation

(3) Standby shutoff valve motor removal

(4) Standby shutoff valve motor installation

(5) Leading edge standby shutoff valve test.

B. The leading edge standby shutoff valve is referred to as the "valve", and the standby shutoff valve motor is referred to as the "valve motor" in this procedure.

C. You can replace the valve motor with the valve installed in the standby flight controls hydraulic module.

TASK 27-81-91-000-801

2. Leading Edge Standby Shutoff Valve Removal

(Figure 201)

A. General

(1) This task gives the steps to remove the leading edge standby shutoff valve [13] assembly from the hydraulic module housing [10].

NOTE: You can remove the leading edge standby shutoff valve with the leading edge standby shutoff valve motor installed with it.

B. References

Reference	Title
29-09-00-860-802	Hydraulic Reservoirs Depressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Tools/Equipment

Reference	Description
STD-1154	Container - 5 Gallon (19 Liter)

D. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
211	Flight Compartment - Left
212	Flight Compartment - Right

E. Prepare for the Removal

SUBTASK 27-81-91-860-011



WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-81-91-860-001

(2) Make sure that there is no power on hydraulic system B. To remove it, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 27-81-91-860-002

(3) Do this task: Hydraulic Reservoirs Depressurization, TASK 29-09-00-860-802.

SUBTASK 27-81-91-860-003

(4) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	7	C00210	FLIGHT CONTROL FLAP SHUTOFF VALVES

F. Leading Edge Standby Shutoff Valve Removal

SUBTASK 27-81-91-020-001

(1) Remove the leading edge standby shutoff valve [13].

(a) Disconnect the electrical connector [1] from the leading edge standby shutoff valve motor [2].

(b) After you disconnect the electrical connector(s), examine the connector(s) for corrosion.

NOTE: The inspection of the electrical connector(s) can be done after the removal step or before the installation step of the electrical connector(s). This is an operator option.

(c) If there was corrosion, refer to SWPM 20-60-03, Paragraph 2 to correct the problem.

(d) Remove the lockwire that attaches to the screws [5].

(e) Remove the screws [5] and washers [6] that attach the leading edge standby shutoff valve motor [2] to the hydraulic module housing [10].

(f) Put a 5 gallon (19 liter) container, STD-1154, below the leading edge standby shutoff valve [13] to catch hydraulic fluid that drains from the leading edge standby shutoff valve [13].

(g) Carefully lift the leading edge standby shutoff valve [13] straight up to remove it from the standby flight controls hydraulic module housing [10].

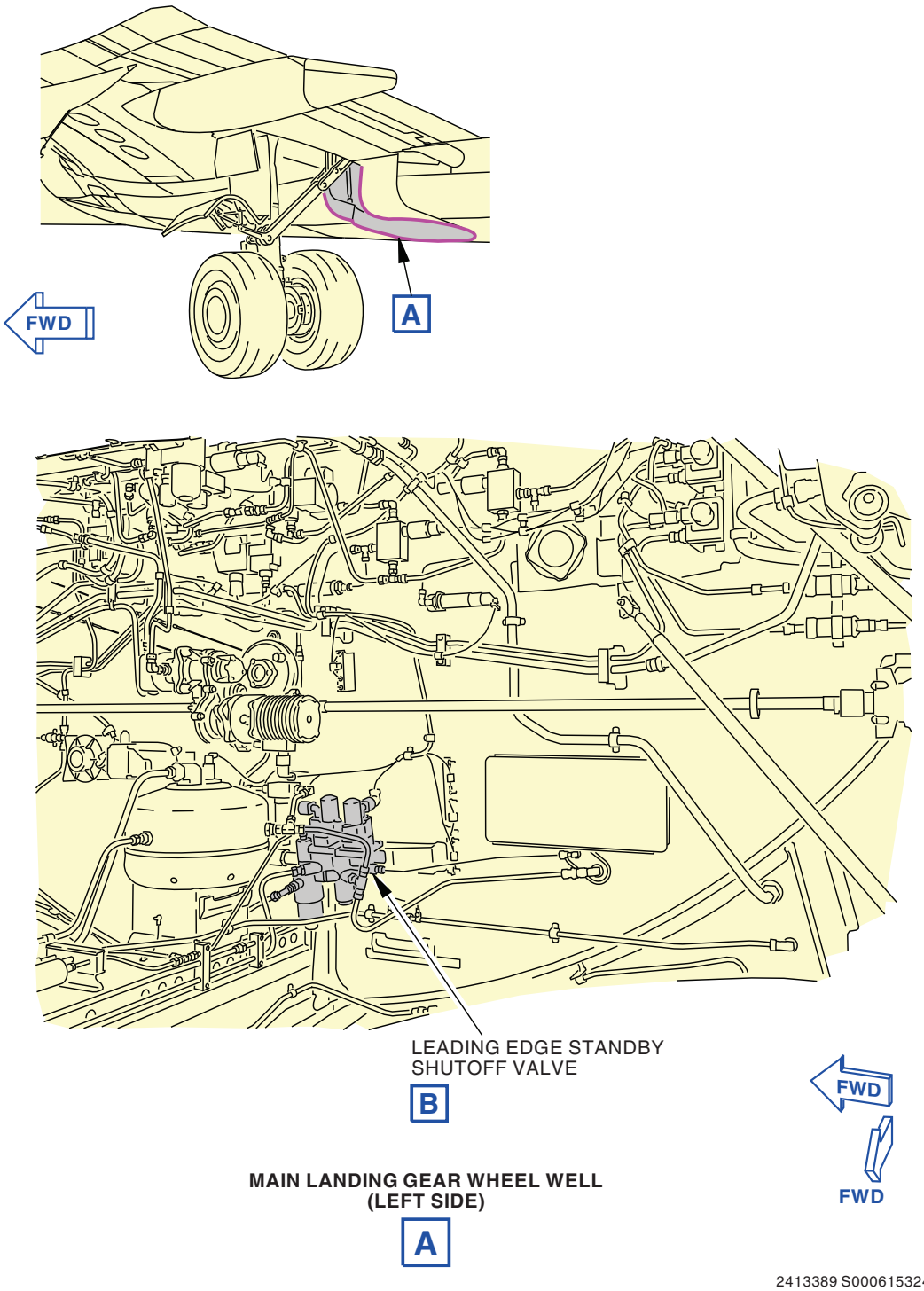
(h) Remove the backup rings [12], O-rings [11], backup rings [8], and O-ring [9] from the leading edge standby shutoff valve [13].

1) Discard the O-ring [9] and O-rings [11].

(i) Put a cover on the hydraulic module housing [10] to keep contamination out of the open leading edge standby shutoff valve motor [2] hole.

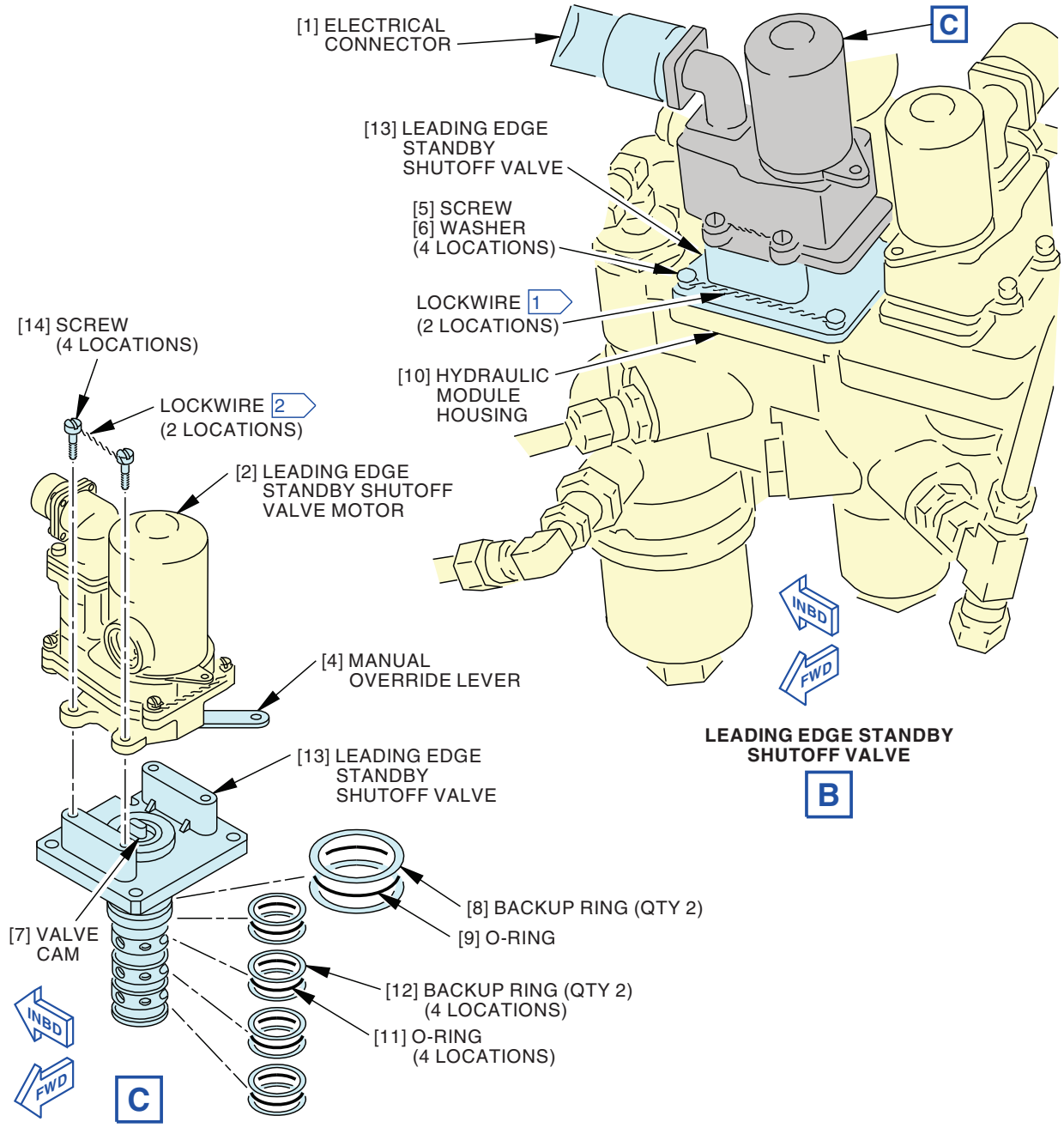
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Leading Edge Standby Shutoff Valve Installation
Figure 201/27-81-91-990-801 (Sheet 1 of 2)

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Leading Edge Standby Shutoff Valve Installation
Figure 201/27-81-91-990-801 (Sheet 2 of 2)

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TASK 27-81-91-400-801

3. Leading Edge Standby Shutoff Valve Installation

(Figure 201)

A. General

(1) This task gives the steps to install the leading edge standby shutoff valve [13] to the hydraulic module housing [10].

B. References

Reference	Title
12-12-00-610-801	Hydraulic Reservoir Servicing (P/B 301)
20-10-44-400-801	Lockwire, Cotter Pins, and Lockrings - Installation (P/B 401)

C. Consumable Materials

Reference	Description	Specification
D00054	Fluid - Hydraulic, Assembly Lubricant, MCS 352B	
D00153	Fluid - Hydraulic Fluid, Fire Resistant (Interchangeable And Intermixable With BMS 3-11 Type V)	BMS3-11 Type IV
G01912	Lockwire - MS20995NC32, Monel - 0.032 Inch (0.8128 mm) Diameter	NASM20995
G50171	Compound - Corrosion Inhibiting Compound, Interior Application - D5026NS or ZC-026	

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
8	Backup ring	27-81-91-02-020	PTF ALL
9	O-ring	27-81-91-02-015	PTF ALL
11	O-ring	27-81-91-02-025	PTF ALL
12	Backup ring	27-81-91-02-030	PTF ALL
13	Leading edge standby shutoff valve	27-81-91-02-035	PTF ALL

E. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
211	Flight Compartment - Left
212	Flight Compartment - Right

F. Leading Edge Standby Shutoff Valve Installation

SUBTASK 27-81-91-420-001

(1) Install the backup rings [8], O-ring [9], backup rings [12], and O-rings [11] on the leading edge standby shutoff valve [13].

(a) Use new O-ring [9] and O-rings [11].

(b) Apply hydraulic fluid, D00153, or MCS 352B fluid, D00054, to the backup rings [8], O-ring [9], backup rings [12], and O-rings [11].

(c) Use new O-ring [9] and O-rings [11].

SUBTASK 27-81-91-420-002

(2) Install the leading edge standby shutoff valve [13].

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(a) Carefully install the leading edge standby shutoff valve [13] in the hydraulic module housing [10].

(b) Install the screws [5] and washers [6] to attach the leading edge standby shutoff valve [13] to the hydraulic module housing [10].

(c) Install the MS20995NC32 lockwire, G01912, on the screws [5] (TASK 20-10-44-400-801).

(d) If you did not examine the electrical connector(s) at the removal step, do these steps:

NOTE: The inspection of the electrical connector(s) can be done after the removal step or before the installation step of the electrical connector(s). This is an operator option.

1) Before you connect the electrical connector(s), examine the connector(s) for corrosion.

2) If there was corrosion, refer to SWPM 20-60-03, Paragraph 2 to correct the problem.

(e) Apply the D5026NS or ZC-026 compound, G50171, to the electrical connector [1] (SWPM 20-60-03, Paragraph 2).

(f) Connect the electrical connector [1] to the leading edge standby shutoff valve motor [2].

NOTE: Corrosion inhibiting compound is applied to the exterior of the mated electrical connectors after the operational test is completed.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 27-81-91-860-004

(1) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	7	C00210	FLIGHT CONTROL FLAP SHUTOFF VALVES

SUBTASK 27-81-91-610-001

(2) If it is necessary to service the hydraulic reservoirs, do this task: Hydraulic Reservoir Servicing, TASK 12-12-00-610-801.

SUBTASK 27-81-91-720-001

(3) Do this task: Leading Edge Standby Shutoff Valve Test, TASK 27-81-91-720-801.

——— END OF TASK ———

TASK 27-81-91-000-802

4. Standby Shutoff Valve Motor Removal

(Figure 201)

A. General

(1) This task gives the steps to remove the leading edge standby shutoff valve motor [2] from the leading edge standby shutoff valve [13].

B. References

Reference	Title
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left

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(Continued)

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Prepare for the Removal

SUBTASK 27-81-91-860-012

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 27-81-91-860-005

(2) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	7	C00210	FLIGHT CONTROL FLAP SHUTOFF VALVES

E. Standby Shutoff Valve Motor Removal

SUBTASK 27-81-91-020-002

(1) Remove the leading edge standby shutoff valve motor [2].

(a) Disconnect the electrical connector [1] from the leading edge standby shutoff valve motor [2].

(b) After you disconnect the electrical connector(s), examine the connector(s) for corrosion.

NOTE: The inspection of the electrical connector(s) can be done after the removal step or before the installation step of the electrical connector(s). This is an operator option.

(c) If there was corrosion, refer to SWPM 20-60-03, Paragraph 2 to correct the problem.

(d) Set the manual override lever [4] on the leading edge standby shutoff valve motor [2] to POSITION 2.

(e) Remove the lockwire that attaches to the screws [14].

(f) Remove the screws [14] that attach the leading edge standby shutoff valve motor [2] to the leading edge standby shutoff valve [13].

(g) Remove the leading edge standby shutoff valve motor [2] from the leading edge standby shutoff valve [13].

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-81-91-400-802

5. Standby Shutoff Valve Motor Installation

(Figure 201)

A. General

(1) This task gives the steps to install the leading edge standby shutoff valve motor [2] to the leading edge standby shutoff valve [13].

NOTE: You can install the leading edge standby shutoff valve motor with the leading edge standby shutoff valve installed in the hydraulic module housing.

B. References

Reference	Title
20-10-44-400-801	Lockwire, Cotter Pins, and Lockrings - Installation (P/B 401)

C. Consumable Materials

Reference	Description	Specification
G02166	Lockwire - MS20995NC20, Monel - 0.020 Inch (0.508 mm) Diameter	NASM20995
G50171	Compound - Corrosion Inhibiting Compound, Interior Application - D5026NS or ZC-026	

D. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
211	Flight Compartment - Left
212	Flight Compartment - Right

E. Standby Shutoff Valve Motor Installation

SUBTASK 27-81-91-420-003

(1) Install the leading edge standby shutoff valve motor [2].

(a) Make sure that the manual override lever [4] on the leading edge standby shutoff valve motor [2] is set to POSITION 2.

(b) Set the leading edge standby shutoff valve motor [2] in its position on the leading edge standby shutoff valve [13].

(c) Engage the leading edge standby shutoff valve motor [2] with the valve cam [7].

(d) Install the screws [14] to attach the leading edge standby shutoff valve motor [2].

(e) Tighten the screws [14] to 17 in-lb (1.92 N·m) - 20 in-lb (2.26 N·m).

(f) Install the MS20995NC20 lockwire, G02166, on the screws [14] (TASK 20-10-44-400-801).

(g) If you did not examine the electrical connector(s) at the removal step, do these steps:

NOTE: The inspection of the electrical connector(s) can be done after the removal step or before the installation step of the electrical connector(s). This is an operator option.

1) Before you connect the electrical connector(s), examine the connector(s) for corrosion.

2) If there was corrosion, refer to SWPM 20-60-03, Paragraph 2 to correct the problem.

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(h) Apply the D5026NS or ZC-026 compound, G50171, to the electrical connector [1] (SWPM 20-60-03, Paragraph 2).

(i) Connect the electrical connector [1] to the leading edge standby shutoff valve motor [2].

NOTE: Corrosion inhibiting compound is applied to the exterior of the mated electrical connectors after the operational test is completed.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 27-81-91-860-006

(1) Set the manual override lever [4] on the leading edge standby shutoff valve motor [2] to POSITION 1.

SUBTASK 27-81-91-860-007

(2) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	7	C00210	FLIGHT CONTROL FLAP SHUTOFF VALVES

SUBTASK 27-81-91-720-002

(3) Do this task: Leading Edge Standby Shutoff Valve Test, TASK 27-81-91-720-801.

——— END OF TASK ———

TASK 27-81-91-720-801

6. Leading Edge Standby Shutoff Valve Test

(Figure 202)

A. References

Reference	Title
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)

B. Consumable Materials

Reference	Description	Specification
G50170	Compound - Corrosion Inhibiting Compound, Soft Film, Exterior Use - Ardrox AV25	

C. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Leading Edge Standby Shutoff Valve Motor Test

SUBTASK 27-81-91-710-001

(1) Do the test for the leading edge standby shutoff valve [13].

(a) Make sure that the manual override lever [4] is at position 1.

(b) To pressurize hydraulic system B, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

(c) Make sure that the flap control lever is in the UP detent.

(d) Set the alternate flaps arm switch, on the P5 flight controls panel, to the ARM position.

1) Make sure that the STANDBY HYD LOW PRESSURE light comes on.

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2) Make sure that the standby hydraulic pump starts.

3) Make sure that the STANDBY HYD LOW PRESSURE light is off.

WARNING MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF THE LEADING EDGE SLATS. THE LEADING EDGE SLATS CAN MOVE QUICKLY. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(e) Set the alternate flaps control switch, on the P5 flight controls panel, to the DOWN position.

1) Make sure that the manual override lever [4] is at position 2.

2) Make sure that the leading edge flaps and slats fully extend.

(f) Set the alternate flaps control switch to the OFF position.

1) Make sure that the leading edge flaps and slats do not move.

WARNING MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF THE LEADING EDGE SLATS. THE LEADING EDGE SLATS CAN MOVE QUICKLY. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(g) Set the alternate flaps arm switch to the OFF position.

1) Make sure that the manual override lever [4] moves to position 1.

2) Make sure that the leading edge flaps and slats fully retract.

NOTE: The leading edge flaps and slats retract because of the pressurization of hydraulic system B.

SUBTASK 27-81-91-210-001

(2) Make sure that there is no hydraulic fluid leakage from the valve hydraulic fittings.

SUBTASK 27-81-91-860-010

(3) To remove power from hydraulic system B, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

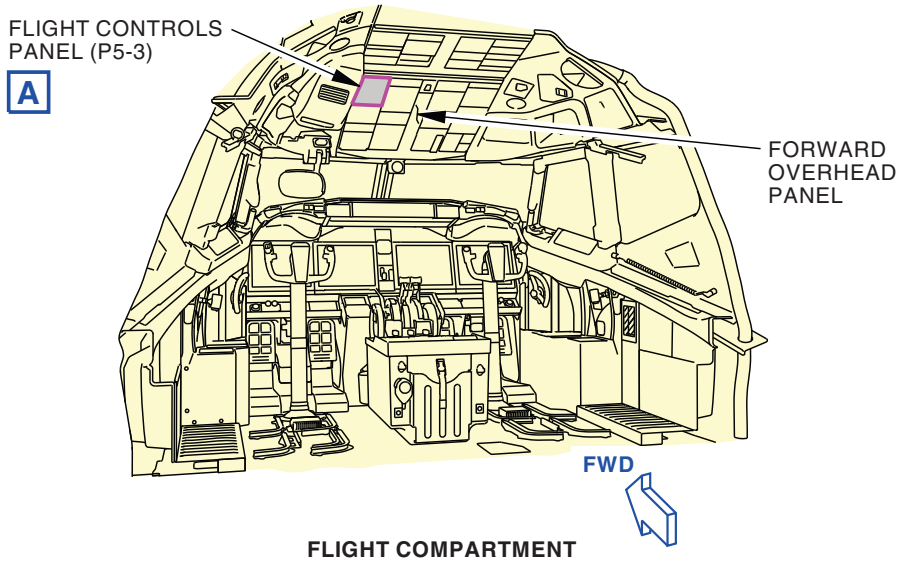
SUBTASK 27-81-91-212-001

(4) Apply the Ardrox AV25 compound, G50170, to the mated electrical connector assembly completely, until the compound drips from the connector.

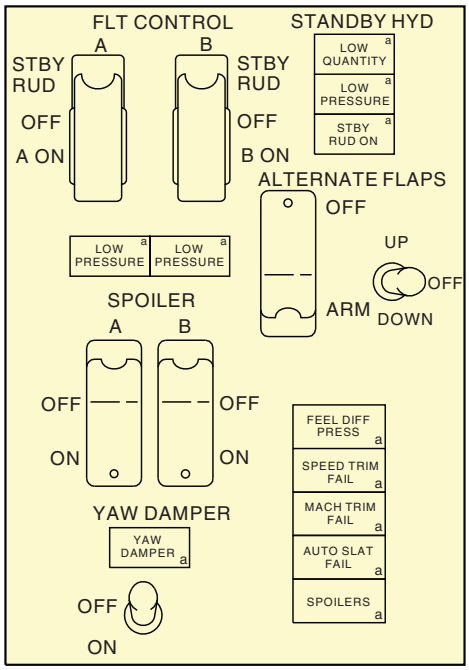
(a) Do not remove the excess corrosion inhibiting compound.

——— END OF TASK ———

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FLIGHT COMPARTMENT



FLIGHT CONTROLS PANEL (P5-3)

A

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Alternate Flaps Control Switches
Figure 202/27-81-91-990-802

ECCN 9E991 BOEING PROPRIETARY - See title page for details

LEADING EDGE AUTOSLAT SYSTEM - ADJUSTMENT/TEST

1. General

A. This procedure has these tasks:

(1) Test of the autoslat system indicator light

(2) Operational test of the leading edge autoslat system.

TASK 27-83-00-720-801

2. Autoslat System Indicator Light - Functional Test

(Figure 501)

A. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

B. Do a test of the autoslat system indicator light

SUBTASK 27-83-00-860-004

(1) Make sure that these circuit breakers are closed:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
E	5	C01204	SMYD-1 CMPTR DC

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
B	5	C01206	SMYD-2 CMPTR DC

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	14	C01070	FLIGHT CONTROL AUTOSLAT DC 2
C	14	C01068	FLIGHT CONTROL AUTOSLAT DC 1

SUBTASK 27-83-00-820-001

(2) Make sure that the AUTOSLAT FAIL light, on the P5-3 overhead panel, is off.

SUBTASK 27-83-00-720-001

(3) Push and release the AUTOSLAT FAIL light switch.

(a) Make sure that the AUTOSLAT FAIL light comes on when you push the light switch then it goes off when you release.

SUBTASK 27-83-00-720-002

(4) Put the LIGHTS switch in the TEST position and make sure that these lights come on (Figure 501):

(a) The MASTER CAUTION lights.

(b) The FLT CONT light.

(c) The AUTOSLAT FAIL light.

SUBTASK 27-83-00-860-005

(5) Put the LIGHTS switch in the BRT position.

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SUBTASK 27-83-00-720-003

(6) Push the MASTER CAUTION light switch and make sure that these lights go off:

(a) The MASTER CAUTION lights.

(b) The FLT CONT light.

(c) The AUTOSLAT FAIL light.

SUBTASK 27-83-00-720-004

(7) Push the master caution annunciator panel.

(a) Make sure that the AUTOSLAT FAIL light stays off.

SUBTASK 27-83-00-720-005

(8) Make sure that the autoslat fault display works correctly when autoslat 1 is disabled:

(a) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	14	C01068	FLIGHT CONTROL AUTOSLAT DC 1

(b) Push the master caution annunciator panel and make sure that these lights come on:

1) The MASTER CAUTION lights.

2) The FLT CONT light.

3) The AUTOSLAT FAIL light.

(c) Push the MASTER CAUTION switch light and make sure that these lights go off:

1) The MASTER CAUTION lights.

2) The FLT CONT light.

3) The AUTOSLAT FAIL light.

SUBTASK 27-83-00-720-006

(9) Make sure that the autoslat fault display works correctly when autoslat 1 and 2 are disabled:

(a) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	14	C01070	FLIGHT CONTROL AUTOSLAT DC 2

(b) Push the master caution annunciator panel and make sure that these lights come on:

1) The MASTER CAUTION lights.

2) The FLT CONT light.

3) The AUTOSLAT FAIL light.

(c) Push the MASTER CAUTION light switch and make sure that these actions occur:

1) The MASTER CAUTION lights go off.

2) The FLT CONT light goes off.

3) The AUTOSLAT FAIL light stays on.

SUBTASK 27-83-00-720-007

(10) Make sure that the autoslat fault display works properly when autoslat 2 is disabled:

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(a) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	14	C01068	FLIGHT CONTROL AUTOSLAT DC 1

(b) Make sure that the AUTOSLAT FAIL light goes off.

(c) Push the master caution annunciator panel and make sure that these lights come on:

1) The MASTER CAUTION lights.
2) The FLT CONT light.
3) The AUTOSLAT FAIL light.

(d) Push the MASTER CAUTION light switch and make sure that these lights go off:

1) The MASTER CAUTION lights.
2) The FLT CONT light.
3) The AUTOSLAT FAIL light.

SUBTASK 27-83-00-860-006

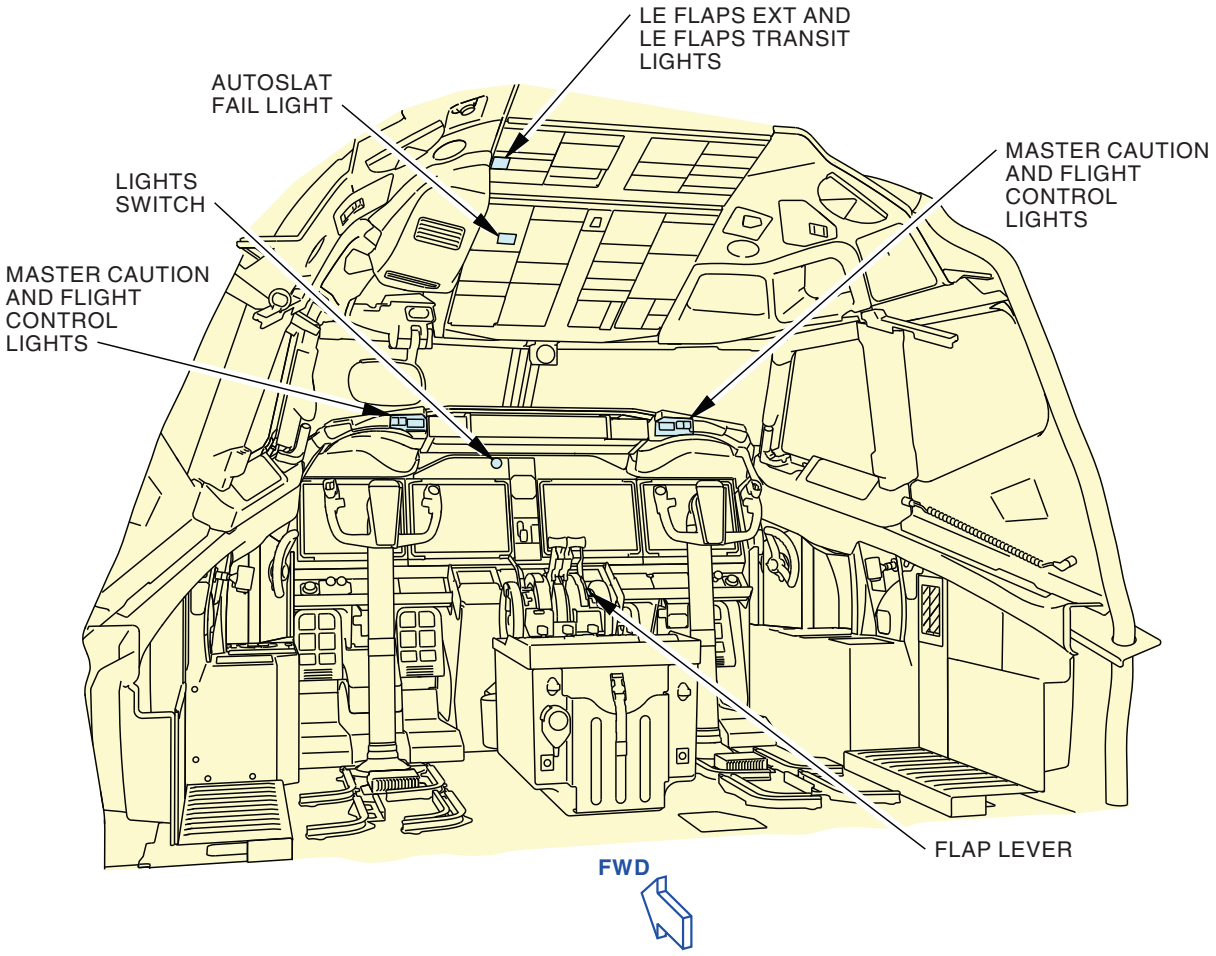
(11) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	14	C01070	FLIGHT CONTROL AUTOSLAT DC 2

——— END OF TASK ———

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FLIGHT COMPARTMENT

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Leading Edge Autoslat System Test
Figure 501/27-83-00-990-802

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TASK 27-83-00-710-801

3. Leading Edge Autoslat System - Functional Test

(Figure 502)

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-09-00-840-802	Return the Airplane Systems Back to Their Normal on Ground Condition (P/B 201)
32-09-00-860-801	Put the Airplane in the Air Mode (P/B 201)
32-09-00-860-802	Return the Airplane to the Ground Mode (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1917	Fixture - Test, Angle of Attack Sensor, ROSEMOUNT AOA's Part #: J34002-19 Supplier: 81205 Opt Part #: A34012-24 Supplier: 81205 Opt Part #: J34002-18 Supplier: 81205

C. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
211	Flight Compartment - Left
212	Flight Compartment - Right
500	Left Wing
600	Right Wing

D. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

E. Prepare for the Procedure

NOTE: There is no need to simulate airspeed during this test.

SUBTASK 27-83-00-860-007

WARNING

MAKE SURE ALL PERSONS AND EQUIPMENT ARE CLEAR OF THE FLIGHT CONTROLS SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

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(WARNING PRECEDES)

WARNING OBEY THE PROCEDURE THAT PUTS THE AIRPLANE IN THE AIR MODE. IF YOU DO THE PROCEDURE INCORRECTLY, INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) For the hydraulic system B, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 27-83-00-010-001

(2) Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

F. Autoslat Operation Test

SUBTASK 27-83-00-860-008

(1) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
E	4	C01392	STICK SHAKER LEFT

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
C	3	C01072	HEATERS ALPHA VANE LEFT
D	3	C01071	HEATERS ALPHA VANE RIGHT

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
B	6	C01393	STICK SHAKER RIGHT

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	14	C01070	FLIGHT CONTROL AUTOSLAT DC 2
C	14	C01068	FLIGHT CONTROL AUTOSLAT DC 1

SUBTASK 27-83-00-840-002

WARNING KEEP PERSONS AND EQUIPMENT AWAY FROM THE LEADING EDGE SLATS WHEN OPERATE THIS TEST. THE LEADING EDGE SLATS WILL MOVE. INJURY TO PERSONNEL OR DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Make sure that there are no personnel or equipment near the leading edge slats.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-83-00-820-002

WARNING MAKE SURE THAT THE HEATER CIRCUIT BREAKERS ABOVE ARE OPEN AND THE AOA SENSOR HEAT IS OFF BEFORE YOU MOVE THE AOA SENSOR. IF THE HEATER CIRCUIT BREAKERS ARE CLOSED, INJURY TO PERSONS CAN OCCUR.

(3) Turn the Left AOA sensor counterclockwise and the right AOA sensor clockwise to 30 degrees (Figure 502).

NOTE: You can use Method 1 or Method 2 to measure the AOA sensor angle.

NOTE: For the duration of this test, airspeed must not be simulated.

(a) Method 1 to measure the AOA sensor angle:

1) Install the angle of attack probe test fixture, SPL-1917, on the AOA sensor.

NOTE: The test fixture allows for accurate measurement of the AOA sensor angle.

(b) Method 2 to measure the AOA sensor angle:

1) Do these steps on each Stall Management Yaw Damper (SMYD) to determine when the AOA sensor angles show 30 degrees airplane nose up:

a) Push the ON/OFF switch on the SMYD.

b) Push the down arrow switch on the SMYD until the display shows GROUND TESTS?.

c) Push the YES switch.

d) Push the down arrow switch on the SMYD until the display shows ANALOG INPUTS?.

e) Push the YES switch.

<1> Make sure the display shows AOA SENSOR.

f) Push the YES switch.

g) The display will show AOA and the angle of the AOA sensor.

SUBTASK 27-83-00-860-009

WARNING MAKE SURE ALL PERSONS AND EQUIPMENT ARE CLEAR OF THE FLIGHT CONTROLS SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(4) Move the flaps to the 1 position.

(a) Make sure that the leading edge slats move to the flaps 1 (mid-extend) position.

SUBTASK 27-83-00-860-010

(5) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	14	C01068	FLIGHT CONTROL AUTOSLAT DC 1

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-83-00-840-003

WARNING MAKE SURE THAT YOU DO THE STEPS TO PREPARE THE SYSTEMS FOR THE AIR MODE CORRECTLY. IF YOU DO NOT FOLLOW THESE INSTRUCTIONS, INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT CAN OCCUR.

(6) Do this task: Put the Airplane in the Air Mode, TASK 32-09-00-860-801.

WARNING MAKE SURE ALL PERSONS AND EQUIPMENT ARE CLEAR OF THE FLIGHT CONTROLS SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(a) Make sure that the slats move to the full extend position.

(b) Make sure that LE FLAPS EXT displays and LE FLAPS TRANSIT not displays on the Engine Display of the main instrument panel.

SUBTASK 27-83-00-860-011

(7) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	14	C01068	FLIGHT CONTROL AUTOSLAT DC 1

WARNING MAKE SURE ALL PERSONS AND EQUIPMENT ARE MOVED AWAY FROM THE CONTROL SURFACES BEFORE YOU START MOVEMENT. THIS WILL PREVENT INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(a) Make sure that the slats move to the flaps 1 (mid-extend) position.

SUBTASK 27-83-00-860-012

(8) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	14	C01070	FLIGHT CONTROL AUTOSLAT DC 2

WARNING MAKE SURE ALL PERSONS AND EQUIPMENT ARE MOVED AWAY FROM THE CONTROL SURFACES BEFORE YOU START MOVEMENT. THIS WILL PREVENT INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(a) Make sure that the slats move to the full extend position.

(b) Make sure that LE FLAPS EXT displays and LE FLAPS TRANSIT not displays on the Engine Display of the main instrument panel.

SUBTASK 27-83-00-860-013

(9) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	14	C01070	FLIGHT CONTROL AUTOSLAT DC 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING MAKE SURE ALL PERSONS AND EQUIPMENT ARE CLEAR OF THE FLIGHT CONTROLS SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(a) Make sure that the slats move to the flaps 1 (mid-extend) position.

SUBTASK 27-83-00-080-003

(10) If installed, remove the angle of attack probe test fixture, SPL-1917, from the AOA sensor.

SUBTASK 27-83-00-820-003

(11) Turn the Left and right AOA sensors to 0 degrees (Figure 502).

SUBTASK 27-83-00-820-004

(12) Put the flap lever in the UP position.

SUBTASK 27-83-00-860-014

(13) Do this task: Return the Airplane to the Ground Mode, TASK 32-09-00-860-802.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 27-83-00-840-004

(1) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
E	4	C01392	STICK SHAKER LEFT

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
C	3	C01072	HEATERS ALPHA VANE LEFT
D	3	C01071	HEATERS ALPHA VANE RIGHT

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
B	6	C01393	STICK SHAKER RIGHT

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	14	C01070	FLIGHT CONTROL AUTOSLAT DC 2
C	14	C01068	FLIGHT CONTROL AUTOSLAT DC 1

SUBTASK 27-83-00-840-005

(2) Do this task: Return the Airplane Systems Back to Their Normal on Ground Condition, TASK 32-09-00-840-802.

SUBTASK 27-83-00-840-006

(3) Push the ON/OFF switch on the SMYD to turn off the SMYD display.

(a) Push the YES switch to make sure that the display is off.

SUBTASK 27-83-00-840-007

(4) If hydraulic power is not necessary, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

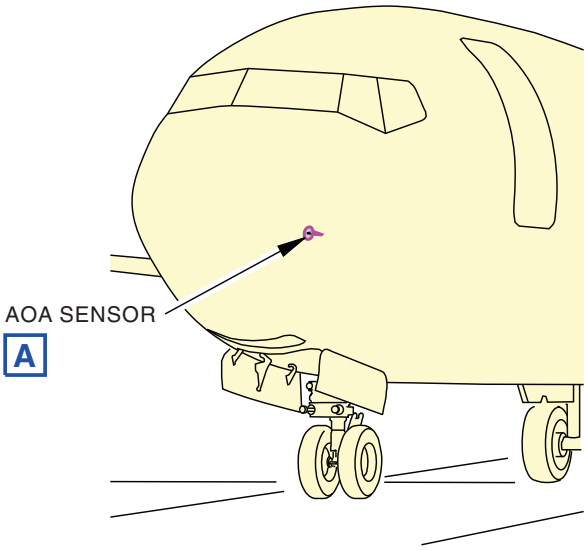
SUBTASK 27-83-00-940-001

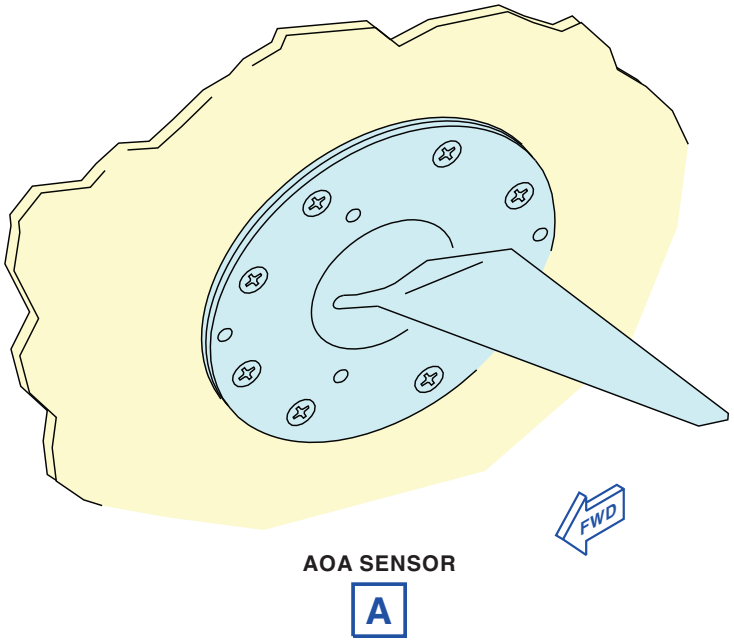
(5) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details





NOTE:
LEFT AOA SENSOR IS SHOWN, RIGHT AOA SENSOR IS OPPOSITE.

2413393 S00061532480_V2

Angle of Attack (AOA) Sensor
Figure 502/27-83-00-990-803

ECCN 9E991 BOEING PROPRIETARY - See title page for details

AUTOSLAT CONTROL VALVE - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Removal of the autoslat control valve

(2) Installation of the autoslat control valve.

B. There are solenoids installed on the autoslat control valve. It is not necessary to remove the solenoids to remove the control valve.

TASK 27-83-21-000-801

2. Autoslat Control Valve Removal

(Figure 401)

A. References

Reference	Title
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
SWPM 20-60-03	Special Protection of Electrical Connectors

B. Location Zones

Zone	Area
192	Lower Wing-To-Body Fairing - Under Wing Box

C. Access Panels

Number	Name/Location
192BR	ECS Ram Air Inlet Mixing Duct Panel - Forward

D. Prepare for the Removal

SUBTASK 27-83-21-480-001

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins installed on the main and nose landing gear. Do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 27-83-21-860-001

(2) Do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 27-83-21-010-001

(3) To gain access to the autoslat control valve,

Remove this access panel:

Number	Name/Location
192BR	ECS Ram Air Inlet Mixing Duct Panel - Forward

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Autoslat Control Valve Removal

SUBTASK 27-83-21-860-002

(1) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	14	C01070	FLIGHT CONTROL AUTOSLAT DC 2
C	14	C01068	FLIGHT CONTROL AUTOSLAT DC 1

SUBTASK 27-83-21-020-001

(2) Disconnect the electrical connector [6] and electrical connector [7] from the autoslat control valve [1].

(a) After you disconnect the electrical connector(s), examine the connector(s) for corrosion.

NOTE: You can do the inspection of the electrical connector(s) after the removal or before the installation steps of the electrical connector(s). This is an operator option.

(b) If there was corrosion, refer to (SWPM 20-60-03) to correct the problem.

SUBTASK 27-83-21-480-002

(3) Put a bucket or other applicable container to collect hydraulic fluid below the autoslat control valve [1].

SUBTASK 27-83-21-020-002

(4) Disconnect the hydraulic line (C2) [4], hydraulic line [5], hydraulic line (C1) [8], hydraulic line [9], hydraulic line [10], and hydraulic line [11] from the autoslat control valve [1].

SUBTASK 27-83-21-020-003

(5) Remove the bolts [2], washers [3], and the autoslat control valve [1].

SUBTASK 27-83-21-480-003

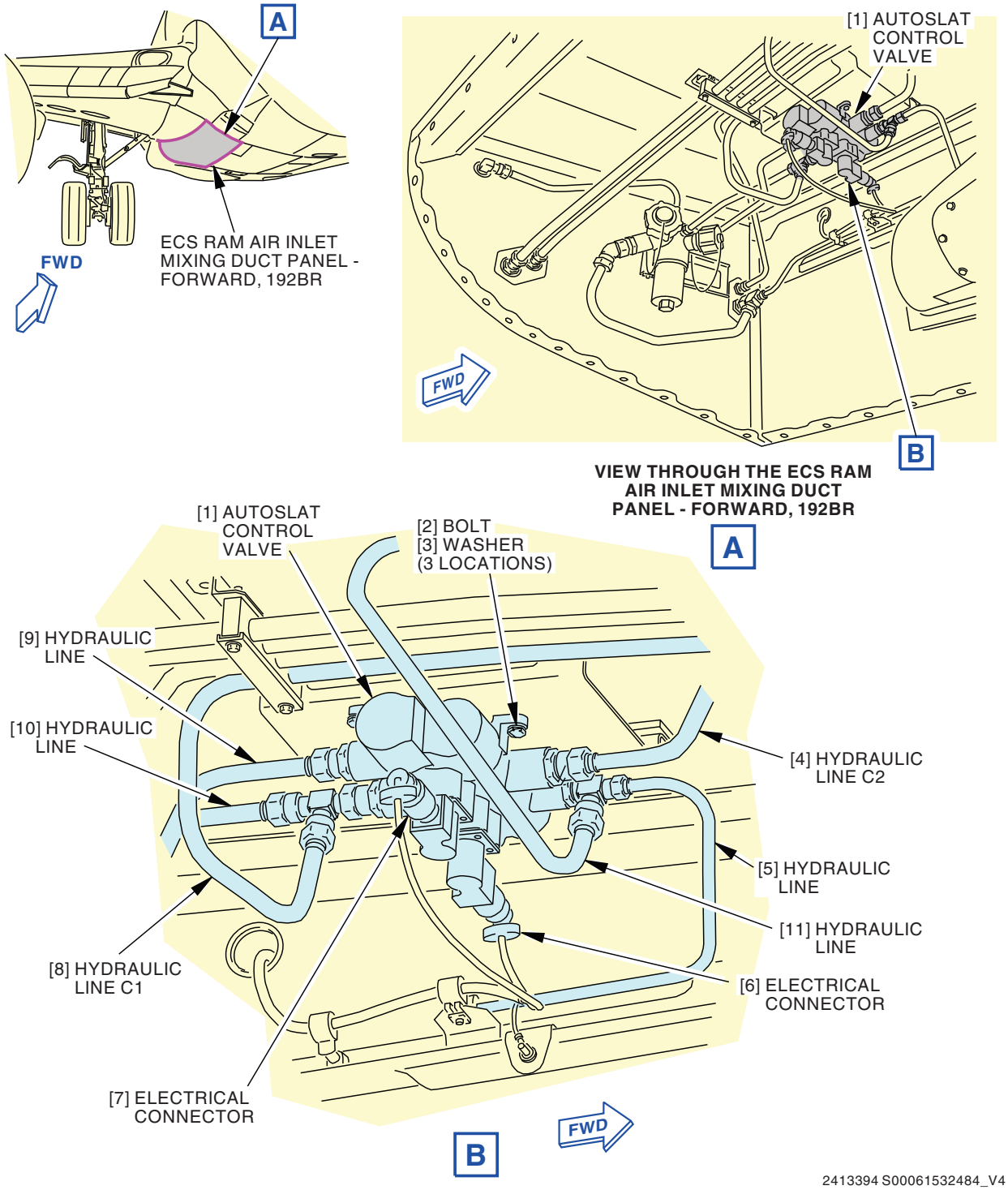
(6) Install caps on the hydraulic line (C2) [4], hydraulic line [5], hydraulic line (C1) [8], hydraulic line [9], hydraulic line [10], and hydraulic line [11].

SUBTASK 27-83-21-480-004

(7) Install plugs in the hydraulic ports on the autoslat control valve [1].

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Autoslat Control Valve Installation
Figure 401/27-83-21-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-83-21-400-801

3. Autoslat Control Valve Installation

(Figure 401)

A. References

Reference	Title
12-12-00-610-801	Hydraulic Reservoir Servicing (P/B 301)
27-83-00-710-801	Leading Edge Autoslat System - Functional Test (P/B 501)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
SWPM 20-60-03	Special Protection of Electrical Connectors

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659

C. Consumable Materials

Reference	Description	Specification
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95
G50171	Compound - Corrosion Inhibiting Compound, Interior Application - D5026NS or ZC-026	

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Autoslat control valve	27-83-21-01-005	PTF ALL

E. Location Zones

Zone	Area
192	Lower Wing-To-Body Fairing - Under Wing Box

F. Access Panels

Number	Name/Location
192BR	ECS Ram Air Inlet Mixing Duct Panel - Forward

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Prepare for the Installation

SUBTASK 27-83-21-480-005

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins installed on the main and nose landing gear. Do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 27-83-21-860-003

(2) Make sure that there is no hydraulic power. If there is, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

H. Autoslat Control Valve Installation

SUBTASK 27-83-21-420-001

(1) Install the autoslat control valve [1]:

(a) Make sure that the mating surface of the autoslat control valve [1] and the mounting bracket are clean.

(b) Apply sealant, A00247 on the bolts [2] threads.

(c) Put the autoslat control valve [1] in its position on the bracket.

(d) Install the bolts [2] and washers [3] that attach the autoslat control valve [1].

(e) With the intrinsically safe approved bonding meter, COM-1550, make sure that the bonding resistance between the autoslat control valve [1] and the structure is less than 0.0025 ohms.

(f) Remove the caps from the hydraulic line (C2) [4], hydraulic line [5], hydraulic line (C1) [8], and hydraulic line [9].

(g) Remove the plugs from the ports on the autoslat control valve [1].

SUBTASK 27-83-21-420-002

(2) Connect the hydraulic line (C2) [4], hydraulic line (C1) [8], hydraulic line [9], and hydraulic line [10] to the autoslat control valve [1].

(a) Tighten the B-nuts on hydraulic line (C2) [4], hydraulic line (C1) [8], hydraulic line [9], and hydraulic line [10] to 500 ±25 in-lb (56 ±3 N·m).

SUBTASK 27-83-21-420-004

(3) Connect the hydraulic line [5] and hydraulic line [11] to the autoslat control valve [1].

(a) Tighten the B-nuts on hydraulic line [5] and hydraulic line [11] to 170 ±9 in-lb (19 ±1 N·m).

SUBTASK 27-83-21-210-001

(4) If you did not examine the electrical connector(s) at the removal step, do these steps:

NOTE: You can do the inspection of the electrical connector(s) after the removal or before the installation steps of the electrical connector(s). This is an operator option.

(a) Before you connect the electrical connector(s), examine the connector(s) for corrosion.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING

DO THE STEPS BELOW IF THE AIRPLANE OPERATES AT AIRPORTS WHERE DEICING FLUID THAT CONTAIN POTASSIUM FORMATE IS USED. ALSO DO THE STEPS FOR ALL AIRPLANES THAT YOU FOUND CORROSION IN THE ELECTRICAL CONNECTORS IN THE MAIN WHEEL WELL. THE ELECTRICAL CONNECTOR IS PART OF A SYSTEM THAT IS NECESSARY FOR SAFE FLIGHT.

(b) If there was corrosion, refer to (SWPM 20-60-03) to correct the problem.

SUBTASK 27-83-21-800-001

(5) Apply the D5026NS or ZC-026 compound, G50171 to the connectors (SWPM 20-60-03).

SUBTASK 27-83-21-420-003

(6) Connect the electrical connector [6] and electrical connector [7] to the autoslat control valve [1].

SUBTASK 27-83-21-860-004

(7) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	14	C01070	FLIGHT CONTROL AUTOSLAT DC 2
C	14	C01068	FLIGHT CONTROL AUTOSLAT DC 1

SUBTASK 27-83-21-610-001

(8) If it is necessary for the hydraulic system, do this task: Hydraulic Reservoir Servicing, TASK 12-12-00-610-801.

SUBTASK 27-83-21-740-001

(9) Do this task: Leading Edge Autoslat System - Functional Test, TASK 27-83-00-710-801.

SUBTASK 27-83-21-210-002

(10) Make sure that there are no hydraulic fluid leaks at the autoslat control valve [1].

I. Put the Airplane Back to Its Usual Condition

SUBTASK 27-83-21-940-001

(1) Install this access panel:

Number	Name/Location
192BR	ECS Ram Air Inlet Mixing Duct Panel - Forward

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

AUTOSLAT CONTROL VALVE - INSPECTION/CHECK

1. General

A. This task does a check for internal hydraulic leakage in the autoslat valve.

TASK 27-83-21-790-801

2. Autoslat Valve Internal Hydraulic Leakage Check

A. General

(1) This procedure does an internal hydraulic leakage check for the leading edge auto slat valves. The purpose of this leakage test is to identify high leakage valve.

(2) You must measure the amount of leakage of hydraulic fluid during different conditions of operation to measure the fluid flow rate. It will be necessary to disconnect the return hydraulic line.

B. References

Reference	Title
12-12-00-610-801	Hydraulic Reservoir Servicing (P/B 301)
27-83-00-710-801	Leading Edge Autoslat System - Functional Test (P/B 501)
27-83-21-000-801	Autoslat Control Valve Removal (P/B 401)
27-83-21-400-801	Autoslat Control Valve Installation (P/B 401)
29-00-00-800-801	Hydraulic Power System - Deactivation (P/B 201)
29-00-00-800-802	Hydraulic Power System - Activation (P/B 201)
29-09-00-860-801	Hydraulic Reservoirs Pressurization (P/B 201)
29-09-00-860-802	Hydraulic Reservoirs Depressurization (P/B 201)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-09-00-860-801	Put the Airplane in the Air Mode (P/B 201)
32-09-00-860-802	Return the Airplane to the Ground Mode (P/B 201)

C. Tools/Equipment

Reference	Description
STD-6505	Container - Measuring, 500 cc, Calibrated in cc (ml)

D. Location Zones

Zone	Area
192	Lower Wing-To-Body Fairing - Under Wing Box

E. Access Panels

Number	Name/Location
192BR	ECS Ram Air Inlet Mixing Duct Panel - Forward

F. Prepare for the Internal Leakage Check

SUBTASK 27-83-21-860-006

(1) If the trailing edge flaps are not in the retracted position, then do these steps:

(a) Supply system B hydraulic power. To supply hydraulic power, do this task:
Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

(b) Move the flap control lever to the FLAP UP position to retract the trailing edge flaps.

(c) Remove system B hydraulic power. To remove hydraulic power, do this task:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 27-83-21-860-007

(2) Make sure the flap control lever is at the FLAP UP position.

SUBTASK 27-83-21-860-008

(3) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	14	C01070	FLIGHT CONTROL AUTOSLAT DC 2
C	14	C01068	FLIGHT CONTROL AUTOSLAT DC 1

SUBTASK 27-83-21-010-002

(4) To gain access to the autoslat control valve,

Remove this access panel:

Number	Name/Location
192BR	ECS Ram Air Inlet Mixing Duct Panel - Forward

G. Autoslat Valve Internal Leakage Check

SUBTASK 27-83-21-860-009

WARNING

KEEP PERSONNEL AND EQUIPMENT AWAY FROM THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(1) Supply system B hydraulic power. To supply hydraulic power, do this task:

Hydraulic Power System - Activation, TASK 29-00-00-800-802

SUBTASK 27-83-21-860-010

(2) Move the flap control lever to the 15-unit detent position to extend the leading and trailing edge flaps and slats.

NOTE: The flap control lever operates the control cable input to the flap control unit. The flap control unit controls the leading edge flap and slat control valve to provide the hydraulic power to the autoslat valve.

SUBTASK 27-83-21-860-011

(3) Remove system B hydraulic power. To remove hydraulic power, do this task:

Hydraulic Power System - Deactivation, TASK 29-00-00-800-801.

SUBTASK 27-83-21-860-012

(4) Remove the pressure from the system B reservoir. To remove the pressure, do this task:

Hydraulic Reservoirs Depressurization, TASK 29-09-00-860-802.

NOTE: It is necessary to remove the pressure from the reservoir. When you disconnect a hydraulic line below the level of the reservoir, all the hydraulic fluid will flow and drain out of the reservoir if pressure is applied.

SUBTASK 27-83-21-790-001

(5) Put a measuring container, STD-6505 under the return line on the autoslat valve.

SUBTASK 27-83-21-020-004

(6) Disconnect the return line from the autoslat valve. Use the applicable steps from this task

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Autoslat Control Valve Removal, TASK 27-83-21-000-801.

NOTE: The return line goes to a tee, with one line to the hydraulic reservoir, and one line to the leading edge UCM valve.

SUBTASK 27-83-21-790-002

(7) Install a cap or plug on the hydraulic line that goes to the tee.

SUBTASK 27-83-21-860-013

(8) Pressurize the system B hydraulic reservoir. To pressurize the reservoir, do this task:

Hydraulic Reservoirs Pressurization, TASK 29-09-00-860-801.

SUBTASK 27-83-21-860-014

(9) Supply system B hydraulic power. To supply hydraulic power, do this task:

Hydraulic Power System - Activation, TASK 29-00-00-800-802.

SUBTASK 27-83-21-790-003

(10) Measure the leakage from the autoslat valve with the solenoids not energized:

(a) Measure and record the leakage rate from the return port on the autoslat valve.

NOTE: The leakage rate must not be more than 40 cc/minute.

SUBTASK 27-83-21-860-015

(11) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	14	C01070	FLIGHT CONTROL AUTOSLAT DC 2
C	14	C01068	FLIGHT CONTROL AUTOSLAT DC 1

SUBTASK 27-83-21-860-016

(12) Do this task: Put the Airplane in the Air Mode, TASK 32-09-00-860-801.

SUBTASK 27-83-21-860-017

(13) Move the flap control lever to the 30-unit detent position.

NOTE: A small amount of hydraulic fluid will leak from the autoslat valve when the autoslat valve solenoids are energized. The solenoids will be energized for approximately 1/2 second, when the trailing edge flaps are between the 15 and 25 unit positions.

SUBTASK 27-83-21-790-004

(14) Measure the leakage from the autoslat valve after the solenoids have been cycled:

(a) Measure and record the leakage rate from the return port on the autoslat valve again.

NOTE: The leakage rate must not be more than 40 cc/minute.

SUBTASK 27-83-21-860-018

(15) Move the flap control lever to the UP position to retract the leading and trailing edge flaps and slats.

SUBTASK 27-83-21-860-019

(16) Do this task Return the Airplane to the Ground Mode, TASK 32-09-00-860-802.

SUBTASK 27-83-21-860-020

(17) Remove system B hydraulic power. To remove hydraulic power, do this task:

Hydraulic Power System - Deactivation, TASK 29-00-00-800-801

SUBTASK 27-83-21-860-021

(18) Remove the pressure from the system B reservoir. To remove the pressure, do this task:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Hydraulic Reservoirs Depressurization, TASK 29-09-00-860-802.

H. Return the Airplane to the Usual Condition

SUBTASK 27-83-21-410-001

(1) Remove the cap or plug from the hydraulic line that goes to the hydraulic reservoir.

SUBTASK 27-83-21-410-002

(2) Re-connect the return line to the autoslat valve. Do the applicable steps from this task:
Autoslat Control Valve Installation, TASK 27-83-21-400-801.

SUBTASK 27-83-21-610-002

(3) If it is necessary for the hydraulic reservoir, do this task:
Hydraulic Reservoir Servicing, TASK 12-12-00-610-801.

SUBTASK 27-83-21-720-001

(4) Do this task:
Leading Edge Autoslat System - Functional Test, TASK 27-83-00-710-801.

SUBTASK 27-83-21-210-003

(5) Make sure there are no hydraulic fluid leaks at the autoslat control valve.

SUBTASK 27-83-21-410-003

(6) Install this access panel:

Number	Name/Location
192BR	ECS Ram Air Inlet Mixing Duct Panel - Forward

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

LEADING EDGE FLAP AND SLAT POSITION INDICATING SYSTEM - ADJUSTMENT/TEST

1. General

A. This procedure contains an operational test of the leading edge flap and slat position indicating system.

TASK 27-88-00-710-801

2. Leading Edge Flap and Slat Position Indicating System - Operational Test

(Figure 501)

A. General

(1) The leading edge flap and slat position indicating system must be correctly adjusted before you do the operational test. The leading edge proximity sensors and targets are used to find the leading edge flap and slat position.

(2) The leading edge slats move to the extend position for flap lever positions 1-unit detent through 25-unit detent. The leading edge slats move to the full extend position at the 30-unit detent flap lever position.

B. References

Reference	Title
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
512	Left Wing - Krueger Flap No. 2
513	Left Wing - Krueger Flap No. 1
522	Left Wing - Slat No. 4
523	Left Wing - Slat No. 3
524	Left Wing - Slat No. 2
525	Left Wing - Slat No. 1
612	Right Wing - Krueger Flap No. 3
613	Right Wing - Krueger Flap No. 4
622	Right Wing - Slat No. 5
623	Right Wing - Slat No. 6
624	Right Wing - Slat No. 7
625	Right Wing - Slat No. 8

D. Prepare for the Operational Test

SUBTASK 27-88-00-860-008

WARNING KEEP PERSONNEL AND EQUIPMENT AWAY FROM THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(1) To pressurize hydraulic system B, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Leading Edge Flap and Slat Position Indicating System - Operational Test

SUBTASK 27-88-00-710-002

(1) Move the flap control lever to the FLAP UP position.

(a) Make sure each flap and slat is in the retract position and align with the wing structure.

(b) Make sure the amber LE FLAPS TRANSIT light and green LE FLAPS EXT light on the engine indication display of the captains inboard display unit (DU) on panel P1, are off.

(c) Make sure all the position lights on the leading edge flap/slat position annunciator, P5-12, on the aft overhead panel are off.

(d) Set the master dim and test lights switch, on the P1 panel, to the DIM position.

(e) Push the TEST switch below the leading edge flap/slat position annunciator, P5-12, on the aft overhead panel.

(f) Make sure all the annunciator lights are on.

(g) Release the TEST switch.

(h) Make sure all the annunciator lights go off.

SUBTASK 27-88-00-710-003

CAUTION

MAKE SURE THAT THE INBOARD FAN COWL PANEL AND THE INBOARD AND OUTBOARD THRUST REVERSERS ARE CLOSED AND IN THE STOWED POSITION OR REMOVED BEFORE YOU EXTEND THE LEADING EDGE FLAPS AND SLATS. THERE IS NOT SUFFICIENT CLEARANCE FOR THE FLAPS AND SLATS TO EXTEND IF THE INBOARD FAN COWL PANEL AND THE INBOARD AND OUTBOARD THRUST REVERSERS ARE IN THE OPEN POSITION. THIS CAN CAUSE DAMAGE TO EQUIPMENT.

(2) Move the flap control lever to the 1-unit detent position.

(a) When the leading edge flaps and slats start to move to the commanded 1-unit detent position, look for these conditions:

1) Make sure the amber LE FLAPS TRANSIT light on the engine indication display of the captain's inboard DU on panel P1, comes on.

2) Make sure these lights come on in the leading edge flap/slat position annunciator, P5-12, on the aft overhead panel.

a) Make sure the annunciator LE DEVICES FLAPS 1 through 4, amber TRANSIT lights, comes on.

b) Make sure the annunciator LE DEVICES SLATS 1 through 8, amber TRANSIT lights, comes on.

(b) When the leading edge flaps and slats stop at the commanded 1-unit detent position, look for these conditions:

1) Make sure the amber LE FLAPS TRANSIT light, on the captain's inboard DU engine indication display, goes off.

2) Make sure the green LE FLAPS EXT light, on the captain's inboard DU engine indication display, comes on.

3) Make sure the annunciator LE DEVICES FLAPS 1 through 4, amber TRANSIT lights, go off.

4) Make sure the annunciator LE DEVICES SLATS 1 through 8, amber TRANSIT lights, go off.

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5) Make sure the annunciator LE DEVICES FLAPS 1 through 4, green EXT lights, come on.

6) Make sure the annunciator LE DEVICES SLATS 1 through 8, green EXT lights, come on.

7) Make sure that the leading edge flaps 1 through 4 are at the extend position.

8) Make sure that the leading edge slats 1 through 8 are at the extend position.

SUBTASK 27-88-00-710-004

(3) Move the flap control lever to the 5-unit detent position.

(a) Make sure the annunciator LE DEVICES FLAPS 1 through 4, green EXT lights, are on.

(b) Make sure the annunciator LE DEVICES SLATS 1 through 8, green EXT lights, are on.

(c) Make sure leading edge slats 1 through 8 remain at the extend position.

SUBTASK 27-88-00-710-006

(4) Move the flap control lever to the 30-unit detent position.

(a) Do these checks while the leading edge slats move to the commanded 30-unit detent position:

1) Make sure the green LE FLAPS EXT light, on the captain's inboard DU engine indication display, goes off.

2) Make sure the amber LE FLAPS TRANSIT light, on the captain's inboard DU engine indication display, comes on.

3) Make sure the annunciator LE DEVICES SLATS 1 through 8, green EXT lights, go off.

4) Make sure the annunciator LE DEVICES SLATS 1 through 8, amber TRANSIT lights, come on.

(b) Do these checks when the leading edge slats stop at the commanded 30-unit detent position:

1) Make sure the amber LE FLAPS TRANSIT light, on the captain's inboard DU engine indication display, goes off.

2) Make sure the green LE FLAPS EXT light, on the captain's inboard DU engine indication display, comes on.

3) Make sure the annunciator LE DEVICES FLAPS 1 through 4, green EXT lights, are on.

4) Make sure the annunciator LE DEVICES SLATS 1 through 8, amber TRANSIT lights, go off.

5) Make sure the annunciator LE DEVICES SLATS 1 through 8, green FULL EXT lights, come on.

6) Make sure that the leading edge slats 1 through 8 move from the extend position to the full extend position.

SUBTASK 27-88-00-710-007

(5) Move the flap control lever to the 40-unit detent position.

(a) Make sure the green LE FLAPS EXT light, on the captain's inboard DU engine indication display, remains on.

(b) Make sure the annunciator LE DEVICES FLAPS 1 through 4, green EXT lights, remain on.

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(c) Make sure the annunciator LE DEVICES SLATS 1 through 8, green FULL EXT lights, remain on.

(d) Make sure that the leading edge slats 1 through 8 remain at the full extend position.

(e) Make sure that the leading edge flaps 1 through 4 remain at the extend position.

SUBTASK 27-88-00-710-009

(6) Move the flap control lever to the 30-unit detent position.

(a) Make sure the green LE FLAPS EXT light, on the captain's inboard DU engine indication display remains on.

(b) Make sure the annunciator LE DEVICES FLAPS 1 through 4, green EXT lights, remain on.

(c) Make sure the annunciator LE DEVICES SLATS 1 through 8, green FULL EXT lights, remain on.

(d) Make sure that the leading edge slats 1 through 8 remains at the full extend position.

(e) Make sure that the leading edge flaps 1 through 4 remains at the extend position.

SUBTASK 27-88-00-710-011

(7) Move the flap control lever to the 25-unit detent position.

(a) Do these checks while the leading edge slats move to the commanded 25-unit detent position:

1) Make sure the green LE FLAPS EXT light, on the captain's inboard DU engine indication display, goes off.

2) Make sure the amber LE FLAPS TRANSIT light, on the captain's inboard DU engine indication display, comes on.

3) Make sure the annunciator LE DEVICES FLAPS 1 through 4, green EXT lights, remain on.

4) Make sure the annunciator LE DEVICES SLATS 1 through 8, green FULL EXT lights, go off.

5) Make sure the annunciator LE DEVICES SLATS 1 through 8, amber TRANSIT lights, come on.

6) Make sure that the leading edge flaps 1 through 4 remain at the extend position.

7) Make sure that the leading edge slats 1 through 8 move from the full extend position to the extend position.

(b) Do these checks when the leading edge slats stop at the commanded 25-unit detent position:

1) Make sure the amber LE FLAPS TRANSIT light, on the captain's inboard DU engine indication display, goes off.

2) Make sure the green LE FLAPS EXT light, on the captain's inboard DU engine indication display, comes on.

3) Make sure the annunciator LE DEVICES FLAPS 1 through 4, green EXT lights, remain on.

4) Make sure the annunciator LE DEVICES SLATS 1 through 8, amber TRANSIT lights, go off.

5) Make sure the annunciator LE DEVICES SLATS 1 through 8, green EXT lights, come on.

6) Make sure that the leading edge flaps 1 through 4 remain at the extend position.

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7) Make sure that the leading edge slats 1 through 8 are at the extend position.

SUBTASK 27-88-00-710-012

(8) Move the flap control lever to the 1-unit detent position.

(a) Make sure the green LE FLAPS EXT light, on the captain's inboard DU engine indication display remains on.

(b) Make sure the annunciator LE DEVICES FLAPS 1 through 4, green EXT lights, remain on.

(c) Make sure the annunciator LE DEVICES SLATS 1 through 8, green EXT lights, remain on.

(d) Make sure that the leading edge flaps 1 through 4 remain at the extend position.

(e) Make sure that the leading edge slats 1 through 8 remain at the extend position.

SUBTASK 27-88-00-710-013

(9) Move the flap control lever to the FLAP UP detent position.

(a) Make sure the leading edge flaps and slats move to the retract position.

1) Make sure the green LE FLAPS EXT light, on the captain's inboard DU engine indication display, goes off.

2) Make sure the amber LE FLAPS TRANSIT light on the engine indication display of the captain's inboard DU on panel P1, comes on.

3) Make sure the annunciator LE DEVICES FLAPS 1 through 4, green EXT lights, goes off.

4) Make sure the annunciator LE DEVICES SLATS 1 through 8, green EXT lights, goes off.

5) Make sure the annunciator LE DEVICES FLAPS 1 through 4, amber TRANSIT lights, come on.

6) Make sure the annunciator LE DEVICES SLATS 1 through 8, amber TRANSIT lights, come on.

(b) Do these checks when the leading edge flaps and slats are at the retract position:

1) Make sure the annunciator LE DEVICES FLAPS 1 through 4, amber TRANSIT lights, goes off.

2) Make sure the annunciator LE DEVICES SLATS 1 through 8, amber TRANSIT lights, goes off.

3) Make sure that all annunciator lights are off.

4) Make sure the amber LE FLAPS TRANSIT light on the engine indication display of the captain's inboard DU on panel P1, goes off.

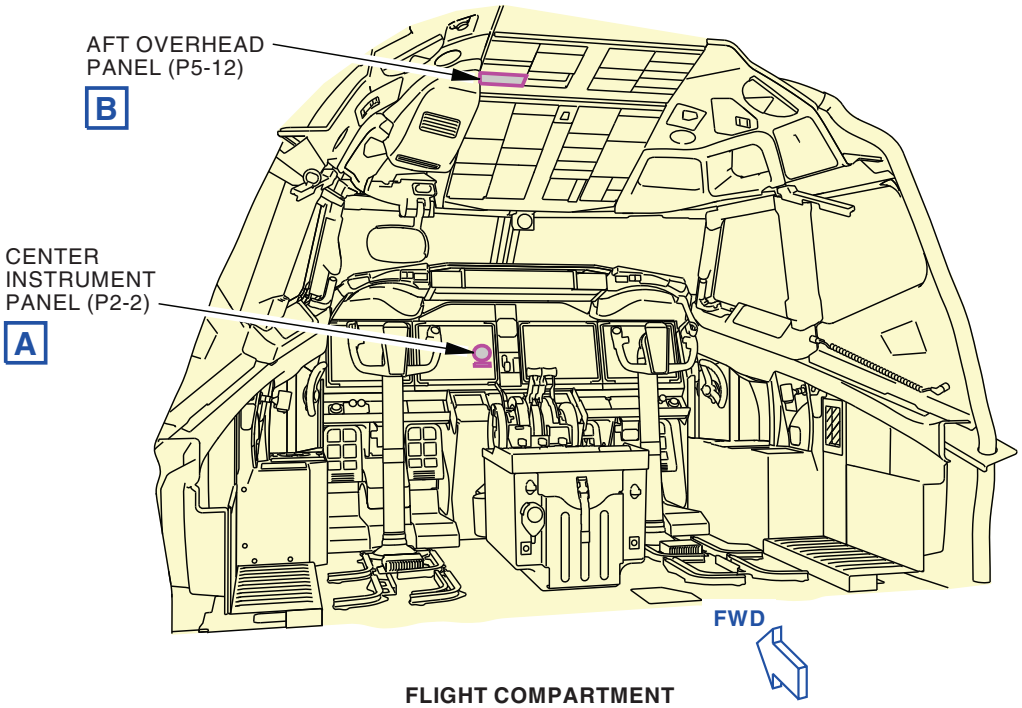
F. Put the Airplane Back to Its Usual Condition

SUBTASK 27-88-00-860-009

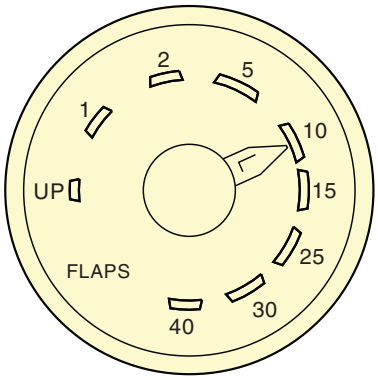
(1) Remove hydraulic power. This is the task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

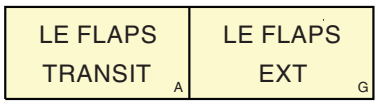
END OF TASK

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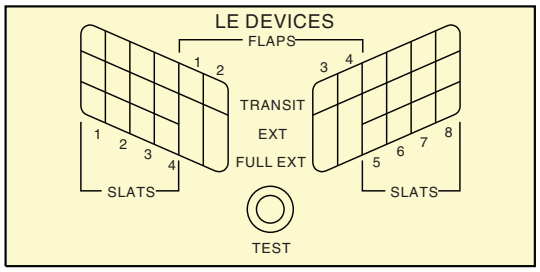
FLIGHT COMPARTMENT





CENTER INSTRUMENT PANEL (P2-2)

A



AFT OVERHEAD PANEL (P5-12)

B

2413395 S00061532493_V2

Leading Edge Flap and Slat Position Indicating Test
Figure 501/27-88-00-990-802

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LEADING EDGE FLAP AND SLAT PROXIMITY SENSOR - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the leading edge slat proximity sensor

(2) An installation of the leading edge slat proximity sensor

(3) A removal of the leading edge flap proximity sensor

(4) An installation of the leading edge flap proximity sensor.

TASK 27-88-11-000-801

2. Leading Edge Slat Proximity Sensor Removal

(Figure 401)

A. References

Reference	Title
27-81-00-480-801	Leading Edge Flap and Slat Locks Installation (P/B 201)
27-81-00-860-803	Leading Edge Flaps and Slats Extension (P/B 201)
57-41-02-000-801	Leading Edge Access Panel Removal (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1734	Lockout Set - Leading Edge Slats and Krueger Flaps Actuators Part #: C27051-29 Supplier: 81205

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
521	Left Wing - Leading Edge to Front Spar
522	Left Wing - Slat No. 4
523	Left Wing - Slat No. 3
524	Left Wing - Slat No. 2
525	Left Wing - Slat No. 1
621	Right Wing - Leading Edge to Front Spar
622	Right Wing - Slat No. 5
623	Right Wing - Slat No. 6
624	Right Wing - Slat No. 7
625	Right Wing - Slat No. 8

D. Access Panels

Number	Name/Location
521CB	Lower Leading Edge Access Panel
521DB	Lower Leading Edge Access Panel
521FB	Lower Leading Edge Access Panel
521JB	Lower Leading Edge Access Panel

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
521KB	Lower Leading Edge Access Panel
521LB	Lower Leading Edge Access Panel
521MB	Lower Leading Edge Access Panel
521QB	Lower Leading Edge Access Panel
521RB	Lower Leading Edge Access Panel
521SB	Lower Leading Edge Access Panel
521TB	Lower Leading Edge Access Panel
521WB	Lower Leading Edge Access Panel
521XB	Lower Leading Edge Access Panel
521YB	Lower Leading Edge Access Panel
521ZB	Lower Leading Edge Access Panel
621CB	Lower Leading Edge Access Panel
621DB	Lower Leading Edge Access Panel
621FB	Lower Leading Edge Access Panel
621JB	Lower Leading Edge Access Panel
621KB	Lower Leading Edge Access Panel
621LB	Lower Leading Edge Access Panel
621PB	Lower Leading Edge Access Panel
621QB	Lower Leading Edge Access Panel
621RB	Lower Leading Edge Access Panel
621SB	Lower Leading Edge Access Panel
621VB	Lower Leading Edge Access Panel
621WB	Lower Leading Edge Access Panel
621XB	Lower Leading Edge Access Panel
621YB	Lower Leading Edge Access Panel

E. Prepare for the Removal

SUBTASK 27-88-11-860-001

WARNING

MAKE SURE PERSONS AND EQUIPMENT ARE CLEAR OF THE LEADING EDGE AND TRAILING EDGE CONTROL SURFACES. THE LEADING EDGE AND TRAILING EDGE CONTROL SURFACES CAN EXTEND AND RETRACT QUICKLY. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

CAUTION

MAKE SURE THAT THE INBOARD FAN COWL PANEL AND THE INBOARD AND OUTBOARD THRUST REVERSERS ARE CLOSED, AND IN THE STOWED POSITION, OR REMOVED BEFORE YOU EXTEND THE LEADING EDGE FLAPS AND SLATS. THERE IS NOT SUFFICIENT CLEARANCE FOR THE FLAPS AND SLATS TO EXTEND IF THE INBOARD FAN COWL PANEL AND THE INBOARD AND OUTBOARD THRUST REVERSERS ARE IN THE OPEN POSITION. THIS CAN CAUSE DAMAGE TO EQUIPMENT.

(1) Extend the leading edge flaps and slats to the full extend (30-unit) position with primary control, do this task: Leading Edge Flaps and Slats Extension, TASK 27-81-00-860-803.

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SUBTASK 27-88-11-010-002

(2) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	8	C00211	FLIGHT CONTROL FSEU DC 1
A	9	C01468	FLIGHT CONTROL FSEU DC 2

SUBTASK 27-88-11-480-001

WARNING MAKE SURE THAT YOU INSTALL THE LEADING EDGE FLAP AND SLAT ACTUATORS LOCKOUT SET TO PREVENT ACCIDENTAL OPERATION. THE LEADING EDGE FLAP AND SLAT CAN MOVE QUICKLY. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(3) Install the lockout set, SPL-1734, on the actuators of the adjacent leading edge flaps and slats, do this task: Leading Edge Flap and Slat Locks Installation, TASK 27-81-00-480-801.

SUBTASK 27-88-11-010-001

(4) Remove the applicable access panels to get access to the slat proximity sensor [26] (TASK 57-41-02-000-801):

(a) For the left wing, open these access panels:

Number	Name/Location
521CB	Lower Leading Edge Access Panel
521DB	Lower Leading Edge Access Panel
521FB	Lower Leading Edge Access Panel
521JB	Lower Leading Edge Access Panel
521KB	Lower Leading Edge Access Panel
521LB	Lower Leading Edge Access Panel
521MB	Lower Leading Edge Access Panel
521QB	Lower Leading Edge Access Panel
521RB	Lower Leading Edge Access Panel
521SB	Lower Leading Edge Access Panel
521TB	Lower Leading Edge Access Panel
521WB	Lower Leading Edge Access Panel
521XB	Lower Leading Edge Access Panel
521YB	Lower Leading Edge Access Panel
521ZB	Lower Leading Edge Access Panel

(b) For the right wing, open these access panels:

Number	Name/Location
621CB	Lower Leading Edge Access Panel
621DB	Lower Leading Edge Access Panel
621FB	Lower Leading Edge Access Panel
621JB	Lower Leading Edge Access Panel
621KB	Lower Leading Edge Access Panel
621LB	Lower Leading Edge Access Panel
621PB	Lower Leading Edge Access Panel
621QB	Lower Leading Edge Access Panel
621RB	Lower Leading Edge Access Panel
621SB	Lower Leading Edge Access Panel

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
621VB	Lower Leading Edge Access Panel
621WB	Lower Leading Edge Access Panel
621XB	Lower Leading Edge Access Panel
621YB	Lower Leading Edge Access Panel

F. Leading Edge Slat Proximity Sensor Removal

SUBTASK 27-88-11-020-002

(1) Do these steps to remove the inboard proximity sensor [26] of the slat (with the slats 1 through 8 in the extended position and/or with the slats 1 and 8 in the retracted position) (View B, Figure 401):

(a) Disconnect the wires at the splice.

(b) For the slats 1 and 8 (in the extended position), remove the two screw [29] that attach the proximity sensor [26] to the rib and remove the shim [27], aluminum spacer [28], and washer [30] that is installed under the nuts [31].

(c) For the slats 2, 3, 6 and 7 (in the extended position), remove the two screws [32] that attach the proximity sensor [26] to the rib and remove the shim [27], aluminum spacer [34], and washer [30] that is installed under the nuts [31].

1) Measure the thickness of the shim [27] and the thickness of the aluminum spacer [34], and keep record of it for easier completion of the sensor-target clearance required during proximity sensor [26] installation task.

(d) For the slats 4 and 5 (in the extended position), remove the two screws [33] that attach the proximity sensor [26] to the rib and remove the shim [27] and washer [30] that is installed under the nuts [31].

1) Measure the thickness of the shim [27] and keep record of it for easier completion of the sensor-target clearance required during the proximity sensor [26] installation task.

(e) For the slats 1 and 8 (in the retracted position), remove the two screws [29] that attach the proximity sensor [26] to the rib, and the aluminum spacer [28]:

1) Remove the bolt [37], shim [27], washer [35], and nut [36] from the aluminum spacer [34], if necessary.

(f) Remove the inboard proximity sensor [26].

SUBTASK 27-88-11-020-004

(2) Do these steps to remove the outboard aluminum spacer [28] or aluminum spacer [34] of the slats 1 through 8 (in the extended or flap lever 5 position) (View C, Figure 401):

(a) Disconnect the wires at the splice.

(b) For the slat 1 through 8 (in the extended or flap lever 5 position) remove the (aluminum pigmented) cap screws [52]], aluminum shim [27], and aluminum spacer [28] or aluminum spacer [34].

NOTE: The P-clamp [51] gets disconnected when the cap screw [52] are removed.

1) Measure the thickness of the aluminum shim [27] and the thickness of the aluminum aluminum spacer [28] or aluminum spacer [34], and keep record of it for easier completion of the sensor-target clearance required during the proximity sensor [26] installation task.

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(c) For the slats 1 and 8 (in the retracted position), remove the cap screw [38] and aluminum spacer [28].

(d) Remove the outboard proximity sensor [26].

SUBTASK 27-88-11-020-005

(3) Do these steps to remove the retract proximity sensor [24] (View D and View E, Figure 401):

NOTE: The retract proximity sensors [24] are installed on slats 2, 3, 4, 5, 6 and 7.

(a) Remove the slat actuator door [23] from the slat actuator to get access to the mid retract proximity sensor [24]:

1) Remove the nuts [19], washers [18], and bolts [17].

2) Remove the nuts [22], washers [21], and bolts [20] that attach the slat actuator door [23] and the bracket to the slat actuator.

3) Remove the slat actuator door [23].

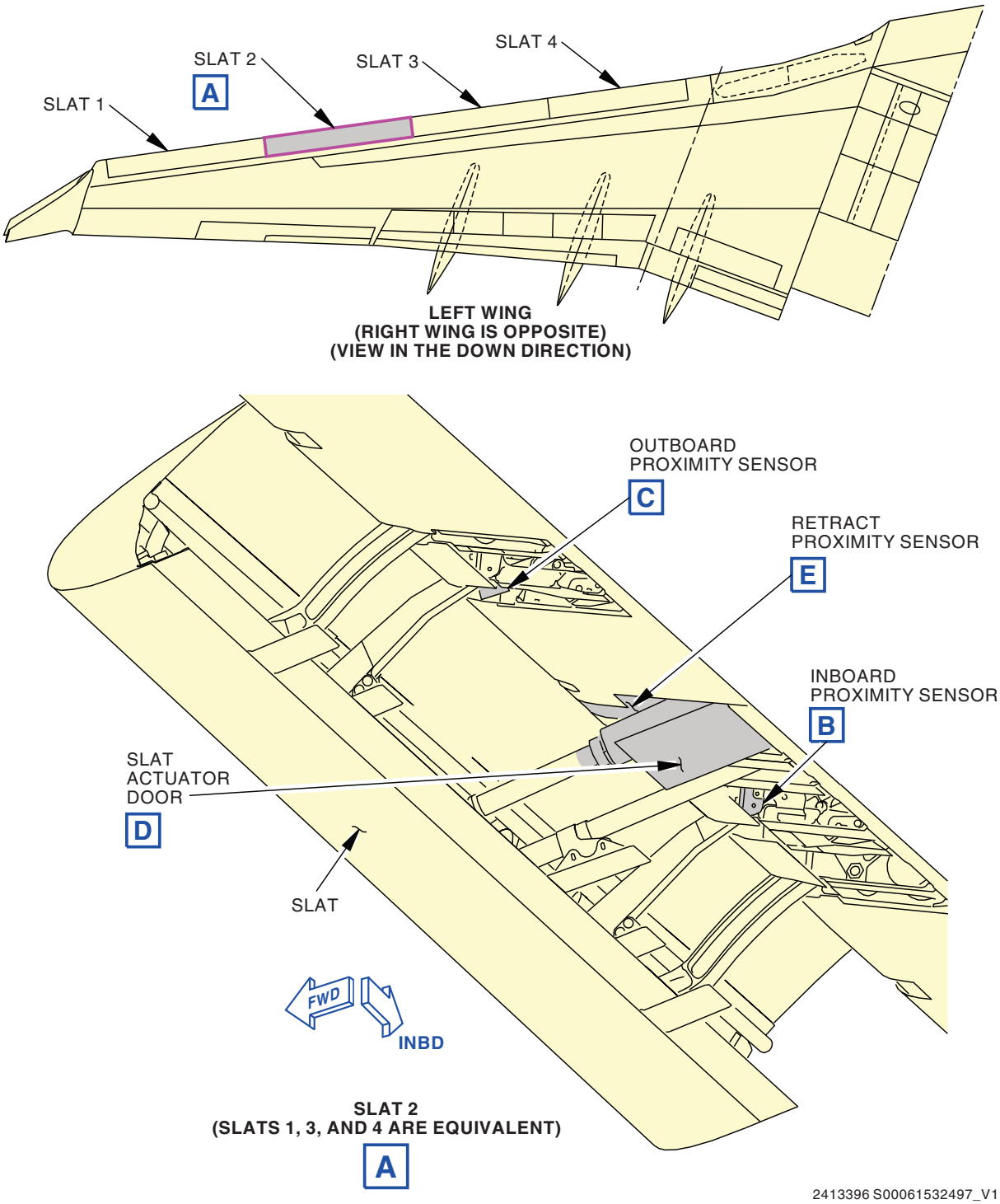
4) Temporarily install the bolts [20], washers [21], and nuts [22] to attach the bracket to the slat actuator.

(b) Disconnect the electrical connector from the retract proximity sensor [24].

(c) Remove the fastener [25] to remove the retract proximity sensor [24].

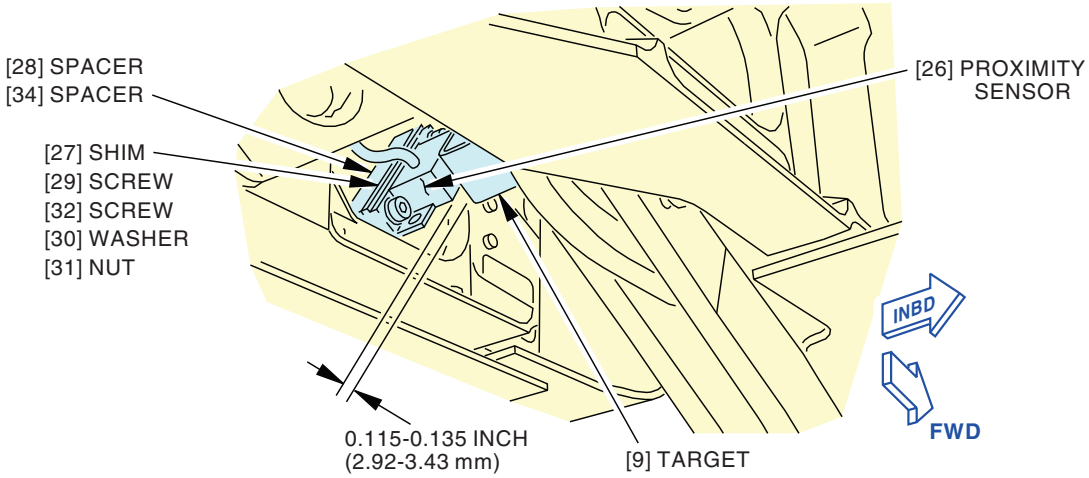
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



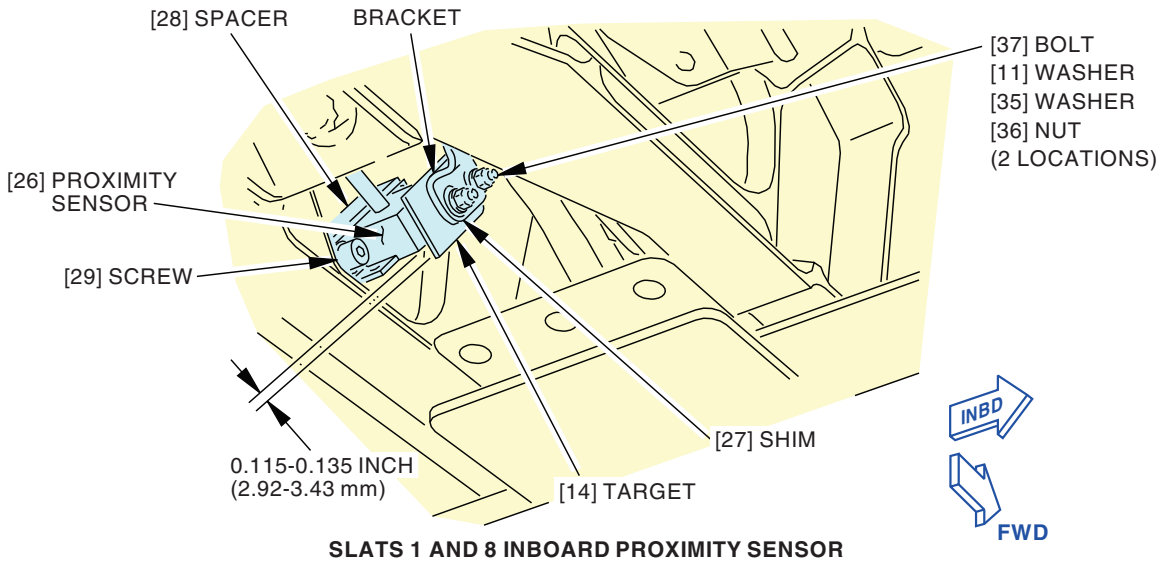
Leading Edge Slat Proximity Sensor Installation
Figure 401/27-88-11-990-801 (Sheet 1 of 6)

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SLAT 1 INBOARD PROXIMITY SENSOR
(SLATS 2, 3, 6, 7 AND 8 INBOARD PROXIMITY SENSORS ARE EQUIVALENT)
(SLAT IN THE EXTEND POSITION)

B



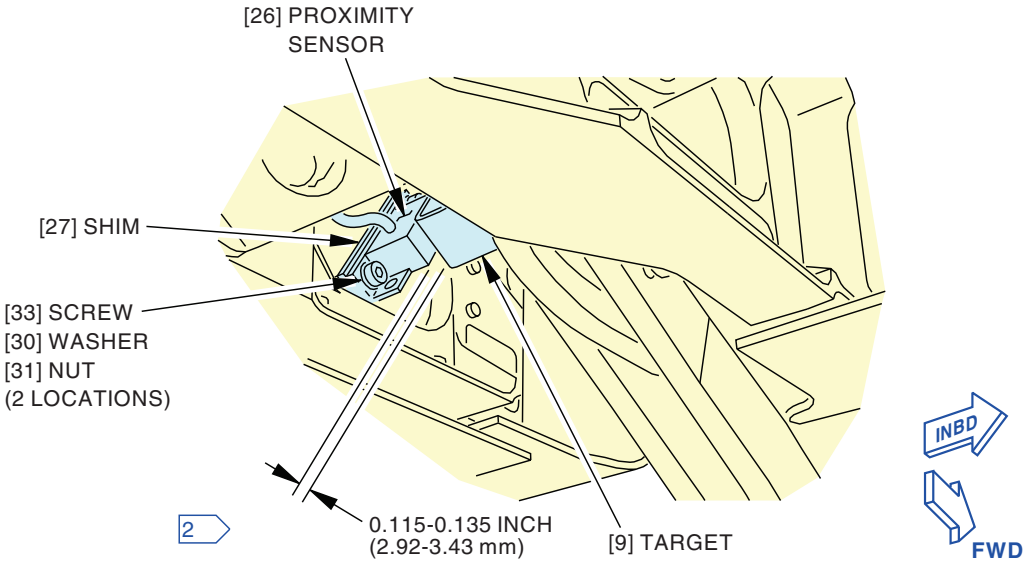
SLATS 1 AND 8 INBOARD PROXIMITY SENSOR
(SLAT IN THE RETRACT POSITION)

B

2413397 S00061532498_V2

Leading Edge Slat Proximity Sensor Installation
Figure 401/27-88-11-990-801 (Sheet 2 of 6)

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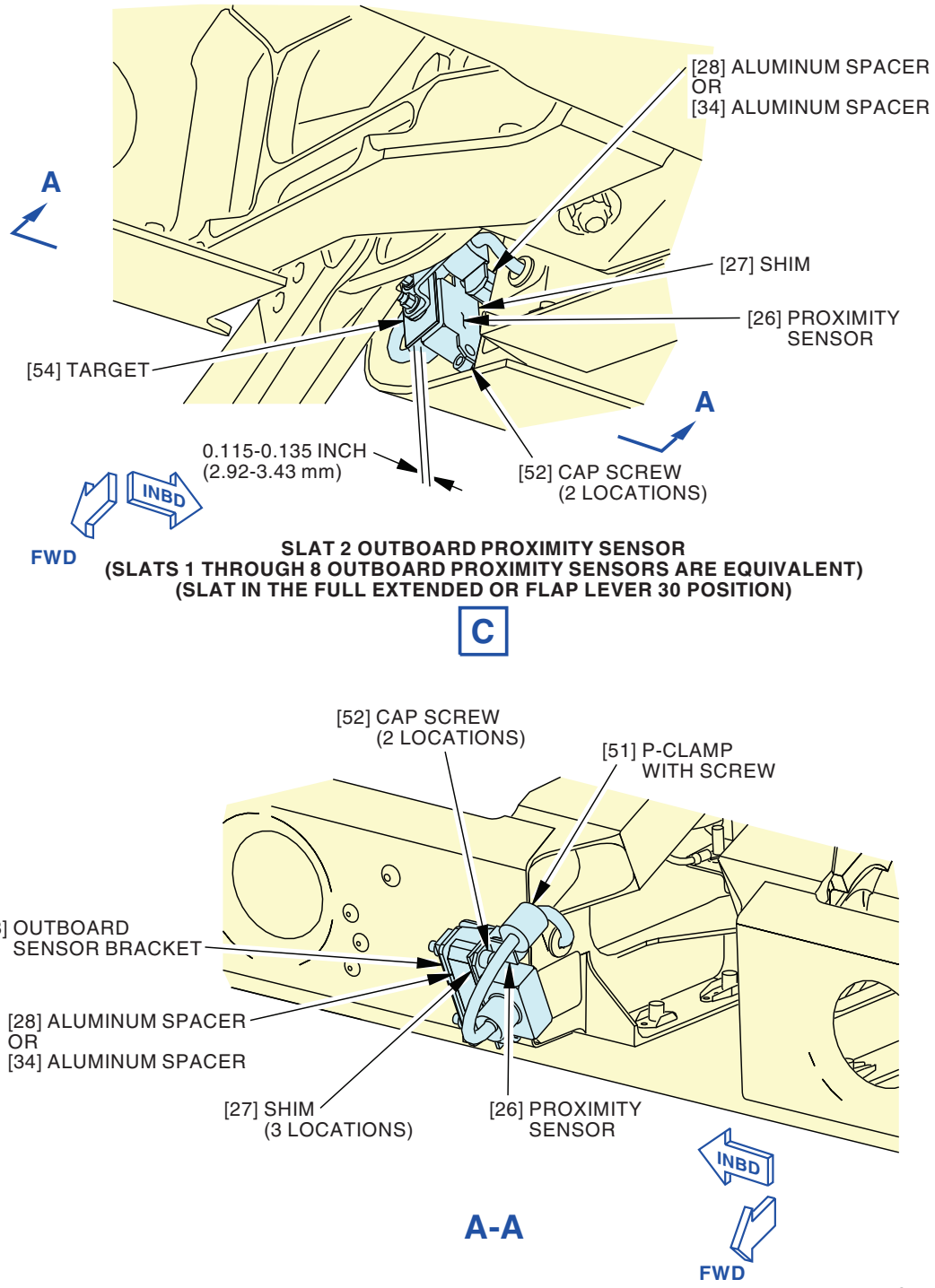
SLATS 4 AND 5 INBOARD PROXIMITY SENSOR (SLAT IN THE EXTEND POSITION)

B

2413398 S00061532499_V1

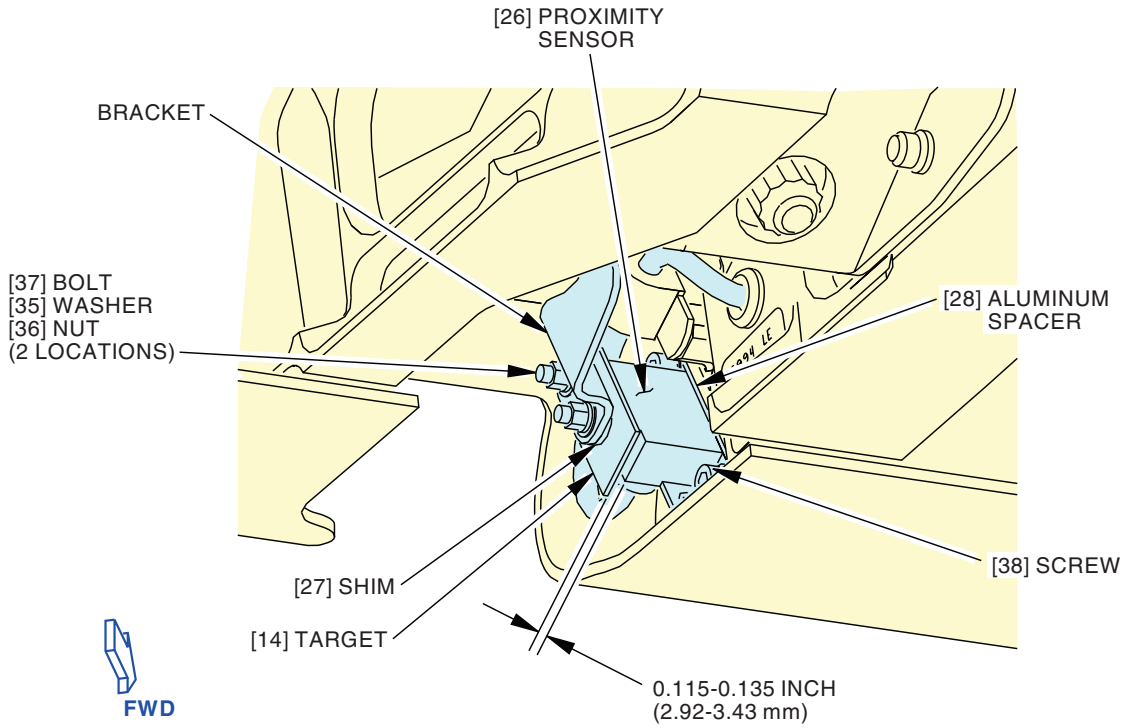
Leading Edge Slat Proximity Sensor Installation
Figure 401/27-88-11-990-801 (Sheet 3 of 6)

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Leading Edge Slat Proximity Sensor Installation
Figure 401/27-88-11-990-801 (Sheet 4 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



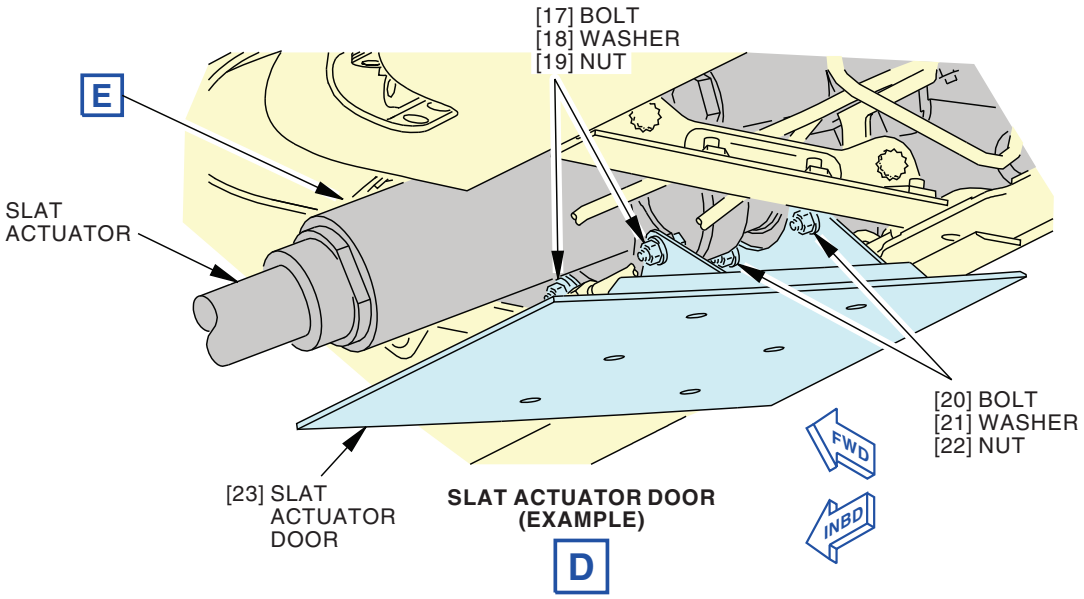
SLATS 1 AND 8 OUTBOARD PROXIMITY SENSOR (SLAT IN THE RETRACT POSITION)

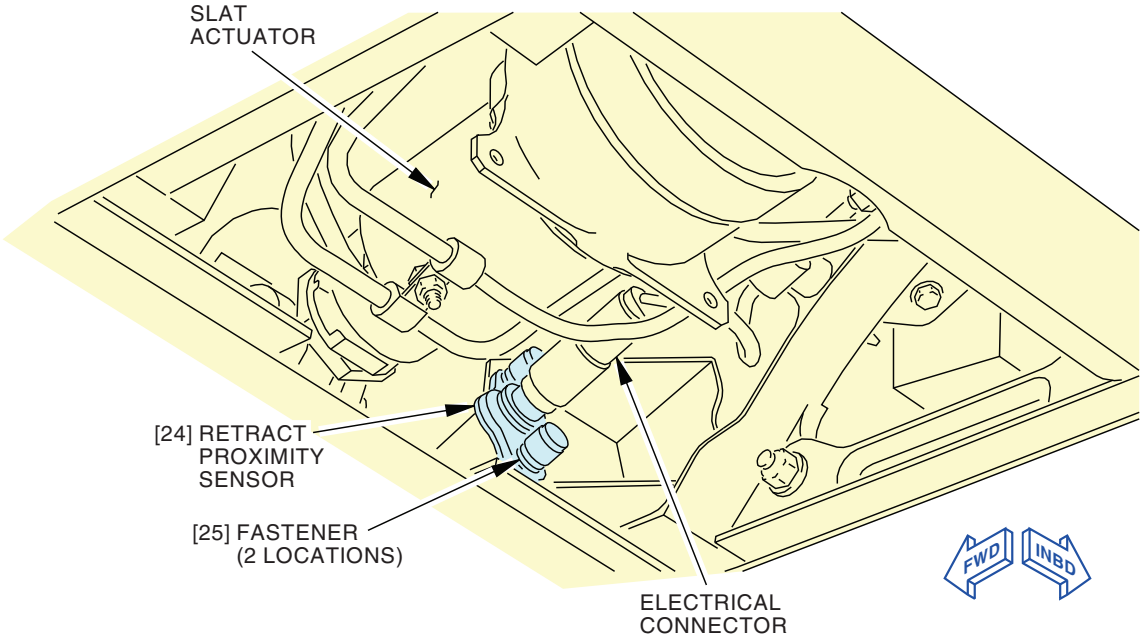
C

2413401 S00061532502_V1

Leading Edge Slat Proximity Sensor Installation
Figure 401/27-88-11-990-801 (Sheet 5 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details





SLAT 2 RETRACT PROXIMITY SENSOR
(SLATS 3, 4, 5, 6 AND 7 RETRACT PROXIMITY SENSORS ARE EQUIVALENT)

E

2413402 S00061532503_V1

Leading Edge Slat Proximity Sensor Installation
Figure 401/27-88-11-990-801 (Sheet 6 of 6)

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TASK 27-88-11-400-801

3. Leading Edge Slat Proximity Sensor Installation

(Figure 401)

A. References

Reference	Title
27-81-00-080-801	Leading Edge Flap and Slat Locks Removal (P/B 201)
27-81-00-480-801	Leading Edge Flap and Slat Locks Installation (P/B 201)
27-81-00-860-804	Leading Edge Flaps and Slats Retraction (P/B 201)
27-88-11-700-801	Leading Edge Flap and Slat Proximity Sensor Test (P/B 501)
57-41-02-400-801	Leading Edge Access Panel Installation (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1734	Lockout Set - Leading Edge Slats and Krueger Flaps Actuators Part #: C27051-29 Supplier: 81205

C. Consumable Materials

Reference	Description	Specification
A00165	Adhesive - Neoprene	BMS5-7 Type I
A01076	Adhesive - Synthetic Rubber	BAC5010 Type 93 (BMS5-95 Class B)
B00065	Alcohol - Denatured, Ethyl (Ethanol)	AMS 3002 (Supersedes O-A-396)
B50073	Alcohol - Isopropyl	ASTM D 770
C50069	Coating - Chemical And Solvent Resistant Finish - (BAC 702 White Color, Gloss)	BMS10-11 Type II
D50326	Lubricant - PTFE Release Agent - Miller-Stephenson MS-122XD (Replaces MS-122DF and MS-122V Lubricants)	
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)
G50237	Compound - Corrosion Inhibiting, Non-drying - Cor-Ban 27L	BMS3-38
G50262	Wiper - Cleaning	BMS15-5
G50335	Agent - Non-Peelable Parting (Henkel Loctite - Frekote 44-NC Mold Release).	BAC5000

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
24	Retract proximity sensor	27-88-11-02-060	PTF ALL
26	Proximity sensor	27-88-11-01-060	PTF ALL

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E. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
521	Left Wing - Leading Edge to Front Spar
522	Left Wing - Slat No. 4
523	Left Wing - Slat No. 3
524	Left Wing - Slat No. 2
525	Left Wing - Slat No. 1
621	Right Wing - Leading Edge to Front Spar
622	Right Wing - Slat No. 5
623	Right Wing - Slat No. 6
624	Right Wing - Slat No. 7
625	Right Wing - Slat No. 8

F. Access Panels

Number	Name/Location
521CB	Lower Leading Edge Access Panel
521DB	Lower Leading Edge Access Panel
521FB	Lower Leading Edge Access Panel
521JB	Lower Leading Edge Access Panel
521KB	Lower Leading Edge Access Panel
521LB	Lower Leading Edge Access Panel
521MB	Lower Leading Edge Access Panel
521QB	Lower Leading Edge Access Panel
521RB	Lower Leading Edge Access Panel
521SB	Lower Leading Edge Access Panel
521TB	Lower Leading Edge Access Panel
521WB	Lower Leading Edge Access Panel
521XB	Lower Leading Edge Access Panel
521YB	Lower Leading Edge Access Panel
521ZB	Lower Leading Edge Access Panel
621CB	Lower Leading Edge Access Panel
621DB	Lower Leading Edge Access Panel
621FB	Lower Leading Edge Access Panel
621JB	Lower Leading Edge Access Panel
621KB	Lower Leading Edge Access Panel
621LB	Lower Leading Edge Access Panel
621PB	Lower Leading Edge Access Panel
621QB	Lower Leading Edge Access Panel
621RB	Lower Leading Edge Access Panel
621SB	Lower Leading Edge Access Panel
621VB	Lower Leading Edge Access Panel
621WB	Lower Leading Edge Access Panel
621XB	Lower Leading Edge Access Panel
621YB	Lower Leading Edge Access Panel

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Leading Edge Slat Proximity Sensor Installation

SUBTASK 27-88-11-010-003

(1) Make sure that these circuit breakers are open and have safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	8	C00211	FLIGHT CONTROL FSEU DC 1
A	9	C01468	FLIGHT CONTROL FSEU DC 2

SUBTASK 27-88-11-420-008

(2) Make sure that the lockout set, SPL-1734, is installed on the actuators of the adjacent leading edge flaps and slats (TASK 27-81-00-480-801).

SUBTASK 27-88-11-420-002

(3) Do these steps to install the inboard proximity sensor [26] of the slat (View B, Figure 401):

WARNING USE NITRILE GLOVES FOR SKIN PROTECTION WHEN YOU USE COR-BAN 27L. IF IT GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER. GET MEDICAL AID. THIS MATERIAL CONTAINS FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(a) Apply a thin coat of Cor-Ban 27L Compound, G50237 (preferred), to the threads of the screws [29] or screw [32] before installation.

1) Use the applicable dry or lube torque value corresponding to the choice of the corrosion inhibitor.

NOTE: Noted series fasteners can be adjusted two additional grip lengths longer or shorter to accommodate the shim variations.

(b) For slat No. 1 or 8 (with the slats in the extended position), install the screws [29], washers [30], shim [27], aluminum spacer [28], and nuts [31] (View B, Figure 401).

1) Use the thickness dimensions of the aluminum shim [27] and aluminum spacer [28] that you recorded during the removal procedure to determine the closest amount of shim [3] you need to install.

a) Use the shim [27], as required, to meet the clearance requirement.

<1> Adjust the sensor to target clearance by adding or removing shim laminations as required.

NOTE: The clearance dimension applies to both the forward and aft targets.

b) Apply coating, C50069, to both sides of aluminum shim [27] (dry before install).

c) Bond the aluminum shim [27] and aluminum spacer [28] to the support bracket as follows:

<1> Lightly sand to remove the gloss from the mating surfaces.

<2> Clean the mating surfaces with alcohol, B00065, or alcohol, B50073.

<3> Wipe clean with a cotton wiper, G00034, wiper, G50262, or equivalent.

<4> Mix adhesive, A00165, fully before you use it.

NOTE: The adhesive must not be used if it is too thick.

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<5> Apply two thin, continuous layers of adhesive, A00165, to each mating surface.

NOTE: The first layer of the adhesive must dry fully before you apply the second layer. The second layer of the adhesive must dry only until it is tacky.

<6> Join the mating surfaces while adhesive, A00165, is tacky.

<7> Apply sufficient pressure to make good contact.

d) Apply adhesive, A01076, for faying surface sealing to the side of the aluminum shim [27] that contacts the proximity sensor [26].

e) Apply adhesive, A01076, to the sensor interface with the aluminum spacer [28] or with the structure, if no aluminum shim [27] is needed.

f) Apply coating, C50069 (dry before install), and Frekote 44-NC non-peelable parting agent, G50335, or MS-122XD lubricant, D50326 (as alternative), to the base of the proximity sensor [26].

2) Find the same color wires and make a splice.

3) Attach the wires together to make a bundle.

4) If the wires between the proximity sensor [26] and the last clamp are loosened during adjustment (rigging), do the steps that follow:

a) Remove all slack from slat sensor wires between the proximity sensor [26] and the last clamping location.

b) Redistribute the slack back to main bundle.

<1> Adjust the clamp size to accommodate the sheep shank area.

<2> Stitch-tie to prevent riding condition.

c) Reclamp if necessary.

(c) For slats No. 2, 3, 6 or 7 (with the slats in the extended position), install the screws [32], washers [30], shim [27], aluminum spacer [34], and nuts [31] (View B, Figure 401).

1) Use the thickness dimensions of the aluminum shim [27] and aluminum spacer [34] that you recorded during the removal procedure to determine the closest amount of shim [3] that you need to install.

a) Use shim [27], as required, to meet the clearance requirement.

<1> Adjust the sensor to target clearance by adding or removing shim laminations as required.

NOTE: The clearance dimension applies to both the forward and aft targets.

b) Bond the shim [27] and aluminum spacer [34] to the support bracket as follows:

<1> Lightly sand to remove the gloss from the mating surfaces.

<2> Clean the mating surfaces with alcohol, B00065, or alcohol, B50073.

<3> Wipe clean with a cotton wiper, G00034, wiper, G50262, or equivalent.

<4> Mix the adhesive, A00165, fully before you use it.

NOTE: The adhesive must not be used if it is too thick.

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<5> Apply two thin, continuous layers of the adhesive, A00165, to each mating surface.

NOTE: The first layer of the adhesive must dry fully before you apply the second layer. The second layer of the adhesive must dry only until it is tacky.

<6> Join the mating surfaces while the adhesive, A00165, is tacky.

<7> Apply sufficient pressure to make good contact.

c) Apply adhesive, A01076, for faying surface sealing to the side of the aluminum shim [27] that contacts the proximity sensor [4].

d) Apply adhesive, A01076, to the sensor interface with the aluminum spacer [34] with the structure, if no aluminum shim [27] is needed.

e) Apply coating, C50069 (dry before install), and Frekote 44-NC non-peelable parting agent, G50335, or MS-122XD lubricant, D50326 (as alternative), to the base of the proximity sensor [26].

2) Find the same color wires and make a splice.

3) Attach the wires together to make a bundle.

4) If the wires between the proximity sensor [26] and the last clamp are loosened during adjustment (rigging), do the steps that follow:

a) Remove all slack from slat sensor wires between the proximity sensor [26] and the last clamping location.

b) Redistribute the slack back to main bundle.

<1> Adjust the clamp size to accommodate the sheep shank area.

<2> Stitch-tie to prevent riding condition.

c) Re-clamp if necessary.

(d) For slat No. 4 or 5 (with the slats in the extended position), install the screws [33], washers [30], shim [3], and nuts [31] (View B, Figure 401).

1) Use the thickness dimensions of the aluminum shim [27] that you recorded during the removal procedure to determine the closest amount of shim [27] you need to install.

a) Use shim [27], as required, to meet the clearance requirement.

<1> Adjust the sensor to target clearance by adding or removing shim laminations as required.

NOTE: The clearance dimension applies to both the forward and aft targets.

b) Bond the shim [27] to the support bracket as follows:

<1> Lightly sand to remove the gloss from the mating surfaces.

<2> Clean the mating surfaces with alcohol, B00065, or alcohol, B50073.

<3> Wipe clean with a cotton wiper, G00034, wiper, G50262, or equivalent.

<4> Mix the adhesive, A00165, fully before you use it.

NOTE: The adhesive must not be used if it is too thick.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

<5> Apply two thin, continuous layers of the adhesive, A00165, to each mating surface.

NOTE: The first layer of the adhesive must dry fully before you apply the second layer. The second layer of the adhesive must dry only until it is tacky.

<6> Join the mating surfaces while the adhesive, A00165,is tacky.

<7> Apply sufficient pressure to make good contact.

c) Apply adhesive, A01076, for faying surface sealing to the side of the aluminum shim [27] that contacts the proximity sensor [26].

d) Apply adhesive, A01076, to the sensor interface with the structure, if no aluminum shim [27] is needed.

e) Apply coating, C50069 (dry before install), and Frekote 44-NC non-peelable parting agent, G50335, or MS-122XD lubricant, D50326 (as alternative), to the base of the proximity sensor [26].

2) Find the same color wires and make a splice.

3) Attach the wires together to make a bundle.

4) If the wires between the proximity sensor [26] and the last clamp are loosened during adjustment (rigging), do the steps that follow:

a) Remove all slack from slat sensor wires between the proximity sensor [26] and the last clamping location.

b) Redistribute the slack back to main bundle.

<1> Adjust the clamp size to accommodate the sheep shank area.

<2> Stitch-tie to prevent riding condition.

c) Re-clamp if necessary.

(e) For the slats 1 and 8 (with the slats in the retracted position), install the screw [29], aluminum spacer [28], bolts [37], washers [35], and nuts [36] if you removed the shim [27] (View B, Figure 401).

1) Use the thickness dimensions of the aluminum shim [27] and aluminum spacer [28] that you recorded during the removal procedure to determine the closest amount of shim [27] you need to install.

a) Use shim [27], as required, to meet the clearance requirement.

<1> Adjust the sensor to target clearance by adding or removing shim laminations as required.

NOTE: The clearance dimension applies to both the forward and aft targets.

b) Bond the shim [27] and aluminum spacer [28] to the support bracket as follows:

<1> Lightly sand to remove the gloss from the mating surfaces.

<2> Clean the mating surfaces with alcohol, B00065, or alcohol, B50073.

<3> Wipe clean with a cotton wiper, G00034, wiper, G50262, or equivalent.

<4> Mix the adhesive, A00165, fully before you use it.

NOTE: The adhesive must not be used if it is too thick.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

<5> Apply two thin, continuous layers of the adhesive, A00165, to each mating surface.

NOTE: The first layer of the adhesive must dry fully before you apply the second layer. The second layer of the adhesive must dry only until it is tacky.

<6> Join the mating surfaces while the adhesive, A00165, is tacky.

<7> Apply sufficient pressure to make good contact.

c) Apply adhesive, A01076, to the sensor interface with the aluminum spacer [28] or with the structure.

d) Apply coating, C50069 (dry before install), and Frekote 44-NC non-peelable parting agent, G50335, or MS-122XD lubricant, D50326 (as alternative), to the base of the proximity sensor [26].

2) Find the same color wires and make a splice.

3) Attach the wires together to make a bundle.

4) If the wires between the proximity sensor [26] and the last clamp are loosened during adjustment (rigging), do the steps that follow:

a) Remove all slack from slat sensor wires between the proximity sensor [26] and the last clamping location.

b) Redistribute the slack back to main bundle.

<1> Adjust the clamp size to accommodate the sheep shank area.

<2> Stitch-tie to prevent riding condition.

c) Re-clamp if necessary.

SUBTASK 27-88-11-400-001

(4) Install the outboard proximity sensor [26] of the slat with the slat in the extended or flap lever 5 position.

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION WHEN YOU USE COR-BAN 27L. IF IT GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER. GET MEDICAL AID. THIS MATERIAL CONTAINS FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(a) Apply a thin layer of Cor-Ban 27L Compound, G50237 on the holes of the proximity sensor [26] and on the threads of the (aluminum pigmented) cap screw [52] before you install them.

1) Use applicable dry or lube torque value corresponding to the choice of the corrosion inhibitor.

(b) Use the thickness dimensions of the shim [27] and aluminum spacer [28] or aluminum spacer [34] that you recorded during the removal procedure to determine the closest amount of shim [27] you need to install.

(c) Install the aluminum shim [27] (as many as necessary to get the permitted clearance) between the proximity sensor [26] and the aluminum spacer [28] or aluminum spacer [34] and fasten them to the outboard sensor bracket [53] with the (aluminum pigmented) cap screw [52].

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1) Adjust the sensor to target clearance by adding or removing shim laminations as required.

NOTE: The clearance dimension applies to both the forward and aft targets.

NOTE: Noted series fasteners can be adjusted two additional grip lengths longer or shorter to accommodate the shim variations.

(d) Apply coating, C50069, to both sides of aluminum shim [27] (dry before install).

(e) Bond the aluminum shim [27] and the aluminum spacer [28] or aluminum spacer [34] to the outboard sensor bracket [53] as follows:

1) Lightly sand to remove the gloss from the mating surfaces.

2) Clean the mating surfaces with alcohol, B00065, or alcohol, B50073.

3) Wipe clean with a cotton wiper, G00034, wiper, G50262, or equivalent.

4) Mix the adhesive, A00165, fully before you use it.

NOTE: The adhesive must not be used if it is too thick.

5) Apply two thin, continuous layers of the adhesive, A00165, to each mating surface.

NOTE: The first layer of the adhesive must dry fully before you apply the second layer. The second layer of the adhesive must dry only until it is tacky.

6) Join the mating surfaces while the adhesive, A00165, is tacky.

7) Apply sufficient pressure to make good contact.

(f) Apply adhesive, A01076, for faying surface sealing to the side of the aluminum shim [27] that contacts the proximity sensor [26].

(g) Apply adhesive, A01076, to the sensor interface with the aluminum spacer [28] or aluminum spacer [34] or with the structure, if no aluminum shim [27] is needed.

(h) Apply coating, C50069(dry before install), and Frekote 44-NC non-peelable parting agent, G50335, or MS-122XD lubricant, D50326 (as alternative), to the base of the proximity sensor [4].

(i) Install the P-clamp [51] with the screw to the aluminum spacer [28] or aluminum spacer [34].

(j) Find the same color wires and make a splice.

(k) Attach the wires together to make a bundle.

(l) If the wires between the proximity sensor [26] and the last clamp are loosened during adjustment (rigging), do the steps that follow:

1) Remove all slack from slat sensor wires between the proximity sensor [26] and the last clamping location.

2) Redistribute the slack back to main bundle.

a) Adjust the clamp size to accommodate the sheep shank area.

b) Stitch-tie to prevent riding condition.

3) Reclamp if necessary.

(m) For the slats 1 and 8 (with the slats in the retracted position), install the screw [38], aluminum spacer [28], bolt [37], washer [35], and nut [36] if you removed the shim [27] (View B, Figure 401).

1) Use the thickness dimensions of the aluminum shim [27] and aluminum spacer [28] that you recorded during the removal procedure to determine the closest amount of shim [27] you need to install.

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a) Use shim [27], as required, to meet the clearance requirement.

<1> Adjust the sensor to target clearance by adding or removing shim laminations as required.

NOTE: The clearance dimension applies to both the forward and aft targets.

b) Bond the shim [27] and aluminum spacer [28] to the support bracket as follows:

<1> Lightly sand to remove the gloss from the mating surfaces.

<2> Clean the mating surfaces with alcohol, B00065, or alcohol, B50073.

<3> Wipe clean with a cotton wiper, G00034, wiper, G50262 or equivalent.

<4> Mix the adhesive, A00165, fully before you use it.

NOTE: The adhesive must not be used if it is too thick.

<5> Apply two thin, continuous layers of the adhesive, A00165 to each mating surface.

NOTE: The first layer of the adhesive must dry fully before you apply the second layer. The second layer of the adhesive must dry only until it is tacky.

<6> Join the mating surfaces while the adhesive, A00165, is tacky.

<7> Apply sufficient pressure to make good contact.

c) Apply adhesive, A01076, to the sensor interface with the aluminum spacer [28] or with the structure.

d) Apply coating, C50069 (dry before install), and Frekote 44-NC non-peelable parting agent, G50335, or MS-122XD lubricant, D50326 (as alternative), to the base of the proximity sensor [26].

2) Find the same color wires and make a splice.

3) Attach the wires together to make a bundle.

4) If the wires between the and the last clamp are loosened during adjustment (rigging), do the steps that follow:

a) Remove all slack from slat sensor wires between the proximity sensor [26] and the last clamping location.

b) Redistribute the slack back to main bundle.

<1> Adjust the clamp size to accommodate the sheep shank area.

<2> Stitch-tie to prevent riding condition.

c) Re-clamp if necessary.

SUBTASK 27-88-11-420-004

(5) Do these steps to install the retract proximity sensor [24]:

NOTE: It is not necessary to adjust a clearance for the retract proximity sensor [24].

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WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION WHEN YOU USE COR-BAN 27L. IF IT GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER. GET MEDICAL AID. THIS MATERIAL CONTAINS FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(a) Apply a thin coat of Cor-Ban 27L Compound, G50237, to the pilot diameter and fay surface of the retract proximity sensor [24].

(b) Install the retract proximity sensor [24] with the fastener [25].

(c) Connect the electrical connector to the retract proximity sensor [24].

(d) Install the slat actuator door [23] to the slat actuator.

1) Remove the nut [22], washer [21], and bolt [20] that attached the bracket to the slat actuator that attach the bracket to the slat actuator.

a) If there are three washers installed on the bolt [20], discard the thicker aluminum washer.

2) Put the slat actuator door [23] in its position on the slat actuator.

3) Install bolt [20],washer [21], and nut [22] that attach the slat actuator door [23] and the bracket to the slat actuator.

4) Install the bolt [17], washer [18], and nut [19] that attach the slat actuator door [23] to the slat actuator.

SUBTASK 27-88-11-220-001

(6) With the leading edge slat in the full extend position, make sure the clearance between the inboard or outboard proximity sensor [26] and the target [54] is 0.115 in. (2.92 mm)– 0.135 in. (3.43 mm).

(a) If the permitted clearance is out of tolerance, adjust the clearance between the proximity sensor [26] and the target [54].

1) Use the required consumable materials that you used during the installation procedure above for the re-installation of the proximity sensor [26], as necessary

2) Remove the (aluminum pigmented) cap screw [52], shim [27], and aluminum spacer [28] or aluminum spacer [34].

3) Add or remove shim [27] as it is necessary to get the correct clearance between the proximity sensor [26] and the target [54].

a) Adjust the sensor to target clearance by adding or removing shim laminations as required.

NOTE: The clearance dimension applies to both the forward and aft targets.

b) Use the thickness dimensions of shim [27] and aluminum spacer [28] or aluminum spacer [34] that you recorded during the removal procedure to determine the closest amount of shim [27] you need to install.

4) Fasten the proximity sensor [26], shim [27] and aluminum spacer [28] or aluminum spacer [34] to the outboard sensor bracket [53] with the (aluminum pigmented) cap screw [52].

a) Use the applicable dry or lube torque value corresponding to the choice of the corrosion inhibitor.

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SUBTASK 27-88-11-860-002

WARNING MAKE SURE PERSONS AND EQUIPMENT ARE CLEAR OF THE LEADING EDGE AND TRAILING EDGE CONTROL SURFACES. THE LEADING EDGE AND TRAILING EDGE CONTROL SURFACES CAN EXTEND AND RETRACT QUICKLY. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(7) Retract the leading edge slat to the extend (flap lever 1-unit detent) position with primary control, do this task: Leading Edge Flaps and Slats Retraction, TASK 27-81-00-860-804.

SUBTASK 27-88-11-220-003

(8) With the slat in the 1-unit detent position, make sure the clearance between the inboard and/or outboard proximity sensor [26] and the target [9] for the inboard sensor or target [54] for the outboard sensor is 0.125 ±0.010 in. (3.175 ±0.254 mm).

(a) If the permitted clearance is out of tolerance, adjust the clearance between the proximity sensor [26] and the target [9] or target [54].

1) Remove the screw [29]

or screw [32] for the inboard sensor, or cap screw [52] for the outboard sensors of the slats, washer [30] and nut [31] on the inboard sensor, shim [27] and aluminum spacer [28] or aluminum spacer [34].

2) Add or remove shim [27] as it is necessary to get the correct clearance between the proximity sensor [26] and the target [9] or target [54].

3) Install the screw [29] or screw [32] for the inboard sensors of the slats or cap screw [52] for the outboard sensors of the slats, washer [30] and nut [31] on the inboard slat, and aluminum spacer [28] or aluminum spacer [34].

(b) Bond the shim [27] and aluminum spacer [28] or aluminum spacer [34] to the bracket with adhesive, A00165.

SUBTASK 27-88-11-410-001

(9) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	8	C00211	FLIGHT CONTROL FSEU DC 1
A	9	C01468	FLIGHT CONTROL FSEU DC 2

SUBTASK 27-88-11-860-003

WARNING MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF THE LEADING EDGE SLATS. THE LEADING EDGE SLATS CAN MOVE QUICKLY. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(10) Retract the leading edge slat to the retracted (flap lever UP) position with primary control, do this task: Leading Edge Flaps and Slats Retraction, TASK 27-81-00-860-804.

SUBTASK 27-88-11-220-005

(11) On leading edge slat 1 or 8, make sure that the clearance between the (inboard or outboard) proximity sensor [26] and the target [14] is 0.125 ±0.010 in. (3.175 ±0.254 mm).

(a) If the permitted clearance is out of tolerance, adjust the clearance between the proximity sensor [26] and the target [14].

1) Remove the nut [36], washer [35], shim [27], and bolt [37] to remove the target [14].

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2) Add or remove shim [27] as it is necessary to get the correct clearance between the proximity sensor [26] and the target [14].

3) Install the bolt [37],target [14], washer [35], and nut [36].

SUBTASK 27-88-11-080-003

WARNING CAREFULLY REMOVE THE LEADING EDGE FLAP AND SLAT ACTUATORS LOCKOUT SET. THE LEADING EDGE FLAPS AND SLATS CAN MOVE QUICKLY IF YOU DO NOT REMOVE THE LOCKS CORRECTLY. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(12) Remove the lockout set, SPL-1734, from the actuators of the leading edge flaps and slats, do this task: Leading Edge Flap and Slat Locks Removal, TASK 27-81-00-080-801.

SUBTASK 27-88-11-860-065

WARNING MAKE SURE PERSONS AND EQUIPMENT ARE CLEAR OF THE LEADING EDGE AND TRAILING EDGE CONTROL SURFACES. THE LEADING EDGE AND TRAILING EDGE CONTROL SURFACES CAN EXTEND AND RETRACT QUICKLY. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(13) Retract the remaining leading edge flaps and slats to the retracted (flap lever UP) position with primary control, do this task: Leading Edge Flaps and Slats Retraction, TASK 27-81-00-860-804.

SUBTASK 27-88-11-710-001

(14) Do this task: Leading Edge Flap and Slat Proximity Sensor Test, TASK 27-88-11-700-801.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 27-88-11-940-001

(1) Install the applicable access panels (TASK 57-41-02-400-801):

(a) For the left wing, close these access panels:

Number	Name/Location
521CB	Lower Leading Edge Access Panel
521DB	Lower Leading Edge Access Panel
521FB	Lower Leading Edge Access Panel
521JB	Lower Leading Edge Access Panel
521KB	Lower Leading Edge Access Panel
521LB	Lower Leading Edge Access Panel
521MB	Lower Leading Edge Access Panel
521QB	Lower Leading Edge Access Panel
521RB	Lower Leading Edge Access Panel
521SB	Lower Leading Edge Access Panel
521TB	Lower Leading Edge Access Panel
521WB	Lower Leading Edge Access Panel
521XB	Lower Leading Edge Access Panel
521YB	Lower Leading Edge Access Panel
521ZB	Lower Leading Edge Access Panel

(b) For the right wing, close these access panels:

Number	Name/Location
621CB	Lower Leading Edge Access Panel

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(Continued)

Number	Name/Location
621DB	Lower Leading Edge Access Panel
621FB	Lower Leading Edge Access Panel
621JB	Lower Leading Edge Access Panel
621KB	Lower Leading Edge Access Panel
621LB	Lower Leading Edge Access Panel
621PB	Lower Leading Edge Access Panel
621QB	Lower Leading Edge Access Panel
621RB	Lower Leading Edge Access Panel
621SB	Lower Leading Edge Access Panel
621VB	Lower Leading Edge Access Panel
621WB	Lower Leading Edge Access Panel
621XB	Lower Leading Edge Access Panel
621YB	Lower Leading Edge Access Panel

——— END OF TASK ———

TASK 27-88-11-000-802

4. Leading Edge Flap Proximity Sensor Removal

(Figure 402)

A. References

Reference	Title
27-81-00-480-801	Leading Edge Flap and Slat Locks Installation (P/B 201)
27-81-00-860-803	Leading Edge Flaps and Slats Extension (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1734	Lockout Set - Leading Edge Slats and Krueger Flaps Actuators Part #: C27051-29 Supplier: 81205

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
512	Left Wing - Krueger Flap No. 2
513	Left Wing - Krueger Flap No. 1
612	Right Wing - Krueger Flap No. 3
613	Right Wing - Krueger Flap No. 4

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. Prepare for the Removal

SUBTASK 27-88-11-860-004

WARNING

MAKE SURE PERSONS AND EQUIPMENT ARE CLEAR OF THE LEADING EDGE AND TRAILING EDGE CONTROL SURFACES. THE LEADING EDGE AND TRAILING EDGE CONTROL SURFACES CAN EXTEND AND RETRACT QUICKLY. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

CAUTION

MAKE SURE THAT YOU CLOSE OR REMOVE THE INBOARD FAN COWL AND THRUST REVERSERS BEFORE YOU RETRACT THE LEADING EDGE FLAPS. THE CLEARANCE IS NOT SUFFICIENT FOR THE FLAPS TO RETRACT WITH THE INBOARD FAN DUCT COWL AND THE THRUST REVERSERS IN THE OPEN POSITION. THIS CAN CAUSE DAMAGE TO EQUIPMENT.

(1) To extend the leading edge flaps, do this task: Leading Edge Flaps and Slats Extension, TASK 27-81-00-860-803.

SUBTASK 27-88-11-010-004

(2) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	8	C00211	FLIGHT CONTROL FSEU DC 1
A	9	C01468	FLIGHT CONTROL FSEU DC 2

SUBTASK 27-88-11-480-004

WARNING

CAREFULLY REMOVE THE LEADING EDGE FLAP ACTUATOR LOCK TO PREVENT ACCIDENTAL OPERATION OF THE LEADING EDGE FLAPS. THE LEADING EDGE FLAPS CAN MOVE QUICKLY. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(3) Install the lockout set, SPL-1734 on the actuator of the adjacent leading edge flap. To do this, do this task: Leading Edge Flap and Slat Locks Installation, TASK 27-81-00-480-801.

E. Leading Edge Flap Proximity Sensor Removal

SUBTASK 27-88-11-020-007

(1) Remove the extend proximity sensor [44A] and/or retract proximity sensor [44B] per Figure 402:

(a) Disconnect the wires at the splice.

(b) To remove the retract proximity sensor [44B], remove the nuts [43], washers [42] (installed under the nut), and the (aluminum pigmented) cap screws [49].

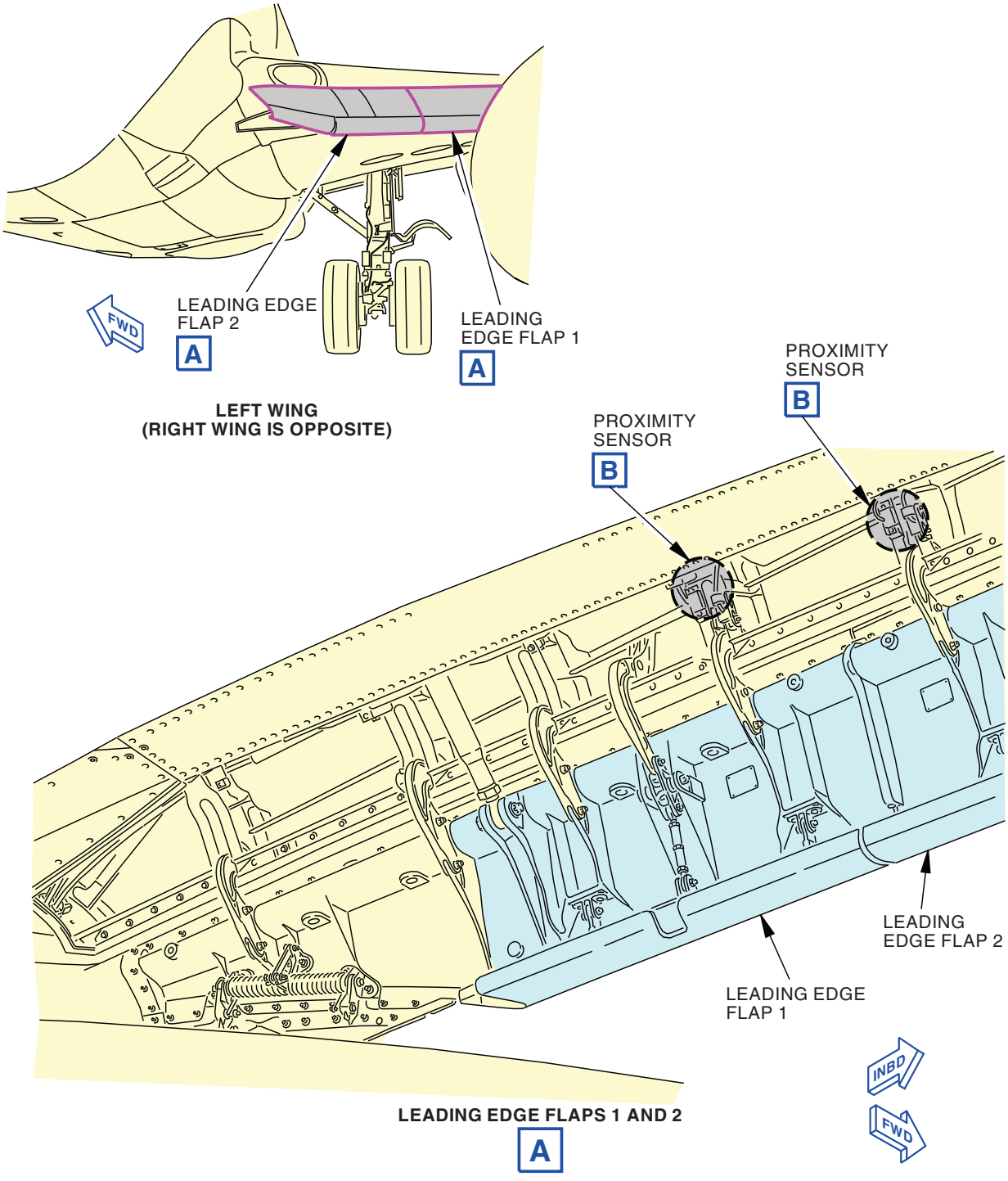
(c) To remove the extend proximity sensor [44A], remove the (aluminum pigmented) cap screws [50] and the shim [46].

1) Measure the thickness of the shim [46] and keep record of it for easier completion of the sensor-target clearance required during the installation task of the extend extend proximity sensor [44A].

(d) Remove the extend proximity sensor [44A] and/or retract proximity sensor [44B].

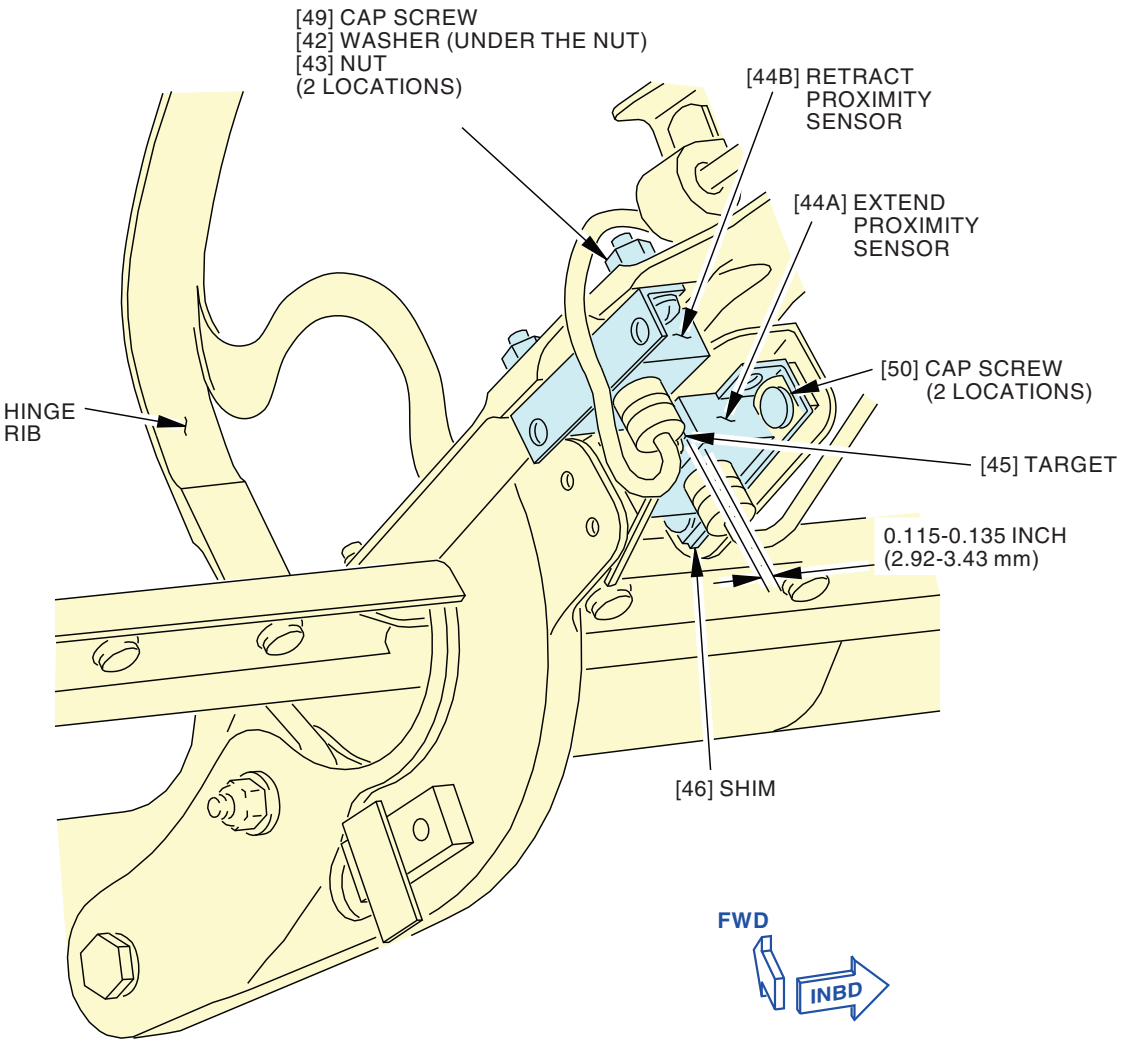
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Leading Edge Flap Proximity Sensor Installation
Figure 402/27-88-11-990-802 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



PROXIMITY SENSOR (EXAMPLE)

B

2413405 S00061532508_V1

Leading Edge Flap Proximity Sensor Installation
Figure 402/27-88-11-990-802 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-88-11-400-802

5. Leading Edge Flap Proximity Sensor Installation

(Figure 402)

A. References

Reference	Title
27-81-00-080-801	Leading Edge Flap and Slat Locks Removal (P/B 201)
27-81-00-860-804	Leading Edge Flaps and Slats Retraction (P/B 201)
27-88-11-700-801	Leading Edge Flap and Slat Proximity Sensor Test (P/B 501)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1734	Lockout Set - Leading Edge Slats and Krueger Flaps Actuators Part #: C27051-29 Supplier: 81205

C. Consumable Materials

Reference	Description	Specification
A01076	Adhesive - Synthetic Rubber	BAC5010 Type 93 (BMS5-95 Class B)
C00259	Coating - Chemical And Solvent Resistant Finish, Corrosion Inhibiting Primer	BMS10-11 Type I
C50069	Coating - Chemical And Solvent Resistant Finish - (BAC 702 White Color, Gloss)	BMS10-11 Type II
D50326	Lubricant - PTFE Release Agent - Miller-Stephenson MS-122XD (Replaces MS-122DF and MS-122V Lubricants)	
G50237	Compound - Corrosion Inhibiting, Non-drying - Cor-Ban 27L	BMS3-38
G50335	Agent - Non-Peelable Parting (Henkel Loctite - Frekote 44-NC Mold Release).	BAC5000

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
44A	Extend proximity sensor	27-88-11-01-130	PTF ALL
44B	Retract proximity sensor	27-88-11-01-130	PTF ALL

E. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
512	Left Wing - Krueger Flap No. 2
513	Left Wing - Krueger Flap No. 1
612	Right Wing - Krueger Flap No. 3
613	Right Wing - Krueger Flap No. 4

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Leading Edge Flap Proximity Sensor Installation

SUBTASK 27-88-11-010-005

(1) Make sure that these circuit breakers are open and have safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	8	C00211	FLIGHT CONTROL FSEU DC 1
A	9	C01468	FLIGHT CONTROL FSEU DC 2

SUBTASK 27-88-11-800-001

(2) Do these steps to install the flap (titanium) extend proximity sensor [44A] or retract proximity sensor [44B] Figure 402:

NOTE: The titanium sensor installed with applicable hardware and consumable is interchangeable with the stainless steel sensor installed with the applicable hardware and consumable.

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION WHEN YOU USE COR-BAN 27L. IF IT GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER. GET MEDICAL AID. THIS MATERIAL CONTAINS FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(a) Apply a thin coat of Cor-Ban 27L Compound, G50237 on the holes of the titanium extend proximity sensor [44A]or , retract proximity sensor [44B], the threads of the (aluminum pigmented) cap screws [49], and cap screws [50].

(b) Apply coating, C50069 (dry before install) or primer, C00259 to the two sides of the aluminum shim [46].

(c) Apply adhesive, A01076 to the faying surface seal on the side that contacts the titanium extend proximity sensor [44A] or retract proximity sensor [44B].

(d) Apply coating, C50069 (dry before install) or primer, C00259 and Frekote 44-NC non-peelable parting agent, G50335, or MS-122XD lubricant, D50326 (as alternative), to the base of the titanium extend proximity sensor [44A] or retract proximity sensor [44B].

(e) Install the retract proximity sensor [44B] with the (aluminum pigmented) cap screws [49], washers [42], and nuts [43].

(f) Install the extend proximity sensor [44A] with the (aluminum pigmented) cap screws [50] and shim [46].

(g) Find the same color wires and make a splice.

(h) Attach the wires together to make a bundle.

SUBTASK 27-88-11-800-002

(3) With the flap in the extend position, make sure the clearance dimension between the extend proximity sensor [44A] and the target [45] is 0.125 ±0.010 in. (3.175 ±0.254 mm).

(a) If the clearance dimension is out of tolerance, adjust the clearance dimension between the extend proximity sensor [44A] and the target [45].

NOTE: The sensor-target clearance of the extend proximity sensor [44A] is adjustable. The sensor-target clearance of the retract proximity sensor [44B] is not adjustable.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Use the necessary materials from the above installation to install the extend proximity sensor [44A] after the adjustment.
2) Remove the (aluminum pigmented) cap screws [50] and shim [46].
3) If necessary, add or remove shims [46] to get the correct clearance dimension between the extend proximity sensor [44A] and the target [45].
   a) Use the thickness dimensions of the shims [46] recorded during the removal procedure to find the necessary quantity of shims [46].
4) Install the (aluminum pigmented) cap screws [50].
   a) Use the applicable dry or lube torque value.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 27-88-11-080-004

WARNING CAREFULLY REMOVE THE LEADING EDGE FLAP ACTUATOR LOCK TO PREVENT ACCIDENTAL OPERATION OF THE LEADING EDGE FLAPS. THE LEADING EDGE FLAPS CAN MOVE QUICKLY. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Remove the lockout set, SPL-1734 from the actuator of the leading edge flap, (TASK 27-81-00-080-801).

SUBTASK 27-88-11-410-002

(2) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	8	C00211	FLIGHT CONTROL FSEU DC 1
A	9	C01468	FLIGHT CONTROL FSEU DC 2

SUBTASK 27-88-11-860-005

WARNING MAKE SURE PERSONS AND EQUIPMENT ARE CLEAR OF THE LEADING EDGE AND TRAILING EDGE CONTROL SURFACES. THE LEADING EDGE AND TRAILING EDGE CONTROL SURFACES CAN EXTEND AND RETRACT QUICKLY. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(3) Retract the leading edge flap to the UP position with primary control, (TASK 27-81-00-860-804).

SUBTASK 27-88-11-710-002

(4) Do this task: Leading Edge Flap and Slat Proximity Sensor Test, TASK 27-88-11-700-801

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

LEADING EDGE FLAP AND SLAT PROXIMITY SENSOR - ADJUSTMENT/TEST

1. General

A. This procedure contains the following tasks:

(1) Leading Edge Flap and Slat Proximity Sensor Test

(2) Leading Edge Flap and Slat Proximity Sensor Test Using a LCR Meter

(3) Leading Edge Flap and Slat Proximity Sensor Test Using a Breakout Box and a LCR Meter.

B. These tasks are functional tests of the proximity switch systems sensors.

NOTE: If available, use B737 FSEU Test/Breakout Box MPN 380-026-02 to do the inductance and resistance tests.

C. These tests are intended to determine a proximity sensor(s) health preventing unnecessary replacement of the Flap/Slat Electronics Unit (FSEU).

D. The proximity sensors provide position sensing for the leading edge flaps and slats systems.

E. The proximity sensors remain installed on the airplane during the test.

TASK 27-88-11-700-801

2. Leading Edge Flap and Slat Proximity Sensor Test

(Figure 501)

A. References

Reference	Title
27-51-00-740-801	FSEU Display and Existing Faults Test (P/B 501)
27-81-00-860-804	Leading Edge Flaps and Slats Retraction (P/B 201)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)

B. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
512	Left Wing - Krueger Flap No. 2
513	Left Wing - Krueger Flap No. 1
522	Left Wing - Slat No. 4
524	Left Wing - Slat No. 2
525	Left Wing - Slat No. 1
612	Right Wing - Krueger Flap No. 3
613	Right Wing - Krueger Flap No. 4
622	Right Wing - Slat No. 5
623	Right Wing - Slat No. 6
624	Right Wing - Slat No. 7
625	Right Wing - Slat No. 8

C. Prepare for the Test

SUBTASK 27-88-11-860-006

(1) Make sure that the flap control lever is in the FLAP UP detent position with the leading edge (LE) flaps and slats in the retract position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) If it is necessary, do this task: Leading Edge Flaps and Slats Retraction, TASK 27-81-00-860-804.

SUBTASK 27-88-11-740-001

(2) Do this task: FSEU Display and Existing Faults Test, TASK 27-51-00-740-801.

SUBTASK 27-88-11-860-008

WARNING

KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(3) Do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

D. Leading Edge Flap and Slat Proximity Sensor Test

SUBTASK 27-88-11-730-001

WARNING

MAKE SURE PERSONS AND EQUIPMENT ARE CLEAR OF THE LEADING EDGE AND TRAILING EDGE CONTROL SURFACES. THE LEADING EDGE AND TRAILING EDGE CONTROL SURFACES CAN EXTEND AND RETRACT QUICKLY. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

CAUTION

MAKE SURE THE INBOARD FAN DUCT COWL AND THE INBOARD AND OUTBOARD THRUST REVERSERS ARE CLOSED OR REMOVED BEFORE YOU EXTEND THE LEADING EDGE FLAPS AND SLATS IN THIS TEST. THERE IS NOT SUFFICIENT CLEARANCE FOR THE FLAPS AND SLATS TO EXTEND IF THE INBOARD FAN DUCT COWL AND THE THRUST REVERSERS ARE IN THE OPEN POSITION. THIS CAN CAUSE DAMAGE TO EQUIPMENT.

(1) Do a test of the leading edge flap and slat proximity sensor:

(a) Push the ON/OFF button on the front panel of the FSEU to turn on the display.

NOTE: The display will show EXISTING FAULTS?.

(b) Push the down arrow until OTHER FUNCTNS? shows on the display.

(c) Push the YES button to select OTHER FUNCTNS? on the FSEU.

(d) Push the down arrow until I/O MONITOR? shows on the display.

(e) Push the YES button to select I/O MONITOR? from the menu.

(f) Push the YES button to select PROX SENSORS? from the I/O MONITOR? menu.

(g) Push the down arrow to step through the proximity sensor states at the LE flaps and slats UP position.

(h) Make sure that these displays and the proximity sensor states show on the FSEU (Table 501):

Table 501/27-88-11-993-801

Display	State
S1 OUTBD	GAP NEAR
S1 INBD	GAP NEAR

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Table 501/27-88-11-993-801 (Continued)

Display	State
S2 OUTBD	GAP FAR
S2 INBD	GAP FAR
S2 RETR	GAP NEAR
S3 OUTBD	GAP FAR
S3 INBD	GAP FAR
S3 RETR	GAP NEAR
S4 OUTBD	GAP FAR
S4 INBD	GAP FAR
S4 RETR	GAP NEAR
S5 OUTBD	GAP FAR
S5 INBD	GAP FAR
S5 RETR	GAP NEAR
S6 OUTBD	GAP FAR
S6 INBD	GAP FAR
S6 RETR	GAP NEAR
S7 OUTBD	GAP FAR
S7 INBD	GAP FAR
S7 RETR	GAP NEAR
S8 OUTBD	GAP NEAR
S8 INBD	GAP NEAR
F1 RETR	GAP NEAR
F1 EXT	GAP FAR
F2 RETR	GAP NEAR
F2 EXT	GAP FAR
F3 RETR	GAP NEAR
F3 EXT	GAP FAR
F4 RETR	GAP NEAR
F4 EXT	GAP FAR

(i) Move the flap control lever to the 1-unit detent position.

1) Make sure that these lights come on during the flaps and slats extension and go off after extension:

a) The LE FLAPS TRANSIT light on the P2-2 center instrument panel.

b) All the LE flaps and slats TRANSIT lights on the P5-12 aft overhead panel.

2) Push the down arrow on the FSEU to step through the proximity sensor states.

3) Make sure that these displays and the proximity sensor states show on the FSEU (Table 502):

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Table 502/27-88-11-993-802

Display	State
S1 OUTBD	GAP FAR
S1 INBD	GAP NEAR
S2 OUTBD	GAP FAR
S2 INBD	GAP NEAR
S2 RETR	GAP FAR
S3 OUTBD	GAP FAR
S3 INBD	GAP NEAR
S3 RETR	GAP FAR
S4 OUTBD	GAP FAR
S4 INBD	GAP NEAR
S4 RETR	GAP FAR
S5 OUTBD	GAP FAR
S5 INBD	GAP NEAR
S5 RETR	GAP FAR
S6 OUTBD	GAP FAR
S6 INBD	GAP NEAR
S6 RETR	GAP FAR
S7 OUTBD	GAP FAR
S7 INBD	GAP NEAR
S7 RETR	GAP FAR
S8 OUTBD	GAP FAR
S8 INBD	GAP NEAR
F1 RETR	GAP FAR
F1 EXT	GAP NEAR
F2 RETR	GAP FAR
F2 EXT	GAP NEAR
F3 RETR	GAP FAR
F3 EXT	GAP NEAR
F4 RETR	GAP FAR
F4 EXT	GAP NEAR

4) Make sure that the LE FLAPS EXT light on the P2-2 center instrument panel is on.

5) Make sure that the LE flaps and slats EXT lights on the P5-12 aft overhead panel are on.

(j) Move the flap control lever to the 30-unit detent position.

1) Make sure that these lights come on during the flaps and slats extension and go off after extension:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

a) The LE FLAPS TRANSIT light on the P2 center instrument panel.

b) All the LE slats TRANSIT lights on the P5-12 aft overhead panel.

2) Push the down arrow on the FSEU to step through the proximity sensor states.

3) Make sure that these displays and the proximity sensor states show on the FSEU (Table 503):

Table 503/27-88-11-993-803

Display	State
S1 OUTBD	GAP NEAR
S1 INBD	GAP FAR
S2 OUTBD	GAP NEAR
S2 INBD	GAP FAR
S2 RETR	GAP FAR
S3 OUTBD	GAP NEAR
S3 INBD	GAP FAR
S3 RETR	GAP FAR
S4 OUTBD	GAP NEAR
S4 INBD	GAP FAR
S4 RETR	GAP FAR
S5 OUTBD	GAP NEAR
S5 INBD	GAP FAR
S5 RETR	GAP FAR
S6 OUTBD	GAP NEAR
S6 INBD	GAP FAR
S6 RETR	GAP FAR
S7 OUTBD	GAP NEAR
S7 INBD	GAP FAR
S7 RETR	GAP FAR
S8 OUTBD	GAP NEAR
S8 INBD	GAP FAR
F1 RETR	GAP FAR
F1 EXT	GAP NEAR
F2 RETR	GAP FAR
F2 EXT	GAP NEAR
F3 RETR	GAP FAR
F3 EXT	GAP NEAR
F4 RETR	GAP FAR
F4 EXT	GAP NEAR

ECCN 9E991 BOEING PROPRIETARY - See title page for details

4) Make sure that the LE FLAPS EXT light on the P2-2 center instrument pane is on.

5) Make sure that the LE flaps EXTEND and slats FULL EXTEND lights on the P5-12 aft overhead panel are on.

E. Put the Airplane Back to Its Usual Condition

SUBTASK 27-88-11-860-009

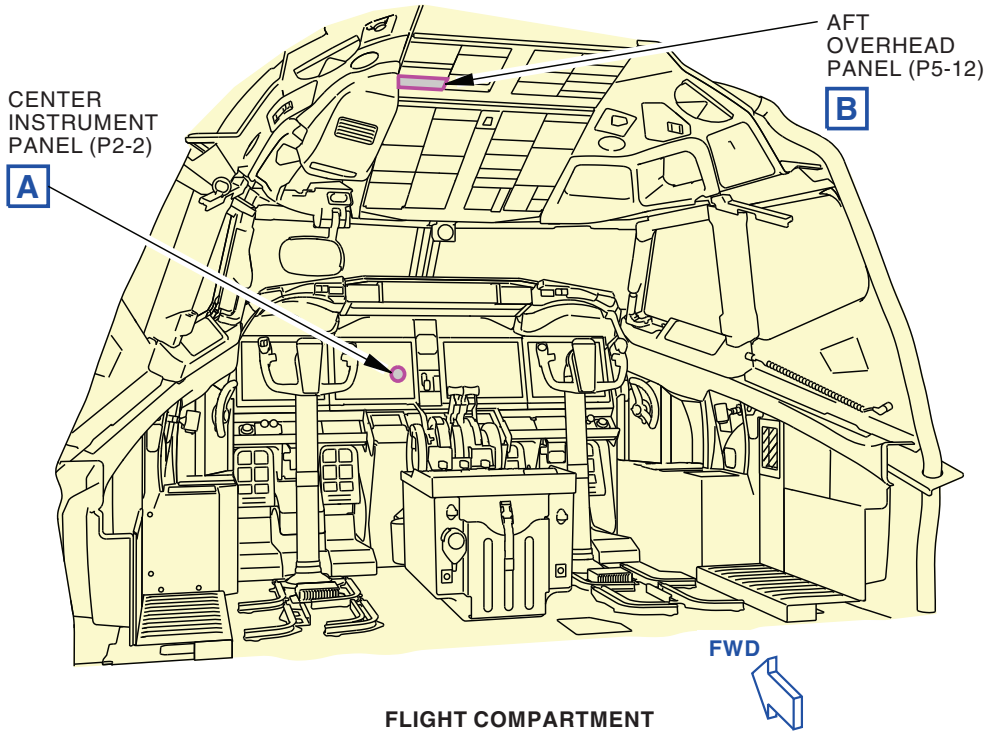
(1) Move the flap control lever to the FLAP UP detent to retract the leading edge flaps and slats.

SUBTASK 27-88-11-860-010

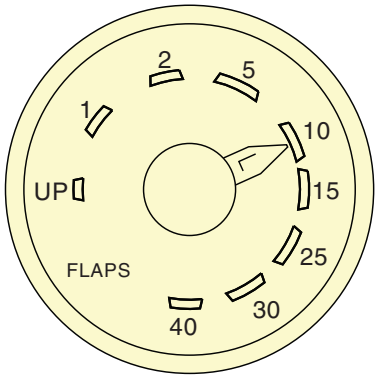
(2) Do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

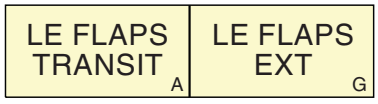
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



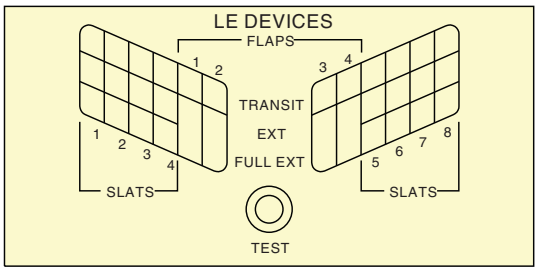
FLIGHT COMPARTMENT





CENTER INSTRUMENT PANEL (P2-2)

A



AFT OVERHEAD PANEL (P5-12)

B

2413406 S00061532512_V2

Leading Edge Flap and Slat Proximity Sensor Test
Figure 501/27-88-11-990-803

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-88-11-760-801

3. Leading Edge Flap and Slat Proximity Sensor Test Using a LCR Meter

(Figure 502)

A. General

(1) This test will measure the resistance and inductance of the 3-wire proximity sensor and wire bundle. And compare the values to a known range. This test also examines the target to the sensor condition.

(2) Obey all applicable warnings, cautions, and maintenance practice instructions when necessary to measure the Proximity Sensor in the test.

(3) Use an LCR meter (Inductance, Capacitance, Resistance), COM-1741 that has a Universal Serial Bus (USB) port to record the data with a PC automatically. For example 878B or 879B have a USB port. Use the specified instructions included with the LCR meter (Inductance, Capacitance, Resistance), COM-1741.

(4) The Sensor Test Gauge Kit (proximity sensor test set, SPL-1690) consists of six go-gauges, sensor de-actuator tool and an instruction booklet. Use the go-gauges in place of a sensor, if you suspect the sensor is faulty. Refer to 27-88 IFIM if necessary.

B. References

Reference	Title
27-51-00-740-801	FSEU Display and Existing Faults Test (P/B 501)
27-51-01-000-801	Flap/Slat Electronics Unit Removal (P/B 401)
27-51-01-400-801	Flap/Slat Electronics Unit Installation (P/B 401)
27-81-00-860-803	Leading Edge Flaps and Slats Extension (P/B 201)
27-81-00-860-804	Leading Edge Flaps and Slats Retraction (P/B 201)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
WDM 27-81-11	Wiring Diagram Manual
WDM 27-81-12	Wiring Diagram Manual
WDM 27-81-21	Wiring Diagram Manual
WDM 27-81-22	Wiring Diagram Manual

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1741	Meter - LCR (Inductance, Capacitance, Resistance; Use COM-20443 for OH bleed loop testing) Part #: 878B Supplier: 08098 Part #: 879B Supplier: 08098 Part #: 880 Supplier: 08098 Part #: U1733P Supplier: 1LQK8 Opt Part #: 875B Supplier: 08098 Opt Part #: 878A Supplier: 08098 Opt Part #: 879 Supplier: 08098

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Reference	Description
COM-1793	Multimeter - Digital/Analog (or equivalent meter meets task requirements) Part #: 117 Supplier: 89536 Part #: 260-8XPI Supplier: 55026 Part #: 287 Supplier: 89536 Part #: 289 Supplier: 89536 Part #: 87V Supplier: 89536 Part #: FLUKE 27 II Supplier: 89536 Part #: FLUKE-77-4 Supplier: 89536 Part #: WA2025 Supplier: 4MJL8 Opt Part #: 187 Supplier: 89536 Opt Part #: 189 Supplier: 89536 Opt Part #: 21 Supplier: 89536 Opt Part #: 27 Supplier: 89536 Opt Part #: 77 SERIES III Supplier: 89536 Opt Part #: 87 Supplier: 89536 Opt Part #: FLUKE 27 Supplier: 89536 Opt Part #: MODEL 27 Supplier: 89536
SPL-1690	Actuators/Deactuators Set - Proximity Sensor Part #: 8-758-01 Supplier: 08748 Part #: A27092-139 Supplier: 81205 Part #: A27092-146 Supplier: 81205 Opt Part #: A27092-106 Supplier: 81205
STD-6481	Voltmeter - Digital

D. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right

E. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

F. Prepare for the Test

SUBTASK 27-88-11-860-014

WARNING MAKE SURE PERSONS AND EQUIPMENT ARE CLEAR OF THE LEADING EDGE AND TRAILING EDGE CONTROL SURFACES. THE LEADING EDGE AND TRAILING EDGE CONTROL SURFACES CAN EXTEND AND RETRACT QUICKLY. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

CAUTION MAKE SURE THAT YOU CLOSE OR REMOVE THE INBOARD FAN COWL AND THRUST REVERSERS BEFORE YOU RETRACT THE LEADING EDGE FLAPS. THE CLEARANCE IS NOT SUFFICIENT FOR THE FLAPS TO RETRACT WITH THE INBOARD FAN DUCT COWL AND THE THRUST REVERSERS IN THE OPEN POSITION. THIS CAN CAUSE DAMAGE TO EQUIPMENT.

(1) Make sure that the flaps and slats are in the retracted position.

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(a) If the flaps and slats are not retracted, do this task: Leading Edge Flaps and Slats Retraction, TASK 27-81-00-860-804.

SUBTASK 27-88-11-740-002

(2) Do this task: FSEU Display and Existing Faults Test, TASK 27-51-00-740-801.

SUBTASK 27-88-11-860-041

WARNING

KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(3) Do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 27-88-11-730-003

WARNING

MAKE SURE PERSONS AND EQUIPMENT ARE CLEAR OF THE LEADING EDGE AND TRAILING EDGE CONTROL SURFACES. THE LEADING EDGE AND TRAILING EDGE CONTROL SURFACES CAN EXTEND AND RETRACT QUICKLY. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

CAUTION

MAKE SURE THE INBOARD FAN DUCT COWL AND THE INBOARD AND OUTBOARD THRUST REVERSERS ARE CLOSED OR REMOVED BEFORE YOU EXTEND THE LEADING EDGE FLAPS AND SLATS IN THIS TEST. THERE IS NOT SUFFICIENT CLEARANCE FOR THE FLAPS AND SLATS TO EXTEND IF THE INBOARD FAN DUCT COWL AND THE THRUST REVERSERS ARE IN THE OPEN POSITION. THIS CAN CAUSE DAMAGE TO EQUIPMENT.

(4) Do a test of the leading edge flap and slat proximity sensor:

(a) Push the ON/OFF button on the front panel of the Flap/Slat Electronics Unit (FSEU) to turn on the display.

NOTE: The display will show EXISTING FAULTS?.

(b) Push the down arrow until OTHER FUNCTNS? shows on the display.

(c) Push the YES button to select OTHER FUNCTNS? on the FSEU.

(d) Push the down arrow until I/O MONITOR? shows on the display.

(e) Push the YES button to select I/O MONITOR? from the menu.

(f) Push the YES button to select PROX SENSORS? from the I/O MONITOR? menu.

(g) Push the down arrow to step through the proximity sensor states at the LE flaps and slats UP position.

(h) Make sure that these displays and the proximity sensor states show on the FSEU (Table 501).

SUBTASK 27-88-11-860-011

(5) Remove electrical power from the system for the test.

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(a) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	6	C00566	FLIGHT CONTROL FLAP LOAD RELIEF
A	7	C00210	FLIGHT CONTROL FLAP SHUTOFF VALVES
A	8	C00211	FLIGHT CONTROL FSEU DC 1
A	9	C01468	FLIGHT CONTROL FSEU DC 2
A	11	C01443	FC TE FLAP POS/SKEW SNSR & IND RIGHT
A	12	C00213	FC TE FLAP POS/SKEW SNSR & IND LEFT

SUBTASK 27-88-11-010-006

(6) Open this access door:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 27-88-11-020-008

(7) Do this task to remove the FSEU: Flap/Slat Electronics Unit Removal, TASK 27-51-01-000-801.

(a) Keep the FSEU for installation.

(b) Install blanking caps on the removed FSEU rear Aeronautical Radio Incorporated (ARINC) 629 bus connectors.

SUBTASK 27-88-11-210-001

(8) Examine the ARINC connector pins for damage or corrosion.

(a) Repair as necessary.

G. Measure the Proximity Sensor and Wiring Installation Resistance with the Flaps and Slats in the Retracted Position

SUBTASK 27-88-11-760-001

(1) Refer to the applicable WDM (WDM 27-81-11, WDM 27-81-12, WDM 27-81-21,WDM 27-81-22) for the 3-wire sensor to FSEU ARINC connector pin callout.

SUBTASK 27-88-11-760-048

(2) Connect the LCR meter (Inductance, Capacitance, Resistance), COM-1741 (a digital voltmeter, STD-6481, or digital/analog multimeter, COM-1793, may be used to measure the resistance).

(a) First do the Red to Yellow measurement.

(b) Then the Blue to Yellow measurement.

(c) Alternate connections as needed.

SUBTASK 27-88-11-760-050

(3) Supply power to the LCR meter (Inductance, Capacitance, Resistance), COM-1741.

SUBTASK 27-88-11-760-049

(4) Set the LCR meter (Inductance, Capacitance, Resistance), COM-1741, to measure series resistance (Rs) at 120 Hz.

SUBTASK 27-88-11-700-001

(5) Measure the resistance at the ARINC connector pins on the E1-1 rack for each sensor under test and record in the log sheet (Figure 502).

(a) The “DATA HOLD” button is used to "capture" your current measured result.

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(b) To make a new measurement, the, "DATA HOLD" must be released, by pressing the button once, then the new measurement can be made.

(c) Record the resistance of sensors in the appropriate space on log sheet. Values to be recorded in hundredths (two numbers right of the decimal point), for example XX.XX or XXX.XX.

SUBTASK 27-88-11-760-051

(6) Compare the measured resistance value to Table 505.

Table 504/27-88-11-993-810 GO-GAUGES TABLE

BOEING SENSOR PART NUMBER	CRANE SENSOR PART NUMBER	TARGET NEAR INDUCTANCE (WITH GO-GAUGE)	TARGET FAR INDUCTANCE (WITH GO-GAUGE)	COIL RESISTANCE RED-YELLOW (TARGET FAR)	RESISTANCE BLUE-YELLOW (TARGET FAR)
10-61226-29	1-899-29	38.0 mH minimum	36.0-37.5 mH	27.0-39.0 Ω	344.5-351.5 Ω
10-61226-30	1-899-30	38.0 mH minimum	36.0-37.5 mH	27.0-33.0 Ω	344.5-351.5 Ω
10-61226-45	80-207-01	38.0 mH minimum	36.0-37.5 mH	27.0-33.0 Ω	344.5-351.5 Ω
10-61226-47	80-207-02	38.0 mH minimum	36.0-37.5 mH	27.0-33.0 Ω	344.5-351.5 Ω

SUBTASK 27-88-11-760-053

(7) If the measured resistance value is less than 5.00 ohms, this indicates a short circuit condition.

(a) If there is a short circuit, inspect the associated wiring and connectors for contamination or damage.

SUBTASK 27-88-11-760-054

(8) If the resistance value as measured between the Red and Yellow leads is greater than 100.00 ohms, this indicates a loose contact or wire damage or a defective sensor.

(a) If there is loose contact or wire damage, inspect the associated wiring and connectors for contamination or damage.

SUBTASK 27-88-11-760-055

(9) If the resistance value as measured between the Blue and Yellow leads is greater than 351.50 ohms, this indicates a loose contact or wire damage or a defective sensor.

(a) If there is loose contact or wire damage, inspect the associated wiring and connectors for contamination or damage.

H. Measure the Inductance at the ARINC Connector Pins on the E1-1 Rack for each Sensor Under Test with the Flaps and Slats in the Retracted Position

SUBTASK 27-88-11-760-002

(1) Refer to the applicable WDM (WDM 27-81-11, WDM 27-81-12, WDM 27-81-21,WDM 27-81-22) for the 3-wire sensor to FSEU ARINC connector pin callout.

SUBTASK 27-88-11-760-052

(2) Connect the LCR meter (Inductance, Capacitance, Resistance), COM-1741, for Red to Blue measurement.

SUBTASK 27-88-11-760-056

(3) Supply power to the LCR meter (Inductance, Capacitance, Resistance), COM-1741.

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SUBTASK 27-88-11-760-057

(4) Set the LCR meter (Inductance, Capacitance, Resistance), COM-1741, to read series inductance (Ls) at 1 kHz.

SUBTASK 27-88-11-760-058

(5) Measure the inductance for the targets with the flaps and slats in the retracted position and record in the log sheet for each sensor (Figure 502).

(a) The “DATA HOLD” button is used to "capture" your current measured result.

(b) To make a new measurement, the, "DATA HOLD" must be released, by pressing the button once, then the new measurement can be made.

(c) Record the inductance of sensors in the appropriate space on log sheet. Values to be recorded in hundredths (two numbers right of the decimal point), for example XX.XX or XXX.XX.

(d) If you suspect a faulty sensor, use the applicable, Sensor Test Gauge Kit – Go-Gauge (proximity sensor test set, SPL-1690) to simulate a configuration change of the sensors current position to a target FAR or a target NEAR position.

(e) Compare the measured inductance values to Table 504 only if using GO-GAUGES.

SUBTASK 27-88-11-760-059

(6) Make sure that the inductance value for the sensor under test is within the range of Table 505.

Table 505/27-88-11-993-811 SENSOR'S NOMINAL VALUES

VALUE	SENSOR CONDITION
Three-wire Sensors	
Less than 40 mH	Target FAR
40 mH - 50 mH	Correctly rigged NEAR sensor
Greater than 50 mH	Target may be rigged too near to the sensor
Resistance on aircraft (Red-Yellow resistance plus Blue-Yellow resistance): ≤ 390 Ω	

NOTE: Sensor inductance values will vary between sensor positions because of differences in operation and design requirements. In some condition, the inductance value cannot be achieved. Inductance values outside of the nominal range are permitted if the minimum value of 38.00 mH (Target NEAR) is obtained and no system anomaly is observed.

I. Measure the Inductance at the ARINC Connector Pins on the E1-1 Rack for each Sensor Under Test with the Flaps and Slats in the Extended Position

SUBTASK 27-88-11-410-008

(1) Do this task to install the FSEU: Flap/Slat Electronics Unit Installation, TASK 27-51-01-400-801.

(a) Remove the blanking caps on the FSEU rear ARINC 629 bus connectors.

SUBTASK 27-88-11-860-054

(2) Restore electrical power.

(a) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	6	C00566	FLIGHT CONTROL FLAP LOAD RELIEF

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(Continued)

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	7	C00210	FLIGHT CONTROL FLAP SHUTOFF VALVES
A	8	C00211	FLIGHT CONTROL FSEU DC 1
A	9	C01468	FLIGHT CONTROL FSEU DC 2
A	11	C01443	FC TE FLAP POS/SKEW SNSR & IND RIGHT
A	12	C00213	FC TE FLAP POS/SKEW SNSR & IND LEFT

SUBTASK 27-88-11-860-055

WARNING MAKE SURE PERSONS AND EQUIPMENT ARE CLEAR OF THE LEADING EDGE AND TRAILING EDGE CONTROL SURFACES. THE LEADING EDGE AND TRAILING EDGE CONTROL SURFACES CAN EXTEND AND RETRACT QUICKLY. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

CAUTION MAKE SURE THAT YOU CLOSE OR REMOVE THE INBOARD FAN COWL AND THRUST REVERSERS BEFORE YOU RETRACT THE LEADING EDGE FLAPS. THE CLEARANCE IS NOT SUFFICIENT FOR THE FLAPS TO RETRACT WITH THE INBOARD FAN DUCT COWL AND THE THRUST REVERSERS IN THE OPEN POSITION. THIS CAN CAUSE DAMAGE TO EQUIPMENT.

(3) Move the flaps and slats to the extended position for the test.

(a) Do this task to extend the leading edge flaps: Leading Edge Flaps and Slats Extension, TASK 27-81-00-860-803.

1) Move the flap control lever to detent position 1, 2, 5, 10, 15, and 25.

2) Stop for 2 seconds at each detent position.

SUBTASK 27-88-11-010-010

(4) Do this task to remove the FSEU: Flap/Slat Electronics Unit Removal, TASK 27-51-01-000-801.

(a) Keep the FSEU for installation.

(b) Install blanking caps on the removed FSEU rear ARINC 629 bus connectors.

SUBTASK 27-88-11-860-056

(5) Remove electrical power from the system for the test.

(a) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	6	C00566	FLIGHT CONTROL FLAP LOAD RELIEF
A	7	C00210	FLIGHT CONTROL FLAP SHUTOFF VALVES
A	8	C00211	FLIGHT CONTROL FSEU DC 1
A	9	C01468	FLIGHT CONTROL FSEU DC 2
A	11	C01443	FC TE FLAP POS/SKEW SNSR & IND RIGHT
A	12	C00213	FC TE FLAP POS/SKEW SNSR & IND LEFT

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SUBTASK 27-88-11-760-034

(6) Refer to the applicable WDM (WDM 27-81-11, WDM 27-81-12, WDM 27-81-21,WDM 27-81-22) for the 3-wire sensor to FSEU ARINC connector pin callout.

SUBTASK 27-88-11-760-035

(7) Connect the LCR meter (Inductance, Capacitance, Resistance), COM-1741, for Red to Blue measurement.

SUBTASK 27-88-11-760-036

(8) Supply power to the LCR meter (Inductance, Capacitance, Resistance), COM-1741.

SUBTASK 27-88-11-760-037

(9) Set the LCR meter (Inductance, Capacitance, Resistance), COM-1741, to read series inductance (Ls) at 1 kHz.

SUBTASK 27-88-11-760-038

(10) Measure the inductance for the targets with the flaps and slats in the extended position and record in the log sheet for each sensor (Figure 502).

(a) The “DATA HOLD” button is used to "capture" your current measured result.

(b) To make a new measurement, the, "DATA HOLD" must be released, by pressing the button once, then the new measurement can be made.

(c) Record the inductance of sensors in the appropriate space on log sheet. Values to be recorded in hundredths (two numbers right of the decimal point), for example XX.XX or XXX.XX.

(d) If you suspect a faulty sensor, use the applicable, Sensor Test Gauge Kit – Go-Gauge (proximity sensor test set, SPL-1690) to simulate a configuration change of the sensors current position to a target FAR or a target NEAR position.

(e) Compare the measured inductance value to the range in Table 504 only if using GO-GAUGES.

SUBTASK 27-88-11-760-039

(11) Make sure that the inductance value for the sensor under test is within the range of Table 505.

J. Measure the Inductance at the ARINC Connector Pins on the E1-1 Rack for each Sensor Under Test with the Flaps and Slats in the Full Extended Position

SUBTASK 27-88-11-410-004

(1) Do this task to install the FSEU: Flap/Slat Electronics Unit Installation, TASK 27-51-01-400-801.

(a) Remove the blanking caps on the FSEU rear ARINC 629 bus connectors.

SUBTASK 27-88-11-860-028

(2) Restore electrical power.

(a) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	6	C00566	FLIGHT CONTROL FLAP LOAD RELIEF
A	7	C00210	FLIGHT CONTROL FLAP SHUTOFF VALVES
A	8	C00211	FLIGHT CONTROL FSEU DC 1
A	9	C01468	FLIGHT CONTROL FSEU DC 2
A	11	C01443	FC TE FLAP POS/SKEW SNSR & IND RIGHT

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(Continued)

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	12	C00213	FC TE FLAP POS/SKEW SNSR & IND LEFT

SUBTASK 27-88-11-860-029

WARNING

MAKE SURE PERSONS AND EQUIPMENT ARE CLEAR OF THE LEADING EDGE AND TRAILING EDGE CONTROL SURFACES. THE LEADING EDGE AND TRAILING EDGE CONTROL SURFACES CAN EXTEND AND RETRACT QUICKLY. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

CAUTION

MAKE SURE THAT YOU CLOSE OR REMOVE THE INBOARD FAN COWL AND THRUST REVERSERS BEFORE YOU RETRACT THE LEADING EDGE FLAPS. THE CLEARANCE IS NOT SUFFICIENT FOR THE FLAPS TO RETRACT WITH THE INBOARD FAN DUCT COWL AND THE THRUST REVERSERS IN THE OPEN POSITION. THIS CAN CAUSE DAMAGE TO EQUIPMENT.

(3) Move the flaps and slats to the full extended position for the test.

(a) Do this task to extend the leading edge flaps: Leading Edge Flaps and Slats Extension, TASK 27-81-00-860-803.

1) Stop for 2 seconds at each detent position until detent position 30.

SUBTASK 27-88-11-010-008

(4) Do this task to remove the FSEU: Flap/Slat Electronics Unit Removal, TASK 27-51-01-000-801.

(a) Keep the FSEU for installation.

(b) Install blanking caps on the removed FSEU rear ARINC 629 bus connectors.

SUBTASK 27-88-11-860-030

(5) Remove electrical power from the system for the test.

(a) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	6	C00566	FLIGHT CONTROL FLAP LOAD RELIEF
A	7	C00210	FLIGHT CONTROL FLAP SHUTOFF VALVES
A	8	C00211	FLIGHT CONTROL FSEU DC 1
A	9	C01468	FLIGHT CONTROL FSEU DC 2
A	11	C01443	FC TE FLAP POS/SKEW SNSR & IND RIGHT
A	12	C00213	FC TE FLAP POS/SKEW SNSR & IND LEFT

SUBTASK 27-88-11-760-003

(6) Refer to the applicable WDM (WDM 27-81-11, WDM 27-81-12, WDM 27-81-21,WDM 27-81-22) for the 3-wire sensor to FSEU ARINC connector pin callout.

SUBTASK 27-88-11-760-040

(7) Connect the LCR meter (Inductance, Capacitance, Resistance), COM-1741, for Red to Blue measurement.

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SUBTASK 27-88-11-760-041

(8) Supply power to the LCR meter (Inductance, Capacitance, Resistance), COM-1741.

SUBTASK 27-88-11-760-042

(9) Set the LCR meter (Inductance, Capacitance, Resistance), COM-1741, to read series inductance (Ls) at 1 kHz.

SUBTASK 27-88-11-760-043

(10) Measure the inductance for the targets with the flaps and slats in the full extended position and record in the log sheet for each sensor (Figure 502).

(a) The “DATA HOLD” button is used to "capture" your current measured result.

(b) To make a new measurement, the, "DATA HOLD" must be released, by pressing the button once, then the new measurement can be made.

(c) Record the inductance of sensors in the appropriate space on log sheet. Values to be recorded in hundredths (two numbers right of the decimal point), for example XX.XX or XXX.XX.

(d) If you suspect a faulty sensor, use the applicable, Sensor Test Gauge Kit – Go-Gauge (proximity sensor test set, SPL-1690) to simulate a configuration change of the sensors current position to a target FAR or a target NEAR position.

(e) Compare the measured inductance value to the range in Table 504 only if using GO-GAUGES.

SUBTASK 27-88-11-760-044

(11) Make sure that the inductance value for the sensor under test is within the range of Table 505.

K. Put the Airplane Back to Its Usual Condition

SUBTASK 27-88-11-020-009

(1) Remove the blanking caps on the FSEU rear ARINC 629 bus connectors.

SUBTASK 27-88-11-420-006

(2) Do this task to install the FSEU: Flap/Slat Electronics Unit Installation, TASK 27-51-01-400-801.

SUBTASK 27-88-11-860-013

(3) Restore electrical power.

(a) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	6	C00566	FLIGHT CONTROL FLAP LOAD RELIEF
A	7	C00210	FLIGHT CONTROL FLAP SHUTOFF VALVES
A	8	C00211	FLIGHT CONTROL FSEU DC 1
A	9	C01468	FLIGHT CONTROL FSEU DC 2
A	11	C01443	FC TE FLAP POS/SKEW SNSR & IND RIGHT
A	12	C00213	FC TE FLAP POS/SKEW SNSR & IND LEFT

SUBTASK 27-88-11-730-002

(4) Do this task: Leading Edge Flap and Slat Proximity Sensor Test, TASK 27-88-11-700-801.

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SUBTASK 27-88-11-410-010

(5) Close this access door:

Number	Name/Location
117A	Electronic Equipment Access Door

——— END OF TASK ———

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	DATE:		LOC:			
	A/C:					
	RESISTANCE OHMS (MEASURE AT 120 HZ)			INDUCTANCE mH (MEAS. AT 1 Khz)		
	LE SLATS POSITION	RETRACTED	RETRACTED	RETRACTED	EXTENDED	FULL EXTENDED
SENSOR EQUIP. NO.	SENSOR INSTALLATION	RED-YELLOW (COILS)	BLUE-YELLOW (TC RESISTOR)	RED-BLUE	RED-BLUE	RED-BLUE
S994	LEFT SLAT 1 OB					
S995	LEFT SLAT 1 IB					
S996	LEFT SLAT 2 OB					
S997	LEFT SLAT 2 IB					
S998	LEFT SLAT 3 OB					
S999	LEFT SLAT 3 IB					
S1000	LEFT SLAT 4 OB					
S1001	LEFT SLAT 4 IB					
S1002	RIGHT SLAT 5 OB					
S1003	RIGHT SLAT 5 IB					
S1004	RIGHT SLAT 6 OB					
S1005	RIGHT SLAT 6 IB					
S1006	RIGHT SLAT 7 OB					
S1007	RIGHT SLAT 7 IB					
S1008	RIGHT SLAT 8 OB					
S1009	RIGHT SLAT 8 IB					
	RESISTANCE OHMS (MEASURE AT 120 HZ)			INDUCTANCE mH (MEAS. AT 1 Khz)		
	LE FLAPS POSITION	RETRACTED	RETRACTED	RETRACTED	EXTENDED	FULL EXTENDED
SENSOR EQUIP. NO.	SENSOR INSTALLATION	RED-YELLOW (COILS)	BLUE-YELLOW (TC RESISTOR)	RED-BLUE	RED-BLUE	RED-BLUE
S168	LEFT FLAP 1 RET					
S169	LEFT FLAP 1 EXT					
S170	LEFT FLAP 2 RET					
S171	LEFT FLAP 2 EXT					
S172	RIGHT FLAP 3 RET					
S173	RIGHT FLAP 3 EXT					
S174	RIGHT FLAP 4 RET					
S175	RIGHT FLAP 4 EXT					

2889565 S0000686667_V2

Proximity Sensor Performance Values - Log Sheet
Figure 502/27-88-11-990-805

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TASK 27-88-11-720-801

4. Leading Edge Flap and Slat Proximity Sensor Test Using a Breakout Box and LCR Meter

(Figure 503, Figure 504)

A. General

CAUTION	THE PROXIMITY SENSOR BREAKOUT BOX IS USED ONLY TO EXAMINE BASIC SENSOR HEALTH AND INSTALLATION. DO NOT USE AS A RIGGING TOOL. RIGGING OF SENSORS AND OTHER MAINTENANCE ACTIONS SHOULD BE DONE ONLY IN ACCORDANCE WITH AIRLINE OPERATING AND MAINTENANCE PROCEDURES, APPROVED MAINTENANCE MANUALS, OR DIRECTIVES OF THE APPROPRIATE CERTIFICATION AGENCY.

(1) The test will measure the inductance and resistance of the 3-wire sensors, sensor wires, and connectors. And compare the values to a known range. This test also examines the target to sensor conditions.

(2) This test uses the Crane Sensor Test System made up of the following parts:

(a) Proximity Sensor Breakout Box (BOB) and a LCR Meter.

(b) Sensors Test Gauge Set.

1) The controls on the face of the proximity sensor BOB are used to select the desired sensor for testing.

a) The proximity sensor BOB does not test two-wire sensors. Two-wire sensors, do not need to be tested.

2) The Sensor Test Gauge Kit (proximity sensor test set, SPL-1690) consists of six go-gauges, sensor de-actuator tool and an instruction booklet. Use the go-gauges in place of a sensor, if you suspect the sensor is faulty. Refer to IFIM 27-88 if necessary.

(3) Obey all applicable warnings, cautions, and maintenance practice instructions when necessary to measure the Proximity Sensor in the test.

(4) Use an LCR meter that has a USB port to record the data with a PC automatically. For example 878B or 879B have a USB port. Use the specified instructions included with the LCR meter. Full instructions are included in the BOB's Sensor Test System.

B. References

Reference	Title
27-51-00-740-801	FSEU Display and Existing Faults Test (P/B 501)
27-51-01-000-801	Flap/Slat Electronics Unit Removal (P/B 401)
27-51-01-400-801	Flap/Slat Electronics Unit Installation (P/B 401)
27-81-00-860-803	Leading Edge Flaps and Slats Extension (P/B 201)
27-81-00-860-804	Leading Edge Flaps and Slats Retraction (P/B 201)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

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Reference	Description
COM-1741	Meter - LCR (Inductance, Capacitance, Resistance; Use COM-20443 for OH bleed loop testing) Part #: 878B Supplier: 08098 Part #: 879B Supplier: 08098 Part #: 880 Supplier: 08098 Part #: U1733P Supplier: 1LQK8 Opt Part #: 875B Supplier: 08098 Opt Part #: 878A Supplier: 08098 Opt Part #: 879 Supplier: 08098
COM-1793	Multimeter - Digital/Analog (or equivalent meter meets task requirements) Part #: 117 Supplier: 89536 Part #: 260-8XPI Supplier: 55026 Part #: 287 Supplier: 89536 Part #: 289 Supplier: 89536 Part #: 87V Supplier: 89536 Part #: FLUKE 27 II Supplier: 89536 Part #: FLUKE-77-4 Supplier: 89536 Part #: WA2025 Supplier: 4MJL8 Opt Part #: 187 Supplier: 89536 Opt Part #: 189 Supplier: 89536 Opt Part #: 21 Supplier: 89536 Opt Part #: 27 Supplier: 89536 Opt Part #: 77 SERIES III Supplier: 89536 Opt Part #: 87 Supplier: 89536 Opt Part #: FLUKE 27 Supplier: 89536 Opt Part #: MODEL 27 Supplier: 89536
SPL-1690	Actuators/Deactuators Set - Proximity Sensor Part #: 8-758-01 Supplier: 08748 Part #: A27092-139 Supplier: 81205 Part #: A27092-146 Supplier: 81205 Opt Part #: A27092-106 Supplier: 81205
SPL-14585	Breakout Box - Proximity Sensor Part #: 380-026-02 Supplier: 08748
STD-6481	Voltmeter - Digital

D. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right

E. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

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F. Prepare the Test

SUBTASK 27-88-11-860-015

WARNING MAKE SURE PERSONS AND EQUIPMENT ARE CLEAR OF THE LEADING EDGE AND TRAILING EDGE CONTROL SURFACES. THE LEADING EDGE AND TRAILING EDGE CONTROL SURFACES CAN EXTEND AND RETRACT QUICKLY. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

CAUTION MAKE SURE THAT YOU CLOSE OR REMOVE THE INBOARD FAN COWL AND THRUST REVERSERS BEFORE YOU RETRACT THE LEADING EDGE FLAPS. THE CLEARANCE IS NOT SUFFICIENT FOR THE FLAPS TO RETRACT WITH THE INBOARD FAN DUCT COWL AND THE THRUST REVERSERS IN THE OPEN POSITION. THIS CAN CAUSE DAMAGE TO EQUIPMENT.

(1) Make sure that the flaps and slats are in the retracted position.

(a) If the flaps and slats are not retracted, do this task: Leading Edge Flaps and Slats Retraction, TASK 27-81-00-860-804.

SUBTASK 27-88-11-740-003

(2) Do this task: FSEU Display and Existing Faults Test, TASK 27-51-00-740-801.

SUBTASK 27-88-11-860-042

WARNING KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(3) Do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 27-88-11-730-004

WARNING MAKE SURE PERSONS AND EQUIPMENT ARE CLEAR OF THE LEADING EDGE AND TRAILING EDGE CONTROL SURFACES. THE LEADING EDGE AND TRAILING EDGE CONTROL SURFACES CAN EXTEND AND RETRACT QUICKLY. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

CAUTION MAKE SURE THE INBOARD FAN DUCT COWL AND THE INBOARD AND OUTBOARD THRUST REVERSERS ARE CLOSED OR REMOVED BEFORE YOU EXTEND THE LEADING EDGE FLAPS AND SLATS IN THIS TEST. THERE IS NOT SUFFICIENT CLEARANCE FOR THE FLAPS AND SLATS TO EXTEND IF THE INBOARD FAN DUCT COWL AND THE THRUST REVERSERS ARE IN THE OPEN POSITION. THIS CAN CAUSE DAMAGE TO EQUIPMENT.

(4) Do a test of the leading edge flap and slat proximity sensor:

(a) Push the ON/OFF button on the front panel of the FSEU to turn on the display.

NOTE: The display will show EXISTING FAULTS?.

(b) Push the down arrow until OTHER FUNCTNS? shows on the display.

(c) Push the YES button to select OTHER FUNCTNS? on the FSEU.

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(d) Push the down arrow until I/O MONITOR? shows on the display.

(e) Push the YES button to select I/O MONITOR? from the menu.

(f) Push the YES button to select PROX SENSORS? from the I/O MONITOR? menu.

(g) Push the down arrow to step through the proximity sensor states at the LE flaps and slats UP position.

(h) Make sure that these displays and the proximity sensor states show on the FSEU (Table 501).

SUBTASK 27-88-11-860-016

(5) Remove electrical power from the system for the test.

(a) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	6	C00566	FLIGHT CONTROL FLAP LOAD RELIEF
A	7	C00210	FLIGHT CONTROL FLAP SHUTOFF VALVES
A	8	C00211	FLIGHT CONTROL FSEU DC 1
A	9	C01468	FLIGHT CONTROL FSEU DC 2
A	11	C01443	FC TE FLAP POS/SKEW SNSR & IND RIGHT
A	12	C00213	FC TE FLAP POS/SKEW SNSR & IND LEFT

SUBTASK 27-88-11-010-007

(6) Open this access door:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 27-88-11-020-010

(7) Do this task to remove the FSEU: Flap/Slat Electronics Unit Removal, TASK 27-51-01-000-801.

(a) Keep the FSEU for installation.

(b) Install blanking caps on the removed FSEU rear ARINC 629 bus connectors.

SUBTASK 27-88-11-210-002

(8) Examine the ARINC connector pins for damage or corrosion.

(a) Repair if necessary.

SUBTASK 27-88-11-480-005

(9) Install the proximity sensor breakout box, SPL-14585, in the FSEU location.

(a) Remove the blanking caps from the rear connectors and install the blanking caps on the removed FSEU rear ARINC 629 bus connectors.

NOTE: No calibration or test is necessary for the proximity sensor BOB.

(b) If a fault is suspected in the FSEU, test each pin of the ARINC connector for continuity. and repair as necessary. Set the sensor selector switch to each sensor in its turn. And measure for continuity between the sensor inputs of the ARINC connector and the front panel connectors.

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G. Measure the Proximity Sensor and Wiring Installation Resistance with the Flaps and Slats in the Retracted Position

SUBTASK 27-88-11-480-006

(1) Connect the LCR meter (Inductance, Capacitance, Resistance), COM-1741, leads to the RED and YELLOW jacks on the front of the proximity sensor breakout box, SPL-14585.

(a) A digital voltmeter, STD-6481, or digital/analog multimeter, COM-1793, may be used to measure the resistance.

SUBTASK 27-88-11-860-018

(2) Set the LCR meter (Inductance, Capacitance, Resistance), COM-1741, to measure series resistance (Rs) at 120 Hz.

SUBTASK 27-88-11-860-019

(3) Push the L/C/R button to change to resistance mode.

SUBTASK 27-88-11-860-020

(4) Push and hold the LCR meter (Inductance, Capacitance, Resistance), COM-1741, button for more than one second to change to series mode.

(a) SER is shown in the middle-right side of the LCR meter (Inductance, Capacitance, Resistance), COM-1741, display.

SUBTASK 27-88-11-760-006

(5) Measure the resistance for the targets with the flaps and slats in the retracted position (RED to YELLOW) and record in the log sheet for each sensor (Figure 504).

(a) The “DATA HOLD” button is used to "capture" your current measured result.

(b) To make a new measurement, the, "DATA HOLD" must be released, by pressing the button once, then the new measurement can be made.

(c) Record the resistance of sensors in the appropriate space on log sheet. Values to be recorded in hundredths (two numbers right of the decimal point), for example XX.XX or XXX.XX.

SUBTASK 27-88-11-200-002

(6) Compare the measured resistance values to Table 508.

SUBTASK 27-88-11-860-021

(7) Use the proximity sensor breakout box, SPL-14585, controls to set the next sensor. And again measure the resistance (RED to YELLOW) and record for each of the remaining sensors.

SUBTASK 27-88-11-480-007

(8) Connect the LCR meter (Inductance, Capacitance, Resistance), COM-1741, leads to the BLUE and YELLOW jacks on the front of the proximity sensor breakout box, SPL-14585.

(a) A digital voltmeter, STD-6481, or digital/analog multimeter, COM-1793, may be used to measure the resistance.

SUBTASK 27-88-11-760-007

(9) Measure the resistance for the targets with the flaps and slats in retracted position (BLUE to YELLOW) and record in the log sheet for each sensor (Figure 504).

(a) The “DATA HOLD” button is used to "capture" your current measured result.

(b) To make a new measurement, the, "DATA HOLD" must be released, by pressing the button once, then the new measurement can be made.

(c) Record the resistance of sensors in the appropriate space on log sheet. Values to be recorded in hundredths (two numbers right of the decimal point), for example XX.XX or XXX.XX.

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SUBTASK 27-88-11-210-003

(10) Compare the measured resistance values to Table 508.

SUBTASK 27-88-11-860-032

(11) Use the proximity sensor breakout box, SPL-14585, controls to set the next sensor. And again measure the resistance (BLUE to YELLOW) and record for each of the remaining sensors.

H. Measure the Inductance (RED to BLUE) for each Sensor Under Test with the Flaps and Slats in the Retracted Position

SUBTASK 27-88-11-420-007

(1) Connect the LCR meter (Inductance, Capacitance, Resistance), COM-1741, leads to the RED and BLUE jacks on the front of the proximity sensor breakout box, SPL-14585.

SUBTASK 27-88-11-760-004

(2) Supply power to the LCR meter (Inductance, Capacitance, Resistance), COM-1741.

SUBTASK 27-88-11-720-001

(3) Use Table 506 to set the BOB controls to the appropriate sensor for test.

Table 506/27-88-11-993-806 PROXIMITY SENSOR BREAKOUT BOX (BOB)

SENSOR EQUIP. NO.	PROX SENSOR BOB SETTING	P/N 380-026-02
	SW POSITION	FUNCTION
S1008	1	LE SLAT 8 OUTBD
S994	2	LE SLAT 1 OUTBD
S1009	3	LE SLAT 8 INBD
S995	4	LE SLAT 1 INBD
S1006	5	LE SLAT 7 OUTBD
S996	6	LE SLAT 2 OUTBD
S1007	7	LE SLAT 7 INBD
S997	8	LE SLAT 2 INBD
S1004	9	LE SLAT 6 OUTBD
S998	10	LE SLAT 3 OUTBD
S1005	11	LE SLAT 6 INBD
S999	12	LE SLAT 3 INBD
S1002	13	LE SLAT 5 OUTBD
S1000	14	LE SLAT 4 OUTBD
S1003	15	LE SLAT 5 INBD
S1001	16	LE SLAT 4 INBD
S174	17	LE FLAP 4 RETRACT
S168	18	LE FLAP 1 RETRACT
S175	19	LE FLAP 4 EXTEND
S169	20	LE FLAP 1 EXTEND
S172	21	LE FLAP 3 RETRACT

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Table 506/27-88-11-993-806 PROXIMITY SENSOR BREAKOUT BOX (BOB) (Continued)

SENSOR EQUIP. NO.	PROX SENSOR BOB SETTING	P/N 380-026-02
S170	22	LE FLAP 2 RETRACT
S173	23	LE FLAP 3 EXTEND
S171	24	LE FLAP 2 EXTEND

SUBTASK 27-88-11-860-017

(4) Set the LCR meter (Inductance, Capacitance, Resistance), COM-1741, to read series inductance (Ls) at 1 kHz.

SUBTASK 27-88-11-760-005

(5) Measure the inductance for the targets with the flaps and slats in the retracted position (RED to BLUE) and record in the log sheet for each sensor (Figure 504).

(a) The “DATA HOLD” button is used to "capture" your current measured result.

(b) To make a new measurement, the "DATA HOLD" must be released, by pressing the button once, then the new measurement can be made.

(c) Record the inductance of sensors in the appropriate space on log sheet. Values to be recorded in hundredths (two numbers right of the decimal point), for example XX.XX or XXX.XX.

(d) If you suspect a faulty sensor, use the applicable, Sensor Test Gauge Kit – Go-Gauge (proximity sensor test set, SPL-1690) to simulate a configuration change of the sensors current position to a target FAR or a target NEAR position.

SUBTASK 27-88-11-760-060

(6) Compare the measured inductance values to Table 507 only if you are using the GO-GAUGE.

Table 507/27-88-11-993-809 GO-GAUGES TABLE

BOEING SENSOR PART NUMBER	CRANE SENSOR PART NUMBER	TARGET NEAR INDUCTANCE (GO SIDE) (WITH GO-GAUGE)	TARGET FAR INDUCTANCE (NO GO SIDE) (WITH GO-GAUGE)	COIL RESISTANCE RED-YELLOW (TARGET FAR)	RESISTANCE BLUE-YELLOW (TARGET FAR)
10-61226-29	1-899-29	38.0 mH minimum	36.0-37.5 mH	27.0-39.0 Ω	344.5-351.5 Ω
10-61226-30	1-899-30	38.0 mH minimum	36.0-37.5 mH	27.0-33.0 Ω	344.5-351.5 Ω
10-61226-45	80-207-01	38.0 mH minimum	36.0-37.5 mH	27.0-33.0 Ω	344.5-351.5 Ω
10-61226-47	80-207-02	38.0 mH minimum	36.0-37.5 mH	27.0-33.0 Ω	344.5-351.5 Ω

SUBTASK 27-88-11-200-001

(7) Make sure that the inductance value for the sensor under test is within the range of Table 508.

Table 508/27-88-11-993-807 SENSOR'S NOMINAL VALUES

VALUE	SENSOR CONDITION
Three-wire Sensors	
35 mH*[1]	Target FAR
40 mH - 50 mH	Correctly rigged NEAR sensor
Greater than 50 mH	Target may be rigged too near to the sensor
Resistance on aircraft (Red-Yellow resistance plus Blue-Yellow resistance): ≤ 390 Ω	

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*[1] There is no range (tolerance) for Target FAR. The nominal 35 mH value for Target FAR is left without tolerance because many operator's fleet will behave differently based on a variety of conditions such as flight cycles, operating climate, age, etc.

SUBTASK 27-88-11-860-022

(8) Use the proximity sensor breakout box, SPL-14585, controls to set the next sensor. And again measure the inductance (RED to BLUE) and record for each of the remaining sensors.

I. Measure the Inductance (RED to BLUE) for each Sensor Under Test with the Flaps and Slats in the Extended Position

SUBTASK 27-88-11-080-010

(1) Disconnect and remove the LCR meter (Inductance, Capacitance, Resistance), COM-1741, from the proximity sensor breakout box, SPL-14585.

SUBTASK 27-88-11-080-011

(2) Remove the proximity sensor breakout box, SPL-14585, from the FSEU location.

SUBTASK 27-88-11-410-009

(3) Do this task to install the Flap/Slat Electronics Unit (FSEU): Flap/Slat Electronics Unit Installation, TASK 27-51-01-400-801.

(a) Remove the blanking caps on the FSEU rear ARINC 629 bus connectors.

SUBTASK 27-88-11-860-057

(4) Restore electrical power.

(a) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	6	C00566	FLIGHT CONTROL FLAP LOAD RELIEF
A	7	C00210	FLIGHT CONTROL FLAP SHUTOFF VALVES
A	8	C00211	FLIGHT CONTROL FSEU DC 1
A	9	C01468	FLIGHT CONTROL FSEU DC 2
A	11	C01443	FC TE FLAP POS/SKEW SNSR & IND RIGHT
A	12	C00213	FC TE FLAP POS/SKEW SNSR & IND LEFT

SUBTASK 27-88-11-860-058

WARNING MAKE SURE PERSONS AND EQUIPMENT ARE CLEAR OF THE LEADING EDGE AND TRAILING EDGE CONTROL SURFACES. THE LEADING EDGE AND TRAILING EDGE CONTROL SURFACES CAN EXTEND AND RETRACT QUICKLY. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

CAUTION MAKE SURE THAT YOU CLOSE OR REMOVE THE INBOARD FAN COWL AND THRUST REVERSERS BEFORE YOU RETRACT THE LEADING EDGE FLAPS. THE CLEARANCE IS NOT SUFFICIENT FOR THE FLAPS TO RETRACT WITH THE INBOARD FAN DUCT COWL AND THE THRUST REVERSERS IN THE OPEN POSITION. THIS CAN CAUSE DAMAGE TO EQUIPMENT.

(5) Move the flaps and slats to the extended position for the test.

(a) Do this task to extend the leading edge flaps: Leading Edge Flaps and Slats Extension, TASK 27-81-00-860-803.

1) Move the flap control lever to detent position 1, 2, 5, 10, 15, and 25.

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2) Stop for 2 seconds at each detent position.

SUBTASK 27-88-11-020-013

(6) Do this task to remove the FSEU: Flap/Slat Electronics Unit Removal, TASK 27-51-01-000-801.

(a) Keep the FSEU for installation.

(b) Install blanking caps on the removed FSEU rear ARINC 629 bus connectors.

SUBTASK 27-88-11-860-059

(7) Remove electrical power from the system for the test.

(a) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	6	C00566	FLIGHT CONTROL FLAP LOAD RELIEF
A	7	C00210	FLIGHT CONTROL FLAP SHUTOFF VALVES
A	8	C00211	FLIGHT CONTROL FSEU DC 1
A	9	C01468	FLIGHT CONTROL FSEU DC 2
A	11	C01443	FC TE FLAP POS/SKEW SNSR & IND RIGHT
A	12	C00213	FC TE FLAP POS/SKEW SNSR & IND LEFT

SUBTASK 27-88-11-480-012

(8) Install the proximity sensor breakout box, SPL-14585, in the FSEU location.

SUBTASK 27-88-11-480-013

(9) Connect the LCR meter (Inductance, Capacitance, Resistance), COM-1741, leads to the RED and BLUE jacks on the front of the proximity sensor breakout box, SPL-14585.

SUBTASK 27-88-11-860-060

(10) Supply power to the LCR meter (Inductance, Capacitance, Resistance), COM-1741.

SUBTASK 27-88-11-860-061

(11) Use Table 506 to set the BOB controls to the appropriate sensor for test.

SUBTASK 27-88-11-860-062

(12) Set the LCR meter (Inductance, Capacitance, Resistance), COM-1741, to read series inductance (Ls) at 1 kHz.

SUBTASK 27-88-11-760-045

(13) Measure the inductance for the targets with the flaps and slats in the extended position (RED to BLUE) and record in the log sheet for each of the sensors (Figure 504).

(a) The “DATA HOLD” button is used to "capture" your current measured result.

(b) To make a new measurement, the, "DATA HOLD" must be released, by pressing the button once, then the new measurement can be made.

(c) Record the inductance of sensors in the appropriate space on log sheet. Values to be recorded in hundredths (two numbers right of the decimal point), for example XX.XX or XXX.XX.

(d) If you suspect a faulty sensor, use the applicable, Sensor Test Gauge Kit – Go-Gauge (proximity sensor test set, SPL-1690) to simulate a configuration change of the sensors current position to a target FAR or a target NEAR position.

SUBTASK 27-88-11-760-046

(14) Compare the measured inductance value to the range in Table 508 if using the Go-Gauge.

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SUBTASK 27-88-11-860-063

(15) Make sure that the inductance value for the sensor under test is within the range of Table 508.

SUBTASK 27-88-11-860-064

(16) Use the proximity sensor breakout box, SPL-14585, controls to set the next sensor.

(a) Again measure the inductance (RED to BLUE) and record for each of the remaining sensors.

J. Measure the Inductance (RED to BLUE) for each Sensor Under Test with the Flaps and Slats in the Full Extended Position

SUBTASK 27-88-11-080-005

(1) Disconnect and remove the LCR meter (Inductance, Capacitance, Resistance), COM-1741, from the proximity sensor breakout box, SPL-14585.

SUBTASK 27-88-11-080-006

(2) Remove the proximity sensor breakout box, SPL-14585, from the FSEU location.

SUBTASK 27-88-11-410-005

(3) Do this task to install the FSEU: Flap/Slat Electronics Unit Installation, TASK 27-51-01-400-801.

(a) Remove the blanking caps on the FSEU rear ARINC 629 bus connectors.

SUBTASK 27-88-11-860-033

(4) Restore electrical power.

(a) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	6	C00566	FLIGHT CONTROL FLAP LOAD RELIEF
A	7	C00210	FLIGHT CONTROL FLAP SHUTOFF VALVES
A	8	C00211	FLIGHT CONTROL FSEU DC 1
A	9	C01468	FLIGHT CONTROL FSEU DC 2
A	11	C01443	FC TE FLAP POS/SKEW SNSR & IND RIGHT
A	12	C00213	FC TE FLAP POS/SKEW SNSR & IND LEFT

SUBTASK 27-88-11-860-034

WARNING MAKE SURE PERSONS AND EQUIPMENT ARE CLEAR OF THE LEADING EDGE AND TRAILING EDGE CONTROL SURFACES. THE LEADING EDGE AND TRAILING EDGE CONTROL SURFACES CAN EXTEND AND RETRACT QUICKLY. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

CAUTION MAKE SURE THAT YOU CLOSE OR REMOVE THE INBOARD FAN COWL AND THRUST REVERSERS BEFORE YOU RETRACT THE LEADING EDGE FLAPS. THE CLEARANCE IS NOT SUFFICIENT FOR THE FLAPS TO RETRACT WITH THE INBOARD FAN DUCT COWL AND THE THRUST REVERSERS IN THE OPEN POSITION. THIS CAN CAUSE DAMAGE TO EQUIPMENT.

(5) Move the flaps and slats to the full extended position for the test.

(a) Do this task to extend the leading edge flaps: Leading Edge Flaps and Slats Extension, TASK 27-81-00-860-803.

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1) Stop for 2 seconds at each detent position until detent position 30.

SUBTASK 27-88-11-020-011

(6) Do this task to remove the FSEU: Flap/Slat Electronics Unit Removal, TASK 27-51-01-000-801.

(a) Keep the FSEU for installation.

(b) Install blanking caps on the removed FSEU rear ARINC 629 bus connectors.

SUBTASK 27-88-11-860-035

(7) Remove electrical power from the system for the test.

(a) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	6	C00566	FLIGHT CONTROL FLAP LOAD RELIEF
A	7	C00210	FLIGHT CONTROL FLAP SHUTOFF VALVES
A	8	C00211	FLIGHT CONTROL FSEU DC 1
A	9	C01468	FLIGHT CONTROL FSEU DC 2
A	11	C01443	FC TE FLAP POS/SKEW SNSR & IND RIGHT
A	12	C00213	FC TE FLAP POS/SKEW SNSR & IND LEFT

SUBTASK 27-88-11-480-008

(8) Install the proximity sensor breakout box, SPL-14585, in the FSEU location.

SUBTASK 27-88-11-480-009

(9) Connect the LCR meter (Inductance, Capacitance, Resistance), COM-1741, leads to the RED and BLUE jacks on the front of the proximity sensor breakout box, SPL-14585.

SUBTASK 27-88-11-860-036

(10) Supply power to the LCR meter (Inductance, Capacitance, Resistance), COM-1741.

SUBTASK 27-88-11-860-037

(11) Use Table 506 to set the BOB controls to the appropriate sensor for test.

SUBTASK 27-88-11-860-038

(12) Set the LCR meter (Inductance, Capacitance, Resistance), COM-1741, to read series inductance (Ls) at 1 kHz.

SUBTASK 27-88-11-760-008

(13) Measure the inductance for the targets with the flaps and slats in the full extended position (RED to BLUE) and record in the log sheet for each of the sensors (Figure 504).

(a) The “DATA HOLD” button is used to "capture" your current measured result.

(b) To make a new measurement, the, "DATA HOLD" must be released, by pressing the button once, then the new measurement can be made.

(c) Record the inductance of sensors in the appropriate space on log sheet. Values to be recorded in hundredths (two numbers right of the decimal point), for example XX.XX or XXX.XX.

(d) If you suspect a faulty sensor, use the applicable, Sensor Test Gauge Kit – Go-Gauge (proximity sensor test set, SPL-1690) to simulate a configuration change of the sensors current position to a target FAR or a target NEAR position.

SUBTASK 27-88-11-760-047

(14) Compare the measured inductance value to the range in Table 508 if using the Go-Gauge.

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SUBTASK 27-88-11-200-003

(15) Make sure that the inductance value for the sensor under test is within the range of Table 508.

SUBTASK 27-88-11-860-039

(16) Use the proximity sensor breakout box, SPL-14585, controls to set the next sensor.

(a) Again measure the inductance (RED to BLUE) and record for each of the remaining sensors.

K. Put the Airplane Back to Its Usual Condition

SUBTASK 27-88-11-080-007

(1) Disconnect and remove the LCR meter (Inductance, Capacitance, Resistance), COM-1741, from the proximity sensor breakout box, SPL-14585.

SUBTASK 27-88-11-860-024

(2) Remove the proximity sensor breakout box, SPL-14585, from the FSEU location.

SUBTASK 27-88-11-860-027

(3) Remove the blanking caps on the FSEU rear ARINC 629 bus connectors.

SUBTASK 27-88-11-400-002

(4) Do this task to install the FSEU: Flap/Slat Electronics Unit Installation, TASK 27-51-01-400-801.

SUBTASK 27-88-11-860-026

(5) Restore electrical power.

(a) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	6	C00566	FLIGHT CONTROL FLAP LOAD RELIEF
A	7	C00210	FLIGHT CONTROL FLAP SHUTOFF VALVES
A	8	C00211	FLIGHT CONTROL FSEU DC 1
A	9	C01468	FLIGHT CONTROL FSEU DC 2
A	11	C01443	FC TE FLAP POS/SKEW SNSR & IND RIGHT
A	12	C00213	FC TE FLAP POS/SKEW SNSR & IND LEFT

SUBTASK 27-88-11-720-002

(6) Do this task: Leading Edge Flap and Slat Proximity Sensor Test, TASK 27-88-11-700-801.

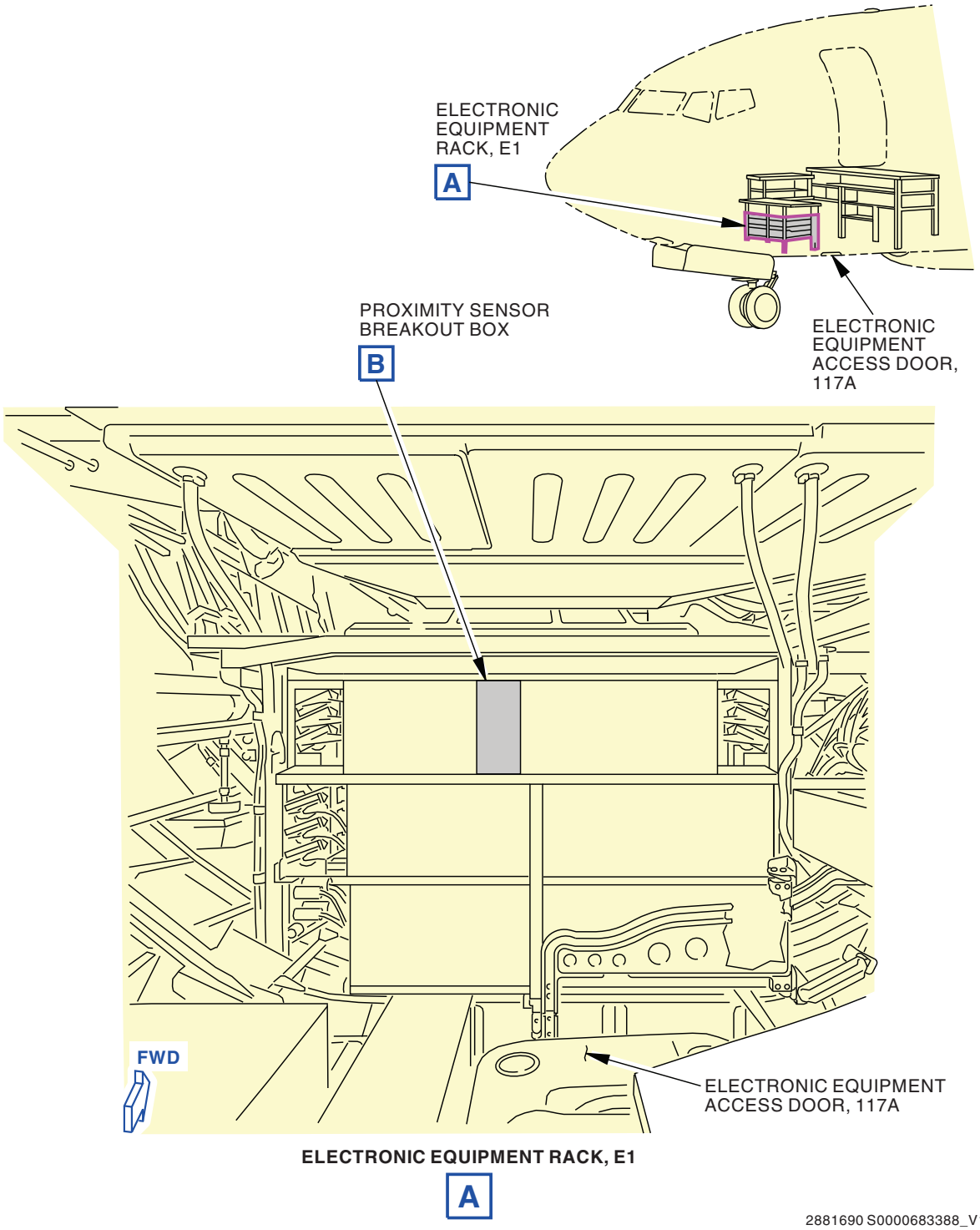
SUBTASK 27-88-11-410-011

(7) Close this access door:

Number	Name/Location
117A	Electronic Equipment Access Door

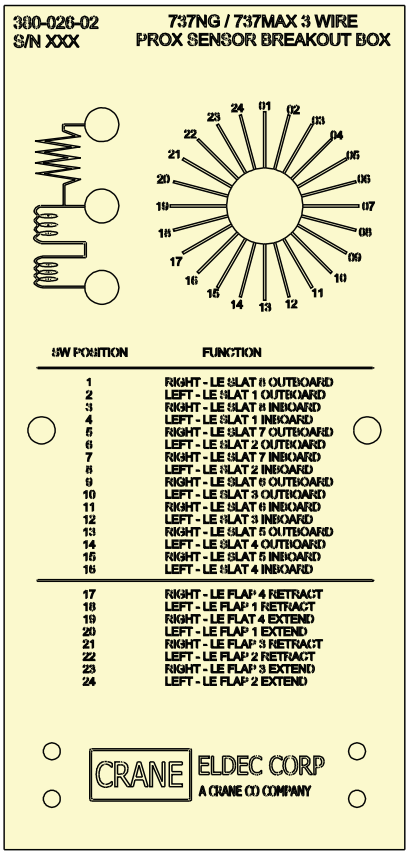
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Proximity Sensor Breakout Box
Figure 503/27-88-11-990-804 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



PROXIMITY SENSOR BREAKOUT BOX

B

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Proximity Sensor Breakout Box
Figure 503/27-88-11-990-804 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

		DATE:		LOC:			
		A/C:					
		RESISTANCE OHMS (MEASURE AT 120 HZ)			INDUCTANCE mH (MEAS. AT 1 Khz)		
		LE SLATS POSITION	RETRACTED	RETRACTED	RETRACTED	EXTENDED	FULL EXTENDED
SENSOR EQUIP. NO.	BREAKOUT BOX SWITCH POSITION	SENSOR INSTALLATION	RED-YELLOW (COILS)	BLUE-YELLOW (TC RESISTOR)	RED-BLUE	RED-BLUE	RED-BLUE
S1008	1	RIGHT SLAT 8 OB					
S994	2	LEFT SLAT 1 OB					
S1009	3	RIGHT SLAT 8 IB					
S995	4	LEFT SLAT 1 IB					
S1006	5	RIGHT SLAT 7 OB					
S996	6	LEFT SLAT 2 OB					
S1007	7	RIGHT SLAT 7 IB					
S997	8	LEFT SLAT 2 IB					
S1004	9	RIGHT SLAT 6 OB					
S998	10	LEFT SLAT 3 OB					
S1005	11	RIGHT SLAT 6 IB					
S999	12	LEFT SLAT 3 IB					
S1002	13	RIGHT SLAT 5 OB					
S1000	14	LEFT SLAT 4 OB					
S1003	15	RIGHT SLAT 5 IB					
S1001	16	LEFT SLAT 4 IB					
		RESISTANCE OHMS (MEASURE AT 120 HZ)			INDUCTANCE mH (MEAS. AT 1 Khz)		
		LE FLAPS POSITION	RETRACTED	RETRACTED	RETRACTED	EXTENDED	FULL EXTENDED
SENSOR EQUIP. NO.	BREAKOUT BOX SWITCH POSITION	SENSOR INSTALLATION	RED-YELLOW (COILS)	BLUE-YELLOW (TC RESISTOR)	RED-BLUE	RED-BLUE	RED-BLUE
S174	17	RIGHT FLAP 4 RET					
S168	18	LEFT FLAP 1 RET					
S175	19	RIGHT FLAP 4 EXT					
S169	20	LEFT FLAP 1 EXT					
S172	21	RIGHT FLAP 3 RET					
S170	22	LEFT FLAP 2 RET					
S173	23	RIGHT FLAP 3 EXT					
S171	24	LEFT FLAP 2 EXT					

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Proximity Sensor Performance Values - Log Sheet
Figure 504/27-88-11-990-806

ECCN 9E991 BOEING PROPRIETARY - See title page for details

P5-12 OVERHEAD LEADING EDGE DEVICE ANNUNCIATOR PANEL - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Removal of the P5-12 Overhead Leading Edge Device Annunciator Panel.

(2) Installation of the P5-12 Overhead Leading Edge Device Annunciator Panel.

TASK 27-88-21-000-801

2. P5-12 Overhead Leading Edge Device Annunciator Panel Removal

(Figure 401)

A. General

(1) This task includes the steps to remove the P5-12 Annunciator Panel.

B. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Prepare for the Removal

SUBTASK 27-88-21-860-001

(1) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
E	7	C01208	STALL WARN ASYM MODE

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	8	C00211	FLIGHT CONTROL FSEU DC 1
A	9	C01468	FLIGHT CONTROL FSEU DC 2

D. P5-12 Annunciator Panel Removal

SUBTASK 27-88-21-020-001

(1) Do these steps to remove the P5-12 annunciator panel [1]:

(a) Release the fasteners [2] on the front of the P5-12 annunciator panel [1].

CAUTION

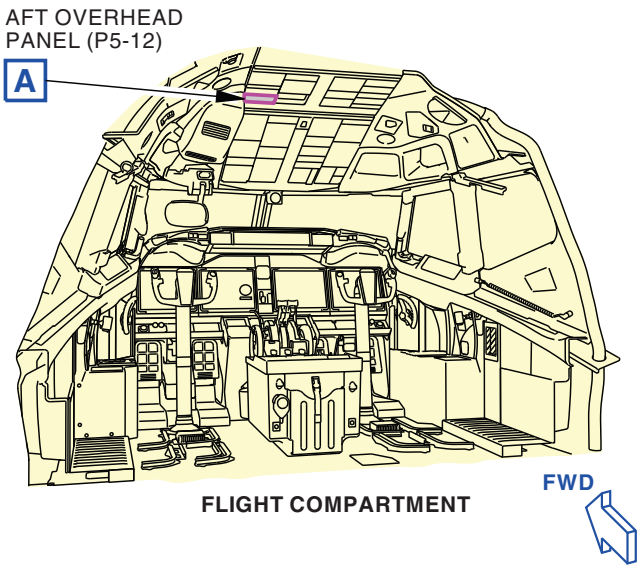
DO NOT APPLY TOO MUCH FORCE. TOO MUCH FORCE WILL CAUSE DAMAGE TO EQUIPMENT.

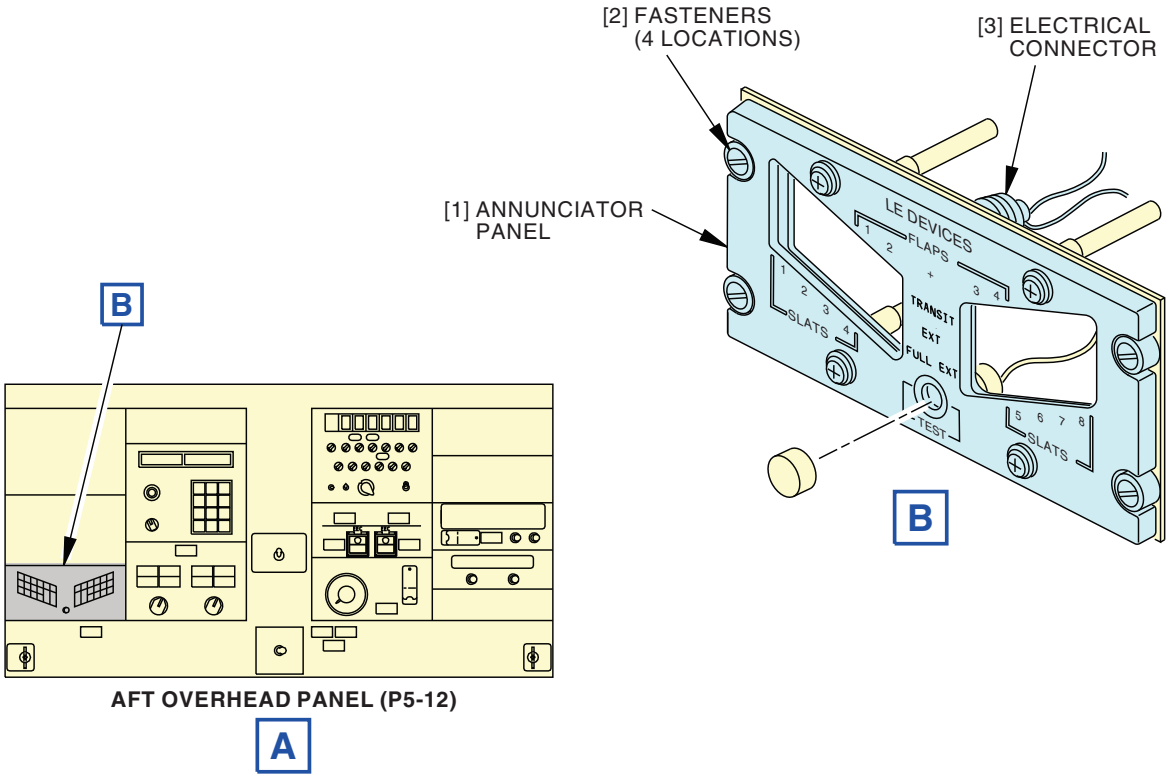
(b) Carefully lower the P5-12 annunciator panel [1] until you can get to the electrical connector [3].

(c) Disconnect the electrical connector [3] from the rear of the panel.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details





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Leading Edge Device Annunciator Panel Installation
Figure 401/27-88-21-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-88-21-400-801

3. P5-12 Overhead Leading Edge Device Annunciator Panel Installation

Figure 401

A. General

(1) This task includes the steps to install the P5-12 Annunciator Panel.

B. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

C. P5-12 Annunciator Panel Installation

SUBTASK 27-88-21-420-001

(1) Do these steps to Install the P5-12 annunciator panel [1]:

(a) Connect the electrical connector [3] to the rear of the P5-12 annunciator panel [1].

CAUTION

DO NOT APPLY TOO MUCH FORCE. TOO MUCH FORCE WILL CAUSE DAMAGE TO EQUIPMENT.

(b) Carefully install the P5-12 annunciator panel [1] into the pilot's P5 Overhead Panel.

(c) Lock the fasteners [2] on the front of the P5-12 annunciator panel [1].

D. P5-12 Annunciator Panel Installation Test

SUBTASK 27-88-21-860-002

(1) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
E	7	C01208	STALL WARN ASYM MODE

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	8	C00211	FLIGHT CONTROL FSEU DC 1
A	9	C01468	FLIGHT CONTROL FSEU DC 2

SUBTASK 27-88-21-710-001

(2) Push test button on P5-12 annunciator panel [1].

(a) Make sure all indicator lights are on.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details